

Cisco Certified Network Associate (CCNA)

**Implement, diagnose and
verify**

IPv4/IPv6 enterprise networks

5 days · 40 hours

38 training + 2 assessment

Current exam v1.1 · announced v2.0 transition

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W · Version 7.0 · 13 September 2026



CCNA

Digital Attendance (Mandatory)

Digital attendance

Use the QR code or attendance link issued on the LMS. Digital attendance and TRAQOM feedback are separate requirements.

TRAQOM feedback

TRAQOM measures training quality and outcomes through feedback surveys. Complete the survey using the provider-issued link.

Assessment attendance

Complete assessment digital attendance at the scheduled assessment block.

About the Trainer · general template

Trainer name

[Trainer name]
[Role and organisation]

Technical experience

[Cisco credentials]
[Routing/switching projects]
[Operational experience]

Delivery / support

[Contact details]
[Availability]
[Class support arrangements]

About the Trainer · named example

Dr Alfred Ang

Named trainer example
Current delivery assignment is confirmed separately by the provider.

Example profile

Demonstrations, guided technical analysis and supervised activities.
Provider confirms credentials and assignment.

Contact / support

enquiry@tertiaryinfotech.com
+65 6100 0613
LMS: lms-tms.tertiaryinfotech.com

Learner Introduction

Your environment

Name, role and network responsibilities.

Example equipment/platform used.

Your diagnostic evidence

Describe a symptom and a command/capture that helped locate it.

Your learning target

Identify one addressing, switching, routing or security task you need to verify.

Ground Rules

Isolated practice

Use synthetic addresses and credentials. Never paste live secrets into AI prompts or public artifacts.

Evidence first

Record commands, outputs and target identity; distinguish simulated evidence from hardware execution.

Collaborative class

Ask questions, respect equipment and peers, and keep activity files organized for review.

Course Outline / Lesson Plan

Days 1-2

Network Fundamentals and
Network Access

Addressing, physical links,
VLANs, trunks, STP and WLANs.

Days 3-4

IP Connectivity, IP Services and
Security

Static/OSPF, NAT, DHCP/DNS,
ACLs and access controls.

Day 5

Automation, integrated verification
and announced v2.0 transition
Written and practical assessment.

Learning Outcomes · TSC ICT-DIT-3009-1.1

A1-A2 requirements

Analyze requirements and select interdependent components using capacity, addressing and path evidence.

A3-A4 implementation

Implement devices and access controls; interpret observable configuration and network state.

A5-A6 operations

Select management tools; report stability and remediation from verified operational evidence.

Briefing for Assessment

Written Assessment

1 hour, open book; knowledge questions aligned to K1-K6. Use current course guide and supplied references.

Practical Performance

1 hour, open book; device/network implementation and verification aligned to A1-A6.

Evidence / submission

Assessment is separate from learning activities and external CCNA practice. Submit using trainer-issued LMS instructions.

Assessment & Funding

Attendance gate

Minimum attendance requirement and assessment eligibility are checked against the current programme requirements.

Assessment criteria

Competency is judged using the approved WSQ assessment criteria; Cisco exam passing rules are separate.

Funding terms

Refer to current official course page and provider eligibility terms. Funding depends on learner eligibility.

LMS Courseware and Assessment

Current courseware

Use the course LMS entry for slides, guide and activities. Verify title and TGS code before downloading.

Private assessments

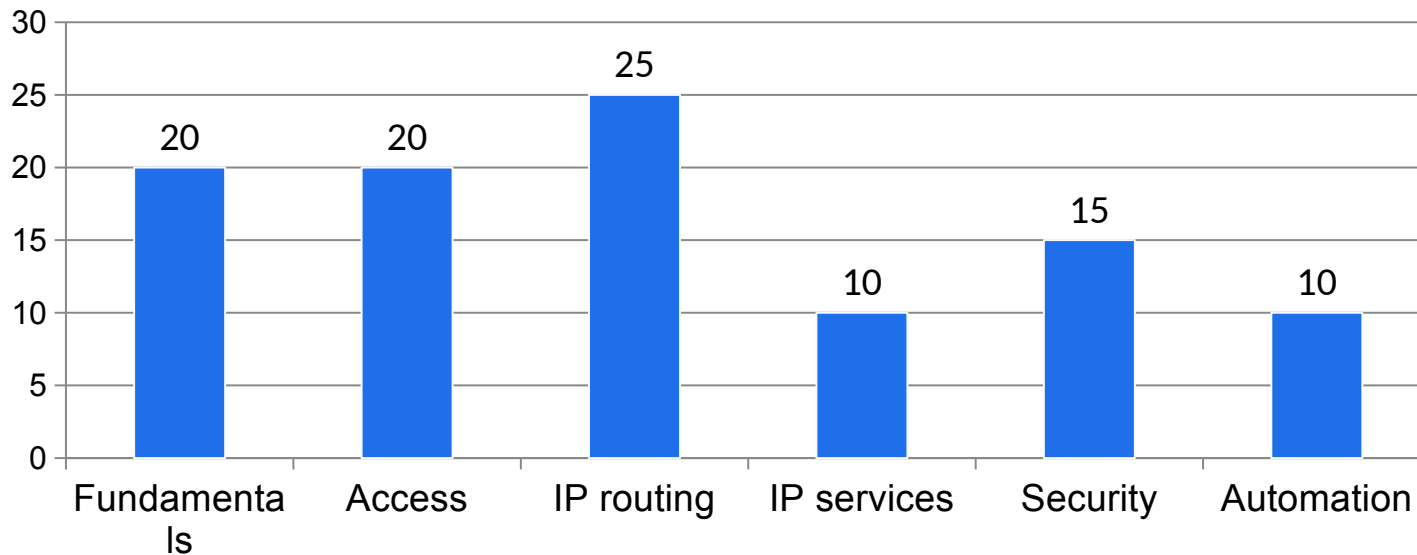
Candidate assessment papers and submission links are trainer-issued. Answer keys remain trainer-only.

Support

<https://lms-tms.tertiaryinfotech.com/>
Report mismatched course links to the trainer.

Current CCNA v1.1 domain model

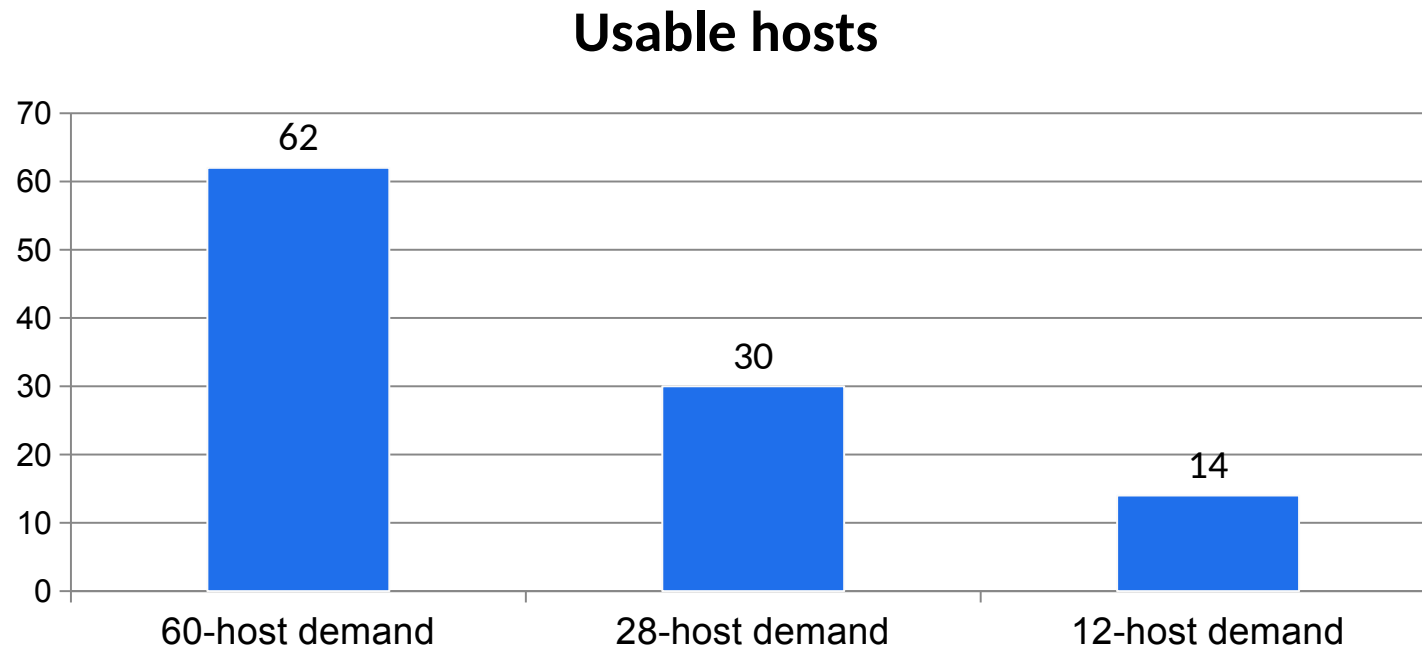
Exam weighting (%)



Network Fundamentals 20%
Network Access 20%
IP Connectivity 25%
IP Services 10%
Security Fundamentals 15%
Automation and
Programmability 10%
Current v1.1 until 2 Feb 2027.

What the data shows: Network Fundamentals 20%

VLSM allocation: demand versus capacity



Usable capacity: 62, 30, 14

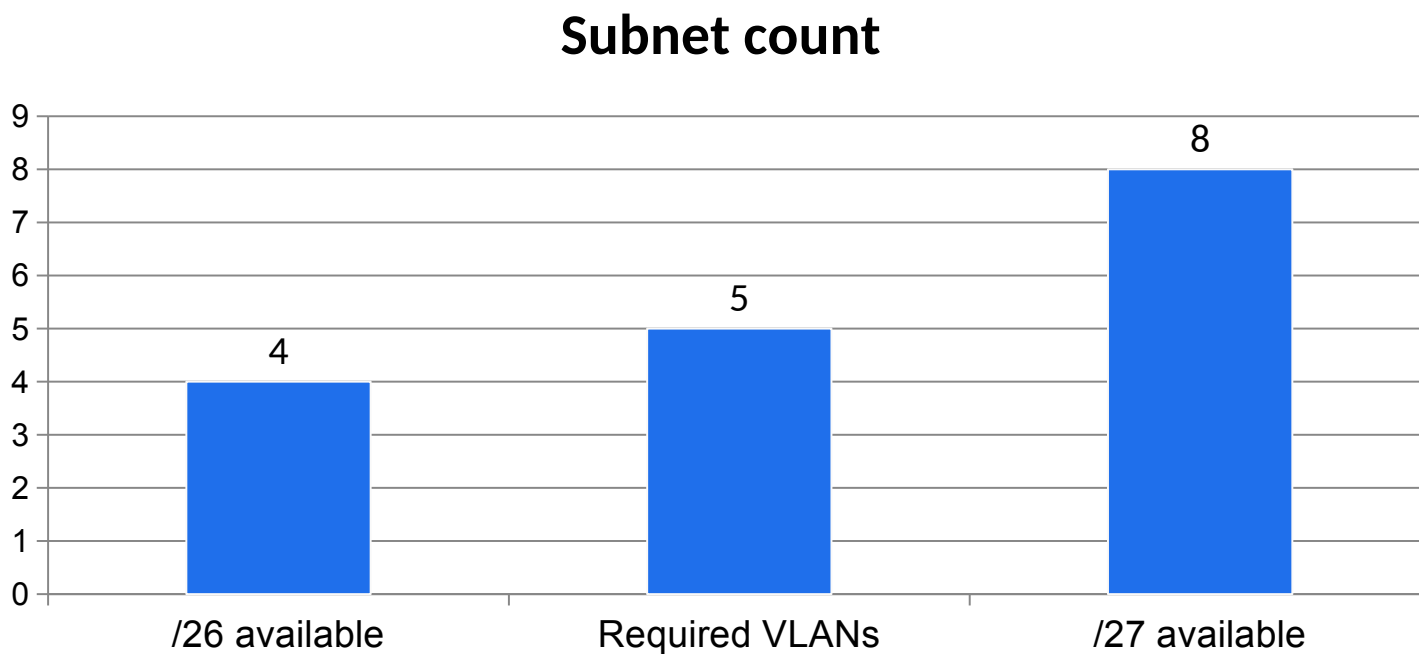
Prefixes: /26, /27, /28

Allocate aligned blocks largest first.

No overlap is permitted.

What the data shows: Usable capacity: 62, 30, 14

Five VLANs: /26 versus /27



A /24 borrowed 2 bits gives 4 subnets

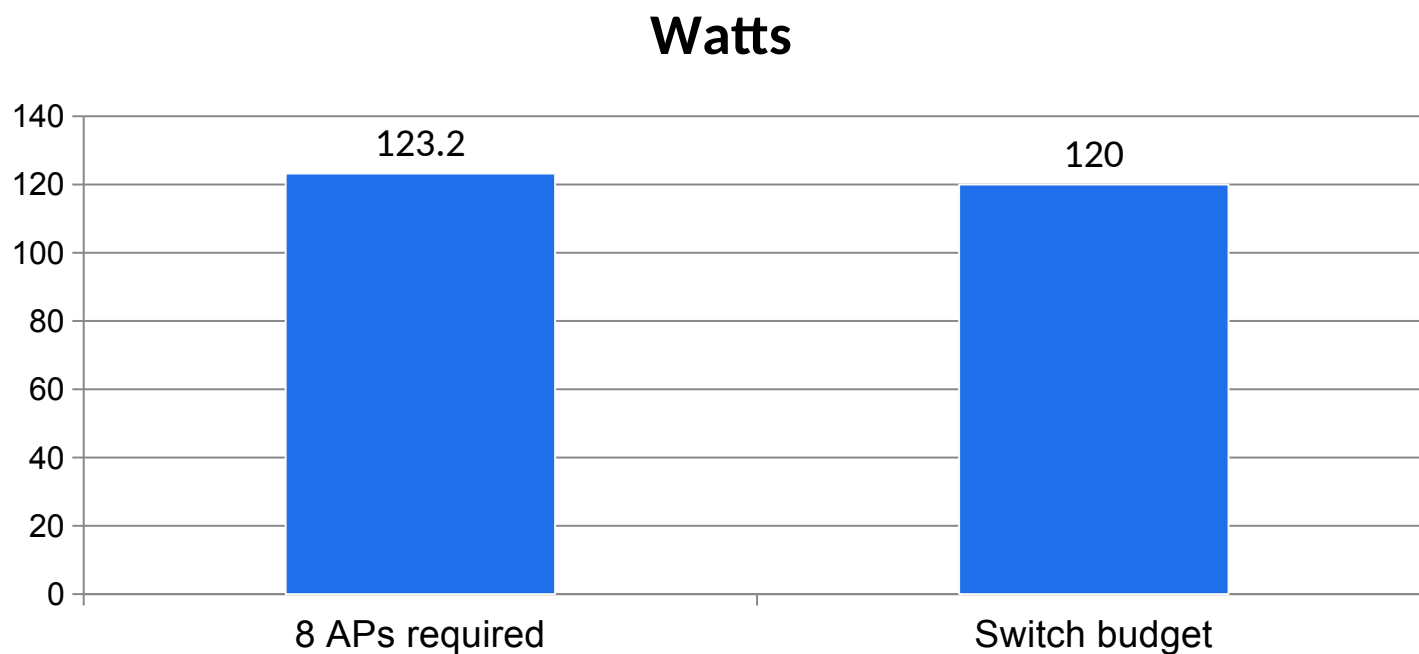
Borrow 3 bits gives 8 subnets

Five equal VLANs require /27

Each block has 30 usable addresses.

What the data shows: A /24 borrowed 2 bits gives 4 subnets

PoE budget: allocated versus available



123.2 W required >120 W available

Shortfall 3.2 W

Check per-port class plus total budget.

Link-up is not proof of delivered power.

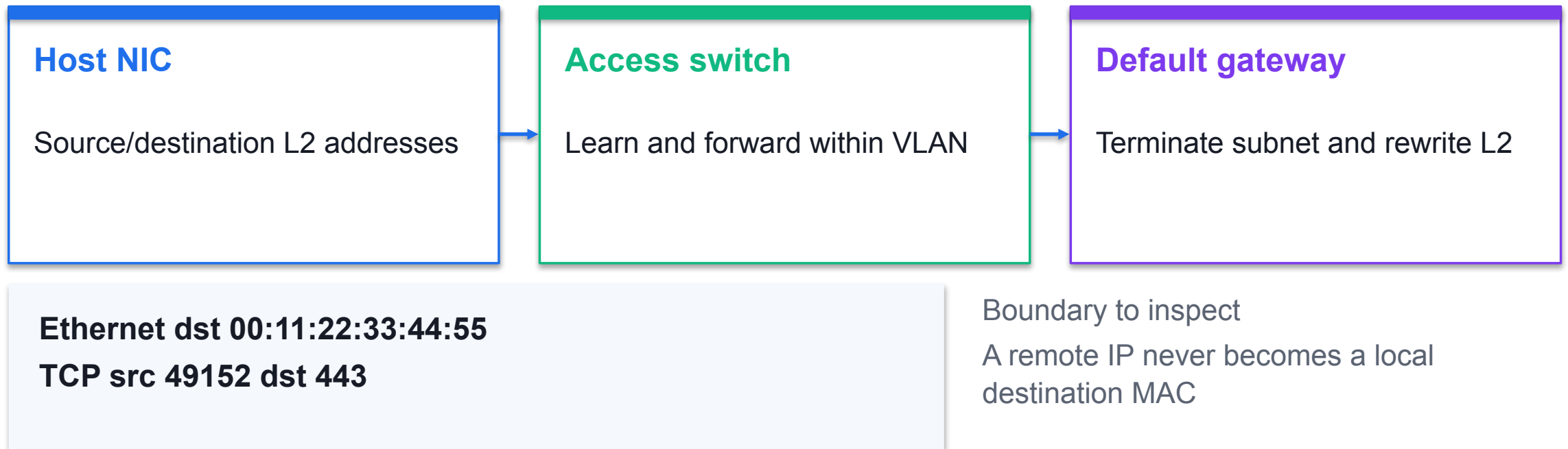
What the data shows: 123.2 W required >120 W available

Network Fundamentals

Network Fundamentals

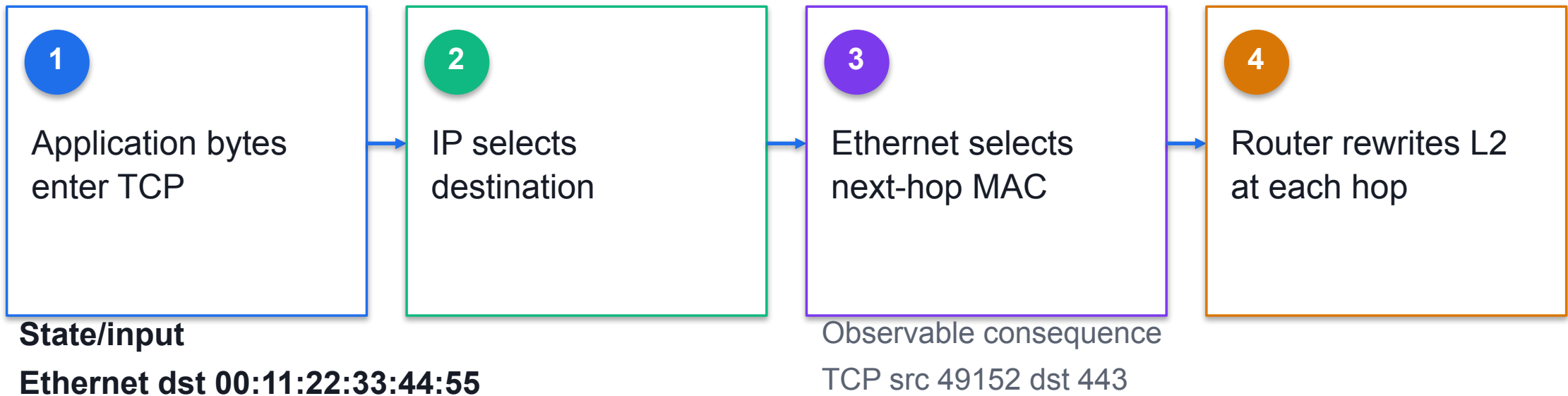
Named boundaries -> runtime trace -> exact contract -> failure/control
-> acceptance evidence

Frame and packet boundaries · boundaries



Capture both sides of R1: IP endpoints persist without NAT; source and destination MAC change

Frame and packet boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Frame and packet boundaries · exact contract

Ethernet dst 00:11:22:33:44:55
IPv4 src 192.0.2.10 dst 198.51.100.20
TCP src 49152 dst 443

INPUT / IDENTITY

Host NIC · Application bytes enter TCP

DECISION / STATE

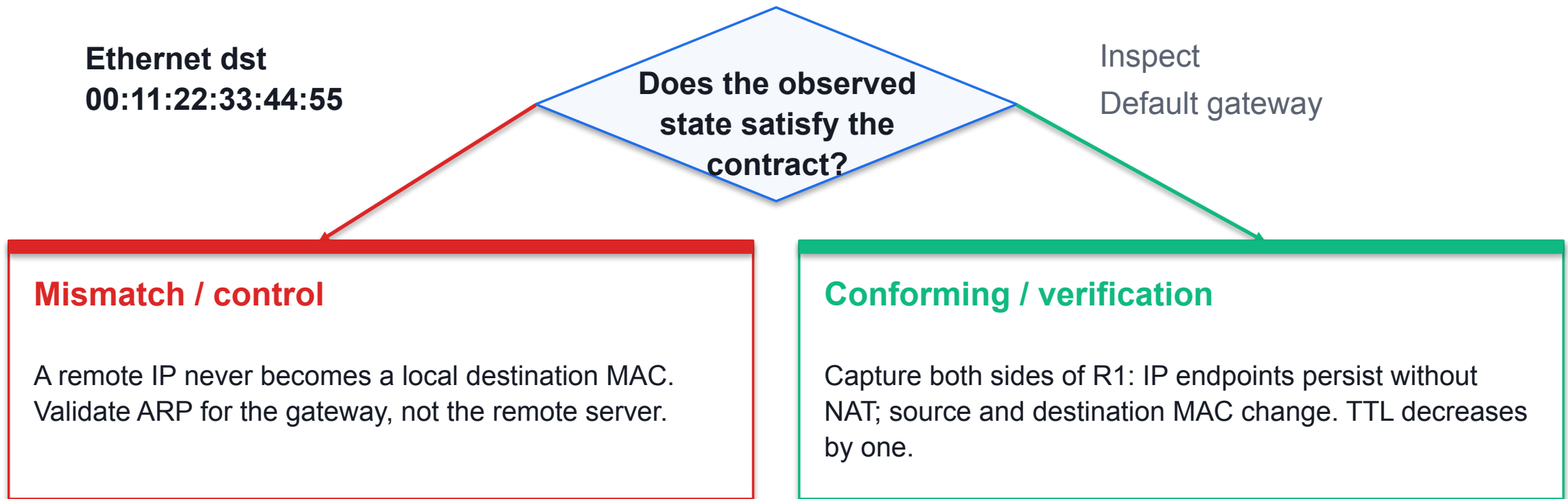
Ethernet selects next-hop MAC

READBACK / RESULT

Router rewrites L2 at each hop

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Frame and packet boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

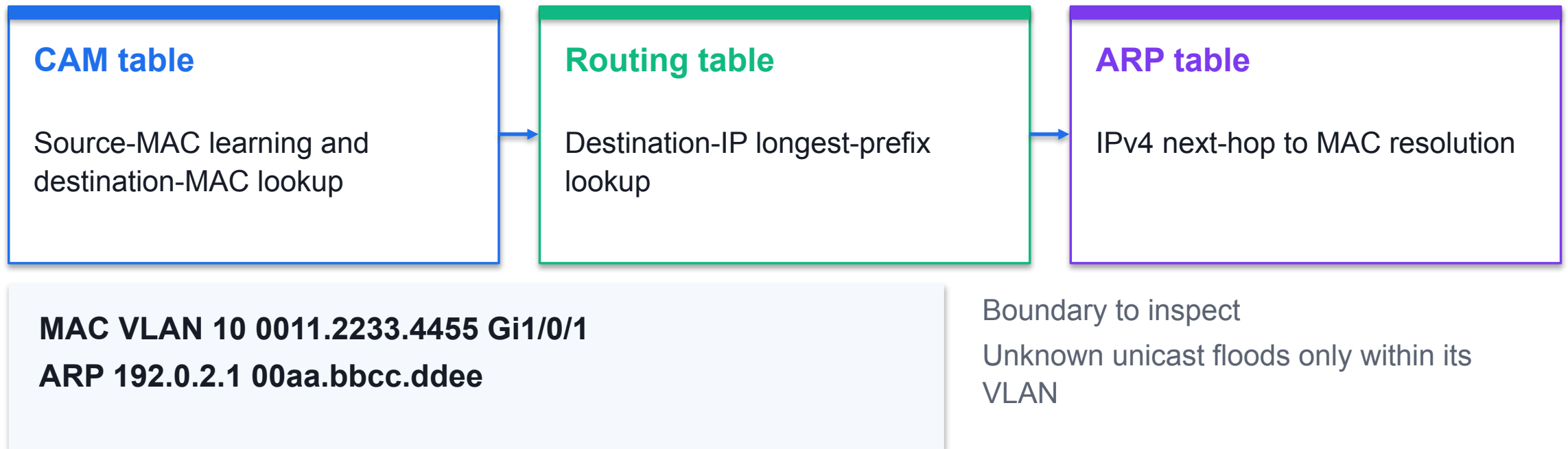
Frame and packet boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Host NIC -> Default gateway	Ethernet dst 00:11:22:33:44:55
Mechanism / state	Router rewrites L2 at each hop	TCP src 49152 dst 443
Result / boundary	Capture both sides of R1: IP endpoints persist without NAT; source and destination MAC change. TTL decreases by one.	A remote IP never becomes a local destination MAC. Validate ARP for the gateway, not the remote server.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

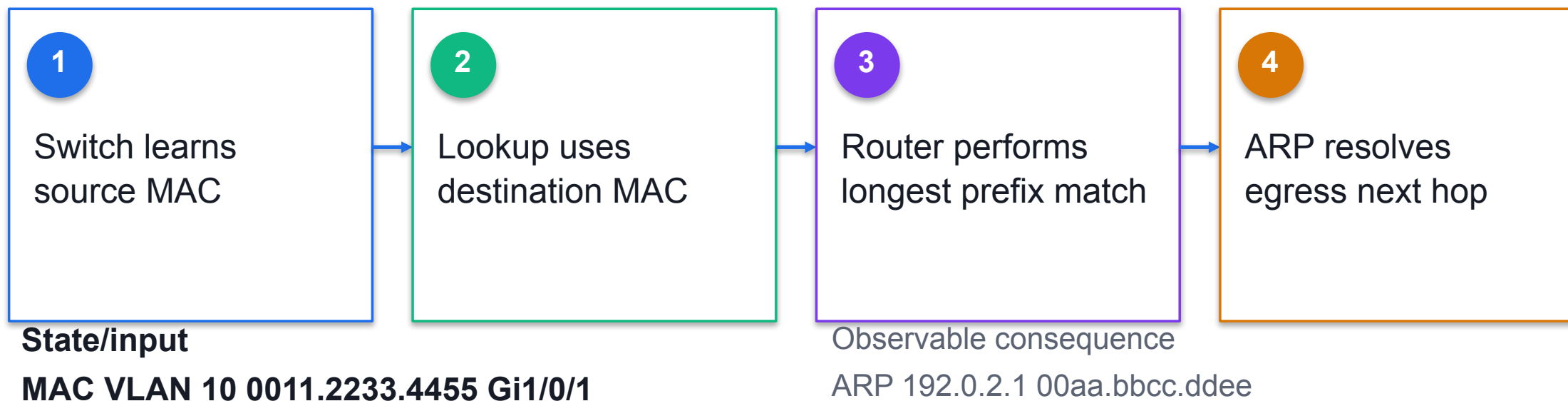
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Router and switch forwarding · boundaries



show mac address-table; show ip route; show ip arp

Router and switch forwarding · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Router and switch forwarding · exact contract

MAC VLAN 10 0011.2233.4455 Gi1/0/1
Route 198.51.100.0/24 via 192.0.2.1
ARP 192.0.2.1 00aa.bbcc.ddee

INPUT / IDENTITY

CAM table · Switch learns source MAC

DECISION / STATE

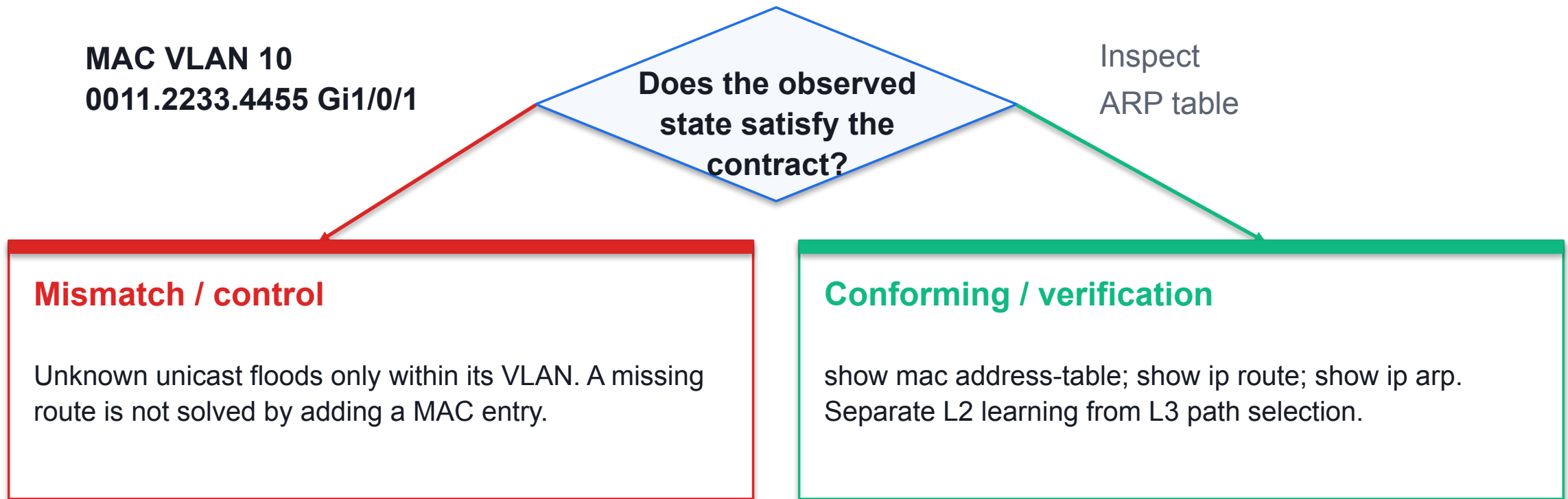
Router performs longest prefix match

READBACK / RESULT

ARP resolves egress next hop

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Router and switch forwarding · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

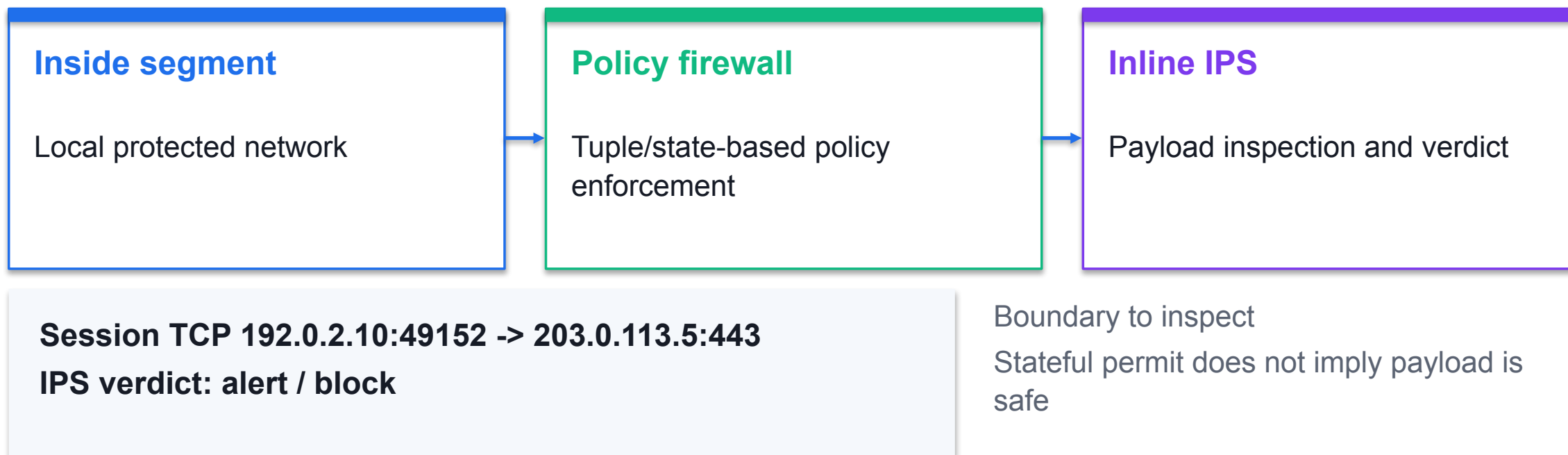
Router and switch forwarding · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	CAM table -> ARP table	MAC VLAN 10 0011.2233.4455 Gi1/0/1
Mechanism / state	ARP resolves egress next hop	ARP 192.0.2.1 00aa.bbcc.ddee
Result / boundary	show mac address-table; show ip route; show ip arp. Separate L2 learning from L3 path selection.	Unknown unicast floods only within its VLAN. A missing route is not solved by adding a MAC entry.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

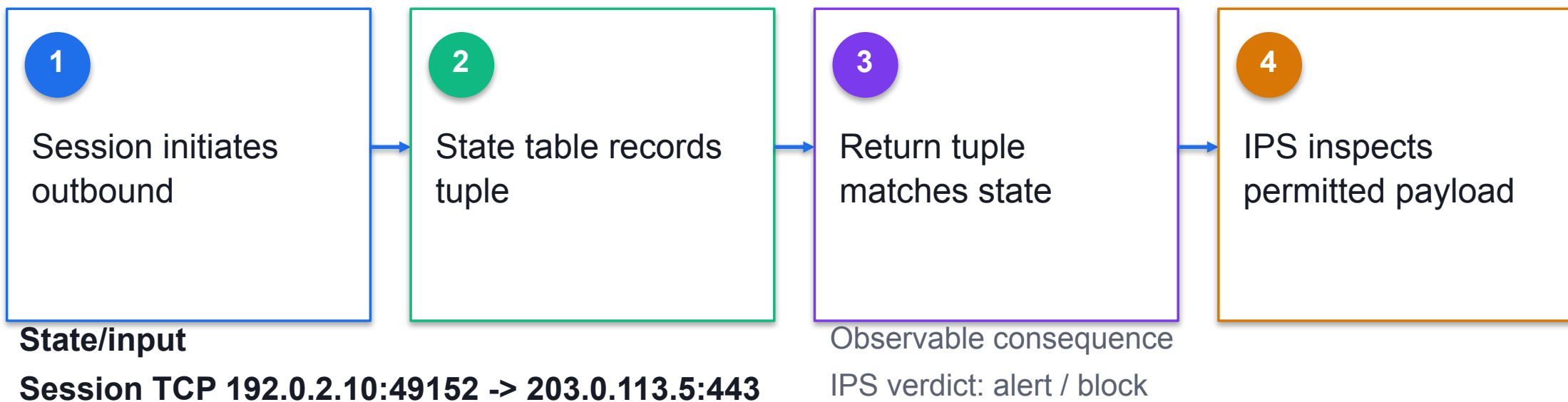
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Firewall and IPS control points · boundaries



Compare policy hit counters, session entries and IPS event timestamps

Firewall and IPS control points · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Firewall and IPS control points · exact contract

Session TCP 192.0.2.10:49152 -> 203.0.113.5:443
State ESTABLISHED
IPS verdict: alert / block

INPUT / IDENTITY

Inside segment · Session initiates outbound

DECISION / STATE

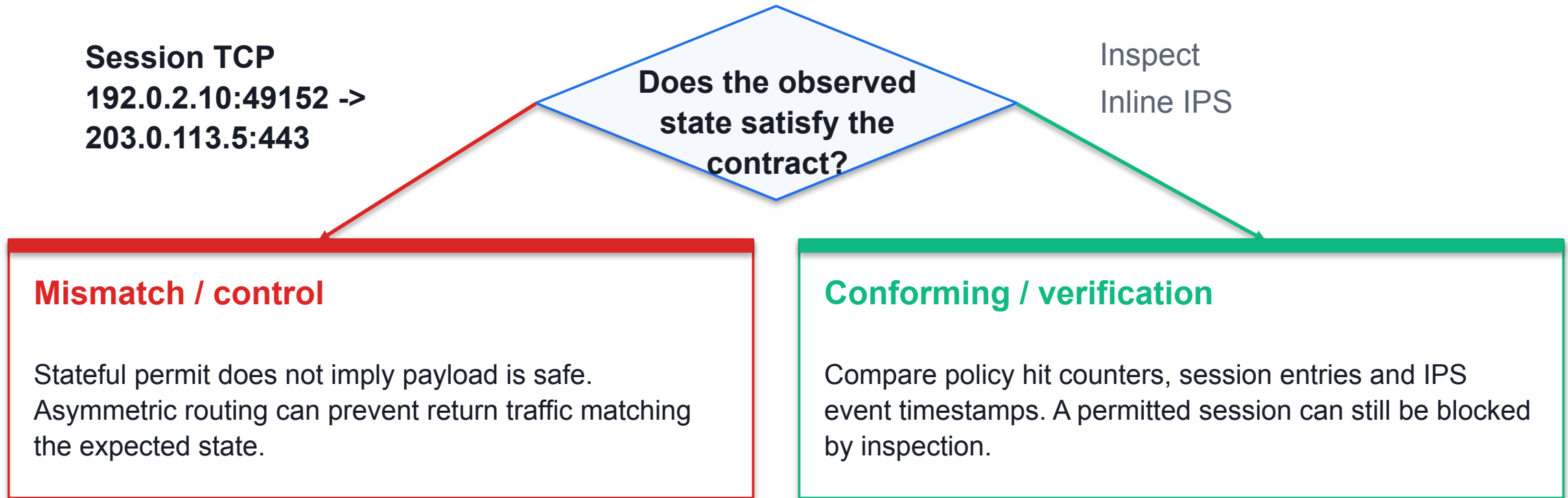
Return tuple matches state

READBACK / RESULT

IPS inspects permitted payload

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Firewall and IPS control points · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

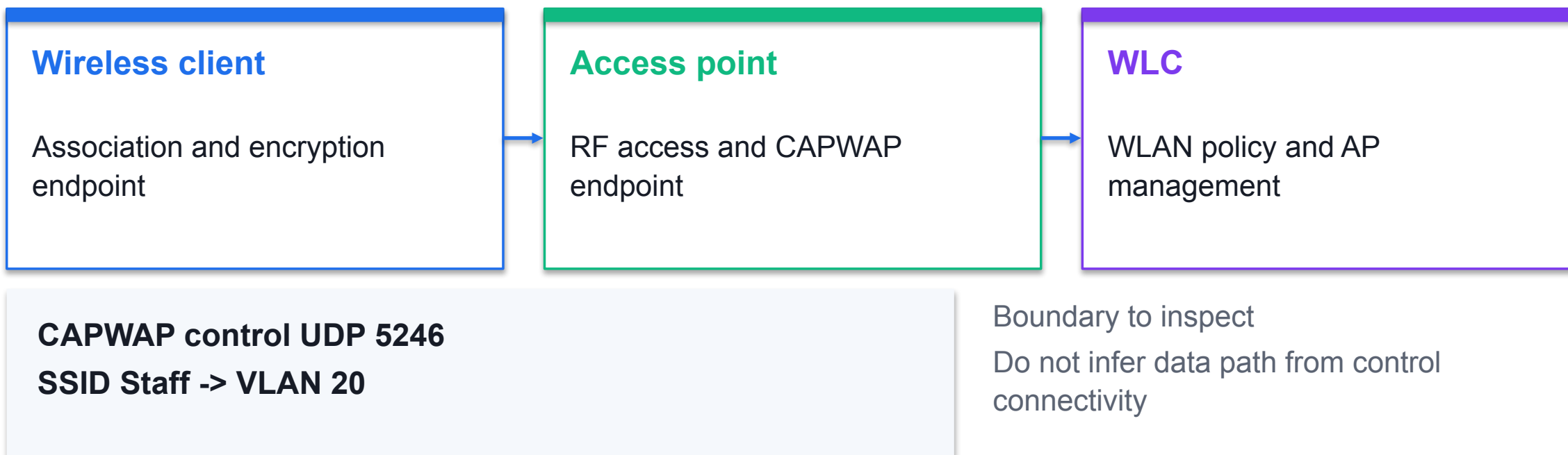
Firewall and IPS control points · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Inside segment -> Inline IPS	Session TCP 192.0.2.10:49152 -> 203.0.113.5:443
Mechanism / state	IPS inspects permitted payload	IPS verdict: alert / block
Result / boundary	Compare policy hit counters, session entries and IPS event timestamps. A permitted session can still be blocked by inspection.	Stateful permit does not imply payload is safe. Asymmetric routing can prevent return traffic matching the expected state.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

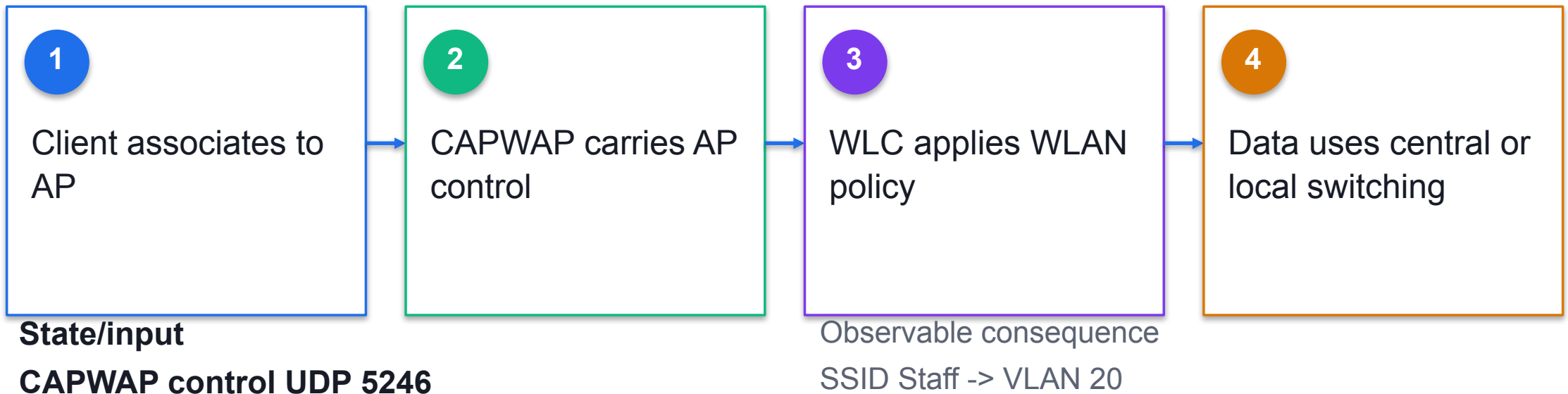
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Wireless controller boundaries · boundaries



Identify AP mode, WLAN-to-VLAN mapping and client IP

Wireless controller boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Wireless controller boundaries · exact contract

CAPWAP control UDP 5246

CAPWAP data UDP 5247

SSID Staff -> VLAN 20

INPUT / IDENTITY

Wireless client · Client associates to AP

DECISION / STATE

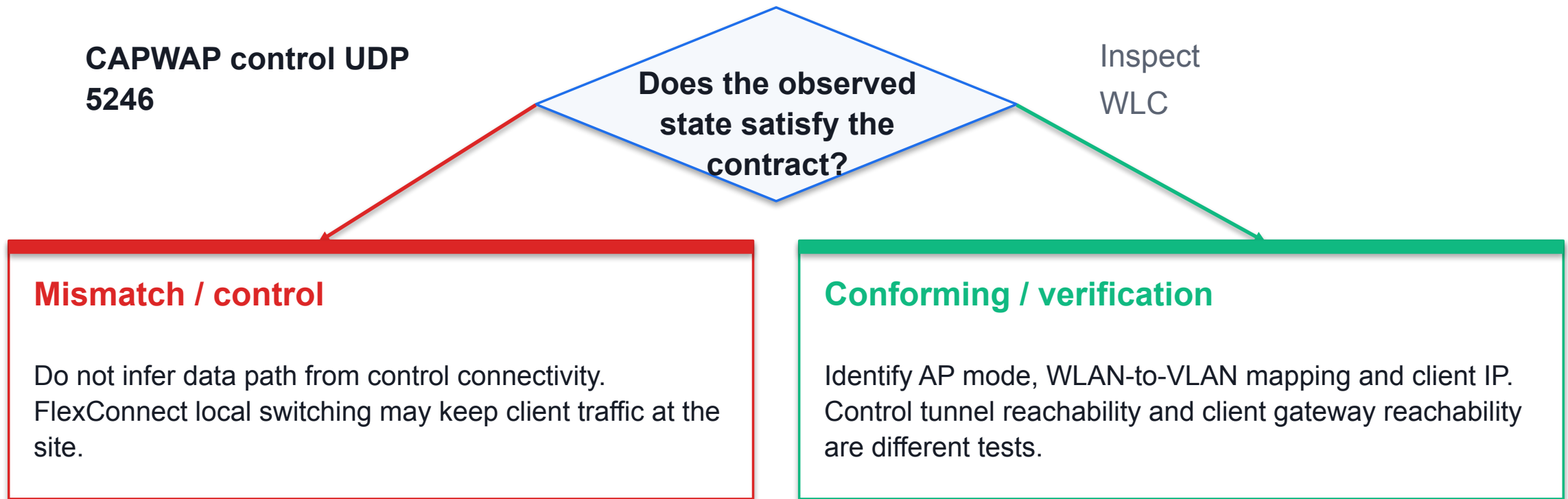
WLC applies WLAN policy

READBACK / RESULT

Data uses central or local switching

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Wireless controller boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

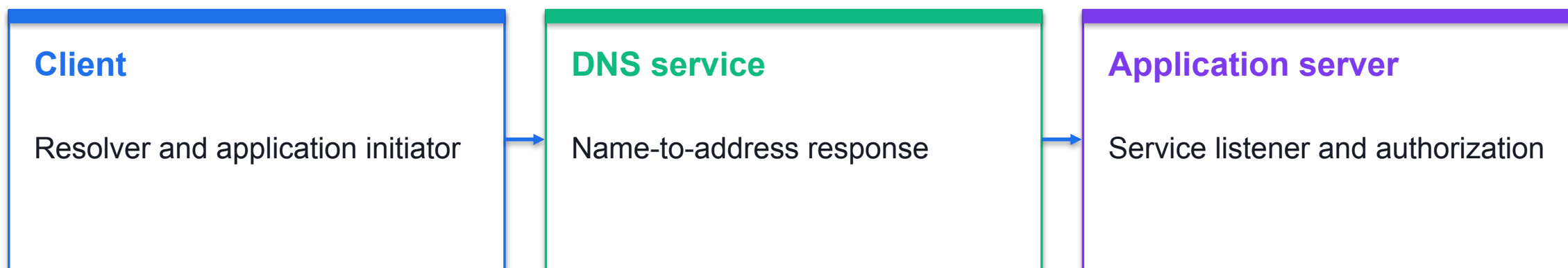
Wireless controller boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Wireless client -> WLC	CAPWAP control UDP 5246
Mechanism / state	Data uses central or local switching	SSID Staff -> VLAN 20
Result / boundary	Identify AP mode, WLAN-to-VLAN mapping and client IP. Control tunnel reachability and client gateway reachability are different tests.	Do not infer data path from control connectivity. FlexConnect local switching may keep client traffic at the site.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Endpoint and server dependencies · boundaries

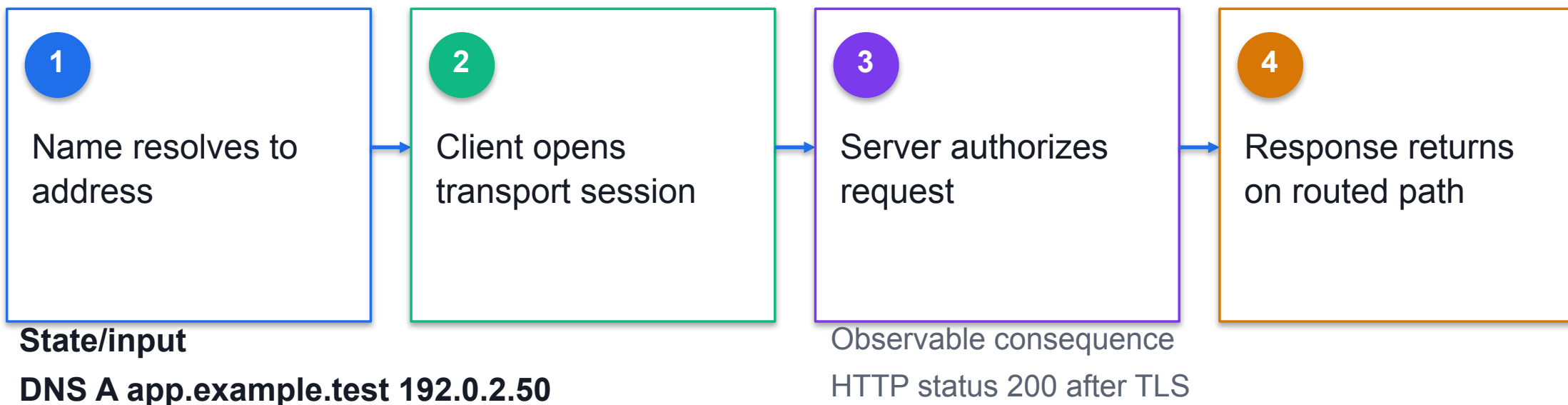


DNS A app.example.test 192.0.2.50
HTTP status 200 after TLS

Boundary to inspect
A successful ping proves neither DNS nor
application authorization

Compare DNS result, TCP connection and HTTP response

Endpoint and server dependencies · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Endpoint and server dependencies · exact contract

DNS A app.example.test 192.0.2.50
TCP SYN dst 443
HTTP status 200 after TLS

INPUT / IDENTITY

Client · Name resolves to address

DECISION / STATE

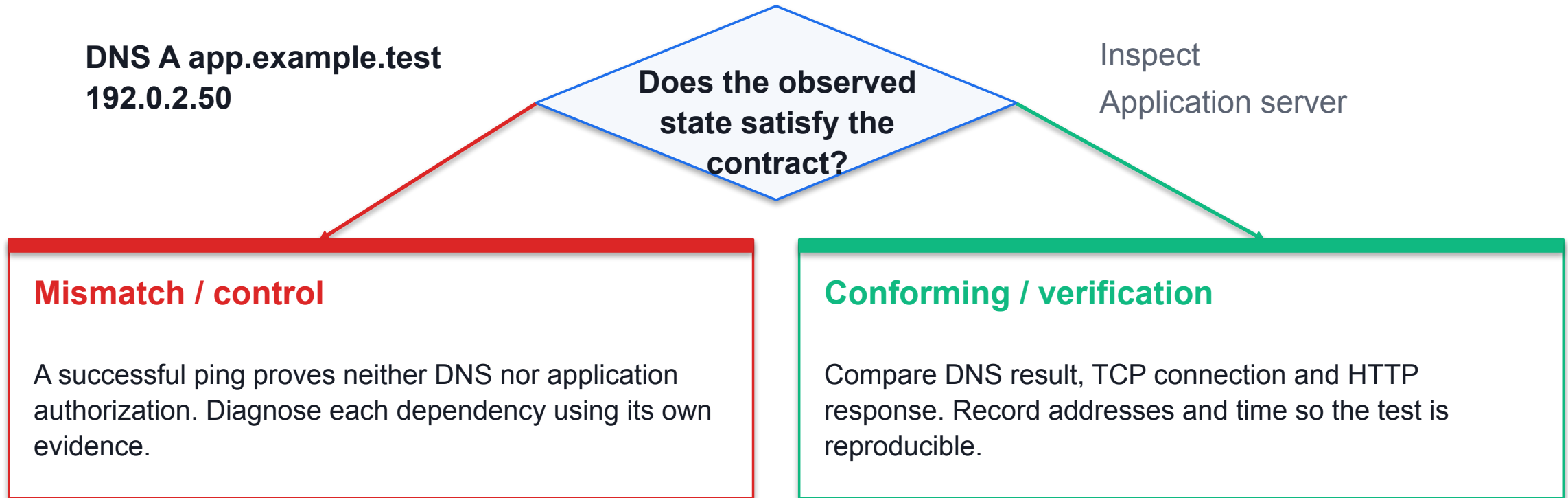
Server authorizes request

READBACK / RESULT

Response returns on routed path

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Endpoint and server dependencies · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

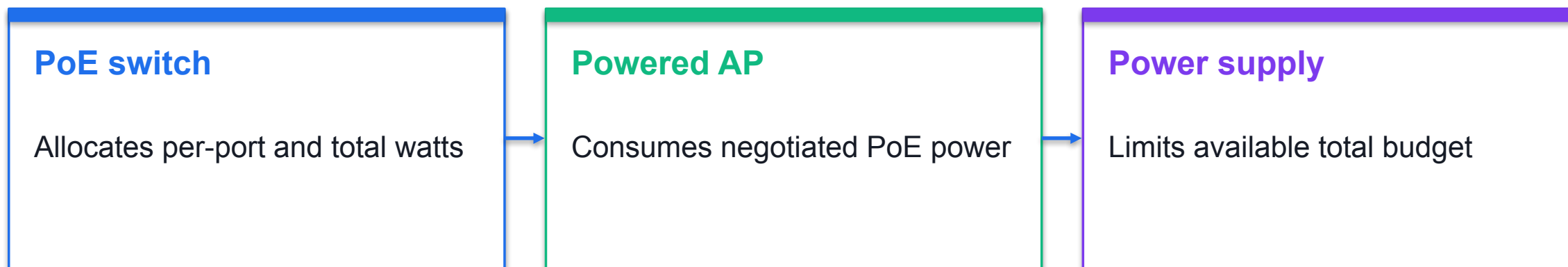
Endpoint and server dependencies · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Client -> Application server	DNS A app.example.test 192.0.2.50
Mechanism / state	Response returns on routed path	HTTP status 200 after TLS
Result / boundary	Compare DNS result, TCP connection and HTTP response. Record addresses and time so the test is reproducible.	A successful ping proves neither DNS nor application authorization. Diagnose each dependency using its own evidence.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

PoE power budgets · boundaries



8 APs x 15.4 W = 123.2 W

Deficit = 3.2 W

Boundary to inspect

A port can have a working data link while power is denied or constrained

show power inline

PoE power budgets · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

PoE power budgets · exact contract

8 APs x 15.4 W = 123.2 W

Switch available budget = 120 W

Deficit = 3.2 W

INPUT / IDENTITY

PoE switch · Device signature detected

DECISION / STATE

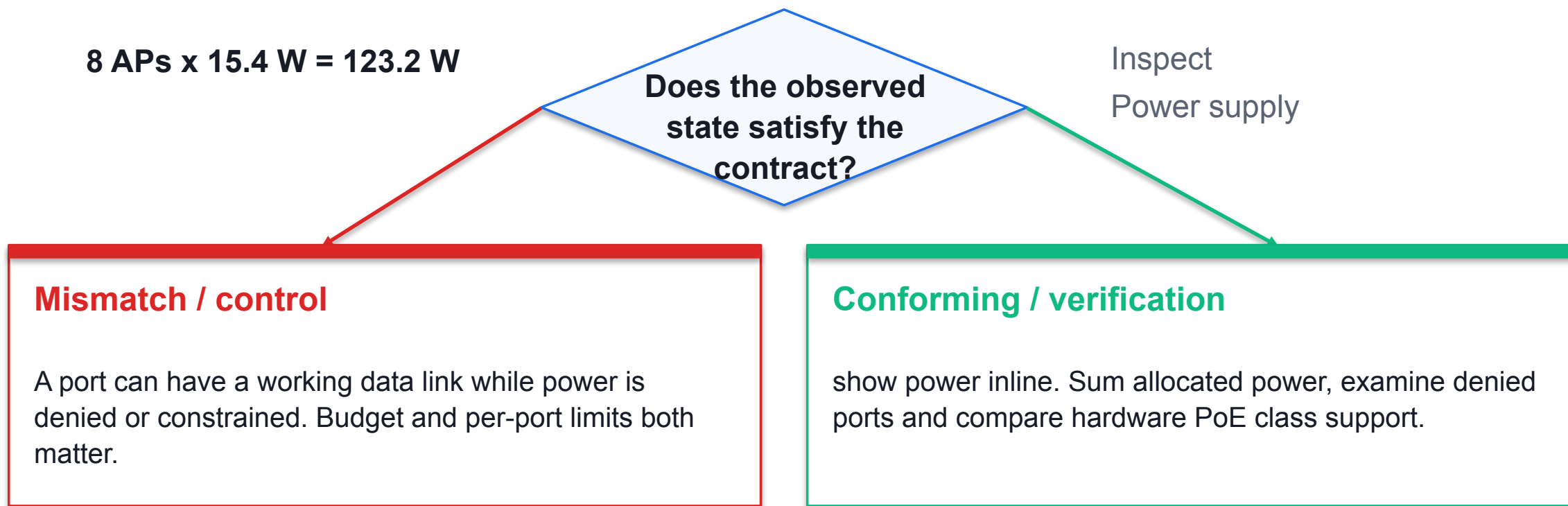
Budget reserves watts

READBACK / RESULT

Link and radio initialize

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

PoE power budgets · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

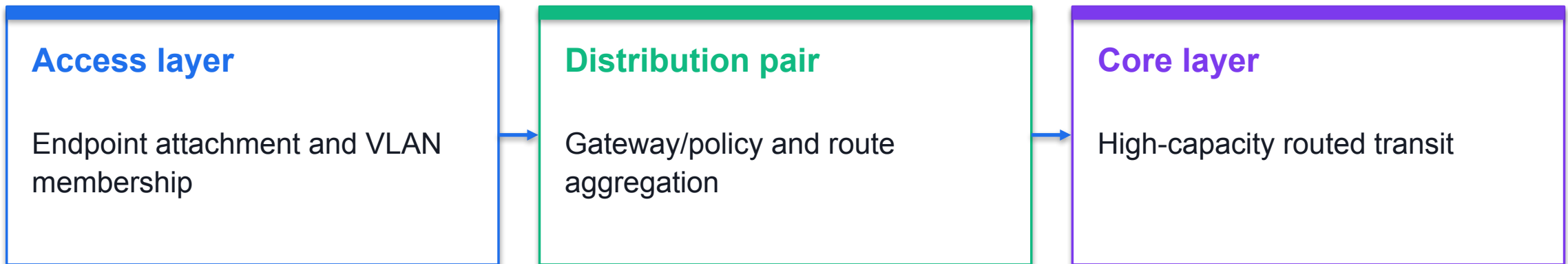
PoE power budgets · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	PoE switch -> Power supply	8 APs x 15.4 W = 123.2 W
Mechanism / state	Link and radio initialize	Deficit = 3.2 W
Result / boundary	show power inline. Sum allocated power, examine denied ports and compare hardware PoE class support.	A port can have a working data link while power is denied or constrained. Budget and per-port limits both matter.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Two-tier and three-tier paths · boundaries

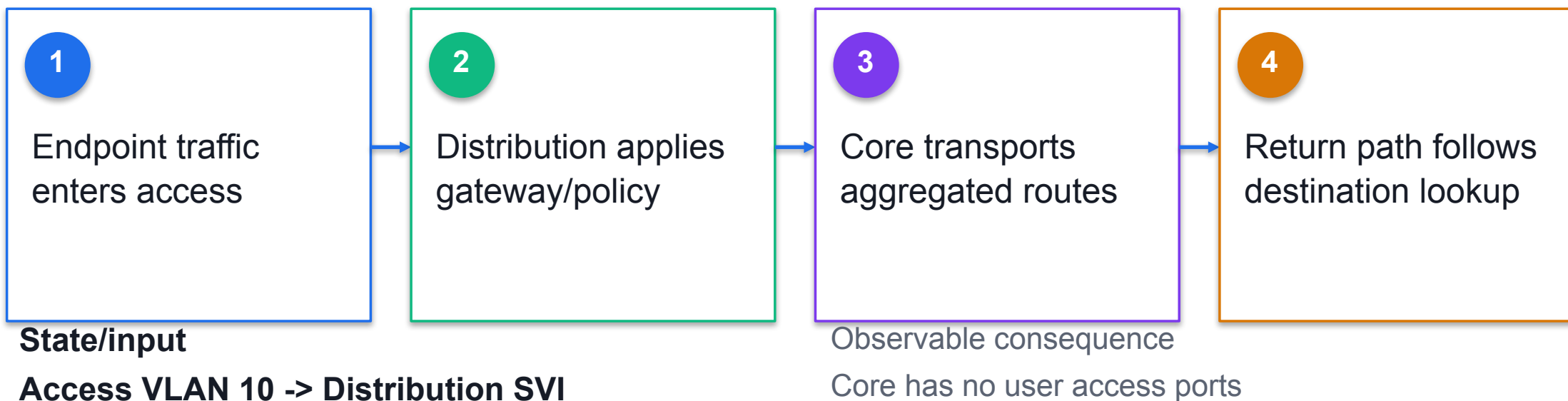


Access VLAN 10 -> Distribution SVI
Core has no user access ports

Boundary to inspect
L2 stretched across distribution increases failure scope

Mark default gateway ownership and L2/L3 boundaries on the topology

Two-tier and three-tier paths · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Two-tier and three-tier paths · exact contract

Access VLAN 10 -> Distribution SVI
Distribution routed uplink -> Core
Core has no user access ports

INPUT / IDENTITY

Access layer · Endpoint traffic enters access

DECISION / STATE

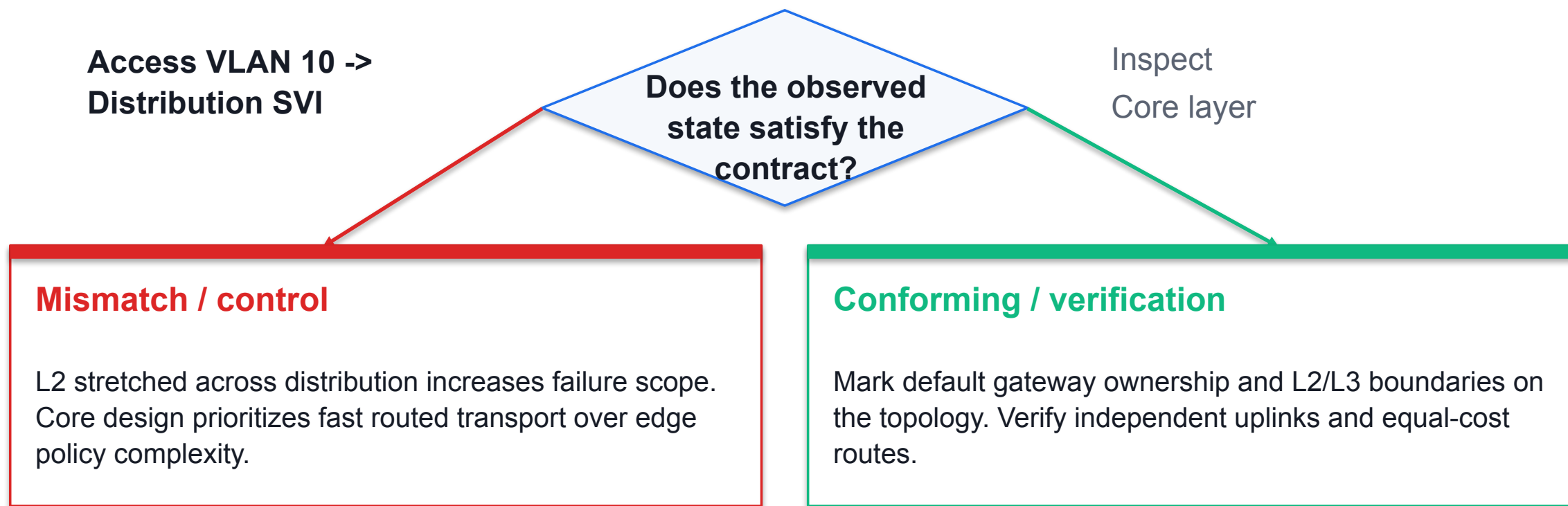
Core transports aggregated routes

READBACK / RESULT

Return path follows destination lookup

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Two-tier and three-tier paths · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

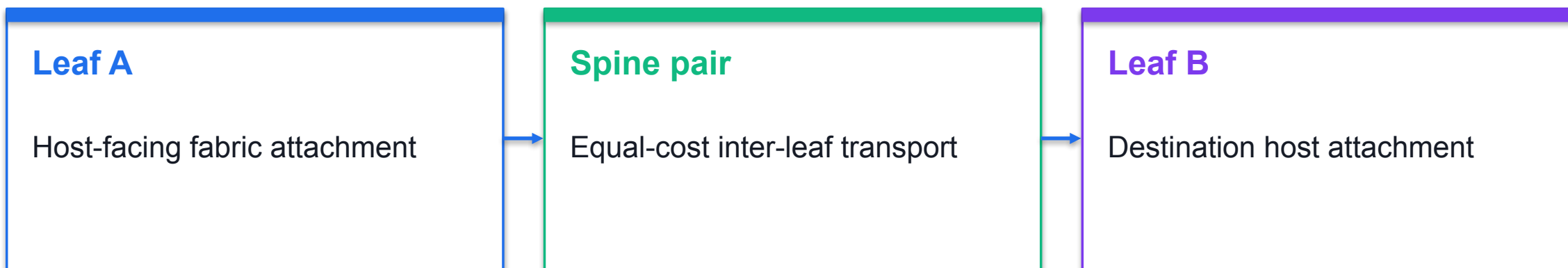
Two-tier and three-tier paths · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Access layer -> Core layer	Access VLAN 10 -> Distribution SVI
Mechanism / state	Return path follows destination lookup	Core has no user access ports
Result / boundary	Mark default gateway ownership and L2/L3 boundaries on the topology. Verify independent uplinks and equal-cost routes.	L2 stretched across distribution increases failure scope. Core design prioritizes fast routed transport over edge policy complexity.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Spine-leaf forwarding · boundaries

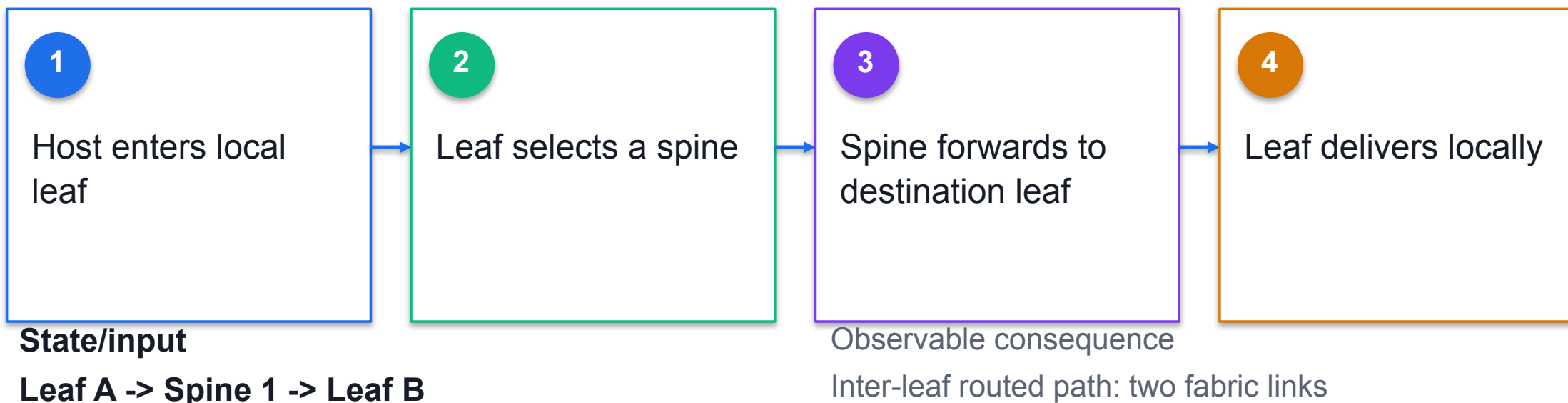


Leaf A -> Spine 1 -> Leaf B
Inter-leaf routed path: two fabric links

Boundary to inspect
Leaf-to-leaf links are not required in a
standard spine-leaf fabric

Count equal-cost paths and hop symmetry

Spine-leaf forwarding · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Spine-leaf forwarding · exact contract

Leaf A -> Spine 1 -> Leaf B

Alternative Leaf A -> Spine 2 -> Leaf B

Inter-leaf routed path: two fabric links

INPUT / IDENTITY

Leaf A · Host enters local leaf

DECISION / STATE

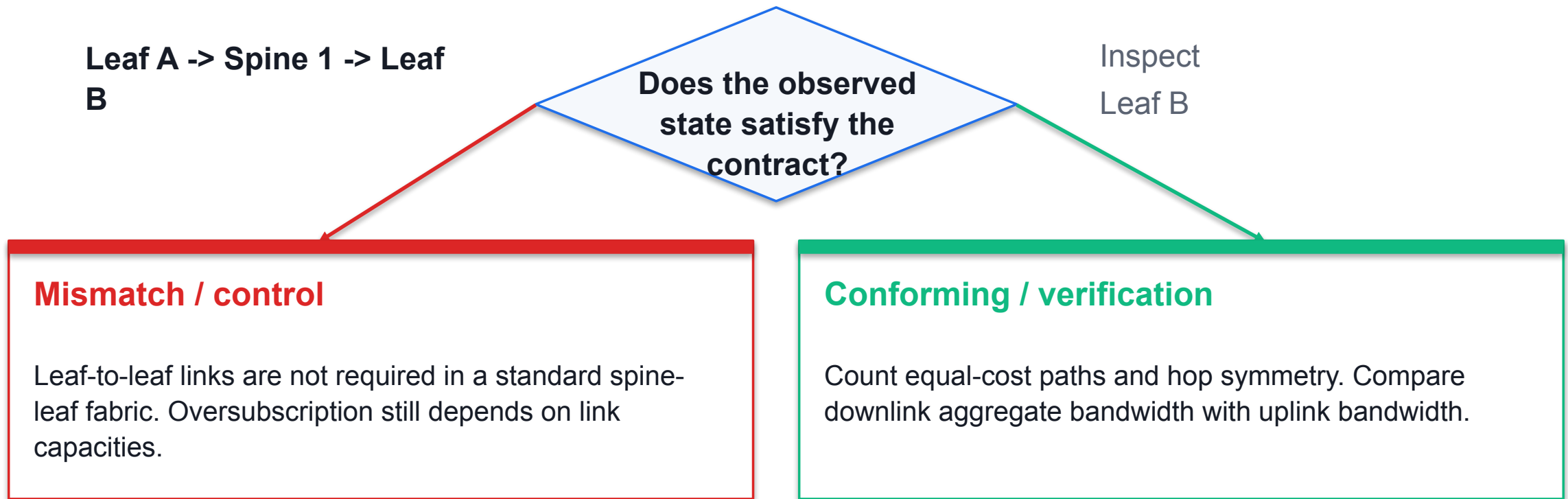
Spine forwards to destination leaf

READBACK / RESULT

Leaf delivers locally

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Spine-leaf forwarding · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

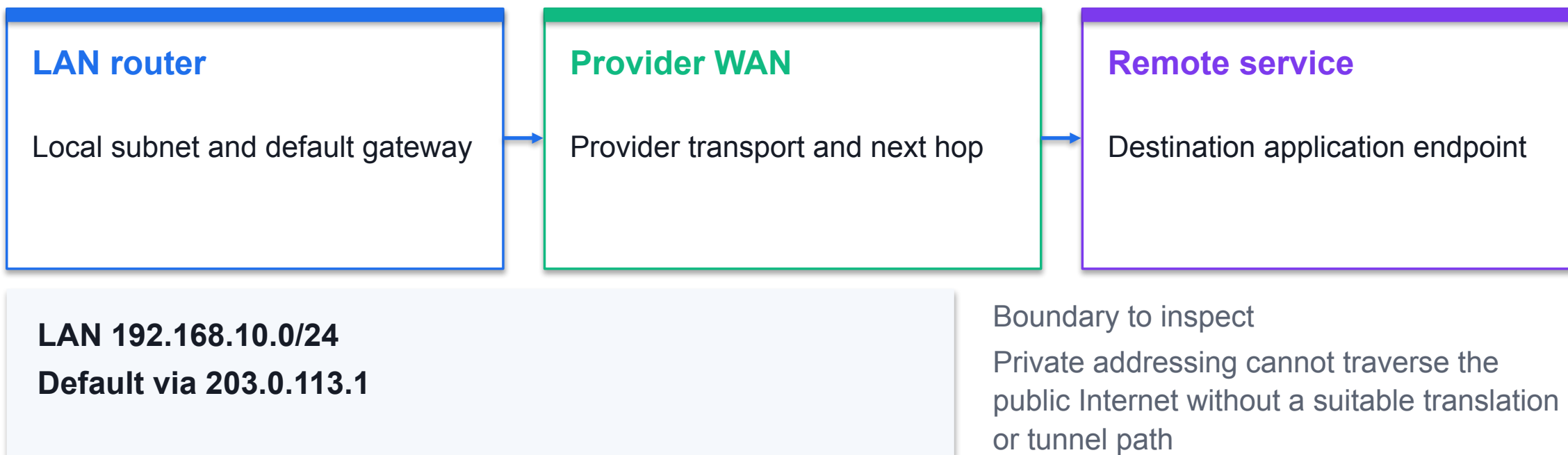
Spine-leaf forwarding · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Leaf A -> Leaf B	Leaf A -> Spine 1 -> Leaf B
Mechanism / state	Leaf delivers locally	Inter-leaf routed path: two fabric links
Result / boundary	Count equal-cost paths and hop symmetry. Compare downlink aggregate bandwidth with uplink bandwidth.	Leaf-to-leaf links are not required in a standard spine-leaf fabric. Oversubscription still depends on link capacities.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

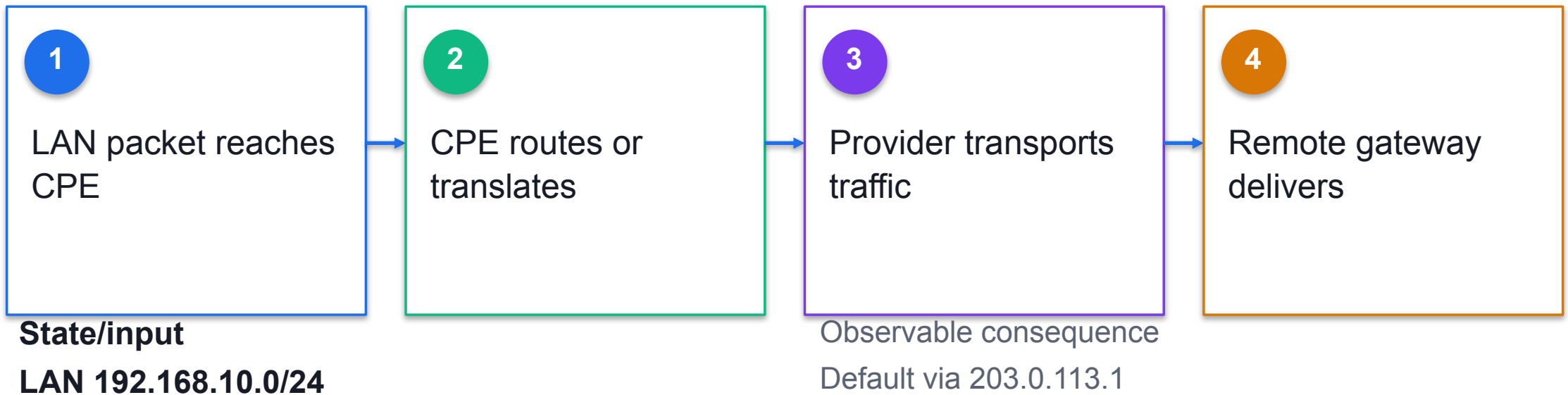
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

WAN and SOHO boundaries · boundaries



Test LAN gateway, provider next hop and a remote address separately; record where loss begins.

WAN and SOHO boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

WAN and SOHO boundaries · exact contract

LAN 192.168.10.0/24
CPE WAN 203.0.113.2/30
Default via 203.0.113.1

INPUT / IDENTITY

LAN router · LAN packet reaches CPE

DECISION / STATE

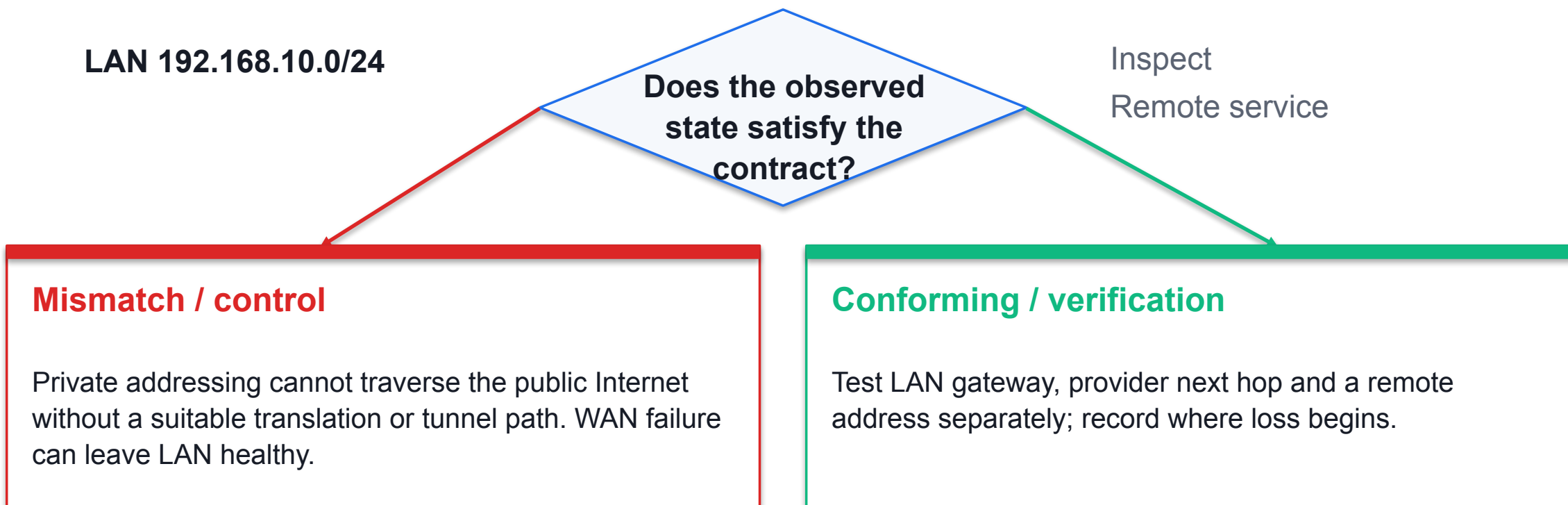
Provider transports traffic

READBACK / RESULT

Remote gateway delivers

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

WAN and SOHO boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

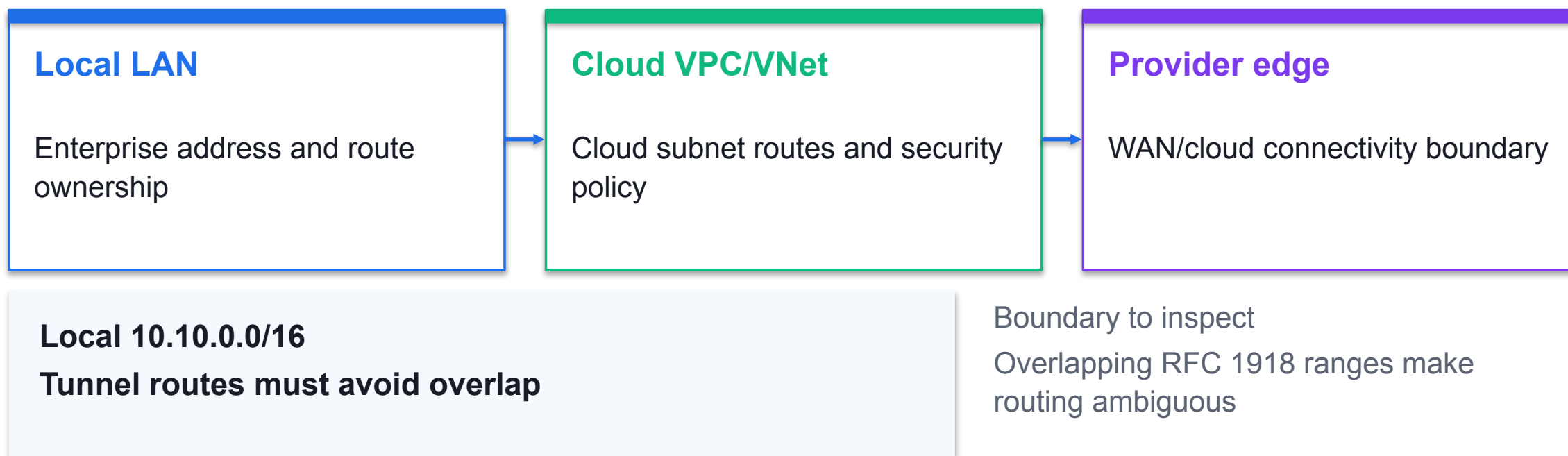
WAN and SOHO boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	LAN router -> Remote service	LAN 192.168.10.0/24
Mechanism / state	Remote gateway delivers	Default via 203.0.113.1
Result / boundary	Test LAN gateway, provider next hop and a remote address separately; record where loss begins.	Private addressing cannot traverse the public Internet without a suitable translation or tunnel path. WAN failure can leave LAN healthy.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

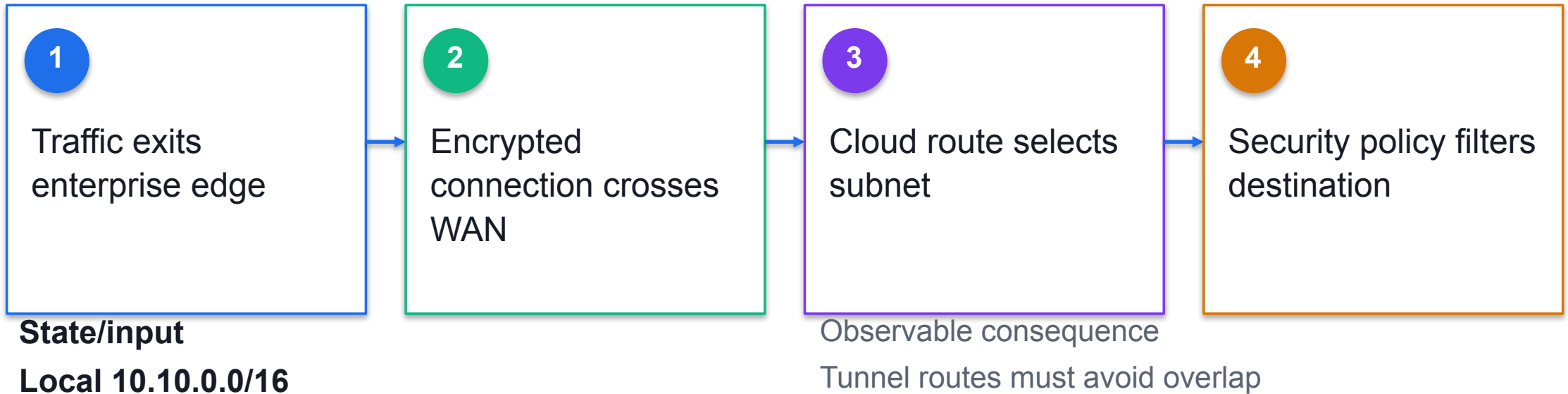
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Cloud and on-premises ownership · boundaries



Check both route tables, tunnel state and cloud security rules

Cloud and on-premises ownership · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Cloud and on-premises ownership · exact contract

Local 10.10.0.0/16

Cloud 10.20.0.0/16

Tunnel routes must avoid overlap

INPUT / IDENTITY

Local LAN · Traffic exits enterprise edge

DECISION / STATE

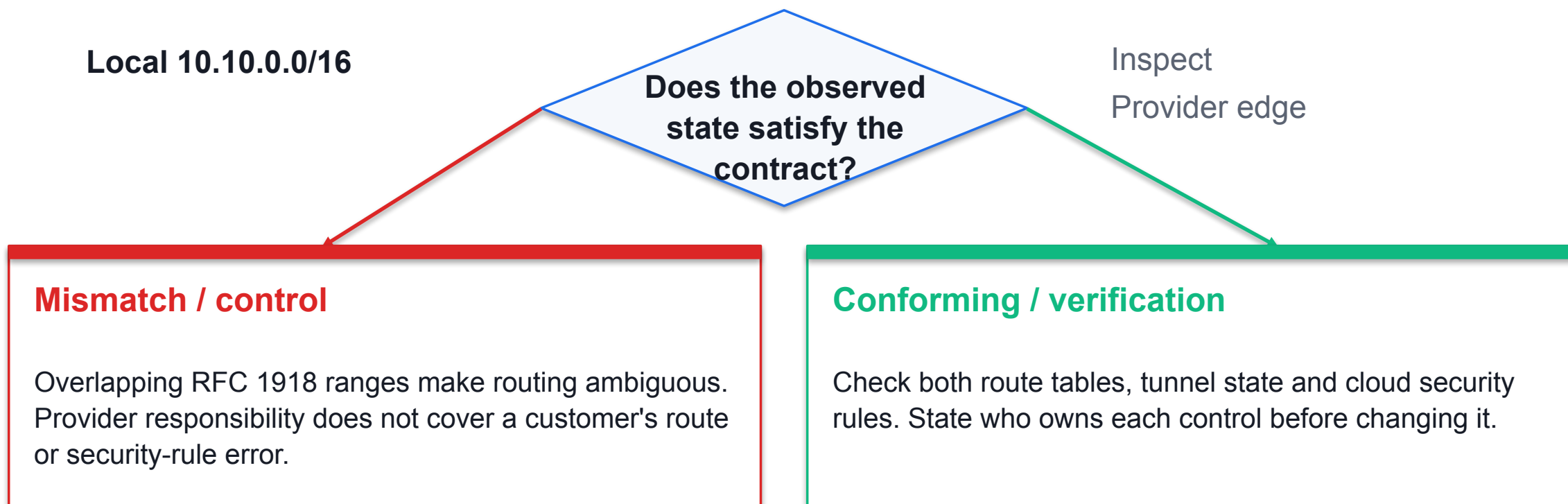
Cloud route selects subnet

READBACK / RESULT

Security policy filters destination

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Cloud and on-premises ownership · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

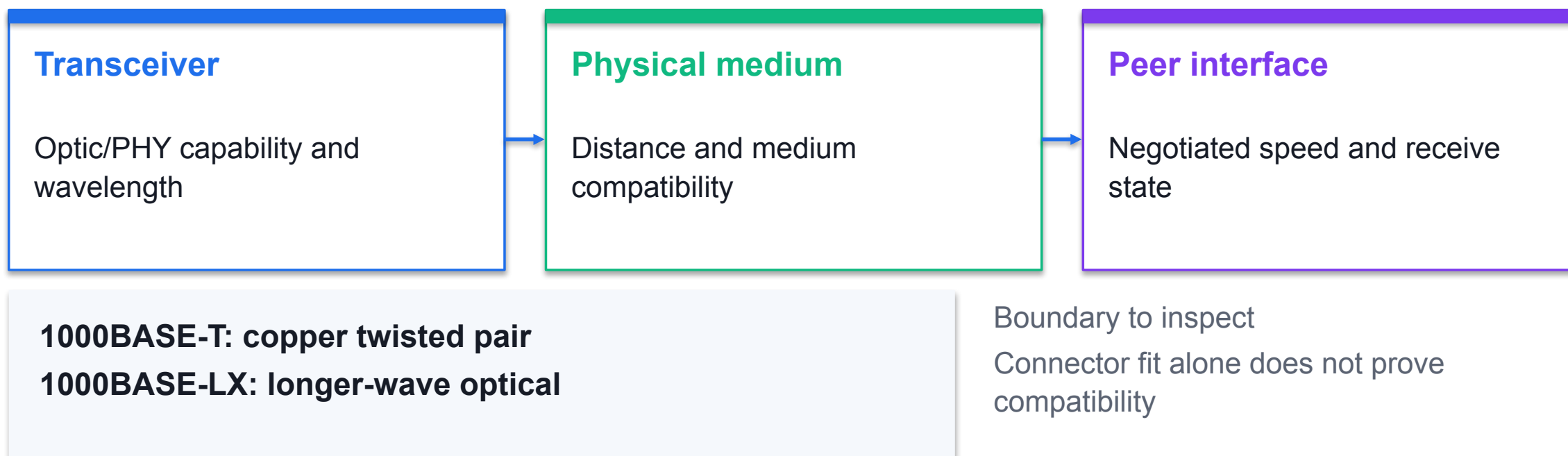
Cloud and on-premises ownership · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Local LAN -> Provider edge	Local 10.10.0.0/16
Mechanism / state	Security policy filters destination	Tunnel routes must avoid overlap
Result / boundary	Check both route tables, tunnel state and cloud security rules. State who owns each control before changing it.	Overlapping RFC 1918 ranges make routing ambiguous. Provider responsibility does not cover a customer's route or security-rule error.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

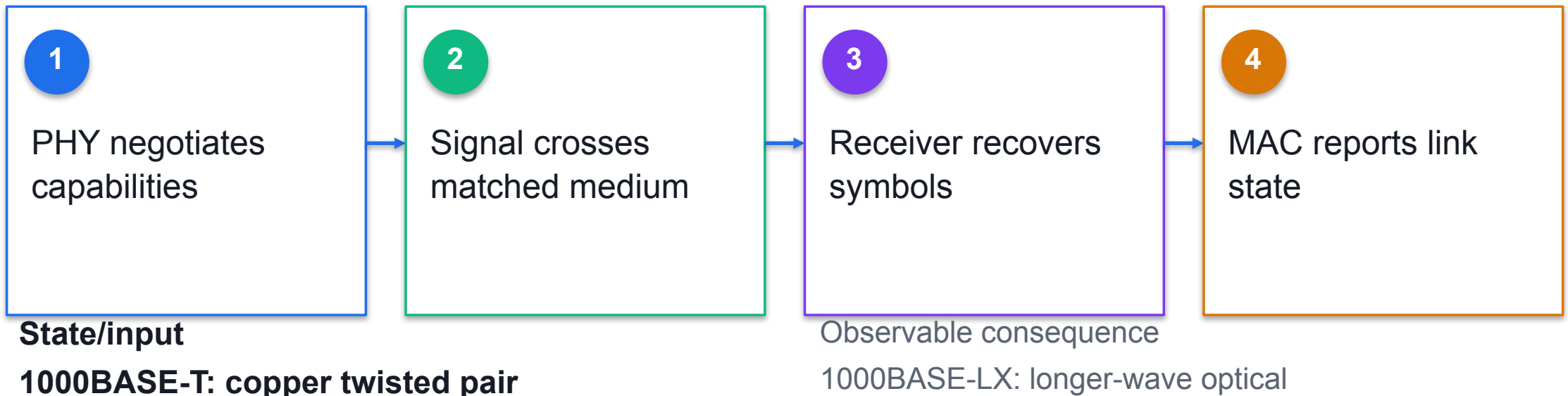
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Copper and optical link contracts · boundaries



Read both interface/transceiver specifications; inspect optical receive power and interface errors.

Copper and optical link contracts · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Copper and optical link contracts · exact contract

1000BASE-T: copper twisted pair
1000BASE-SX: multimode optical
1000BASE-LX: longer-wave optical

INPUT / IDENTITY

Transceiver · PHY negotiates capabilities

DECISION / STATE

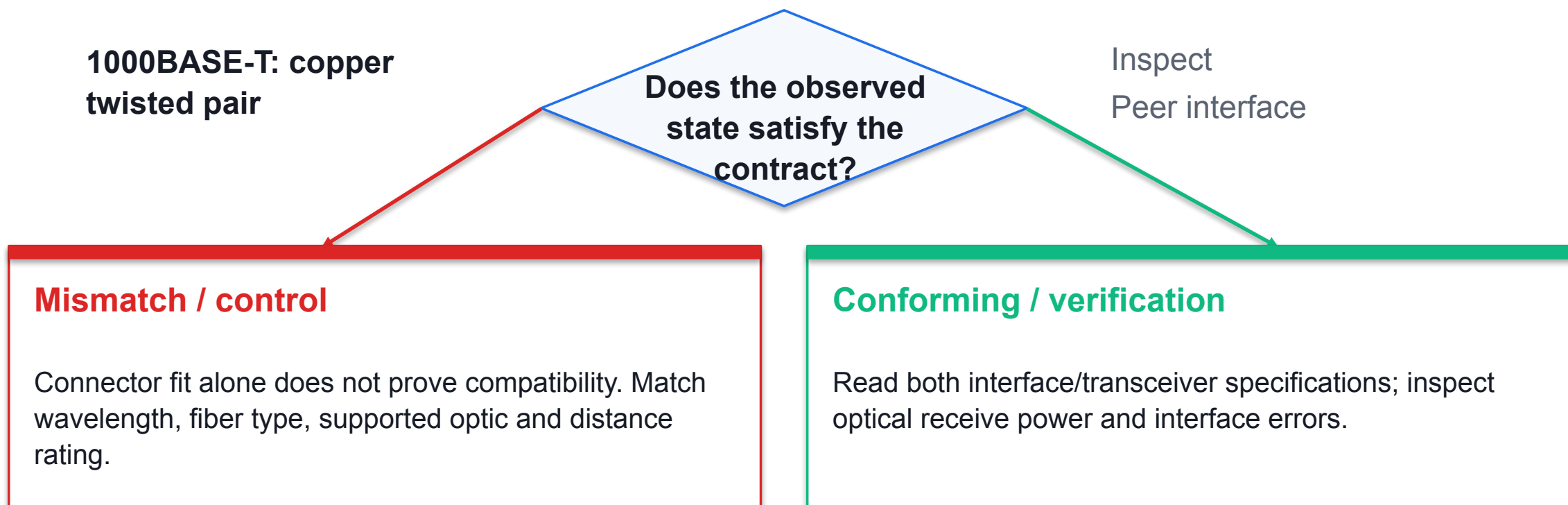
Receiver recovers symbols

READBACK / RESULT

MAC reports link state

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Copper and optical link contracts · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

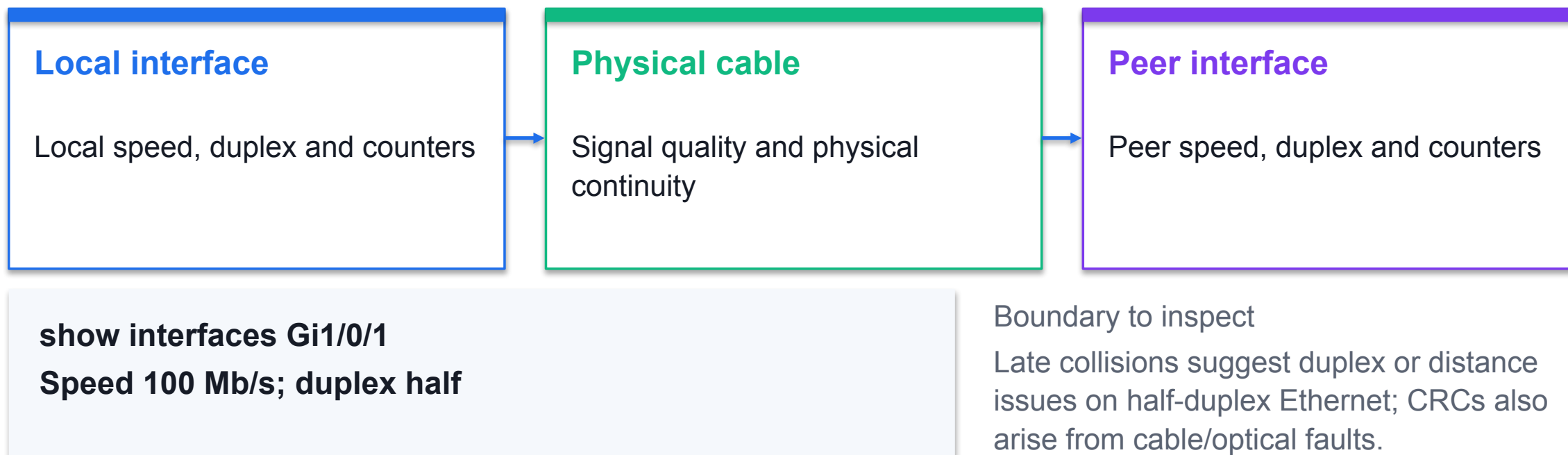
Copper and optical link contracts · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Transceiver -> Peer interface	1000BASE-T: copper twisted pair
Mechanism / state	MAC reports link state	1000BASE-LX: longer-wave optical
Result / boundary	Read both interface/transceiver specifications; inspect optical receive power and interface errors.	Connector fit alone does not prove compatibility. Match wavelength, fiber type, supported optic and distance rating.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

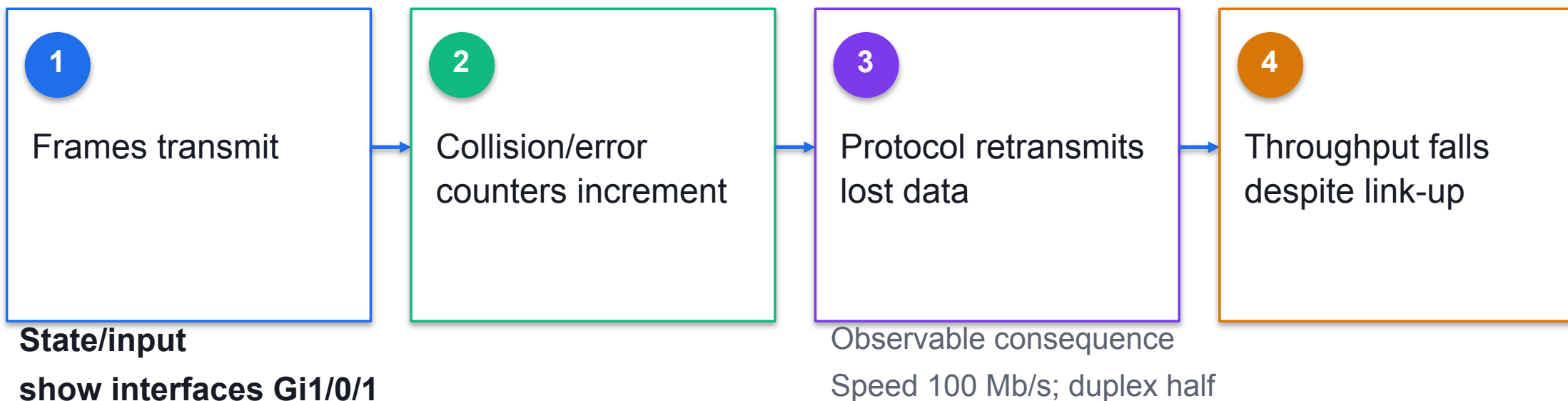
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Duplex and interface errors · boundaries



Compare speed/duplex on both ends and monitor counter deltas after remediation rather than relying on old totals.

Duplex and interface errors · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Duplex and interface errors · exact contract

```
show interfaces Gi1/0/1  
CRC: 240; late collisions: 12  
Speed 100 Mb/s; duplex half
```

INPUT / IDENTITY

Local interface · Frames transmit

DECISION / STATE

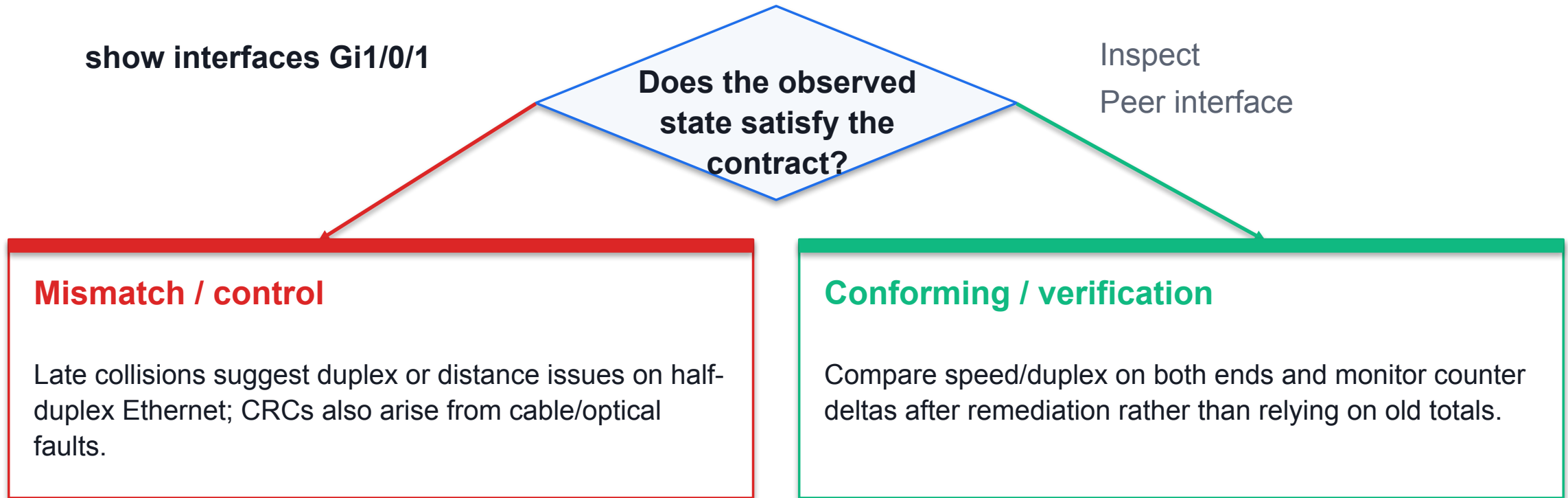
Protocol retransmits lost data

READBACK / RESULT

Throughput falls despite link-up

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Duplex and interface errors · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

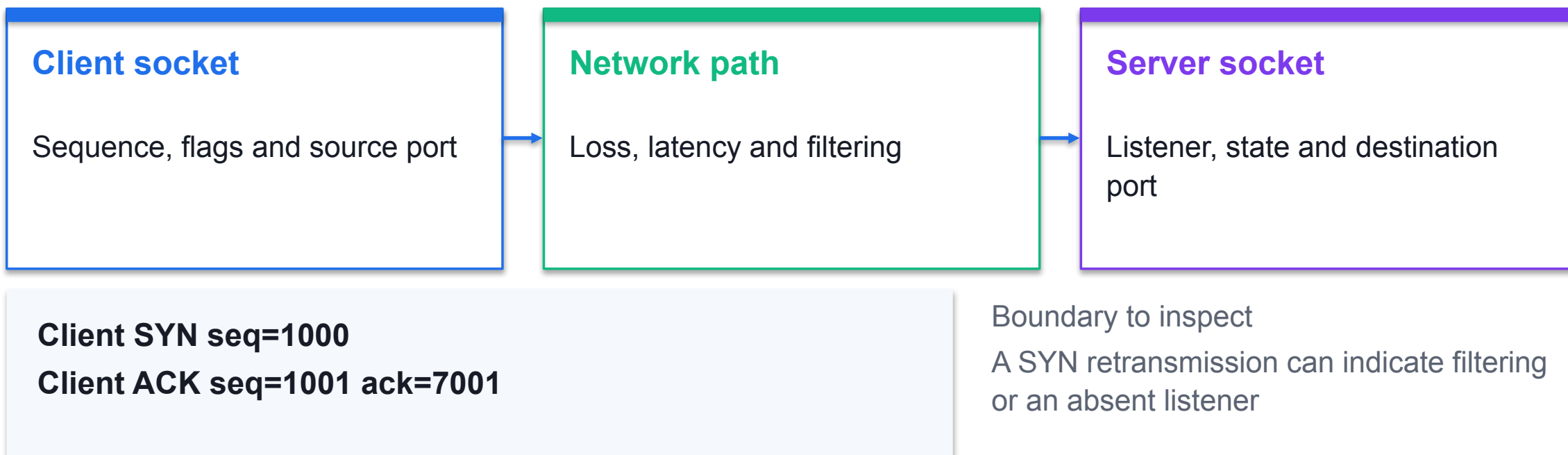
Duplex and interface errors · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Local interface -> Peer interface	show interfaces Gi1/0/1
Mechanism / state	Throughput falls despite link-up	Speed 100 Mb/s; duplex half
Result / boundary	Compare speed/duplex on both ends and monitor counter deltas after remediation rather than relying on old totals.	Late collisions suggest duplex or distance issues on half-duplex Ethernet; CRCs also arise from cable/optical faults.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

TCP session state · boundaries



Read flags and sequence/ack fields in a capture

TCP session state · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

TCP session state · exact contract

Client SYN seq=1000

Server SYN-ACK seq=7000 ack=1001

Client ACK seq=1001 ack=7001

INPUT / IDENTITY

Client socket · SYN proposes sequence

DECISION / STATE

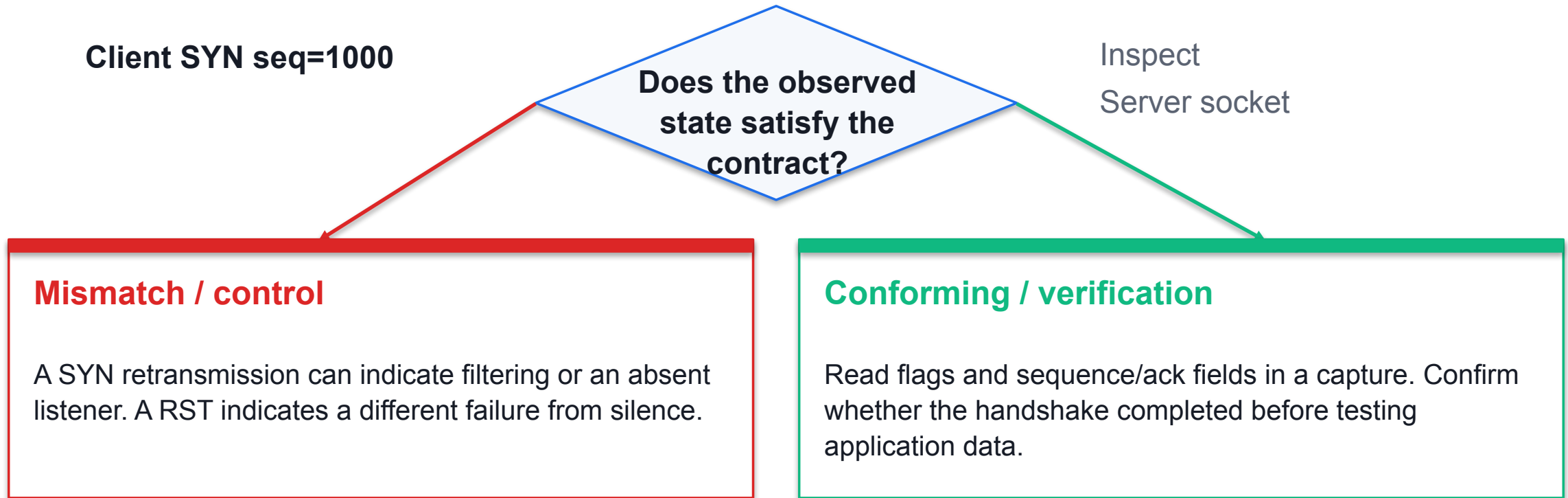
ACK establishes session

READBACK / RESULT

FIN or RST terminates

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

TCP session state · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

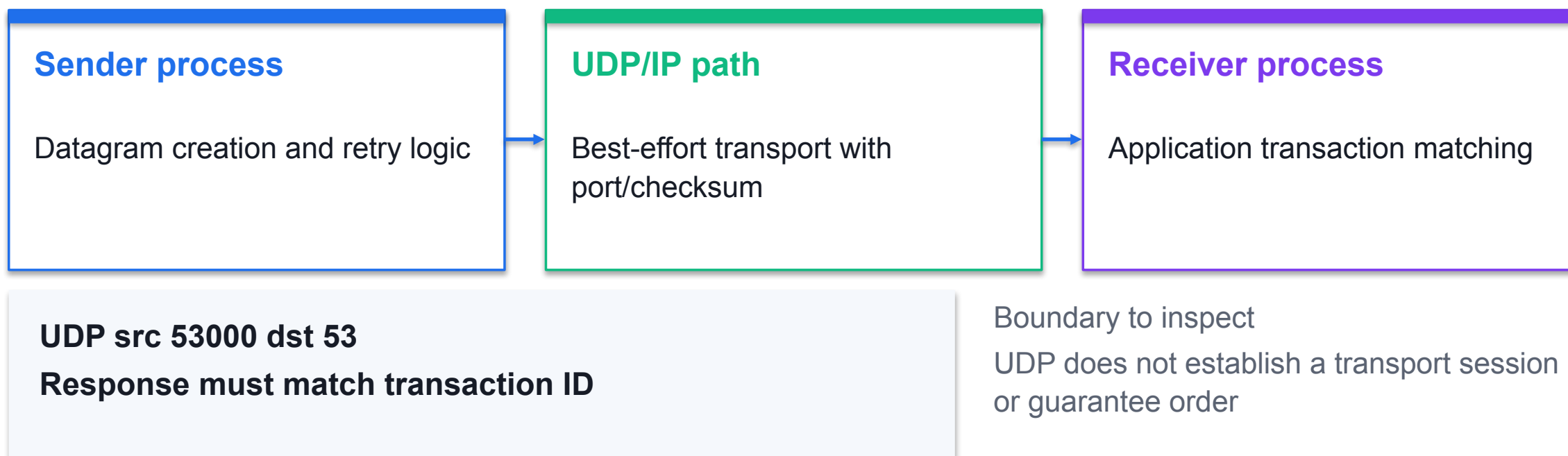
TCP session state · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Client socket -> Server socket	Client SYN seq=1000
Mechanism / state	FIN or RST terminates	Client ACK seq=1001 ack=7001
Result / boundary	Read flags and sequence/ack fields in a capture. Confirm whether the handshake completed before testing application data.	A SYN retransmission can indicate filtering or an absent listener. A RST indicates a different failure from silence.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

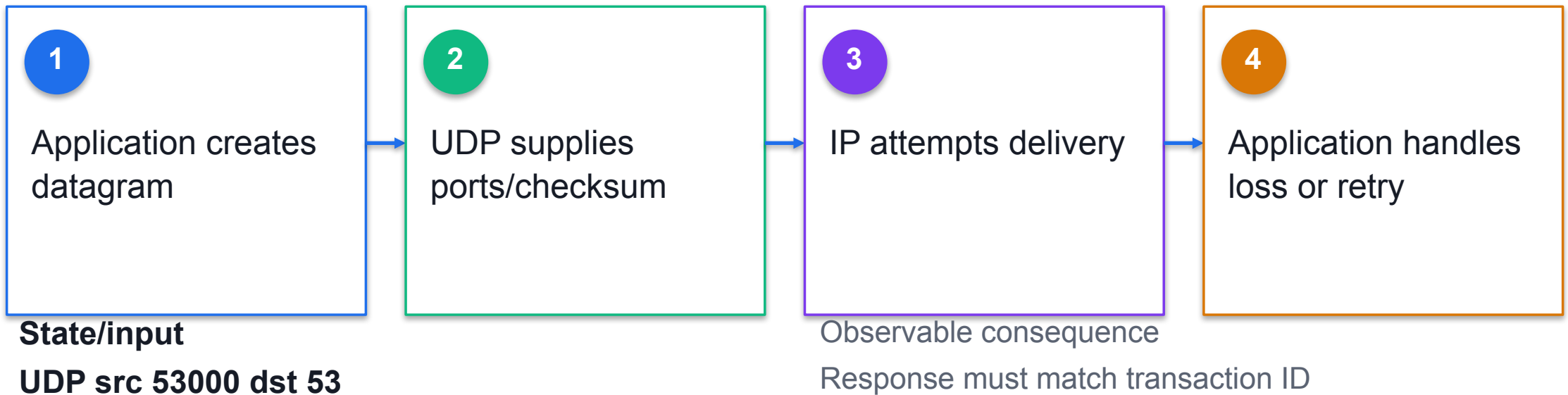
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

UDP delivery and application recovery · boundaries



Correlate request and response using application identifiers, endpoints and timestamps; inspect loss separately from latency.

UDP delivery and application recovery · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

UDP delivery and application recovery · exact contract

UDP src 53000 dst 53

DNS query ID 0x1234

Response must match transaction ID

INPUT / IDENTITY

Sender process · Application creates datagram

DECISION / STATE

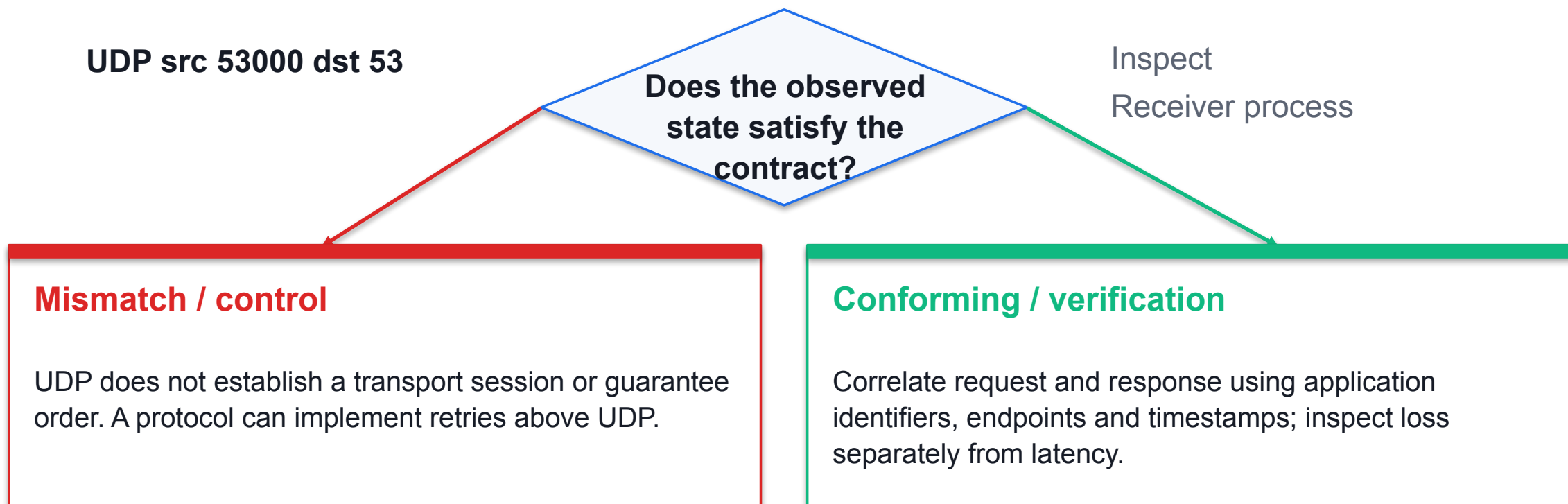
IP attempts delivery

READBACK / RESULT

Application handles loss or retry

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

UDP delivery and application recovery · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

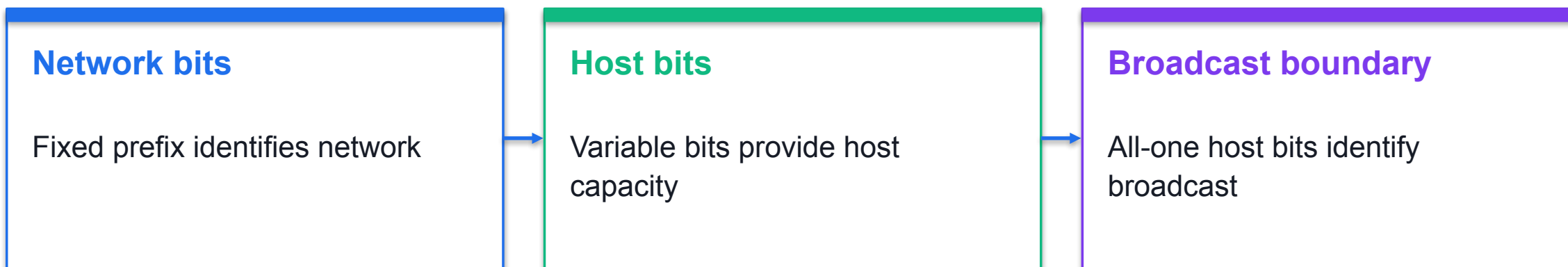
UDP delivery and application recovery · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Sender process -> Receiver process	UDP src 53000 dst 53
Mechanism / state	Application handles loss or retry	Response must match transaction ID
Result / boundary	Correlate request and response using application identifiers, endpoints and timestamps; inspect loss separately from latency.	UDP does not establish a transport session or guarantee order. A protocol can implement retries above UDP.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPv4 subnet boundaries · boundaries



192.0.2.65/26

Network .64; usable .65-.126; broadcast .127

Boundary to inspect

Do not include network or broadcast in a traditional subnet's host allocation

$2^{(32-26)} - 2 = 62$ usable addresses

IPv4 subnet boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPv4 subnet boundaries · exact contract

192.0.2.65/26

Mask 255.255.255.192

Network .64; usable .65-.126; broadcast .127

INPUT / IDENTITY

Network bits · Prefix fixes network portion

DECISION / STATE

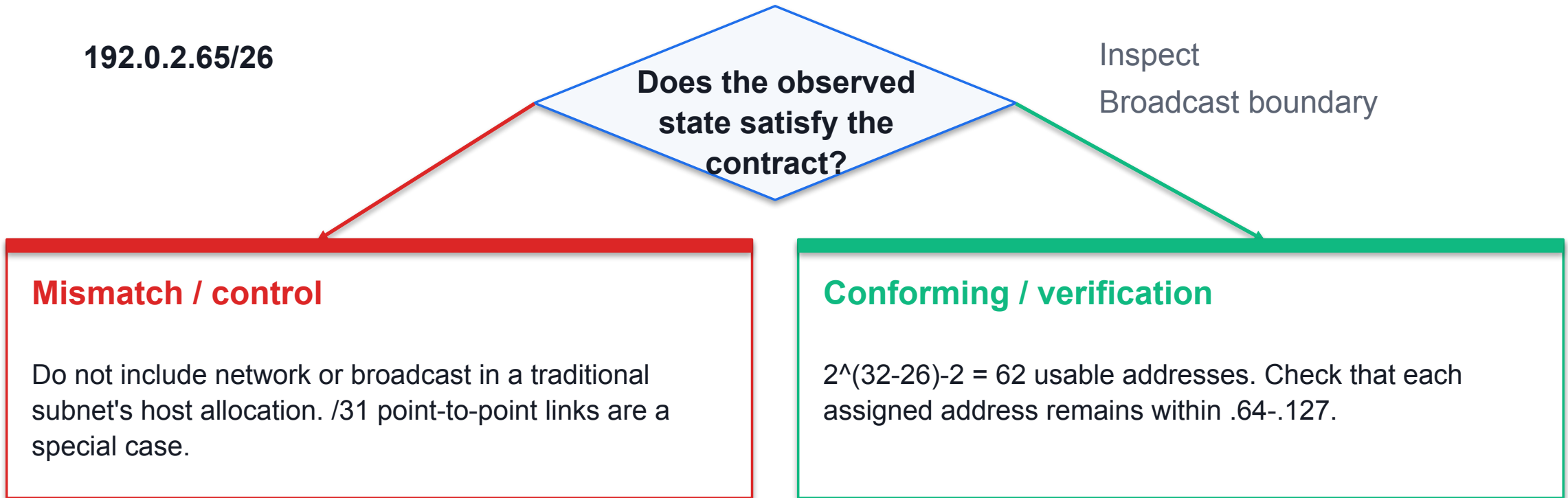
All-zero host identifies network

READBACK / RESULT

All-one host identifies broadcast

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPv4 subnet boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

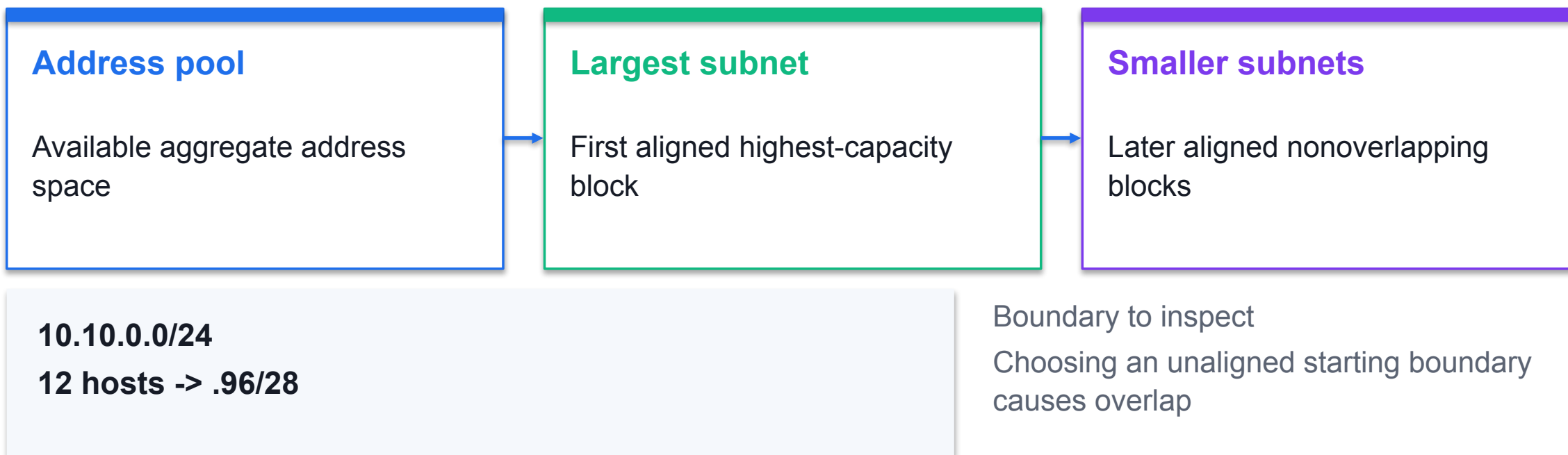
IPv4 subnet boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Network bits -> Broadcast boundary	192.0.2.65/26
Mechanism / state	All-one host identifies broadcast	Network .64; usable .65-.126; broadcast .127
Result / boundary	$2^{(32-26)}-2 = 62$ usable addresses. Check that each assigned address remains within .64-.127.	Do not include network or broadcast in a traditional subnet's host allocation. /31 point-to-point links are a special case.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

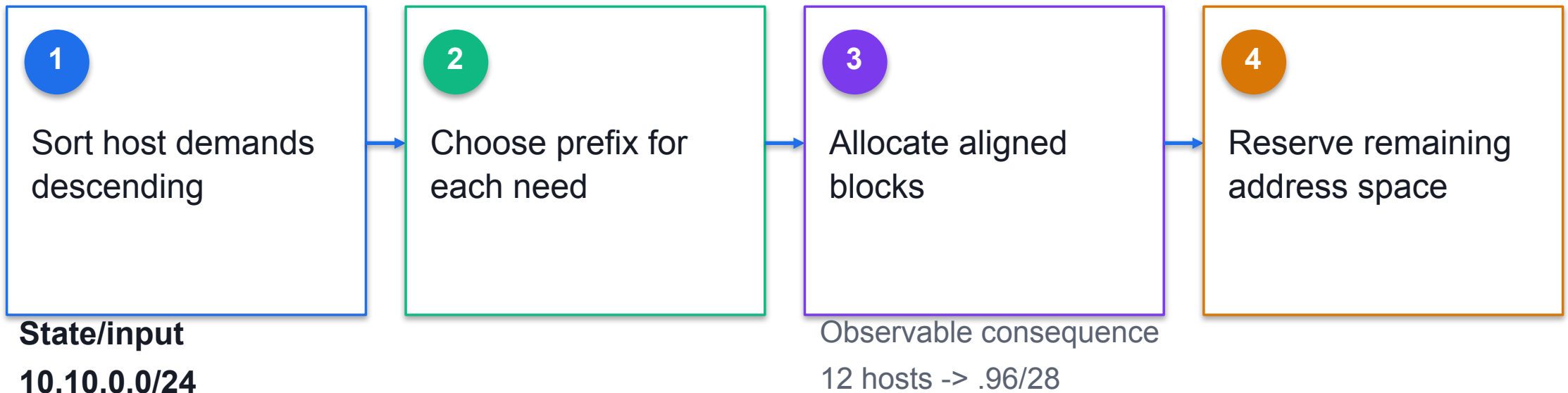
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

VLSM address allocation · boundaries



Prove allocated ranges do not intersect; compare usable host counts 62, 30, 14 with requirements 60, 28, 12.

VLSM address allocation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

VLSM address allocation · exact contract

10.10.0.0/24

60 hosts -> .0/26

28 hosts -> .64/27

12 hosts -> .96/28

INPUT / IDENTITY

Address pool · Sort host demands
descending

DECISION / STATE

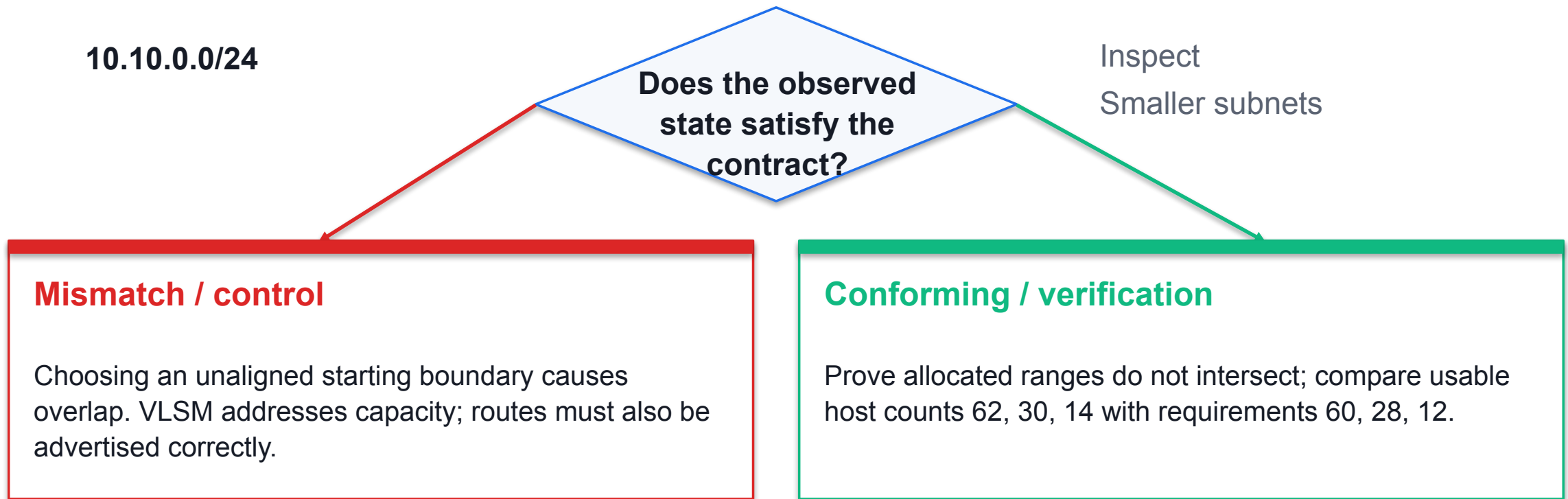
Allocate aligned blocks

READBACK / RESULT

Reserve remaining address space

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

VLSM address allocation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

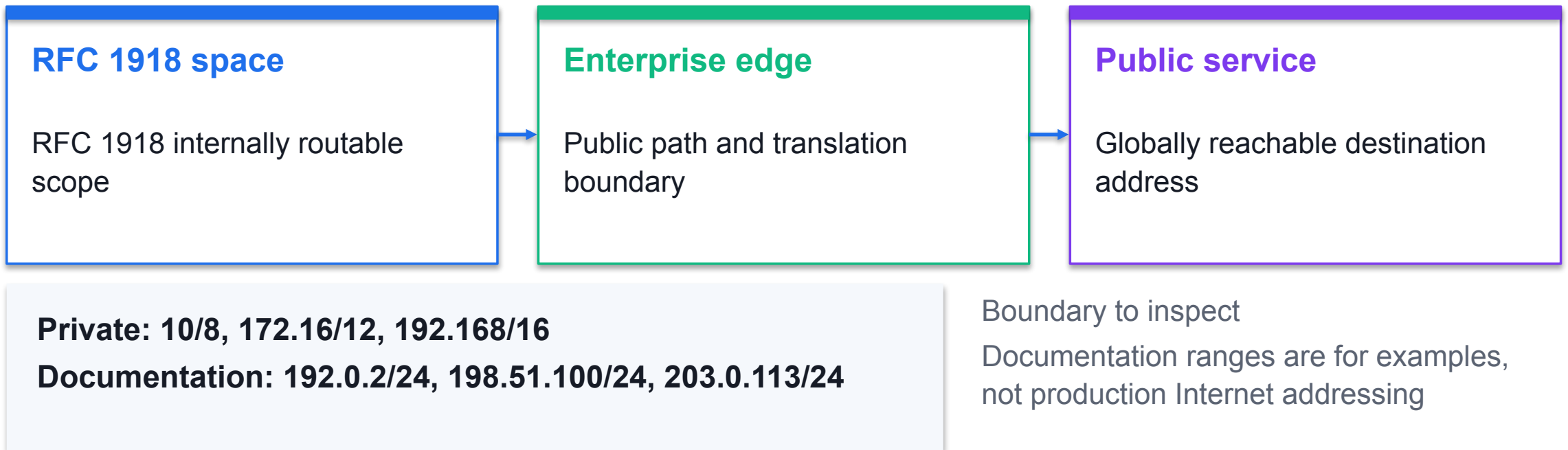
VLSM address allocation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Address pool -> Smaller subnets	10.10.0.0/24
Mechanism / state	Reserve remaining address space	12 hosts -> .96/28
Result / boundary	Prove allocated ranges do not intersect; compare usable host counts 62, 30, 14 with requirements 60, 28, 12.	Choosing an unaligned starting boundary causes overlap. VLSM addresses capacity; routes must also be advertised correctly.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

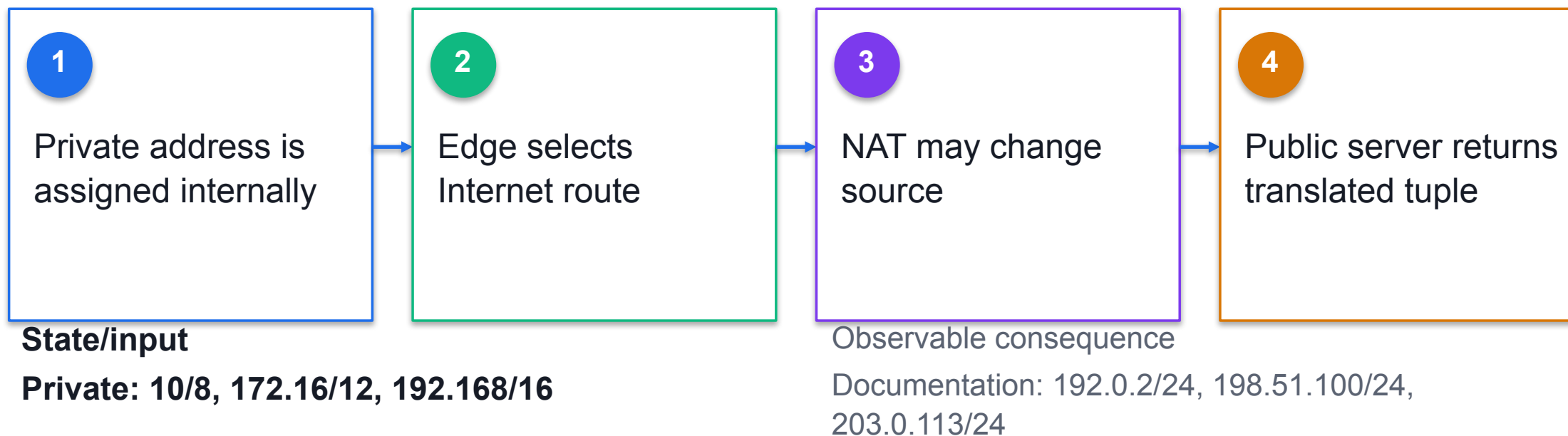
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Private and public addressing · boundaries



Classify each candidate prefix; preserve original and translated addresses in the incident report.

Private and public addressing · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Private and public addressing · exact contract

Private: 10/8, 172.16/12, 192.168/16

Documentation: 192.0.2/24, 198.51.100/24, 203.0.113/24

INPUT / IDENTITY

RFC 1918 space · Private address is assigned internally

DECISION / STATE

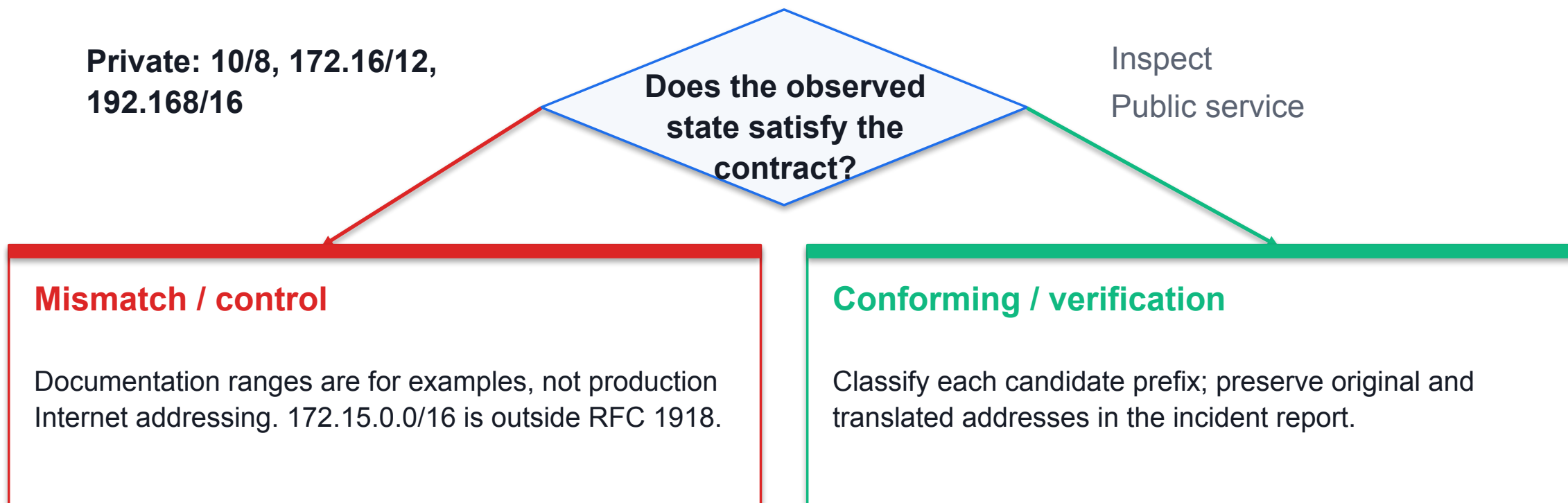
NAT may change source

READBACK / RESULT

Public server returns translated tuple

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Private and public addressing · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

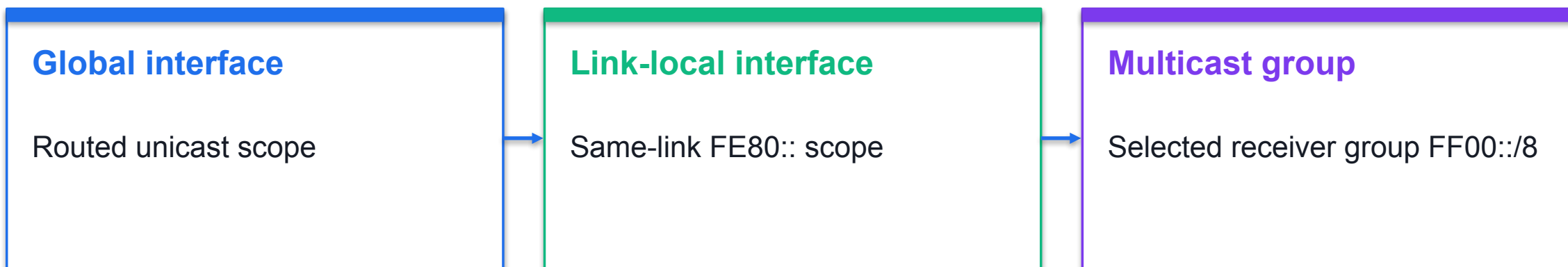
Private and public addressing · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	RFC 1918 space -> Public service	Private: 10/8, 172.16/12, 192.168/16
Mechanism / state	Public server returns translated tuple	Documentation: 192.0.2/24, 198.51.100/24, 203.0.113/24
Result / boundary	Classify each candidate prefix; preserve original and translated addresses in the incident report.	Documentation ranges are for examples, not production Internet addressing. 172.15.0.0/16 is outside RFC 1918.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPv6 address types · boundaries

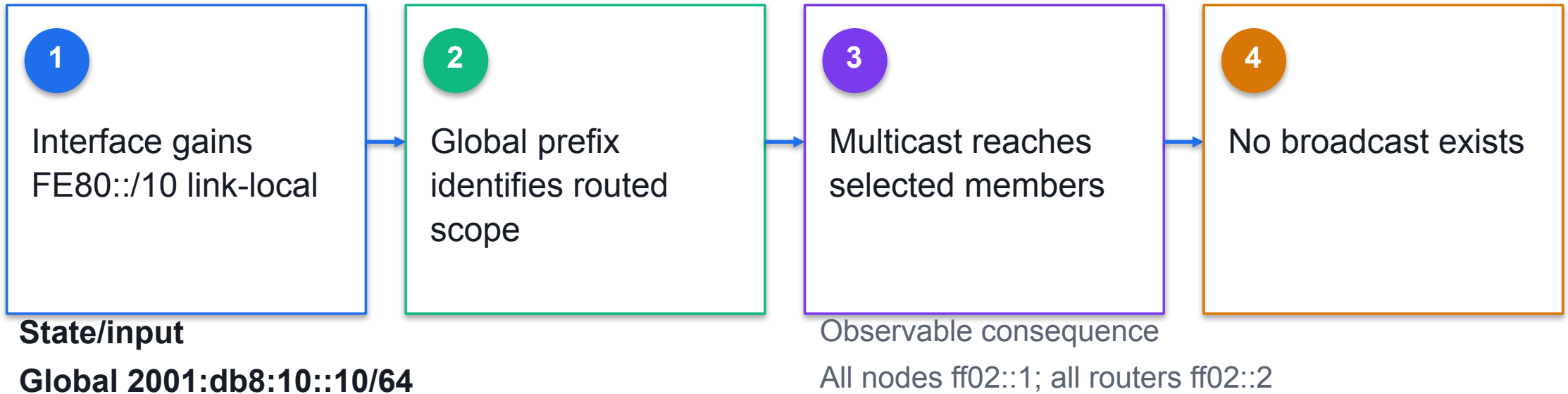


Global 2001:db8:10::10/64
All nodes ff02::1; all routers ff02::2

Boundary to inspect
A link-local next hop needs an outgoing interface scope

Classify global unicast, unique-local FC00::/7, link-local and multicast

IPv6 address types · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPv6 address types · exact contract

Global 2001:db8:10::10/64

Link-local fe80::10

All nodes ff02::1; all routers ff02::2

INPUT / IDENTITY

Global interface · Interface gains
FE80::/10 link-local

DECISION / STATE

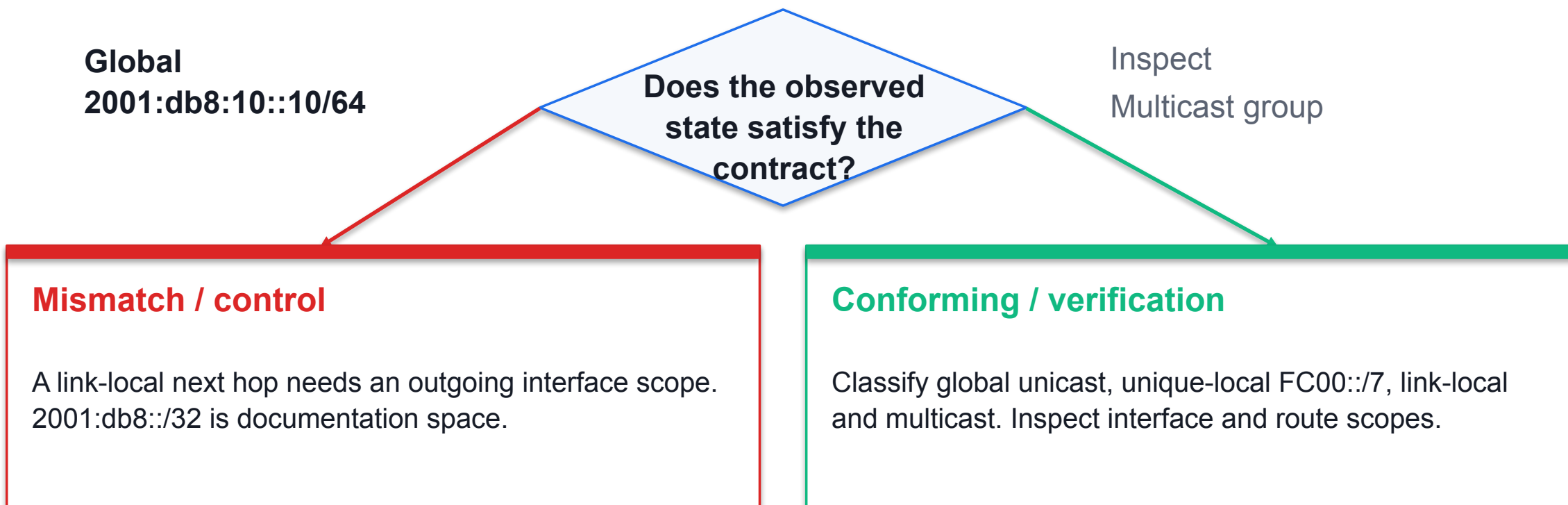
Multicast reaches selected members

READBACK / RESULT

No broadcast exists

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPv6 address types · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

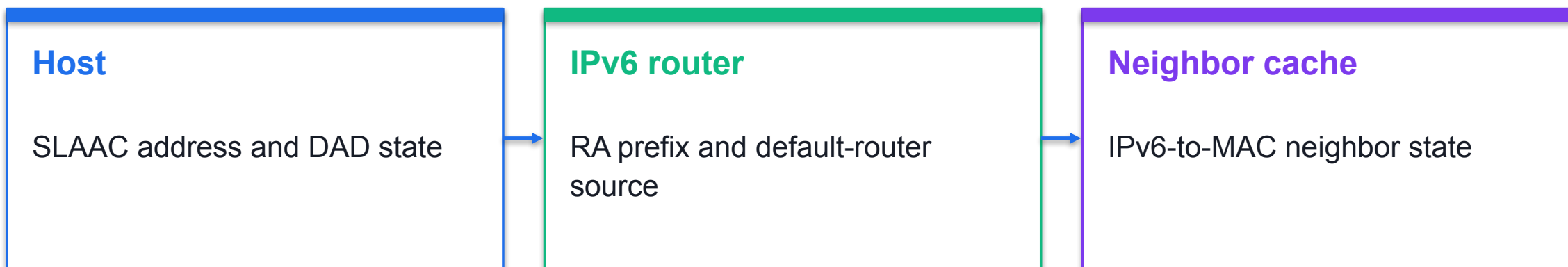
IPv6 address types · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Global interface -> Multicast group	Global 2001:db8:10::10/64
Mechanism / state	No broadcast exists	All nodes ff02::1; all routers ff02::2
Result / boundary	Classify global unicast, unique-local FC00::/7, link-local and multicast. Inspect interface and route scopes.	A link-local next hop needs an outgoing interface scope. 2001:db8::/32 is documentation space.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPv6 neighbor and SLAAC state · boundaries

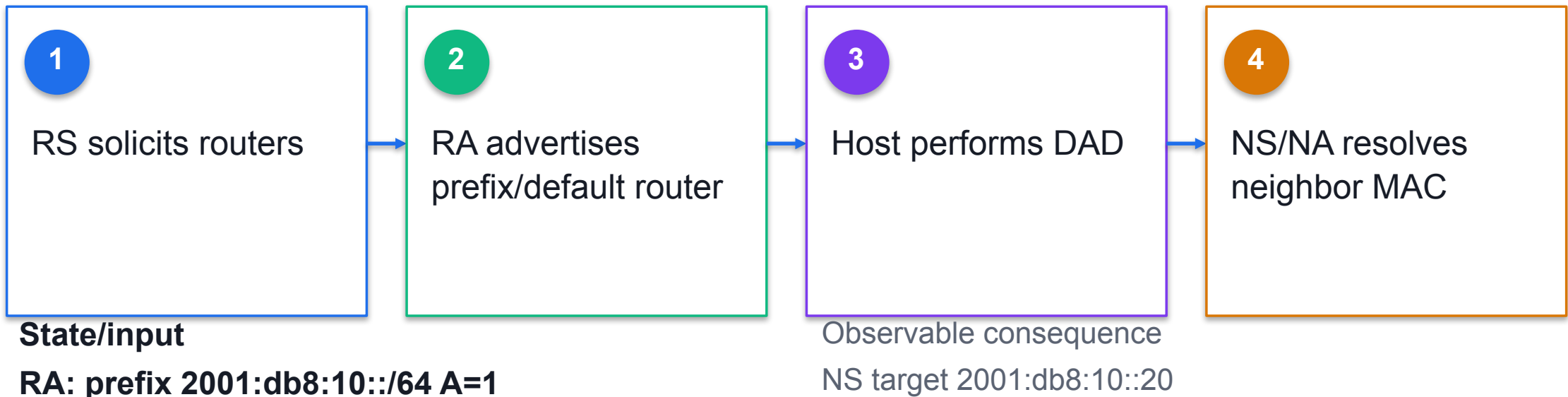


RA: prefix 2001:db8:10::/64 A=1
NS target 2001:db8:10::20

Boundary to inspect
 Blocking necessary ICMPv 6 breaks
 neighbor discovery and path operation

Verify RA prefix, tentative-to-preferred DAD state and neighbor cache entry

IPv6 neighbor and SLAAC state · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPv6 neighbor and SLAAC state · exact contract

RA: prefix 2001:db8:10::/64 A=1
Default router fe80::1
NS target 2001:db8:10::20

INPUT / IDENTITY

Host · RS solicits routers

DECISION / STATE

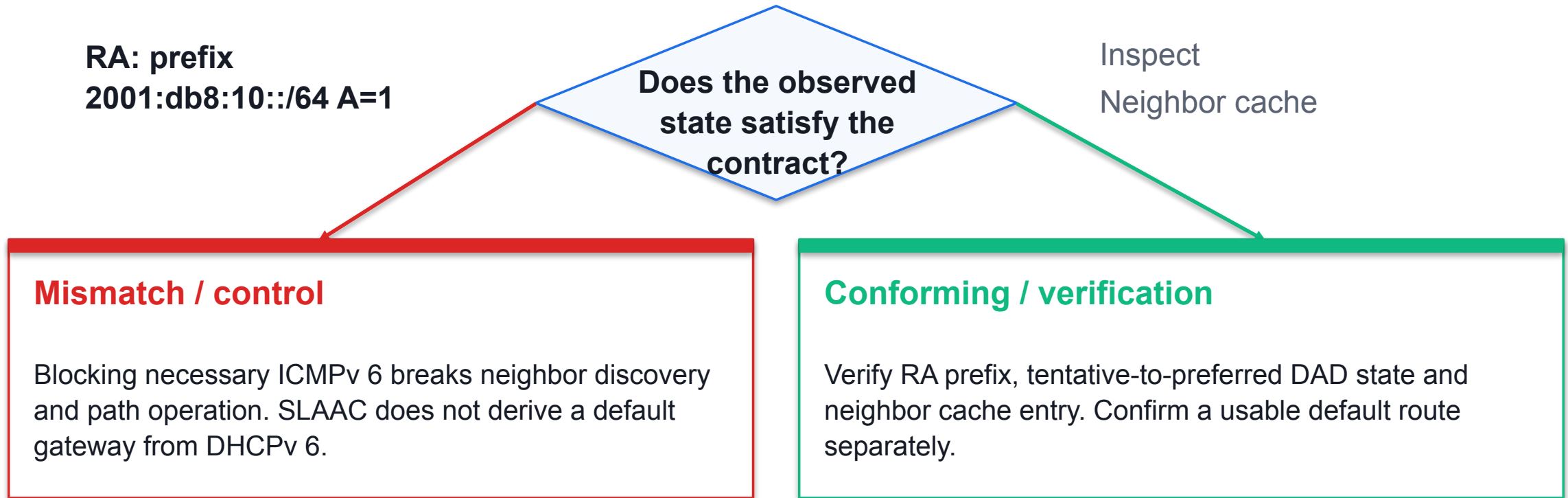
Host performs DAD

READBACK / RESULT

NS/NA resolves neighbor MAC

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPv6 neighbor and SLAAC state · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

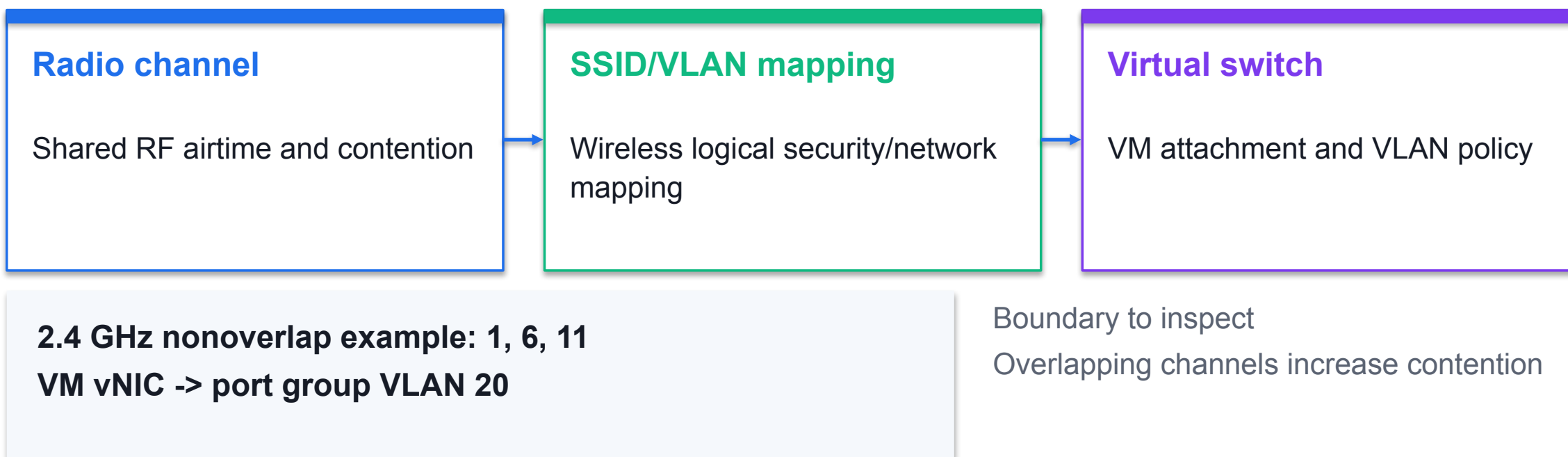
IPv6 neighbor and SLAAC state · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Host -> Neighbor cache	RA: prefix 2001:db8:10::/64 A=1
Mechanism / state	NS/NA resolves neighbor MAC	NS target 2001:db8:10::20
Result / boundary	Verify RA prefix, tentative-to-preferred DAD state and neighbor cache entry. Confirm a usable default route separately.	Blocking necessary ICMPv 6 breaks neighbor discovery and path operation. SLAAC does not derive a default gateway from DHCPv 6.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

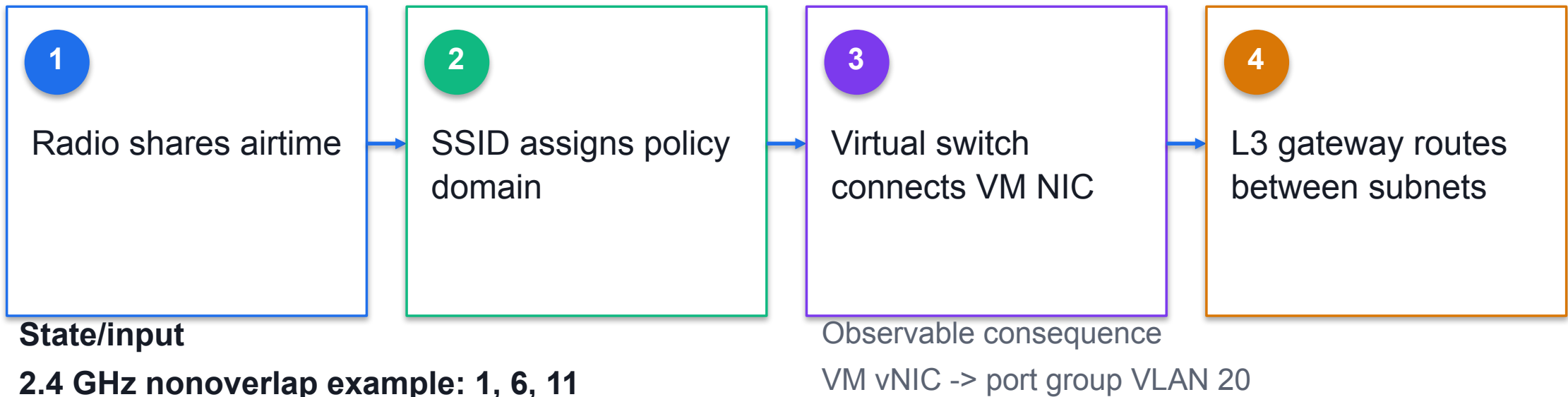
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Wireless channel and virtual boundaries · boundaries



Compare channel utilization, SSID security, VM VLAN tag and host IP/gateway; separate RF evidence from virtual-switch evidence.

Wireless channel and virtual boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Wireless channel and virtual boundaries · exact contract

2.4 GHz nonoverlap example: 1, 6, 11

Staff SSID -> VLAN 20

VM vNIC -> port group VLAN 20

INPUT / IDENTITY

Radio channel · Radio shares airtime

DECISION / STATE

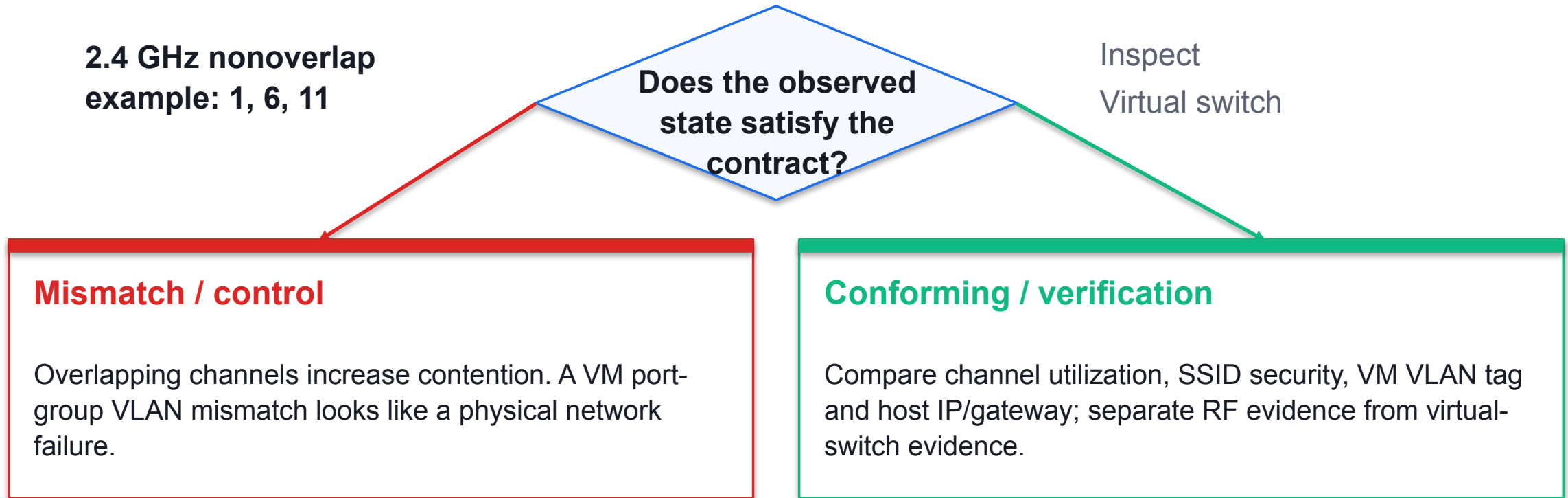
Virtual switch connects VM NIC

READBACK / RESULT

L3 gateway routes between subnets

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Wireless channel and virtual boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

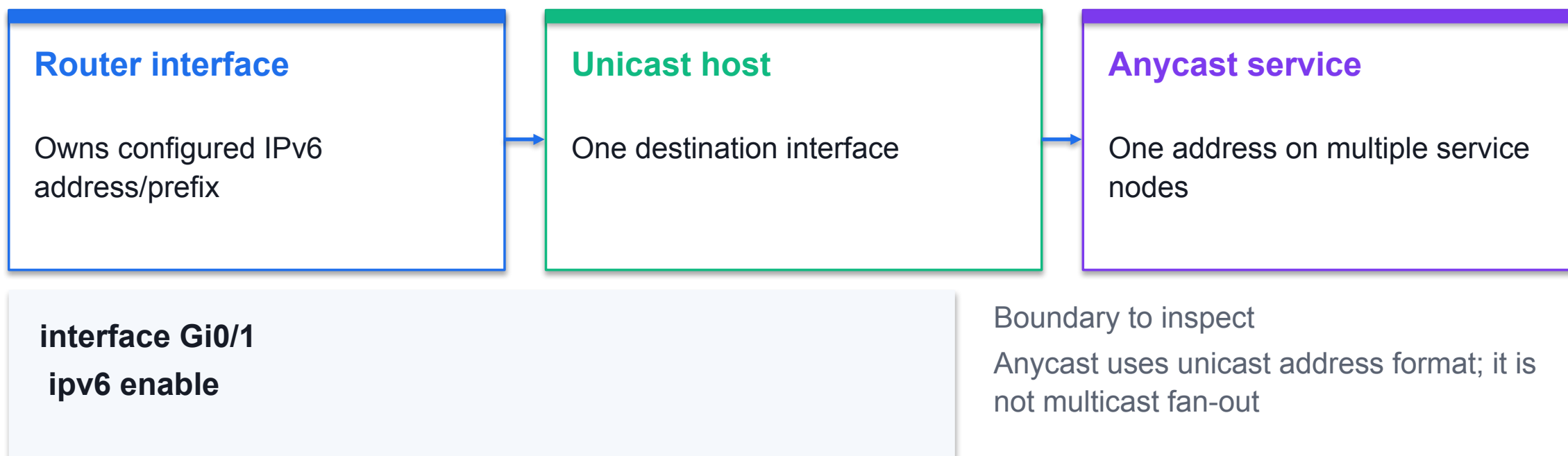
Wireless channel and virtual boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Radio channel -> Virtual switch	2.4 GHz nonoverlap example: 1, 6, 11
Mechanism / state	L3 gateway routes between subnets	VM vNIC -> port group VLAN 20
Result / boundary	Compare channel utilization, SSID security, VM VLAN tag and host IP/gateway; separate RF evidence from virtual-switch evidence.	Overlapping channels increase contention. A VM port-group VLAN mismatch looks like a physical network failure.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

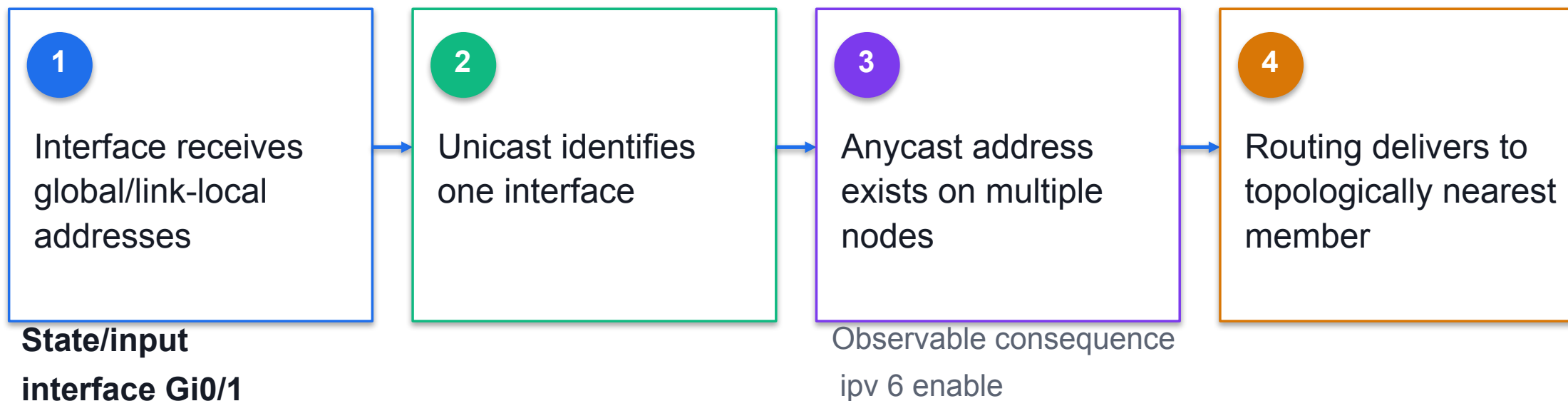
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPv6 interface and anycast contracts · boundaries



show ipv 6 interface brief; compare configured /64 and address scope

IPv6 interface and anycast contracts · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPv6 interface and anycast contracts · exact contract

```
interface Gi0/1
  ipv6 address 2001:db8:10::1/64
  ipv6 address fe80::1 link-local
  ipv6 enable
```

INPUT / IDENTITY

Router interface · Interface receives global/link-local addresses

DECISION / STATE

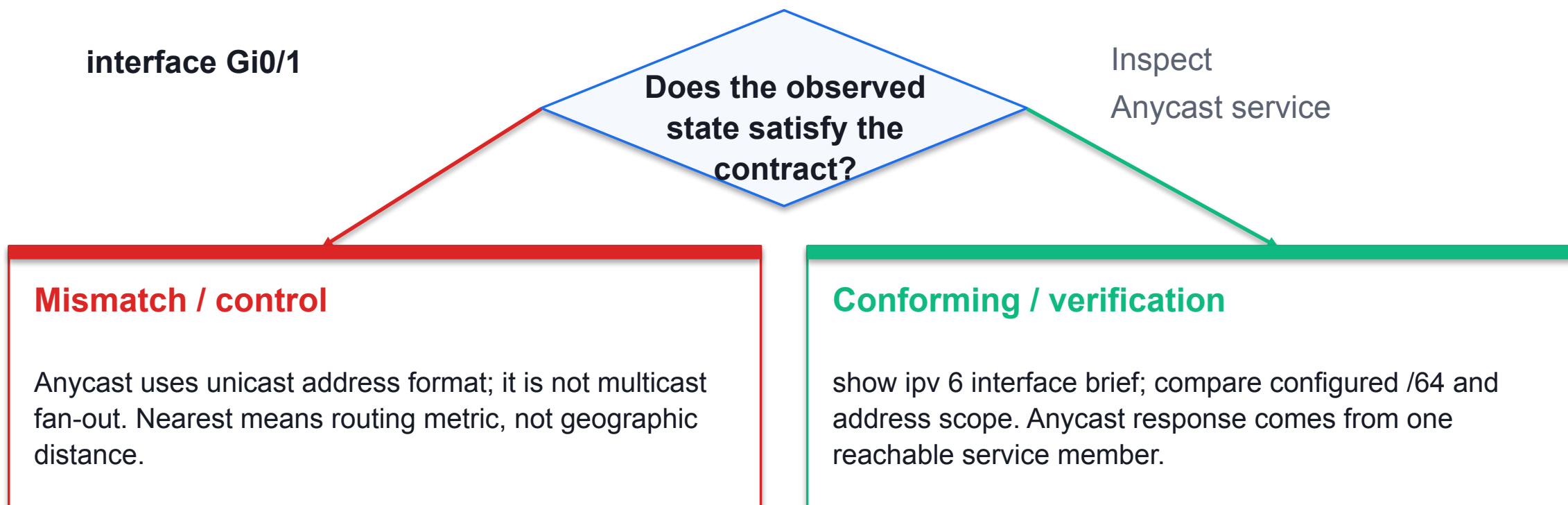
Anycast address exists on multiple nodes

READBACK / RESULT

Routing delivers to topologically nearest member

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPv6 interface and anycast contracts · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

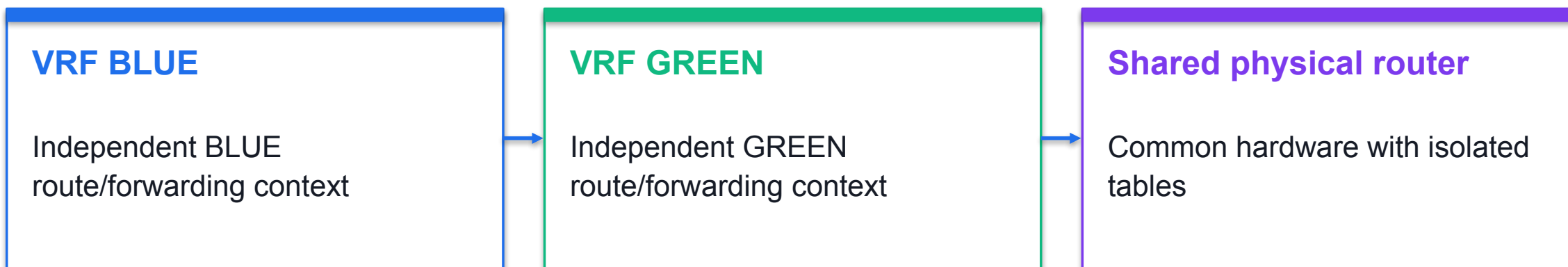
IPv6 interface and anycast contracts · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Router interface -> Anycast service	interface Gi0/1
Mechanism / state	Routing delivers to topologically nearest member	ipv 6 enable
Result / boundary	show ipv 6 interface brief; compare configured /64 and address scope. Anycast response comes from one reachable service member.	Anycast uses unicast address format; it is not multicast fan-out. Nearest means routing metric, not geographic distance.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

VRF routing isolation · boundaries



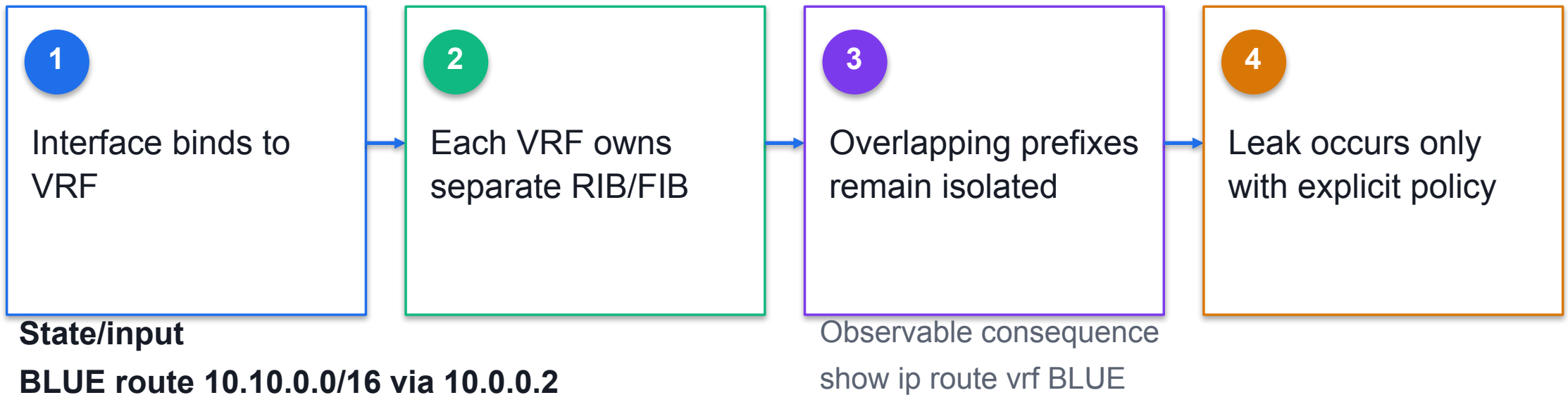
BLUE route 10.10.0.0/16 via 10.0.0.2
show ip route vrf BLUE

Boundary to inspect

Do not inspect the global route table and
assume a VRF-bound interface has the
same paths

Verify interface VRF assignment and VRF-specific route/ARP evidence; same prefix can validly use different next hops.

VRF routing isolation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

VRF routing isolation · exact contract

BLUE route 10.10.0.0/16 via 10.0.0.2
GREEN route 10.10.0.0/16 via 10.0.0.6
show ip route vrf BLUE

INPUT / IDENTITY

VRF BLUE · Interface binds to VRF

DECISION / STATE

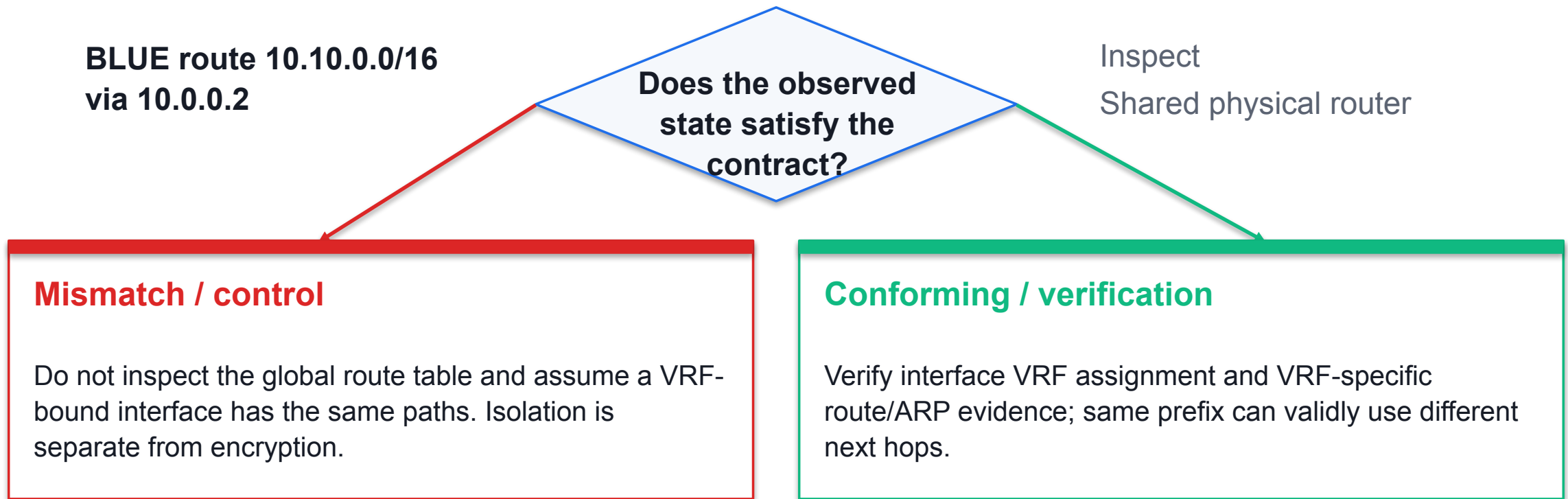
Overlapping prefixes remain isolated

READBACK / RESULT

Leak occurs only with explicit policy

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

VRF routing isolation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

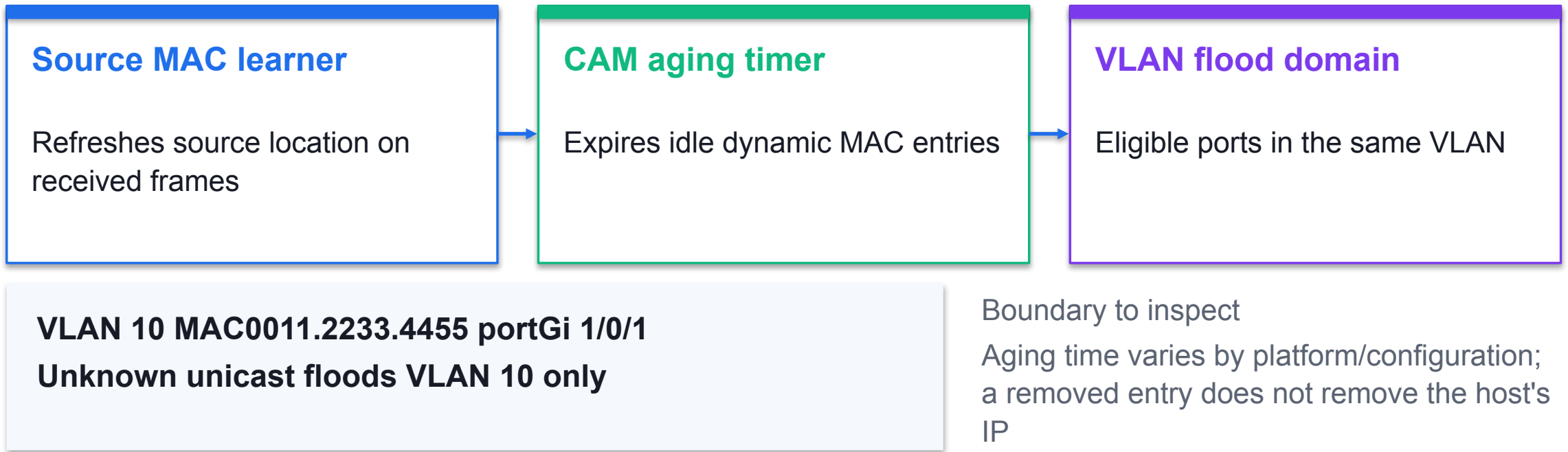
VRF routing isolation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	VRF BLUE -> Shared physical router	BLUE route 10.10.0.0/16 via 10.0.0.2
Mechanism / state	Leak occurs only with explicit policy	show ip route vrf BLUE
Result / boundary	Verify interface VRF assignment and VRF-specific route/ARP evidence; same prefix can validly use different next hops.	Do not inspect the global route table and assume a VRF-bound interface has the same paths. Isolation is separate from encryption.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

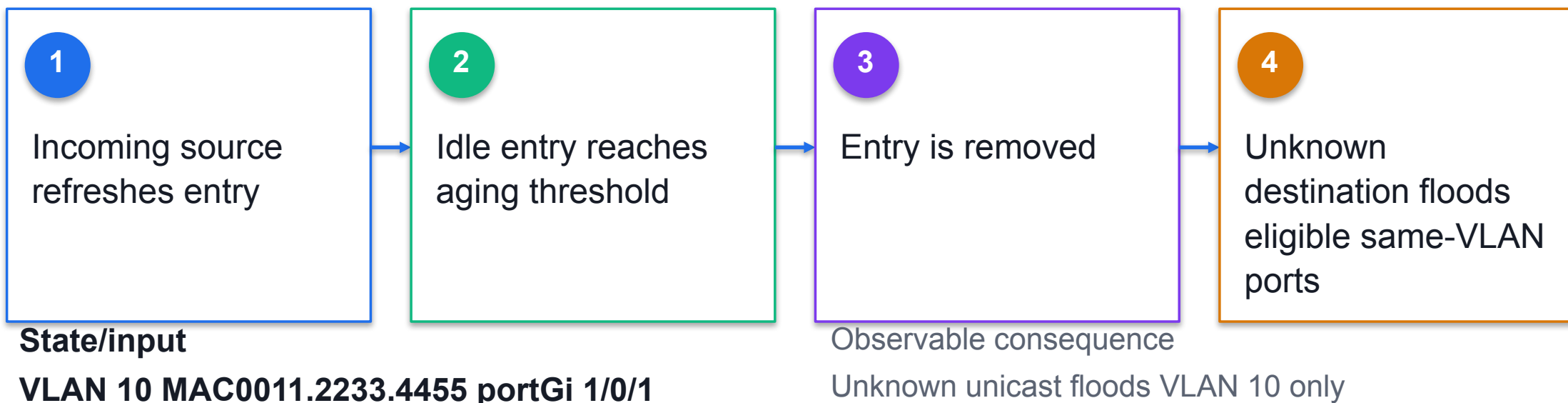
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

MAC aging and unknown-unicast scope · boundaries



show mac address-table aging-time; compare entry before/after inactivity and source relearning; distinguish CAM aging from ARP aging.

MAC aging and unknown-unicast scope · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

MAC aging and unknown-unicast scope · exact contract

VLAN 10 MAC0011.2233.4455 portGi 1/0/1
Dynamic aging timer 300 s example
Unknown unicast floods VLAN 10 only

INPUT / IDENTITY

Source MAC learner · Incoming source refreshes entry

DECISION / STATE

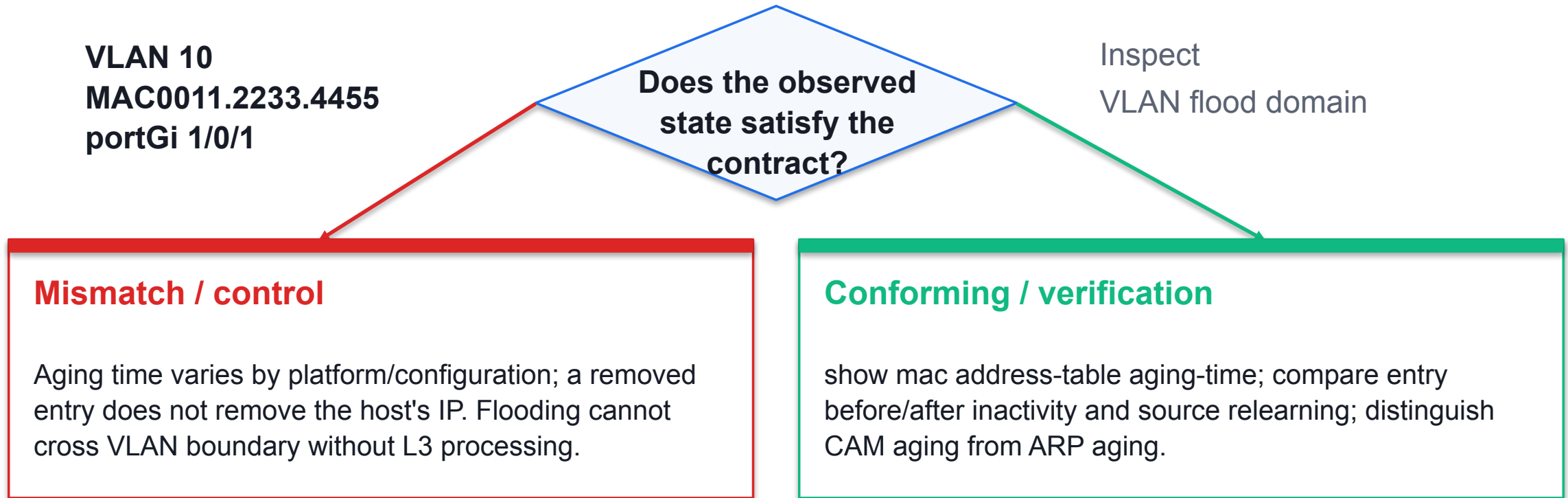
Entry is removed

READBACK / RESULT

Unknown destination floods eligible same-VLAN ports

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

MAC aging and unknown-unicast scope · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

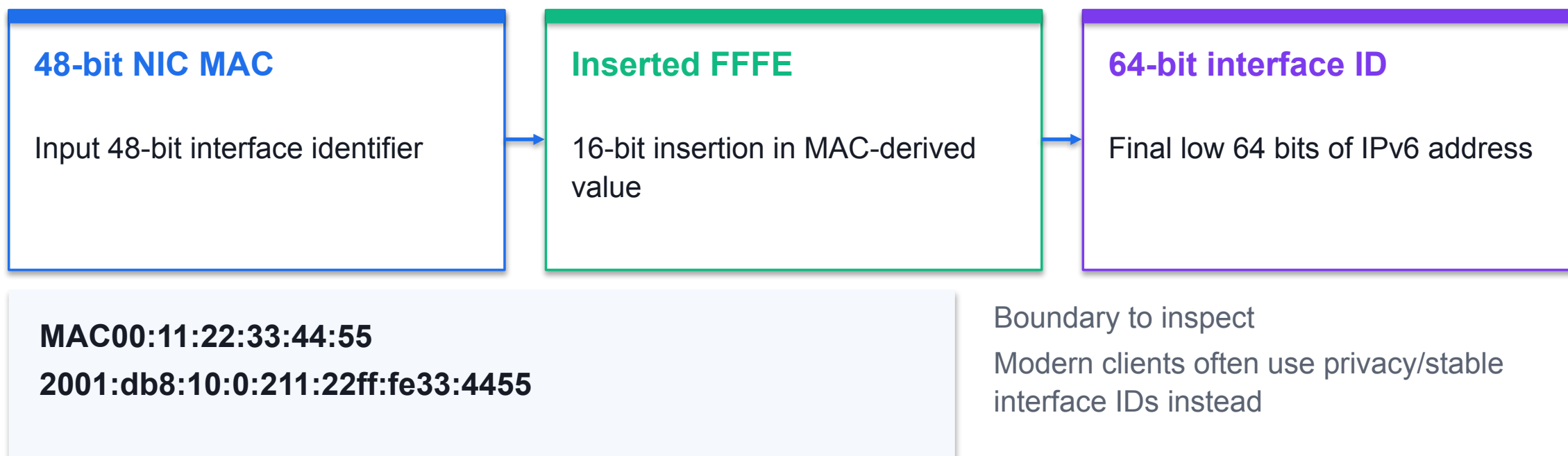
MAC aging and unknown-unicast scope · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Source MAC learner -> VLAN flood domain	VLAN 10 MAC0011.2233.4455 portGi 1/0/1
Mechanism / state	Unknown destination floods eligible same-VLAN ports	Unknown unicast floods VLAN 10 only
Result / boundary	show mac address-table aging-time; compare entry before/after inactivity and source relearning; distinguish CAM aging from ARP aging.	Aging time varies by platform/configuration; a removed entry does not remove the host's IP. Flooding cannot cross VLAN boundary without L3 processing.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

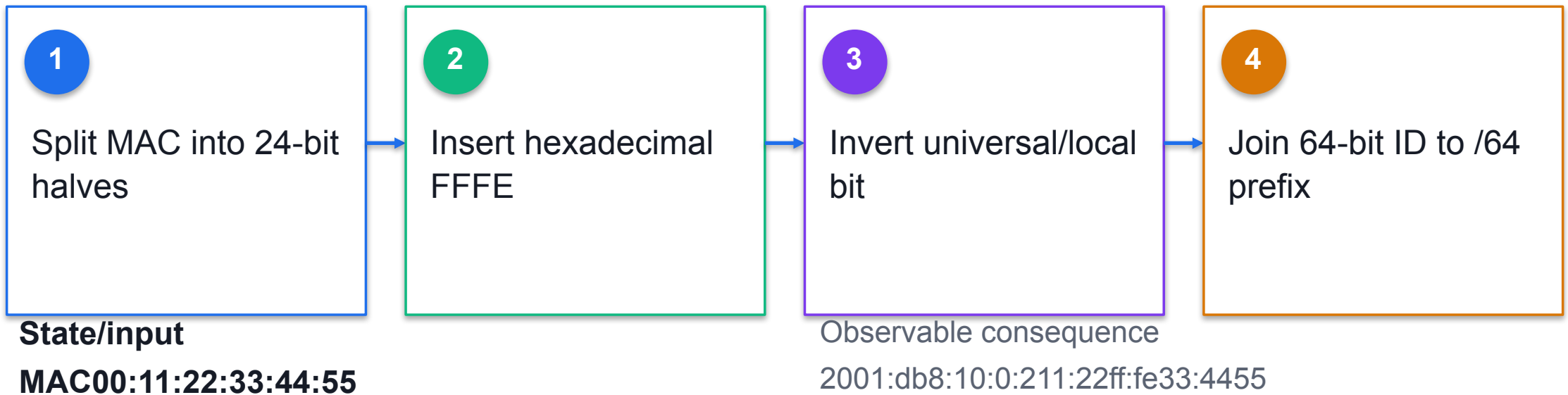
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Modified EUI-64 address construction · boundaries



Prove 00 xor 02 = 02 and exact inserted FFFE placement; compare prefix plus 64-bit interface identifier.

Modified EUI-64 address construction · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Modified EUI-64 address construction · exact contract

MAC00:11:22:33:44:55

ID0211:22ff:fe33:4455

2001:db8:10:0:211:22ff:fe33:4455

INPUT / IDENTITY

48-bit NIC MAC · Split MAC into 24-bit halves

DECISION / STATE

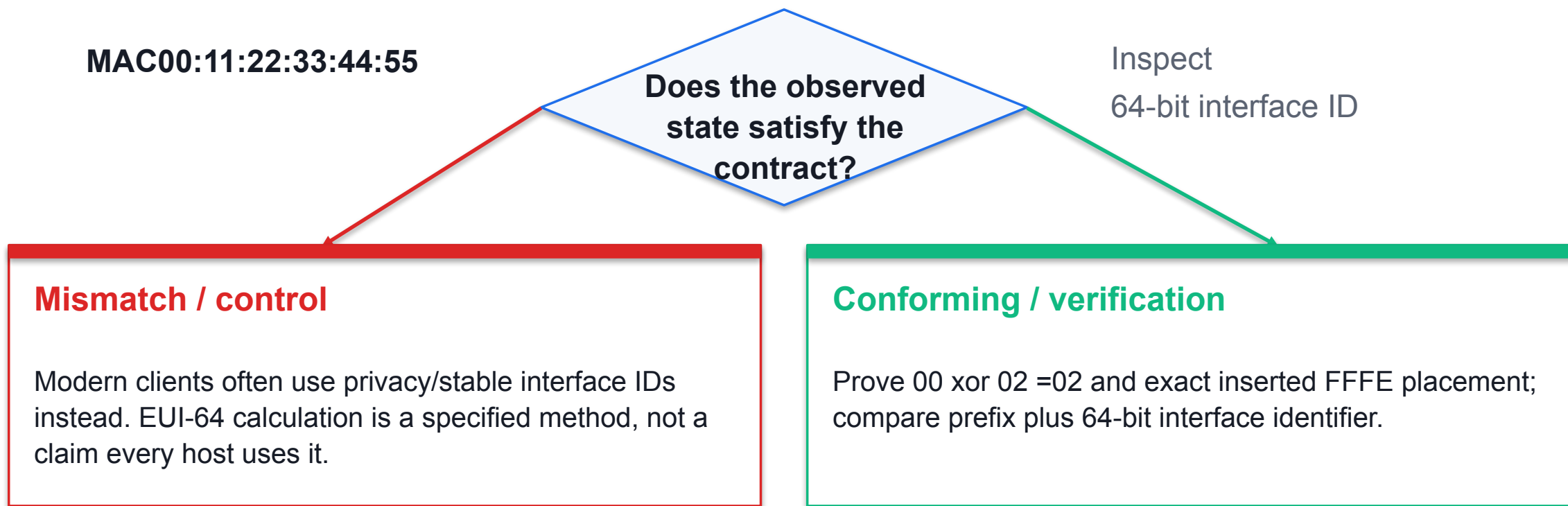
Invert universal/local bit

READBACK / RESULT

Join 64-bit ID to /64 prefix

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Modified EUI-64 address construction · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

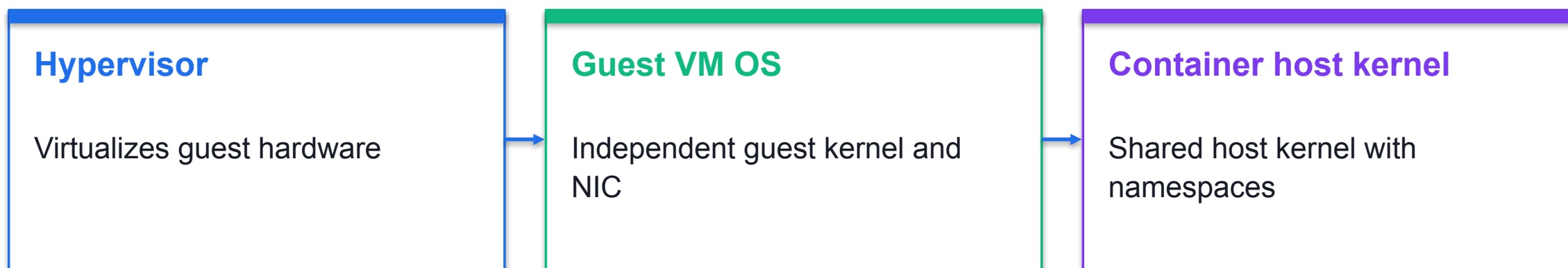
Modified EUI-64 address construction · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	48-bit NIC MAC -> 64-bit interface ID	MAC00:11:22:33:44:55
Mechanism / state	Join 64-bit ID to /64 prefix	2001:db8:10:0:211:22ff:fe33:4455
Result / boundary	Prove 00 xor 02 =02 and exact inserted FFFE placement; compare prefix plus 64-bit interface identifier.	Modern clients often use privacy/stable interface IDs instead. EUI-64 calculation is a specified method, not a claim every host uses it.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Hypervisor VM and container isolation · boundaries

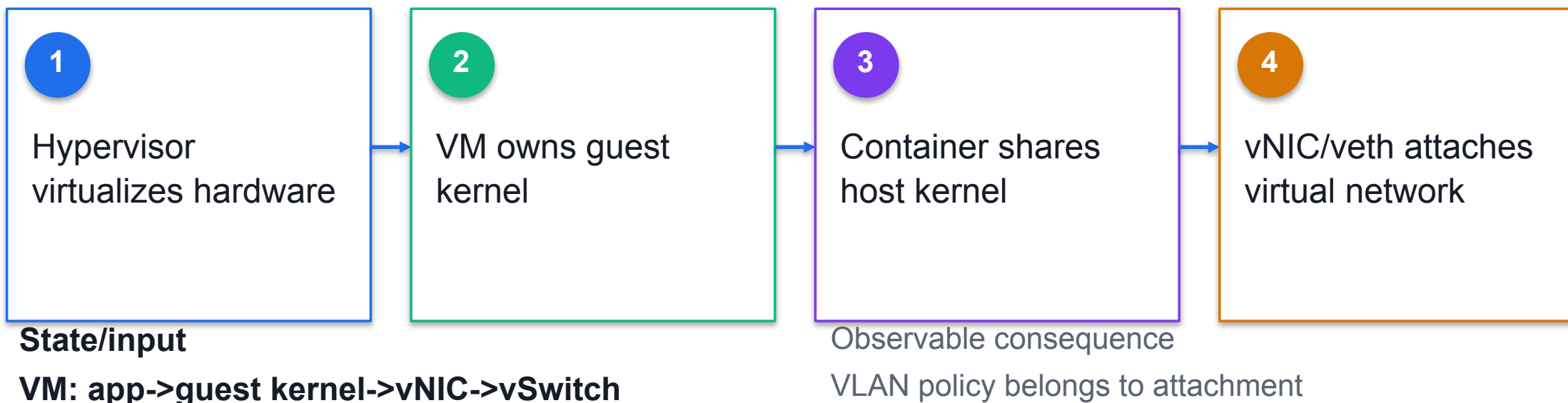


VM: app->guest kernel->vNIC->vSwitch
VLAN policy belongs to attachment

Boundary to inspect
 A container is not a separate guest kernel

Trace address, virtual NIC/bridge, VLAN and gateway; compare host versus guest responsibilities during a connectivity incident.

Hypervisor VM and container isolation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Hypervisor VM and container isolation · exact contract

VM: app->guest kernel->vNIC->vSwitch

Container: app->shared kernel->veth/bridge

VLAN policy belongs to attachment

INPUT / IDENTITY

Hypervisor · Hypervisor virtualizes hardware

DECISION / STATE

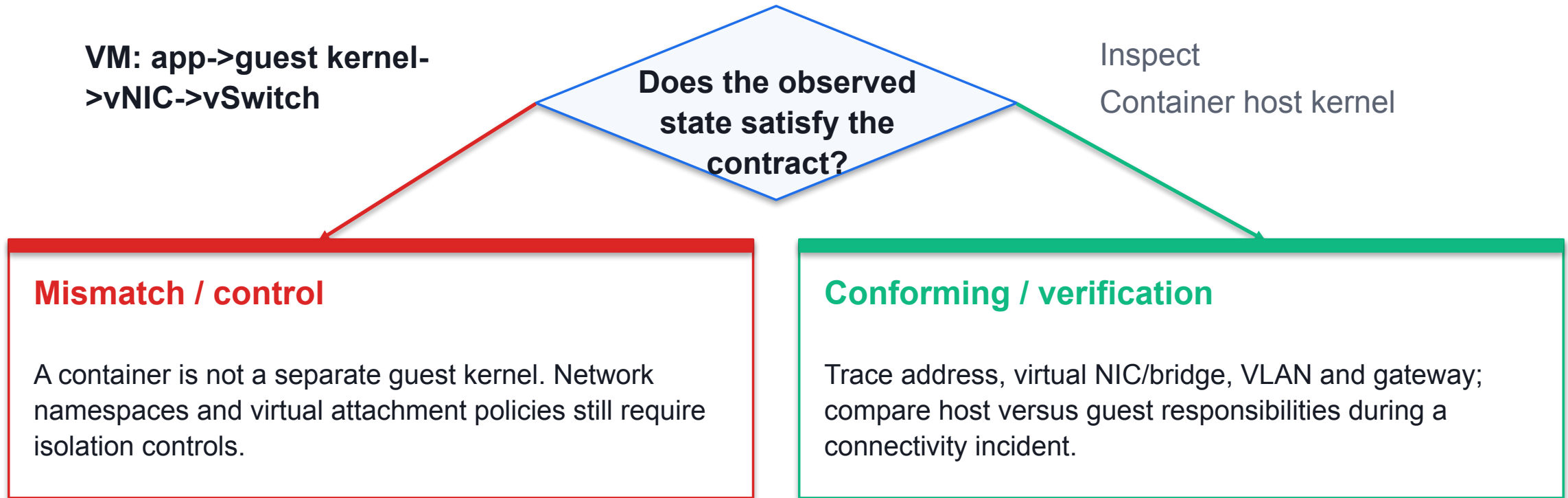
Container shares host kernel

READBACK / RESULT

vNIC/veth attaches virtual network

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Hypervisor VM and container isolation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

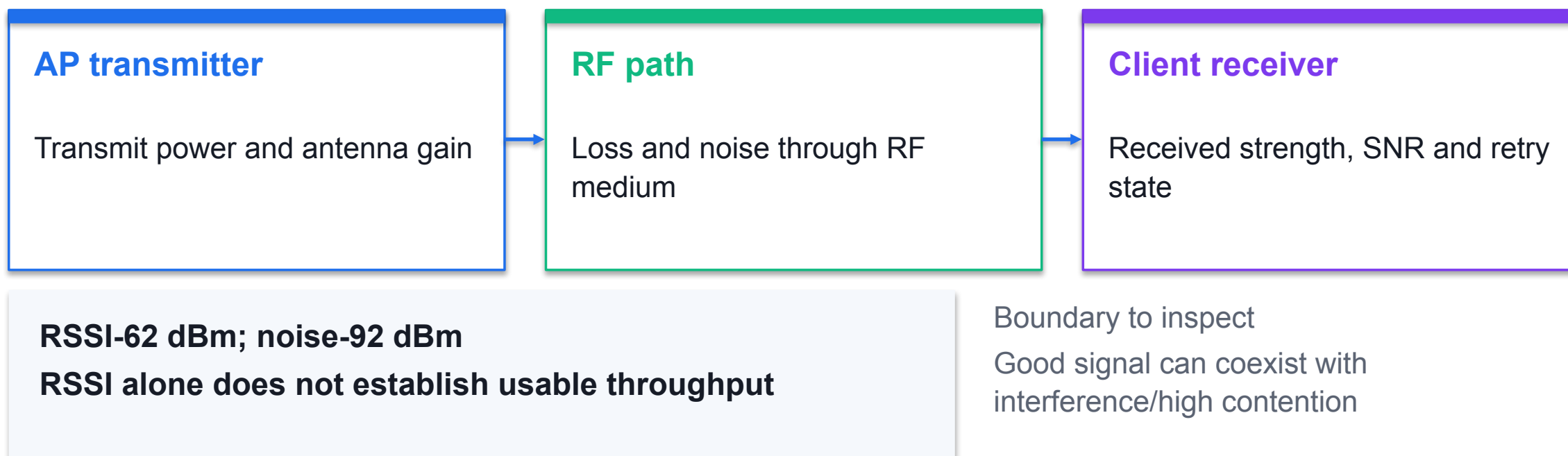
Hypervisor VM and container isolation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Hypervisor -> Container host kernel	VM: app->guest kernel->vNIC->vSwitch
Mechanism / state	vNIC/veth attaches virtual network	VLAN policy belongs to attachment
Result / boundary	Trace address, virtual NIC/bridge, VLAN and gateway; compare host versus guest responsibilities during a connectivity incident.	A container is not a separate guest kernel. Network namespaces and virtual attachment policies still require isolation controls.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

RF link-budget and interference evidence · boundaries



Compare SNR, utilization, retry rate and client throughput; identify whether issue is signal strength, noise or airtime contention.

RF link-budget and interference evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

RF link-budget and interference evidence · exact contract

RSSI-62 dBm; noise-92 dBm

SNR=30 dB

Channel utilization 75%

RSSI alone does not establish usable throughput

INPUT / IDENTITY

AP transmitter · Transmit power enters antenna

DECISION / STATE

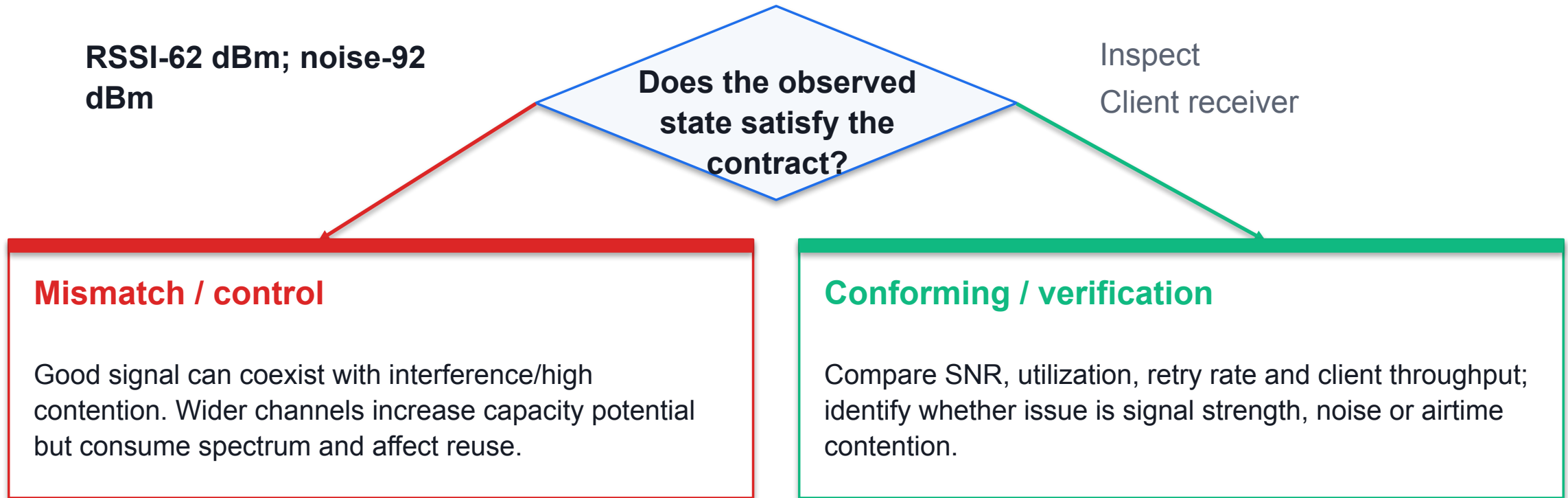
Noise determines signal margin

READBACK / RESULT

Rate adaptation responds to channel conditions

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

RF link-budget and interference evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

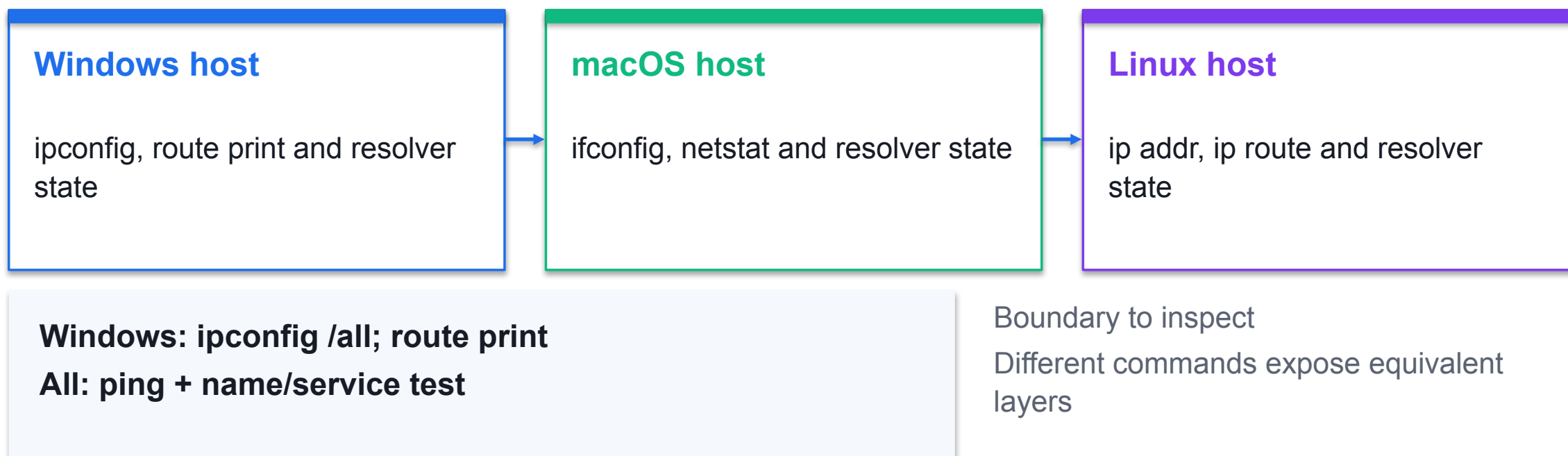
RF link-budget and interference evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	AP transmitter -> Client receiver	RSSI-62 dBm; noise-92 dBm
Mechanism / state	Rate adaptation responds to channel conditions	RSSI alone does not establish usable throughput
Result / boundary	Compare SNR, utilization, retry rate and client throughput; identify whether issue is signal strength, noise or airtime contention.	Good signal can coexist with interference/high contention. Wider channels increase capacity potential but consume spectrum and affect reuse.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

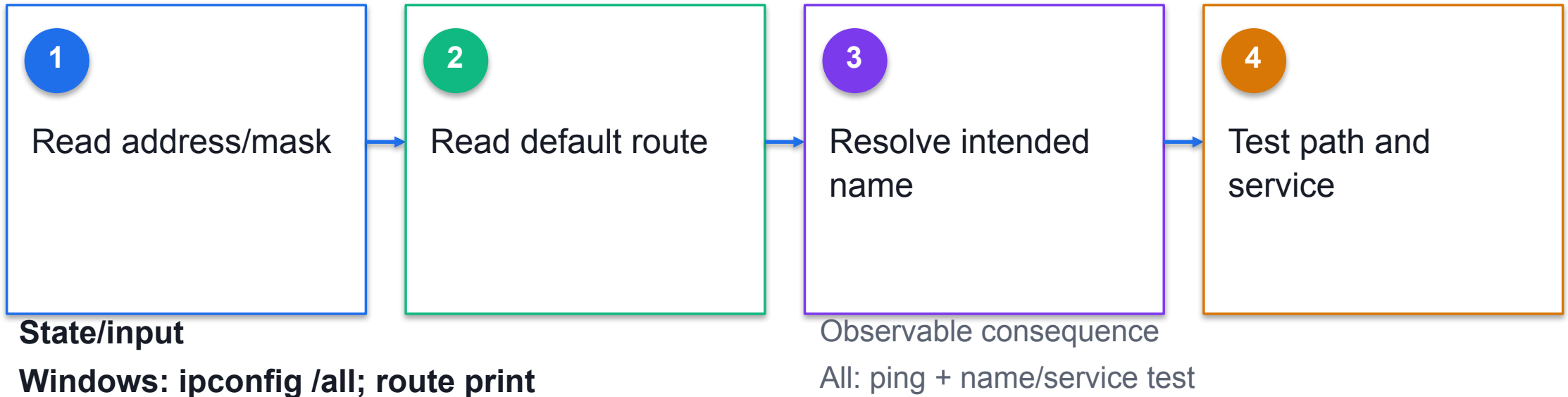
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Cross-platform client diagnostics · boundaries



Build evidence table for address, prefix, gateway, DNS, association/security and first failing path control.

Cross-platform client diagnostics · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Cross-platform client diagnostics · exact contract

Windows: `ipconfig /all; route print`

macOS: `ifconfig; netstat -rn`

Linux: `ip addr; ip route`

All: `ping + name/service test`

INPUT / IDENTITY

Windows host · Read address/mask

DECISION / STATE

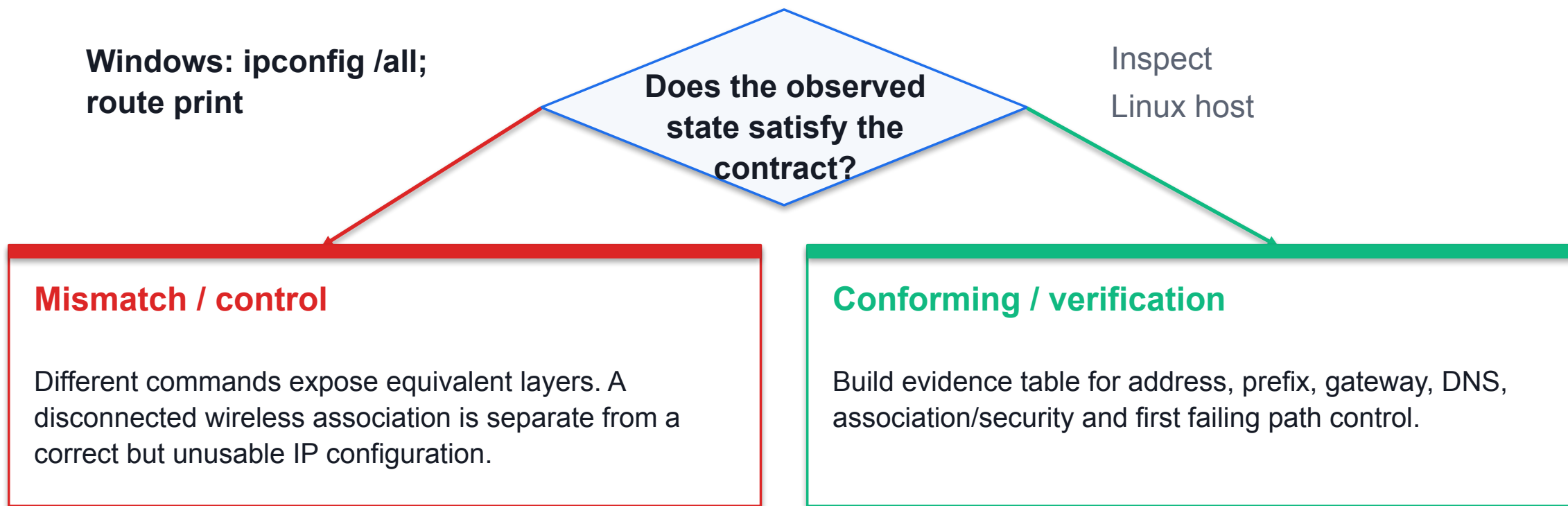
Resolve intended name

READBACK / RESULT

Test path and service

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Cross-platform client diagnostics · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

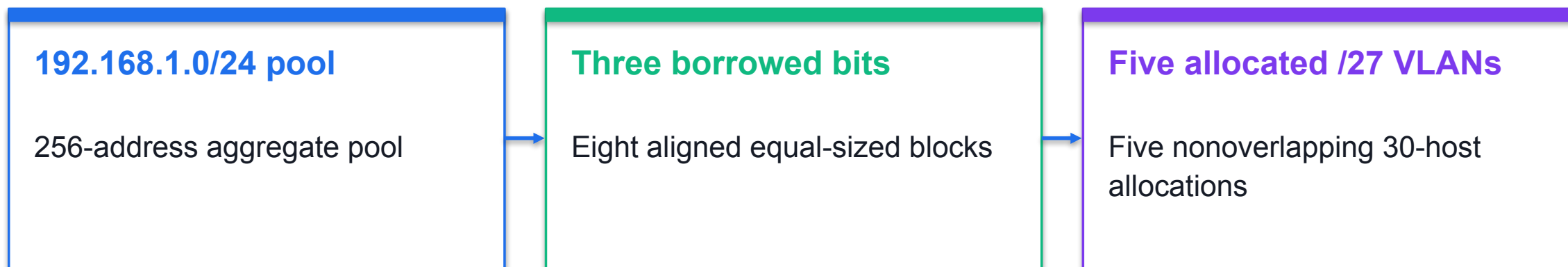
Cross-platform client diagnostics · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Windows host -> Linux host	Windows: ipconfig /all; route print
Mechanism / state	Test path and service	All: ping + name/service test
Result / boundary	Build evidence table for address, prefix, gateway, DNS, association/security and first failing path control.	Different commands expose equivalent layers. A disconnected wireless association is separate from a correct but unusable IP configuration.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Five equal subnets from a /24 · boundaries



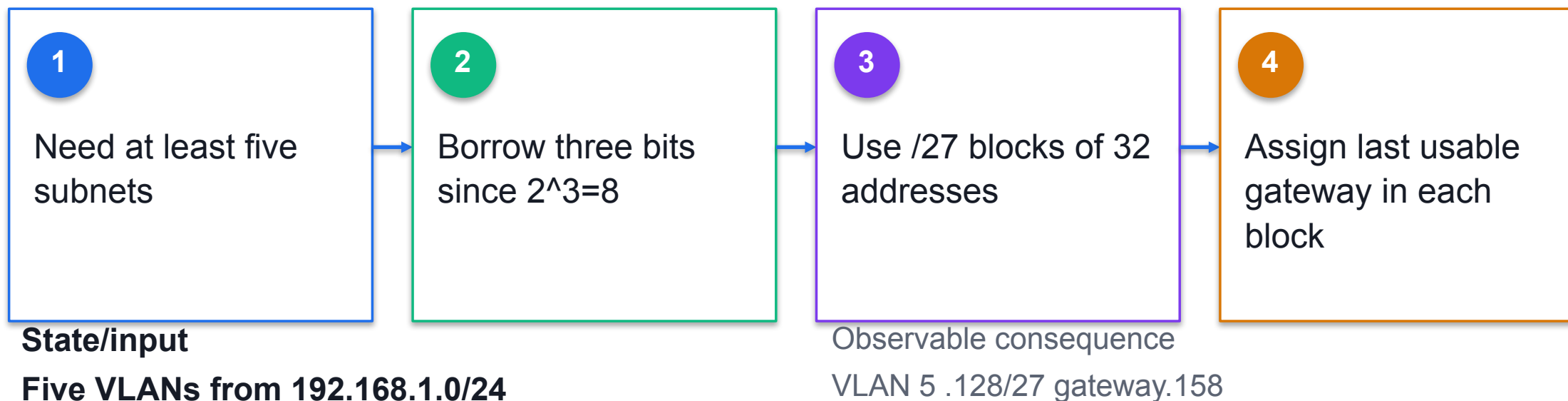
Five VLANs from 192.168.1.0/24
VLAN 5 .128/27 gateway.158

Boundary to inspect

Do not choose /26 for five equal subnets:
only four blocks fit in a /24

Each /27 has 30 usable addresses

Five equal subnets from a /24 · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Five equal subnets from a /24 · exact contract

Five VLANs from 192.168.1.0/24

/26 gives 4 subnets: insufficient

/27 gives 8 subnets: enough

VLAN 1 .0/27 gateway.30; VLAN 2 .32/27 gateway.62

VLAN 3 .64/27 gateway.94; VLAN 4 .96/27 gateway.126

VLAN 5 .128/27 gateway.158

INPUT / IDENTITY

192.168.1.0/24 pool · Need at least five subnets

DECISION / STATE

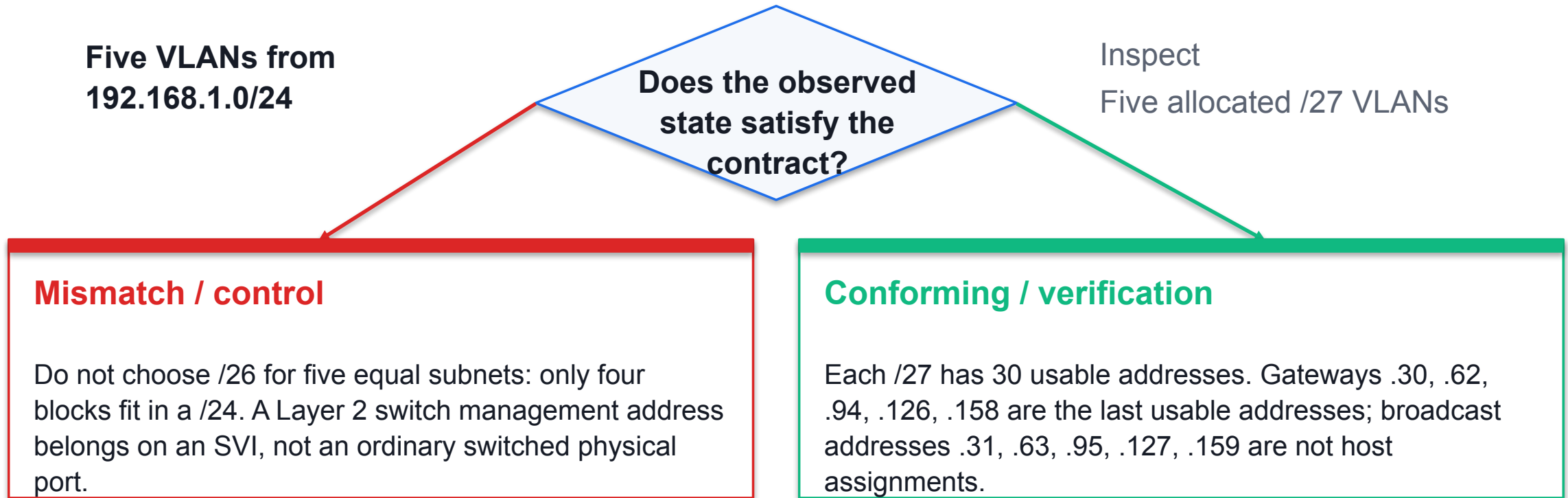
Use /27 blocks of 32 addresses

READBACk / RESULT

Assign last usable gateway in each block

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Five equal subnets from a /24 · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

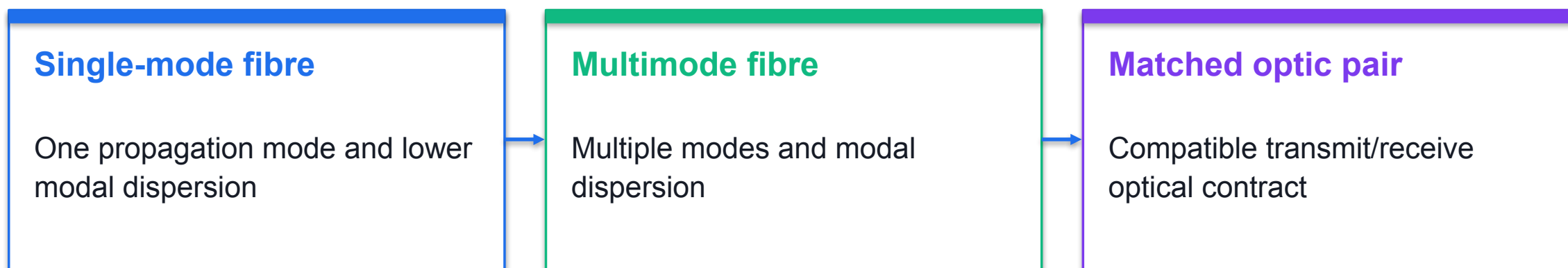
Five equal subnets from a /24 · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	192.168.1.0/24 pool -> Five allocated /27 VLANs	Five VLANs from 192.168.1.0/24
Mechanism / state	Assign last usable gateway in each block	VLAN 5 .128/27 gateway.158
Result / boundary	Each /27 has 30 usable addresses. Gateways .30, .62, .94, .126, .158 are the last usable addresses; broadcast addresses .31, .63, .95, .127, .159 are not host assignments.	Do not choose /26 for five equal subnets: only four blocks fit in a /24. A Layer 2 switch management address belongs on an SVI, not an ordinary switched physical port.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Single-mode versus multimode optical paths · boundaries

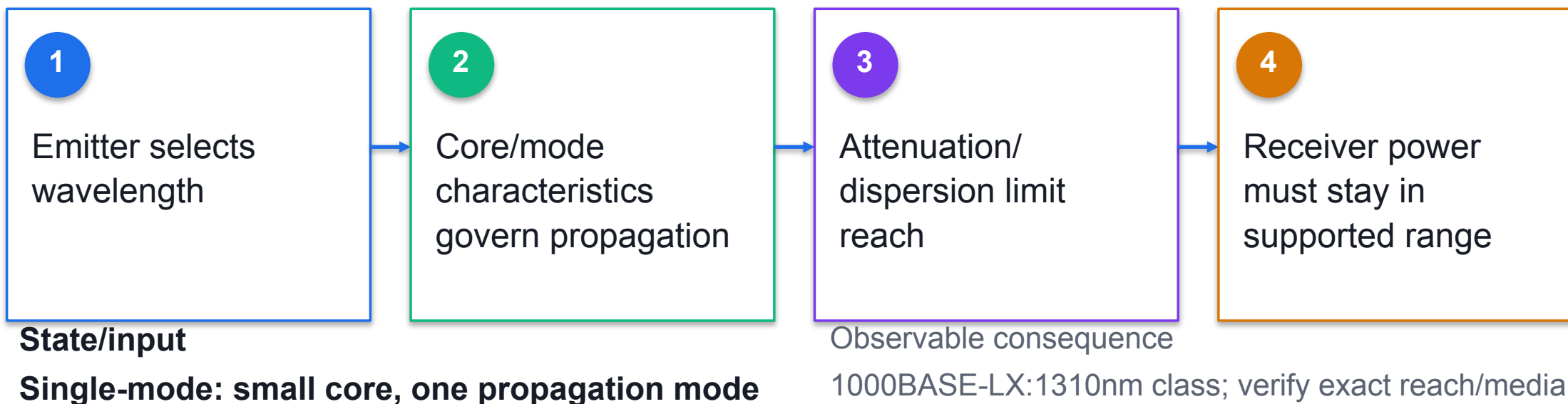


Single-mode: small core, one propagation mode
1000BASE-LX:1310nm class; verify exact reach/media

Boundary to inspect
Fibre jacket colour does not prove specification

Compare both-end wavelength, fibre type, supported distance and receive power; a fitted connector is insufficient evidence.

Single-mode versus multimode optical paths · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Single-mode versus multimode optical paths · exact contract

Single-mode: small core, one propagation mode

Multimode: larger core, multiple modes

1000BASE-SX: short-wave multimode example

1000BASE-LX:1310nm class; verify exact reach/media

INPUT / IDENTITY

Single-mode fibre · Emitter selects wavelength

DECISION / STATE

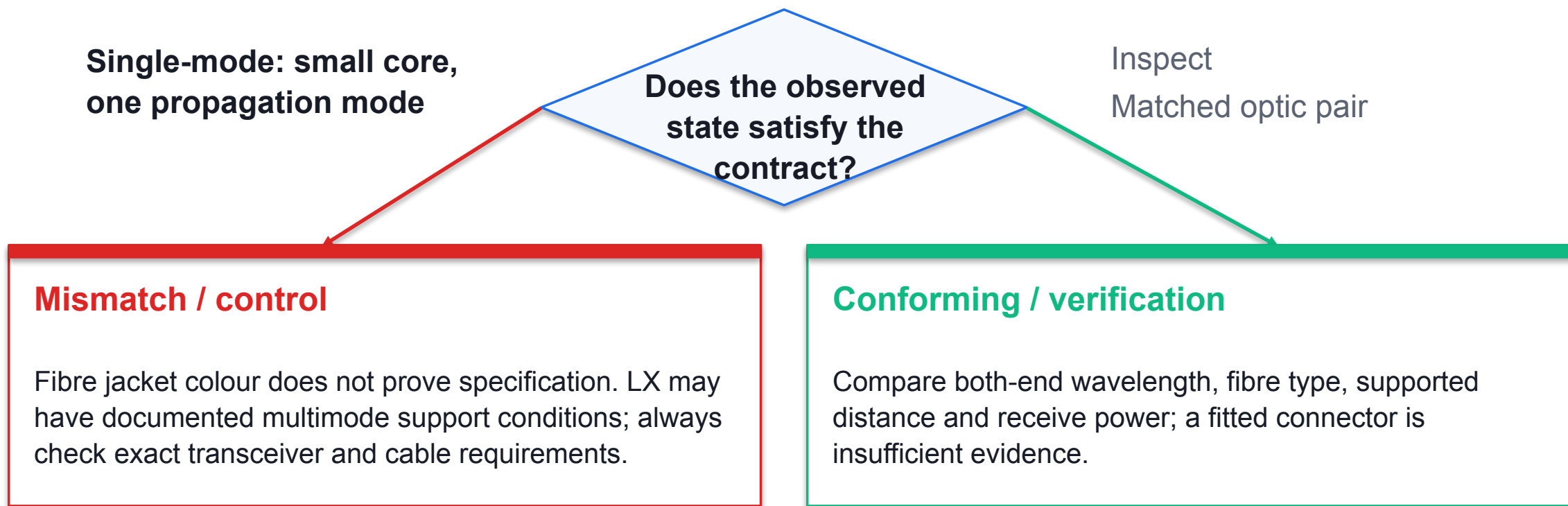
Attenuation/dispersion limit reach

READBACK / RESULT

Receiver power must stay in supported range

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Single-mode versus multimode optical paths · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

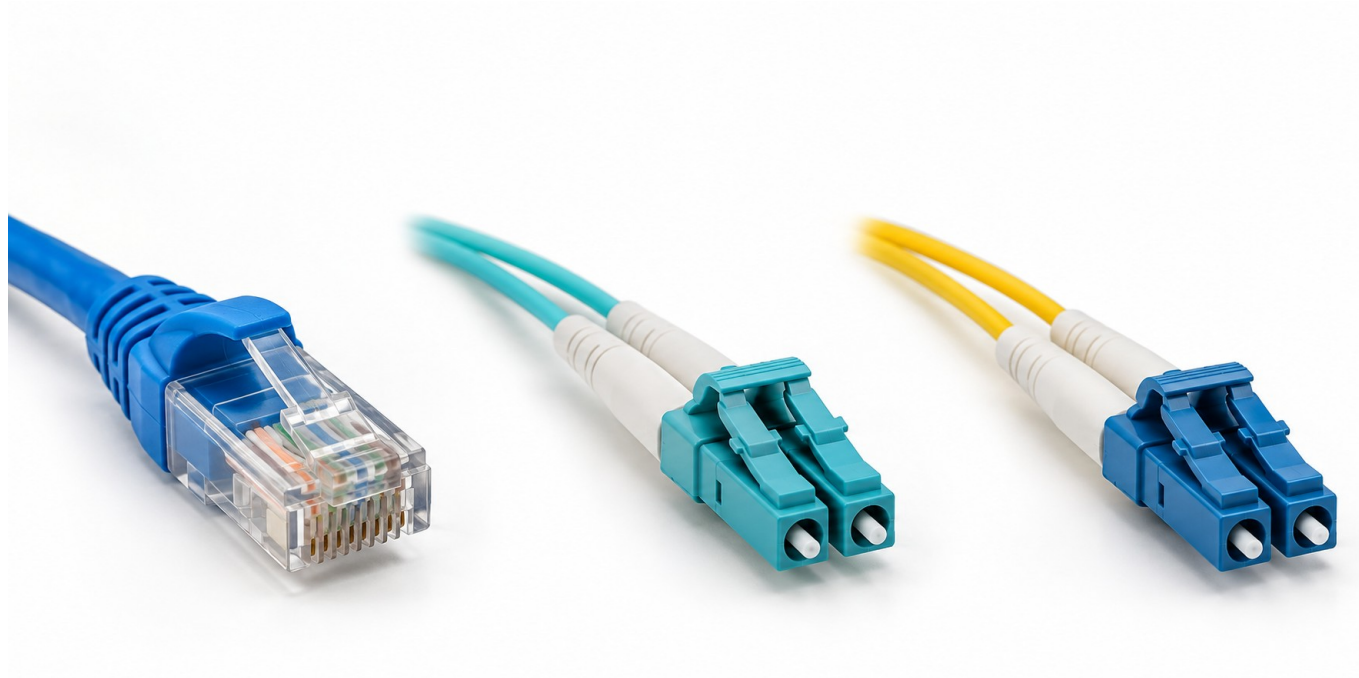
Single-mode versus multimode optical paths · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Single-mode fibre -> Matched optic pair	Single-mode: small core, one propagation mode
Mechanism / state	Receiver power must stay in supported range	1000BASE-LX:1310nm class; verify exact reach/media
Result / boundary	Compare both-end wavelength, fibre type, supported distance and receive power; a fitted connector is insufficient evidence.	Fibre jacket colour does not prove specification. LX may have documented multimode support conditions; always check exact transceiver and cable requirements.

Bridge: Activity 1: addressing / Activity 2: physical and VLAN evidence

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Copper and fibre: inspect the contract



Match both ends

Verify optic support, wavelength, medium, distance and received signal.

Illustration does not verify connector type or fibre performance.

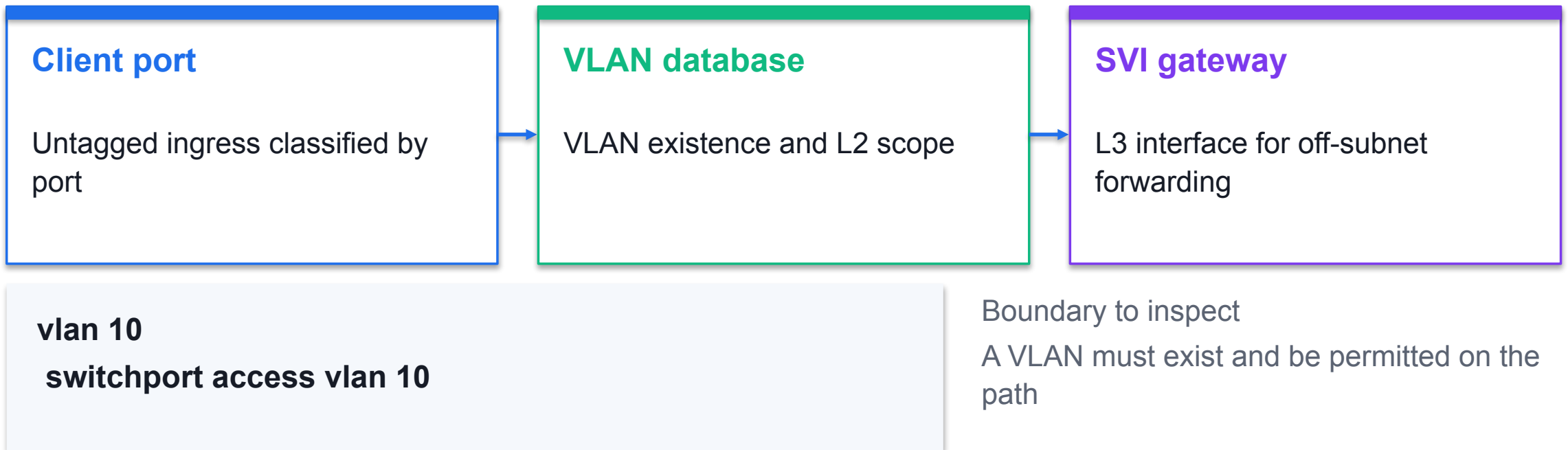
Synthetic cabling illustration; use transceiver specifications and current interface evidence for real identification.

Network Access

Network Access

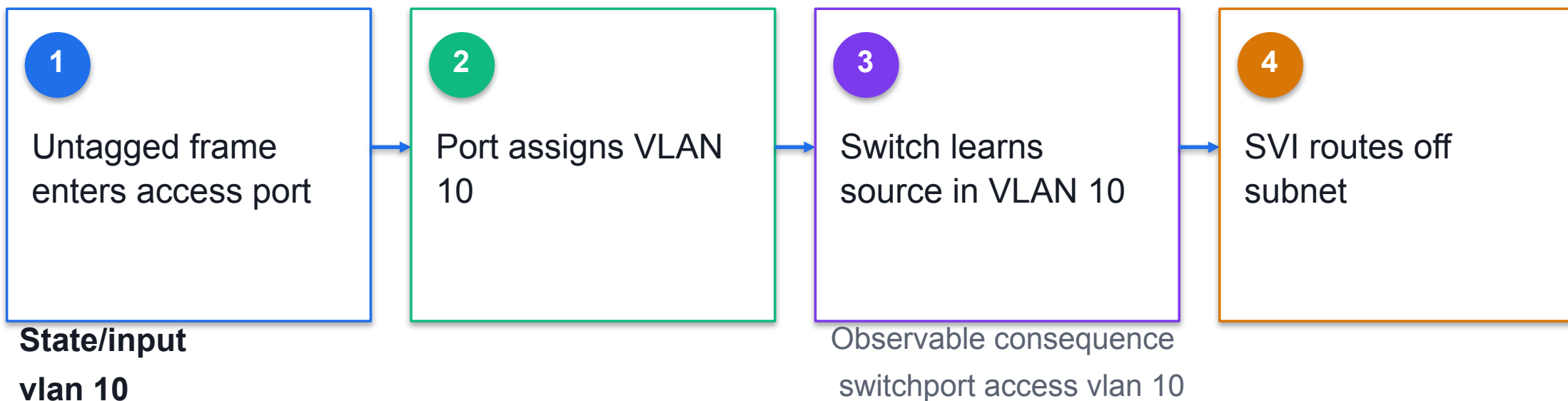
Named boundaries -> runtime trace -> exact contract -> failure/control
-> acceptance evidence

Access VLAN membership · boundaries



show vlan brief; show interfaces switchport

Access VLAN membership · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Access VLAN membership · exact contract

```
vlan 10
name USERS
interface Gi1/0/1
switchport mode access
switchport access vlan 10
```

INPUT / IDENTITY

Client port · Untagged frame enters access port

DECISION / STATE

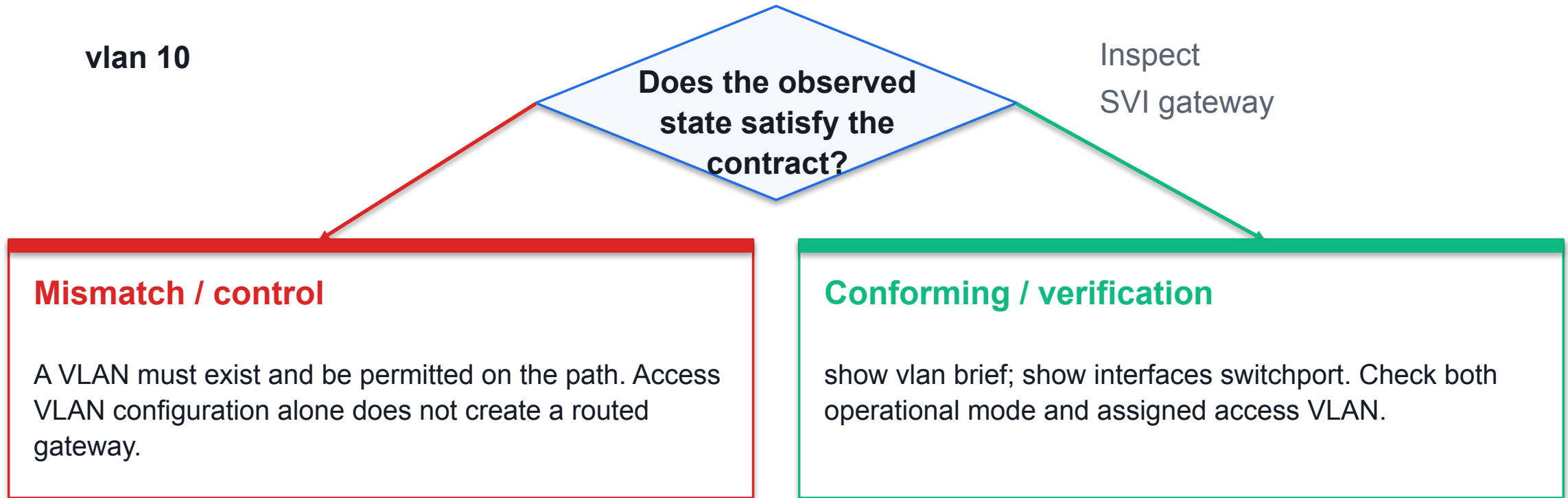
Switch learns source in VLAN 10

READBACK / RESULT

SVI routes off subnet

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Access VLAN membership · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

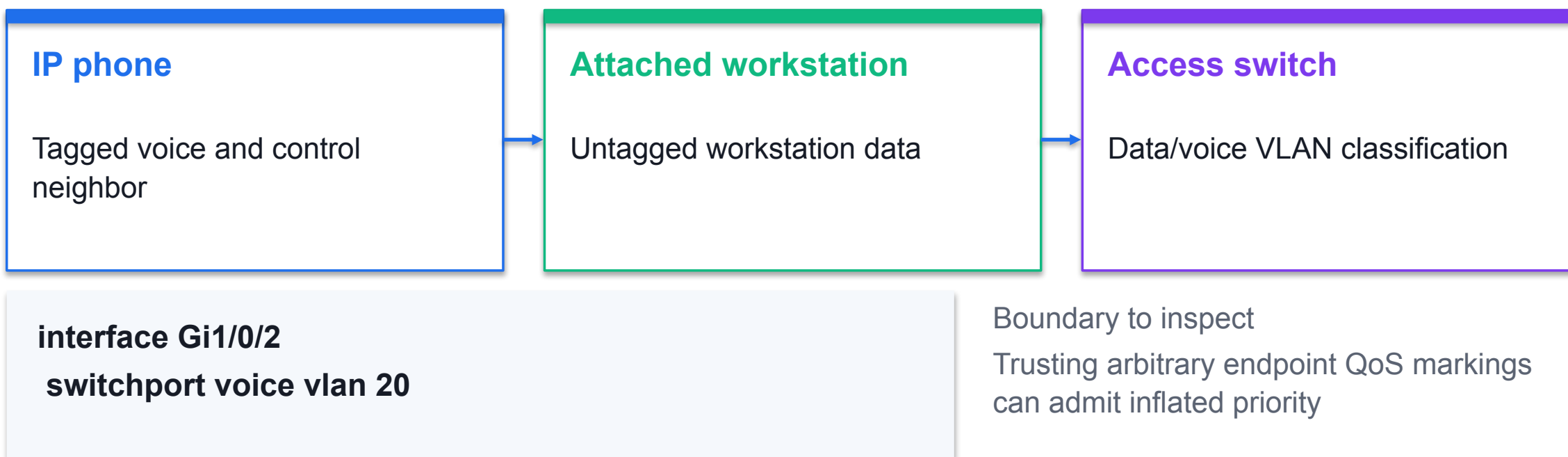
Access VLAN membership · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Client port -> SVI gateway	vlan 10
Mechanism / state	SVI routes off subnet	switchport access vlan 10
Result / boundary	show vlan brief; show interfaces switchport. Check both operational mode and assigned access VLAN.	A VLAN must exist and be permitted on the path. Access VLAN configuration alone does not create a routed gateway.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

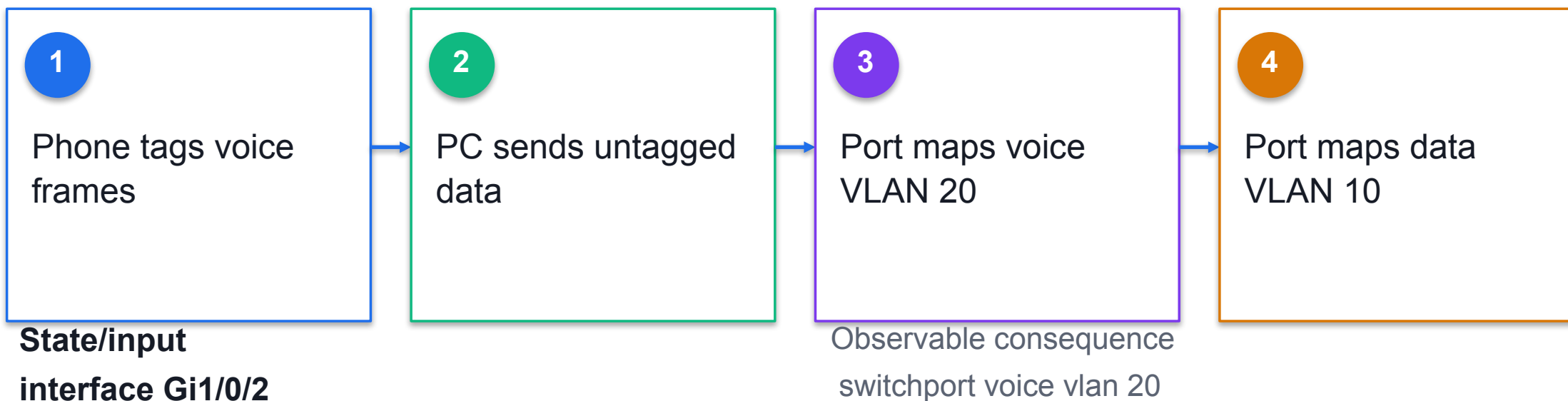
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Voice and data separation · boundaries



Inspect voice VLAN, LLDP/CDP phone advertisement and the separate DHCP/IP ranges for phone and workstation.

Voice and data separation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Voice and data separation · exact contract

```
interface Gi1/0/2
switchport mode access
switchport access vlan 10
switchport voice vlan 20
```

INPUT / IDENTITY

IP phone · Phone tags voice frames

DECISION / STATE

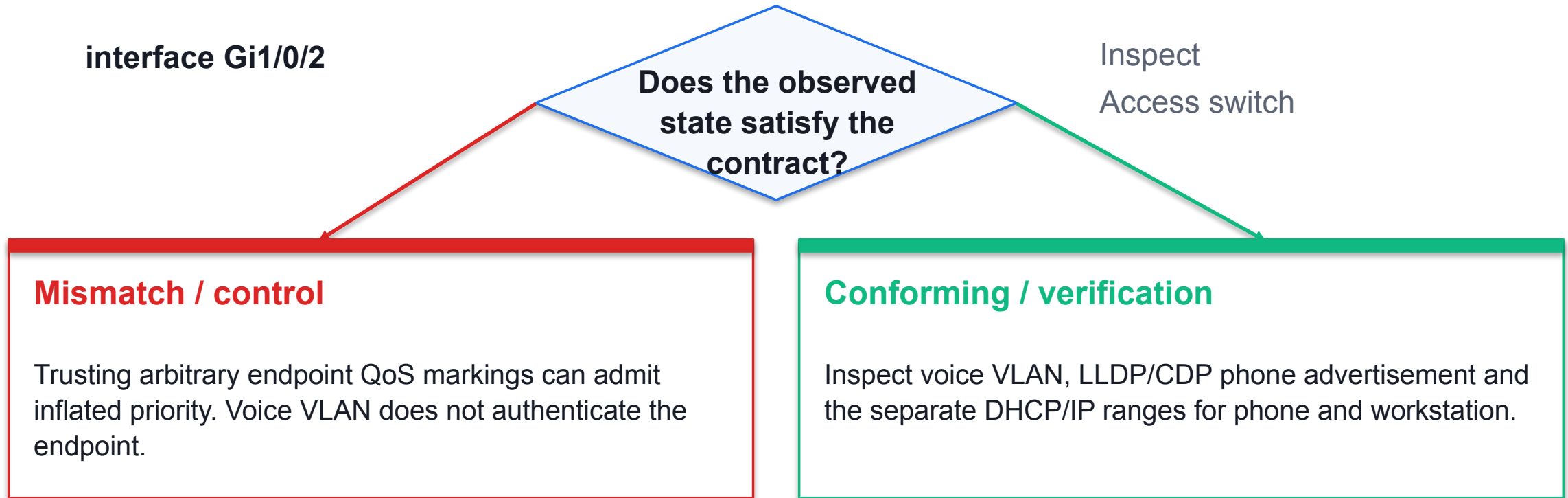
Port maps voice VLAN 20

READBACK / RESULT

Port maps data VLAN 10

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Voice and data separation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

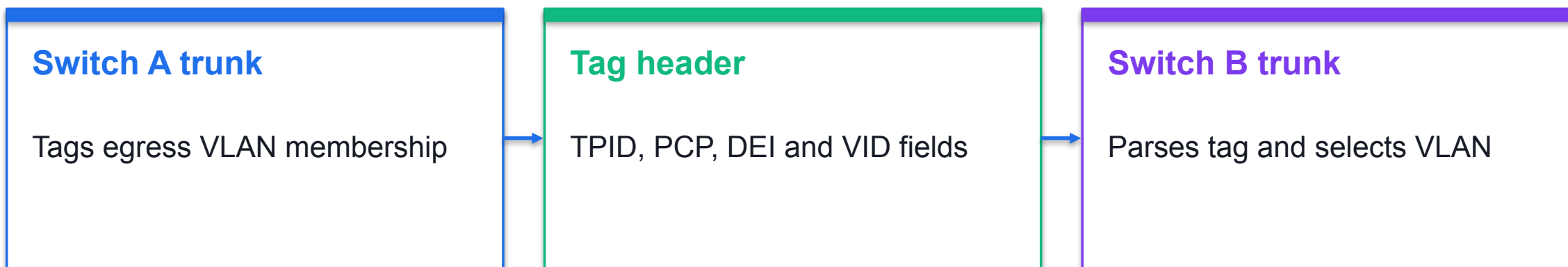
Voice and data separation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	IP phone -> Access switch	interface Gi1/0/2
Mechanism / state	Port maps data VLAN 10	switchport voice vlan 20
Result / boundary	Inspect voice VLAN, LLDP/CDP phone advertisement and the separate DHCP/IP ranges for phone and workstation.	Trusting arbitrary endpoint QoS markings can admit inflated priority. Voice VLAN does not authenticate the endpoint.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

802.1Q trunk tagging · boundaries

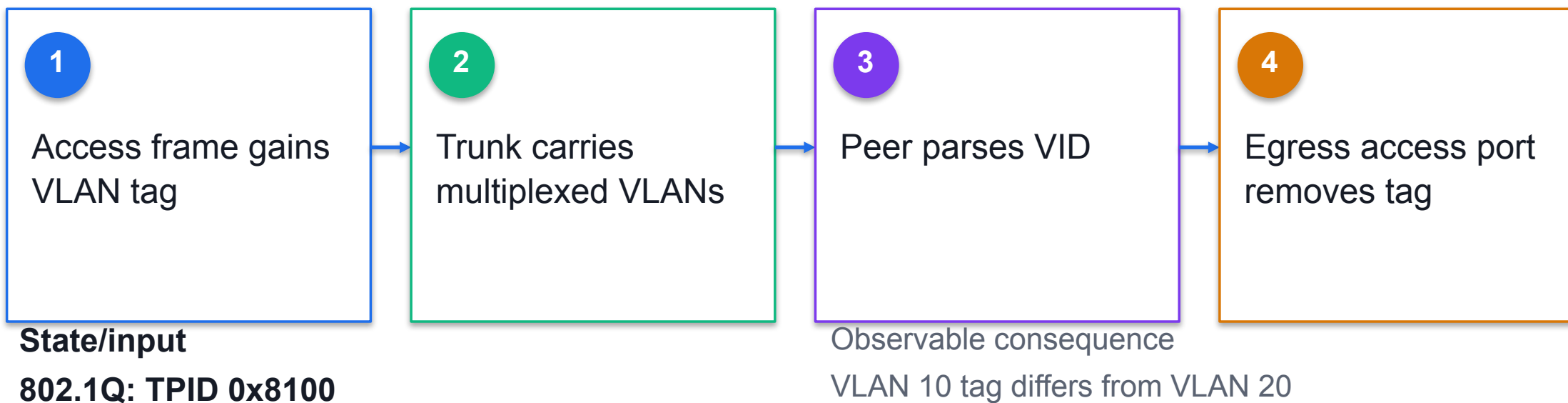


802.1Q: TPID 0x8100
VLAN 10 tag differs from VLAN 20

Boundary to inspect
A trunk can be up while required VLANs are pruned or missing

show interfaces trunk

802.1Q trunk tagging · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

802.1Q trunk tagging · exact contract

802.1Q: TPID 0x8100

PCP 3 bits; DEI 1 bit; VID 12 bits

VLAN 10 tag differs from VLAN 20

INPUT / IDENTITY

Switch A trunk · Access frame gains
VLAN tag

DECISION / STATE

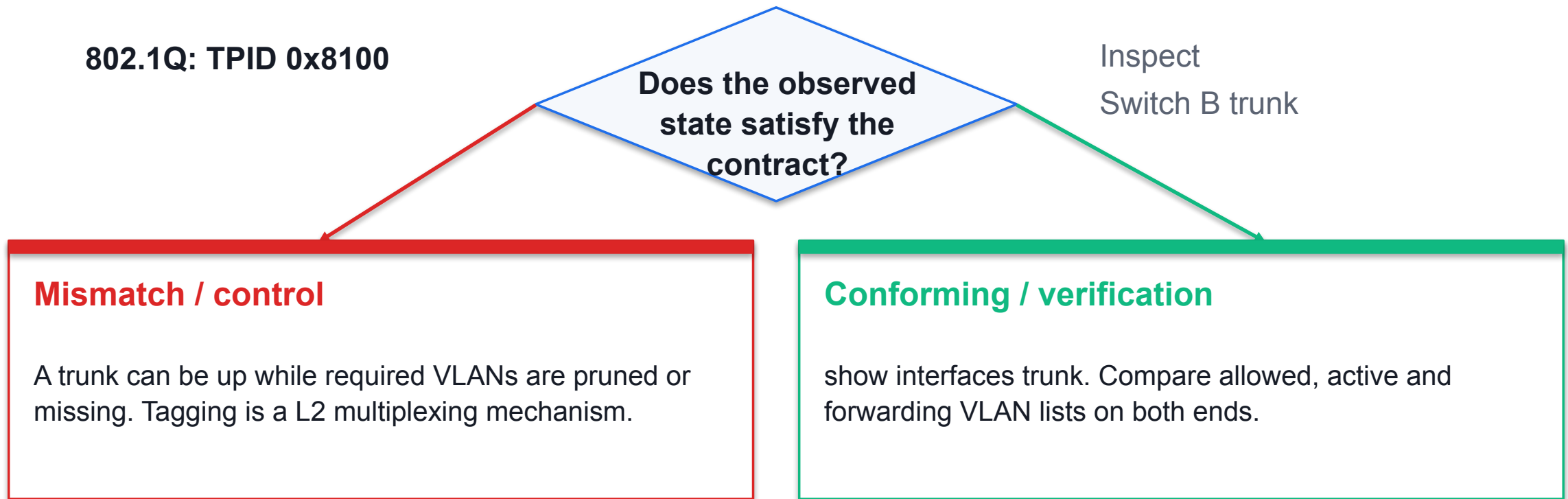
Peer parses VID

READBACK / RESULT

Egress access port removes tag

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

802.1Q trunk tagging · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

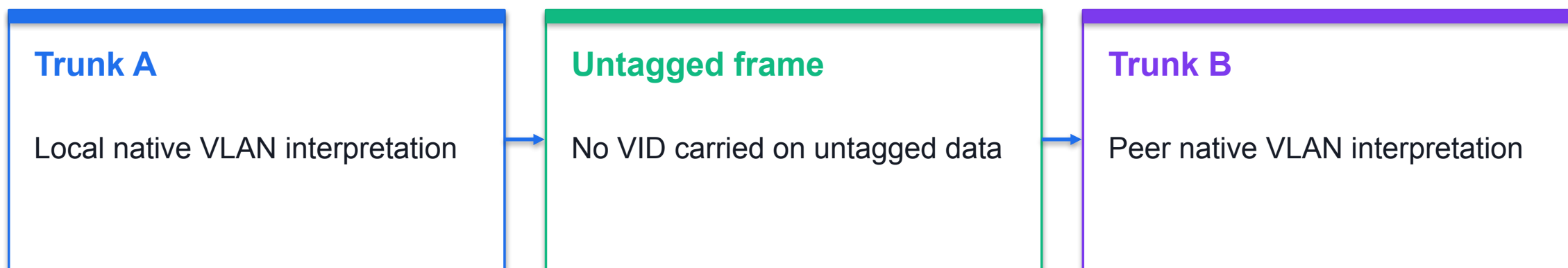
802.1Q trunk tagging · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Switch A trunk -> Switch B trunk	802.1Q: TPID 0x8100
Mechanism / state	Egress access port removes tag	VLAN 10 tag differs from VLAN 20
Result / boundary	show interfaces trunk. Compare allowed, active and forwarding VLAN lists on both ends.	A trunk can be up while required VLANs are pruned or missing. Tagging is a L2 multiplexing mechanism.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Native VLAN consistency · boundaries

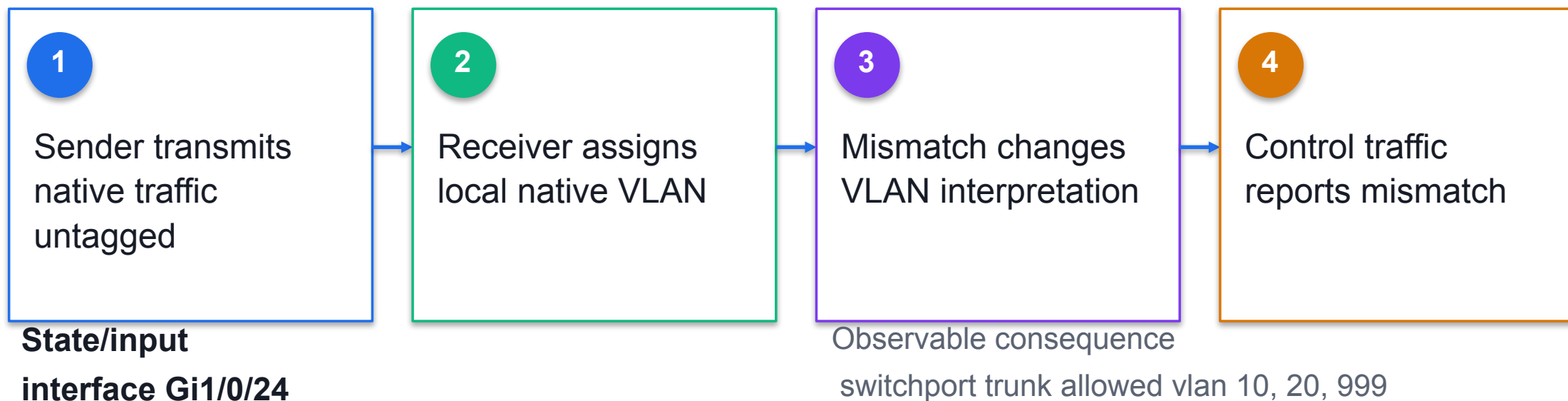


interface Gi1/0/24
switchport trunk allowed vlan 10,20,999

Boundary to inspect
Native VLAN mismatches can join
unintended broadcast domains

show interfaces trunk and CDP native-VLAN warning

Native VLAN consistency · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Native VLAN consistency · exact contract

```
interface Gi1/0/24
switchport mode trunk
switchport trunk native vlan 999
switchport trunk allowed vlan 10,20,999
```

INPUT / IDENTITY

Trunk A · Sender transmits native traffic untagged

DECISION / STATE

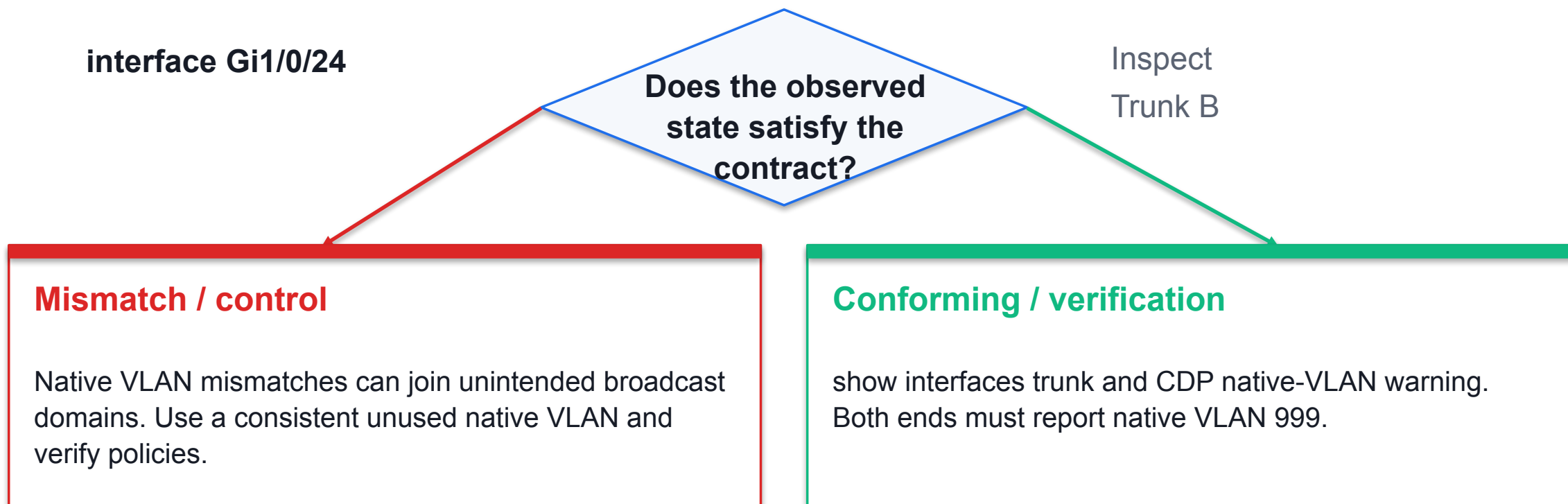
Mismatch changes VLAN interpretation

READBACK / RESULT

Control traffic reports mismatch

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Native VLAN consistency · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

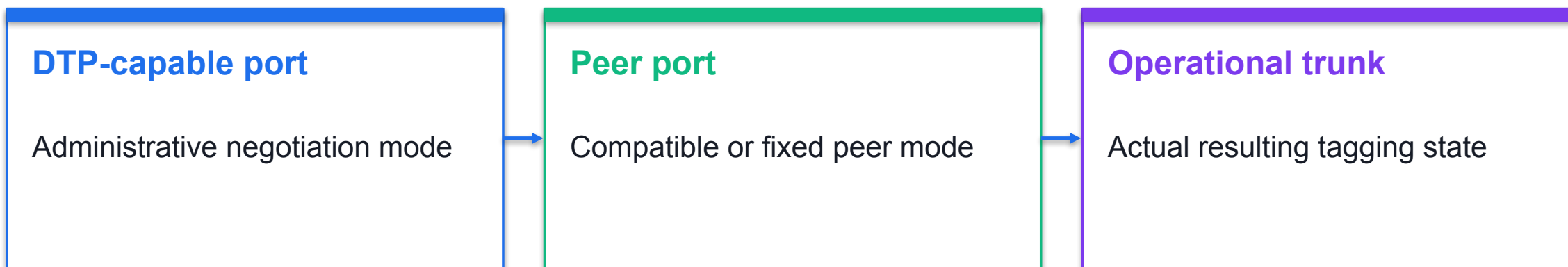
Native VLAN consistency · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Trunk A -> Trunk B	interface Gi1/0/24
Mechanism / state	Control traffic reports mismatch	switchport trunk allowed vlan 10, 20, 999
Result / boundary	show interfaces trunk and CDP native-VLAN warning. Both ends must report native VLAN 999.	Native VLAN mismatches can join unintended broadcast domains. Use a consistent unused native VLAN and verify policies.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Trunk negotiation boundaries · boundaries

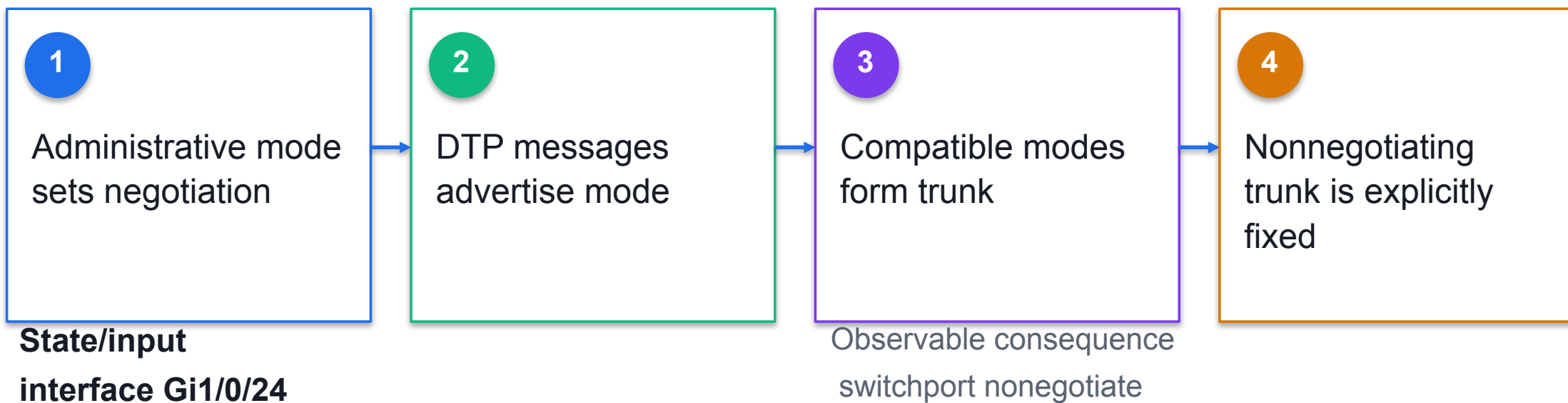


interface Gi1/0/24
switchport nonegotiate

Boundary to inspect
Do not leave user-facing ports dynamically negotiable

Compare administrative/operational modes and DTP status

Trunk negotiation boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Trunk negotiation boundaries · exact contract

```
interface Gi1/0/24
switchport mode trunk
switchport nonegotiate
```

INPUT / IDENTITY

DTP-capable port · Administrative mode sets negotiation

DECISION / STATE

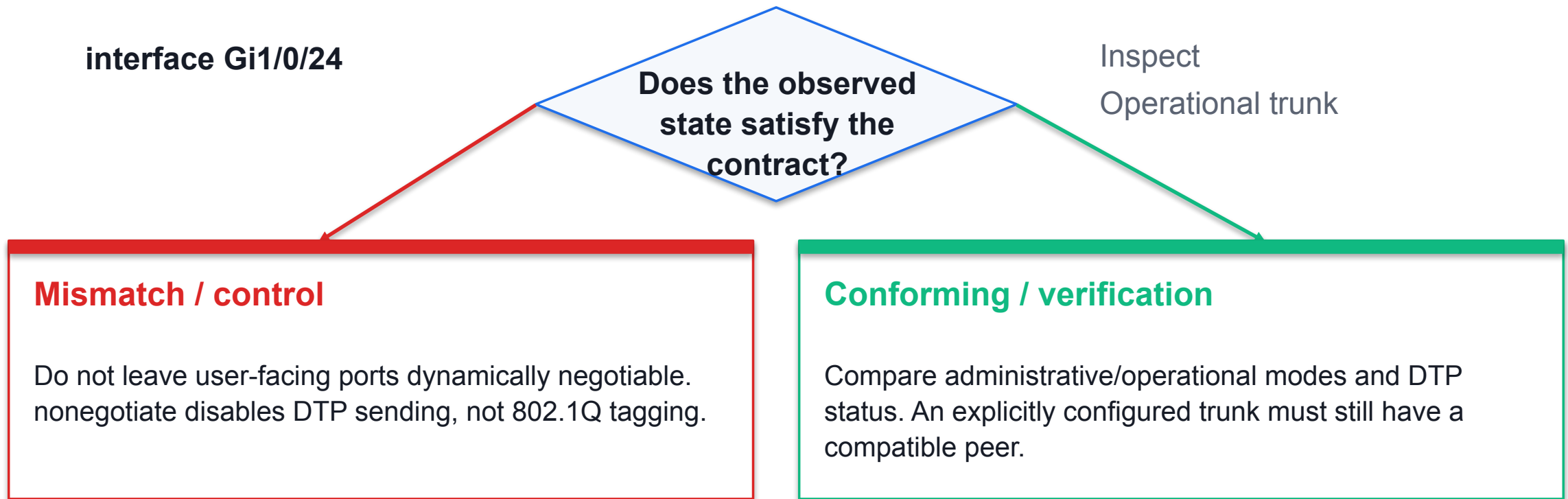
Compatible modes form trunk

READBACK / RESULT

Nonnegotiating trunk is explicitly fixed

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Trunk negotiation boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

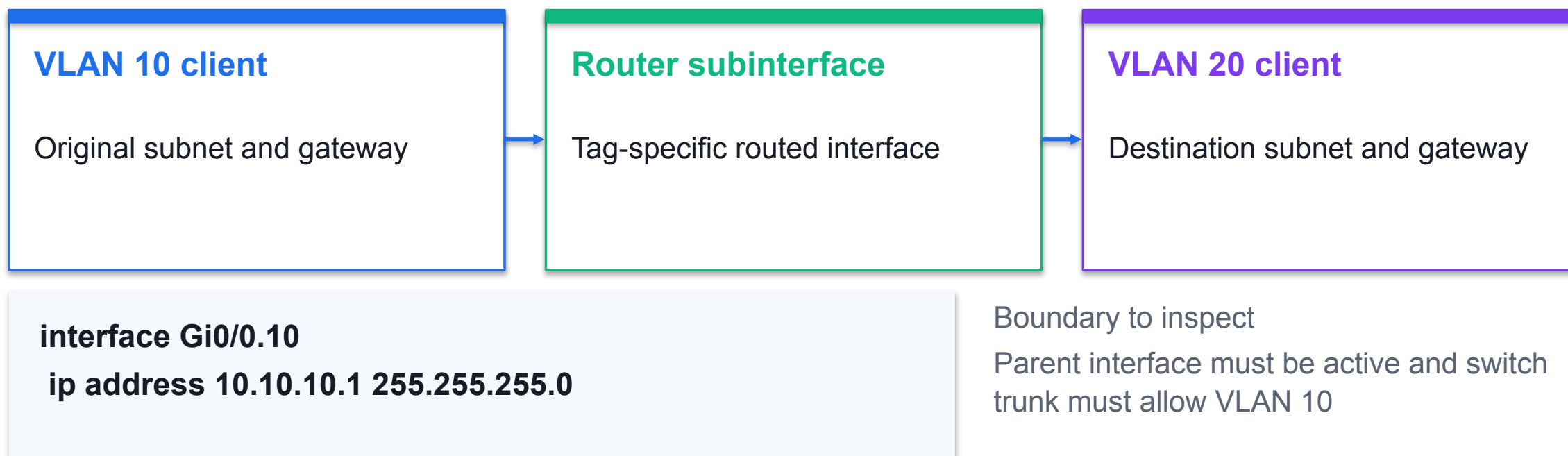
Trunk negotiation boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	DTP-capable port -> Operational trunk	interface Gi1/0/24
Mechanism / state	Nonnegotiating trunk is explicitly fixed	switchport nonegotiate
Result / boundary	Compare administrative/operational modes and DTP status. An explicitly configured trunk must still have a compatible peer.	Do not leave user-facing ports dynamically negotiable. nonegotiate disables DTP sending, not 802.1Q tagging.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

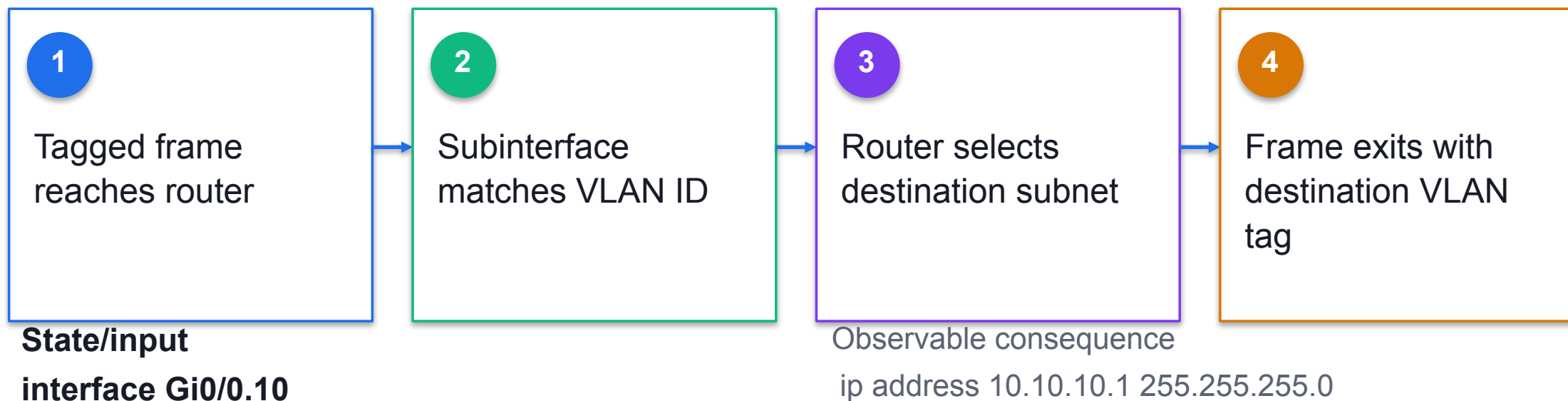
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Router-on-a-stick paths · boundaries



show ip interface brief; ping each subinterface gateway; verify tags and route entries.

Router-on-a-stick paths · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Router-on-a-stick paths · exact contract

```
interface Gi0/0.10
encapsulation dot1Q 10
ip address 10.10.10.1 255.255.255.0
```

INPUT / IDENTITY

VLAN 10 client · Tagged frame reaches router

DECISION / STATE

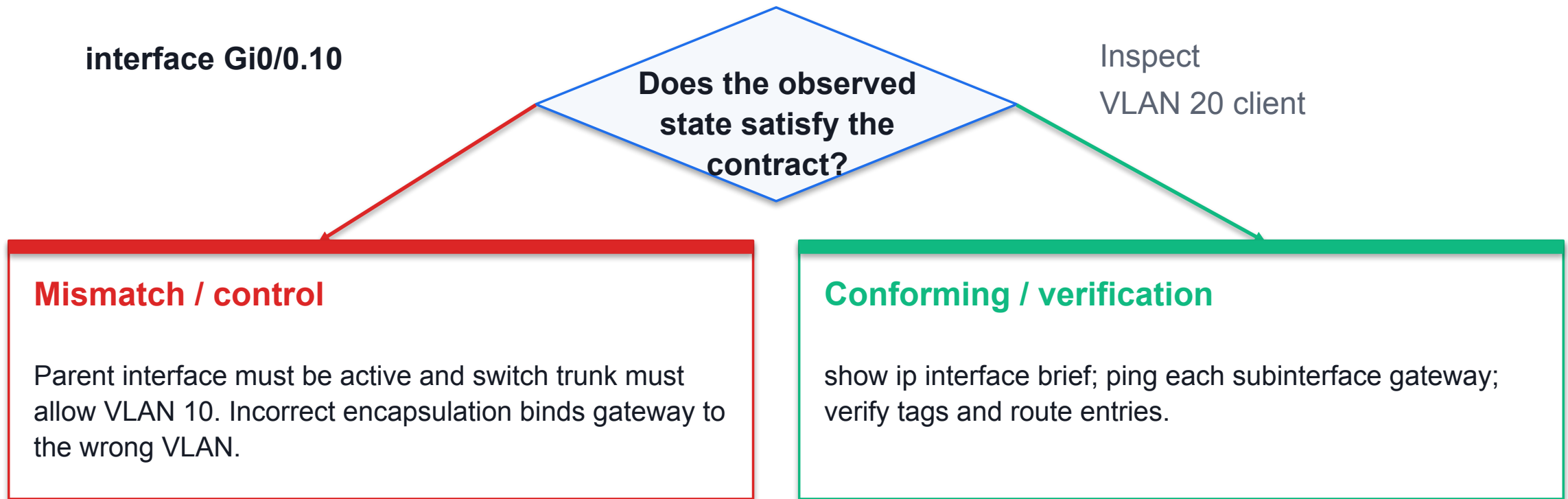
Router selects destination subnet

READBACK / RESULT

Frame exits with destination VLAN tag

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Router-on-a-stick paths · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

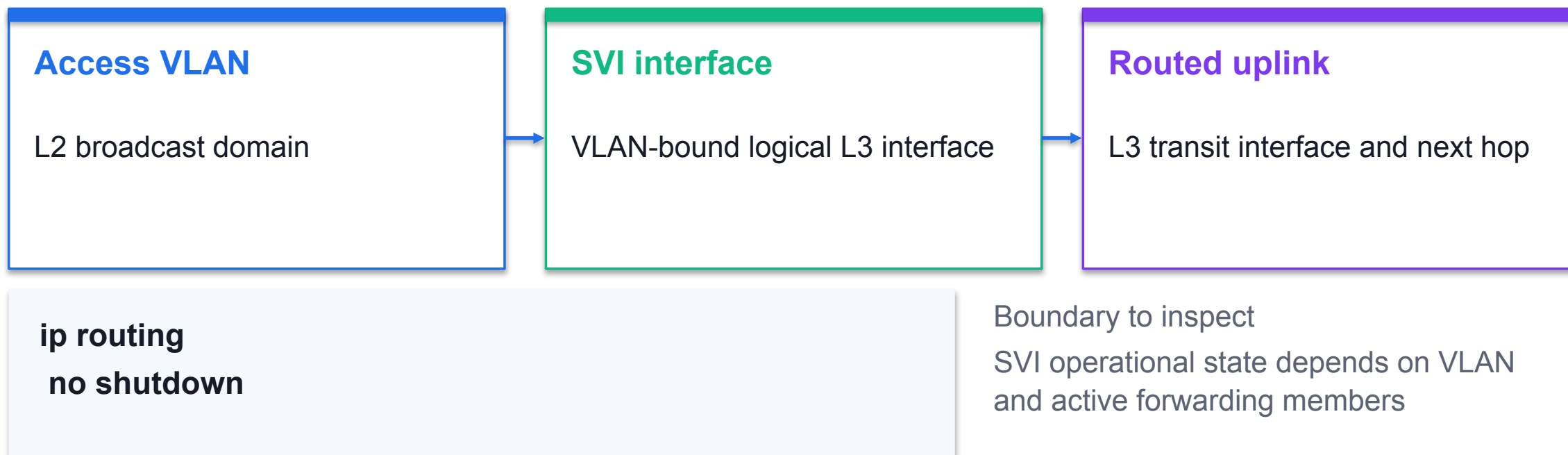
Router-on-a-stick paths · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	VLAN 10 client -> VLAN 20 client	interface Gi0/0.10
Mechanism / state	Frame exits with destination VLAN tag	ip address 10.10.10.1 255.255.255.0
Result / boundary	show ip interface brief; ping each subinterface gateway; verify tags and route entries.	Parent interface must be active and switch trunk must allow VLAN 10. Incorrect encapsulation binds gateway to the wrong VLAN.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

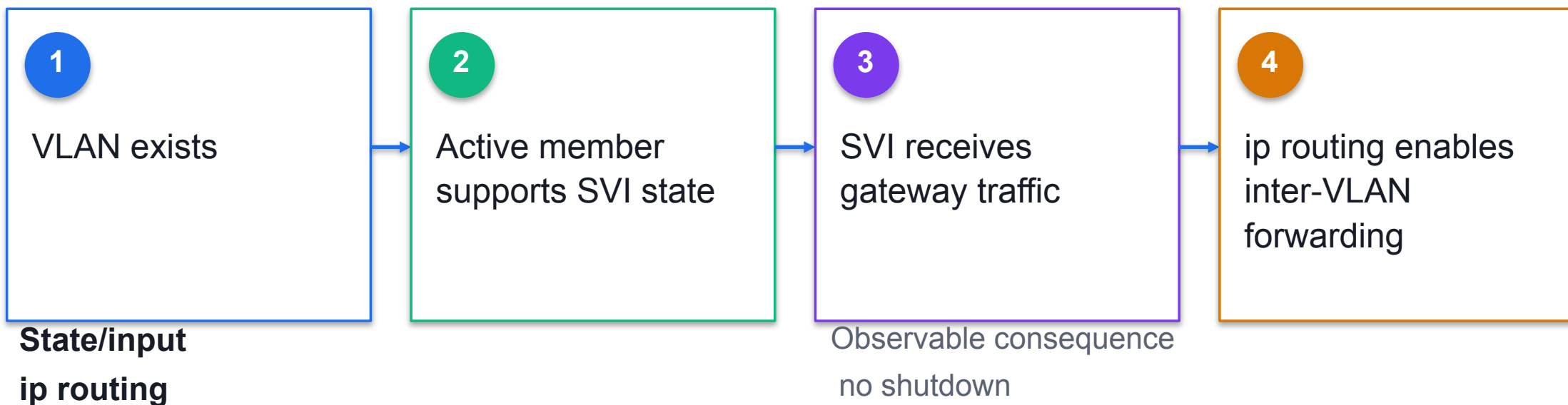
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Multilayer SVI routing · boundaries



show interfaces vlan 10; show ip route connected

Multilayer SVI routing · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Multilayer SVI routing · exact contract

```
ip routing
interface Vlan 10
ip address 10.10.10.1 255.255.255.0
no shutdown
```

INPUT / IDENTITY

Access VLAN · VLAN exists

DECISION / STATE

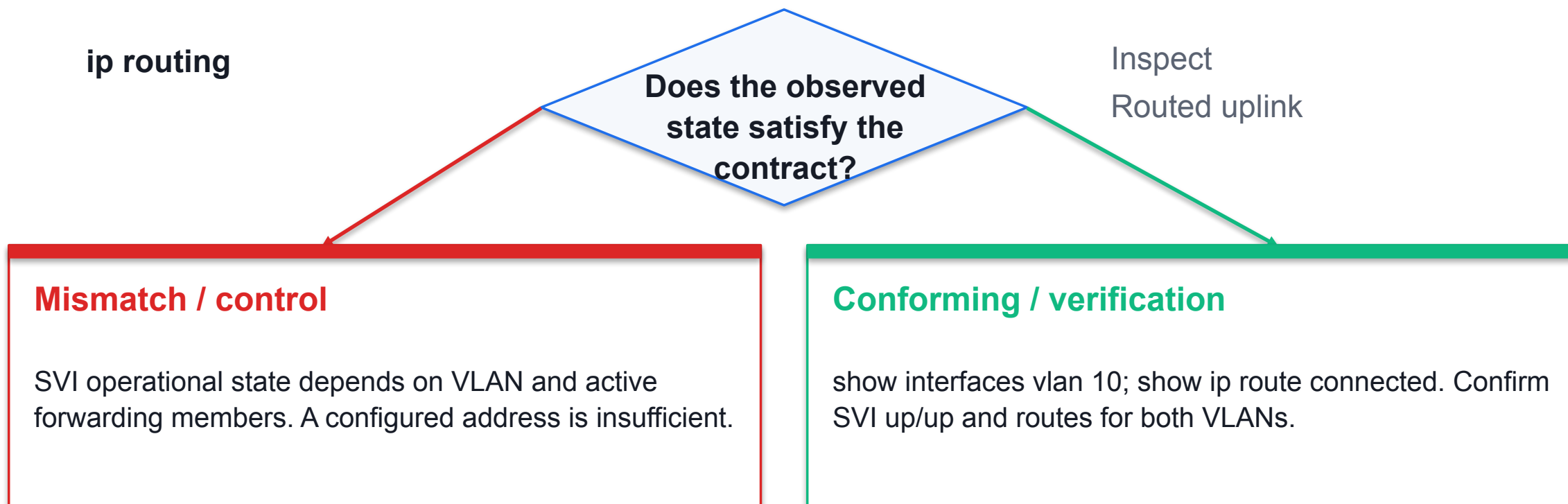
SVI receives gateway traffic

READBACK / RESULT

ip routing enables inter-VLAN forwarding

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Multilayer SVI routing · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

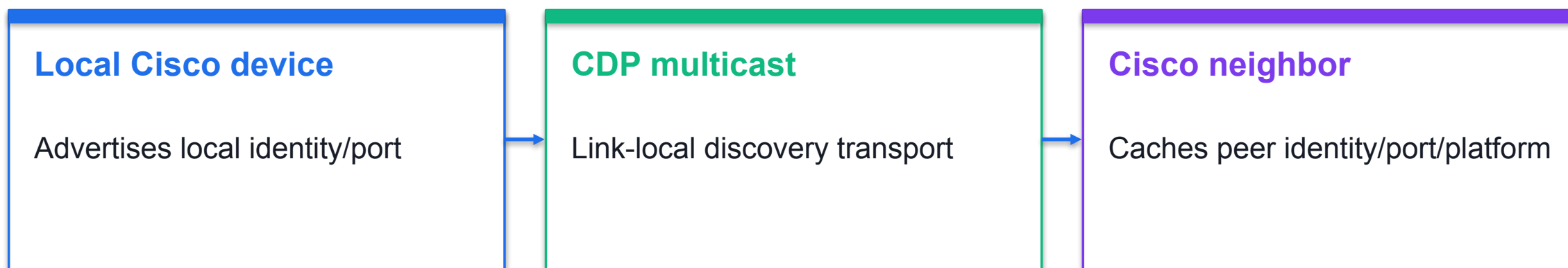
Multilayer SVI routing · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Access VLAN -> Routed uplink	ip routing
Mechanism / state	ip routing enables inter-VLAN forwarding	no shutdown
Result / boundary	show interfaces vlan 10; show ip route connected. Confirm SVI up/up and routes for both VLANs.	SVI operational state depends on VLAN and active forwarding members. A configured address is insufficient.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

CDP neighbor evidence · boundaries

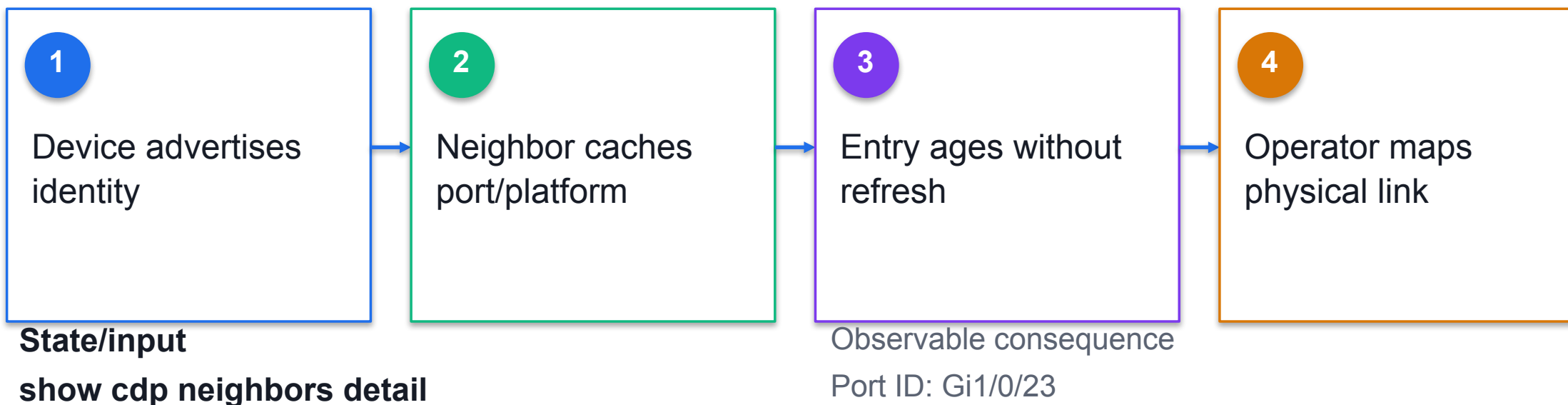


show cdp neighbors detail
Port ID: Gi1/0/23

Boundary to inspect
CDP reveals topology and should be
constrained on untrusted ports

Cross-check local port, peer port and actual cabling

CDP neighbor evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

CDP neighbor evidence · exact contract

show cdp neighbors detail

Device ID: SW2

Interface: Gi1/0/24

Port ID: Gi1/0/23

INPUT / IDENTITY

Local Cisco device · Device advertises identity

DECISION / STATE

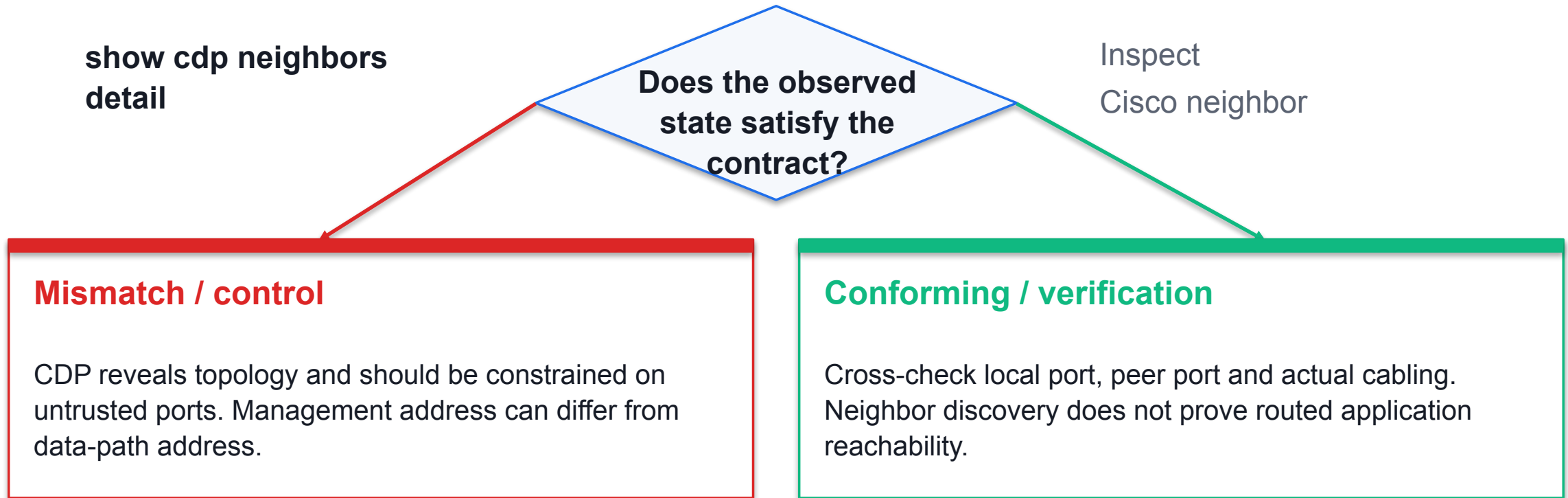
Entry ages without refresh

READBACK / RESULT

Operator maps physical link

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

CDP neighbor evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

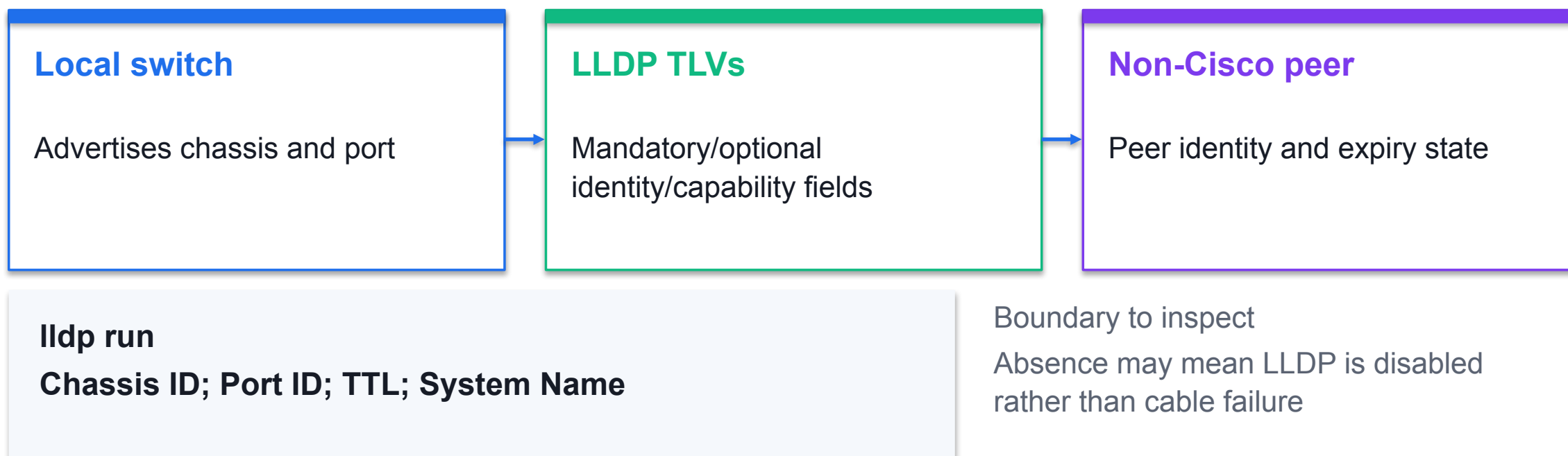
CDP neighbor evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Local Cisco device -> Cisco neighbor	show cdp neighbors detail
Mechanism / state	Operator maps physical link	Port ID: Gi1/0/23
Result / boundary	Cross-check local port, peer port and actual cabling. Neighbor discovery does not prove routed application reachability.	CDP reveals topology and should be constrained on untrusted ports. Management address can differ from data-path address.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

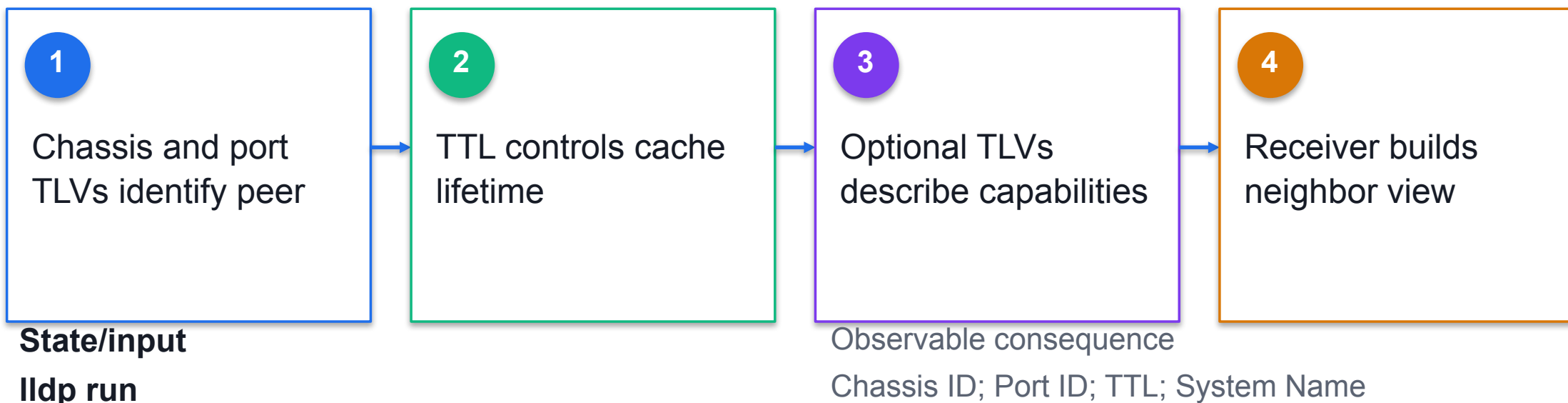
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

LLDP interoperable discovery · boundaries



Compare advertised capabilities and local/remote port IDs; use interface state as independent evidence.

LLDP interoperable discovery · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

LLDP interoperable discovery · exact contract

lldp run
show lldp neighbors detail
Chassis ID; Port ID; TTL; System Name

INPUT / IDENTITY

Local switch · Chassis and port TLVs
identify peer

DECISION / STATE

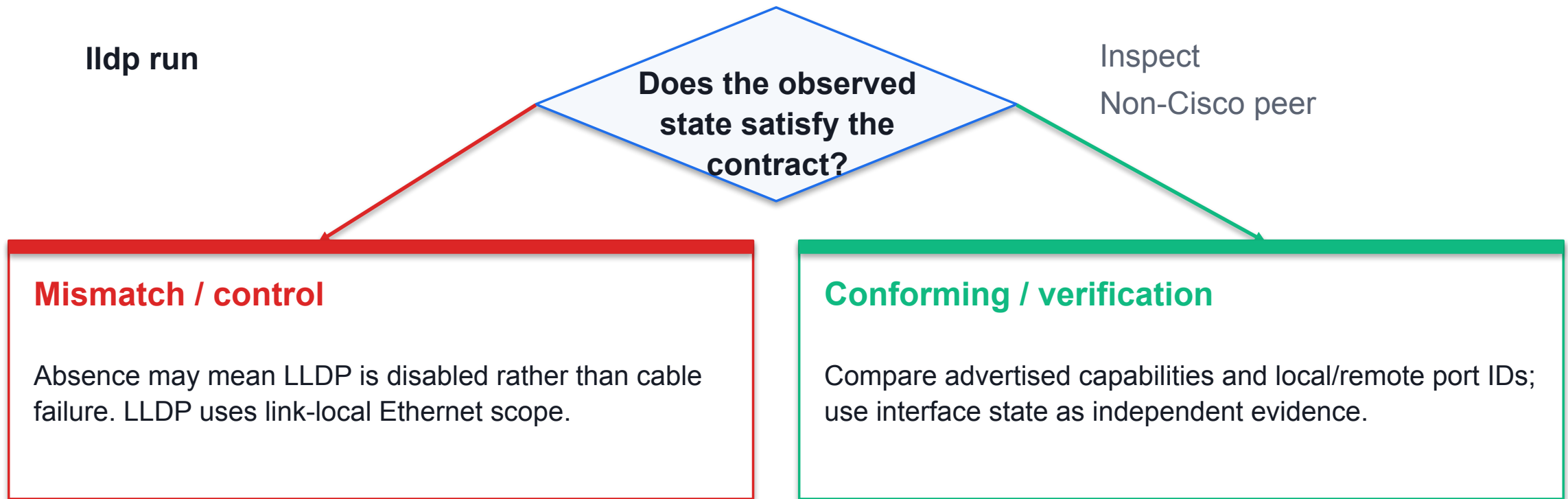
Optional TLVs describe capabilities

READBACK / RESULT

Receiver builds neighbor view

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

LLDP interoperable discovery · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

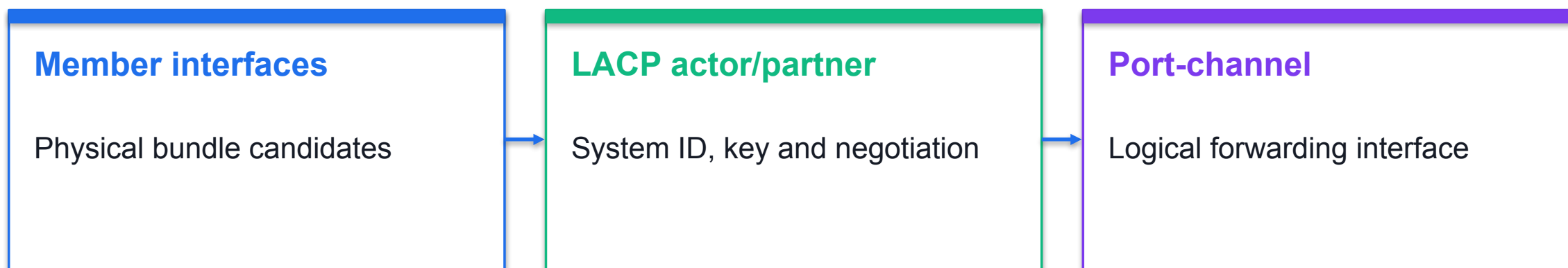
LLDP interoperable discovery · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Local switch -> Non-Cisco peer	lldp run
Mechanism / state	Receiver builds neighbor view	Chassis ID; Port ID; TTL; System Name
Result / boundary	Compare advertised capabilities and local/remote port IDs; use interface state as independent evidence.	Absence may mean LLDP is disabled rather than cable failure. LLDP uses link-local Ethernet scope.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

LACP channel negotiation · boundaries

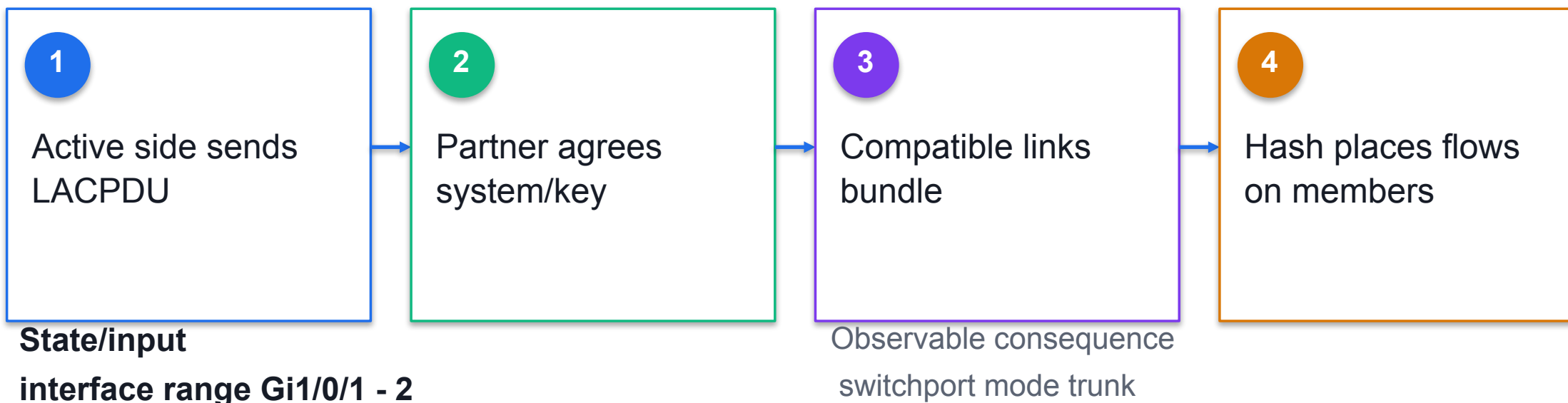


interface range Gi1/0/1 - 2
switchport mode trunk

Boundary to inspect
Passive/passive does not initiate a LACP channel

show etherchannel summary

LACP channel negotiation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

LACP channel negotiation · exact contract

```
interface range Gi1/0/1 - 2
channel-group 1 mode active
interface Port-channel1
switchport mode trunk
```

INPUT / IDENTITY

Member interfaces · Active side sends LACPDU

DECISION / STATE

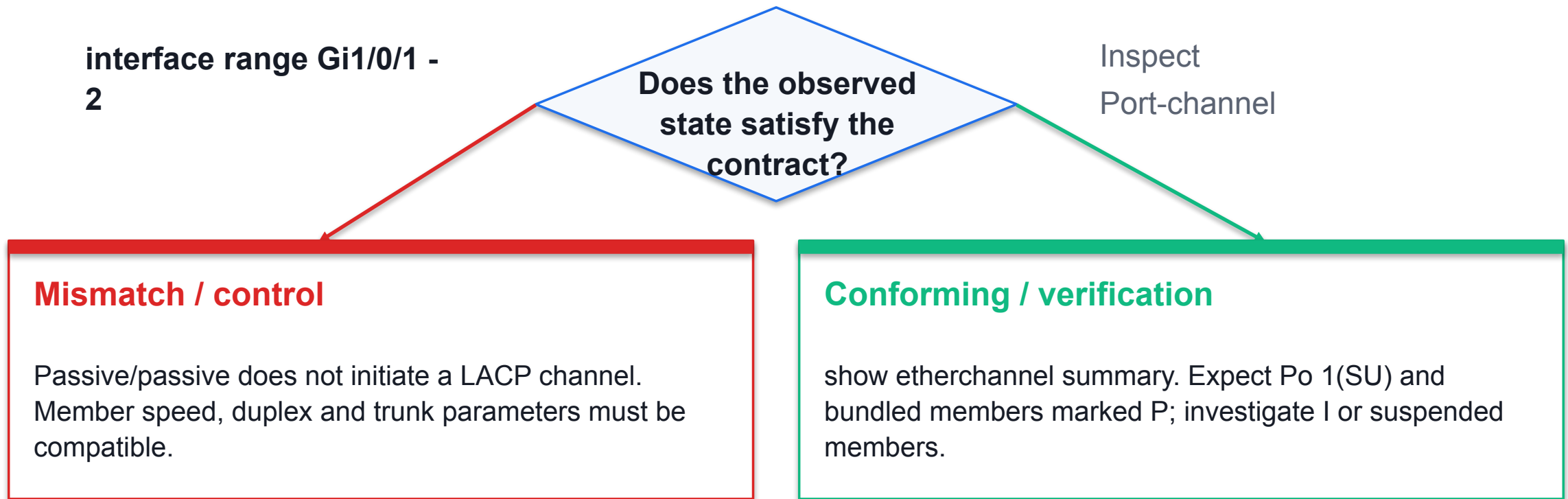
Compatible links bundle

READBACK / RESULT

Hash places flows on members

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

LACP channel negotiation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

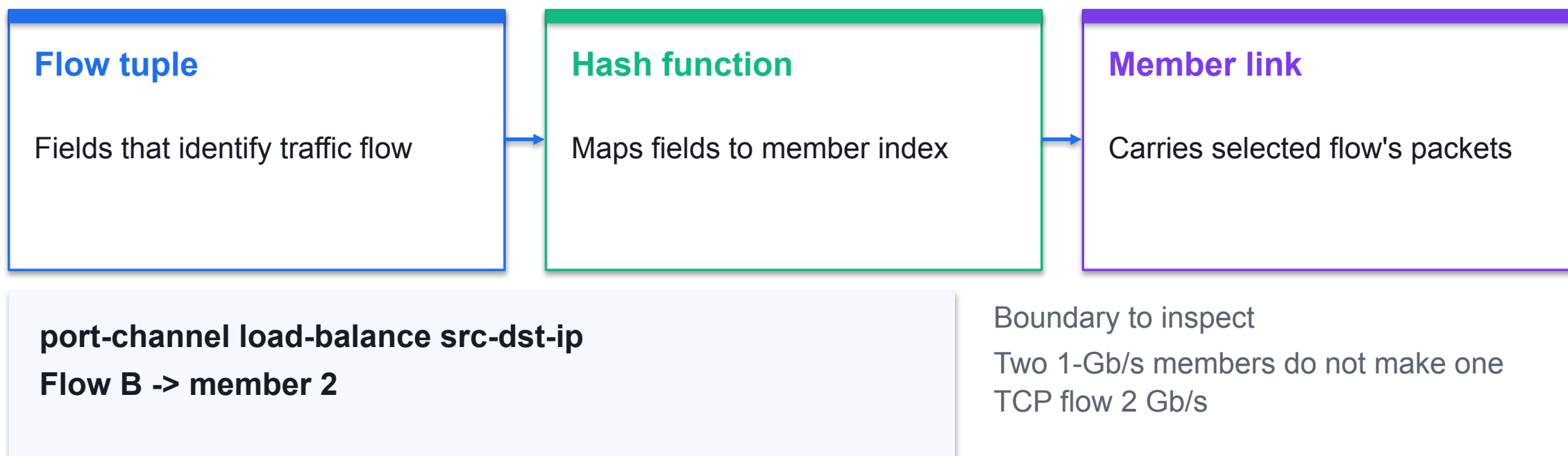
LACP channel negotiation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Member interfaces -> Port-channel	interface range Gi1/0/1 - 2
Mechanism / state	Hash places flows on members	switchport mode trunk
Result / boundary	show etherchannel summary. Expect Po 1(SU) and bundled members marked P; investigate I or suspended members.	Passive/passive does not initiate a LACP channel. Member speed, duplex and trunk parameters must be compatible.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

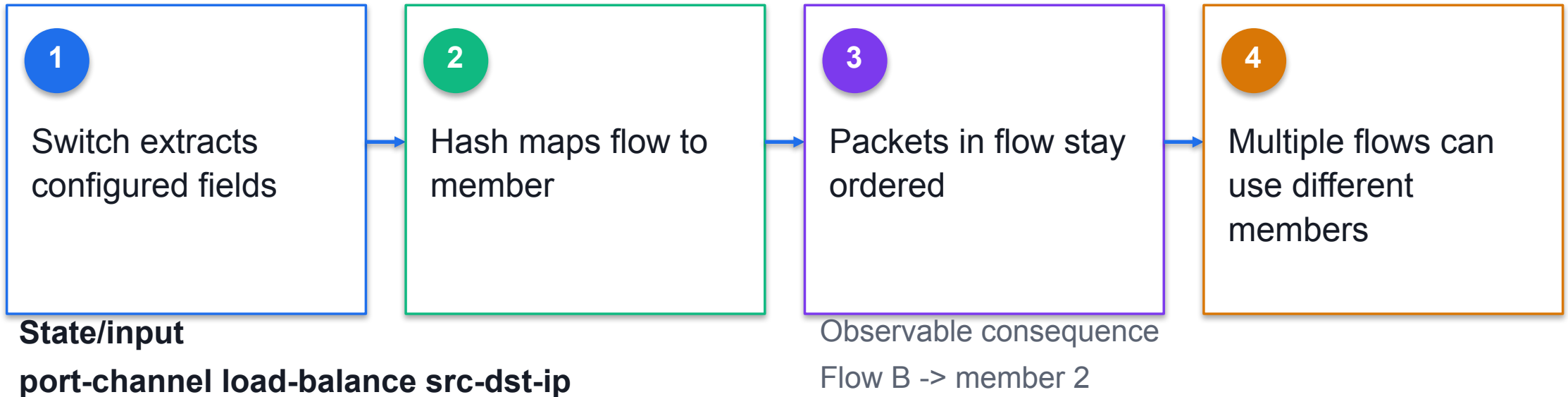
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

EtherChannel load distribution · boundaries



Compare per-member counters across multiple source/destination pairs and interpret aggregate capacity separately.

EtherChannel load distribution · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

EtherChannel load distribution · exact contract

port-channel load-balance src-dst-ip

Flow A -> member 1

Flow B -> member 2

INPUT / IDENTITY

Flow tuple · Switch extracts configured fields

DECISION / STATE

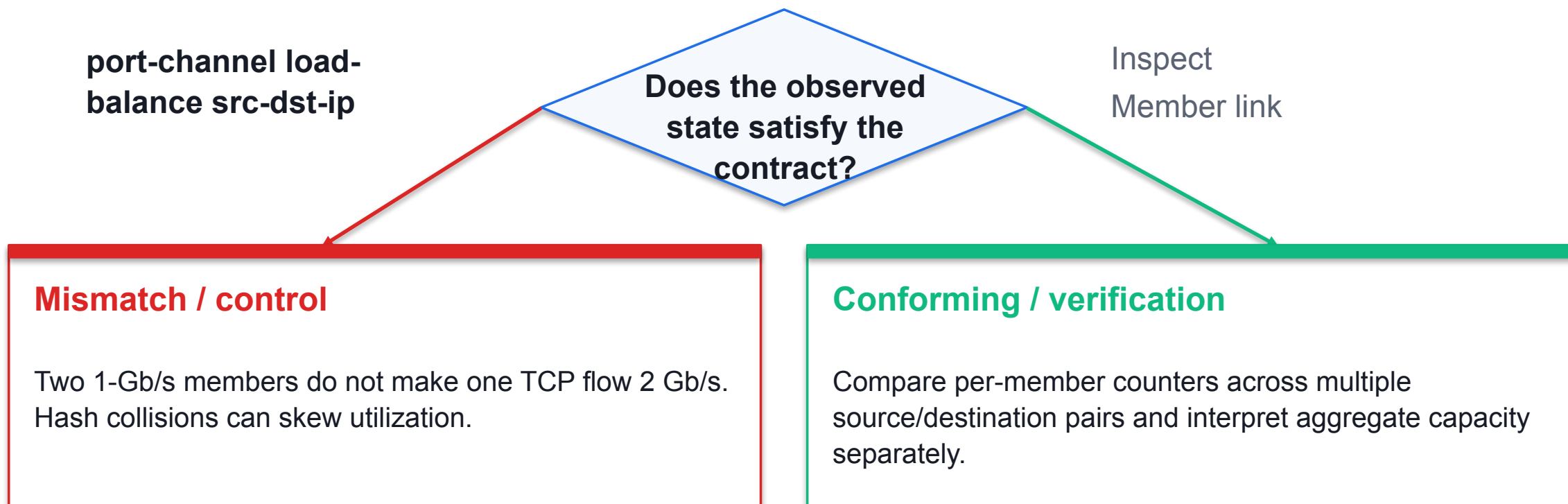
Packets in flow stay ordered

READBACK / RESULT

Multiple flows can use different members

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

EtherChannel load distribution · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

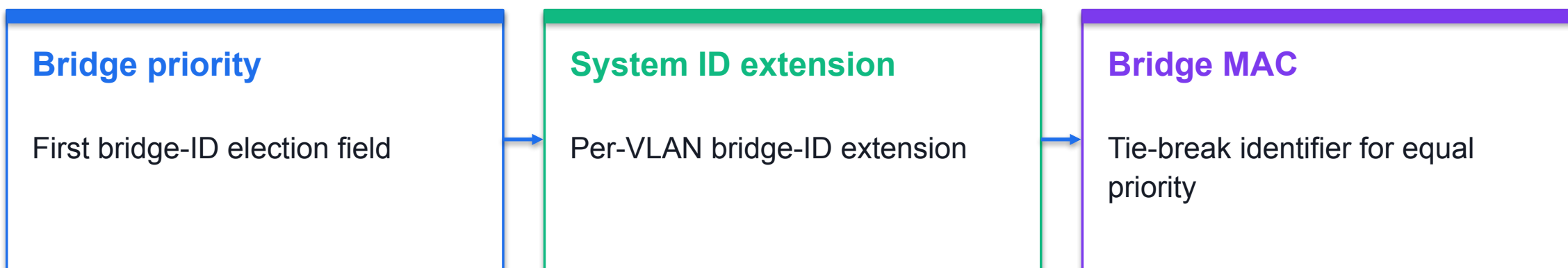
EtherChannel load distribution · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Flow tuple -> Member link	port-channel load-balance src-dst-ip
Mechanism / state	Multiple flows can use different members	Flow B -> member 2
Result / boundary	Compare per-member counters across multiple source/destination pairs and interpret aggregate capacity separately.	Two 1-Gb/s members do not make one TCP flow 2 Gb/s. Hash collisions can skew utilization.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

STP root bridge election · boundaries

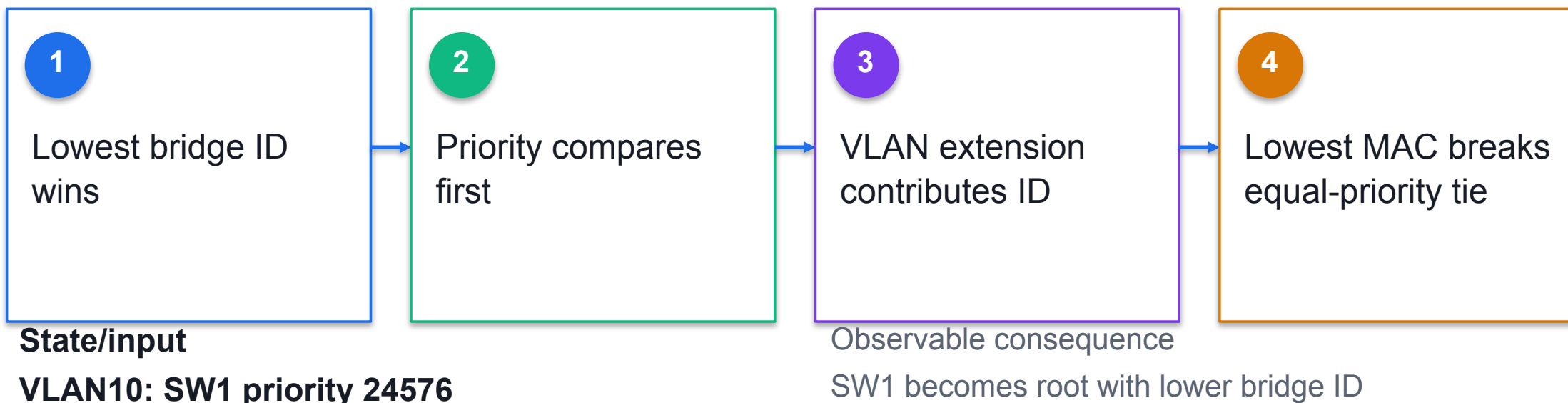


VLAN10: SW1 priority 24576
SW1 becomes root with lower bridge ID

Boundary to inspect
 Unplanned root location can produce poor paths

show spanning-tree vlan 10

STP root bridge election · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

STP root bridge election · exact contract

VLAN10: SW1 priority 24576
SW2 priority 32768
SW1 becomes root with lower bridge ID

INPUT / IDENTITY

Bridge priority · Lowest bridge ID wins

DECISION / STATE

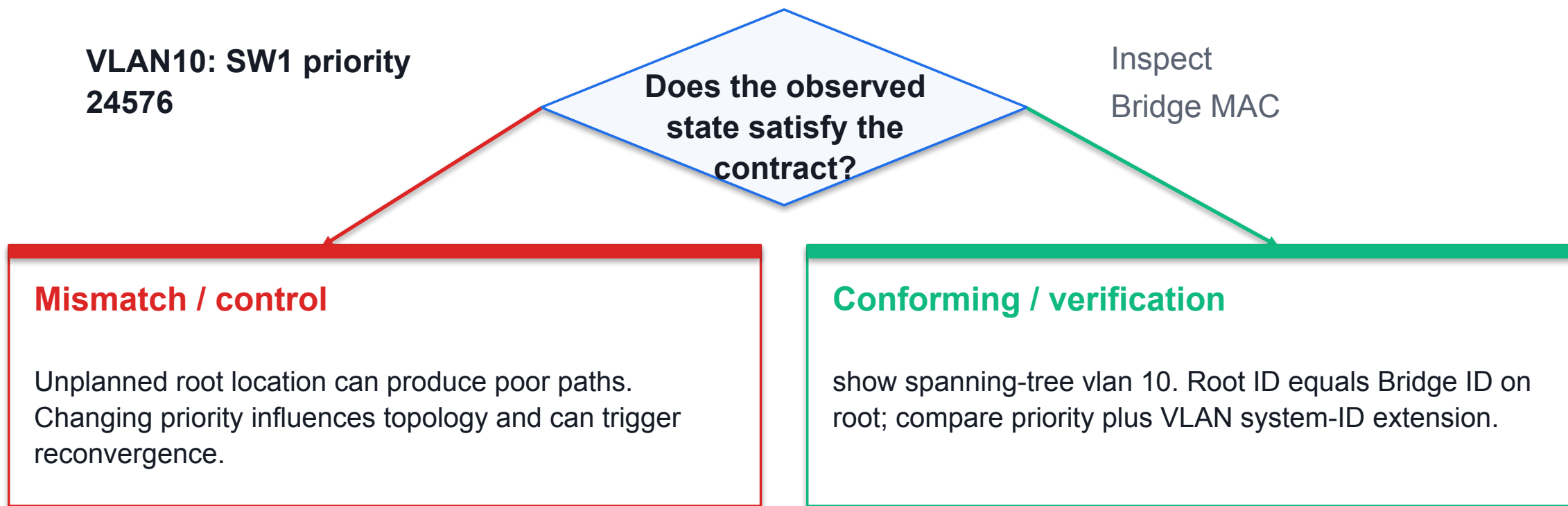
VLAN extension contributes ID

READBACK / RESULT

Lowest MAC breaks equal-priority tie

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

STP root bridge election · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

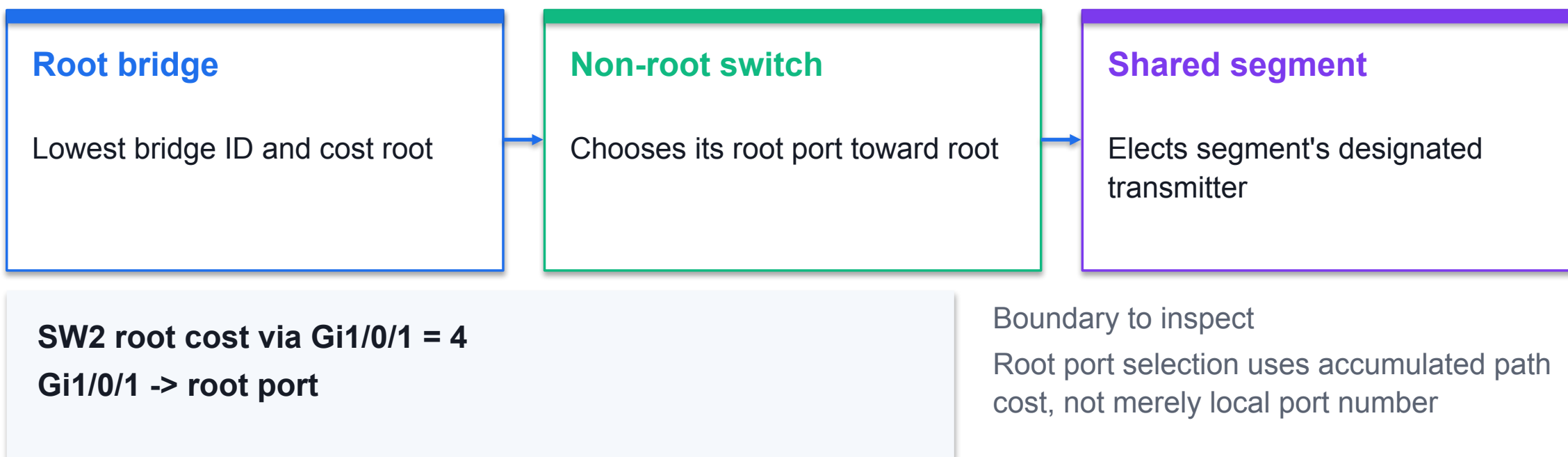
STP root bridge election · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Bridge priority -> Bridge MAC	VLAN10: SW1 priority 24576
Mechanism / state	Lowest MAC breaks equal-priority tie	SW1 becomes root with lower bridge ID
Result / boundary	show spanning-tree vlan 10. Root ID equals Bridge ID on root; compare priority plus VLAN system-ID extension.	Unplanned root location can produce poor paths. Changing priority influences topology and can trigger reconvergence.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

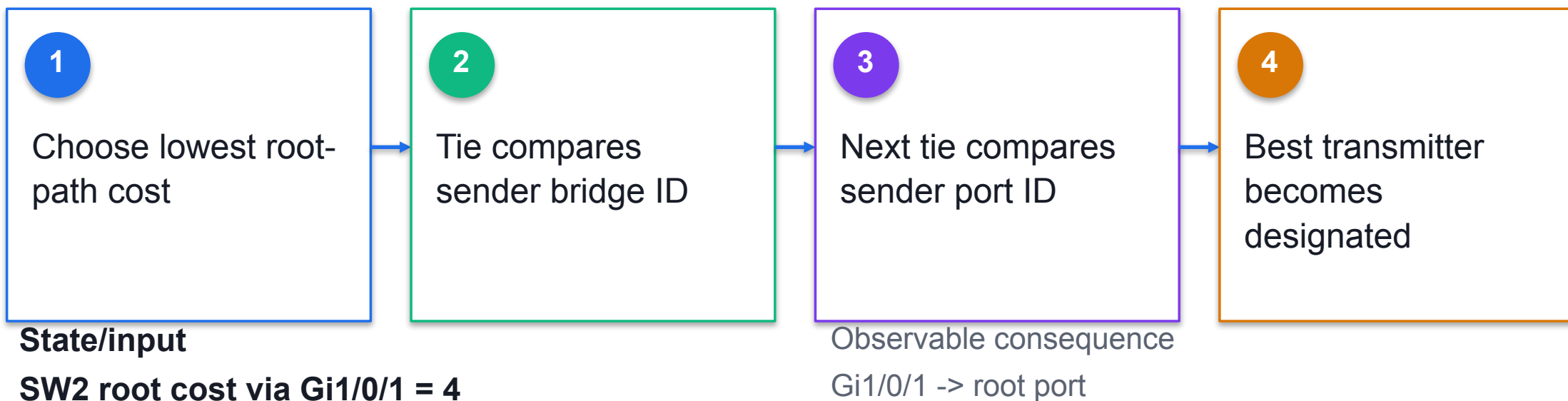
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

STP root and designated ports · boundaries



Annotate each link cost; show spanning-tree reveals root/designated/alternate roles and forwarding state.

STP root and designated ports · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

STP root and designated ports · exact contract

SW2 root cost via Gi1/0/1 = 4
via Gi1/0/2 = 8
Gi1/0/1 -> root port

INPUT / IDENTITY

Root bridge · Choose lowest root-path cost

DECISION / STATE

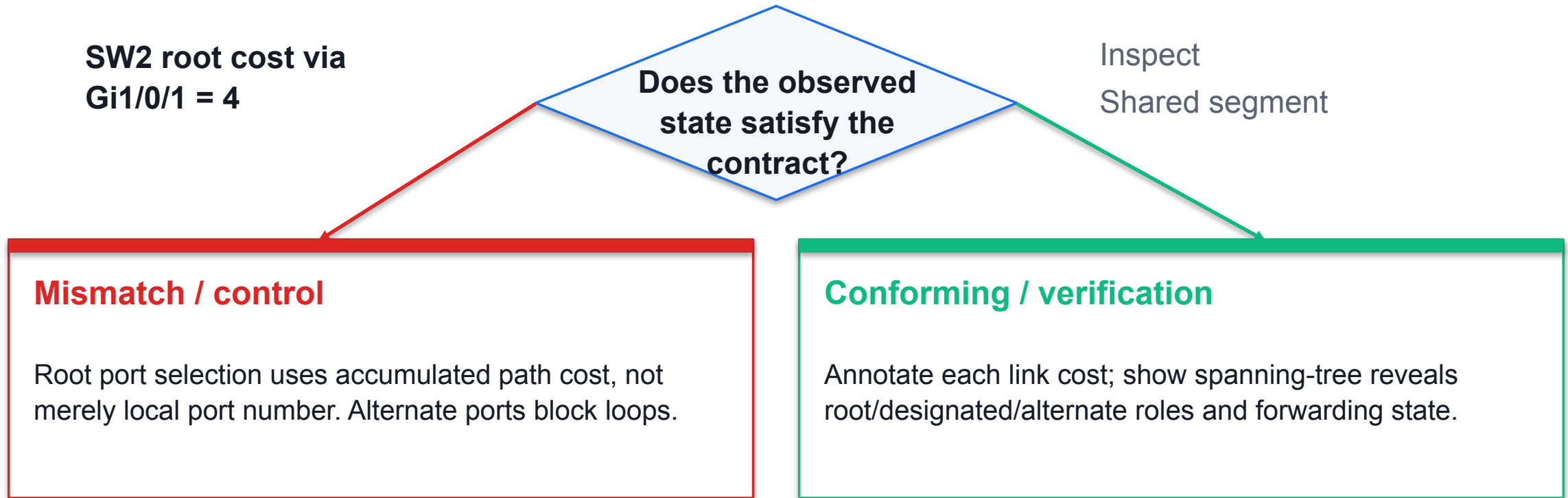
Next tie compares sender port ID

READBACK / RESULT

Best transmitter becomes designated

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

STP root and designated ports · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

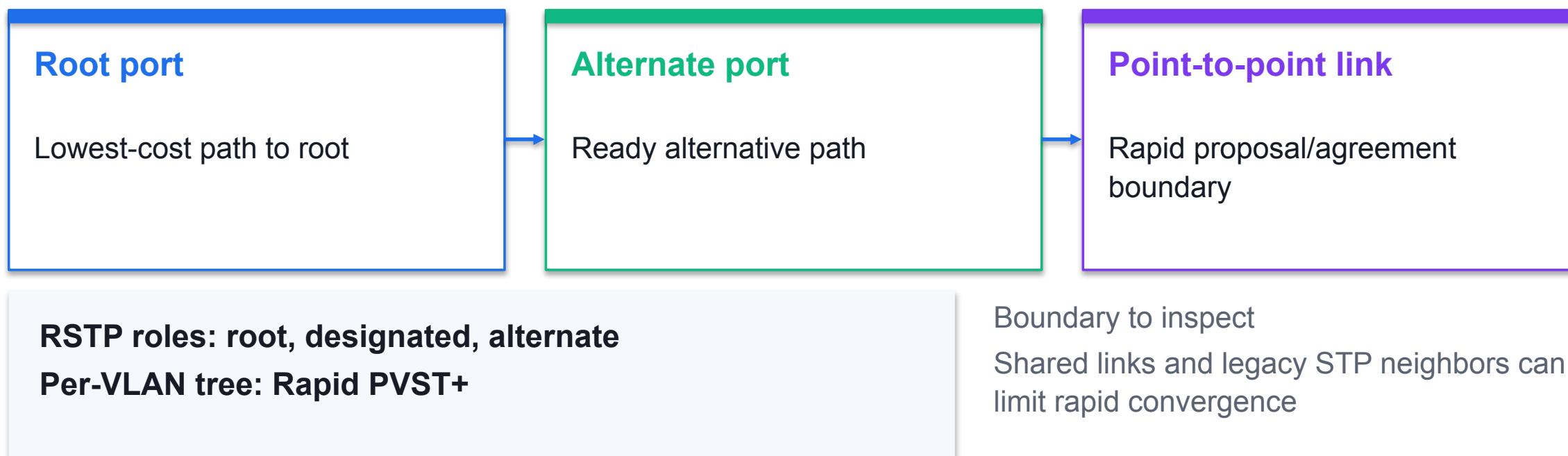
STP root and designated ports · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Root bridge -> Shared segment	SW2 root cost via Gi1/0/1 = 4
Mechanism / state	Best transmitter becomes designated	Gi1/0/1 -> root port
Result / boundary	Annotate each link cost; show spanning-tree reveals root/designated/alternate roles and forwarding state.	Root port selection uses accumulated path cost, not merely local port number. Alternate ports block loops.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

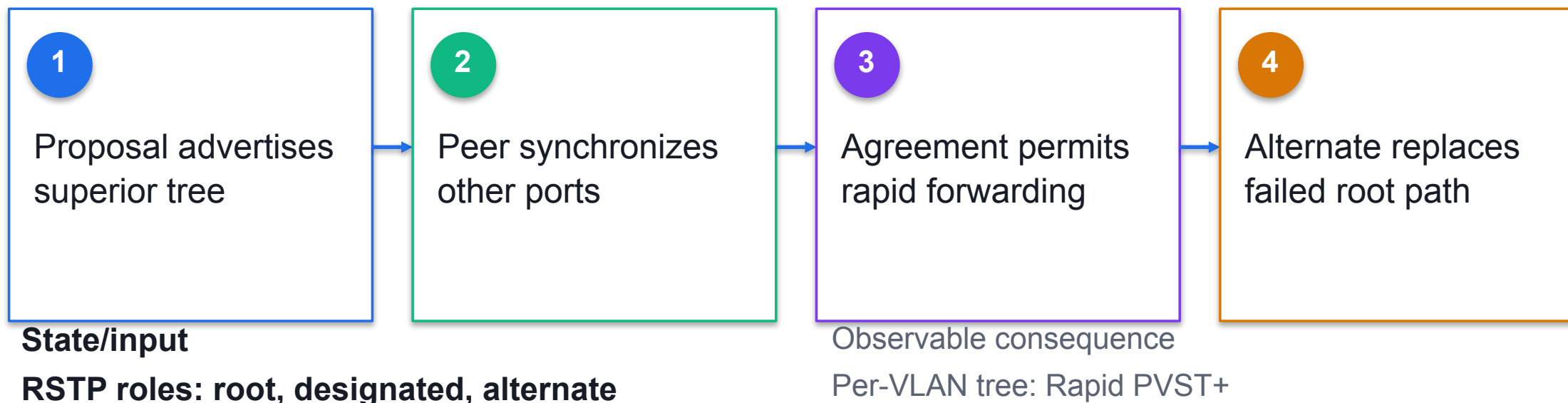
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Rapid PVST convergence · boundaries



show spanning-tree summary and vlan detail; trace role changes after a controlled uplink interruption.

Rapid PVST convergence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Rapid PVST convergence · exact contract

RSTP roles: root, designated, alternate
States: discarding, learning, forwarding
Per-VLAN tree: Rapid PVST+

INPUT / IDENTITY

Root port · Proposal advertises superior tree

DECISION / STATE

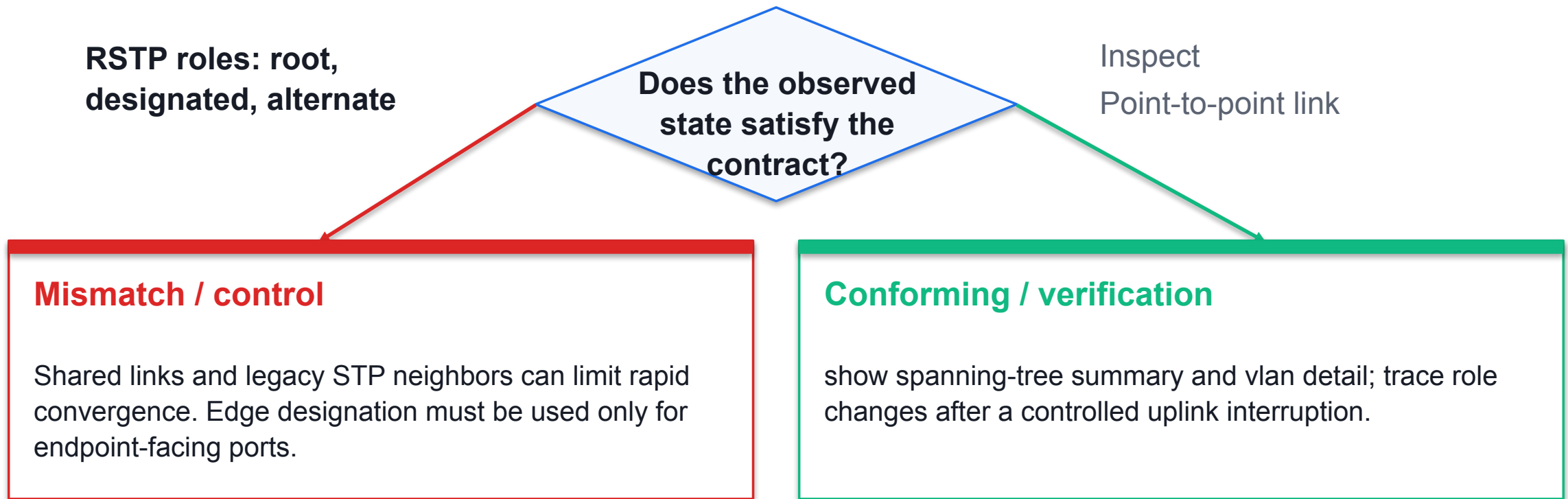
Agreement permits rapid forwarding

READBACK / RESULT

Alternate replaces failed root path

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Rapid PVST convergence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

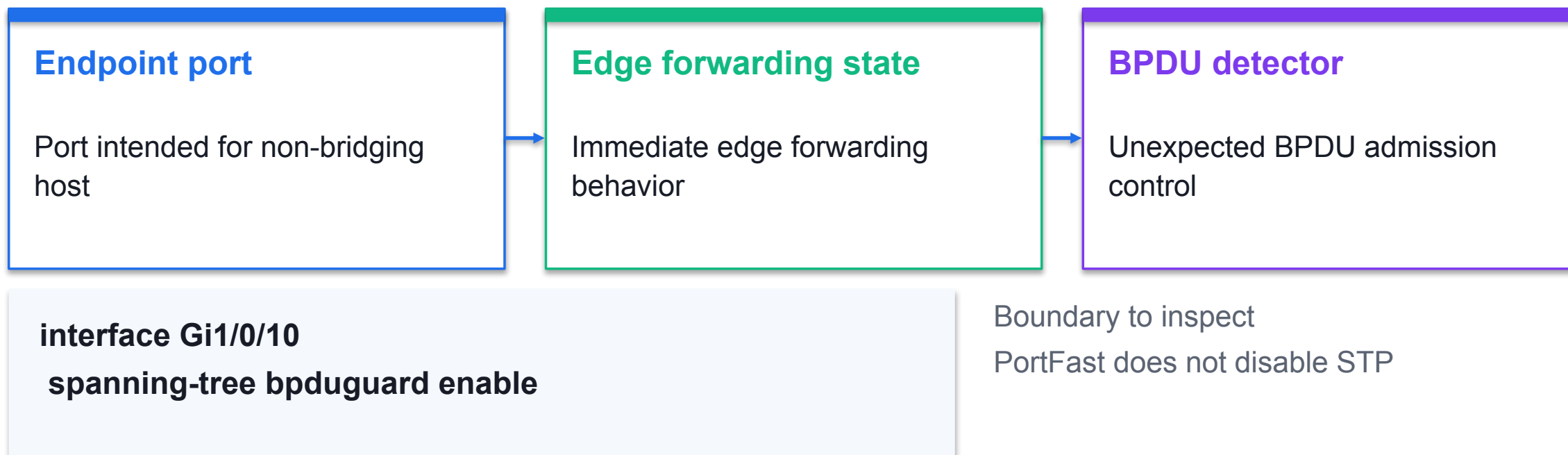
Rapid PVST convergence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Root port -> Point-to-point link	RSTP roles: root, designated, alternate
Mechanism / state	Alternate replaces failed root path	Per-VLAN tree: Rapid PVST+
Result / boundary	show spanning-tree summary and vlan detail; trace role changes after a controlled uplink interruption.	Shared links and legacy STP neighbors can limit rapid convergence. Edge designation must be used only for endpoint-facing ports.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

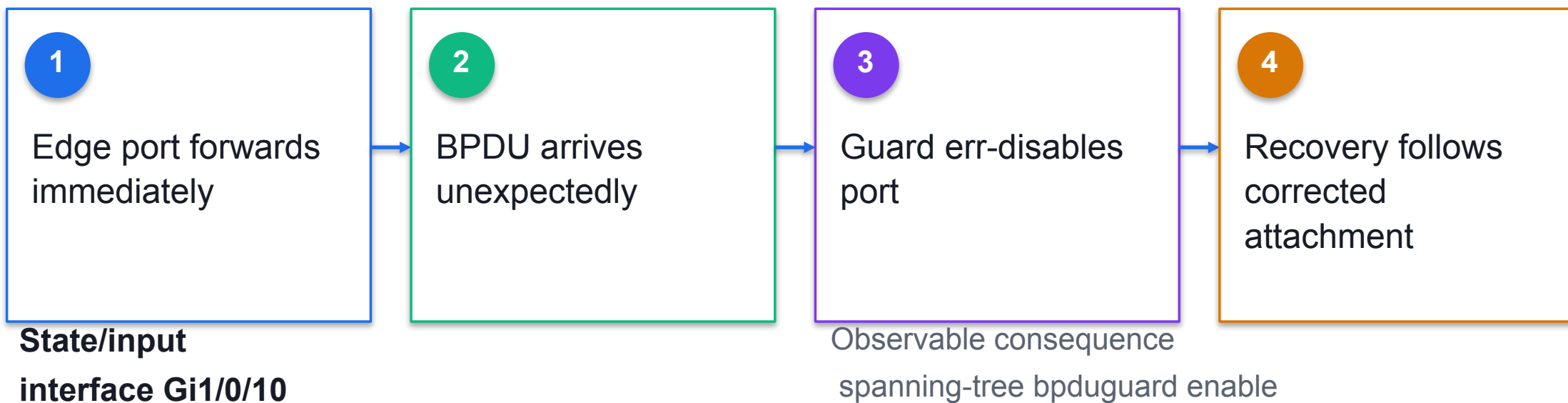
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

PortFast and BPDU guard · boundaries



show interfaces status err-disabled; show logging

PortFast and BPDU guard · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

PortFast and BPDU guard · exact contract

```
interface Gi1/0/10
spanning-tree portfast
spanning-tree bpduguard enable
```

INPUT / IDENTITY

Endpoint port · Edge port forwards immediately

DECISION / STATE

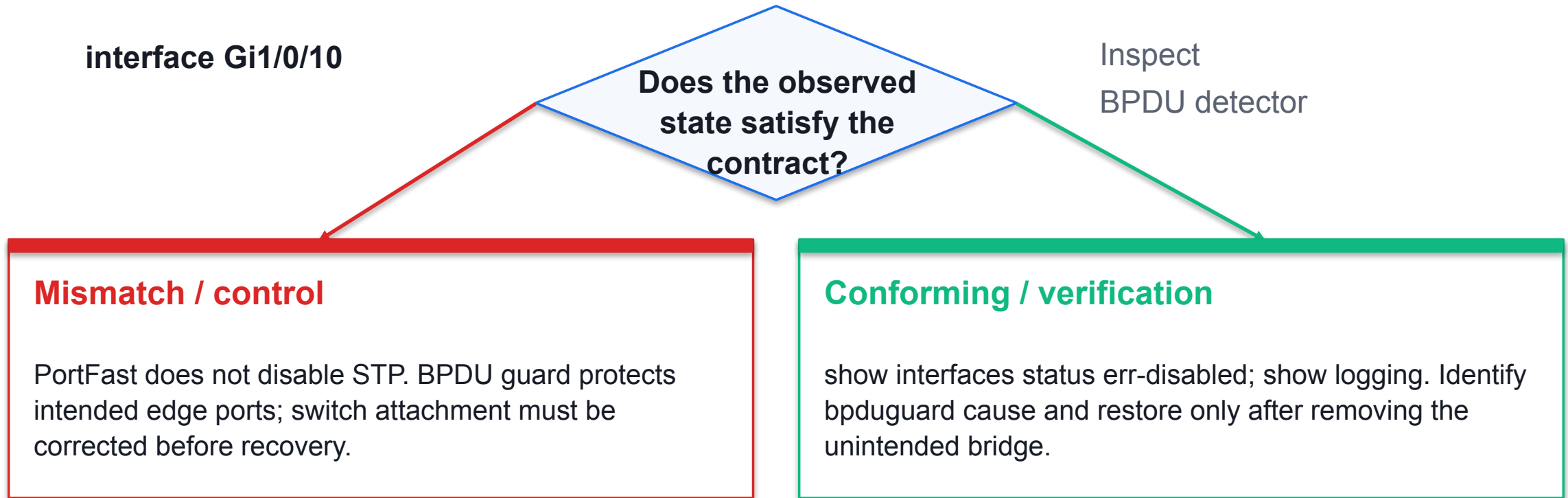
Guard err-disables port

READBACK / RESULT

Recovery follows corrected attachment

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

PortFast and BPDU guard · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

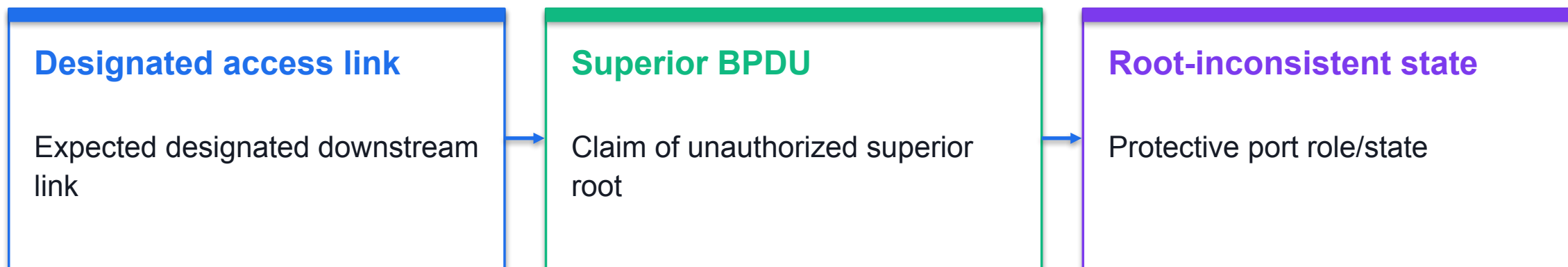
PortFast and BPDU guard · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Endpoint port -> BPDU detector	interface Gi1/0/10
Mechanism / state	Recovery follows corrected attachment	spanning-tree bpduguard enable
Result / boundary	show interfaces status err-disabled; show logging. Identify bpduguard cause and restore only after removing the unintended bridge.	PortFast does not disable STP. BPDU guard protects intended edge ports; switch attachment must be corrected before recovery.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Root guard boundary · boundaries

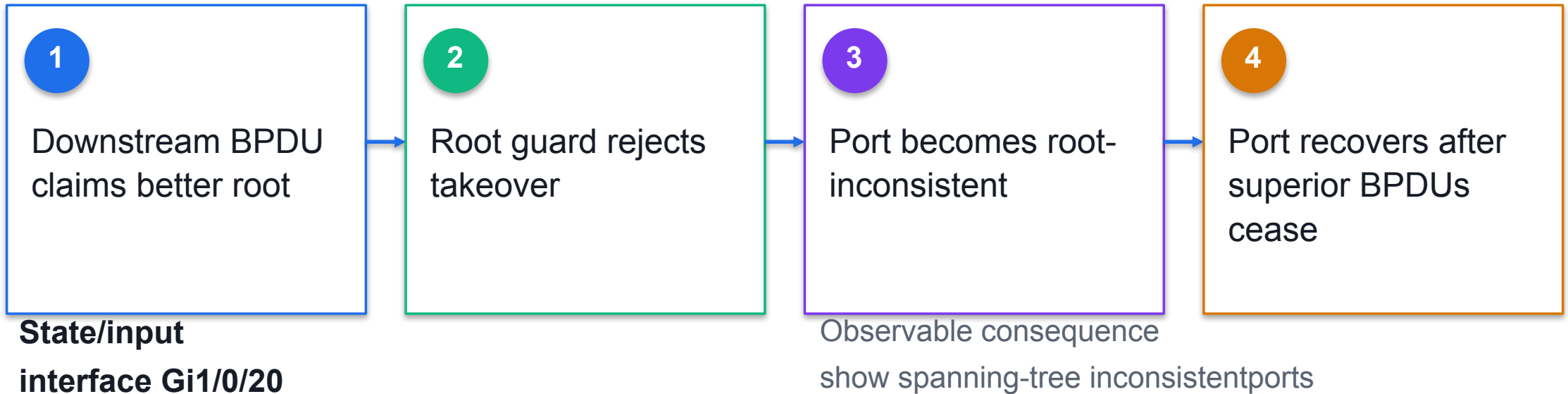


interface Gi1/0/20
show spanning-tree inconsistentports

Boundary to inspect
Do not place root guard where the root port is legitimately expected

Verify root-inconsistent reason and root bridge remains authorized

Root guard boundary · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Root guard boundary · exact contract

```
interface Gi1/0/20
  spanning-tree guard root
  show spanning-tree inconsistentports
```

INPUT / IDENTITY

Designated access link · Downstream
BPDU claims better root

DECISION / STATE

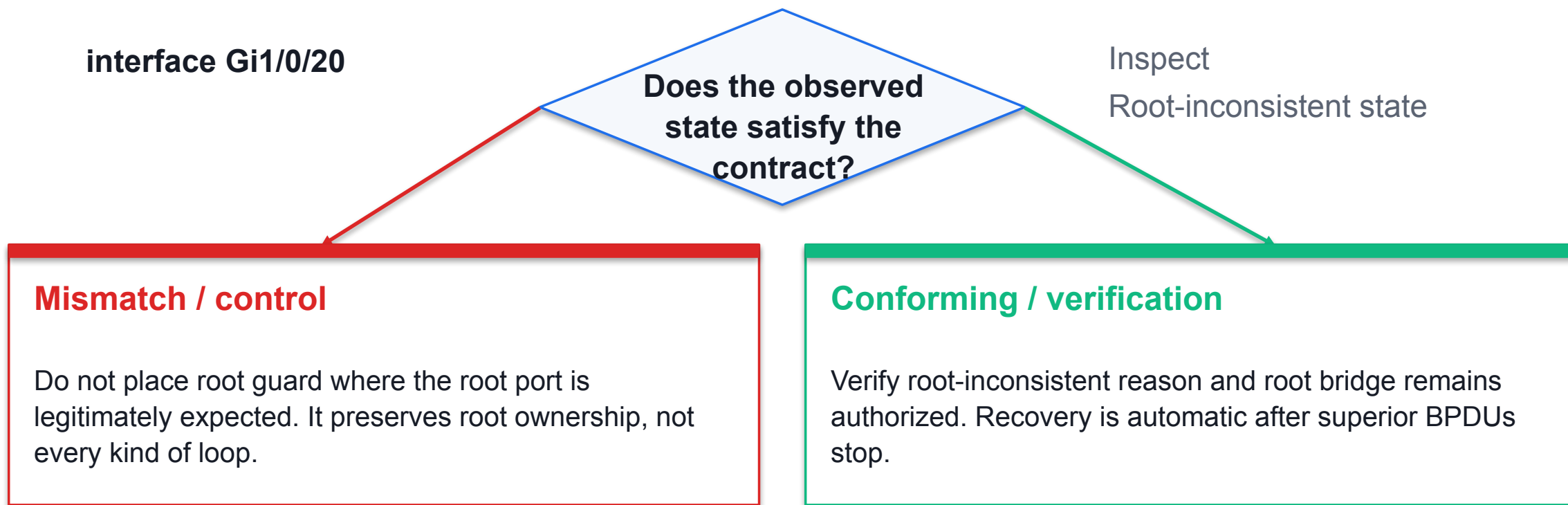
Port becomes root-inconsistent

READBACK / RESULT

Port recovers after superior BPDUs
cease

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Root guard boundary · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

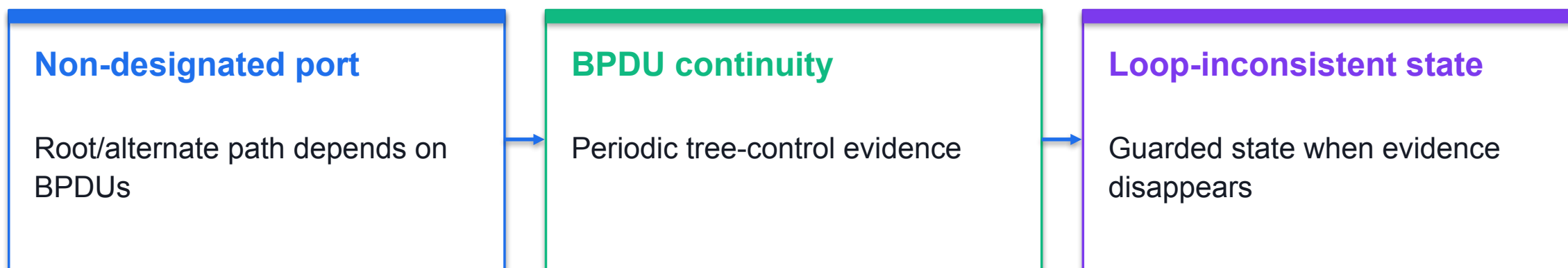
Root guard boundary · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Designated access link -> Root-inconsistent state	interface Gi1/0/20
Mechanism / state	Port recovers after superior BPDUs cease	show spanning-tree inconsistentports
Result / boundary	Verify root-inconsistent reason and root bridge remains authorized. Recovery is automatic after superior BPDUs stop.	Do not place root guard where the root port is legitimately expected. It preserves root ownership, not every kind of loop.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Loop guard and BPDU filter · boundaries

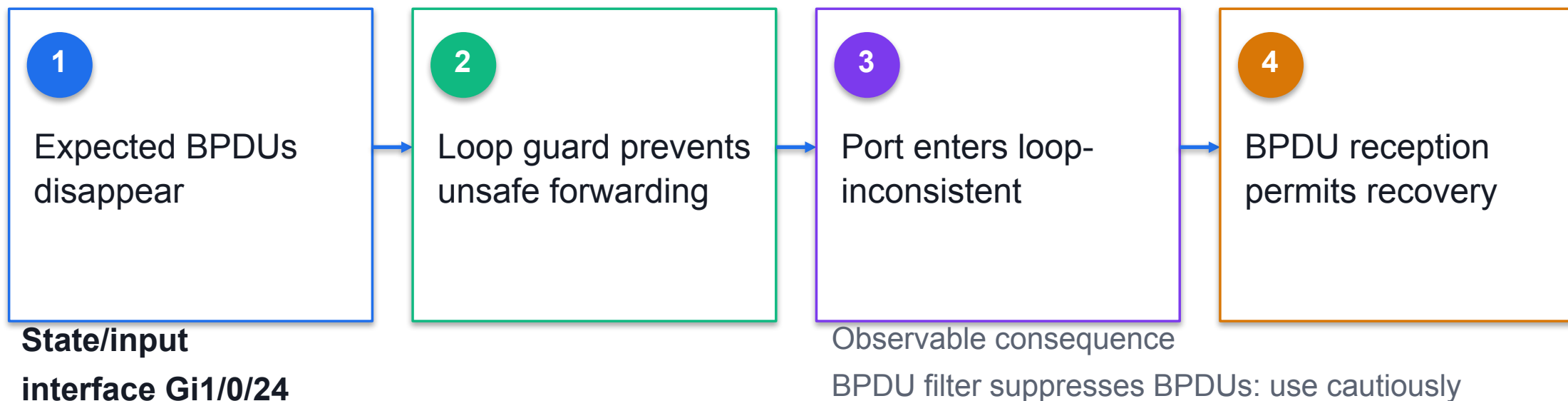


interface Gi1/0/24
BPDU filter suppresses BPDUs: use cautiously

Boundary to inspect
 Interface-level BPDU filtering can effectively remove STP protection and cause loops

show spanning-tree inconsistentports

Loop guard and BPDU filter · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Loop guard and BPDU filter · exact contract

```
interface Gi1/0/24
  spanning-tree guard loop
  BPDU filter suppresses BPDUs: use cautiously
```

INPUT / IDENTITY

Non-designated port · Expected BPDUs disappear

DECISION / STATE

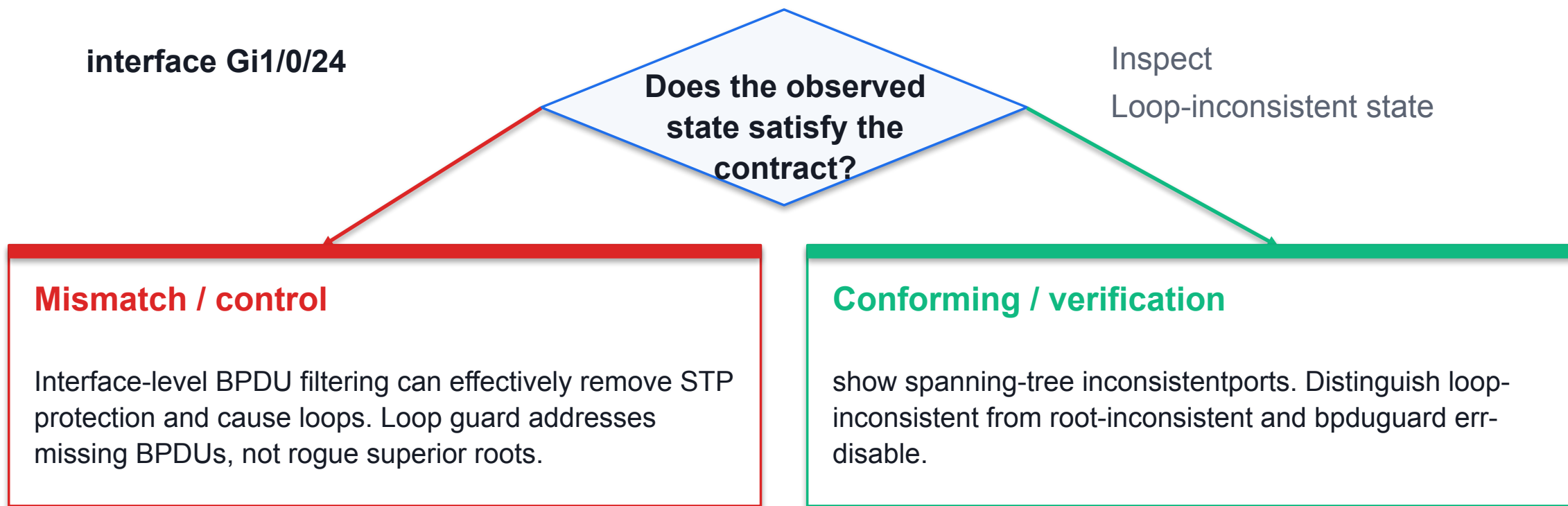
Port enters loop-inconsistent

READBACK / RESULT

BPDU reception permits recovery

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Loop guard and BPDU filter · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

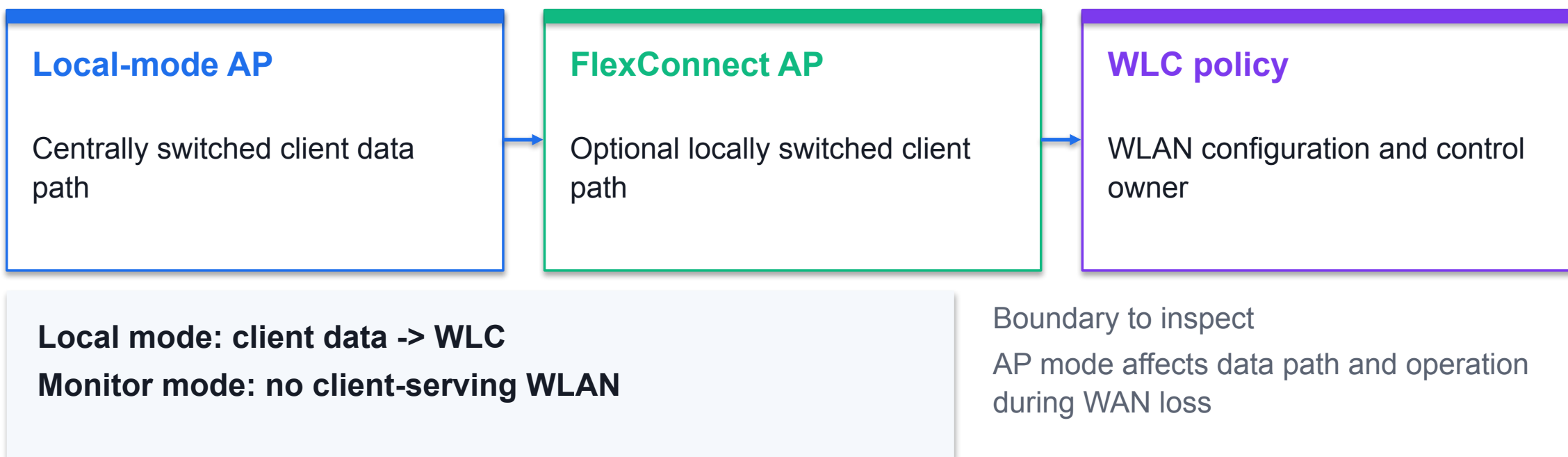
Loop guard and BPDU filter · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Non-designated port -> Loop-inconsistent state	interface Gi1/0/24
Mechanism / state	BPDU reception permits recovery	BPDU filter suppresses BPDUs: use cautiously
Result / boundary	show spanning-tree inconsistentports. Distinguish loop-inconsistent from root-inconsistent and bpduguard err-disable.	Interface-level BPDU filtering can effectively remove STP protection and cause loops. Loop guard addresses missing BPDUs, not rogue superior roots.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

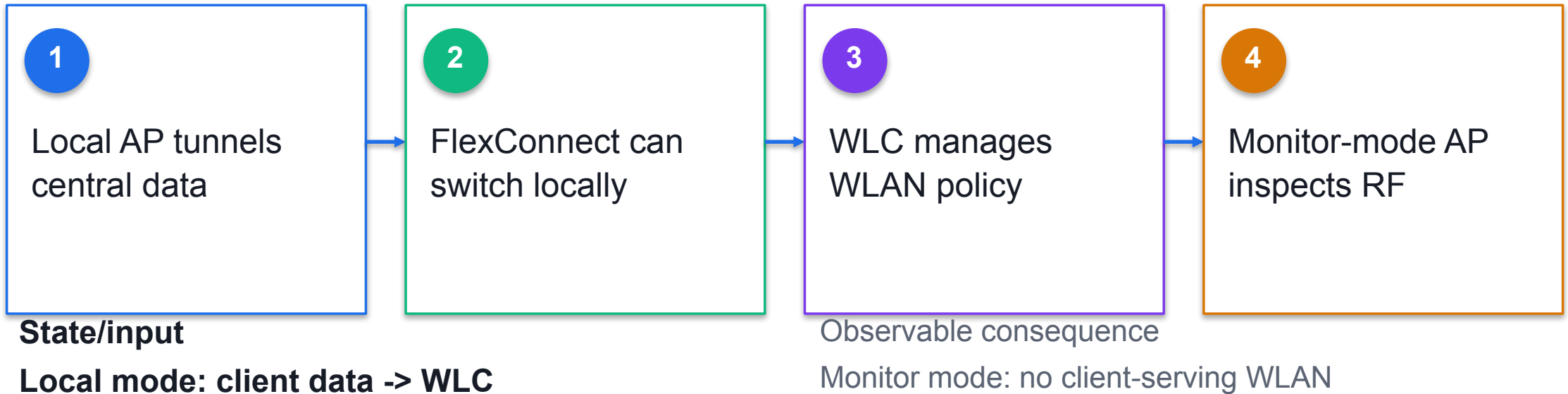
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

AP modes and WLAN transport · boundaries



Identify AP mode and switching policy; compare client gateway location with observed packet path.

AP modes and WLAN transport · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

AP modes and WLAN transport · exact contract

Local mode: client data -> WLC

FlexConnect local: client data -> local VLAN

Monitor mode: no client-serving WLAN

INPUT / IDENTITY

Local-mode AP · Local AP tunnels
central data

DECISION / STATE

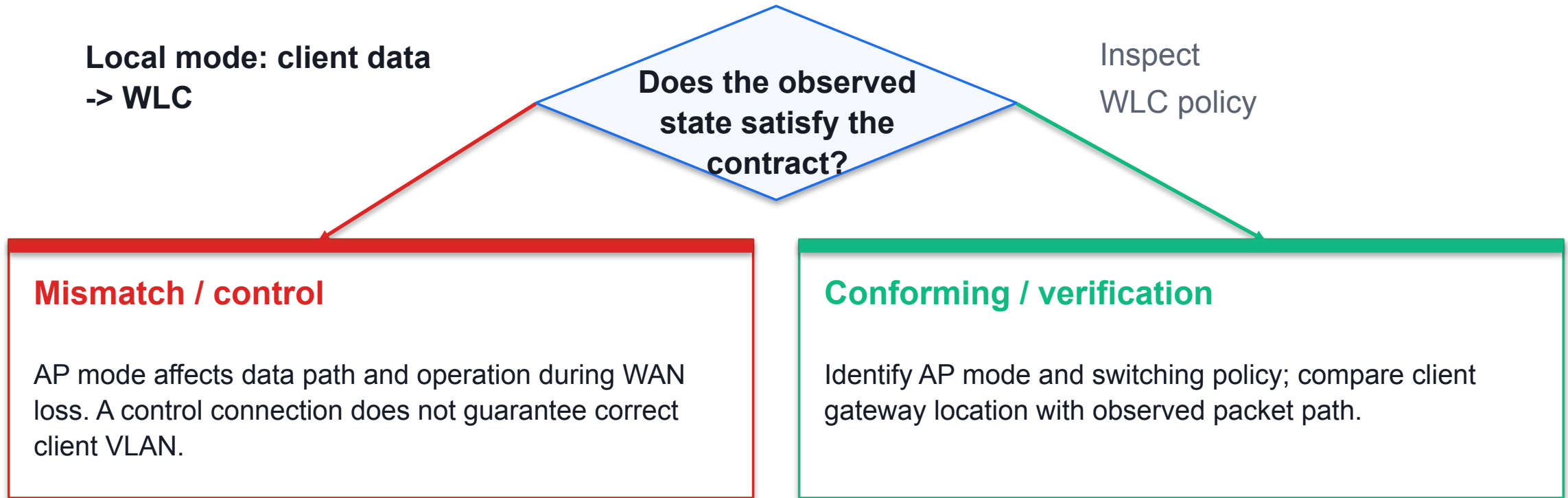
WLC manages WLAN policy

READBACK / RESULT

Monitor-mode AP inspects RF

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

AP modes and WLAN transport · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

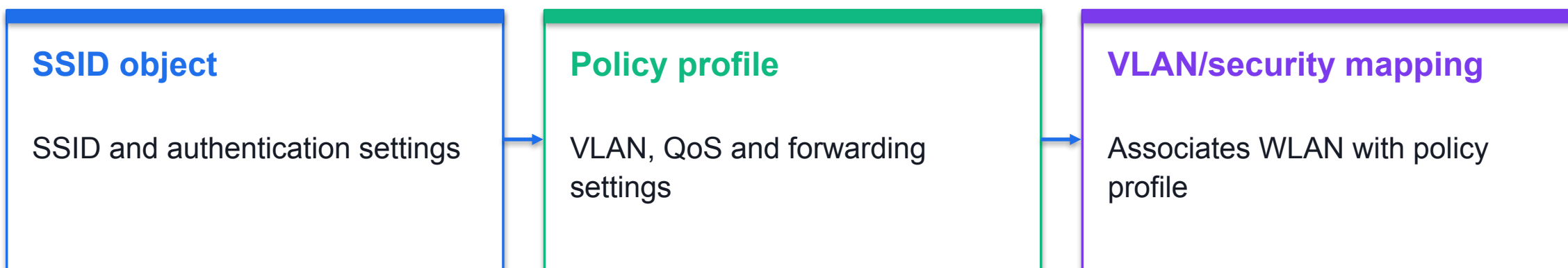
AP modes and WLAN transport · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Local-mode AP -> WLC policy	Local mode: client data -> WLC
Mechanism / state	Monitor-mode AP inspects RF	Monitor mode: no client-serving WLAN
Result / boundary	Identify AP mode and switching policy; compare client gateway location with observed packet path.	AP mode affects data path and operation during WAN loss. A control connection does not guarantee correct client VLAN.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

WLAN GUI policy evidence · boundaries



Catalyst 9800 objects:
Policy Tag WLAN->Profile

Boundary to inspect
A visually correct SSID can map to a wrong policy/VLAN

Read WLAN, policy profile and AP tag together

WLAN GUI policy evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

WLAN GUI policy evidence · exact contract

Catalyst 9800 objects:
WLAN ID + SSID
Policy Profile VLAN 20
Policy Tag WLAN->Profile

INPUT / IDENTITY

SSID object · WLAN defines
SSID/security

DECISION / STATE

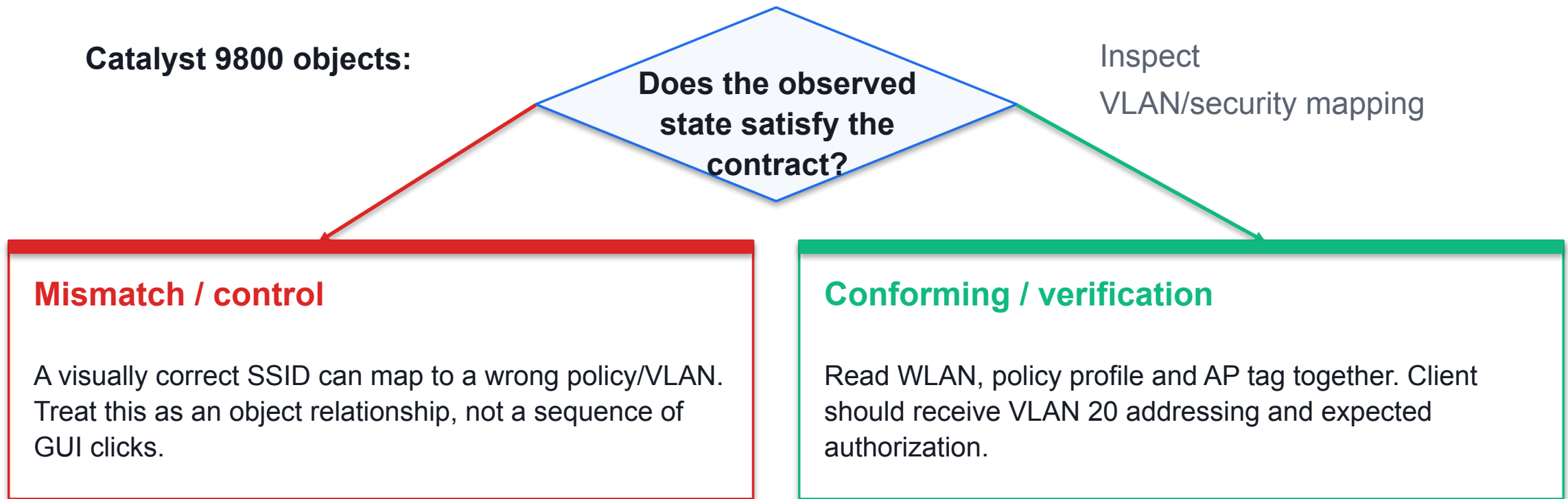
Policy tag associates WLAN and profile

READBACK / RESULT

AP applies assigned tag

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

WLAN GUI policy evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

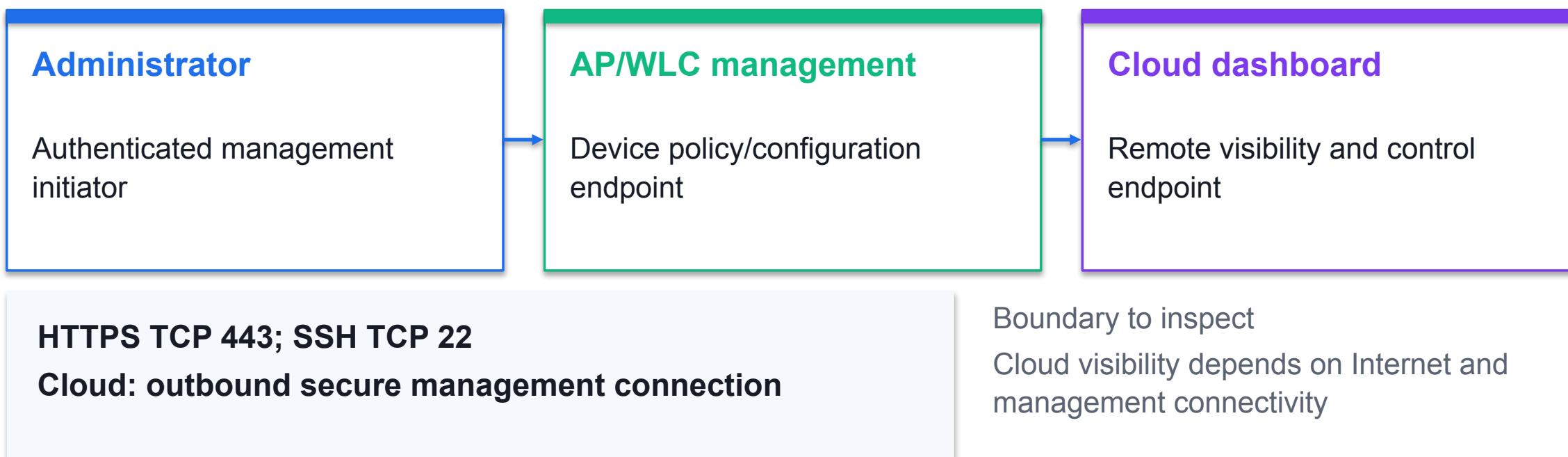
WLAN GUI policy evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	SSID object -> VLAN/security mapping	Catalyst 9800 objects:
Mechanism / state	AP applies assigned tag	Policy Tag WLAN->Profile
Result / boundary	Read WLAN, policy profile and AP tag together. Client should receive VLAN 20 addressing and expected authorization.	A visually correct SSID can map to a wrong policy/VLAN. Treat this as an object relationship, not a sequence of GUI clicks.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

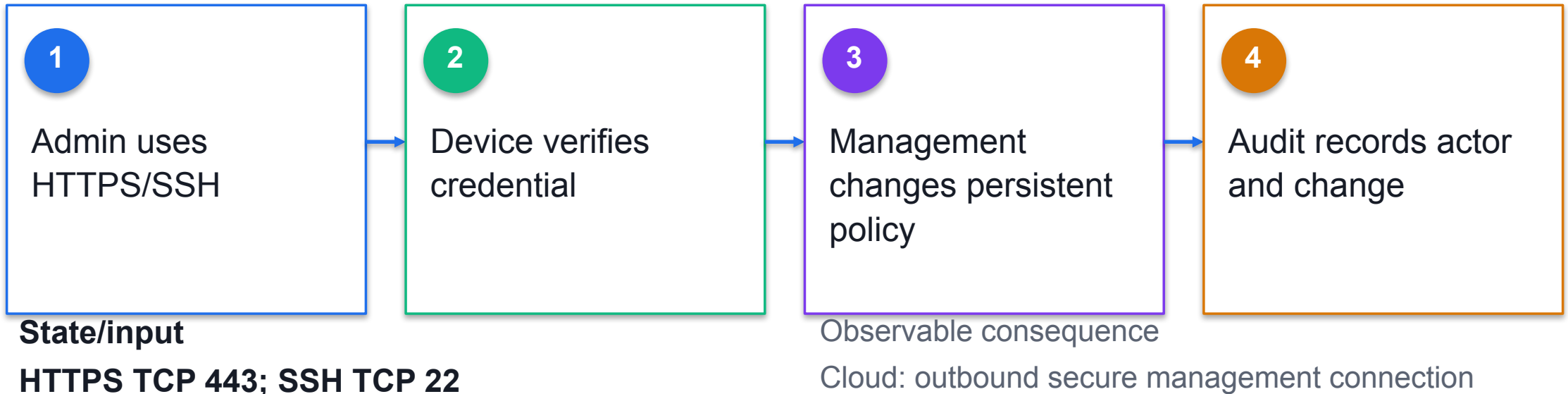
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Wireless management reachability · boundaries



Identify management transport, role permission and audit evidence

Wireless management reachability · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Wireless management reachability · exact contract

HTTPS TCP 443; SSH TCP 22

Console: local out-of-band

Cloud: outbound secure management connection

INPUT / IDENTITY

Administrator · Admin uses HTTPS/SSH

DECISION / STATE

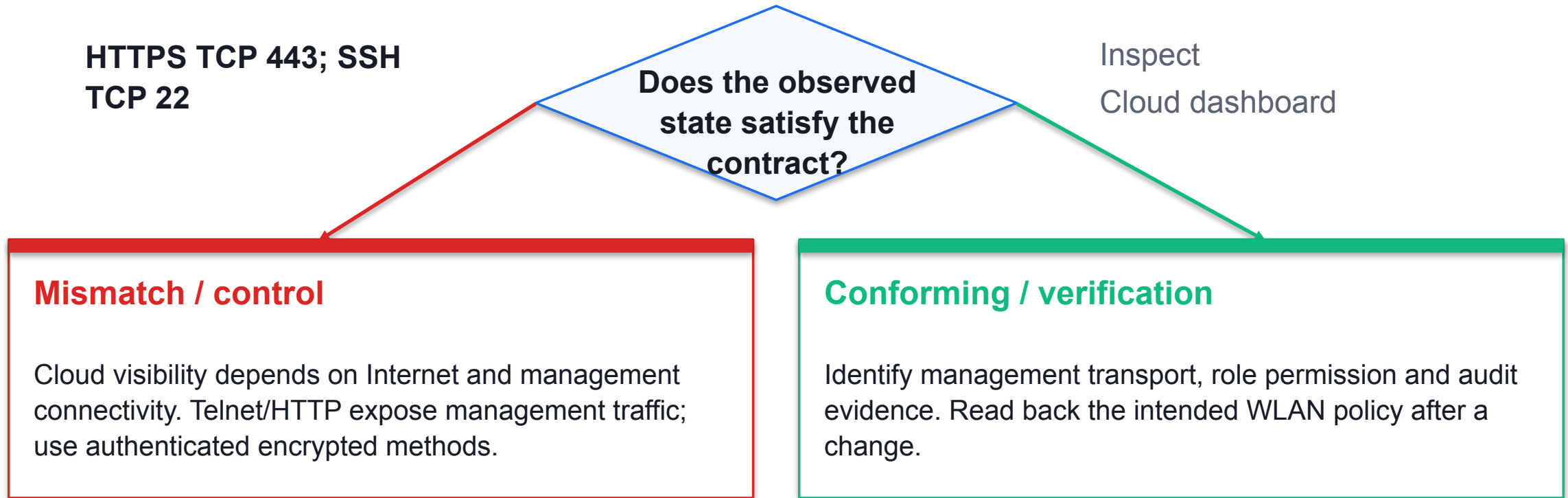
Management changes persistent policy

READBACK / RESULT

Audit records actor and change

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Wireless management reachability · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

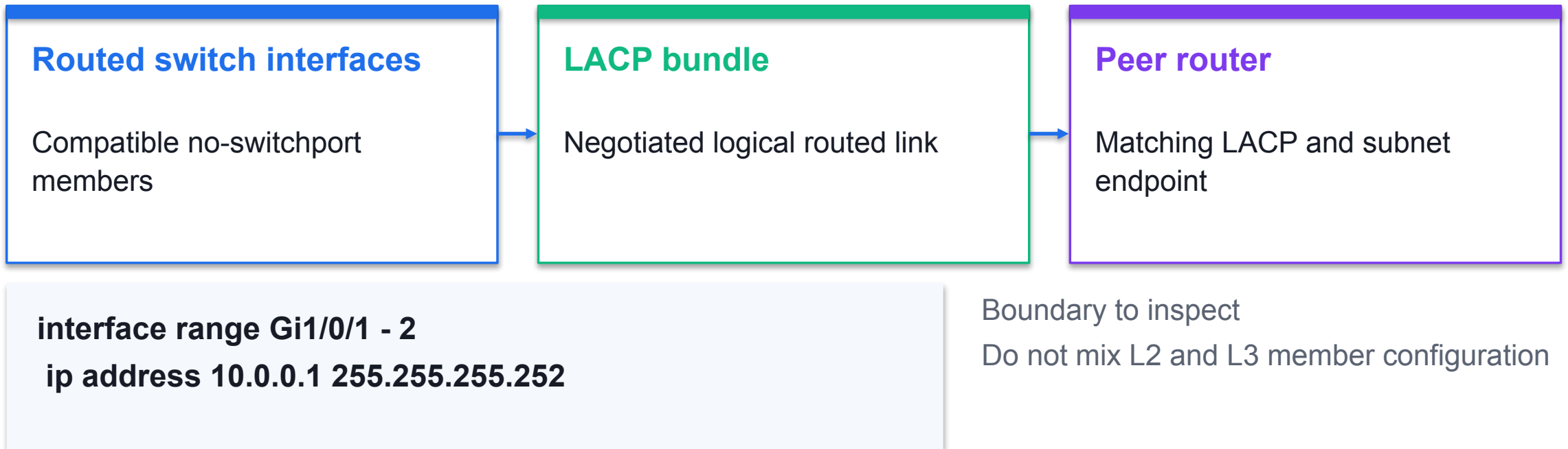
Wireless management reachability · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Administrator -> Cloud dashboard	HTTPS TCP 443; SSH TCP 22
Mechanism / state	Audit records actor and change	Cloud: outbound secure management connection
Result / boundary	Identify management transport, role permission and audit evidence. Read back the intended WLAN policy after a change.	Cloud visibility depends on Internet and management connectivity. Telnet/HTTP expose management traffic; use authenticated encrypted methods.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

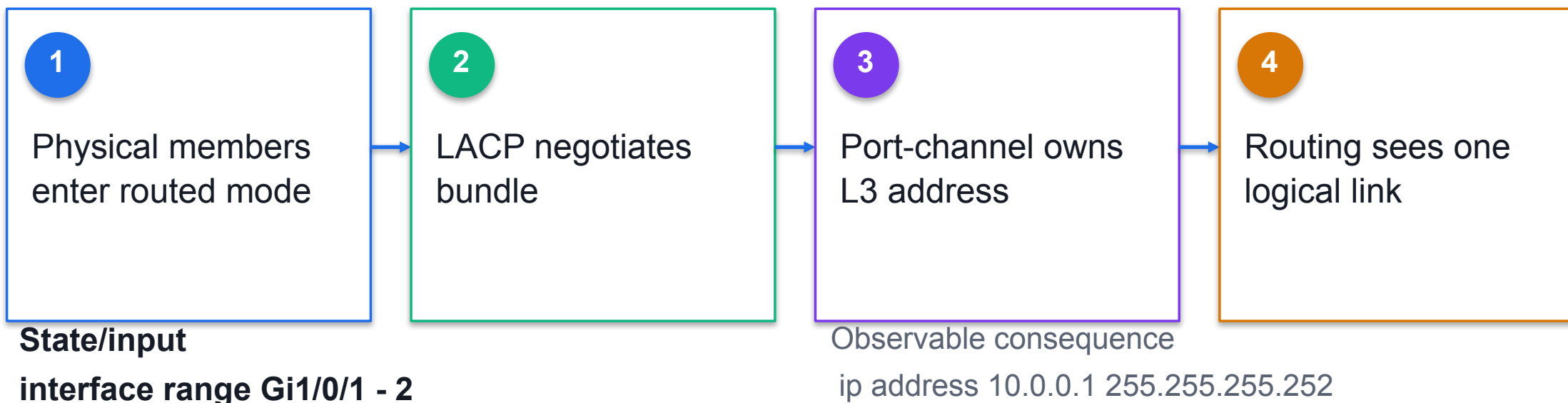
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Layer 3 LACP port-channel · boundaries



show etherchannel summary expects routed R flag and bundled P members; connected subnet attaches to Port-channel 1.

Layer 3 LACP port-channel · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Layer 3 LACP port-channel · exact contract

```
interface range Gi1/0/1 - 2
no switchport
channel-group 1 mode active
interface Port-channel1
no switchport
ip address 10.0.0.1 255.255.255.252
```

INPUT / IDENTITY

Routed switch interfaces · Physical members enter routed mode

DECISION / STATE

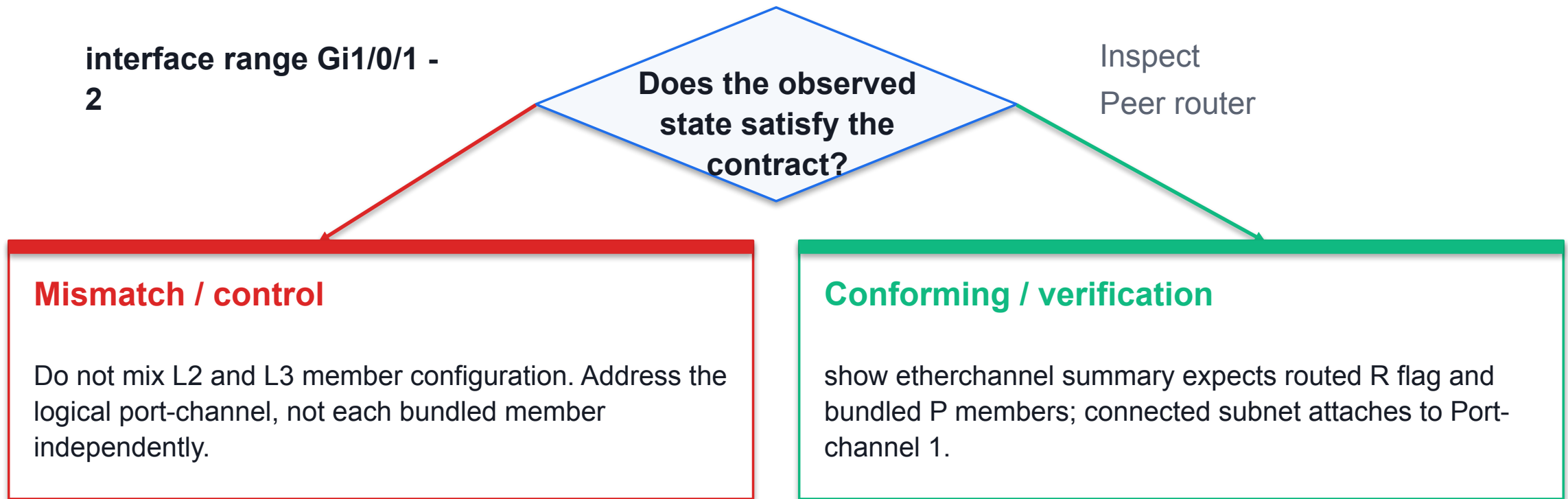
Port-channel owns L3 address

READBACK / RESULT

Routing sees one logical link

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Layer 3 LACP port-channel · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

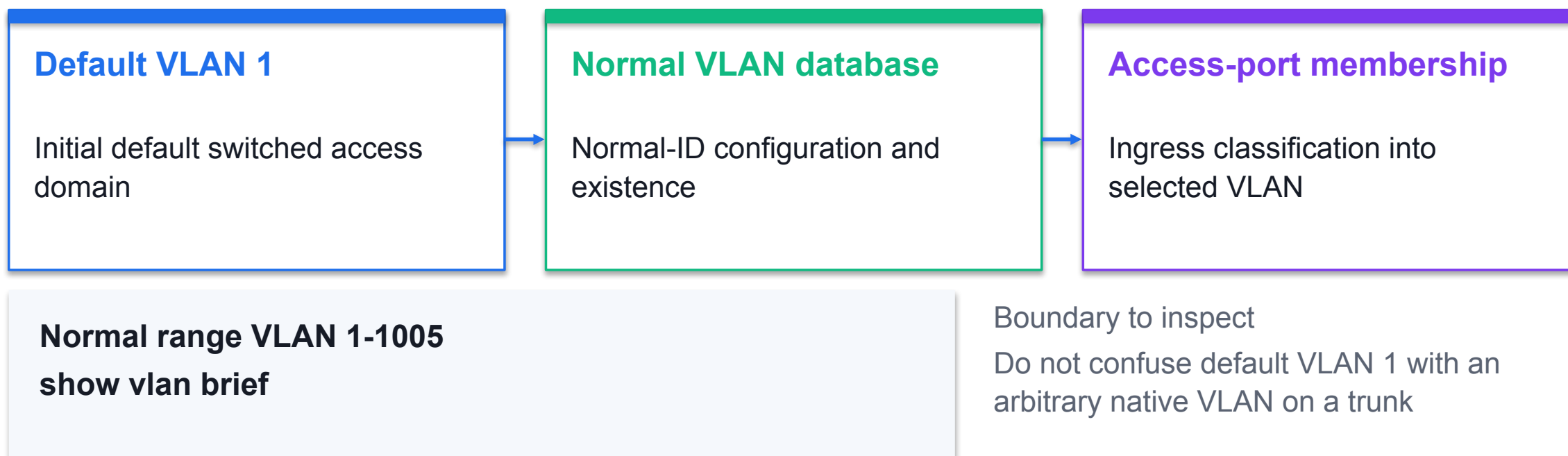
Layer 3 LACP port-channel · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Routed switch interfaces -> Peer router	interface range Gi1/0/1 - 2
Mechanism / state	Routing sees one logical link	ip address 10.0.0.1 255.255.255.252
Result / boundary	show etherchannel summary expects routed R flag and bundled P members; connected subnet attaches to Port-channel 1.	Do not mix L2 and L3 member configuration. Address the logical port-channel, not each bundled member independently.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

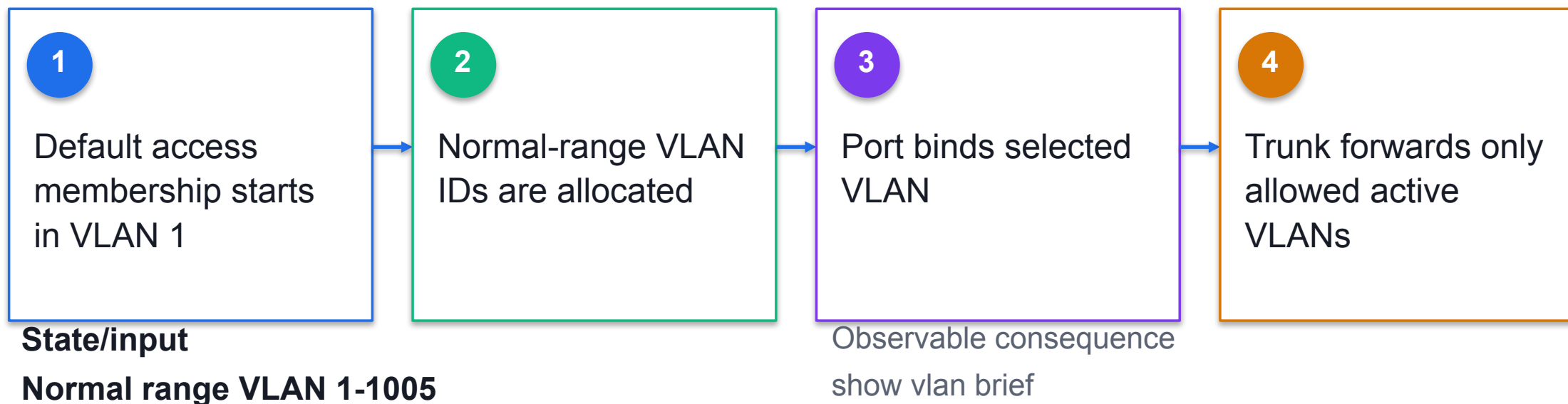
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Default and normal-range VLAN contracts · boundaries



Read VLAN existence, access membership and operational trunk lists; explicitly distinguish default, access, native and management VLAN roles.

Default and normal-range VLAN contracts · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Default and normal-range VLAN contracts · exact contract

Normal range VLAN 1-1005
VLAN 1002-1005 reserved legacy default VLANs
User example VLAN 10
show vlan brief

INPUT / IDENTITY

Default VLAN 1 · Default access membership starts in VLAN 1

DECISION / STATE

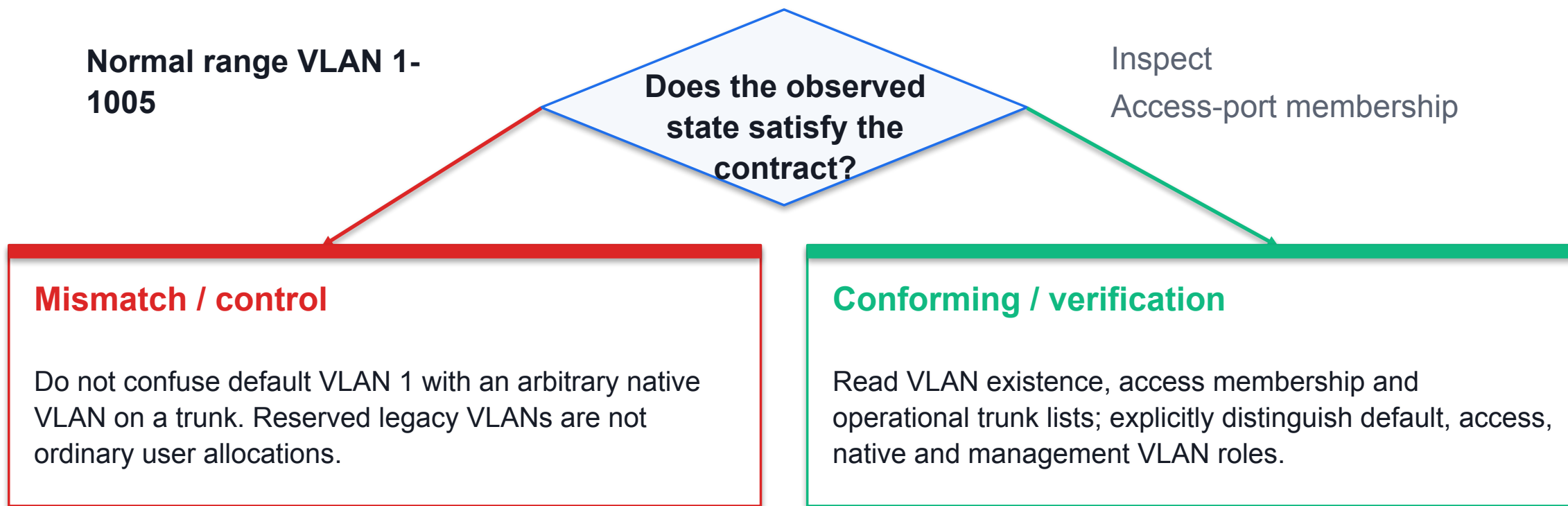
Port binds selected VLAN

READBACK / RESULT

Trunk forwards only allowed active VLANs

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Default and normal-range VLAN contracts · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

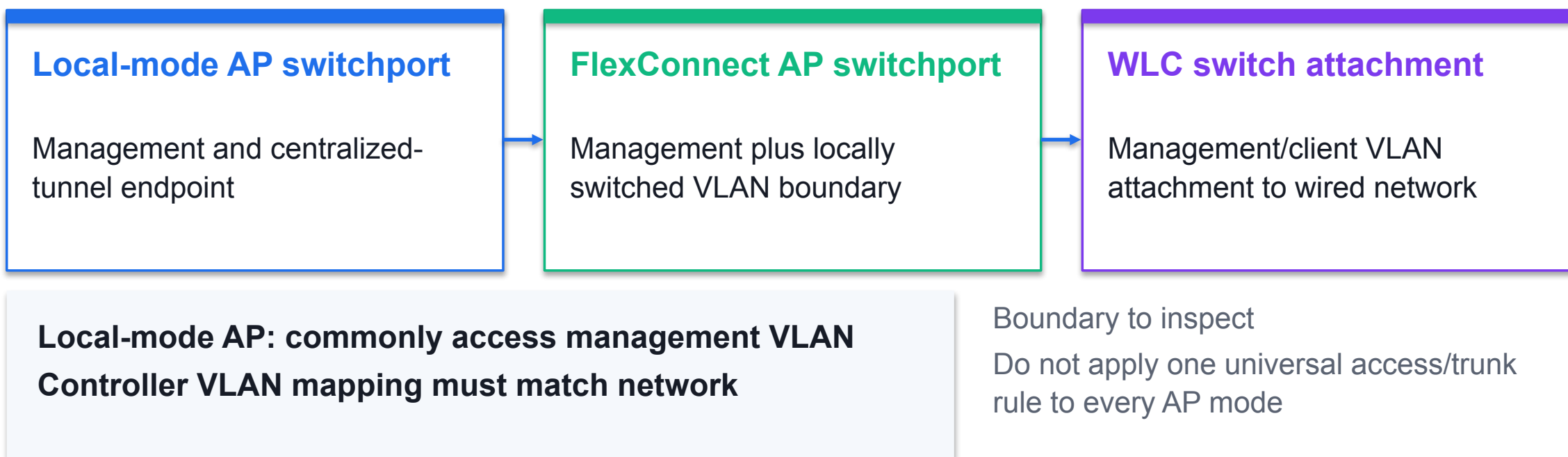
Default and normal-range VLAN contracts · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Default VLAN 1 -> Access-port membership	Normal range VLAN 1-1005
Mechanism / state	Trunk forwards only allowed active VLANs	show vlan brief
Result / boundary	Read VLAN existence, access membership and operational trunk lists; explicitly distinguish default, access, native and management VLAN roles.	Do not confuse default VLAN 1 with an arbitrary native VLAN on a trunk. Reserved legacy VLANs are not ordinary user allocations.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

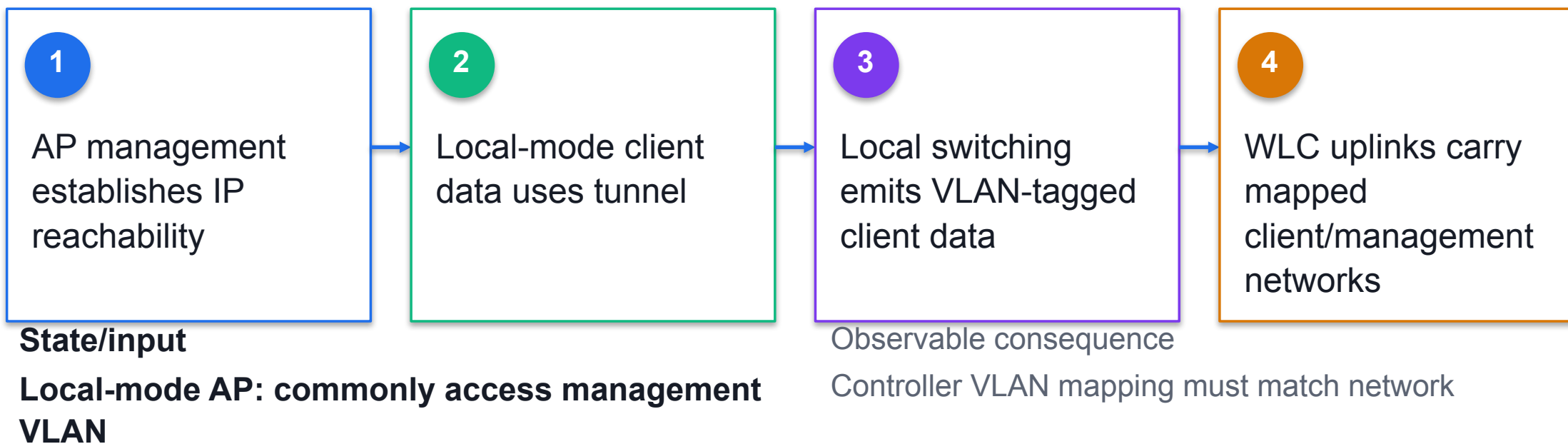
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

AP and WLC physical attachments · boundaries



Compare AP mode, management addressing, trunk allowed/native VLANs, WLC attachment and client gateway location.

AP and WLC physical attachments · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

AP and WLC physical attachments · exact contract

Local-mode AP: commonly access management VLAN
FlexConnect local-switch AP: trunk for client VLANs
WLC: configured trunk/LAG attachment per platform
Controller VLAN mapping must match network

INPUT / IDENTITY

Local-mode AP switchport · AP management establishes IP reachability

DECISION / STATE

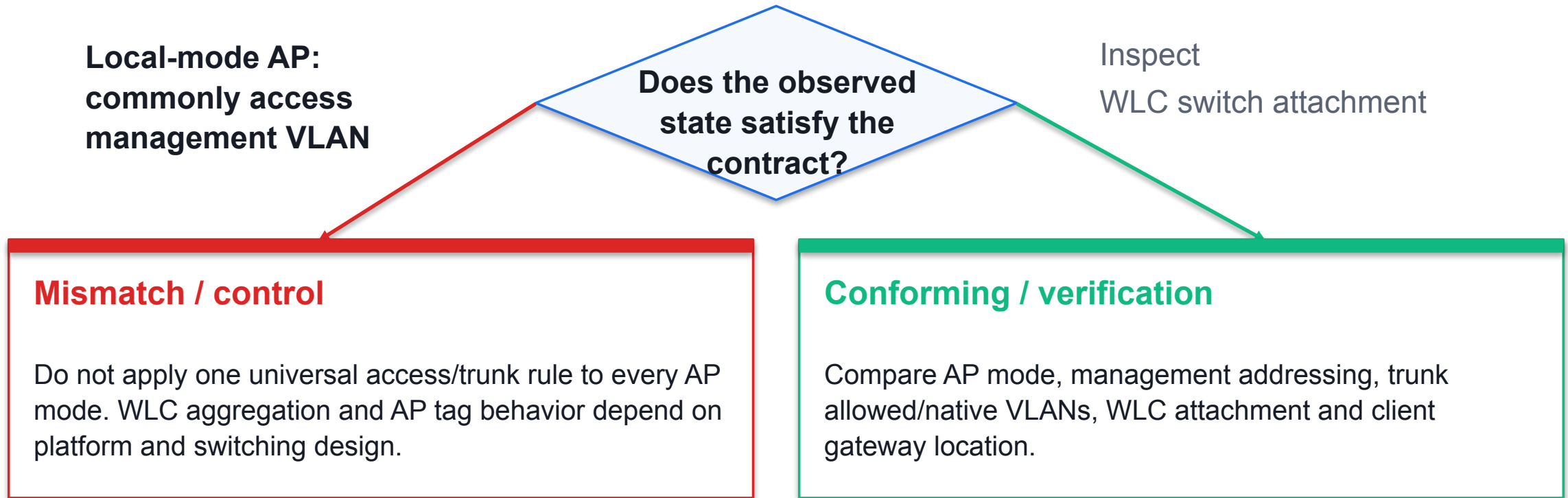
Local switching emits VLAN-tagged client data

READBACK / RESULT

WLC uplinks carry mapped client/management networks

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

AP and WLC physical attachments · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

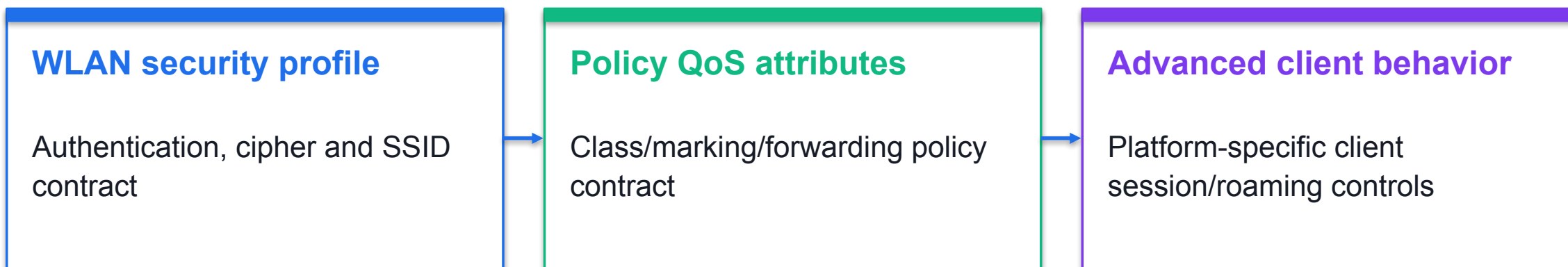
AP and WLC physical attachments · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Local-mode AP switchport -> WLC switch attachment	Local-mode AP: commonly access management VLAN
Mechanism / state	WLC uplinks carry mapped client/management networks	Controller VLAN mapping must match network
Result / boundary	Compare AP mode, management addressing, trunk allowed/native VLANs, WLC attachment and client gateway location.	Do not apply one universal access/trunk rule to every AP mode. WLC aggregation and AP tag behavior depend on platform and switching design.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

WLAN QoS and advanced-policy evidence · boundaries

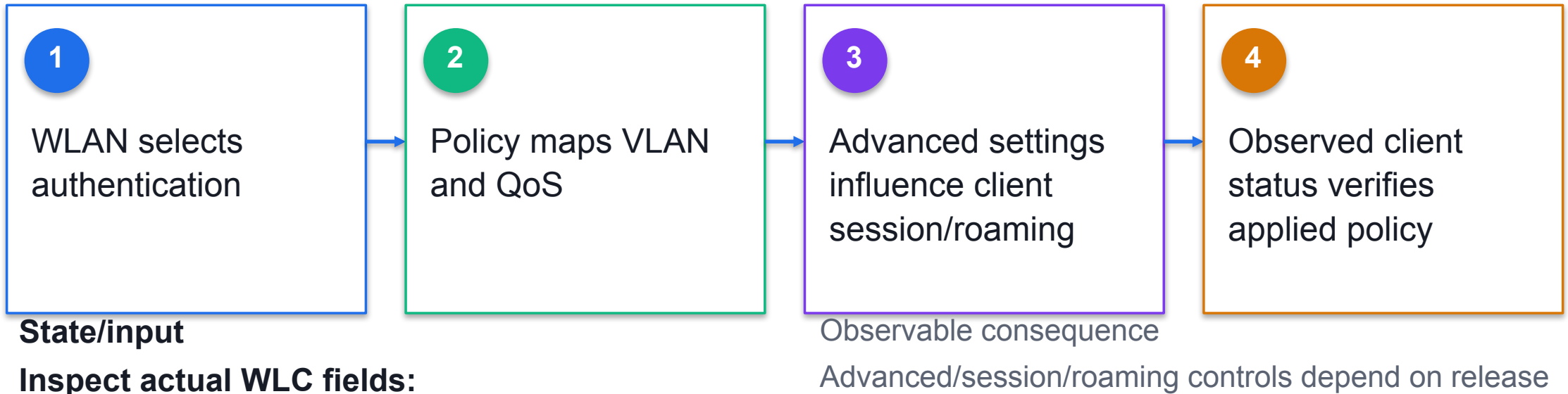


Inspect actual WLC fields:
Advanced/session/roaming controls depend on release

Boundary to inspect
GUI labels and feature availability vary
across WLC versions

Read current controller GUI object relationships, security, QoS and advanced settings; corroborate client VLAN/authentication and relevant policy state.

WLAN QoS and advanced-policy evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

WLAN QoS and advanced-policy evidence · exact contract

Inspect actual WLC fields:
WLAN ID/SSID + WPA2PSK/AES
Policy VLAN + QoS profile
Advanced/session/roaming controls depend on release

INPUT / IDENTITY

WLAN security profile · WLAN selects authentication

DECISION / STATE

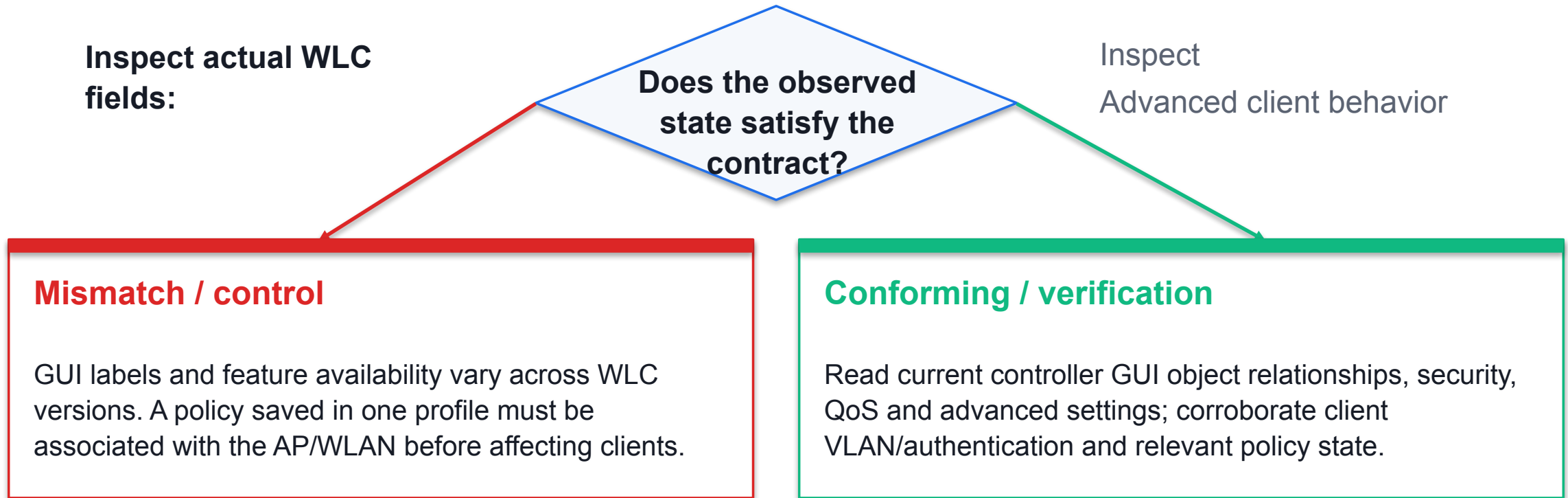
Advanced settings influence client session/roaming

READBACK / RESULT

Observed client status verifies applied policy

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

WLAN QoS and advanced-policy evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

WLAN QoS and advanced-policy evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	WLAN security profile -> Advanced client behavior	Inspect actual WLC fields:
Mechanism / state	Observed client status verifies applied policy	Advanced/session/roaming controls depend on release
Result / boundary	Read current controller GUI object relationships, security, QoS and advanced settings; corroborate client VLAN/authentication and relevant policy state.	GUI labels and feature availability vary across WLC versions. A policy saved in one profile must be associated with the AP/WLAN before affecting clients.

Bridge: Activities 3-5: VLAN paths, STP resilience and wireless policy

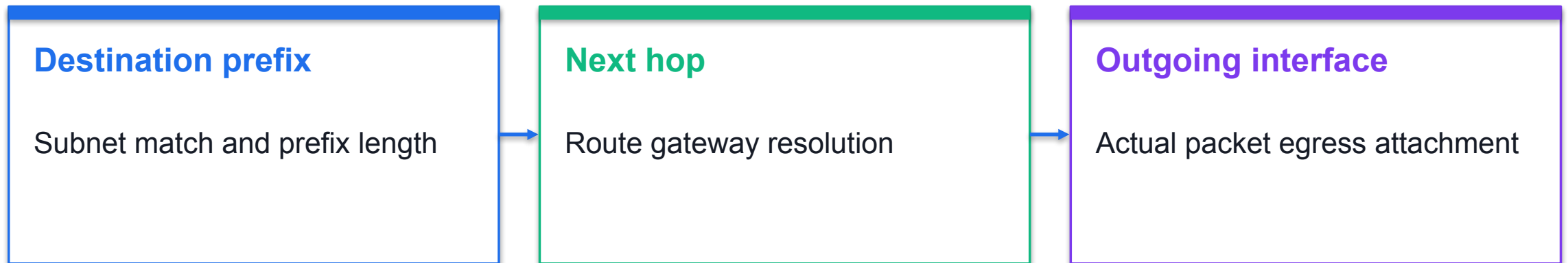
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IP Connectivity

IP Connectivity

Named boundaries -> runtime trace -> exact contract -> failure/control
-> acceptance evidence

Route-table field interpretation · boundaries



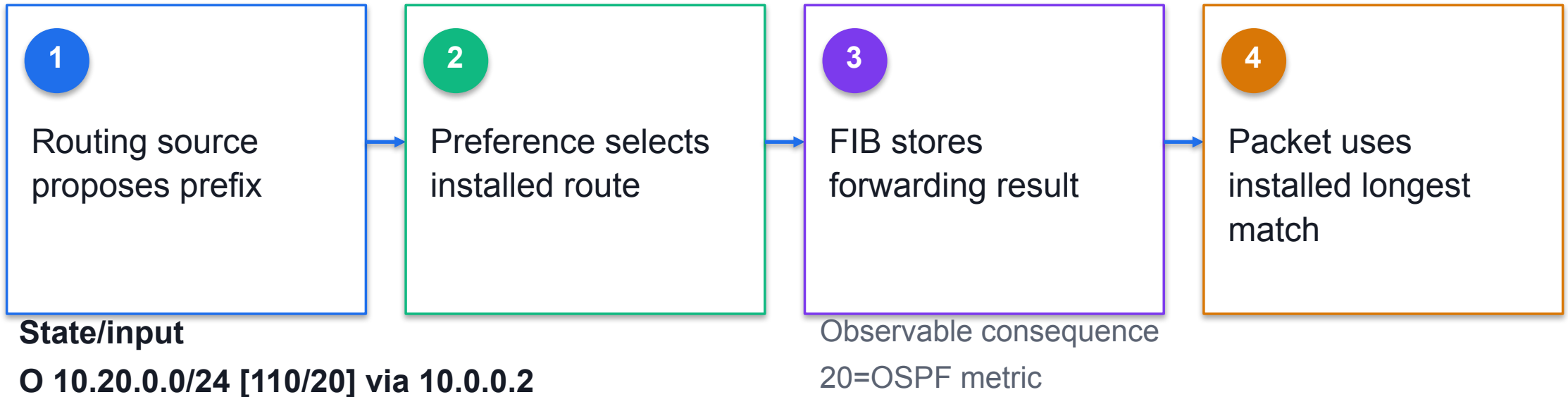
O 10.20.0.0/24 [110/20] via 10.0.0.2
20=OSPF metric

Boundary to inspect

Distance compares sources for the same prefix; it does not override longest prefix match for a packet.

show ip route

Route-table field interpretation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Route-table field interpretation · exact contract

O 10.20.0.0/24 [110/20] via 10.0.0.2
110=administrative distance
20=OSPF metric

INPUT / IDENTITY

Destination prefix · Routing source
proposes prefix

DECISION / STATE

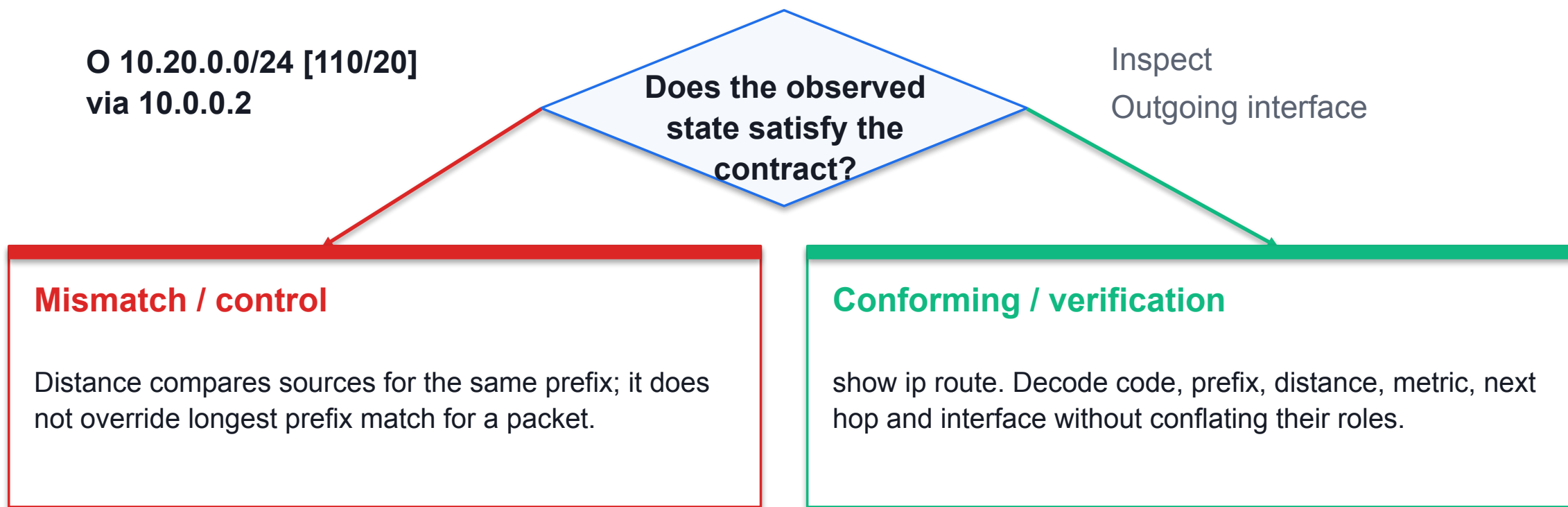
FIB stores forwarding result

READBACK / RESULT

Packet uses installed longest match

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Route-table field interpretation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

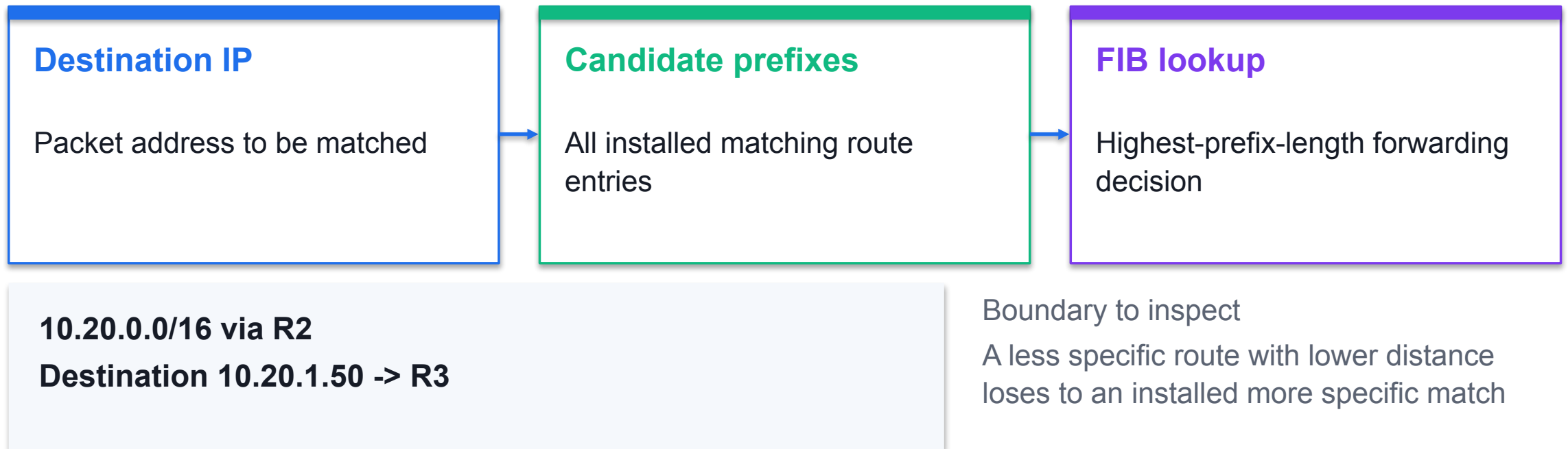
Route-table field interpretation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Destination prefix -> Outgoing interface	O 10.20.0.0/24 [110/20] via 10.0.0.2
Mechanism / state	Packet uses installed longest match	20=OSPF metric
Result / boundary	show ip route. Decode code, prefix, distance, metric, next hop and interface without conflating their roles.	Distance compares sources for the same prefix; it does not override longest prefix match for a packet.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Longest-prefix matching · boundaries



List every matching prefix and choose /24 over /16 and /0; verify using destination-specific route lookup.

Longest-prefix matching · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Longest-prefix matching · exact contract

10.20.0.0/16 via R2

10.20.1.0/24 via R3

Destination 10.20.1.50 -> R3

INPUT / IDENTITY

Destination IP · Compare matching prefixes

DECISION / STATE

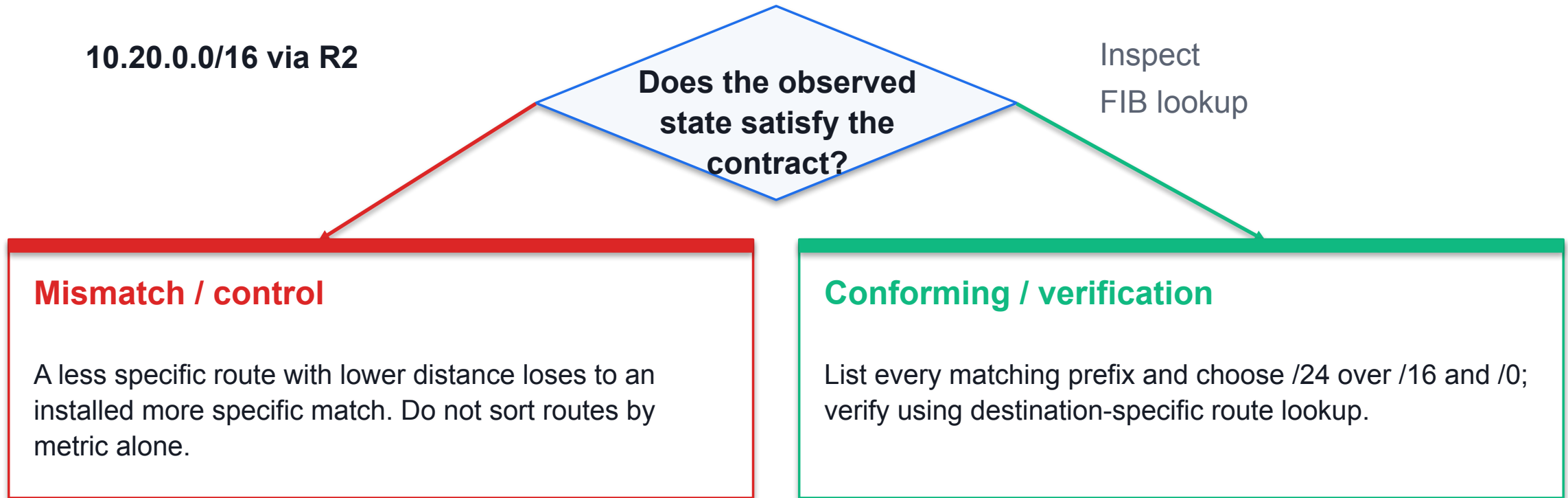
Resolve next hop/interface

READBACK / RESULT

Forward packet or drop

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Longest-prefix matching · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

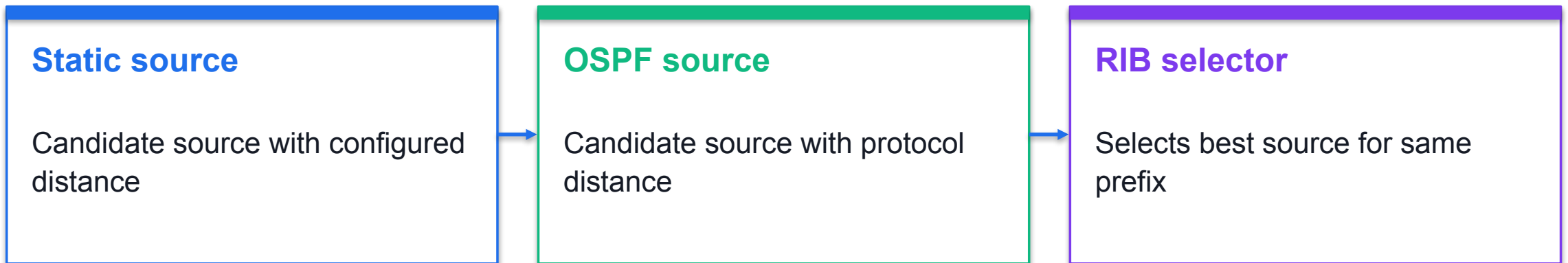
Longest-prefix matching · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Destination IP -> FIB lookup	10.20.0.0/16 via R2
Mechanism / state	Forward packet or drop	Destination 10.20.1.50 -> R3
Result / boundary	List every matching prefix and choose /24 over /16 and /0; verify using destination-specific route lookup.	A less specific route with lower distance loses to an installed more specific match. Do not sort routes by metric alone.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Administrative distance selection · boundaries

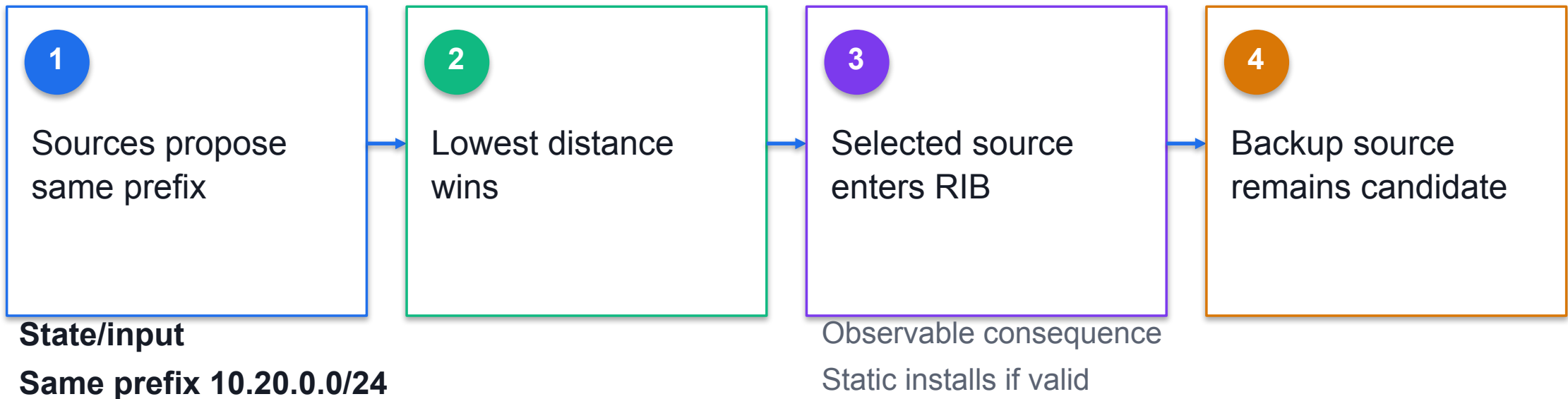


Same prefix 10.20.0.0/24
Static installs if valid

Boundary to inspect
Distance is locally significant and not a bandwidth or hop-count metric

show ip route 10.20.0.0

Administrative distance selection · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Administrative distance selection · exact contract

Same prefix 10.20.0.0/24
Static AD 1 vs OSPF AD 110
Static installs if valid

INPUT / IDENTITY

Static source · Sources propose same prefix

DECISION / STATE

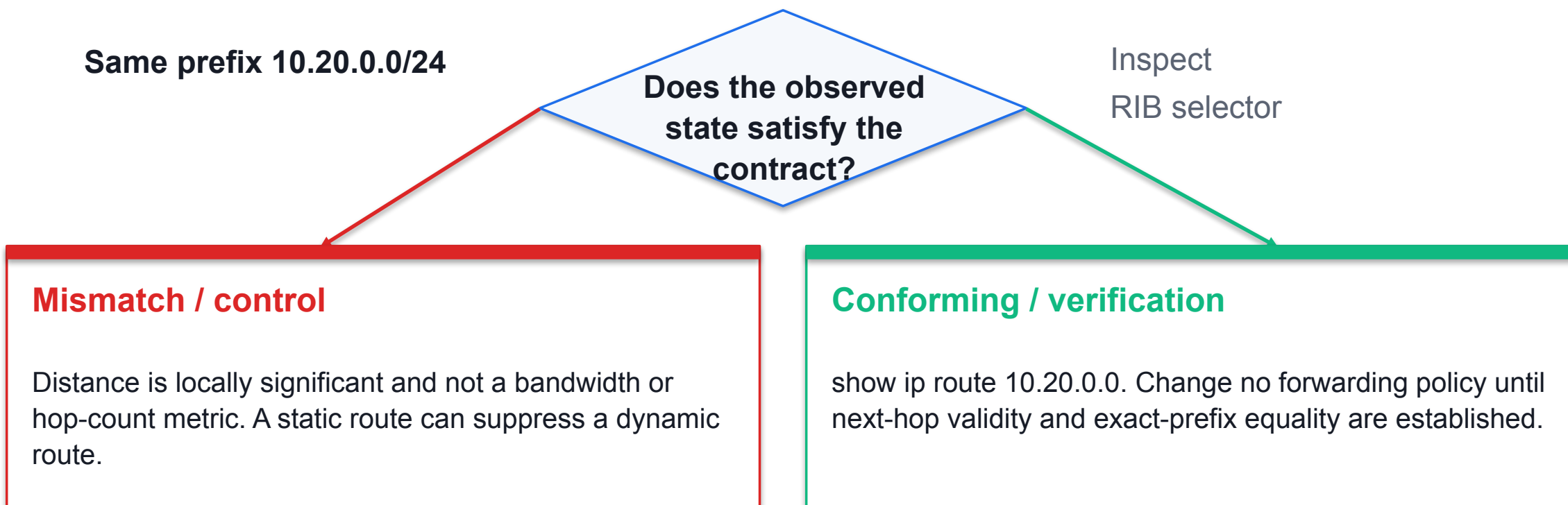
Selected source enters RIB

READBACK / RESULT

Backup source remains candidate

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Administrative distance selection · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

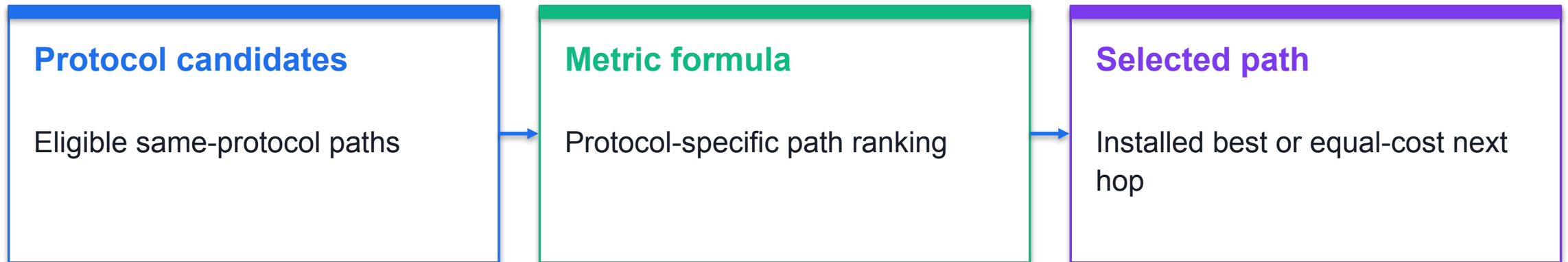
Administrative distance selection · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Static source -> RIB selector	Same prefix 10.20.0.0/24
Mechanism / state	Backup source remains candidate	Static installs if valid
Result / boundary	show ip route 10.20.0.0. Change no forwarding policy until next-hop validity and exact-prefix equality are established.	Distance is locally significant and not a bandwidth or hop-count metric. A static route can suppress a dynamic route.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Metric comparison boundaries · boundaries



OSPF cost 20 vs cost 30 ->20
Metrics across protocols are not directly comparable

Boundary to inspect
OSPF cost and RIP hops measure different things

Explain why metric 20 from OSPF cannot be compared numerically with RIP metric 2 to select a route.

Metric comparison boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Metric comparison boundaries · exact contract

OSPF cost 20 vs cost 30 ->20

RIP hop 2 vs hop 3 ->2

Metrics across protocols are not directly comparable

INPUT / IDENTITY

Protocol candidates · Same protocol evaluates candidates

DECISION / STATE

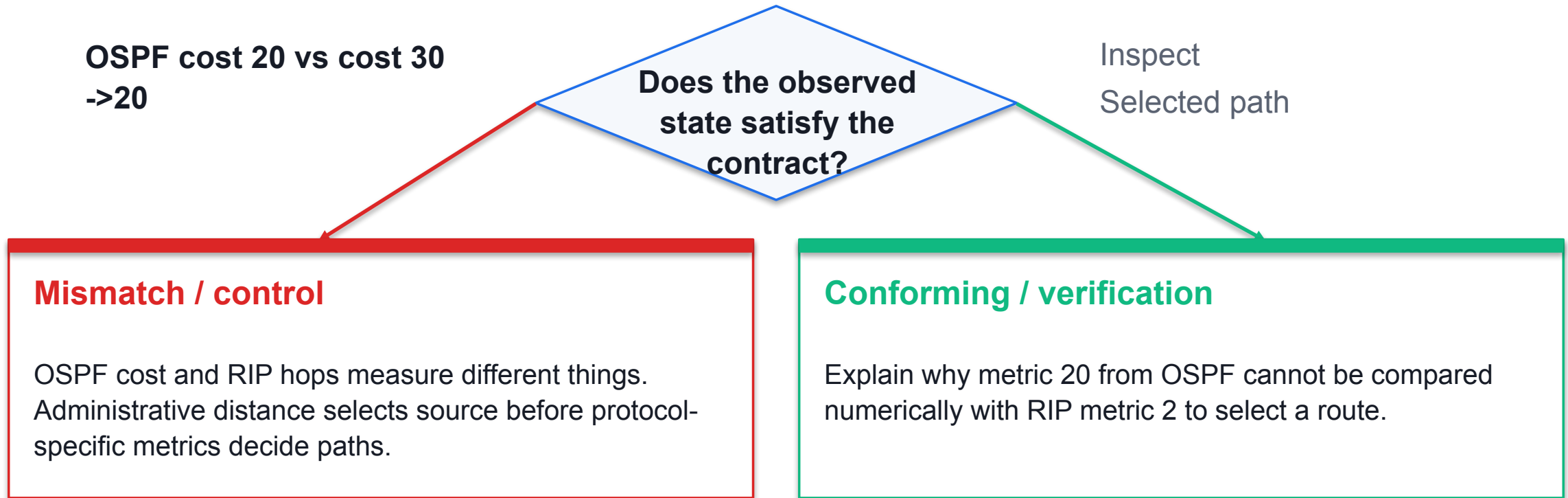
Lowest metric installs

READBACK / RESULT

Equal metrics may permit ECMP

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Metric comparison boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

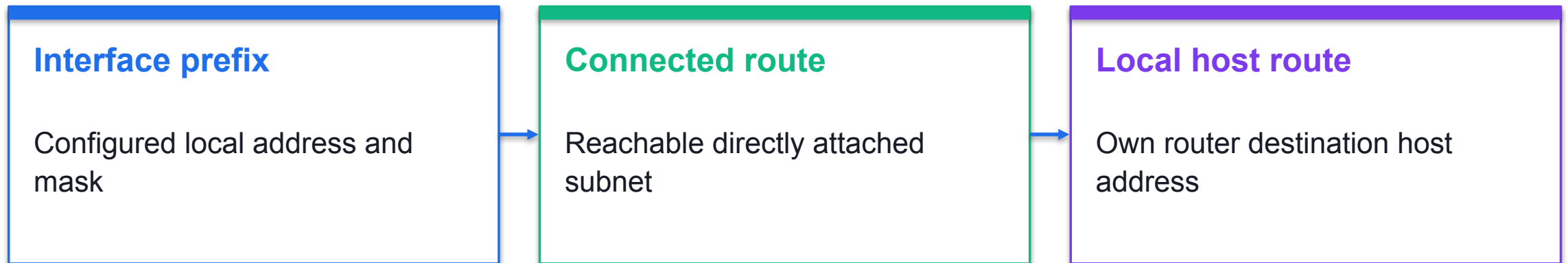
Metric comparison boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Protocol candidates -> Selected path	OSPF cost 20 vs cost 30 ->20
Mechanism / state	Equal metrics may permit ECMP	Metrics across protocols are not directly comparable
Result / boundary	Explain why metric 20 from OSPF cannot be compared numerically with RIP metric 2 to select a route.	OSPF cost and RIP hops measure different things. Administrative distance selects source before protocol-specific metrics decide paths.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Connected and local routes · boundaries

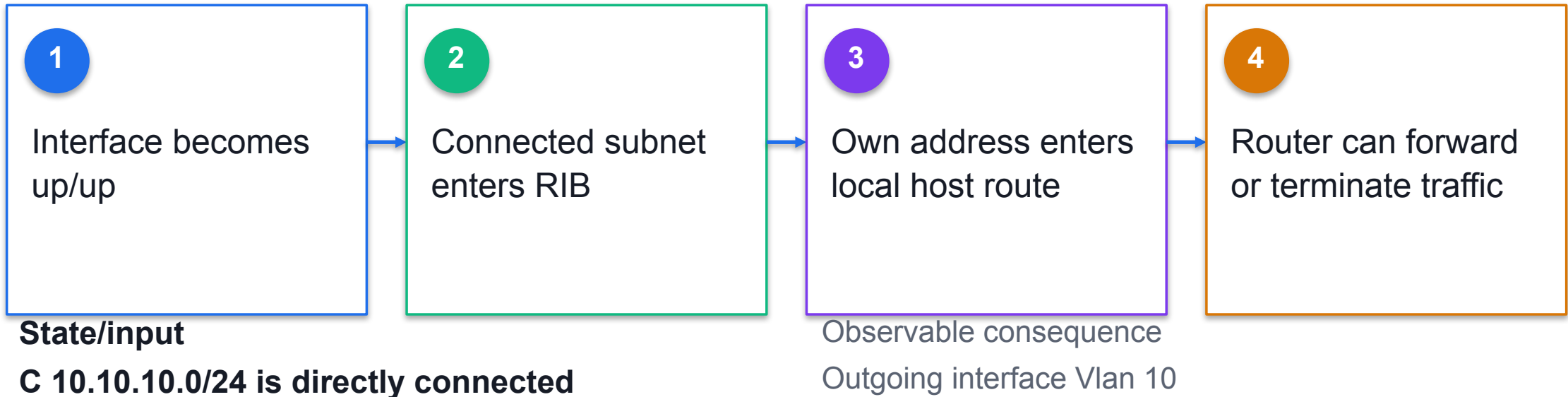


C 10.10.10.0/24 is directly connected
Outgoing interface Vlan 10

Boundary to inspect
A shutdown interface removes connected route eligibility

show ip interface brief; show ip route connected

Connected and local routes · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Connected and local routes · exact contract

C 10.10.10.0/24 is directly connected
L 10.10.10.1/32 is directly connected
Outgoing interface Vlan 10

INPUT / IDENTITY

Interface prefix · Interface becomes up/up

DECISION / STATE

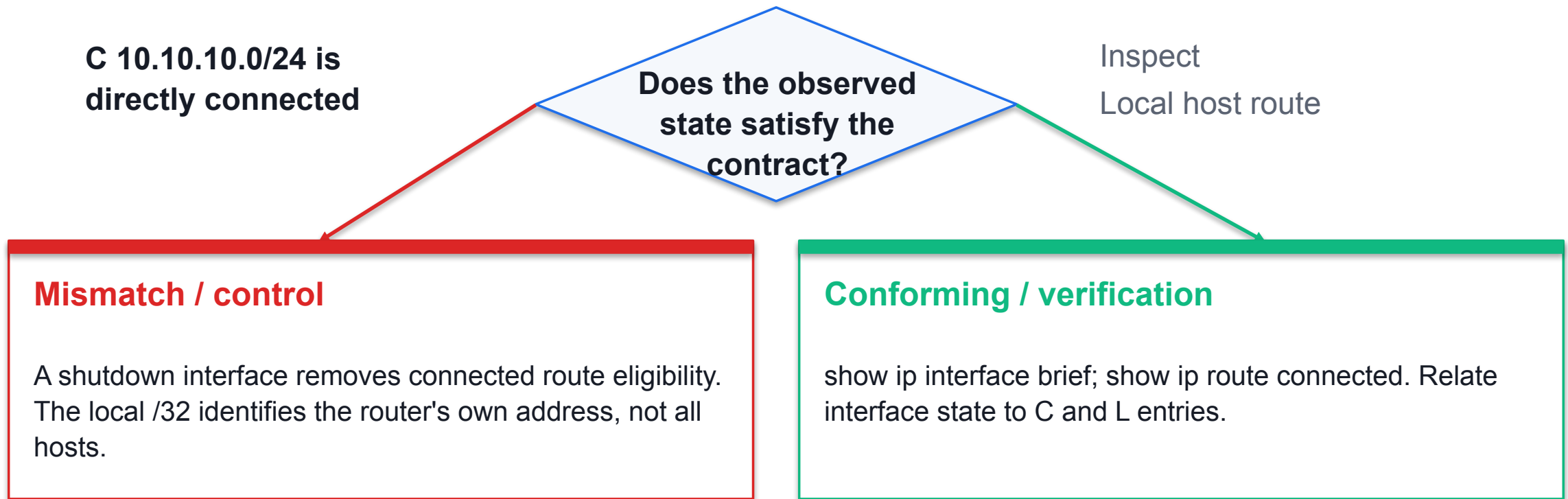
Own address enters local host route

READBACK / RESULT

Router can forward or terminate traffic

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Connected and local routes · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

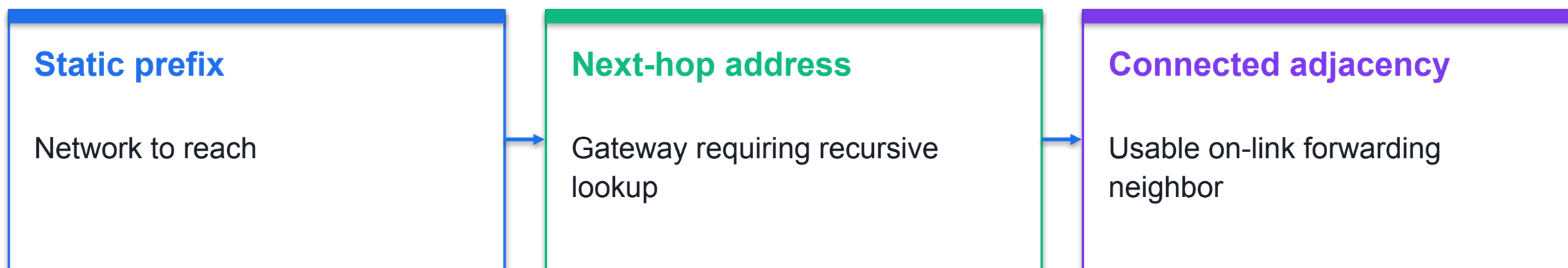
Connected and local routes · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Interface prefix -> Local host route	C 10.10.10.0/24 is directly connected
Mechanism / state	Router can forward or terminate traffic	Outgoing interface Vlan 10
Result / boundary	show ip interface brief; show ip route connected. Relate interface state to C and L entries.	A shutdown interface removes connected route eligibility. The local /32 identifies the router's own address, not all hosts.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Static next-hop recursion · boundaries



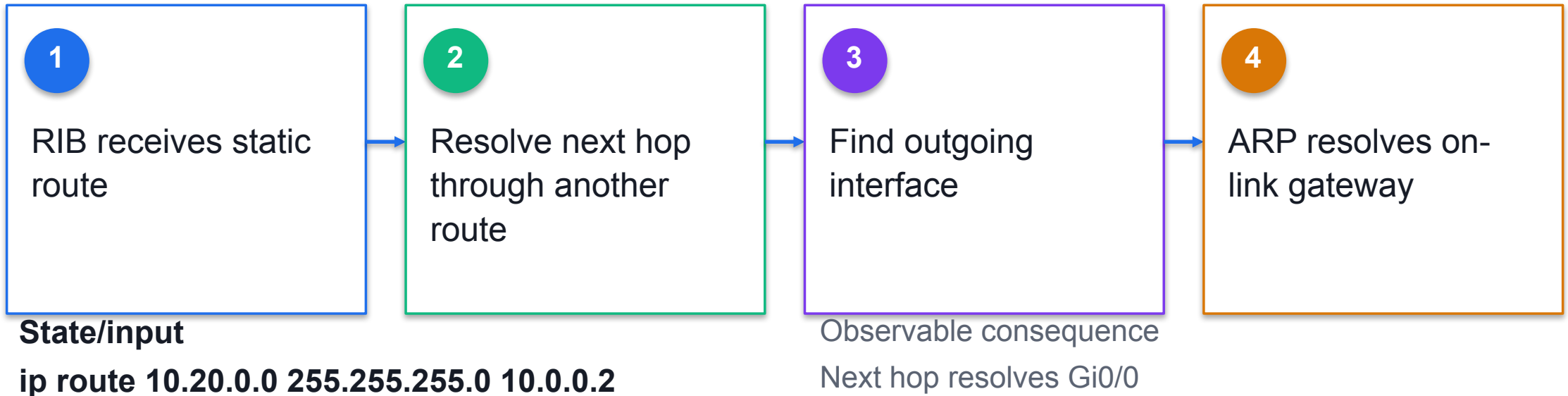
```
ip route 10.20.0.0 255.255.255.0 10.0.0.2
```

Next hop resolves Gi0/0

Boundary to inspect
Recursive resolution must terminate in a usable adjacency

```
show ip route 10.0.0.2; show ip arp
```

Static next-hop recursion · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Static next-hop recursion · exact contract

```
ip route 10.20.0.0 255.255.255.0 10.0.0.2  
10.0.0.0/30 connected Gi0/0  
Next hop resolves Gi0/0
```

INPUT / IDENTITY

Static prefix · RIB receives static route

DECISION / STATE

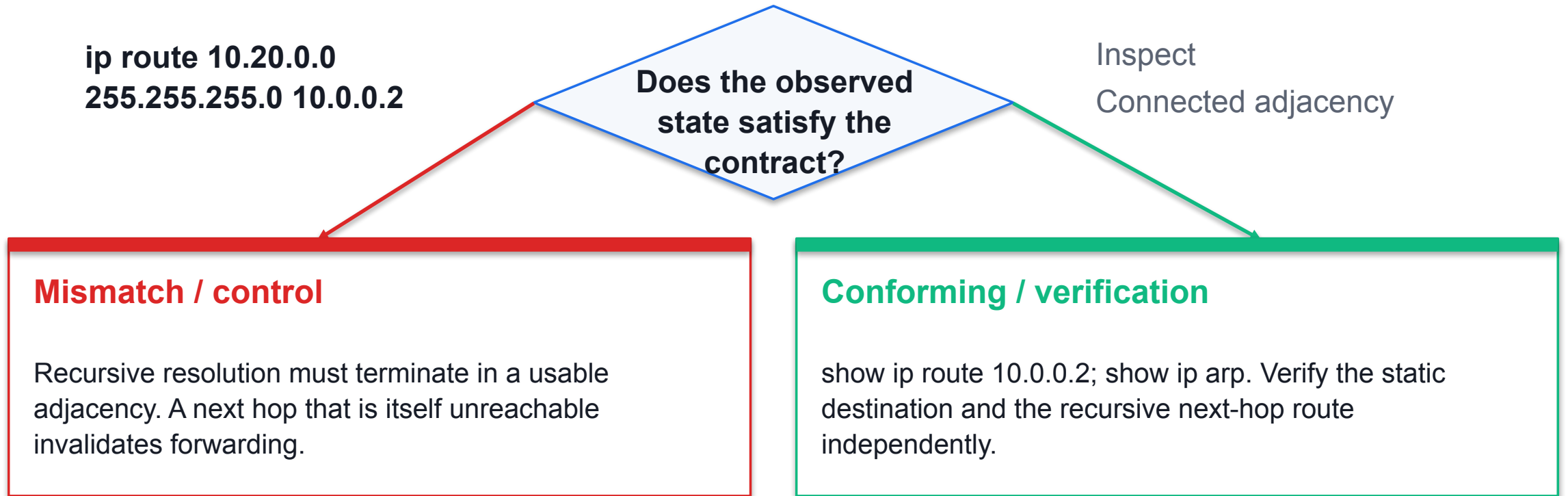
Find outgoing interface

READBACK / RESULT

ARP resolves on-link gateway

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Static next-hop recursion · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

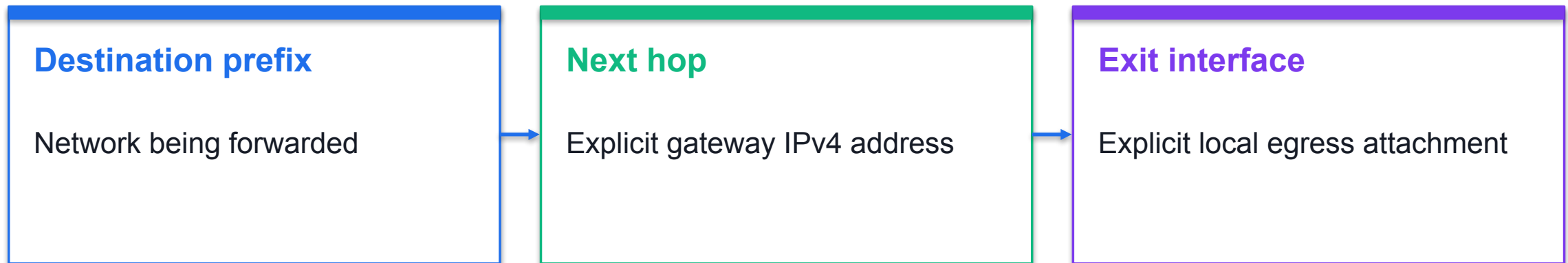
Static next-hop recursion · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Static prefix -> Connected adjacency	ip route 10.20.0.0 255.255.255.0 10.0.0.2
Mechanism / state	ARP resolves on-link gateway	Next hop resolves Gi0/0
Result / boundary	show ip route 10.0.0.2; show ip arp. Verify the static destination and the recursive next-hop route independently.	Recursive resolution must terminate in a usable adjacency. A next hop that is itself unreachable invalidates forwarding.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Fully specified static routes · boundaries

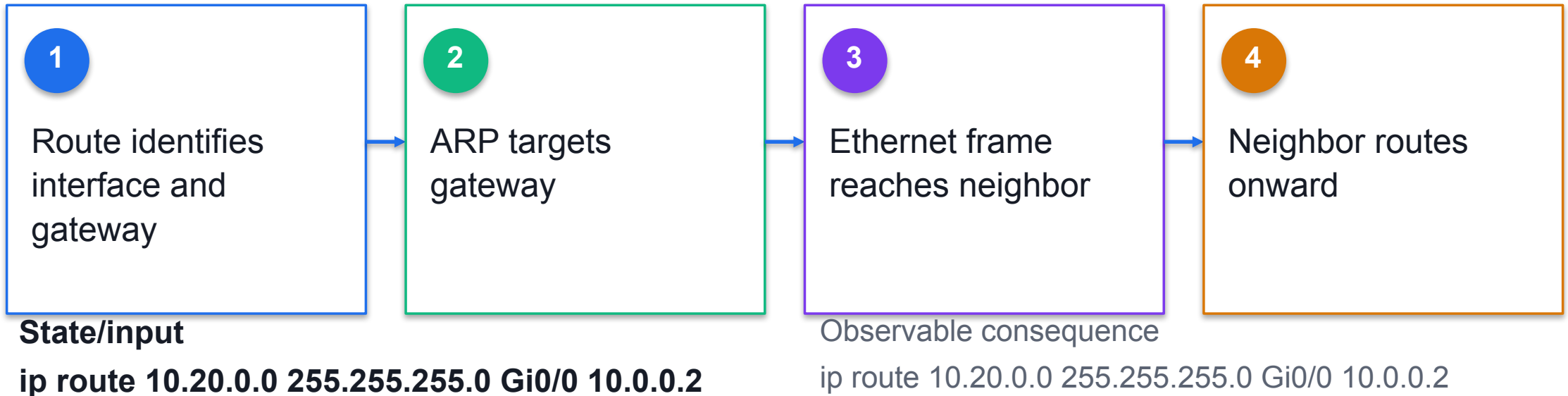


```
ip route 10.20.0.0 255.255.255.0 Gi0/0 10.0.0.2
```

Boundary to inspect
Interface-only static routes on Ethernet can cause ARP for many destinations

Confirm next-hop adjacency and interface state; compare ARP entries before and after fully specified configuration.

Fully specified static routes · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Fully specified static routes · exact contract

```
ip route 10.20.0.0 255.255.255.0 Gi0/0 10.0.0.2
```

INPUT / IDENTITY

Destination prefix · Route identifies interface and gateway

DECISION / STATE

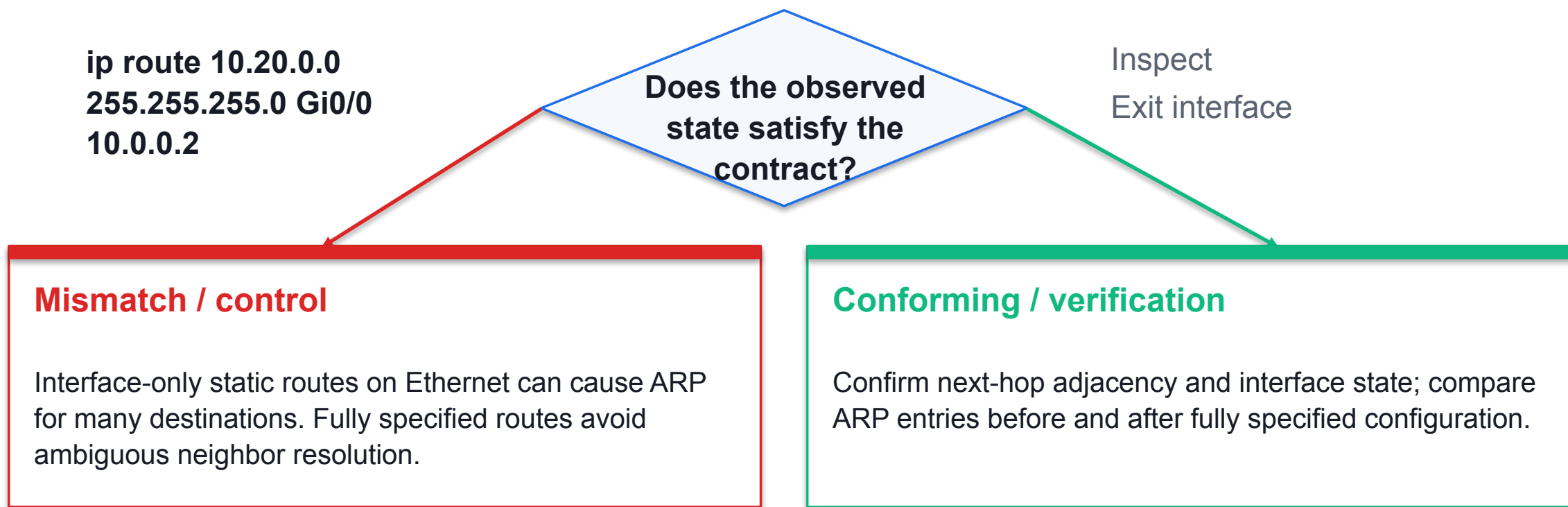
Ethernet frame reaches neighbor

READBACK / RESULT

Neighbor routes onward

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Fully specified static routes · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

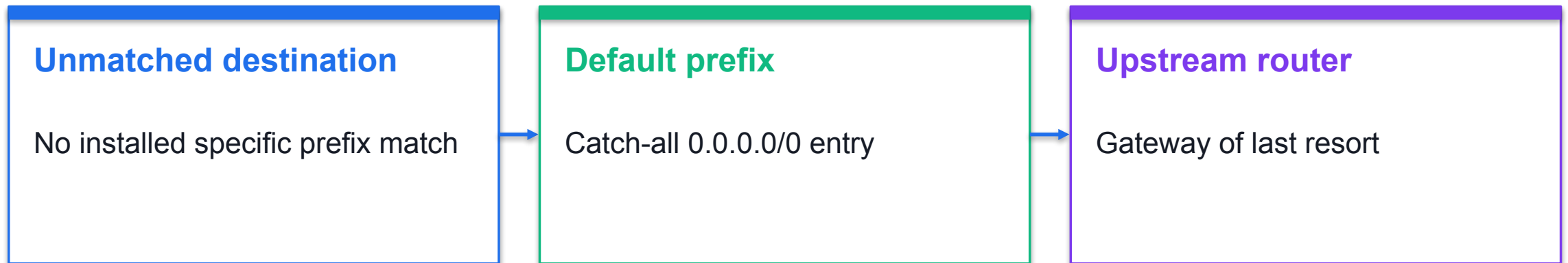
Fully specified static routes · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Destination prefix -> Exit interface	ip route 10.20.0.0 255.255.255.0 Gi0/0 10.0.0.2
Mechanism / state	Neighbor routes onward	ip route 10.20.0.0 255.255.255.0 Gi0/0 10.0.0.2
Result / boundary	Confirm next-hop adjacency and interface state; compare ARP entries before and after fully specified configuration.	Interface-only static routes on Ethernet can cause ARP for many destinations. Fully specified routes avoid ambiguous neighbor resolution.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Default IPv4 routing · boundaries

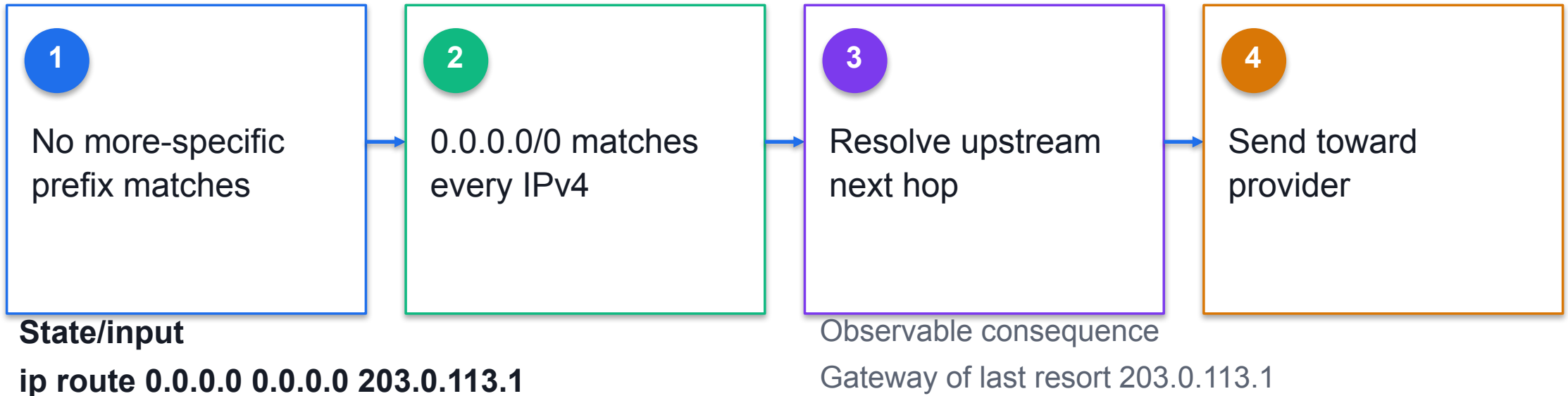


```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
Gateway of last resort 203.0.113.1
```

Boundary to inspect
A default route is not proof the upstream
can reach the destination

```
show ip route 0.0.0.0; traceroute target
```

Default IPv4 routing · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Default IPv4 routing · exact contract

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1  
Gateway of last resort 203.0.113.1
```

INPUT / IDENTITY

Unmatched destination · No more-specific prefix matches

DECISION / STATE

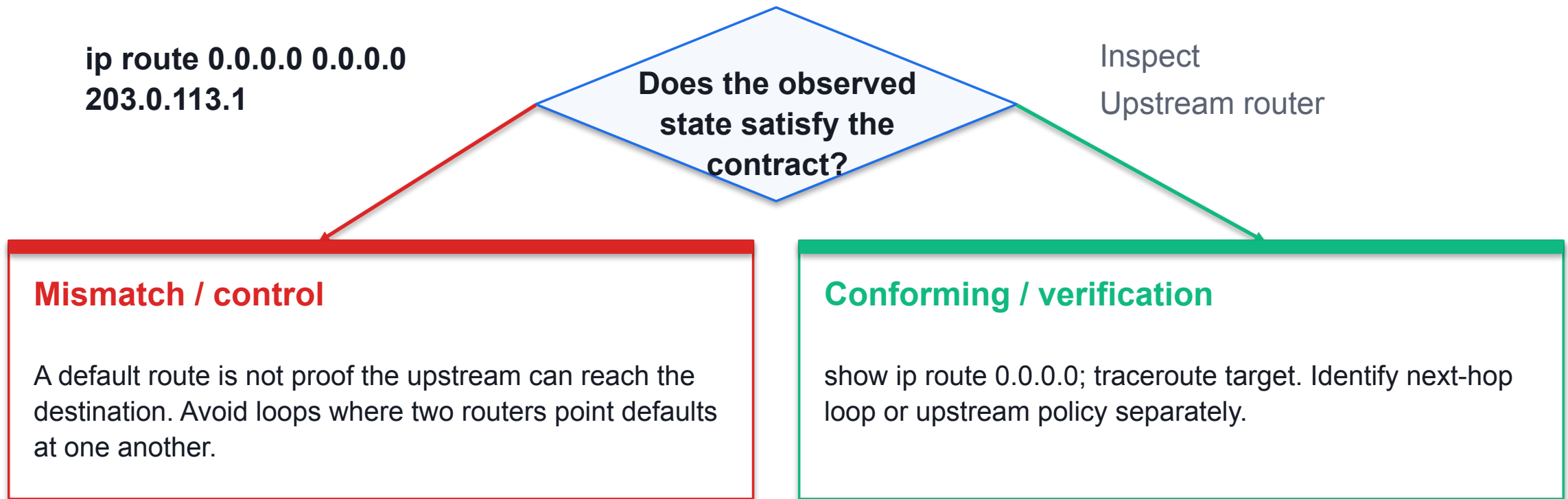
Resolve upstream next hop

READBACK / RESULT

Send toward provider

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Default IPv4 routing · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

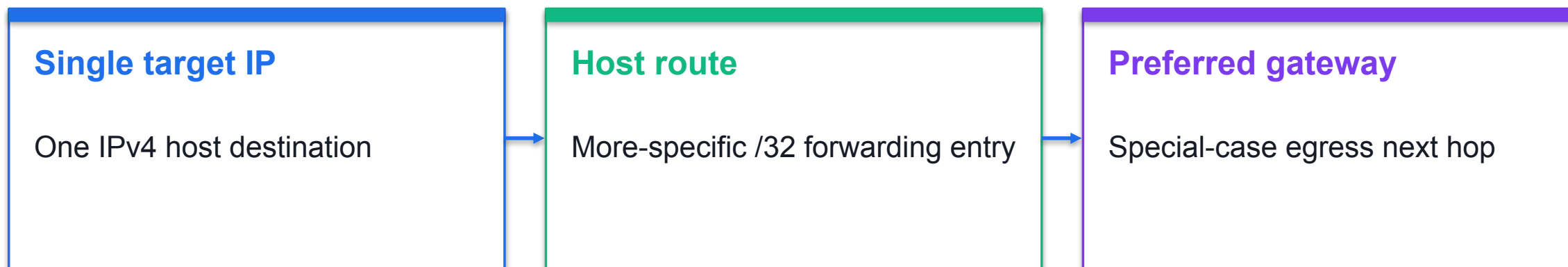
Default IPv4 routing · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Unmatched destination -> Upstream router	ip route 0.0.0.0 0.0.0.0 203.0.113.1
Mechanism / state	Send toward provider	Gateway of last resort 203.0.113.1
Result / boundary	show ip route 0.0.0.0; traceroute target. Identify next-hop loop or upstream policy separately.	A default route is not proof the upstream can reach the destination. Avoid loops where two routers point defaults at one another.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Host-specific static routes · boundaries



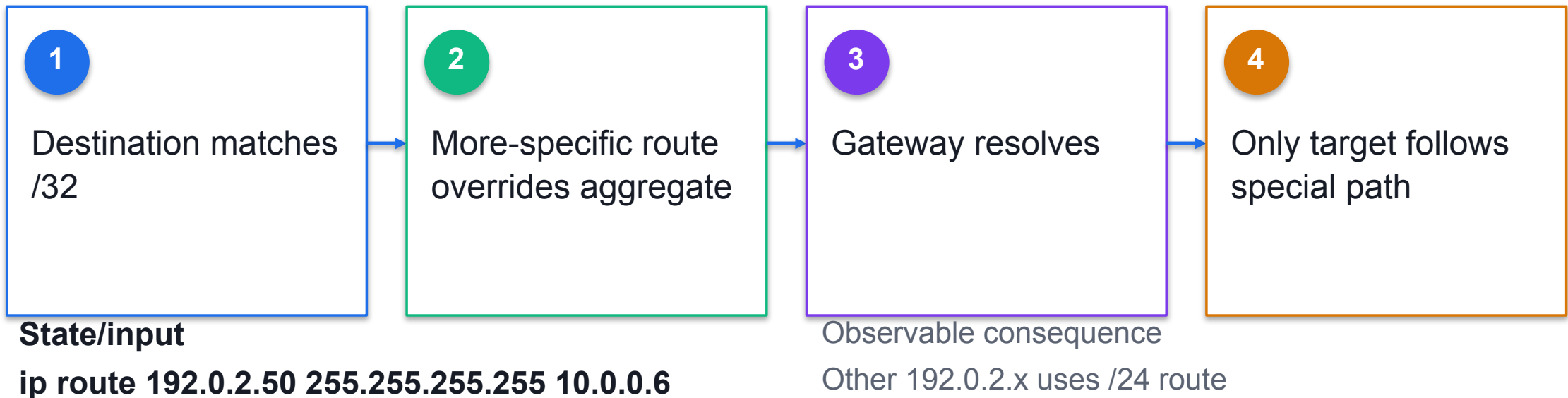
ip route 192.0.2.50 255.255.255.255 10.0.0.6
Other 192.0.2.x uses /24 route

Boundary to inspect

A host route can be used for precise control
but accumulates maintenance complexity

Compare destination .50 with .51

Host-specific static routes · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Host-specific static routes · exact contract

```
ip route 192.0.2.50 255.255.255.255 10.0.0.6  
Other 192.0.2.x uses /24 route
```

INPUT / IDENTITY

Single target IP · Destination matches /32

DECISION / STATE

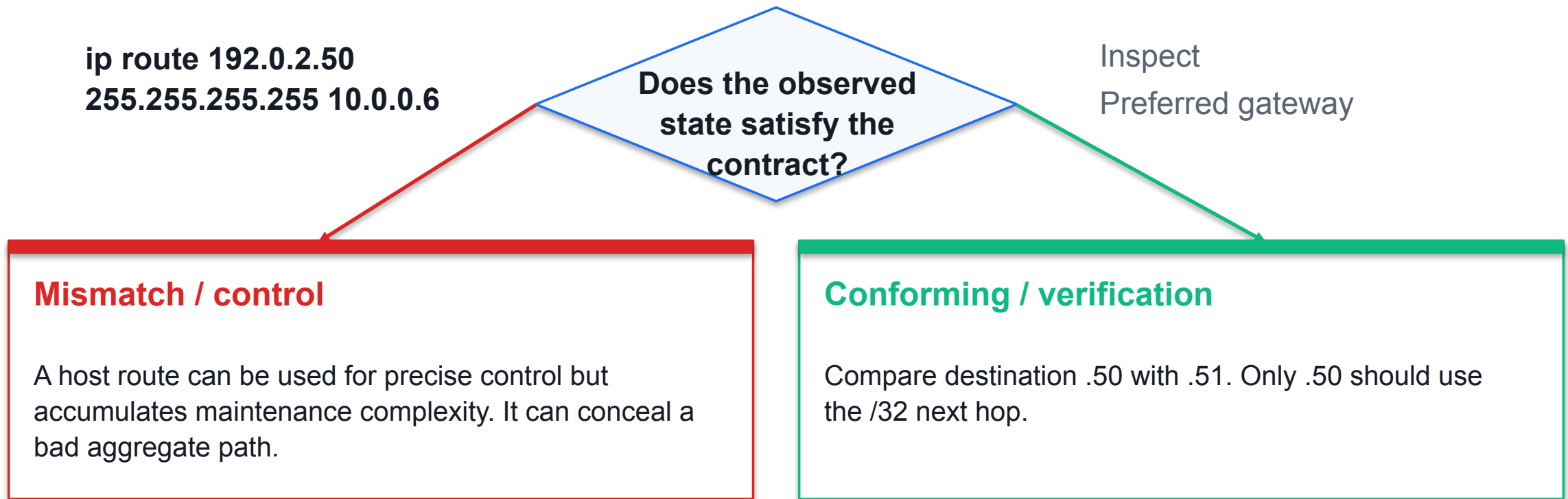
Gateway resolves

READBACK / RESULT

Only target follows special path

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Host-specific static routes · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

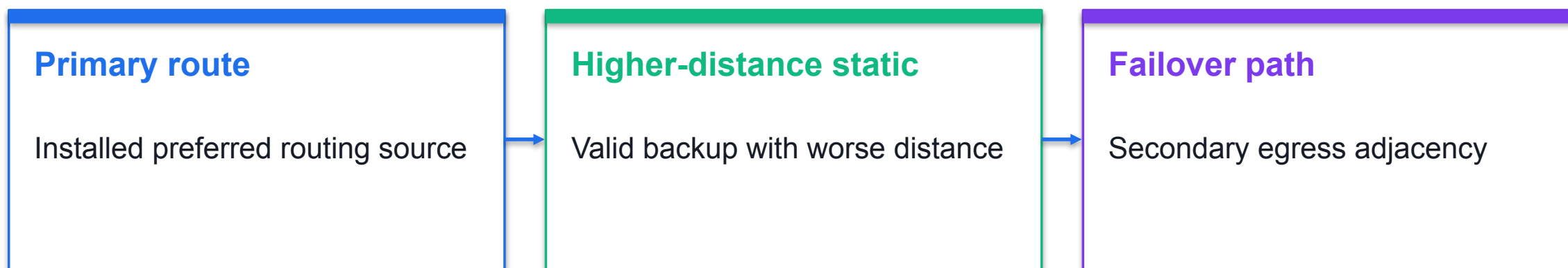
Host-specific static routes · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Single target IP -> Preferred gateway	ip route 192.0.2.50 255.255.255.255 10.0.0.6
Mechanism / state	Only target follows special path	Other 192.0.2.x uses /24 route
Result / boundary	Compare destination .50 with .51. Only .50 should use the /32 next hop.	A host route can be used for precise control but accumulates maintenance complexity. It can conceal a bad aggregate path.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Floating static backup · boundaries

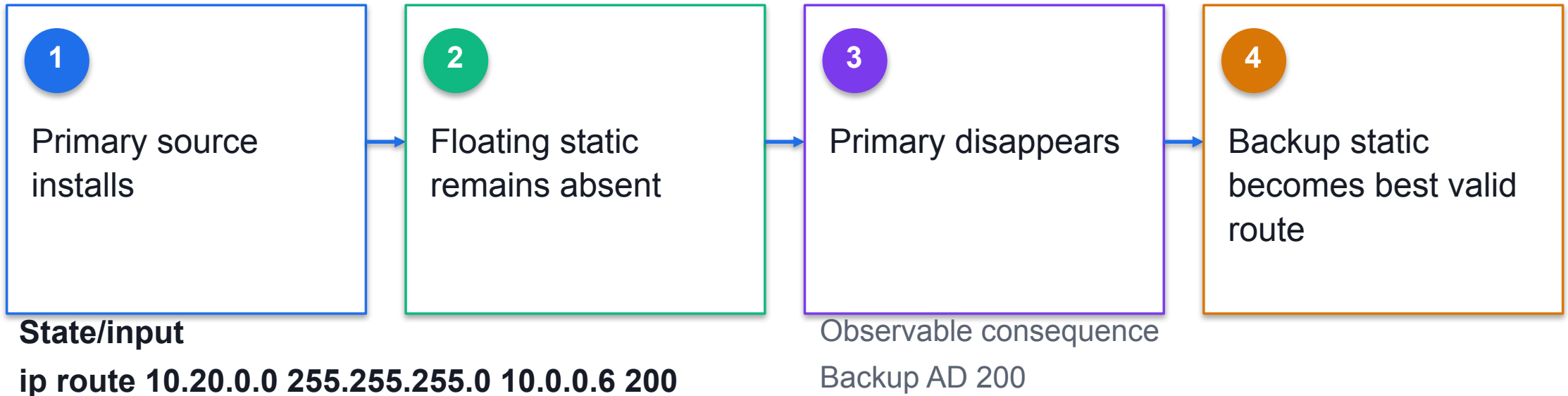


```
ip route 10.20.0.0 255.255.255.0 10.0.0.6 200
Backup AD 200
```

Boundary to inspect
Floating distance must exceed primary
source distance

```
show ip route before/after a controlled primary-link failure
```

Floating static backup · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Floating static backup · exact contract

```
ip route 10.20.0.0 255.255.255.0 10.0.0.6 200  
OSPF primary AD 110  
Backup AD 200
```

INPUT / IDENTITY

Primary route · Primary source installs

DECISION / STATE

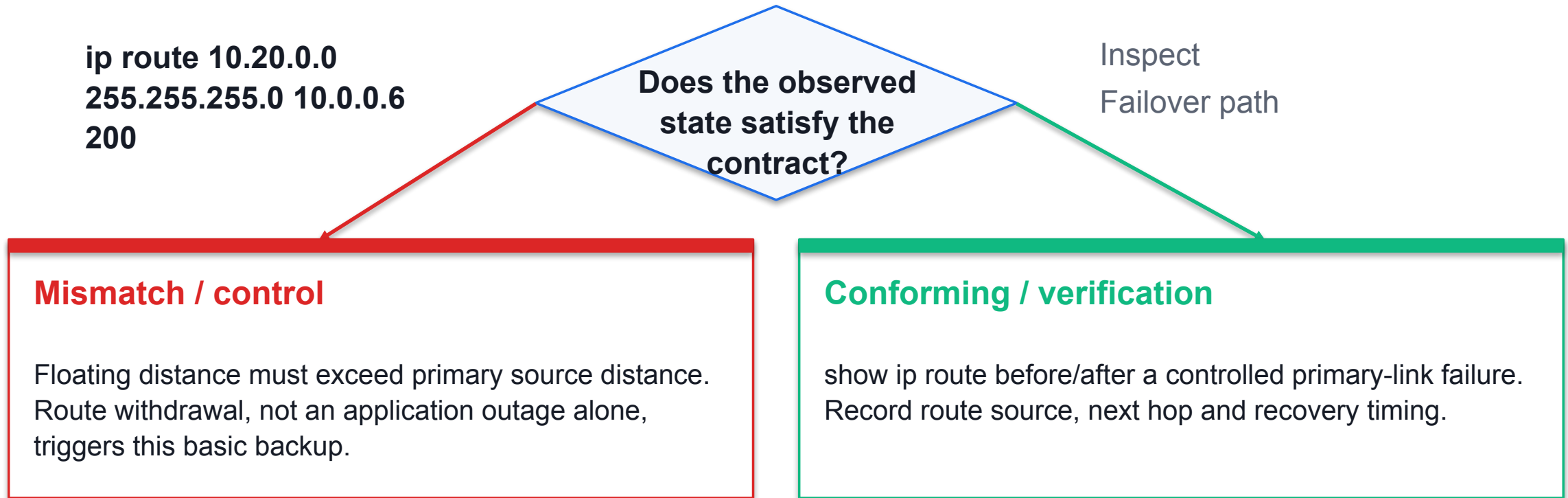
Primary disappears

READBACK / RESULT

Backup static becomes best valid route

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Floating static backup · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

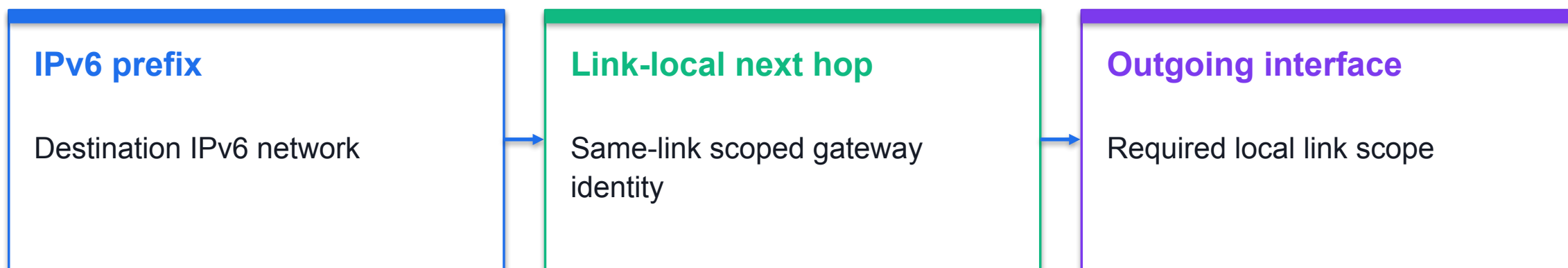
Floating static backup · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Primary route -> Failover path	ip route 10.20.0.0 255.255.255.0 10.0.0.6 200
Mechanism / state	Backup static becomes best valid route	Backup AD 200
Result / boundary	show ip route before/after a controlled primary-link failure. Record route source, next hop and recovery timing.	Floating distance must exceed primary source distance. Route withdrawal, not an application outage alone, triggers this basic backup.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPv6 static-route scope · boundaries



ipv6 unicast-routing

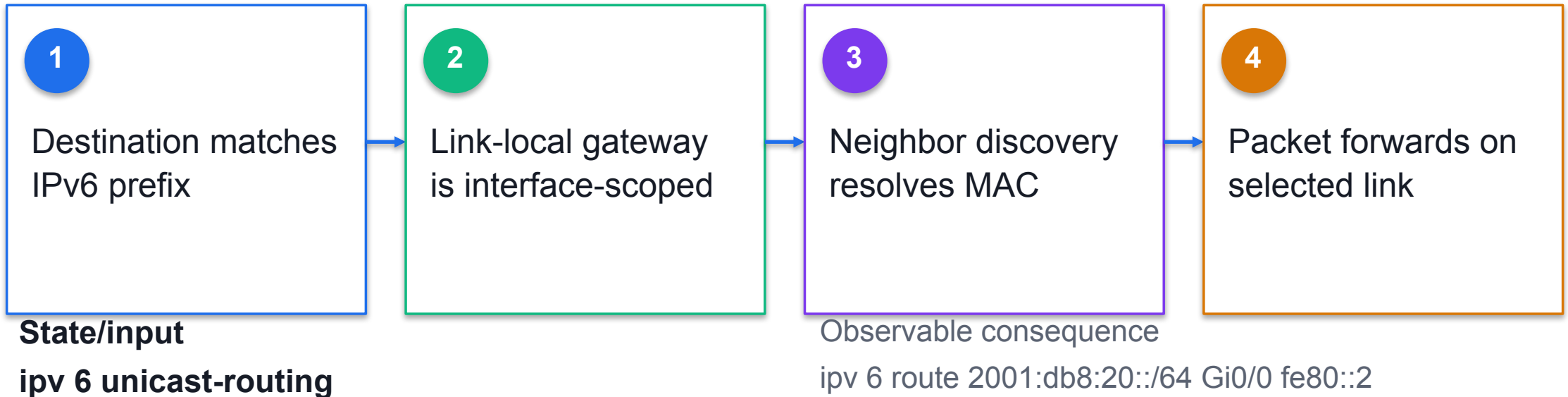
ipv6 route 2001:db8:20::/64 Gi0/0 fe80::2

Boundary to inspect

Link-local addresses can repeat on different links

show ipv 6 route; show ipv 6 neighbors

IPv6 static-route scope · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPv6 static-route scope · exact contract

ipv6 unicast-routing

ipv6 route 2001:db8:20::/64 Gi0/0 fe80::2

INPUT / IDENTITY

IPv6 prefix · Destination matches IPv6 prefix

DECISION / STATE

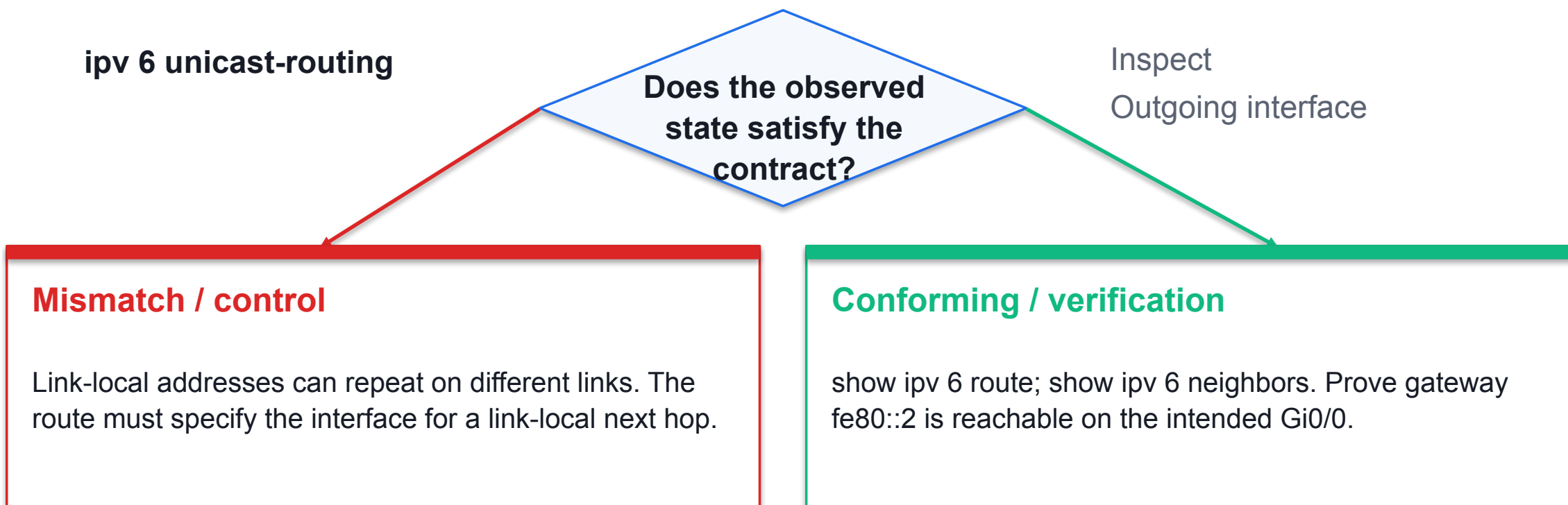
Neighbor discovery resolves MAC

READBACK / RESULT

Packet forwards on selected link

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPv6 static-route scope · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

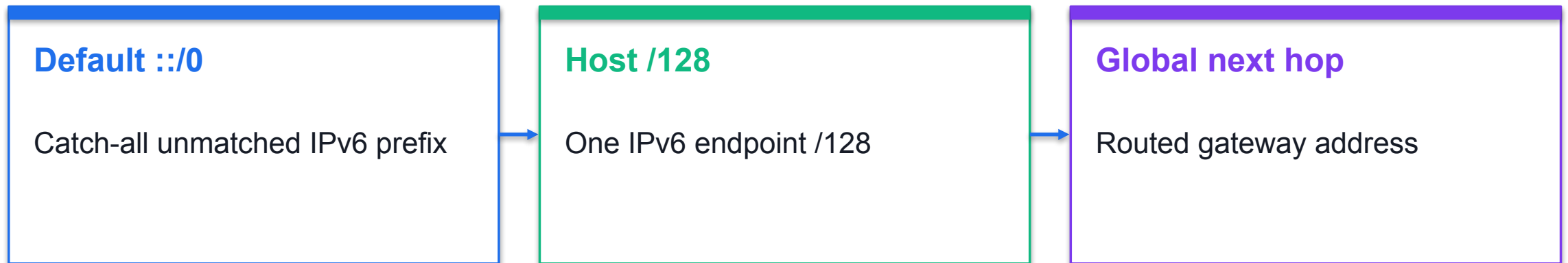
IPv6 static-route scope · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	IPv6 prefix -> Outgoing interface	ipv 6 unicast-routing
Mechanism / state	Packet forwards on selected link	ipv 6 route 2001:db8:20::/64 Gi0/0 fe80::2
Result / boundary	show ipv 6 route; show ipv 6 neighbors. Prove gateway fe80::2 is reachable on the intended Gi0/0.	Link-local addresses can repeat on different links. The route must specify the interface for a link-local next hop.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPv6 default and host routes · boundaries

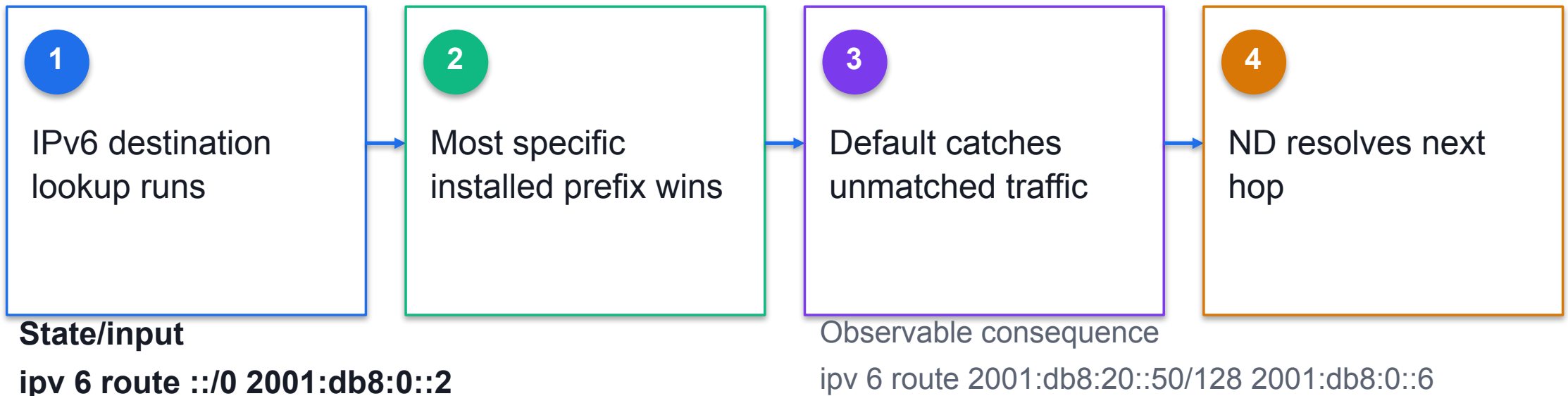


```
ipv6 route ::/0 2001:db8:0::2  
ipv6 route 2001:db8:20::50/128 2001:db8:0::6
```

Boundary to inspect
Do not apply IPv4 broadcast assumptions
to IPv6

```
show ipv 6 route target
```


IPv6 default and host routes · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPv6 default and host routes · exact contract

```
ipv6 route ::/0 2001:db8:0::2  
ipv6 route 2001:db8:20::50/128 2001:db8:0::6
```

INPUT / IDENTITY

Default ::/0 · IPv6 destination lookup runs

DECISION / STATE

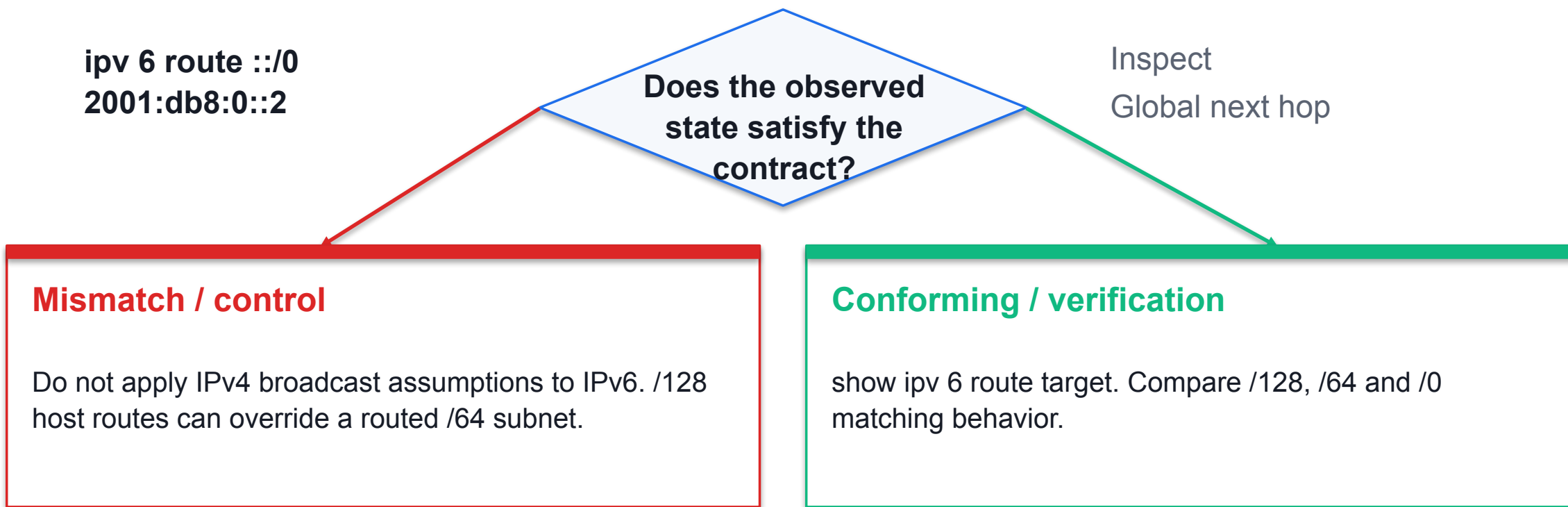
Default catches unmatched traffic

READBACK / RESULT

ND resolves next hop

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPv6 default and host routes · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

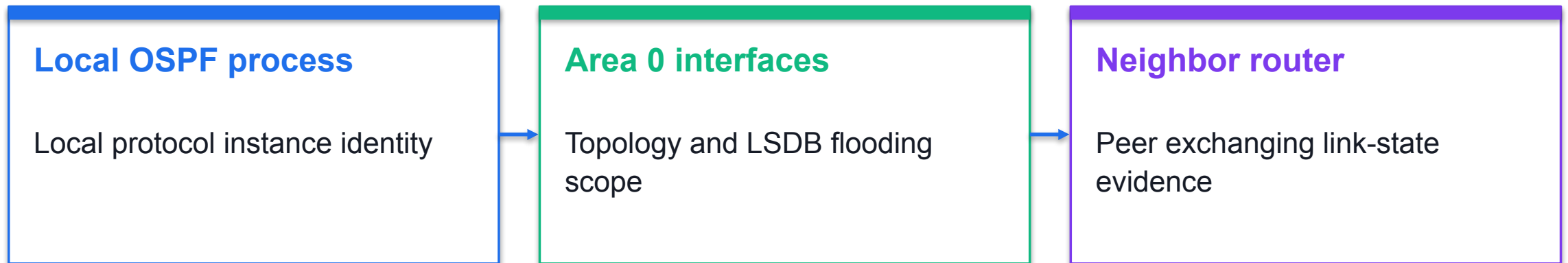
IPv6 default and host routes · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Default ::/0 -> Global next hop	ipv 6 route ::/0 2001:db8:0::2
Mechanism / state	ND resolves next hop	ipv 6 route 2001:db8:20::50/128 2001:db8:0::6
Result / boundary	show ipv 6 route target. Compare /128, /64 and /0 matching behavior.	Do not apply IPv4 broadcast assumptions to IPv6. /128 host routes can override a routed /64 subnet.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF process and area boundaries · boundaries

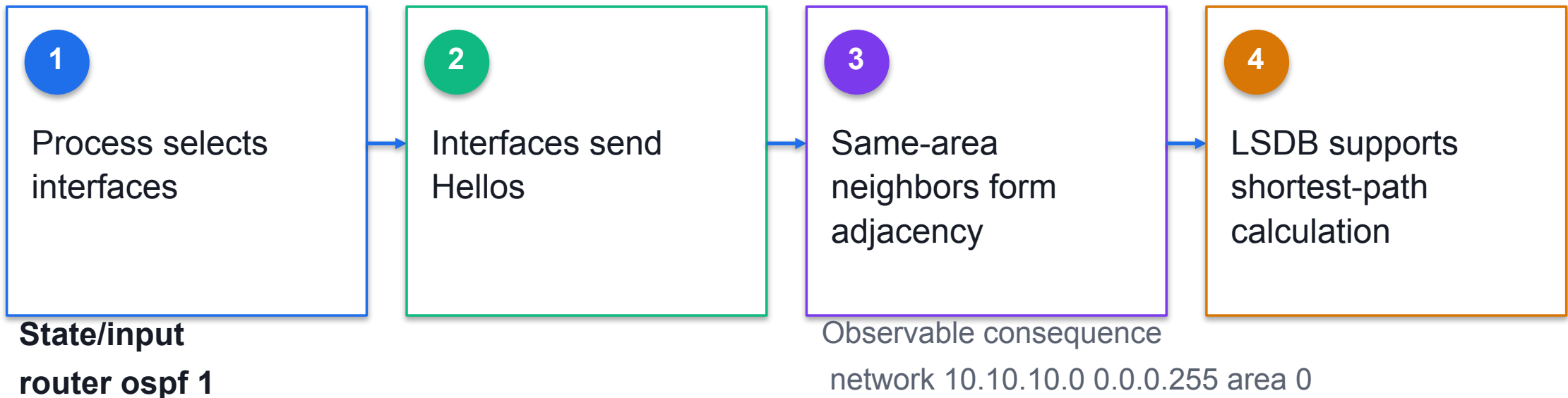


```
router ospf 1
network 10.10.10.0 0.0.0.255 area 0
```

Boundary to inspect
Process ID is locally significant; area ID must match on the link

```
show ip ospf interface brief
```

OSPF process and area boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF process and area boundaries · exact contract

```
router ospf 1
network 10.0.0.0 0.0.0.3 area 0
network 10.10.10.0 0.0.0.255 area 0
```

INPUT / IDENTITY

Local OSPF process · Process selects interfaces

DECISION / STATE

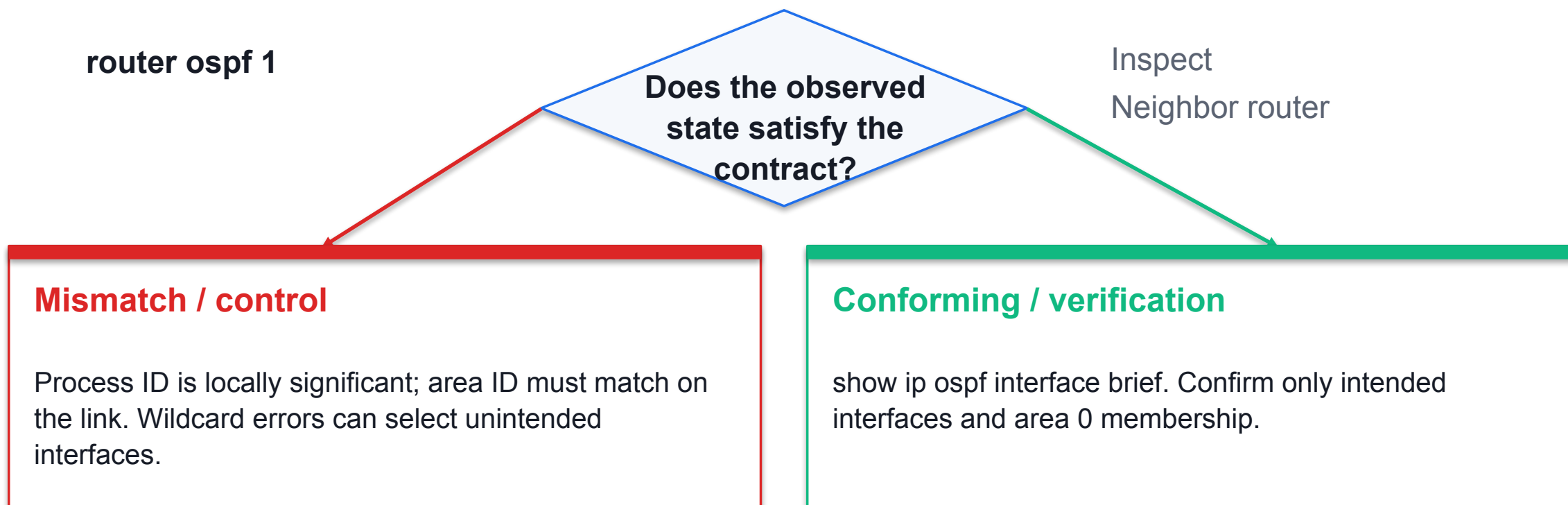
Same-area neighbors form adjacency

READBACK / RESULT

LSDB supports shortest-path calculation

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF process and area boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

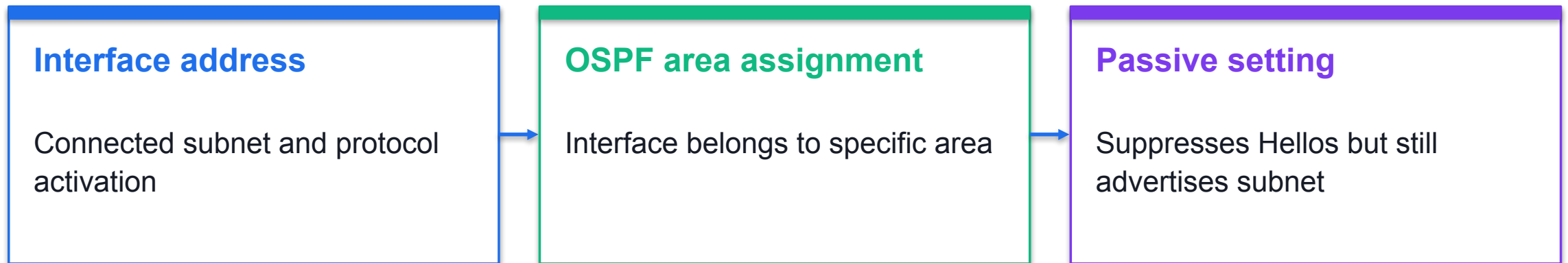
OSPF process and area boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Local OSPF process -> Neighbor router	router ospf 1
Mechanism / state	LSDB supports shortest-path calculation	network 10.10.10.0 0.0.0.255 area 0
Result / boundary	show ip ospf interface brief. Confirm only intended interfaces and area 0 membership.	Process ID is locally significant; area ID must match on the link. Wildcard errors can select unintended interfaces.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF interface activation · boundaries



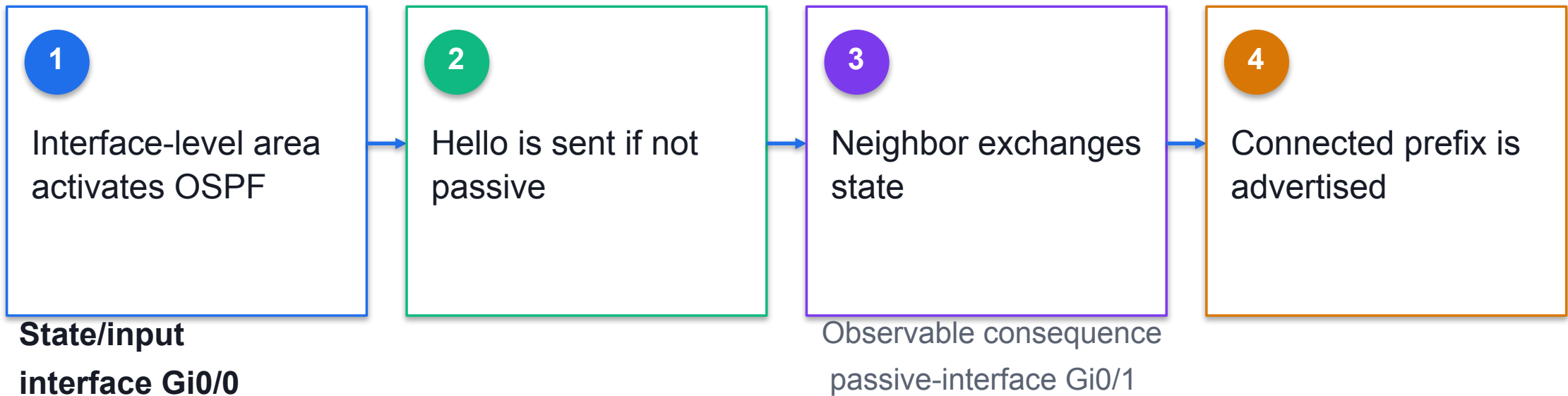
interface Gi0/0
passive-interface Gi0/1

Boundary to inspect

A passive interface advertises its subnet but does not form neighbors there

show ip protocols; show ip ospf interface

OSPF interface activation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF interface activation · exact contract

```
interface Gi0/0  
  ip ospf 1 area 0  
router ospf 1  
  passive-interface Gi0/1
```

INPUT / IDENTITY

Interface address · Interface-level area
activates OSPF

DECISION / STATE

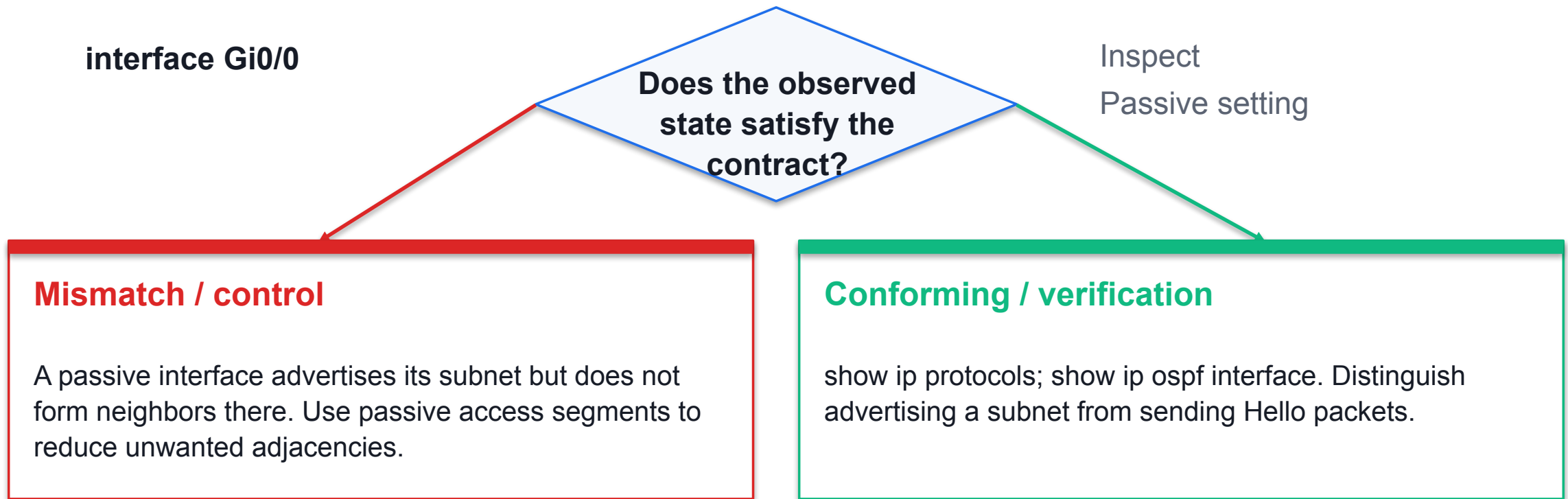
Neighbor exchanges state

READBACK / RESULT

Connected prefix is advertised

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF interface activation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

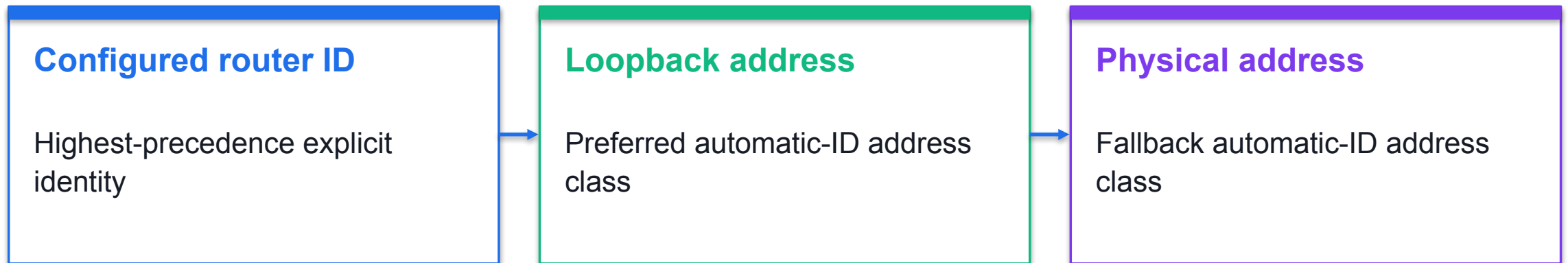
OSPF interface activation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Interface address -> Passive setting	interface Gi0/0
Mechanism / state	Connected prefix is advertised	passive-interface Gi0/1
Result / boundary	show ip protocols; show ip ospf interface. Distinguish advertising a subnet from sending Hello packets.	A passive interface advertises its subnet but does not form neighbors there. Use passive access segments to reduce unwanted adjacencies.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF router ID selection · boundaries

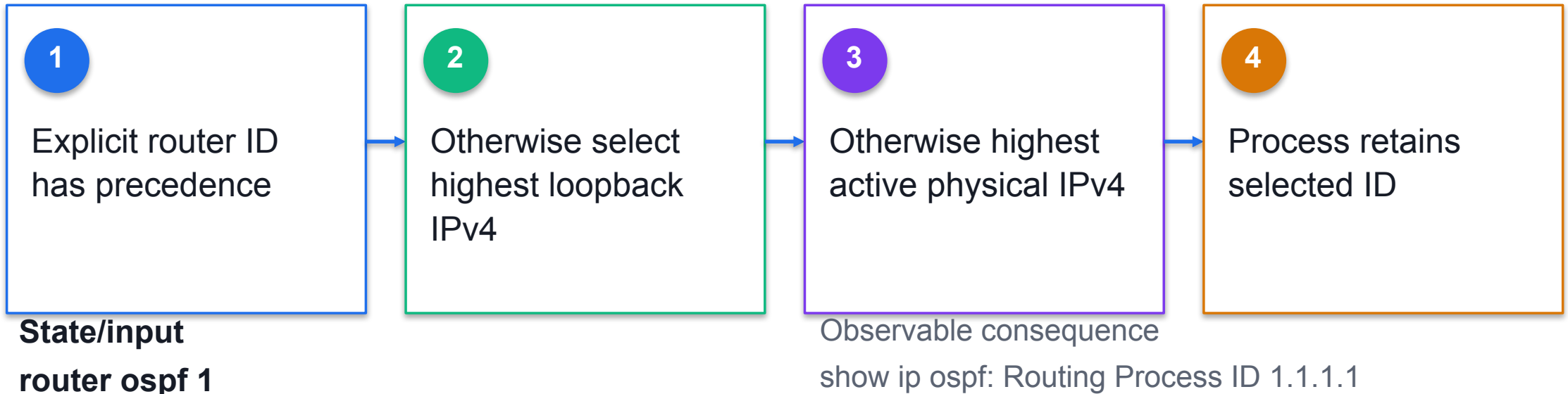


router ospf 1
show ip ospf: Routing Process ID1.1.1.1

Boundary to inspect
Duplicate router IDs disrupt LSDB/neighbor operation

Compare show ip ospf on all routers

OSPF router ID selection · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF router ID selection · exact contract

```
router ospf 1
router-id 1.1.1.1
show ip ospf: Routing Process ID1.1.1.1
```

INPUT / IDENTITY

Configured router ID · Explicit router ID has precedence

DECISION / STATE

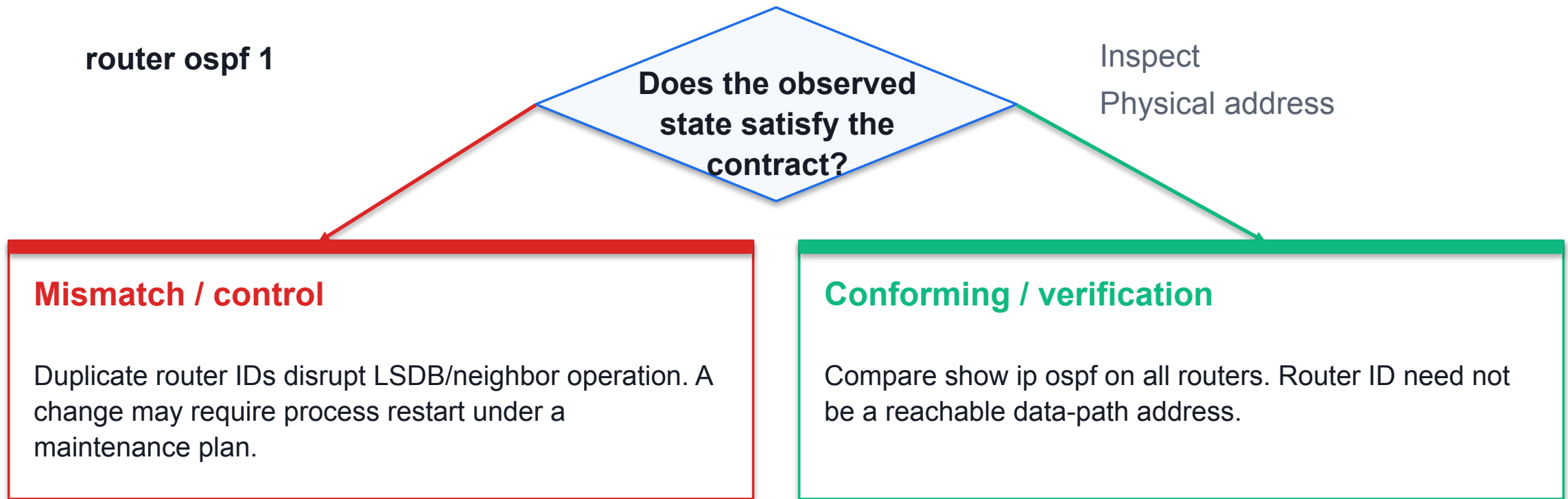
Otherwise highest active physical IPv4

READBACK / RESULT

Process retains selected ID

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF router ID selection · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

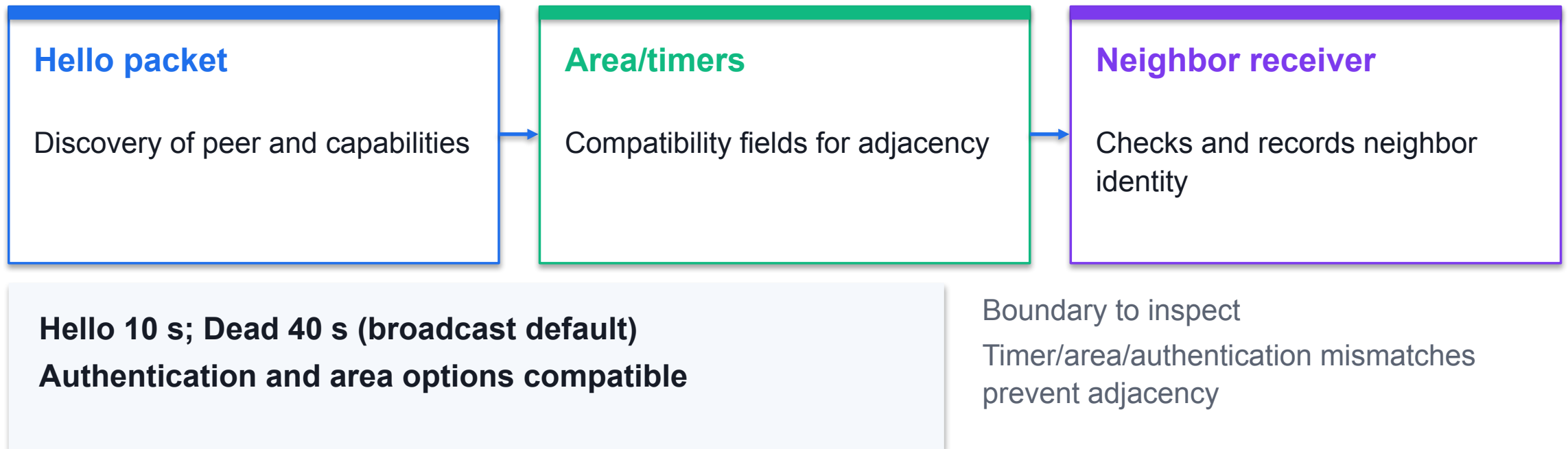
OSPF router ID selection · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Configured router ID -> Physical address	router ospf 1
Mechanism / state	Process retains selected ID	show ip ospf: Routing Process ID 1.1.1.1
Result / boundary	Compare show ip ospf on all routers. Router ID need not be a reachable data-path address.	Duplicate router IDs disrupt LSDB/neighbor operation. A change may require process restart under a maintenance plan.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

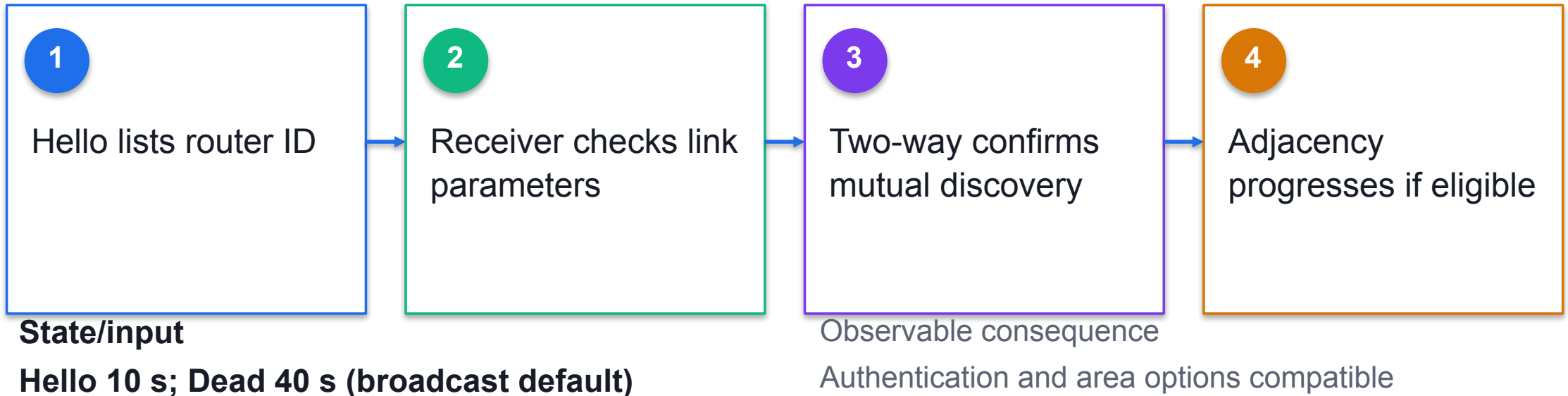
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF Hello compatibility · boundaries



show ip ospf neighbor; inspect interface timers and MTU on both sides

OSPF Hello compatibility · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF Hello compatibility · exact contract

Hello 10 s; Dead 40 s (broadcast default)
Area 0; subnet/mask compatible
Authentication and area options compatible

INPUT / IDENTITY

Hello packet · Hello lists router ID

DECISION / STATE

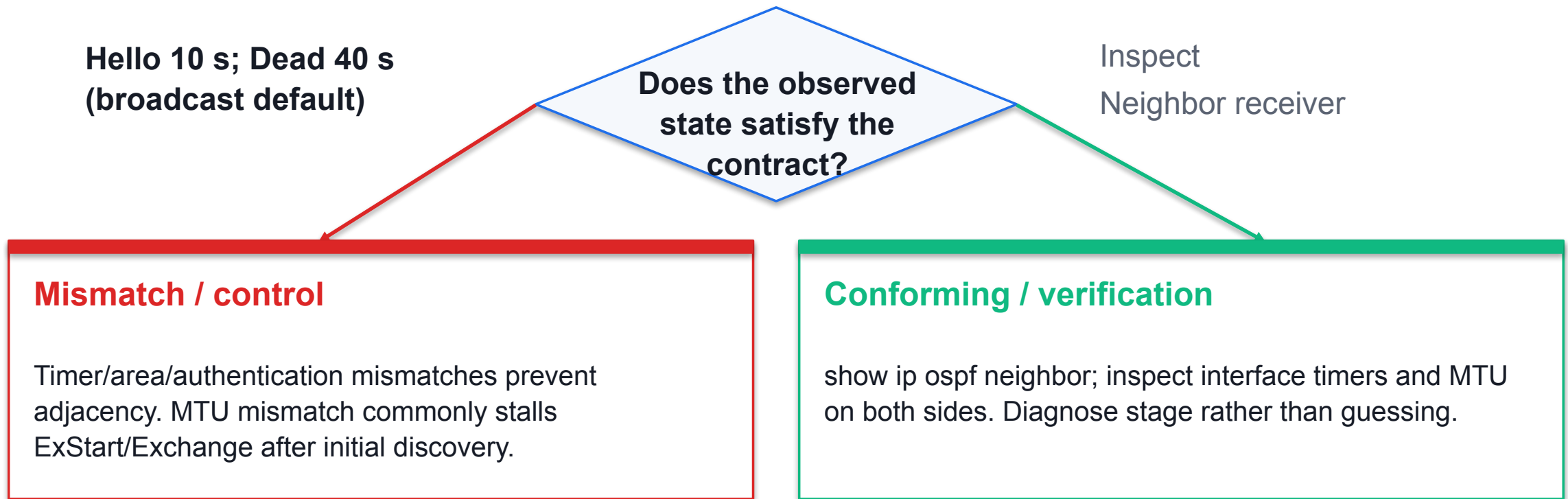
Two-way confirms mutual discovery

READBACK / RESULT

Adjacency progresses if eligible

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF Hello compatibility · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

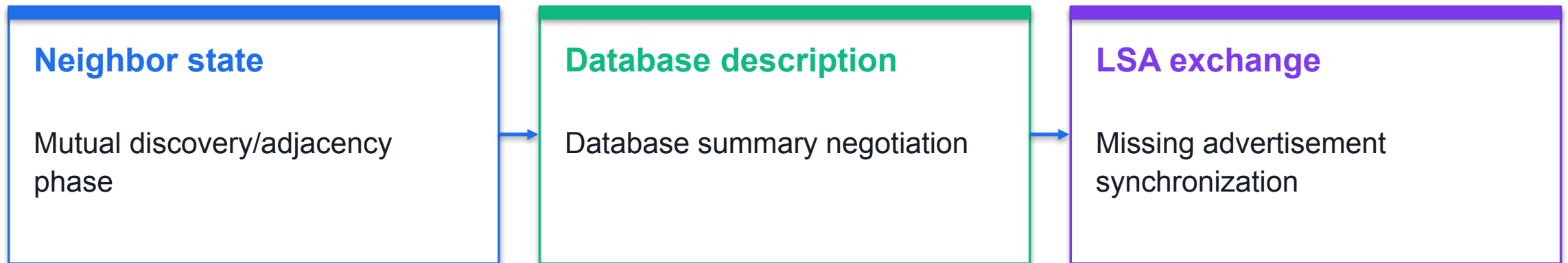
OSPF Hello compatibility · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Hello packet -> Neighbor receiver	Hello 10 s; Dead 40 s (broadcast default)
Mechanism / state	Adjacency progresses if eligible	Authentication and area options compatible
Result / boundary	show ip ospf neighbor; inspect interface timers and MTU on both sides. Diagnose stage rather than guessing.	Timer/area/authentication mismatches prevent adjacency. MTU mismatch commonly stalls ExStart/Exchange after initial discovery.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF neighbor state exchange · boundaries

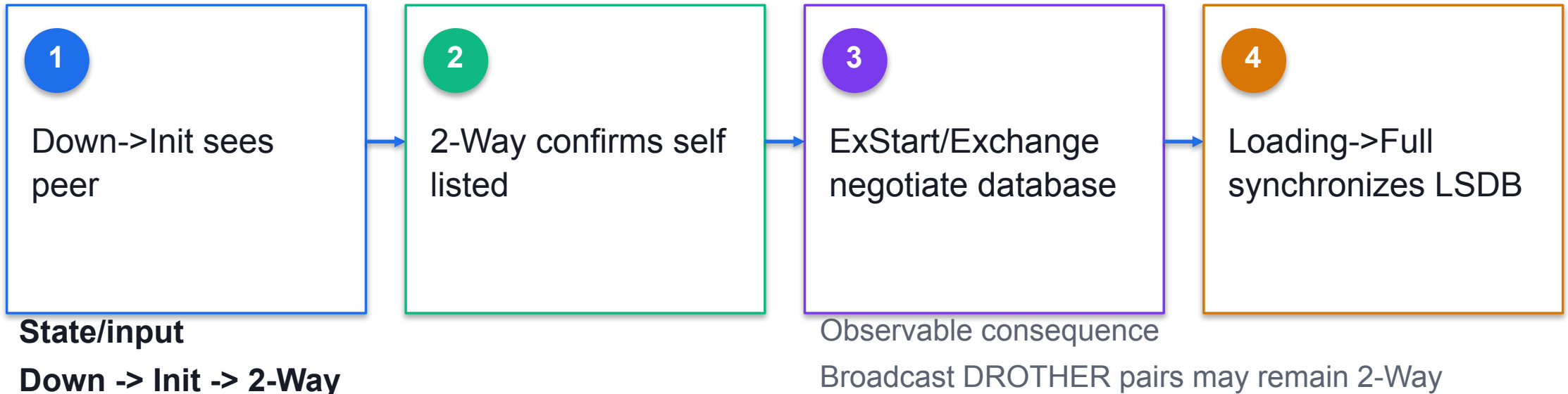


Down -> Init -> 2-Way
Broadcast DROTHER pairs may remain 2-Way

Boundary to inspect
2-Way is not always a failure on broadcast networks

Correlate neighbor role with state

OSPF neighbor state exchange · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF neighbor state exchange · exact contract

Down -> Init -> 2-Way
ExStart -> Exchange -> Loading -> Full
Broadcast DROTHER pairs may remain 2-Way

INPUT / IDENTITY

Neighbor state · Down->Init sees peer

DECISION / STATE

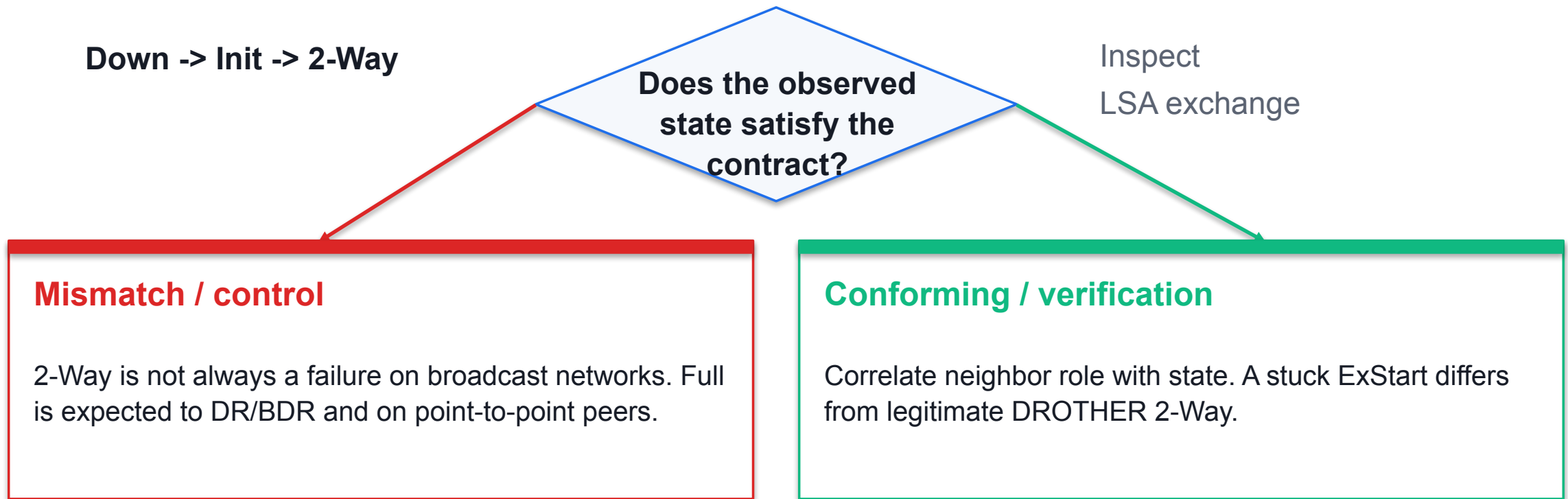
ExStart/Exchange negotiate database

READBACK / RESULT

Loading->Full synchronizes LSDB

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF neighbor state exchange · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

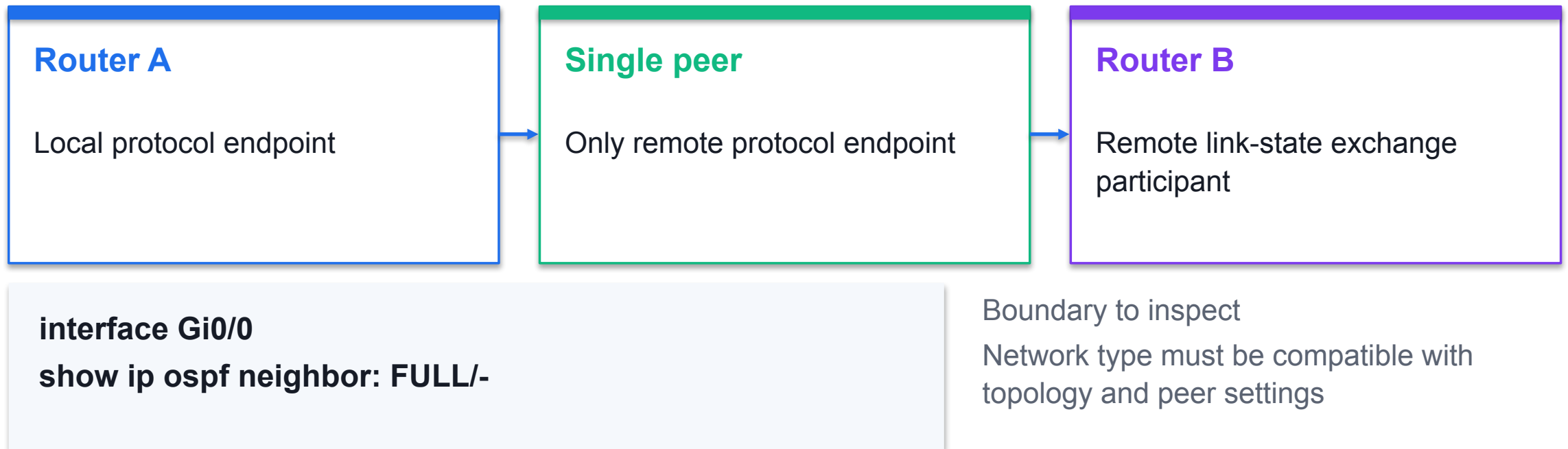
OSPF neighbor state exchange · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Neighbor state -> LSA exchange	Down -> Init -> 2-Way
Mechanism / state	Loading->Full synchronizes LSDB	Broadcast DROTHER pairs may remain 2-Way
Result / boundary	Correlate neighbor role with state. A stuck ExStart differs from legitimate DROTHER 2-Way.	2-Way is not always a failure on broadcast networks. Full is expected to DR/BDR and on point-to-point peers.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

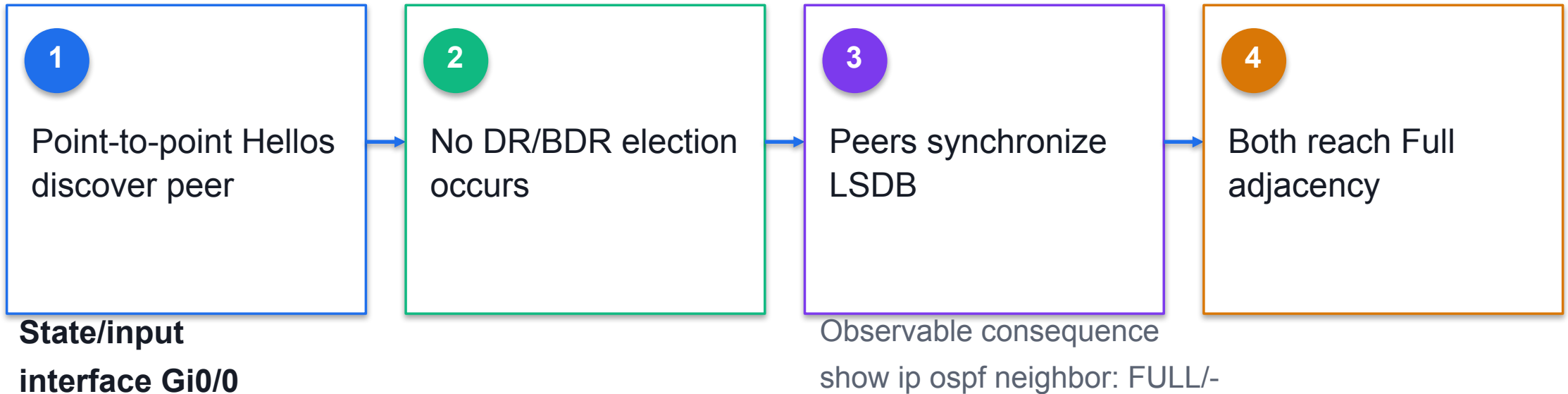
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF point-to-point operation · boundaries



show ip ospf interface verifies POINT_TO_POINT; neighbor should show FULL/- rather than FULL/DR.

OSPF point-to-point operation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF point-to-point operation · exact contract

```
interface Gi0/0
ip ospf network point-to-point
show ip ospf neighbor: FULL/-
```

INPUT / IDENTITY

Router A · Point-to-point Hellos discover peer

DECISION / STATE

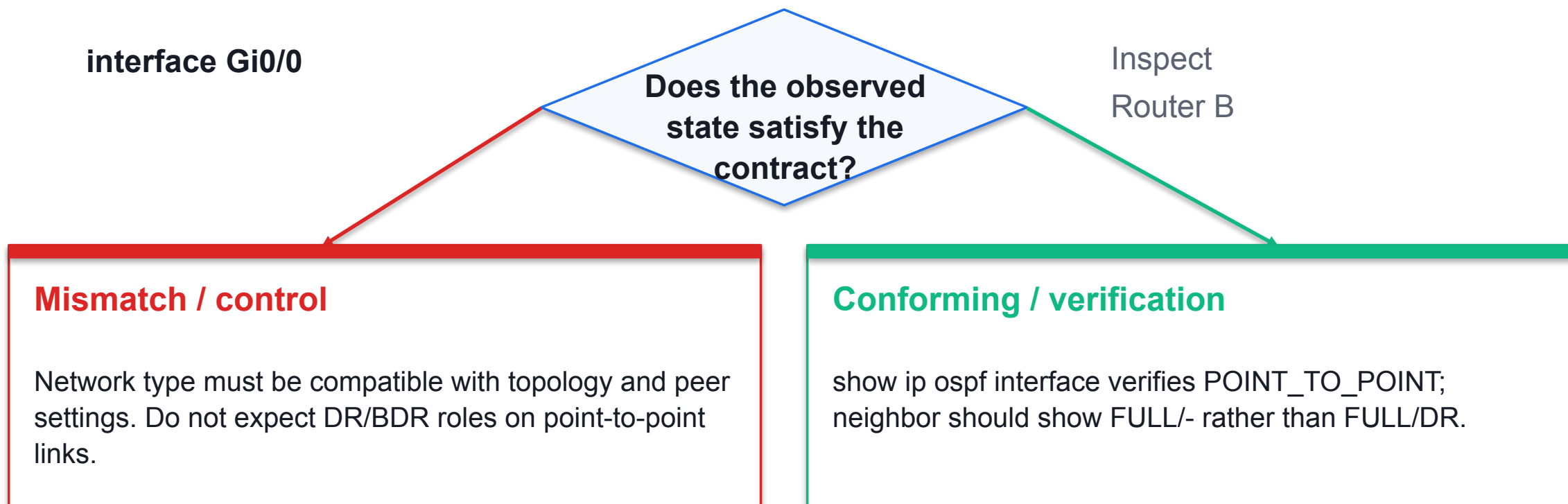
Peers synchronize LSDB

READBACK / RESULT

Both reach Full adjacency

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF point-to-point operation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

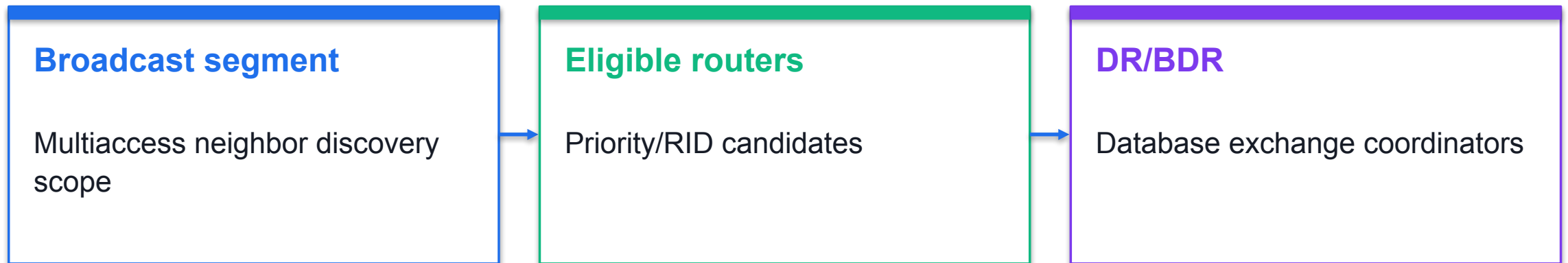
OSPF point-to-point operation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Router A -> Router B	interface Gi0/0
Mechanism / state	Both reach Full adjacency	show ip ospf neighbor: FULL/-
Result / boundary	show ip ospf interface verifies POINT_TO_POINT; neighbor should show FULL/- rather than FULL/DR.	Network type must be compatible with topology and peer settings. Do not expect DR/BDR roles on point-to-point links.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF DR and BDR election · boundaries

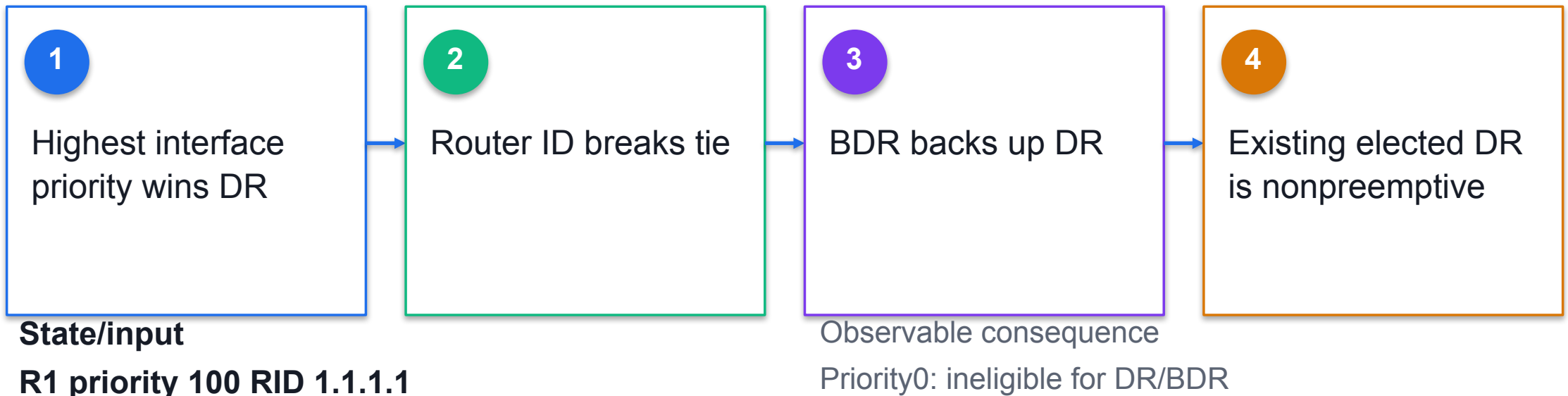


R1 priority 100 RID 1.1.1.1
Priority0: ineligible for DR/BDR

Boundary to inspect
A later higher-priority router does not automatically replace an established DR

show ip ospf interface and neighbor

OSPF DR and BDR election · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF DR and BDR election · exact contract

R1 priority 100 RID 1.1.1.1
R2 priority 50 RID 2.2.2.2
Priority0: ineligible for DR/BDR

INPUT / IDENTITY

Broadcast segment · Highest interface priority wins DR

DECISION / STATE

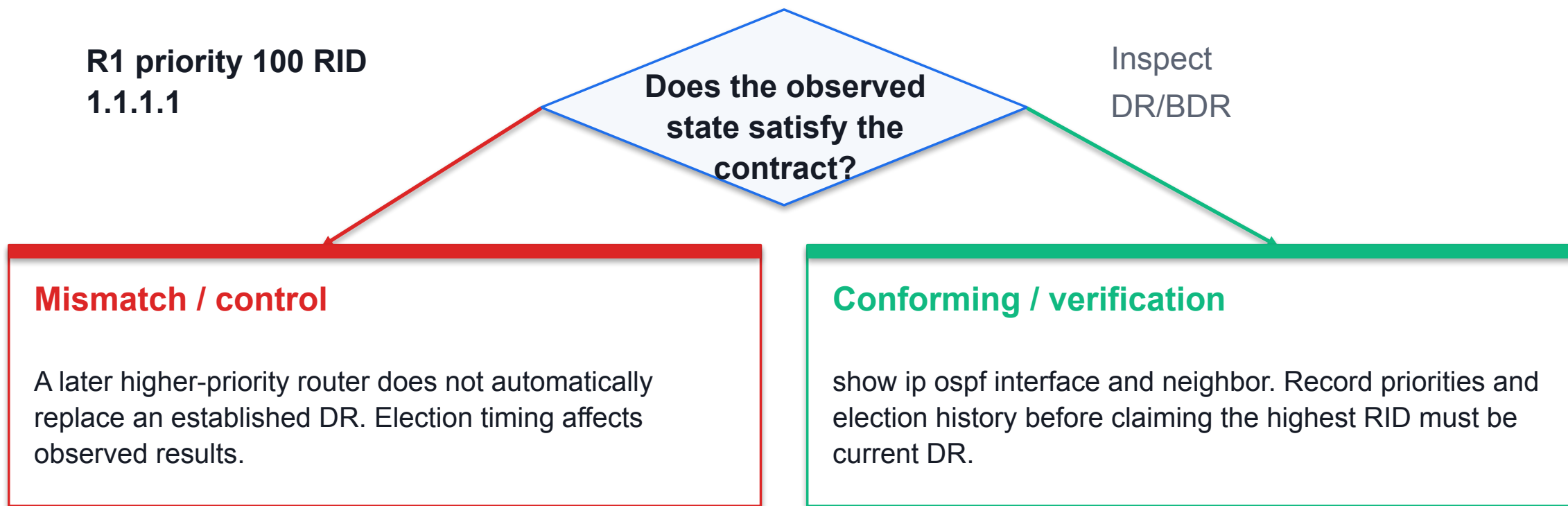
BDR backs up DR

READBACK / RESULT

Existing elected DR is nonpreemptive

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF DR and BDR election · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

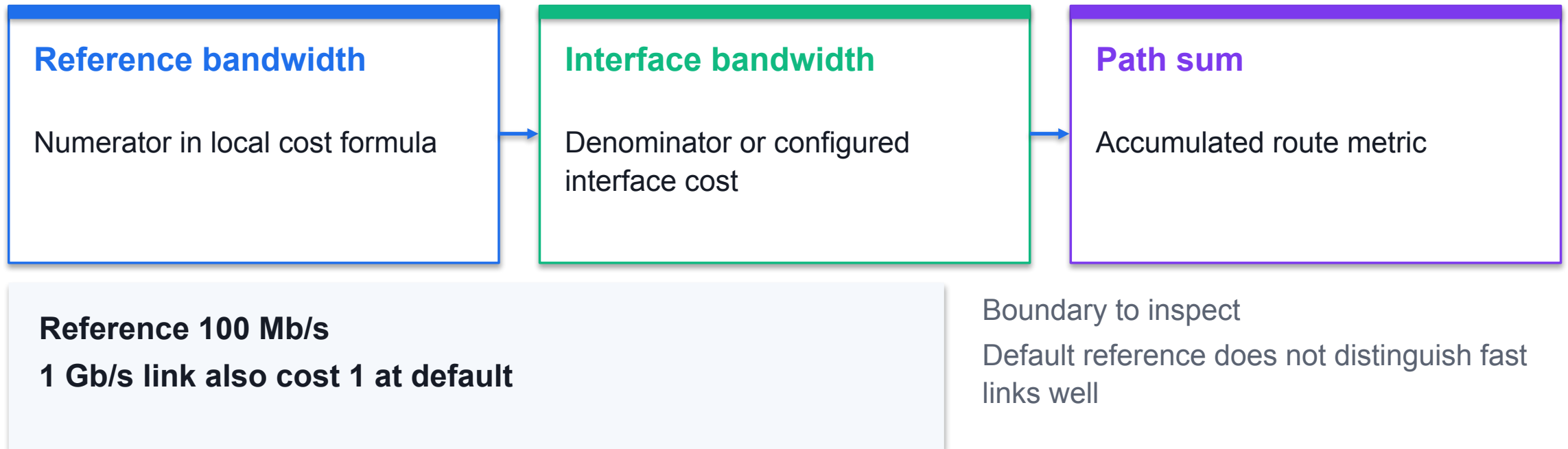
OSPF DR and BDR election · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Broadcast segment -> DR/BDR	R1 priority 100 RID 1.1.1.1
Mechanism / state	Existing elected DR is nonpreemptive	Priority0: ineligible for DR/BDR
Result / boundary	show ip ospf interface and neighbor. Record priorities and election history before claiming the highest RID must be current DR.	A later higher-priority router does not automatically replace an established DR. Election timing affects observed results.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

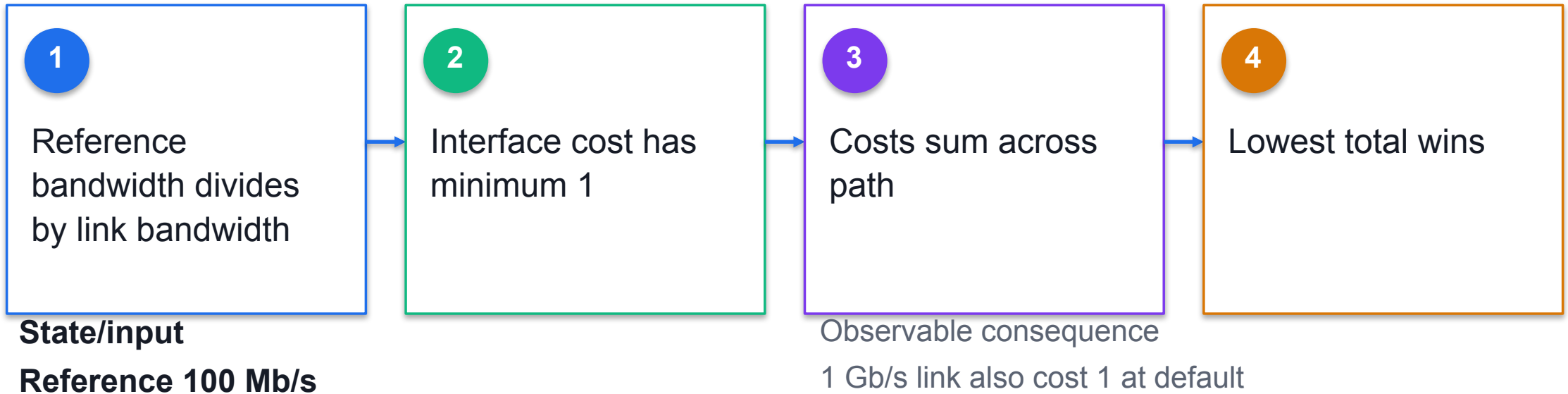
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF cost calculation · boundaries



show ip ospf interface

OSPF cost calculation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF cost calculation · exact contract

Reference 100 Mb/s
100 Mb/s link cost 1
10 Mb/s link cost 10
1 Gb/s link also cost 1 at default

INPUT / IDENTITY

Reference bandwidth · Reference
bandwidth divides by link bandwidth

DECISION / STATE

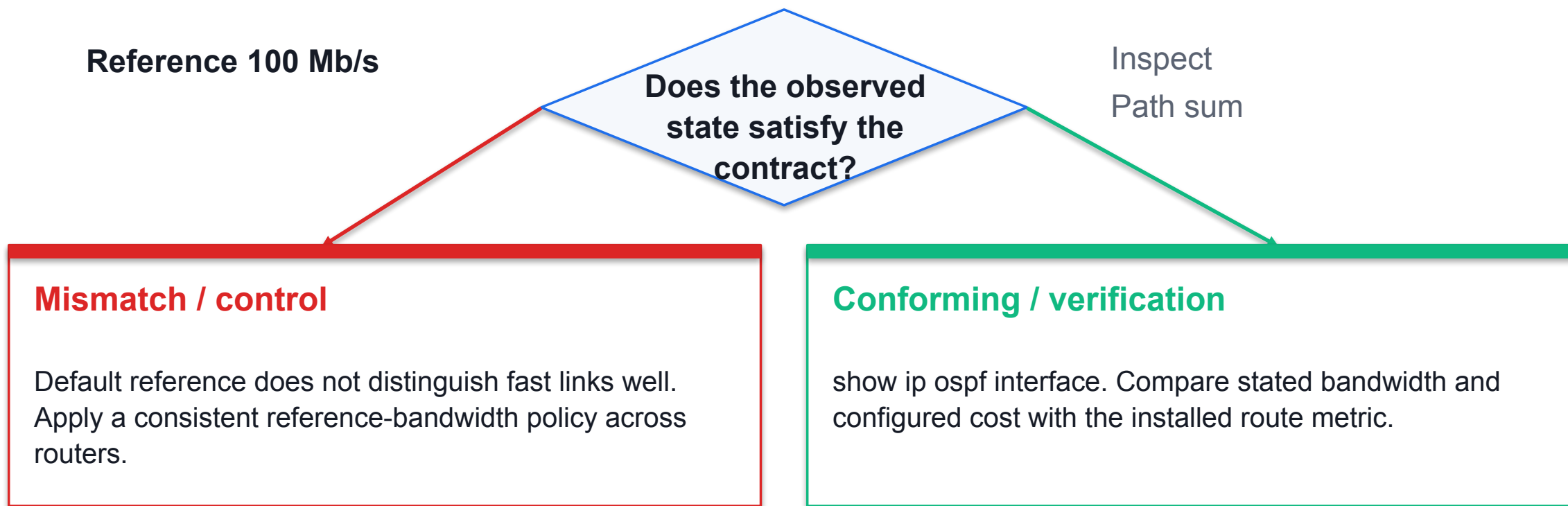
Costs sum across path

READBACK / RESULT

Lowest total wins

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF cost calculation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

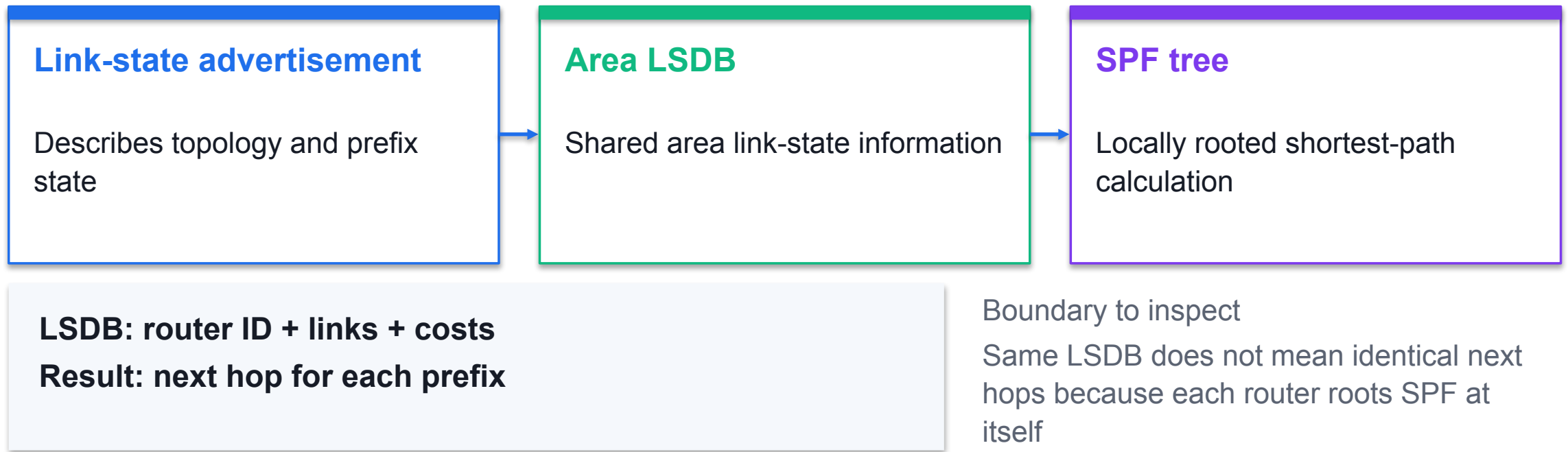
OSPF cost calculation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Reference bandwidth -> Path sum	Reference 100 Mb/s
Mechanism / state	Lowest total wins	1 Gb/s link also cost 1 at default
Result / boundary	show ip ospf interface. Compare stated bandwidth and configured cost with the installed route metric.	Default reference does not distinguish fast links well. Apply a consistent reference-bandwidth policy across routers.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

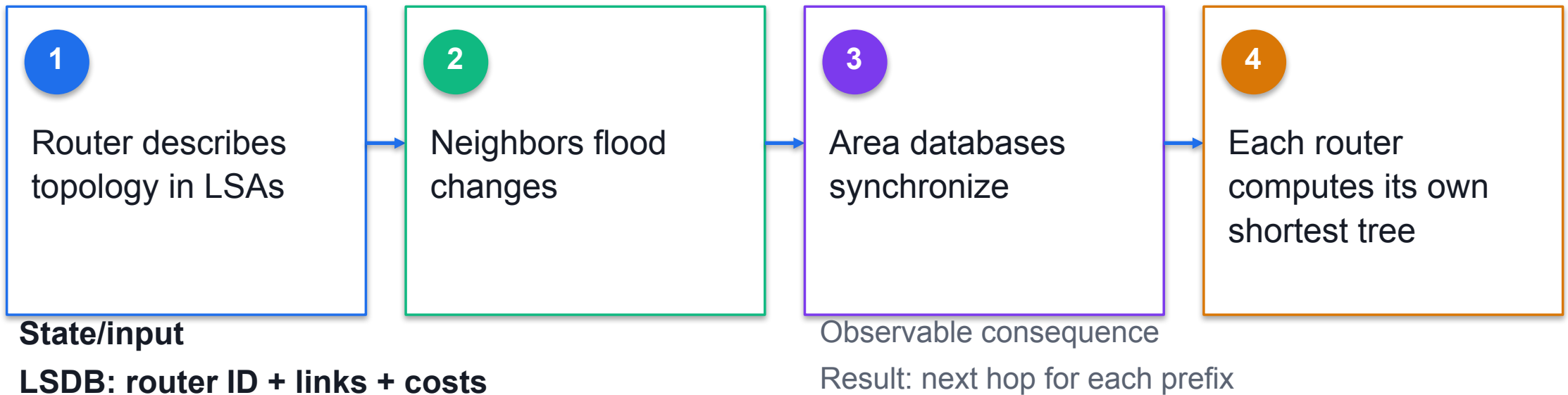
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF LSA and SPF boundaries · boundaries



show ip ospf database and show ip route ospf

OSPF LSA and SPF boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF LSA and SPF boundaries · exact contract

LSDB: router ID + links + costs
SPF root: local router ID
Result: next hop for each prefix

INPUT / IDENTITY

Link-state advertisement · Router describes topology in LSAs

DECISION / STATE

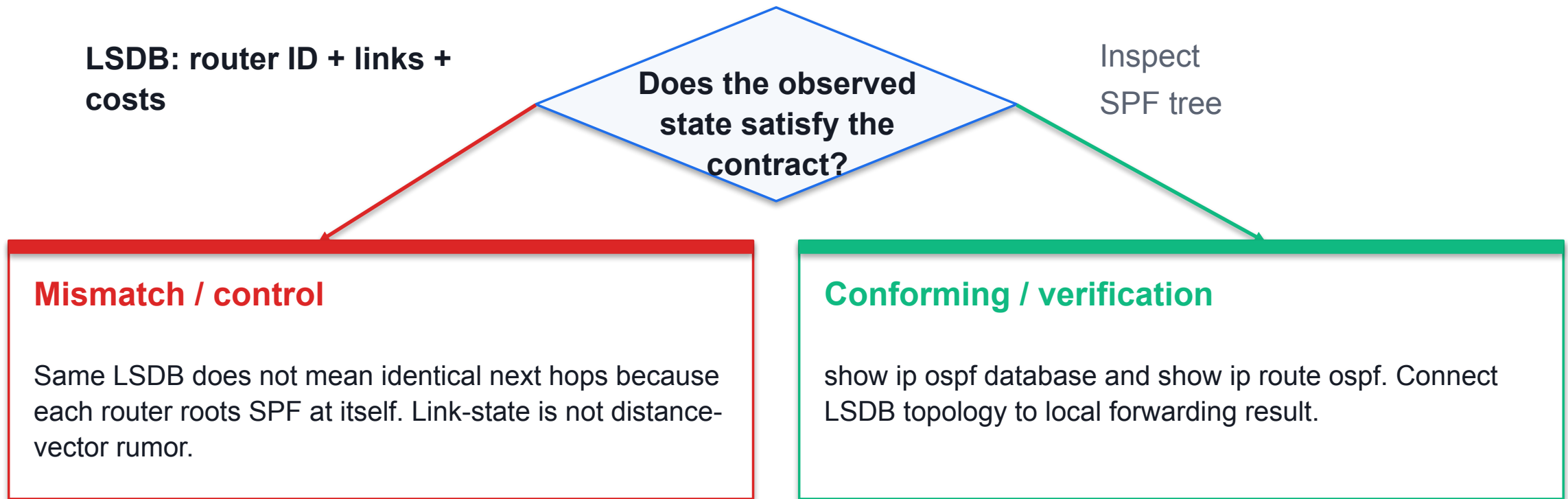
Area databases synchronize

READBACK / RESULT

Each router computes its own shortest tree

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF LSA and SPF boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

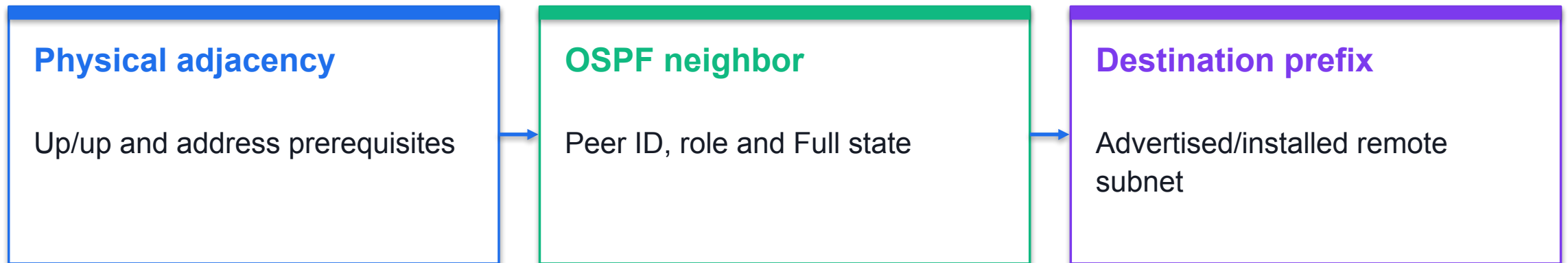
OSPF LSA and SPF boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Link-state advertisement -> SPF tree	LSDB: router ID + links + costs
Mechanism / state	Each router computes its own shortest tree	Result: next hop for each prefix
Result / boundary	show ip ospf database and show ip route ospf. Connect LSDB topology to local forwarding result.	Same LSDB does not mean identical next hops because each router roots SPF at itself. Link-state is not distance-vector rumor.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF failure diagnosis · boundaries



Interface up/up

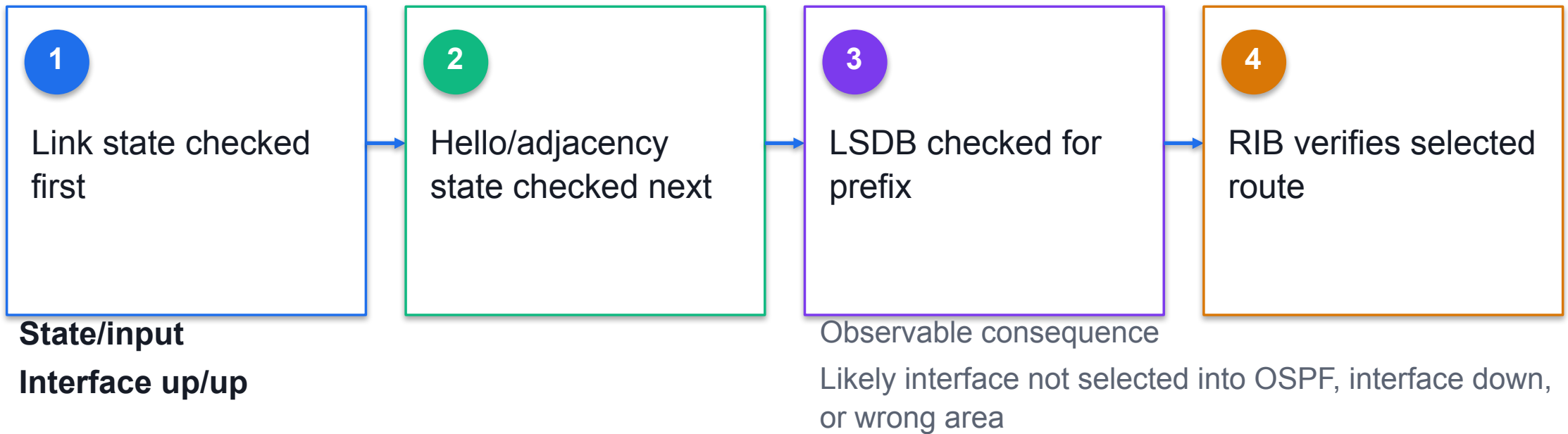
Likely interface not selected into OSPF, interface down, or wrong area

Boundary to inspect

A Full neighbor alone does not prove every needed subnet is advertised

Use show ip ospf neighbor, database and route

OSPF failure diagnosis · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPF failure diagnosis · exact contract

Interface up/up
Neighbor FULL
Prefix absent from LSDB
Likely interface not selected into OSPF, interface down, or wrong area

INPUT / IDENTITY

Physical adjacency · Link state checked first

DECISION / STATE

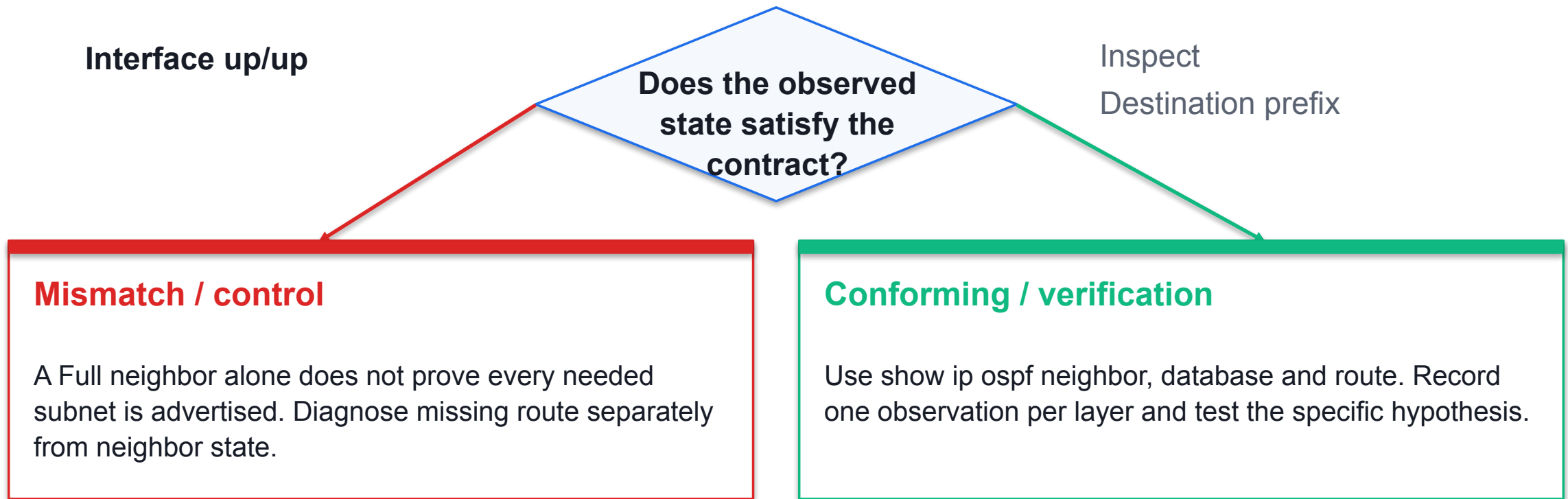
LSDB checked for prefix

READBACK / RESULT

RIB verifies selected route

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPF failure diagnosis · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

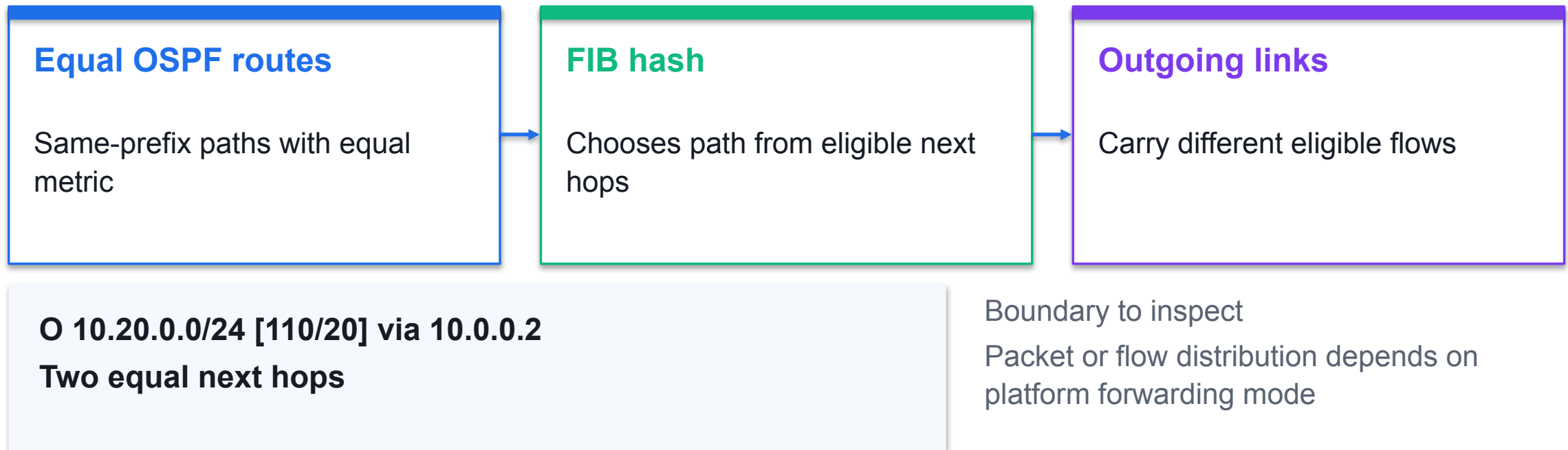
OSPF failure diagnosis · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Physical adjacency -> Destination prefix	Interface up/up
Mechanism / state	RIB verifies selected route	Likely interface not selected into OSPF, interface down, or wrong area
Result / boundary	Use show ip ospf neighbor, database and route. Record one observation per layer and test the specific hypothesis.	A Full neighbor alone does not prove every needed subnet is advertised. Diagnose missing route separately from neighbor state.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

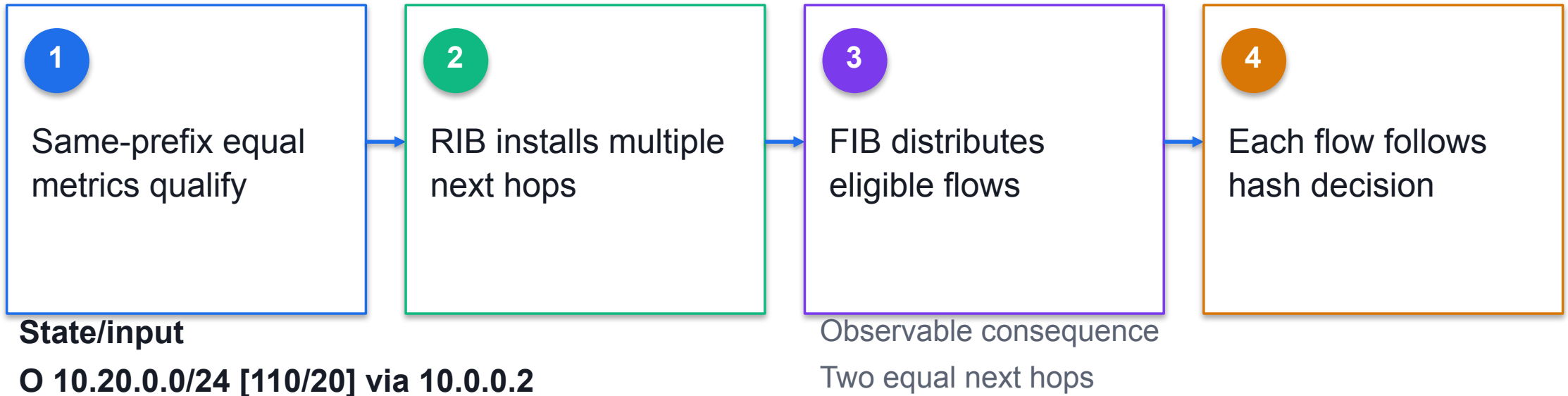
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Equal-cost multipath forwarding · boundaries



Read installed next-hop count and interface counter deltas across several flows; do not infer per-flow doubling.

Equal-cost multipath forwarding · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Equal-cost multipath forwarding · exact contract

**O 10.20.0.0/24 [110/20] via 10.0.0.2
via 10.0.0.6**
Two equal next hops

INPUT / IDENTITY

Equal OSPF routes · Same-prefix equal metrics qualify

DECISION / STATE

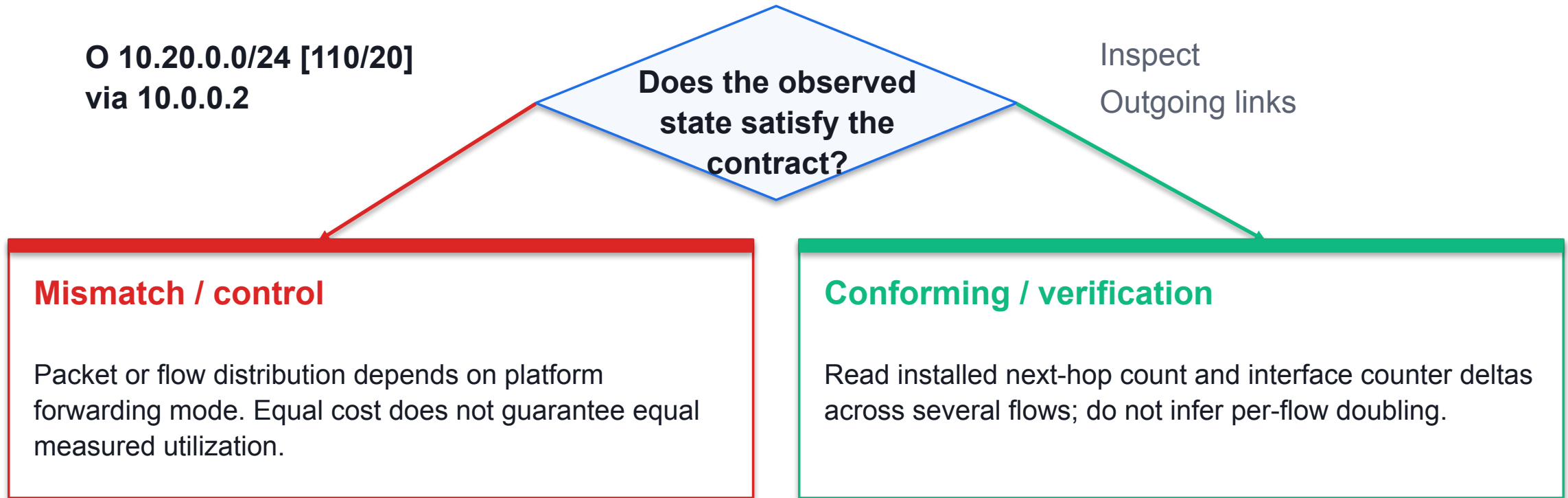
FIB distributes eligible flows

READBACK / RESULT

Each flow follows hash decision

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Equal-cost multipath forwarding · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

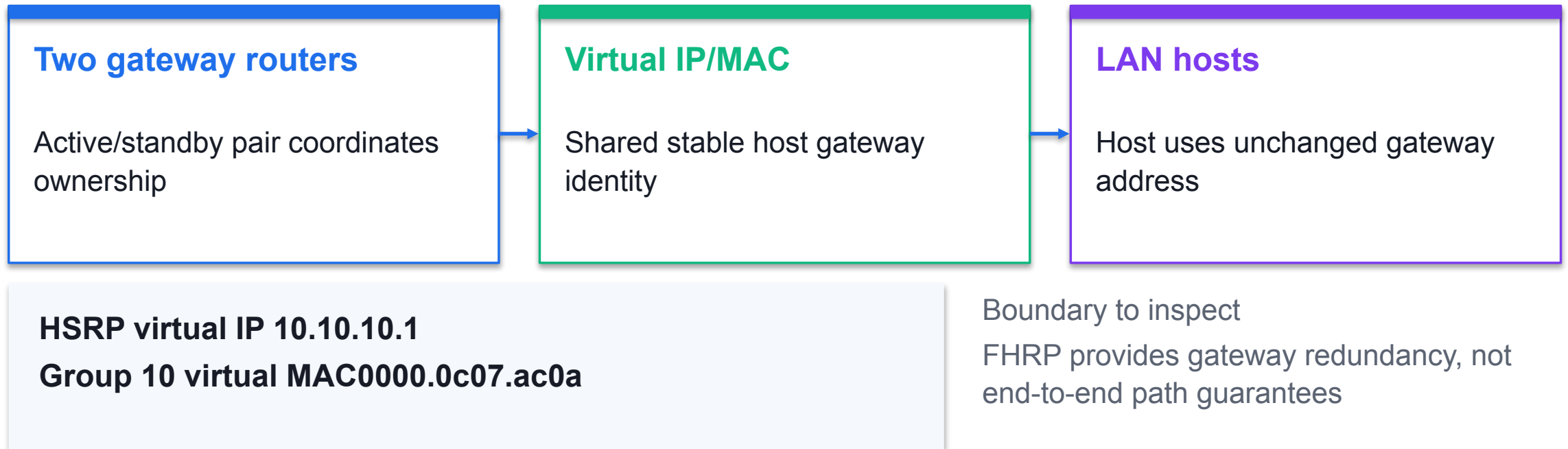
Equal-cost multipath forwarding · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Equal OSPF routes -> Outgoing links	O 10.20.0.0/24 [110/20] via 10.0.0.2
Mechanism / state	Each flow follows hash decision	Two equal next hops
Result / boundary	Read installed next-hop count and interface counter deltas across several flows; do not infer per-flow doubling.	Packet or flow distribution depends on platform forwarding mode. Equal cost does not guarantee equal measured utilization.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

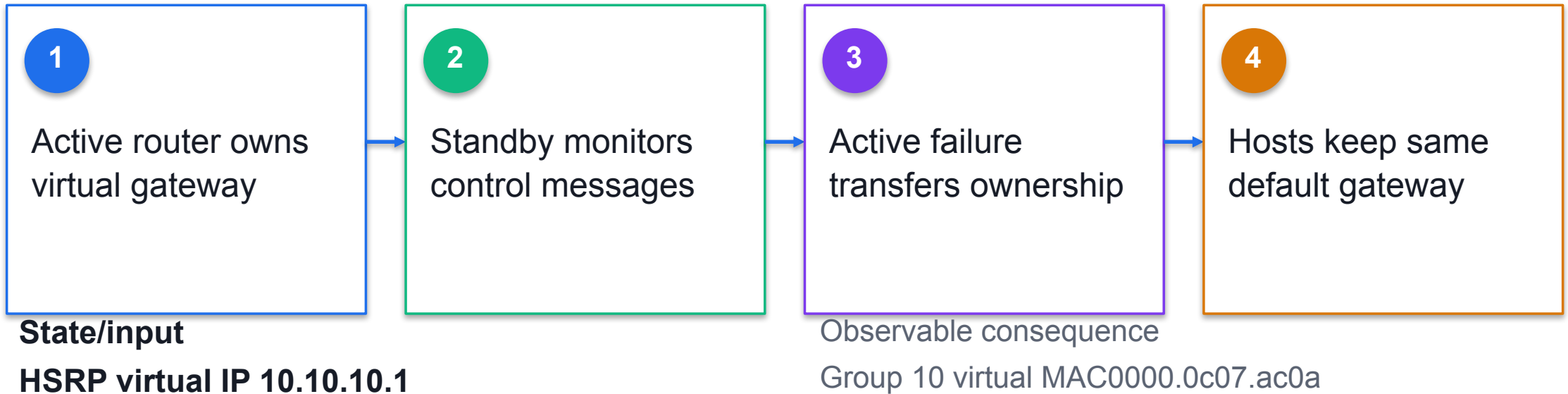
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

FHRP virtual-gateway ownership · boundaries



show standby brief

FHRP virtual-gateway ownership · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

FHRP virtual-gateway ownership · exact contract

HSRP virtual IP 10.10.10.1
R1 physical .2; R2 physical .3
Group 10 virtual MAC0000.0c07.ac0a

INPUT / IDENTITY

Two gateway routers · Active router owns virtual gateway

DECISION / STATE

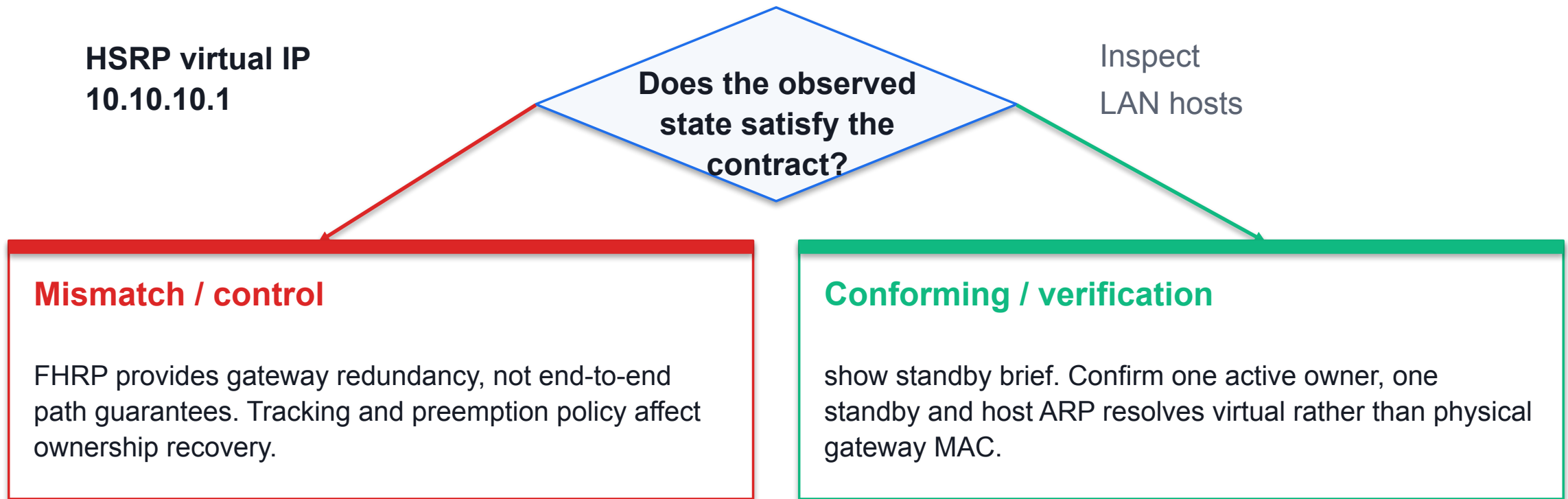
Active failure transfers ownership

READBACK / RESULT

Hosts keep same default gateway

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

FHRP virtual-gateway ownership · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

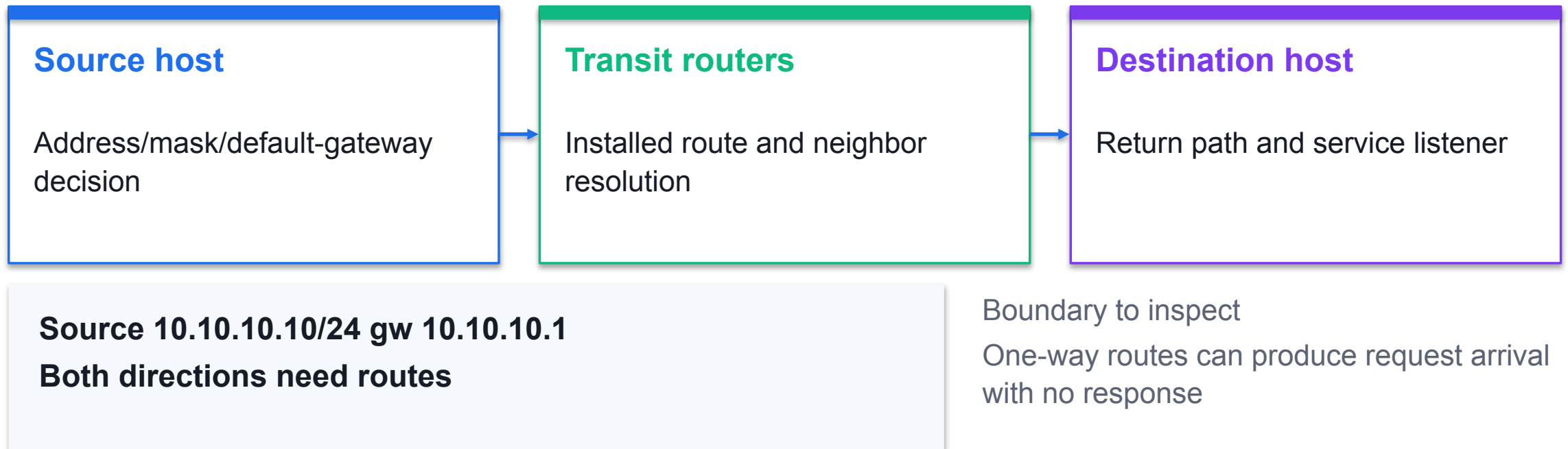
FHRP virtual-gateway ownership · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Two gateway routers -> LAN hosts	HSRP virtual IP 10.10.10.1
Mechanism / state	Hosts keep same default gateway	Group 10 virtual MAC0000.0c07.ac0a
Result / boundary	show standby brief. Confirm one active owner, one standby and host ARP resolves virtual rather than physical gateway MAC.	FHRP provides gateway redundancy, not end-to-end path guarantees. Tracking and preemption policy affect ownership recovery.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

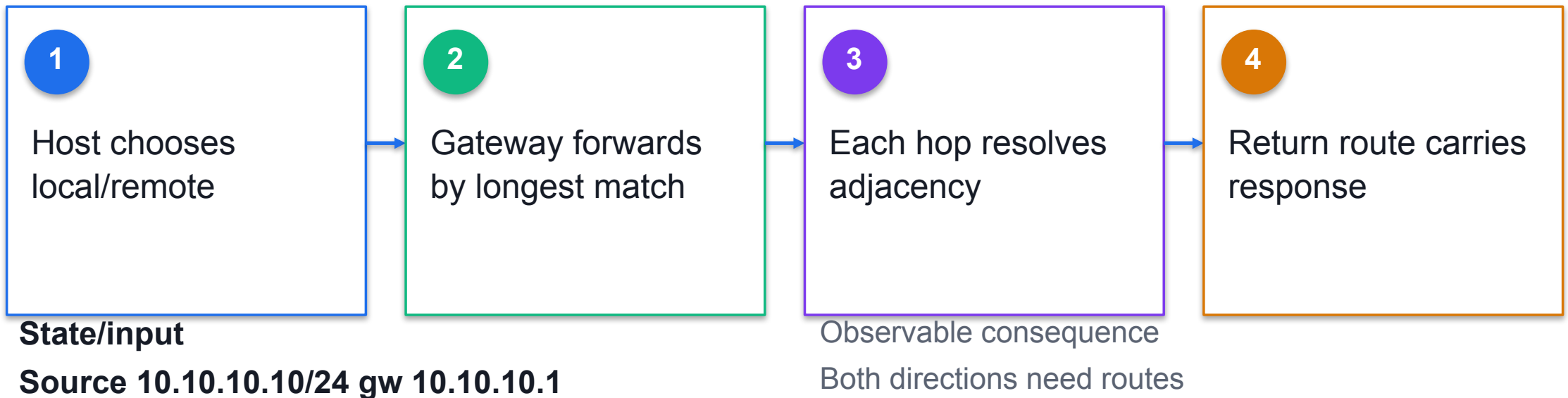
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

End-to-end routed fault isolation · boundaries



Collect source settings, forward route, return route and filtering evidence; isolate first failing control point.

End-to-end routed fault isolation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

End-to-end routed fault isolation · exact contract

Source 10.10.10.10/24 gw 10.10.10.1
Target 10.20.20.20/24 gw 10.20.20.1
Both directions need routes

INPUT / IDENTITY

Source host · Host chooses
local/remote

DECISION / STATE

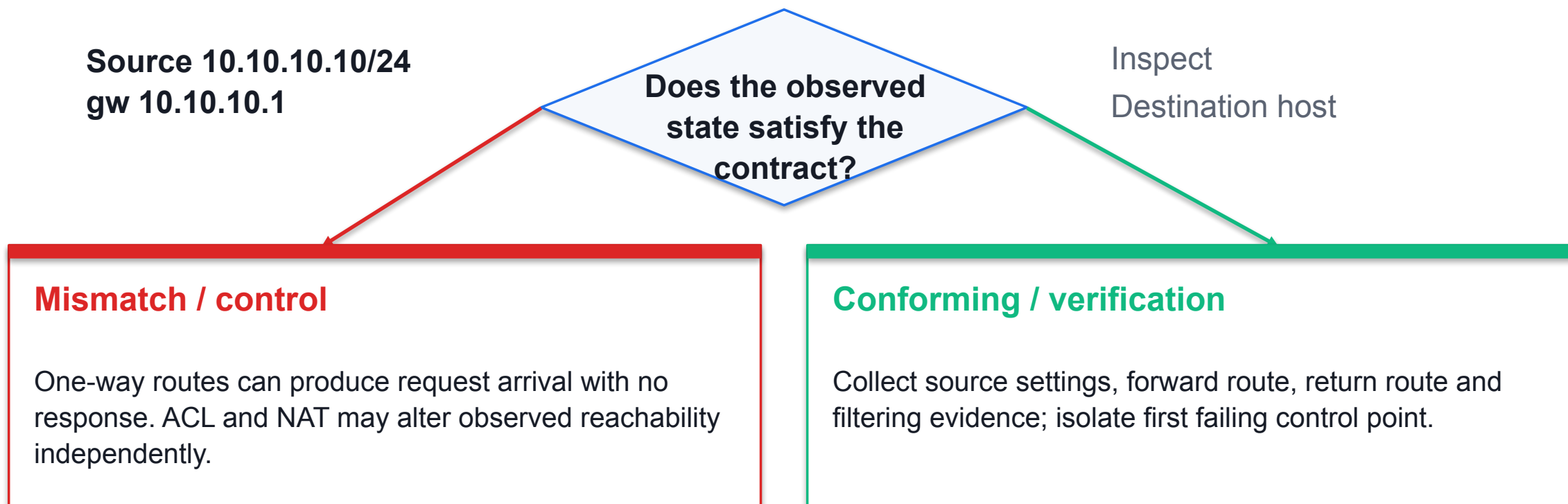
Each hop resolves adjacency

READBACK / RESULT

Return route carries response

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

End-to-end routed fault isolation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

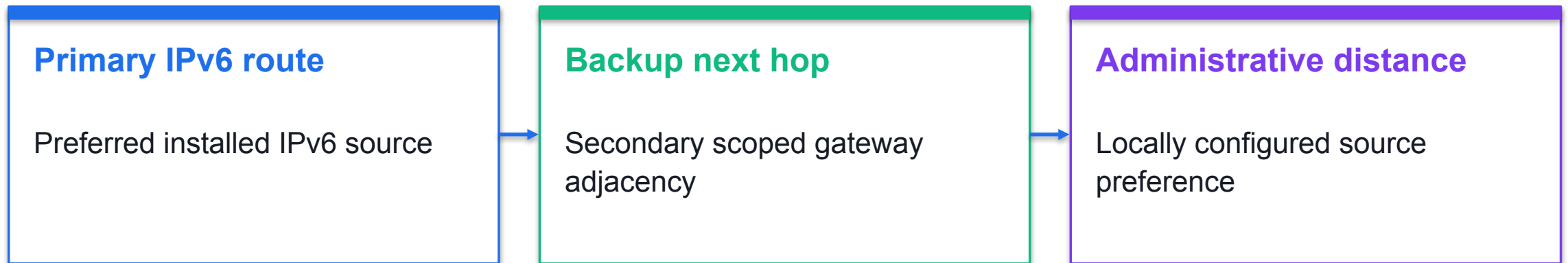
End-to-end routed fault isolation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Source host -> Destination host	Source 10.10.10.10/24 gw 10.10.10.1
Mechanism / state	Return route carries response	Both directions need routes
Result / boundary	Collect source settings, forward route, return route and filtering evidence; isolate first failing control point.	One-way routes can produce request arrival with no response. ACL and NAT may alter observed reachability independently.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPv6 floating static failover · boundaries



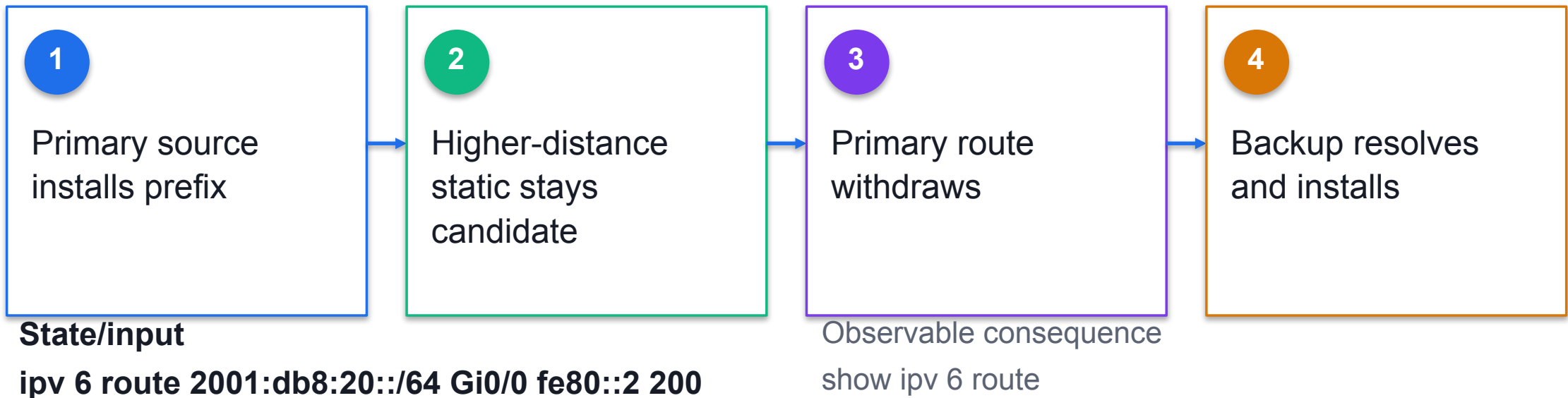
```
ipv6 route 2001:db8:20::/64 Gi0/0 fe80::2 200  
show ipv6 route
```

Boundary to inspect

If primary route remains installed despite remote failure, an untracked floating route may not activate

Compare /64 route source and next hop before/after controlled primary withdrawal; show ipv 6 neighbors confirms backup adjacency.

IPv6 floating static failover · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPv6 floating static failover · exact contract

```
ipv6 route 2001:db8:20::/64 Gi0/0 fe80::2 200
Primary OSPFv3 AD 110
Backup AD 200
show ipv6 route
```

INPUT / IDENTITY

Primary IPv6 route · Primary source
installs prefix

DECISION / STATE

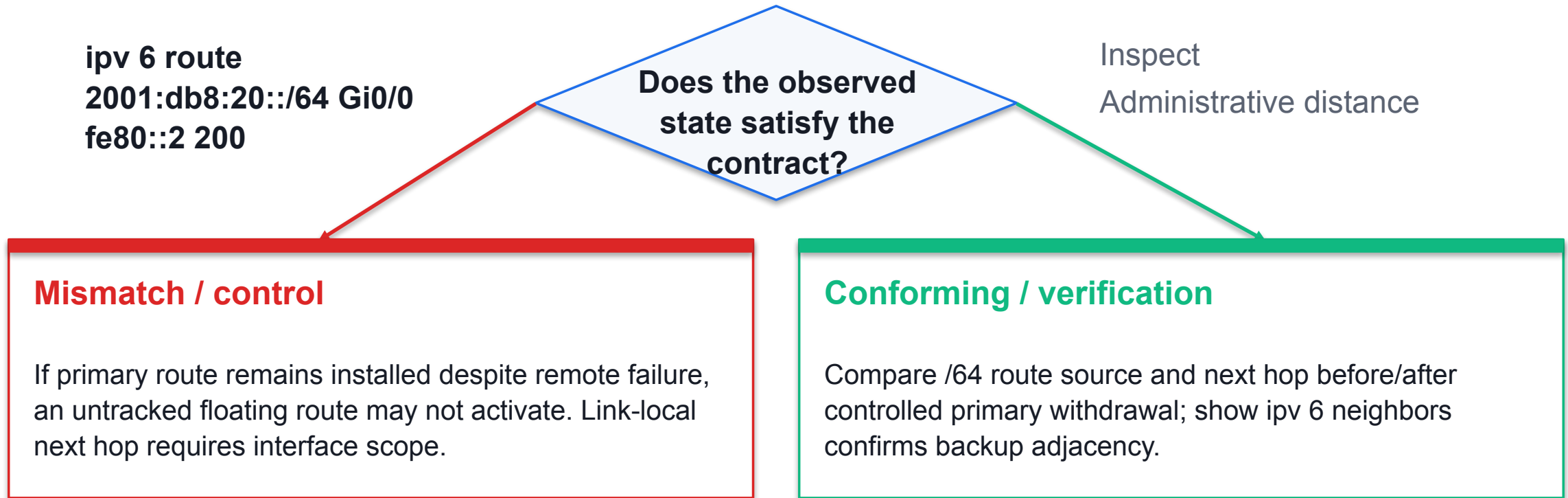
Primary route withdraws

READBACK / RESULT

Backup resolves and installs

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPv6 floating static failover · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

IPv6 floating static failover · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Primary IPv6 route -> Administrative distance	ipv 6 route 2001:db8:20::/64 Gi0/0 fe80::2 200
Mechanism / state	Backup resolves and installs	show ipv 6 route
Result / boundary	Compare /64 route source and next hop before/after controlled primary withdrawal; show ipv 6 neighbors confirms backup adjacency.	If primary route remains installed despite remote failure, an untracked floating route may not activate. Link-local next hop requires interface scope.

Bridge: Activity 6: static/IPv6 routes or Activity 7: OSPF/gateway failover

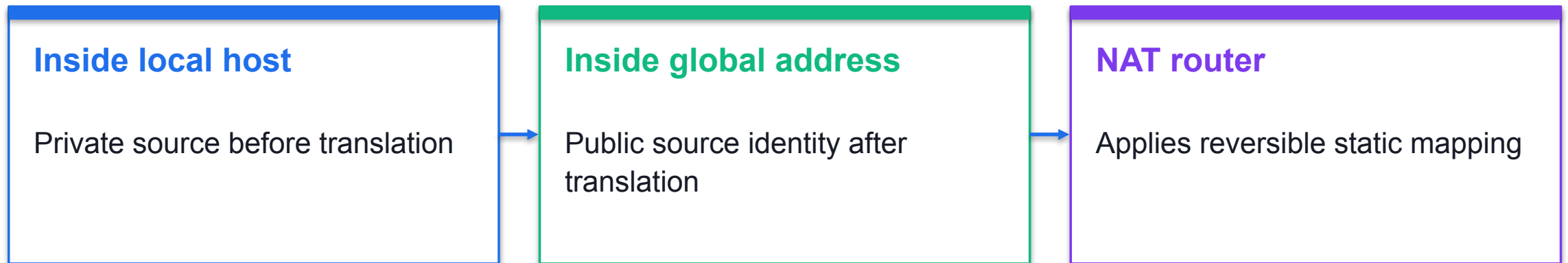
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IP Services

IP Services

Named boundaries -> runtime trace -> exact contract -> failure/control
-> acceptance evidence

Static NAT mapping · boundaries



```
ip nat inside source static 10.10.10.10 203.0.113.10
Inside global 203.0.113.10
```

Boundary to inspect
NAT requires inside/outside interface roles
and routing

show ip nat translations; inspect original/translated endpoints and both route directions.

Static NAT mapping · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Static NAT mapping · exact contract

```
ip nat inside source static 10.10.10.10 203.0.113.10  
Inside local 10.10.10.10  
Inside global 203.0.113.10
```

INPUT / IDENTITY

Inside local host · Inside packet
matches mapping

DECISION / STATE

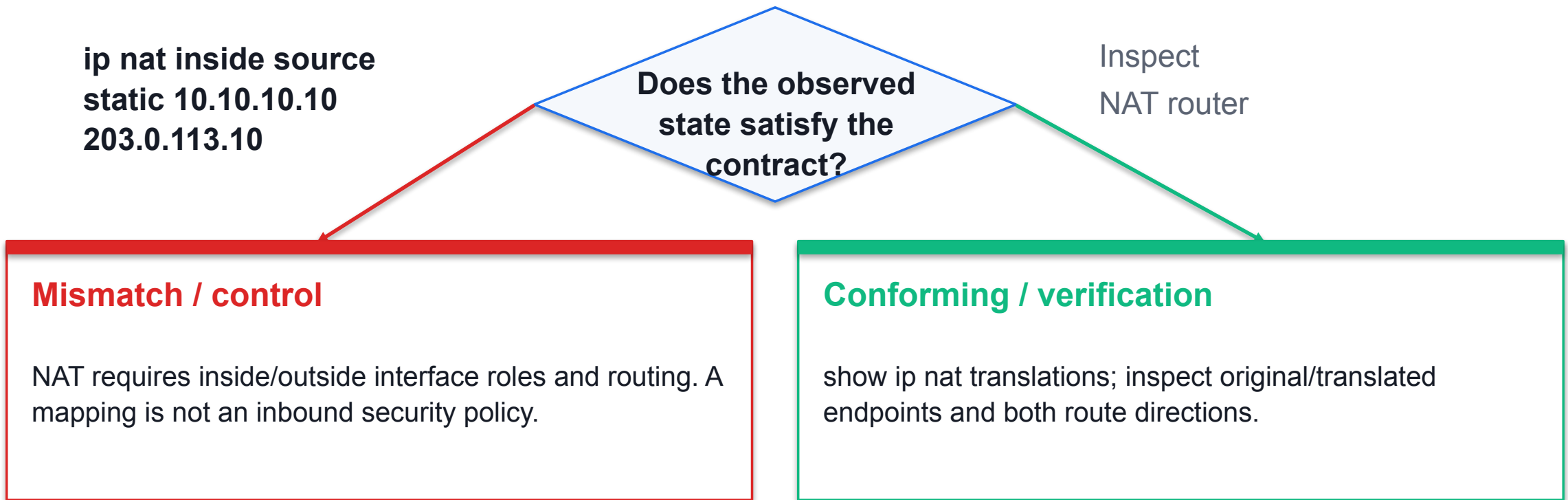
Outside reply matches reverse mapping

READBACK / RESULT

Destination returns to local

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Static NAT mapping · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

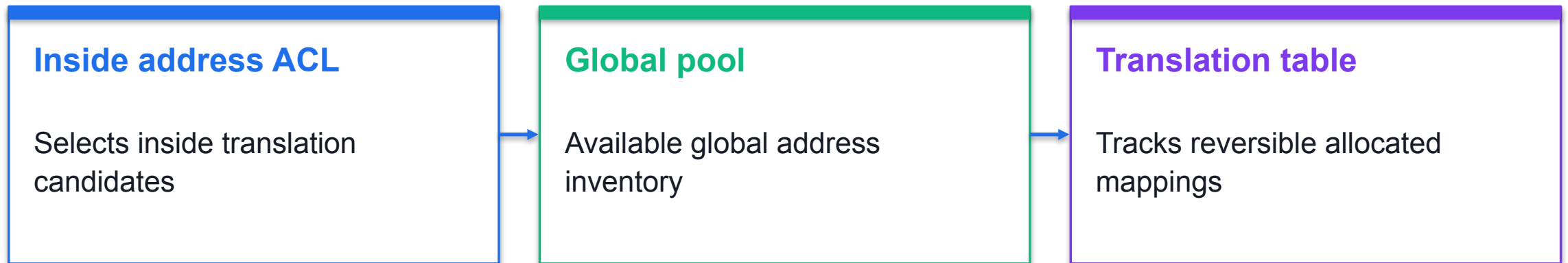
Static NAT mapping · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Inside local host -> NAT router	ip nat inside source static 10.10.10.10 203.0.113.10
Mechanism / state	Destination returns to local	Inside global 203.0.113.10
Result / boundary	show ip nat translations; inspect original/translated endpoints and both route directions.	NAT requires inside/outside interface roles and routing. A mapping is not an inbound security policy.

Bridge: Activity 8: NAT, DHCP, DNS and time

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Dynamic NAT pool allocation · boundaries

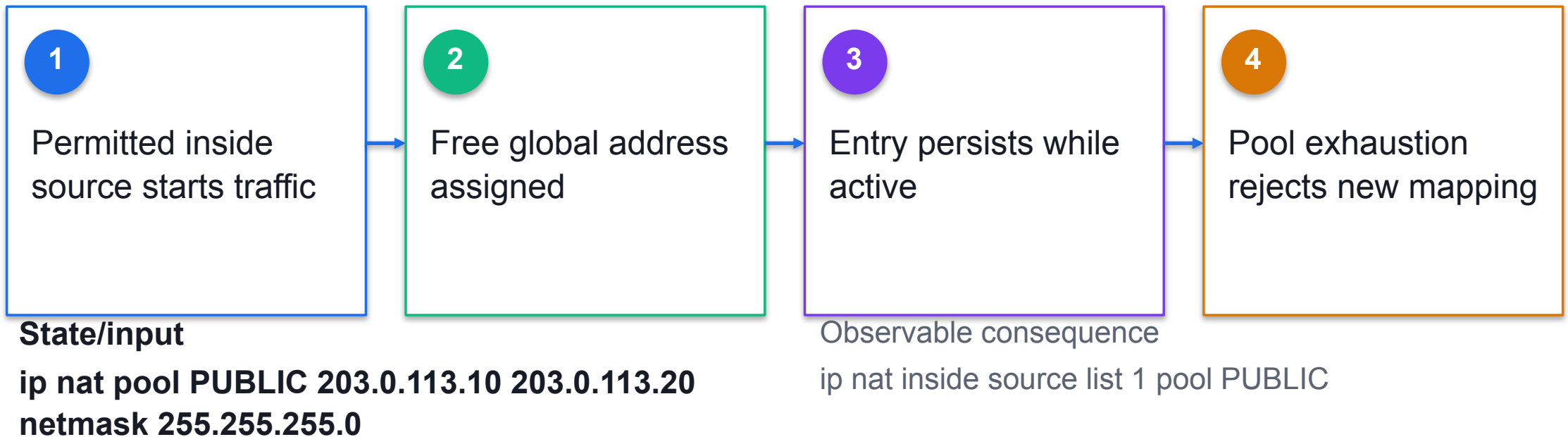


```
ip nat pool PUBLIC 203.0.113.10 203.0.113.20 netmask  
255.255.255.0  
ip nat inside source list 1 pool PUBLIC
```

Boundary to inspect
Pool has 11 addresses; more simultaneous
unique inside mappings can exhaust it

```
show ip nat statistics and translations
```

Dynamic NAT pool allocation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Dynamic NAT pool allocation · exact contract

```
ip nat pool PUBLIC 203.0.113.10 203.0.113.20 netmask  
255.255.255.0  
ip nat inside source list 1 pool PUBLIC
```

INPUT / IDENTITY

Inside address ACL · Permitted inside
source starts traffic

DECISION / STATE

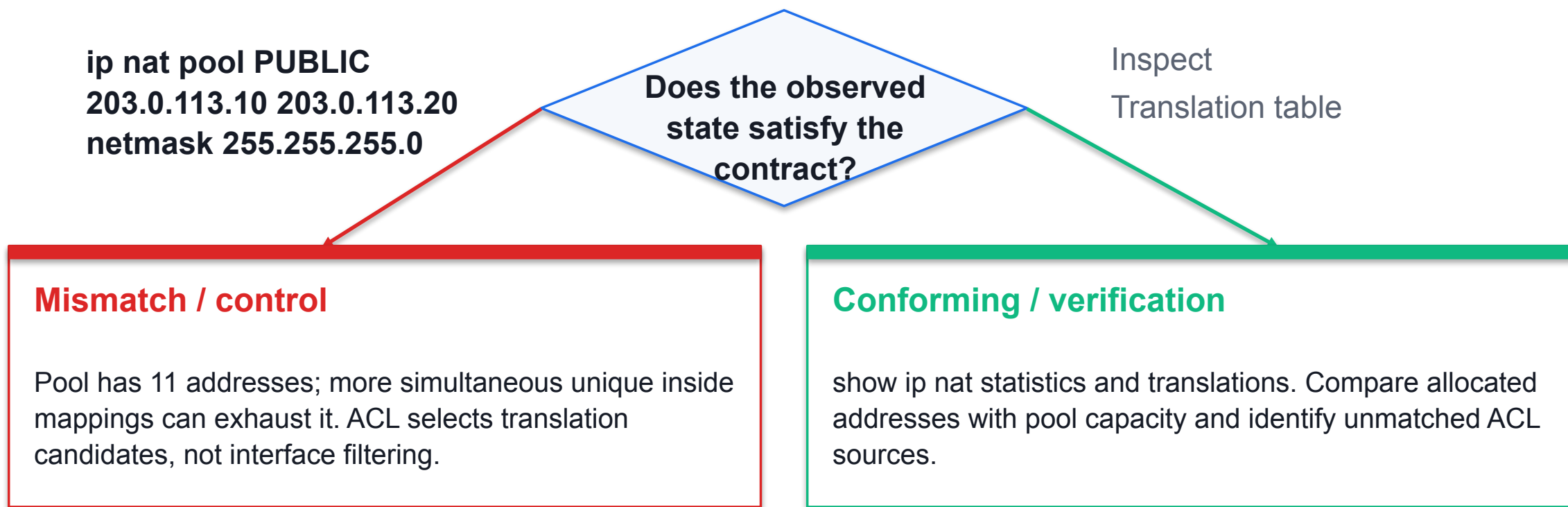
Entry persists while active

READBACK / RESULT

Pool exhaustion rejects new mapping

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Dynamic NAT pool allocation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

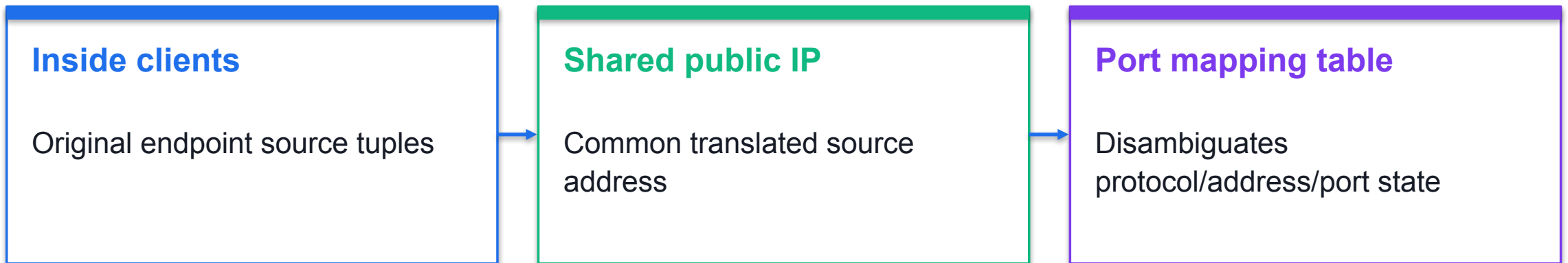
Dynamic NAT pool allocation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Inside address ACL -> Translation table	ip nat pool PUBLIC 203.0.113.10 203.0.113.20 netmask 255.255.255.0
Mechanism / state	Pool exhaustion rejects new mapping	ip nat inside source list 1 pool PUBLIC
Result / boundary	show ip nat statistics and translations. Compare allocated addresses with pool capacity and identify unmatched ACL sources.	Pool has 11 addresses; more simultaneous unique inside mappings can exhaust it. ACL selects translation candidates, not interface filtering.

Bridge: Activity 8: NAT, DHCP, DNS and time

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

PAT port disambiguation · boundaries

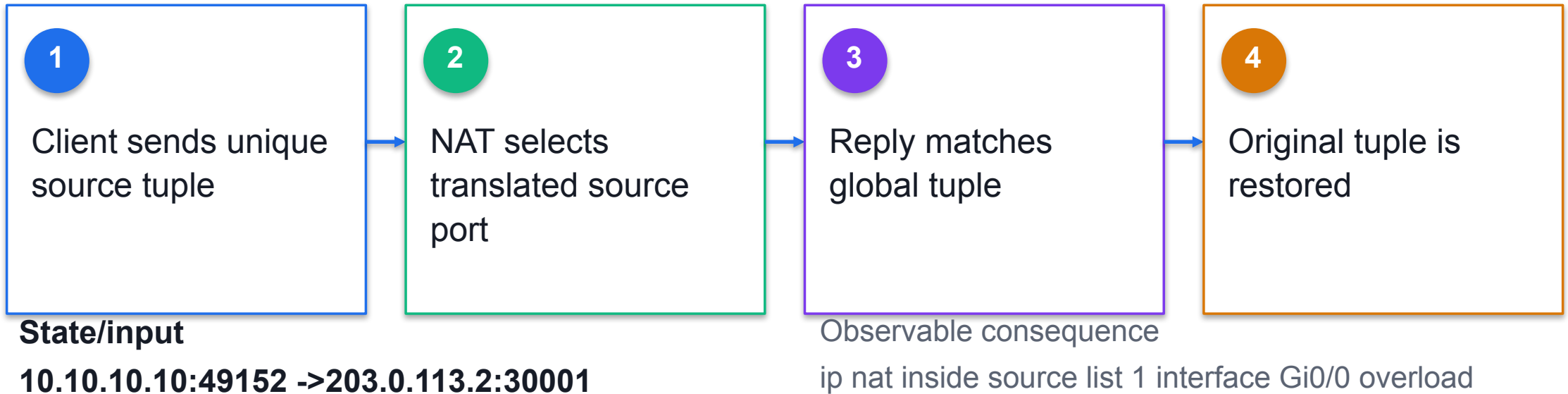


10.10.10.10:49152 -> 203.0.113.2:30001
ip nat inside source list 1 interface Gi0/0 overload

Boundary to inspect
PAT depends on tuple state and protocol behavior

show ip nat translations

PAT port disambiguation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

PAT port disambiguation · exact contract

```
10.10.10.10:49152 ->203.0.113.2:30001  
10.10.10.11:49152 ->203.0.113.2:30002  
ip nat inside source list 1 interface Gi0/0 overload
```

INPUT / IDENTITY

Inside clients · Client sends unique source tuple

DECISION / STATE

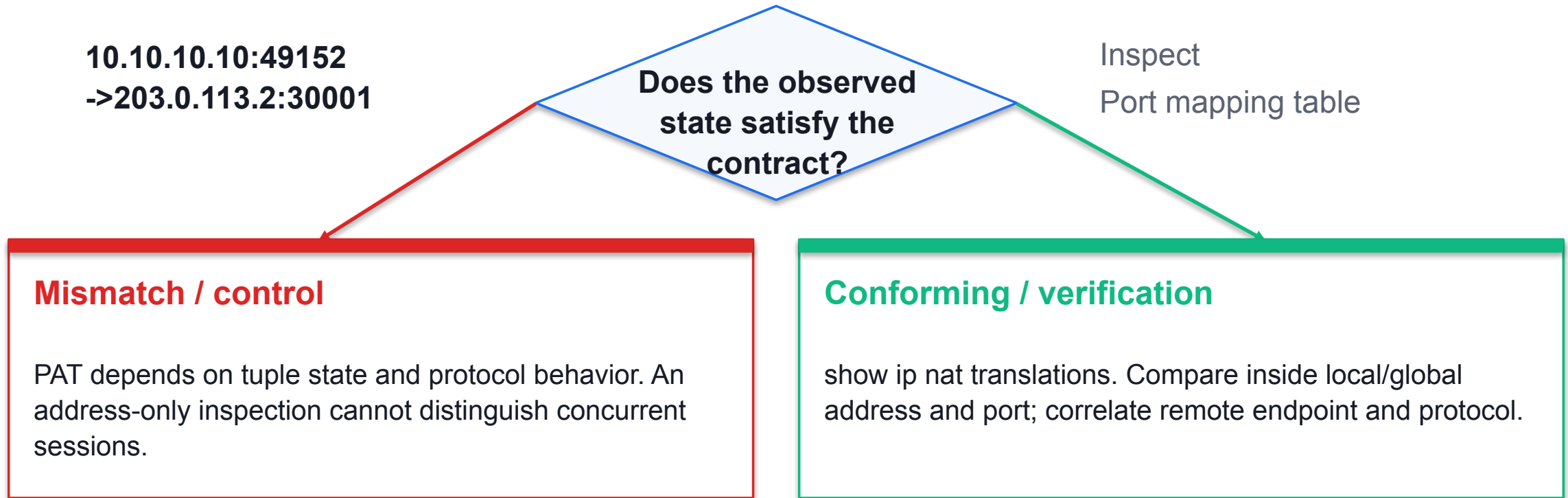
Reply matches global tuple

READBACK / RESULT

Original tuple is restored

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

PAT port disambiguation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

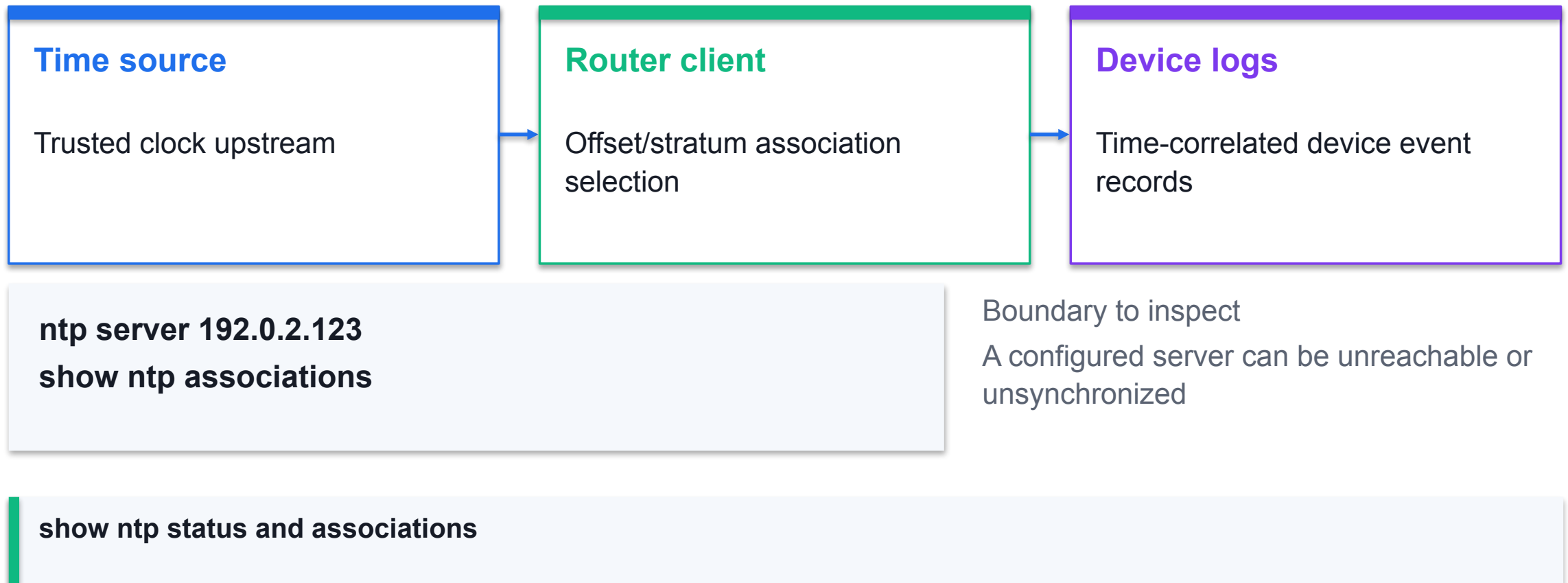
PAT port disambiguation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Inside clients -> Port mapping table	10.10.10.10:49152 ->203.0.113.2:30001
Mechanism / state	Original tuple is restored	ip nat inside source list 1 interface Gi0/0 overload
Result / boundary	show ip nat translations. Compare inside local/global address and port; correlate remote endpoint and protocol.	PAT depends on tuple state and protocol behavior. An address-only inspection cannot distinguish concurrent sessions.

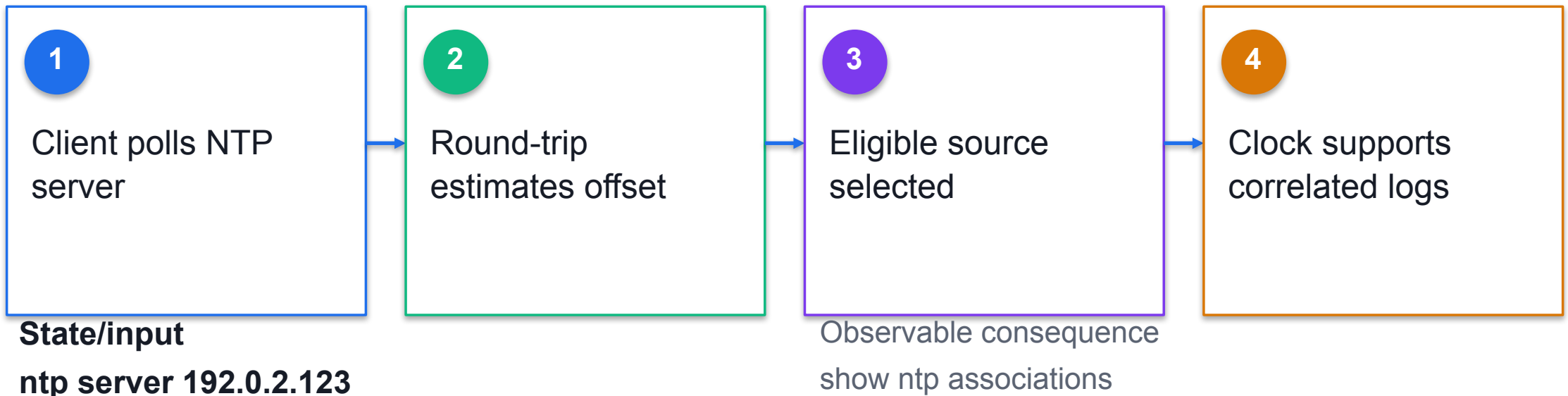
Bridge: Activity 8: NAT, DHCP, DNS and time

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

NTP synchronization hierarchy · boundaries



NTP synchronization hierarchy · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

NTP synchronization hierarchy · exact contract

ntp server 192.0.2.123
Stratum 0 reference clock
Stratum 1 directly synchronized server
Stratum 2 downstream client
show ntp associations

INPUT / IDENTITY

Time source · Client polls NTP server

DECISION / STATE

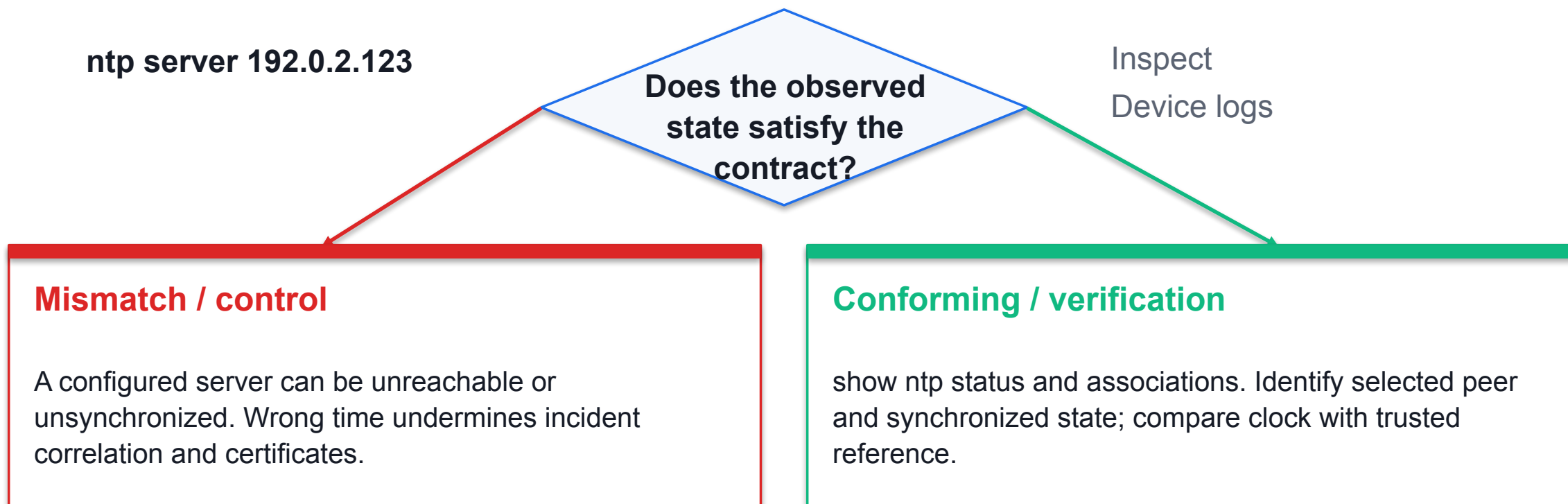
Eligible source selected

READBACK / RESULT

Clock supports correlated logs

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

NTP synchronization hierarchy · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

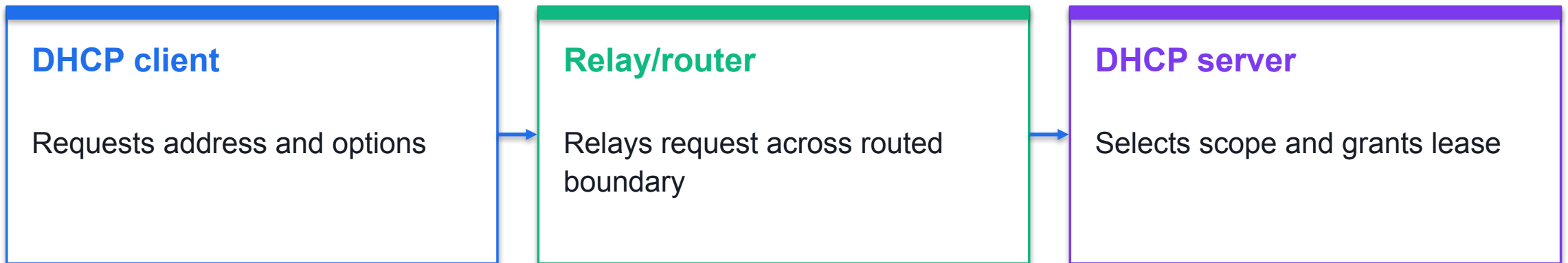
NTP synchronization hierarchy · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Time source -> Device logs	ntp server 192.0.2.123
Mechanism / state	Clock supports correlated logs	show ntp associations
Result / boundary	show ntp status and associations. Identify selected peer and synchronized state; compare clock with trusted reference.	A configured server can be unreachable or unsynchronized. Wrong time undermines incident correlation and certificates.

Bridge: Activity 8: NAT, DHCP, DNS and time

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

DHCP DORA allocation · boundaries

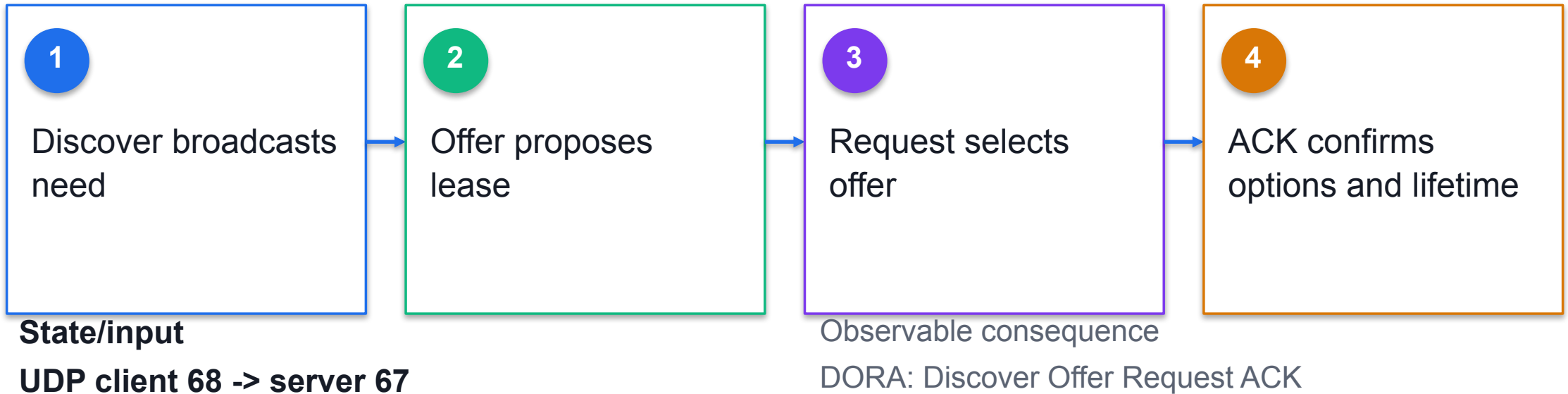


UDP client 68 -> server 67
DORA: Discover Offer Request ACK

Boundary to inspect
A lease without correct router/DNS options
yields partial connectivity

Capture DORA and inspect ipconfig/ip address evidence

DHCP DORA allocation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

DHCP DORA allocation · exact contract

UDP client 68 -> server 67

Options: mask, router, DNS, lease

DORA: Discover Offer Request ACK

INPUT / IDENTITY

DHCP client · Discover broadcasts need

DECISION / STATE

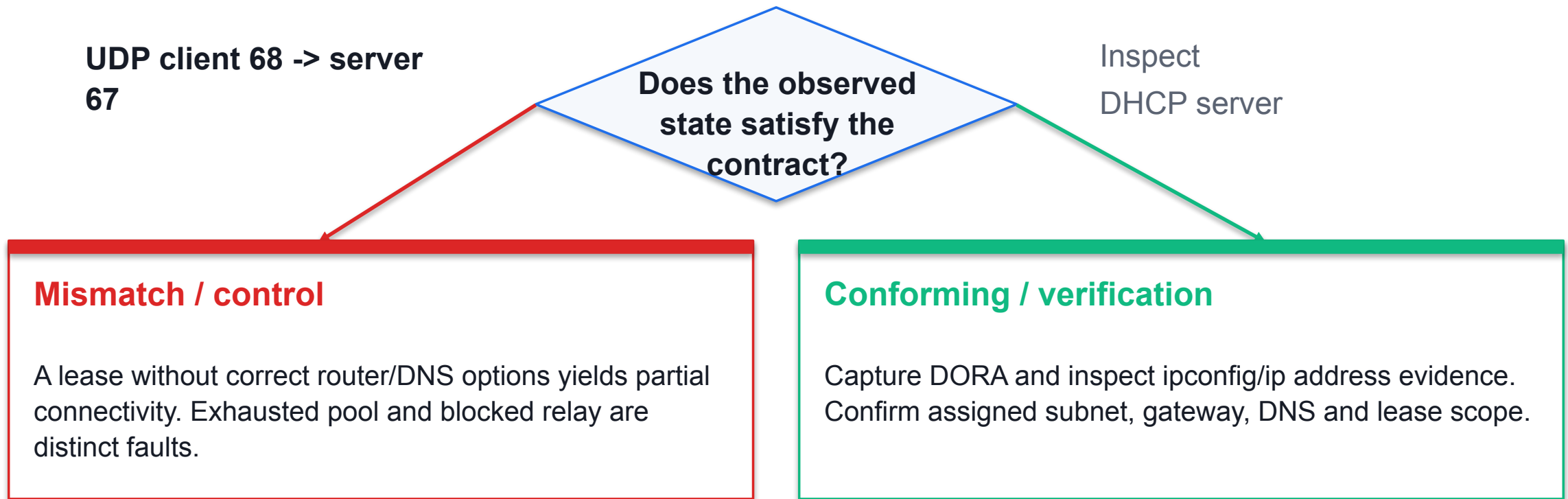
Request selects offer

READBACK / RESULT

ACK confirms options and lifetime

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

DHCP DORA allocation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

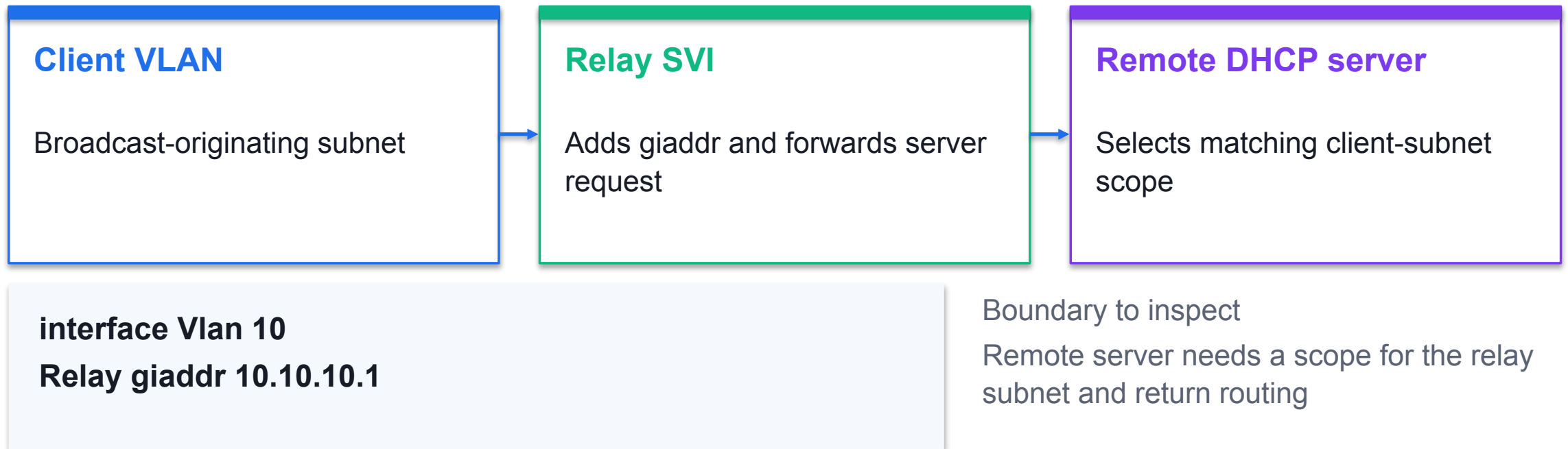
DHCP DORA allocation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	DHCP client -> DHCP server	UDP client 68 -> server 67
Mechanism / state	ACK confirms options and lifetime	DORA: Discover Offer Request ACK
Result / boundary	Capture DORA and inspect ipconfig/ip address evidence. Confirm assigned subnet, gateway, DNS and lease scope.	A lease without correct router/DNS options yields partial connectivity. Exhausted pool and blocked relay are distinct faults.

Bridge: Activity 8: NAT, DHCP, DNS and time

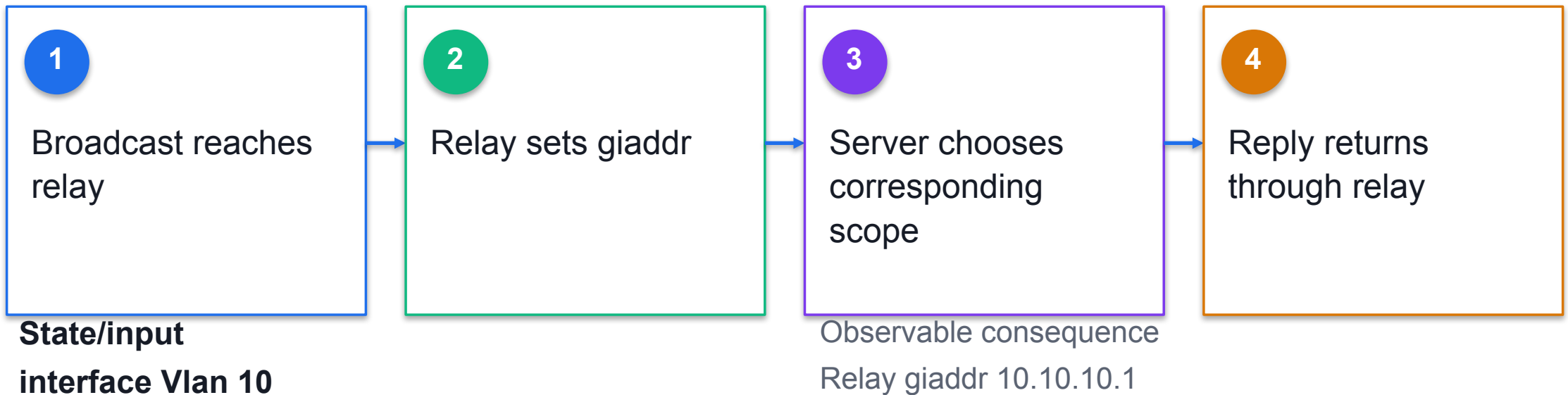
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

DHCP relay and client roles · boundaries



Check giaddr, scope selection and server response; compare client address with VLAN 10 prefix.

DHCP relay and client roles · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

DHCP relay and client roles · exact contract

```
interface Vlan 10
ip helper-address 10.20.0.50
Relay giaddr 10.10.10.1
```

INPUT / IDENTITY

Client VLAN · Broadcast reaches relay

DECISION / STATE

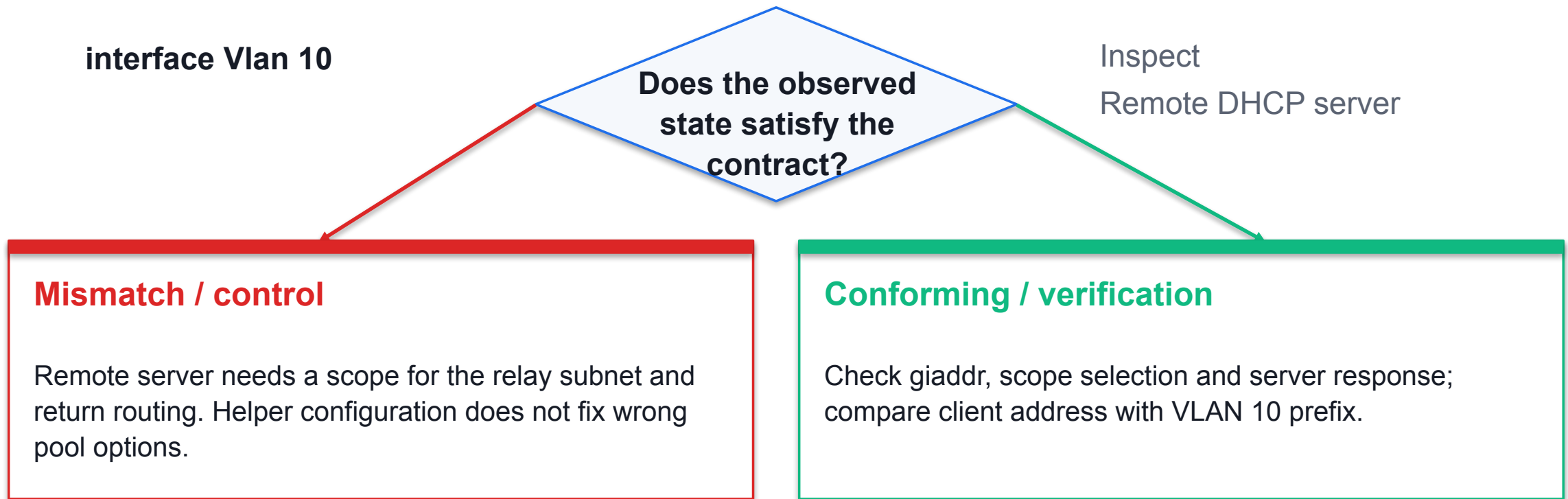
Server chooses corresponding scope

READBACK / RESULT

Reply returns through relay

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

DHCP relay and client roles · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

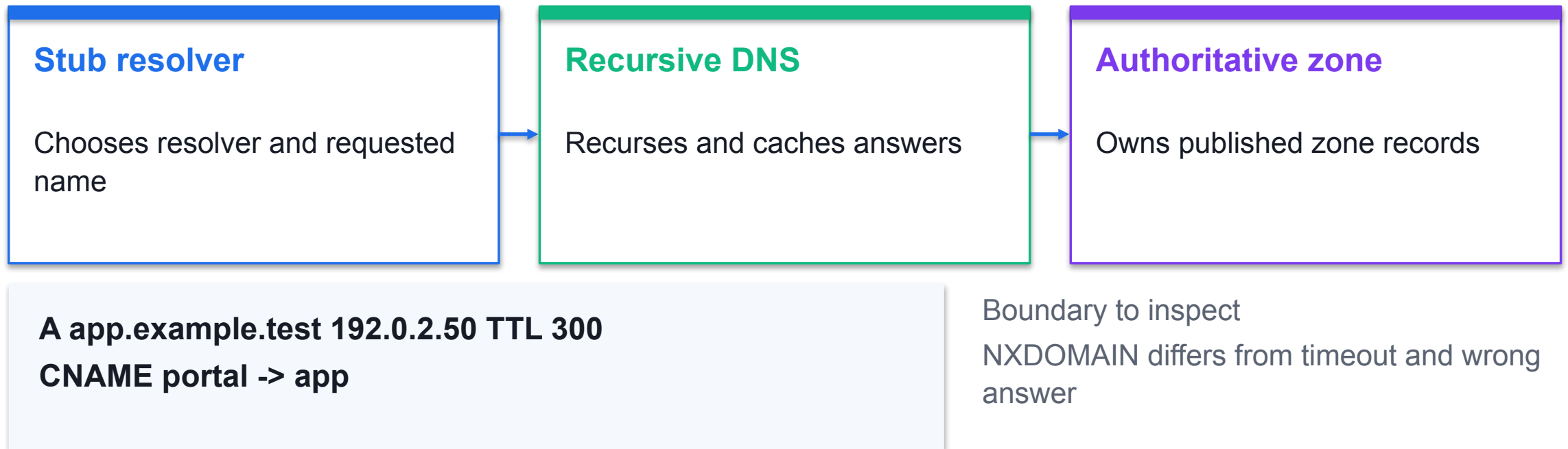
DHCP relay and client roles · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Client VLAN -> Remote DHCP server	interface Vlan 10
Mechanism / state	Reply returns through relay	Relay giaddr 10.10.10.1
Result / boundary	Check giaddr, scope selection and server response; compare client address with VLAN 10 prefix.	Remote server needs a scope for the relay subnet and return routing. Helper configuration does not fix wrong pool options.

Bridge: Activity 8: NAT, DHCP, DNS and time

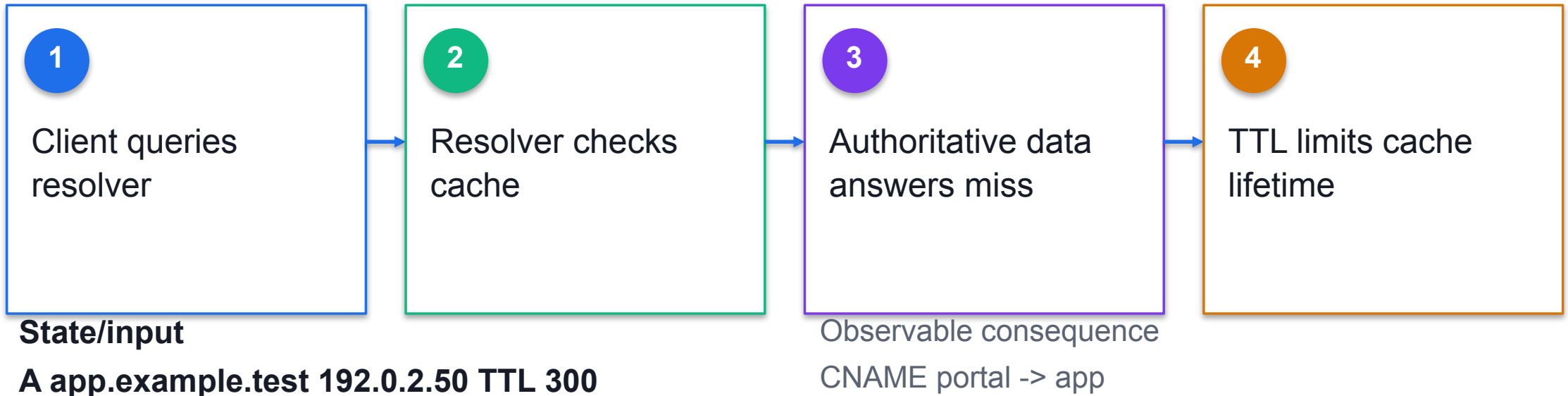
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

DNS record and cache evidence · boundaries



nslookup/dig result: server, status, answer and TTL

DNS record and cache evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

DNS record and cache evidence · exact contract

A app.example.test 192.0.2.50 TTL 300
AAAA app.example.test2001:db8::50
CNAME portal -> app

INPUT / IDENTITY

Stub resolver · Client queries resolver

DECISION / STATE

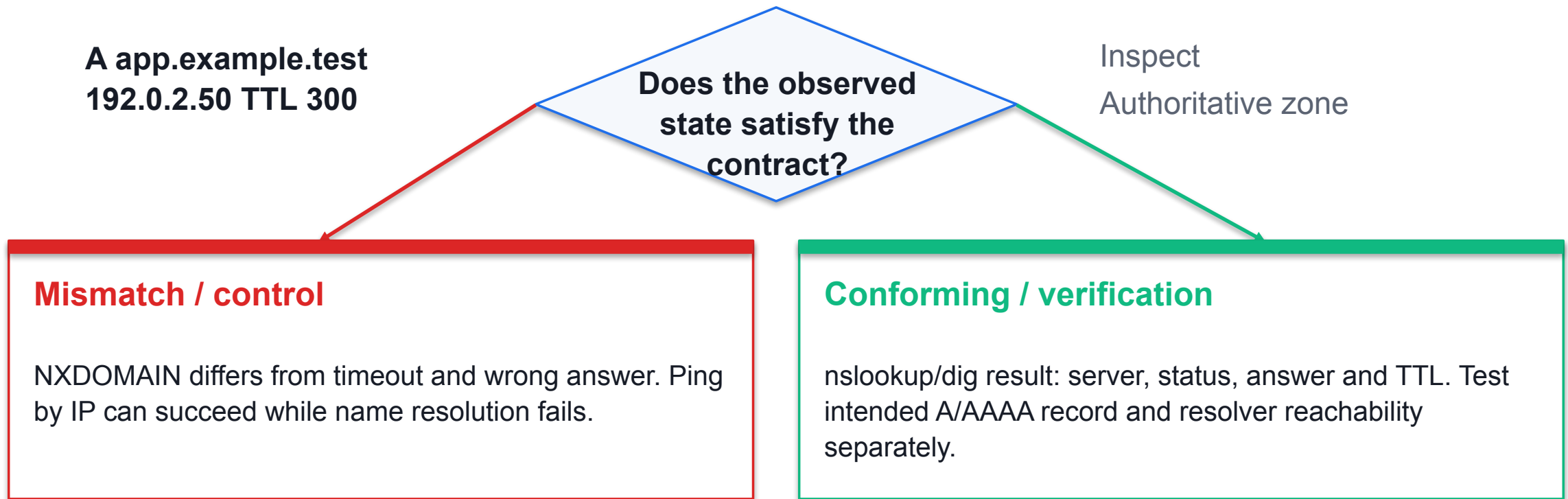
Authoritative data answers miss

READBACK / RESULT

TTL limits cache lifetime

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

DNS record and cache evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

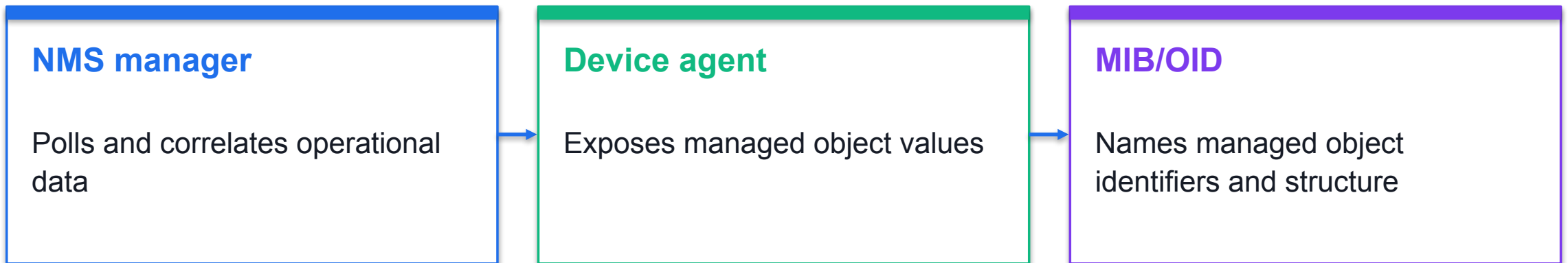
DNS record and cache evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Stub resolver -> Authoritative zone	A app.example.test 192.0.2.50 TTL 300
Mechanism / state	TTL limits cache lifetime	CNAME portal -> app
Result / boundary	nslookup/dig result: server, status, answer and TTL. Test intended A/AAAA record and resolver reachability separately.	NXDOMAIN differs from timeout and wrong answer. Ping by IP can succeed while name resolution fails.

Bridge: Activity 8: NAT, DHCP, DNS and time

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

SNMP manager and agent state · boundaries

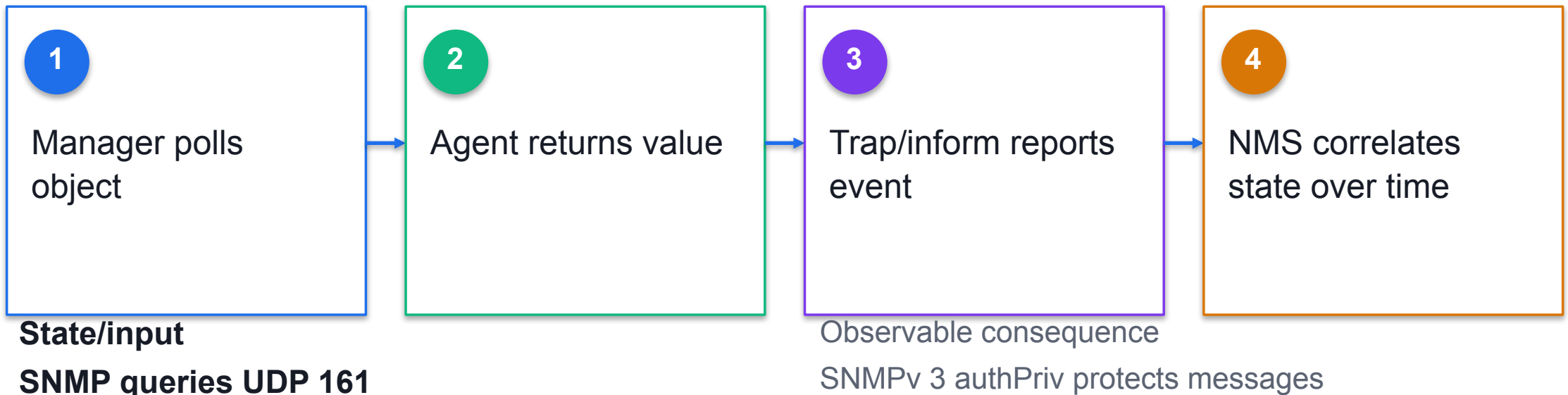


SNMP queries UDP 161
SNMPv 3 authPriv protects messages

Boundary to inspect
SNMPv 1/v 2 c community strings do not
provide modern encryption

Correlate OID value, timestamp, agent reachability and local show output

SNMP manager and agent state · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

SNMP manager and agent state · exact contract

SNMP queries UDP 161
Notifications UDP 162
ifOperStatus: up(1), down(2)
SNMPv 3 authPriv protects messages

INPUT / IDENTITY

NMS manager · Manager polls object

DECISION / STATE

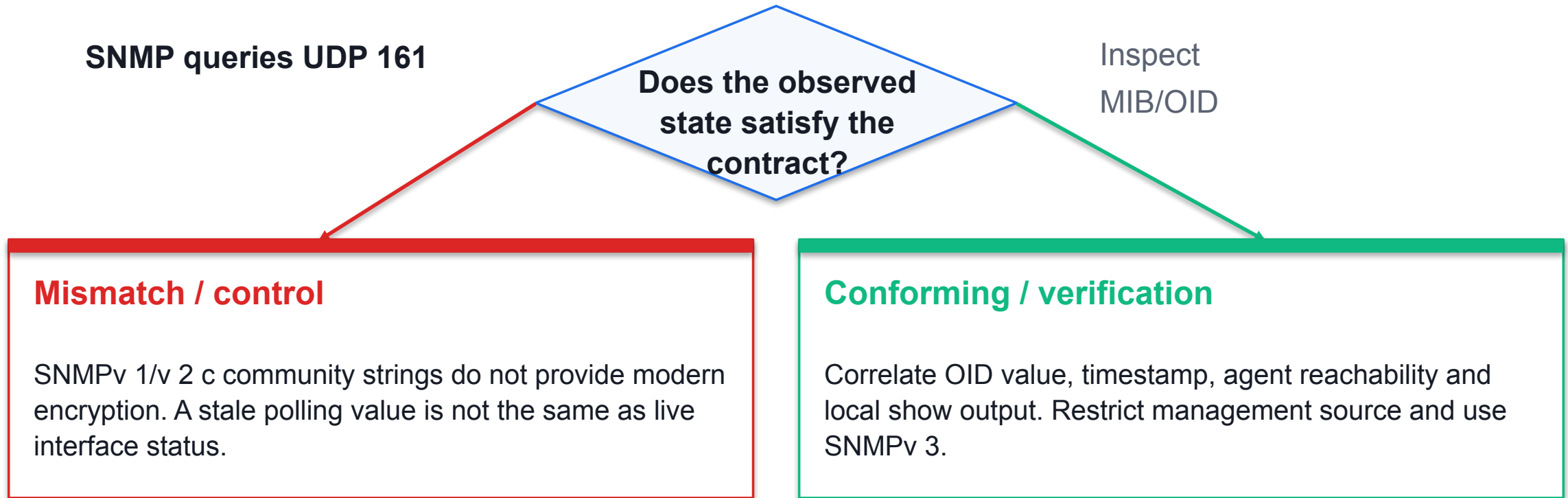
Trap/inform reports event

READBACK / RESULT

NMS correlates state over time

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

SNMP manager and agent state · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

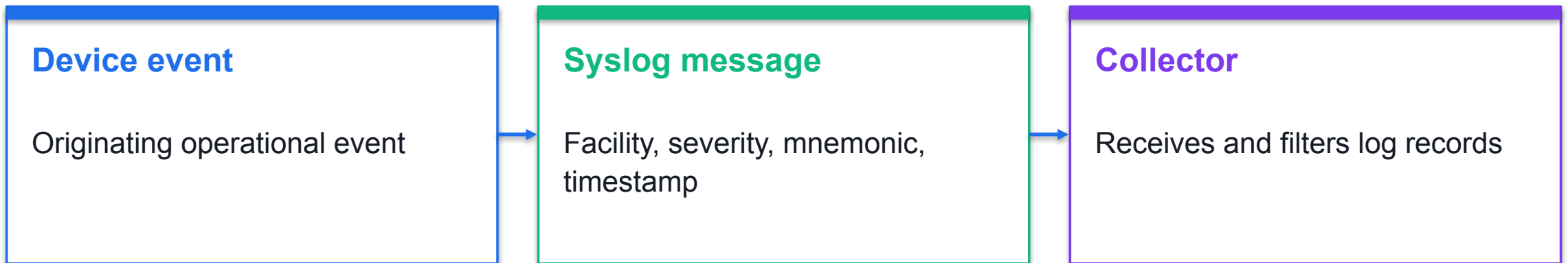
SNMP manager and agent state · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	NMS manager -> MIB/OID	SNMP queries UDP 161
Mechanism / state	NMS correlates state over time	SNMPv 3 authPriv protects messages
Result / boundary	Correlate OID value, timestamp, agent reachability and local show output. Restrict management source and use SNMPv 3.	SNMPv 1/v 2 c community strings do not provide modern encryption. A stale polling value is not the same as live interface status.

Bridge: Activity 8: NAT, DHCP, DNS and time

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Syslog severity and facility · boundaries

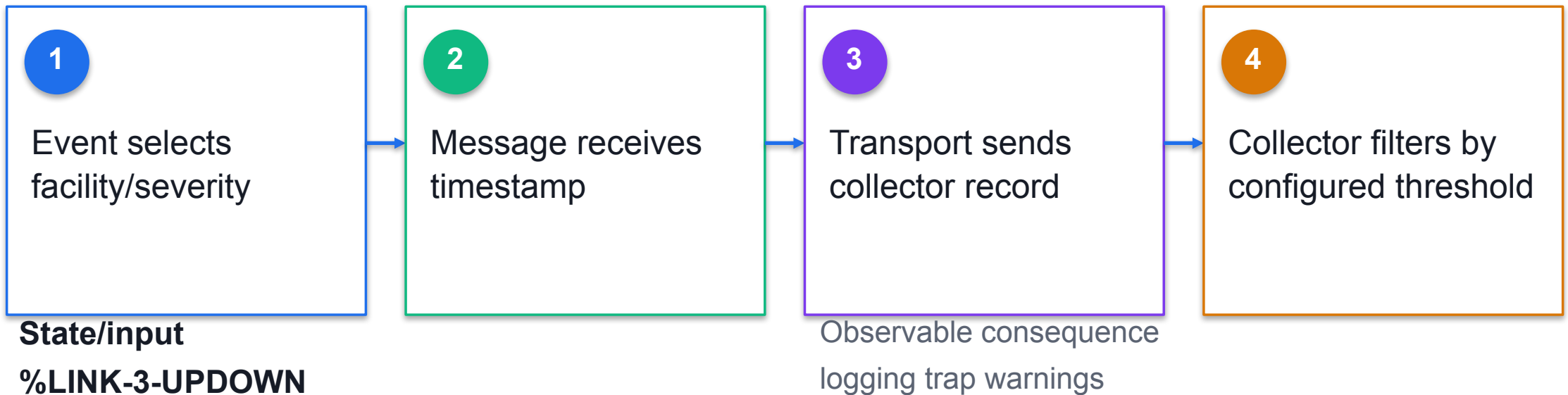


%LINK-3-UPDOWN
logging trap warnings

Boundary to inspect
Threshold 4 includes 0-4, not 5-7

Identify facility, severity and mnemonic

Syslog severity and facility · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Syslog severity and facility · exact contract

%LINK-3-UPDOWN

Severity 0 emergencies ->7 debugging

logging host 192.0.2.60

logging trap warnings

INPUT / IDENTITY

Device event · Event selects
facility/severity

DECISION / STATE

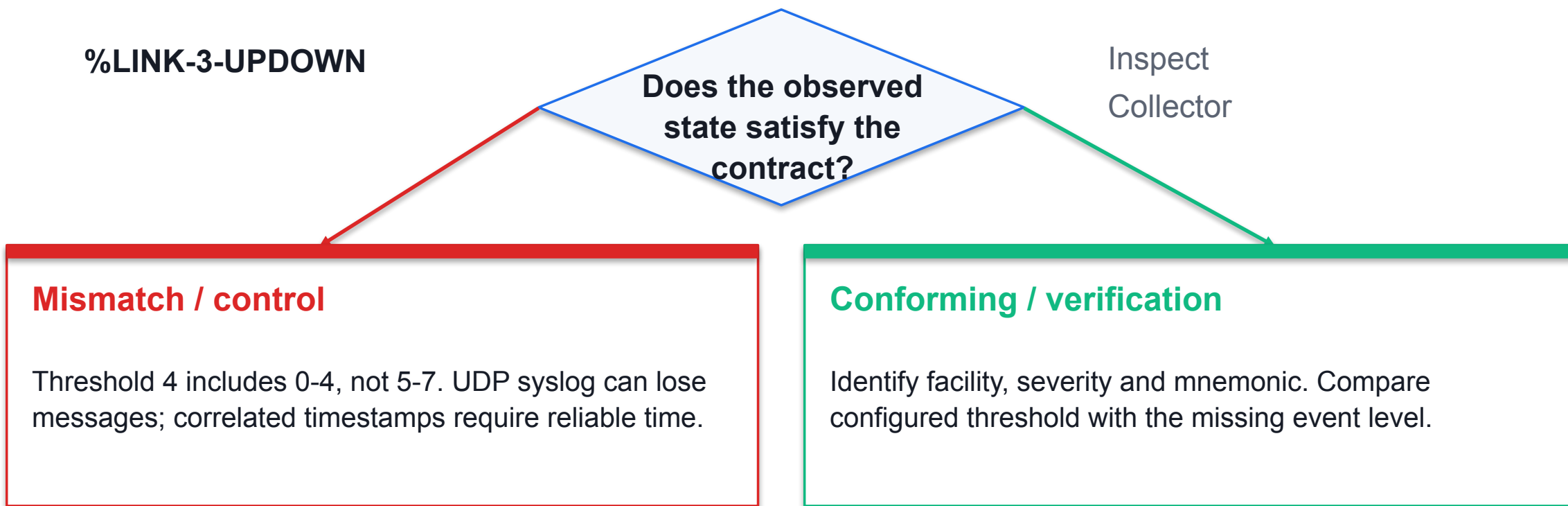
Transport sends collector record

READBACK / RESULT

Collector filters by configured threshold

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Syslog severity and facility · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

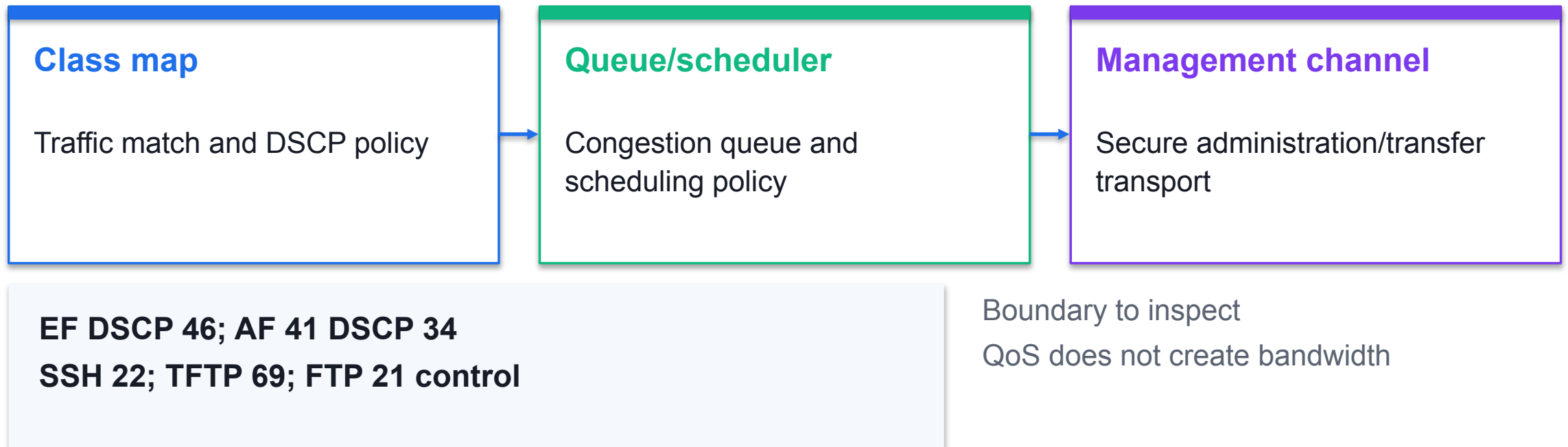
Syslog severity and facility · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Device event -> Collector	%LINK-3-UPDOWN
Mechanism / state	Collector filters by configured threshold	logging trap warnings
Result / boundary	Identify facility, severity and mnemonic. Compare configured threshold with the missing event level.	Threshold 4 includes 0-4, not 5-7. UDP syslog can lose messages; correlated timestamps require reliable time.

Bridge: Activity 8: NAT, DHCP, DNS and time

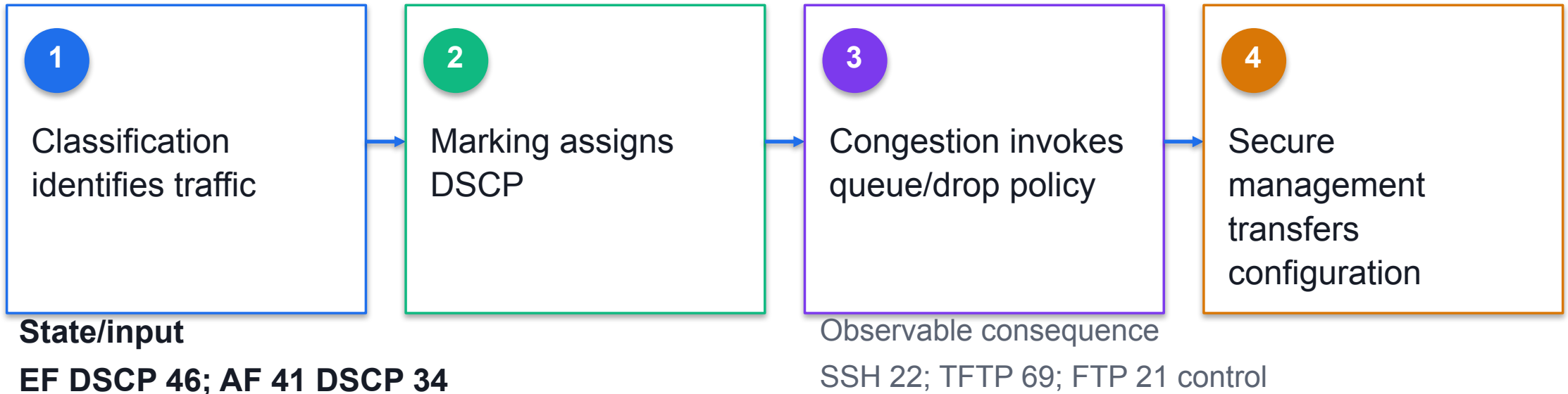
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

QoS classification and management transfer · boundaries



Compare offered rate with policy counters, queue drops and latency

QoS classification and management transfer · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

QoS classification and management transfer · exact contract

EF DSCP 46; AF 41 DSCP 34
Policing can drop/remark excess
Shaping buffers/delays excess
SSH 22; TFTP 69; FTP 21 control

INPUT / IDENTITY

Class map · Classification identifies traffic

DECISION / STATE

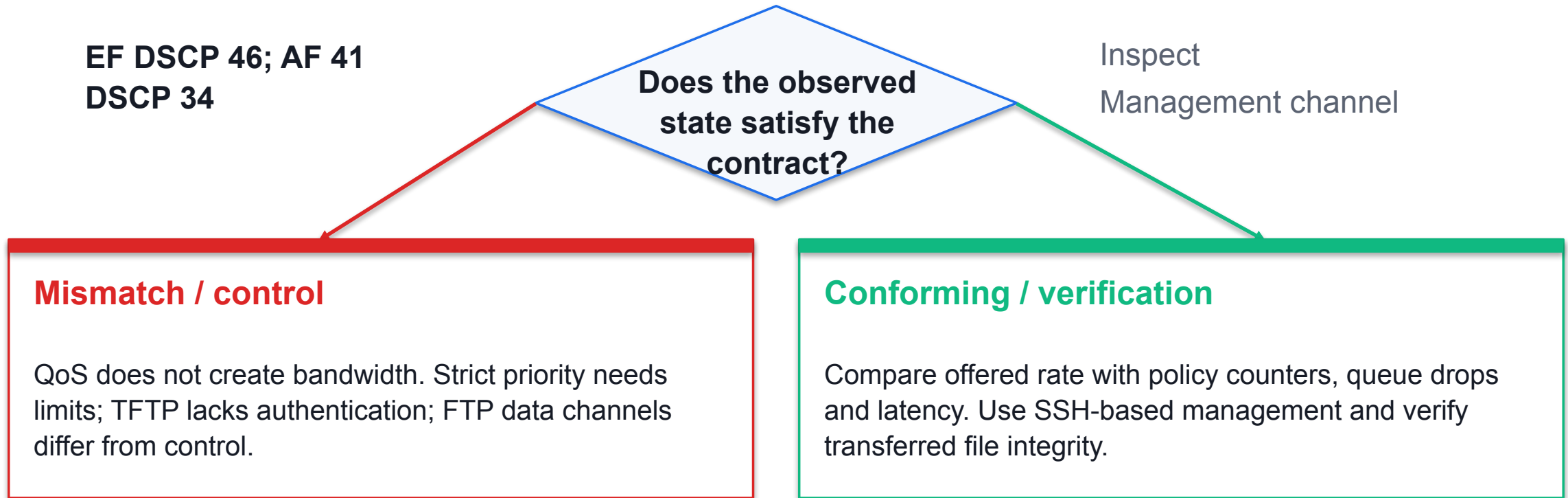
Congestion invokes queue/drop policy

READBACK / RESULT

Secure management transfers configuration

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

QoS classification and management transfer · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

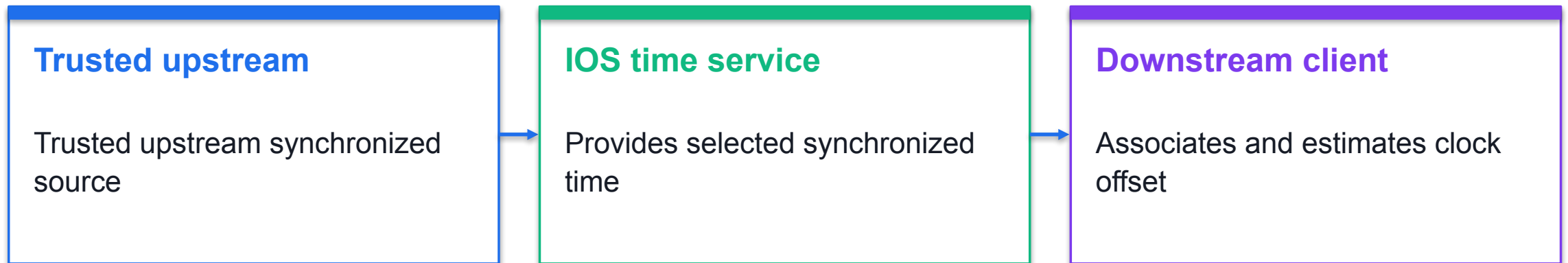
QoS classification and management transfer · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Class map -> Management channel	EF DSCP 46; AF 41 DSCP 34
Mechanism / state	Secure management transfers configuration	SSH 22; TFTP 69; FTP 21 control
Result / boundary	Compare offered rate with policy counters, queue drops and latency. Use SSH-based management and verify transferred file integrity.	QoS does not create bandwidth. Strict priority needs limits; TFTP lacks authentication; FTP data channels differ from control.

Bridge: Activity 8: NAT, DHCP, DNS and time

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

NTP server and client association · boundaries



ntp server 192.0.2.123

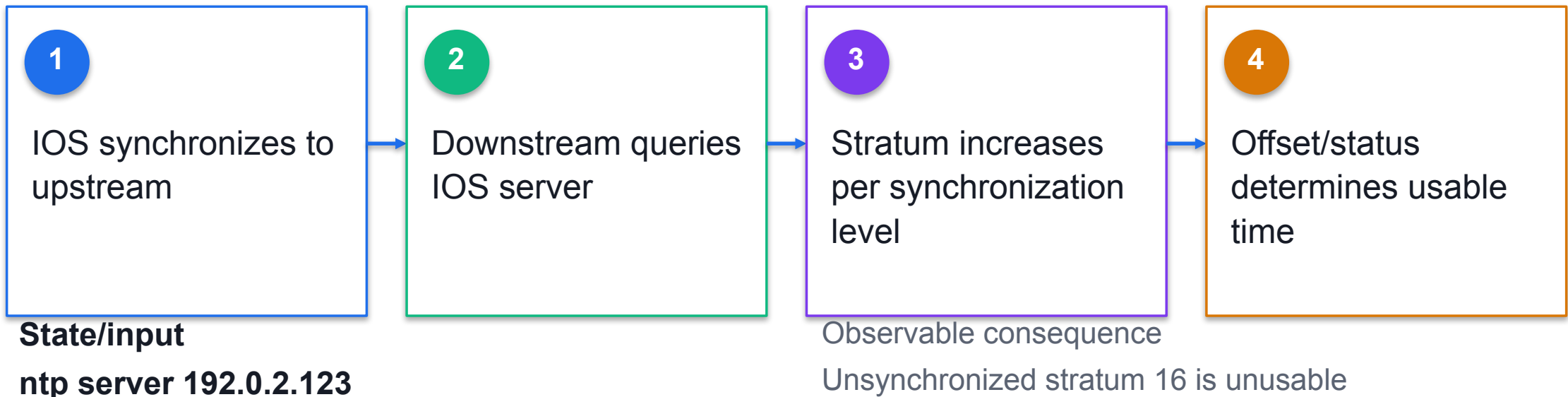
Unsynchronized stratum 16 is unusable

Boundary to inspect

A server's configured source may be unreachable; serving an unsynchronized clock is not evidence of trusted time

Read selected peer, synchronization state, stratum and clock on both server/client; compare log timestamps with trusted reference.

NTP server and client association · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

NTP server and client association · exact contract

```
ntp server 192.0.2.123
show ntp status
Downstream ntp server 10.10.10.1
Unsynchronized stratum 16 is unusable
```

INPUT / IDENTITY

Trusted upstream · IOS synchronizes to upstream

DECISION / STATE

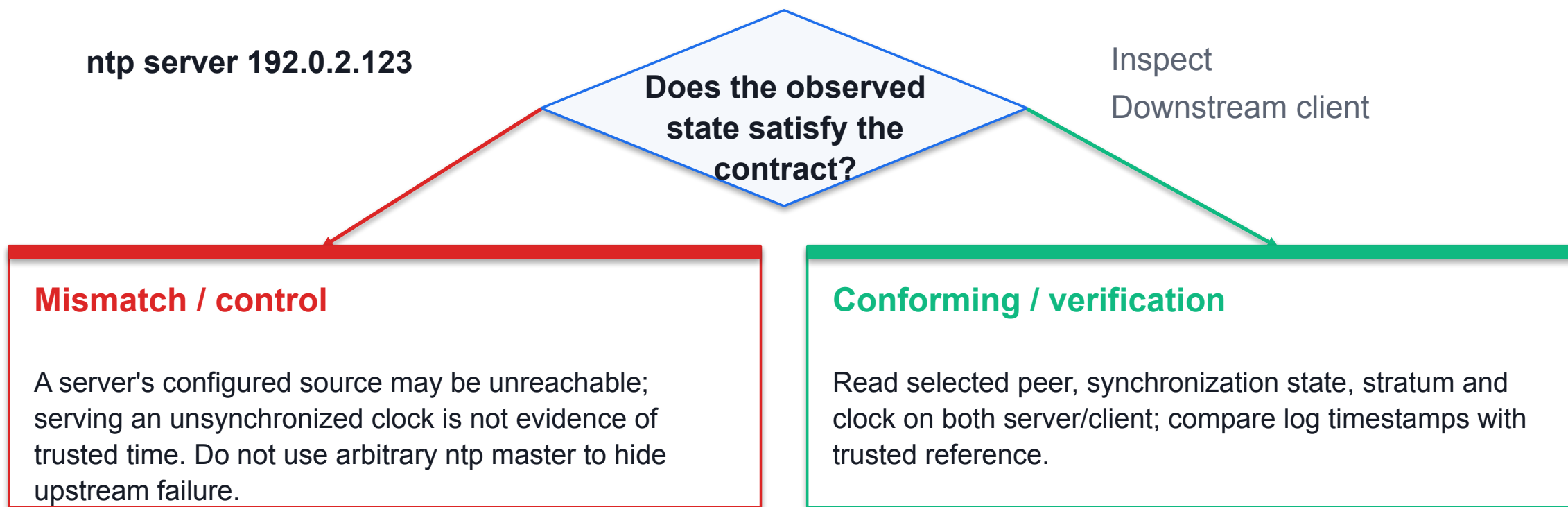
Stratum increases per synchronization level

READBACK / RESULT

Offset/status determines usable time

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

NTP server and client association · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

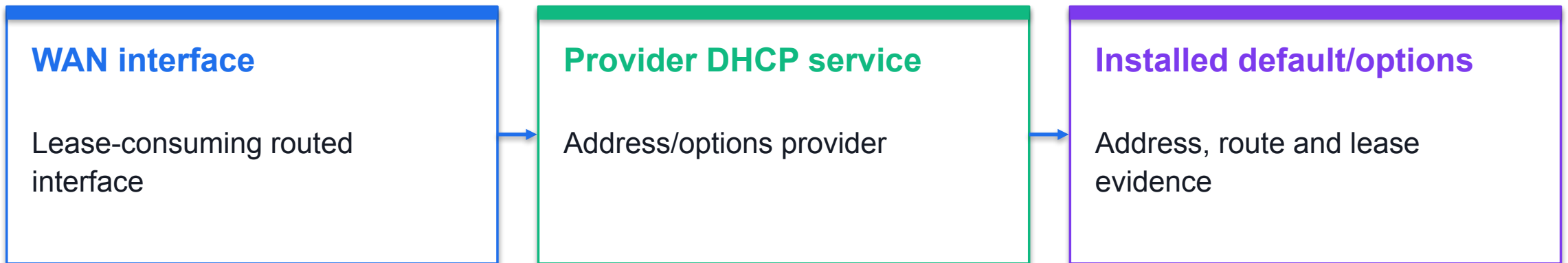
NTP server and client association · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Trusted upstream -> Downstream client	ntp server 192.0.2.123
Mechanism / state	Offset/status determines usable time	Unsynchronized stratum 16 is unusable
Result / boundary	Read selected peer, synchronization state, stratum and clock on both server/client; compare log timestamps with trusted reference.	A server's configured source may be unreachable; serving an unsynchronized clock is not evidence of trusted time. Do not use arbitrary ntp master to hide upstream failure.

Bridge: Activity 8: NAT, DHCP, DNS and time

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IOS DHCP client interface state · boundaries



interface Gi0/0
show ip interface brief

Boundary to inspect
DHCP client role differs from server pool
and relay

Record interface leased address, lease lifetime and received options; inspect route table and provider next-hop reachability.

IOS DHCP client interface state · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IOS DHCP client interface state · exact contract

```
interface Gi0/0
ip address dhcp
no shutdown
show dhcp lease
show ip interface brief
```

INPUT / IDENTITY

WAN interface · Client interface
requests lease

DECISION / STATE

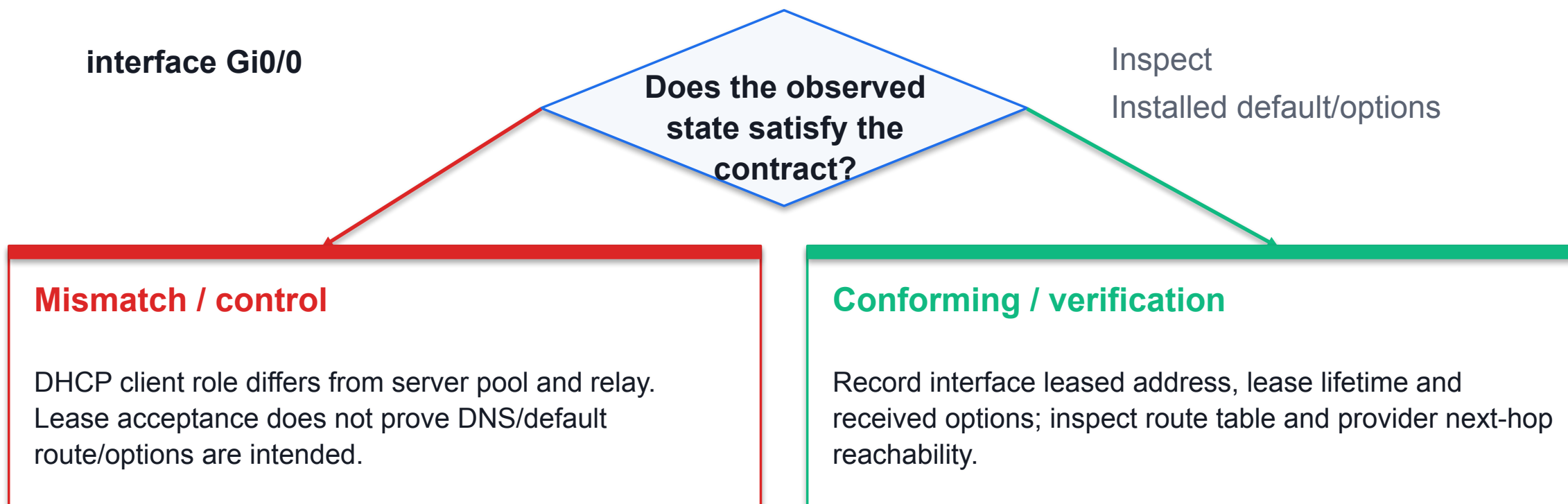
IOS installs leased address

READBACK / RESULT

Lease renews before expiry

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IOS DHCP client interface state · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

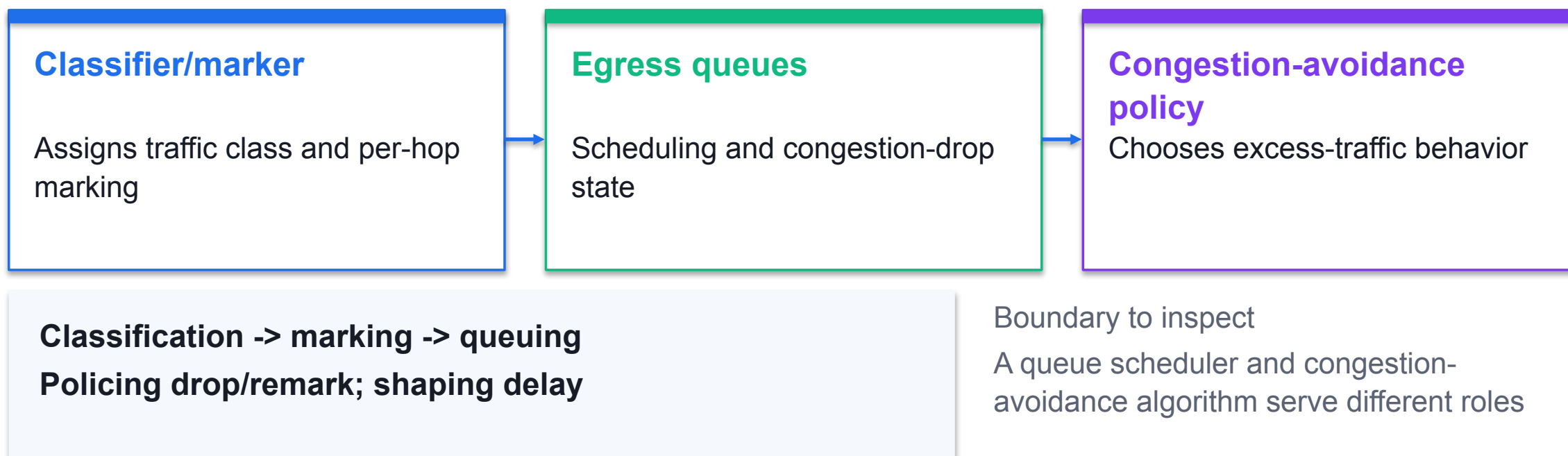
IOS DHCP client interface state · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	WAN interface -> Installed default/options	interface Gi0/0
Mechanism / state	Lease renews before expiry	show ip interface brief
Result / boundary	Record interface leased address, lease lifetime and received options; inspect route table and provider next-hop reachability.	DHCP client role differs from server pool and relay. Lease acceptance does not prove DNS/default route/options are intended.

Bridge: Activity 8: NAT, DHCP, DNS and time

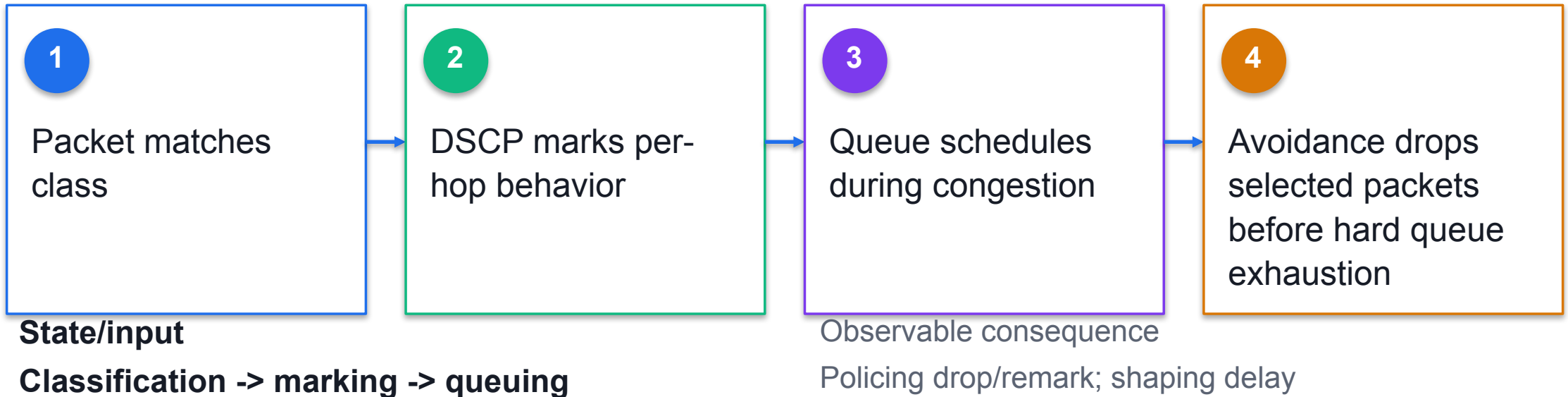
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

QoS congestion decisions · boundaries



Inspect class matches, offered/dropped rates, queue depth and latency; compare baseline with congested test rather than asserting QoS creates bandwidth.

QoS congestion decisions · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

QoS congestion decisions · exact contract

Classification -> marking -> queuing

Tail drop: queue full

WRED: probabilistic early discard

Policing drop/remark; shaping delay

INPUT / IDENTITY

Classifier/marker · Packet matches class

DECISION / STATE

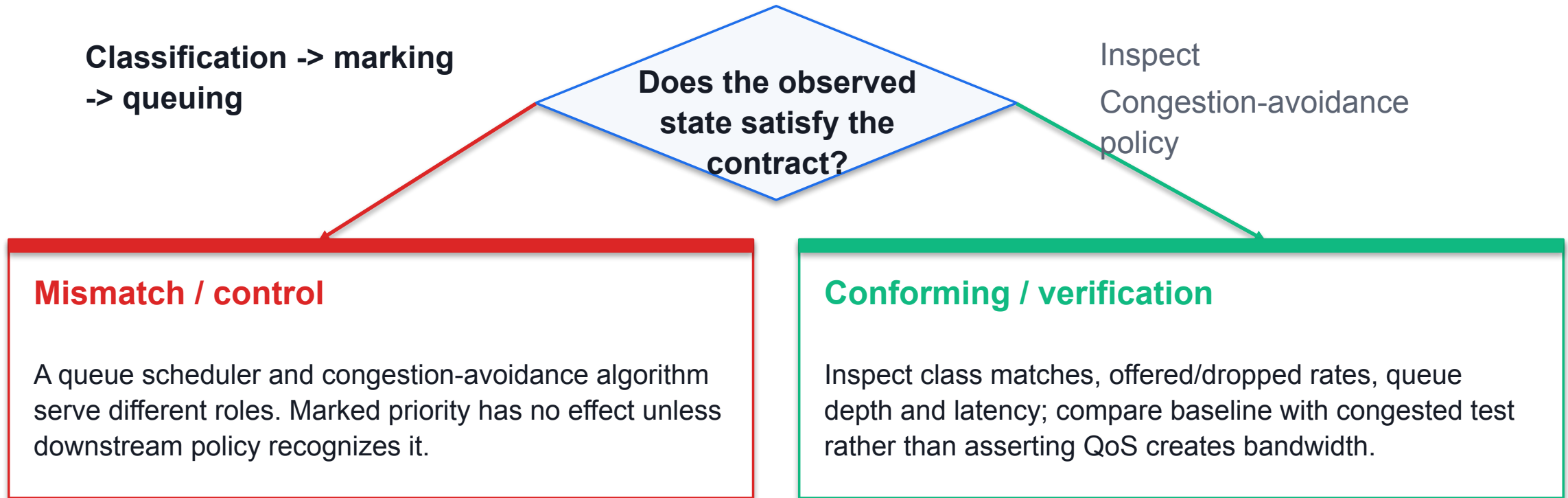
Queue schedules during congestion

READBACK / RESULT

Avoidance drops selected packets before hard queue exhaustion

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

QoS congestion decisions · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

QoS congestion decisions · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Classifier/marker -> Congestion-avoidance policy	Classification -> marking -> queuing
Mechanism / state	Avoidance drops selected packets before hard queue exhaustion	Policing drop/remark; shaping delay
Result / boundary	Inspect class matches, offered/dropped rates, queue depth and latency; compare baseline with congested test rather than asserting QoS creates bandwidth.	A queue scheduler and congestion-avoidance algorithm serve different roles. Marked priority has no effect unless downstream policy recognizes it.

Bridge: Activity 8: NAT, DHCP, DNS and time

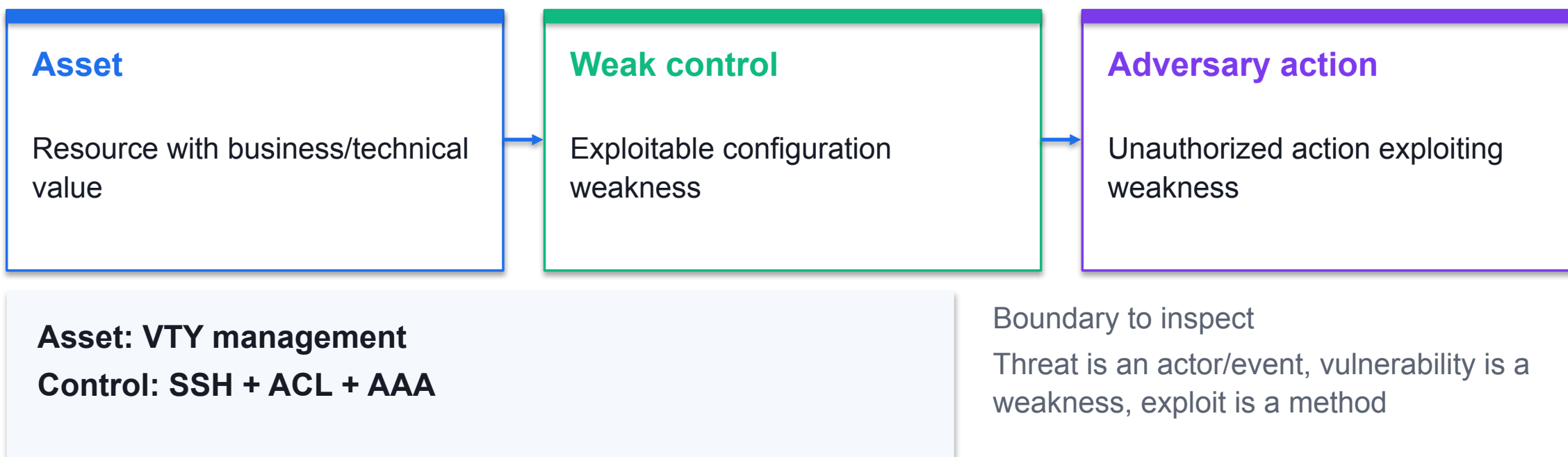
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Security Fundamentals

Security Fundamentals

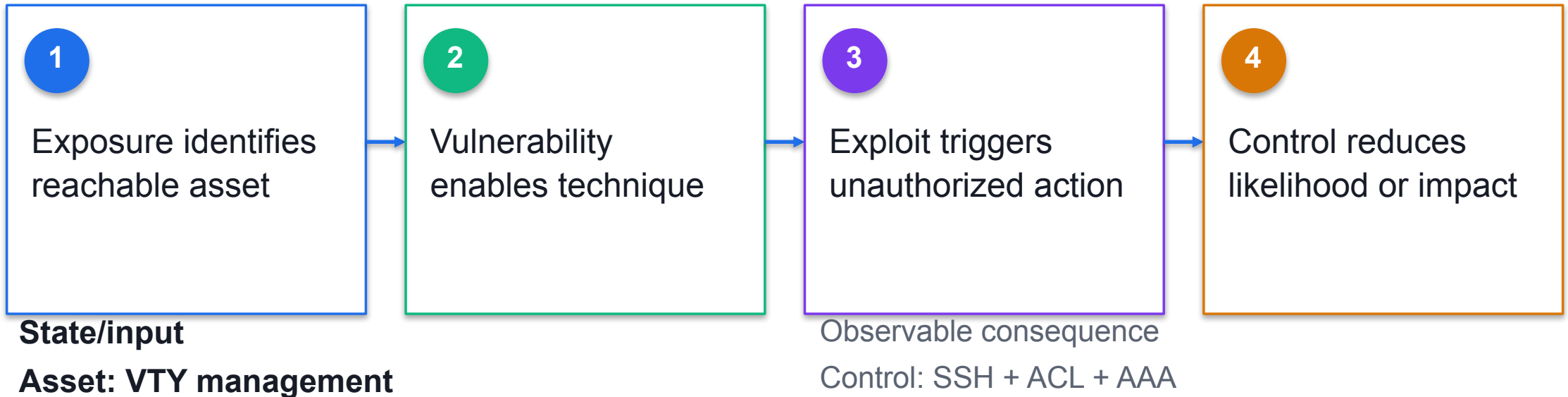
Named boundaries -> runtime trace -> exact contract -> failure/control
-> acceptance evidence

Threat-vulnerability-exploit chain · boundaries



Show observed connection, account/audit event and relevant control; distinguish risk hypothesis from confirmed evidence.

Threat-vulnerability-exploit chain · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Threat-vulnerability-exploit chain · exact contract

Asset: VTY management

Weakness: Telnet + weak password

Exploit: credential interception

Control: SSH + ACL + AAA

INPUT / IDENTITY

Asset · Exposure identifies reachable asset

DECISION / STATE

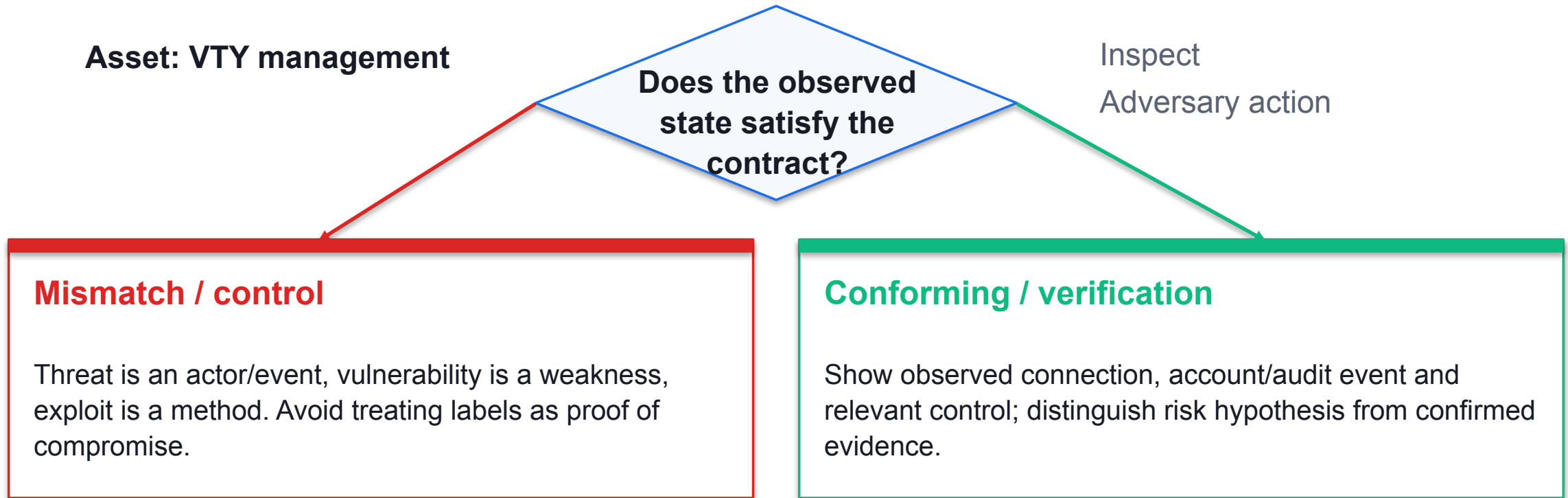
Exploit triggers unauthorized action

READBACK / RESULT

Control reduces likelihood or impact

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Threat-vulnerability-exploit chain · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

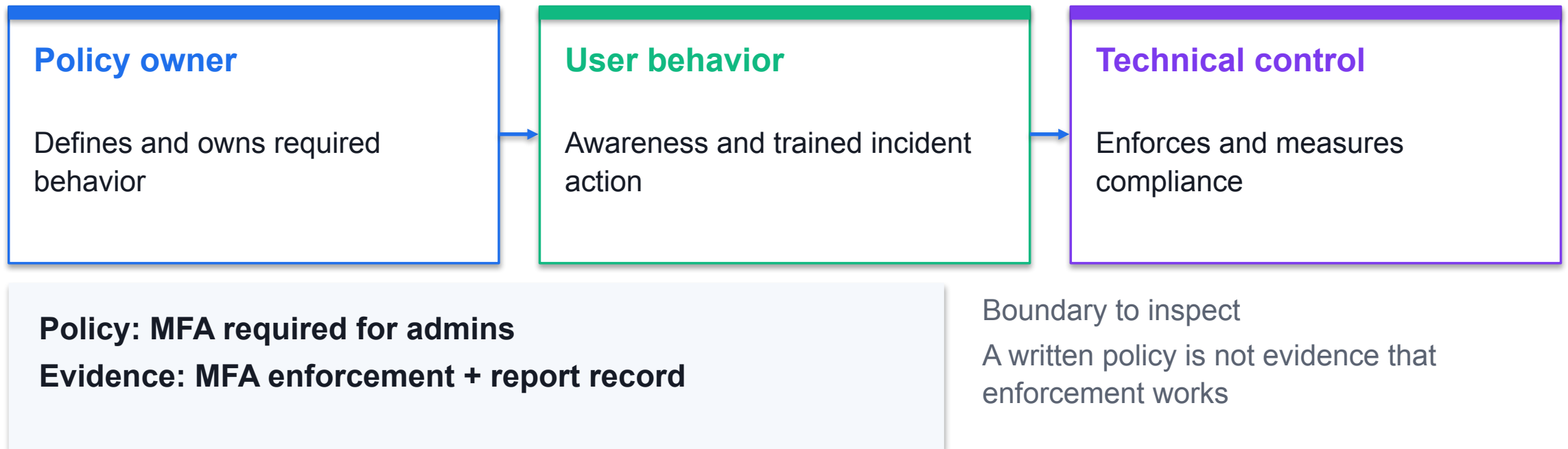
Threat-vulnerability-exploit chain · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Asset -> Adversary action	Asset: VTY management
Mechanism / state	Control reduces likelihood or impact	Control: SSH + ACL + AAA
Result / boundary	Show observed connection, account/audit event and relevant control; distinguish risk hypothesis from confirmed evidence.	Threat is an actor/event, vulnerability is a weakness, exploit is a method. Avoid treating labels as proof of compromise.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Security program evidence · boundaries



Collect policy, control setting, exercise output and remediation owner; track exceptions explicitly.

Security program evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Security program evidence · exact contract

Policy: MFA required for admins

Training: phishing report procedure

Evidence: MFA enforcement + report record

INPUT / IDENTITY

Policy owner · Policy defines requirement

DECISION / STATE

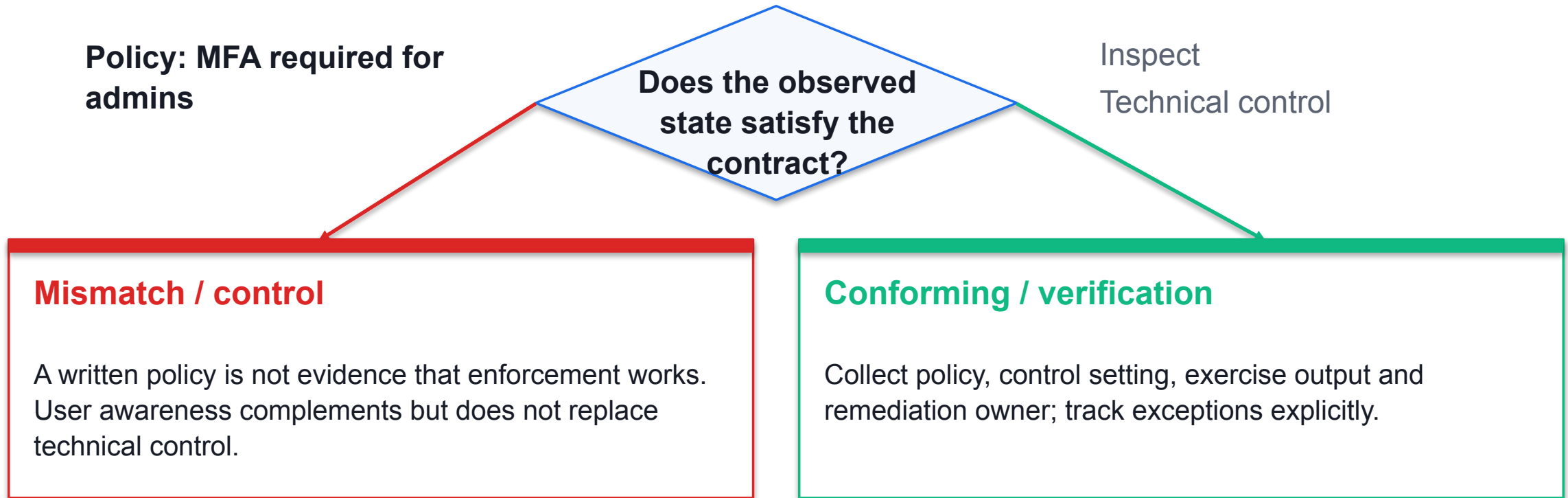
Training rehearses action

READBACK / RESULT

Audit tests whether controls operate

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Security program evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

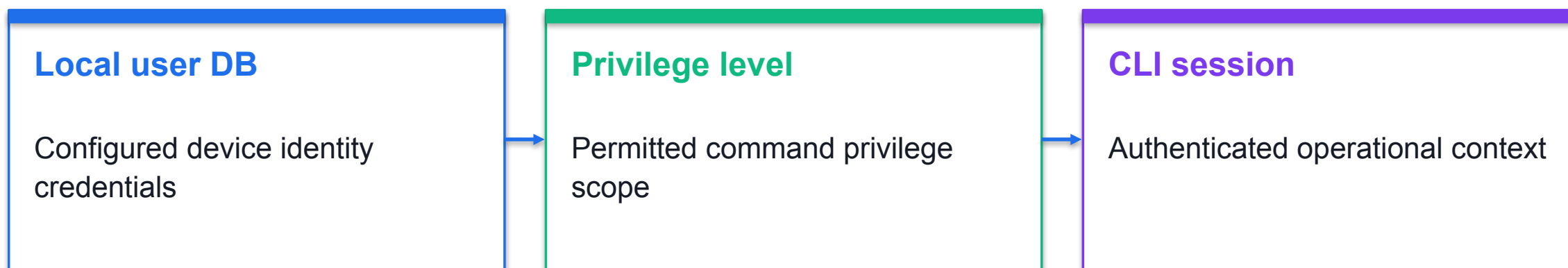
Security program evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Policy owner -> Technical control	Policy: MFA required for admins
Mechanism / state	Audit tests whether controls operate	Evidence: MFA enforcement + report record
Result / boundary	Collect policy, control setting, exercise output and remediation owner; track exceptions explicitly.	A written policy is not evidence that enforcement works. User awareness complements but does not replace technical control.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Local user and enable secrets · boundaries

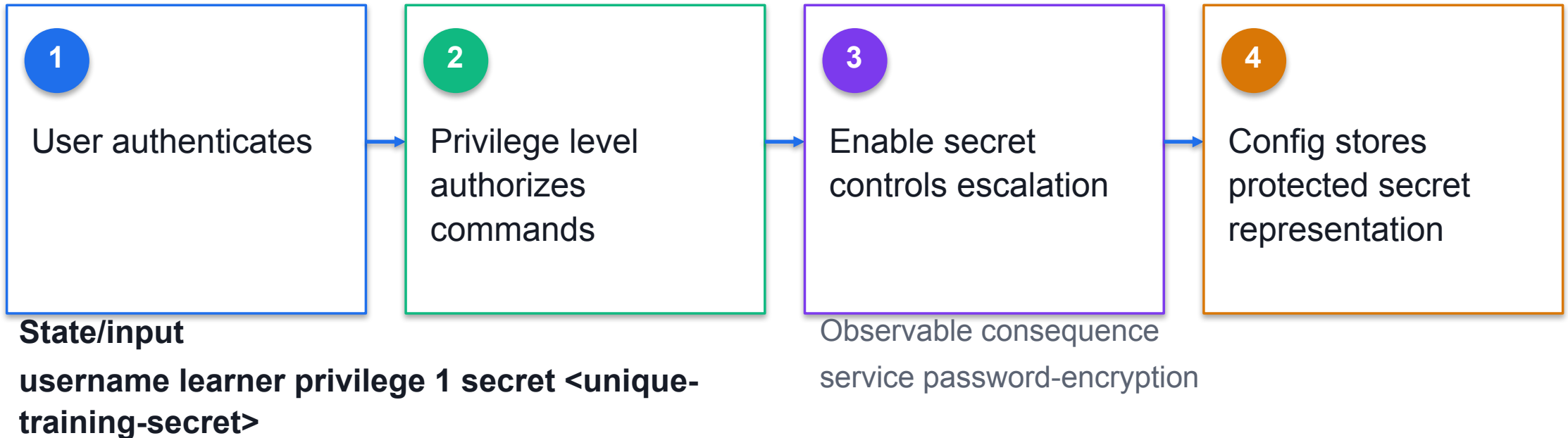


```
username learner privilege 1 secret <unique-training-secret>  
service password-encryption
```

Boundary to inspect
service password-encryption obfuscates
some passwords and is not strong
cryptographic protection

```
show running-config filtered for user/enable settings; test allowed commands without publishing secret values.
```


Local user and enable secrets · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Local user and enable secrets · exact contract

username learner privilege 1 secret <unique-training-secret>

enable secret <unique-training-secret>

service password-encryption

INPUT / IDENTITY

Local user DB · User authenticates

DECISION / STATE

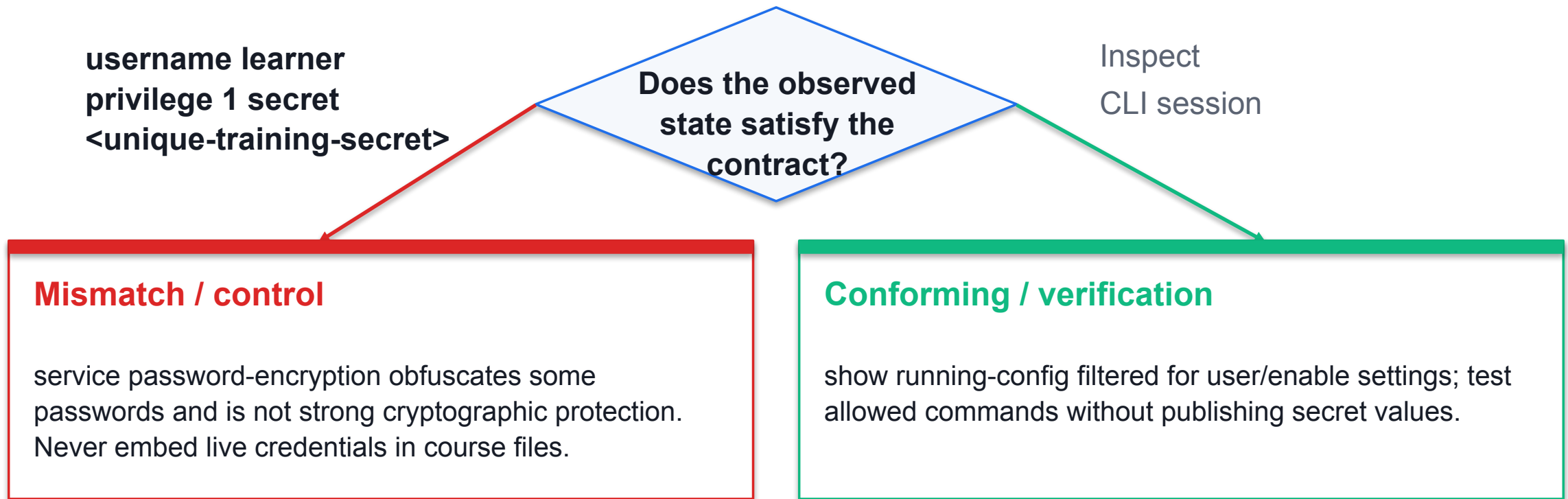
Enable secret controls escalation

READBACK / RESULT

Config stores protected secret representation

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Local user and enable secrets · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

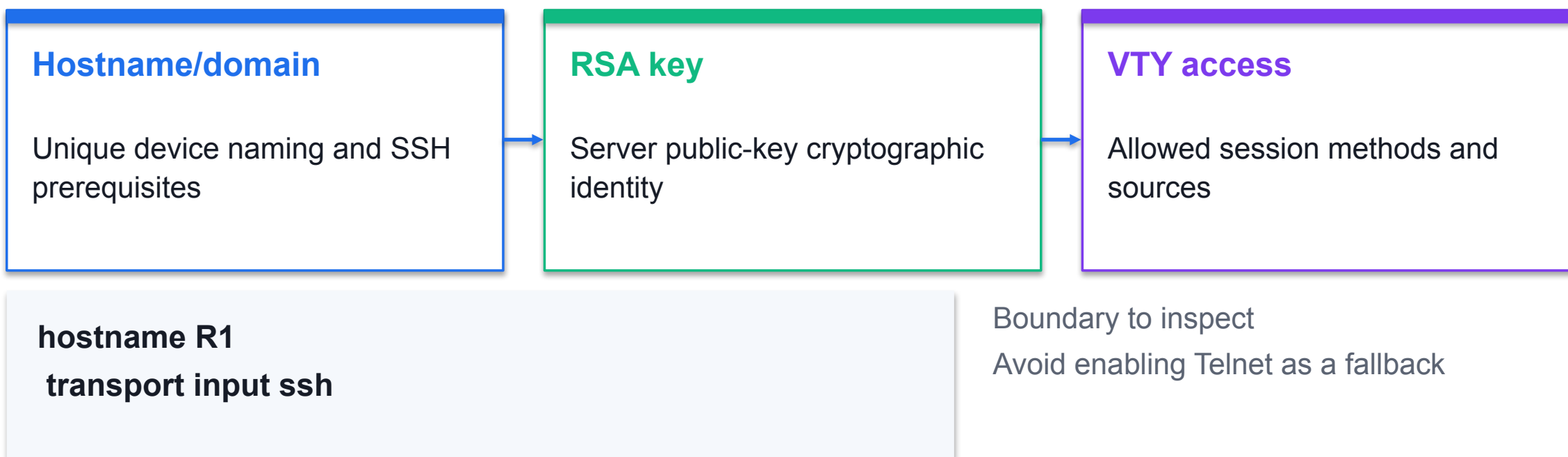
Local user and enable secrets · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Local user DB -> CLI session	username learner privilege 1 secret <unique-training-secret>
Mechanism / state	Config stores protected secret representation	service password-encryption
Result / boundary	show running-config filtered for user/enable settings; test allowed commands without publishing secret values.	service password-encryption obfuscates some passwords and is not strong cryptographic protection. Never embed live credentials in course files.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

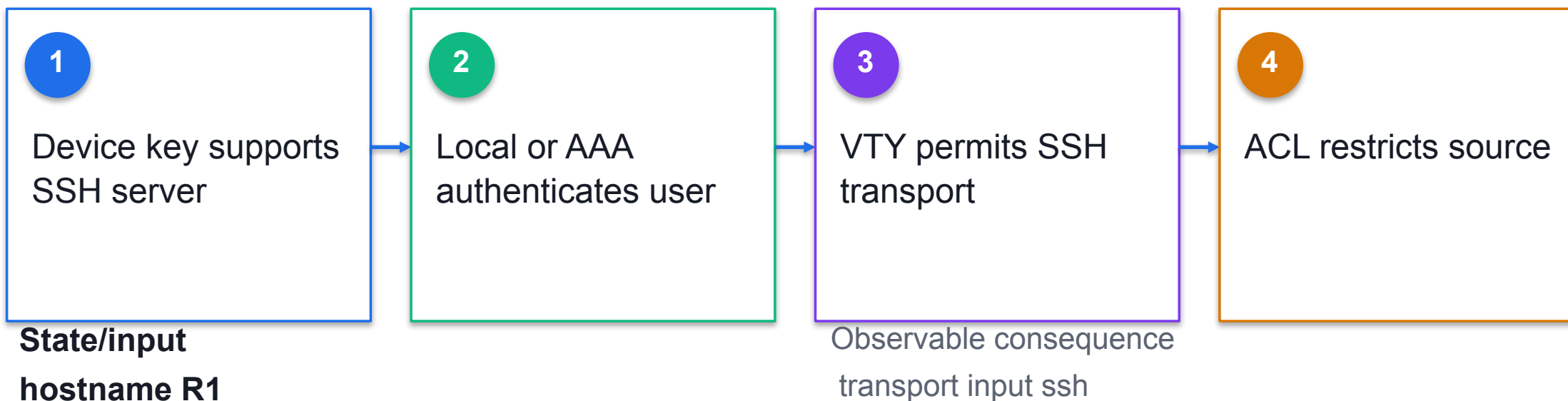
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

SSH management prerequisites · boundaries



show ip ssh; test authenticated SSH from permitted source and rejected access from an unauthorized source.

SSH management prerequisites · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

SSH management prerequisites · exact contract

```
hostname R1
ip domain-name training.example
crypto key generate rsa modulus 2048
line vty 0 4
login local
transport input ssh
```

INPUT / IDENTITY

Hostname/domain · Device key
supports SSH server

DECISION / STATE

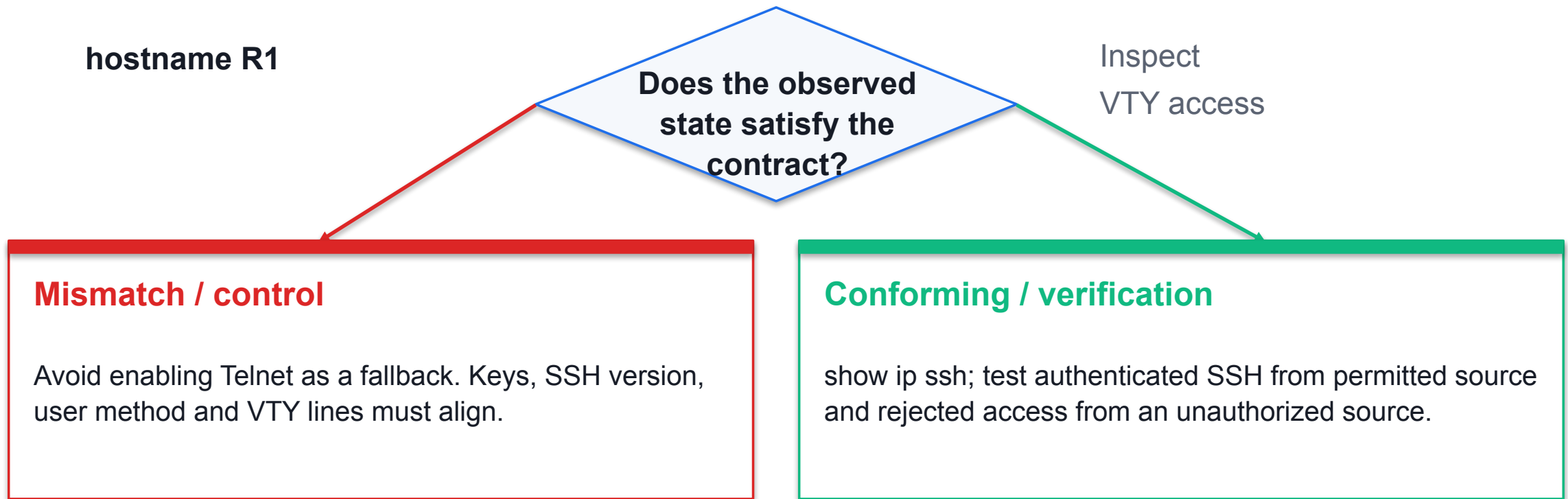
VTY permits SSH transport

READBACK / RESULT

ACL restricts source

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

SSH management prerequisites · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

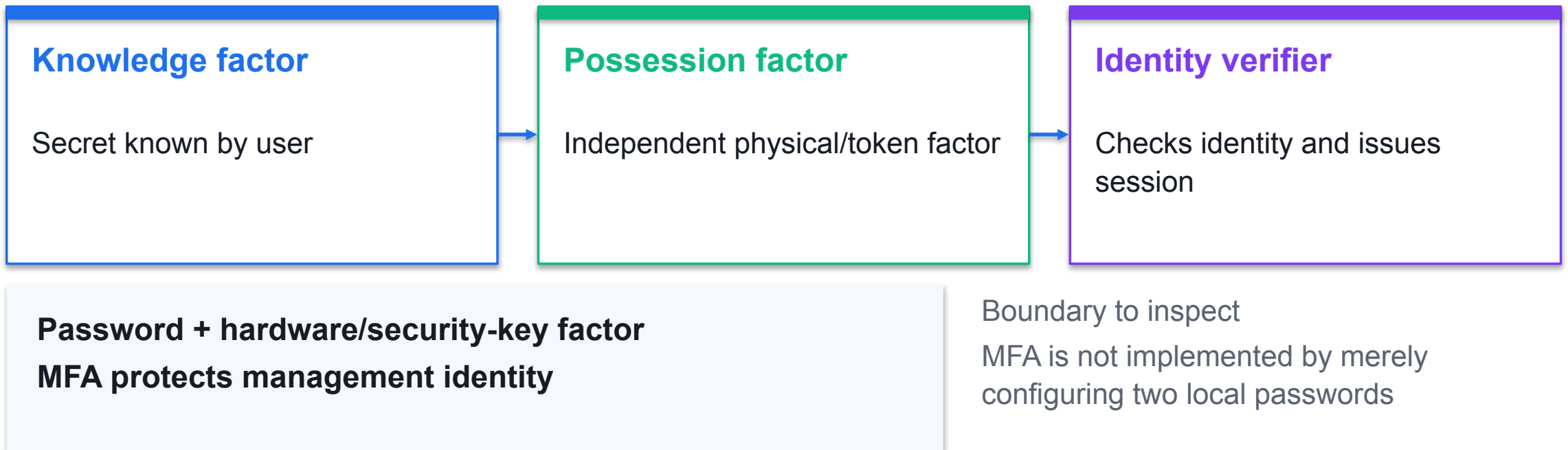
SSH management prerequisites · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Hostname/domain -> VTY access	hostname R1
Mechanism / state	ACL restricts source	transport input ssh
Result / boundary	show ip ssh; test authenticated SSH from permitted source and rejected access from an unauthorized source.	Avoid enabling Telnet as a fallback. Keys, SSH version, user method and VTY lines must align.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

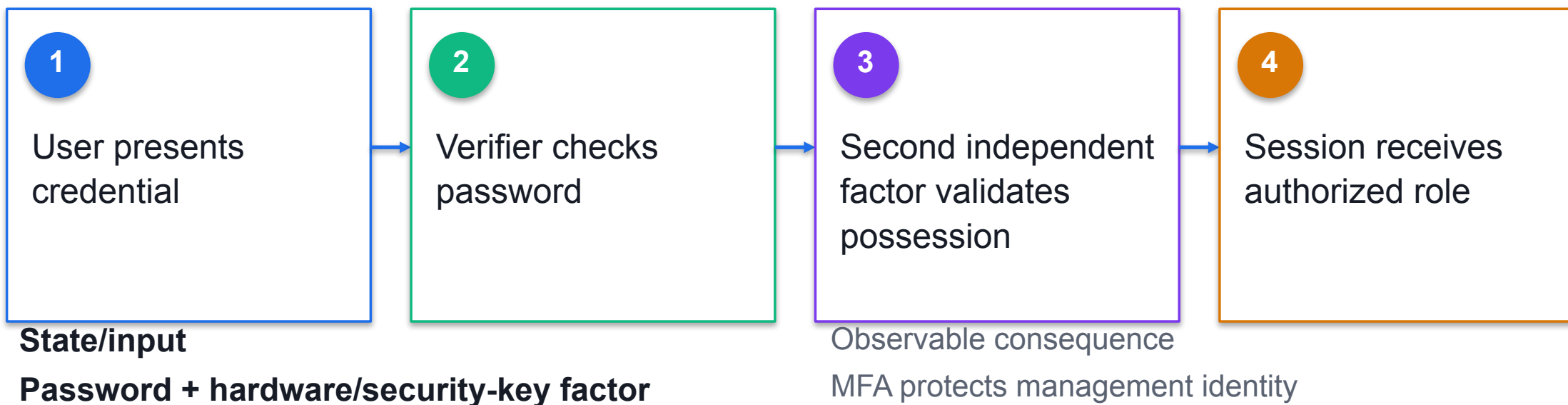
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Password and MFA policy · boundaries



Inspect identity policy, enrollment and successful/failed audit records; identify bypass paths and role scope.

Password and MFA policy · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Password and MFA policy · exact contract

Password + hardware/security-key factor
Two passwords are one factor type
MFA protects management identity

INPUT / IDENTITY

Knowledge factor · User presents credential

DECISION / STATE

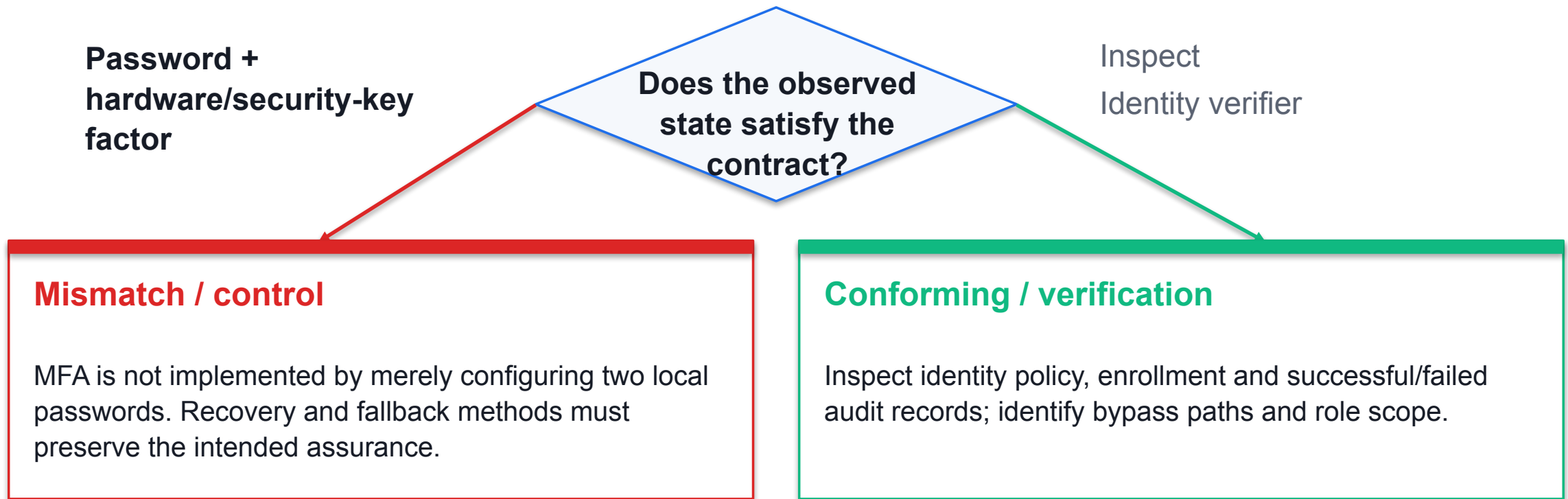
Second independent factor validates possession

READBACK / RESULT

Session receives authorized role

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Password and MFA policy · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

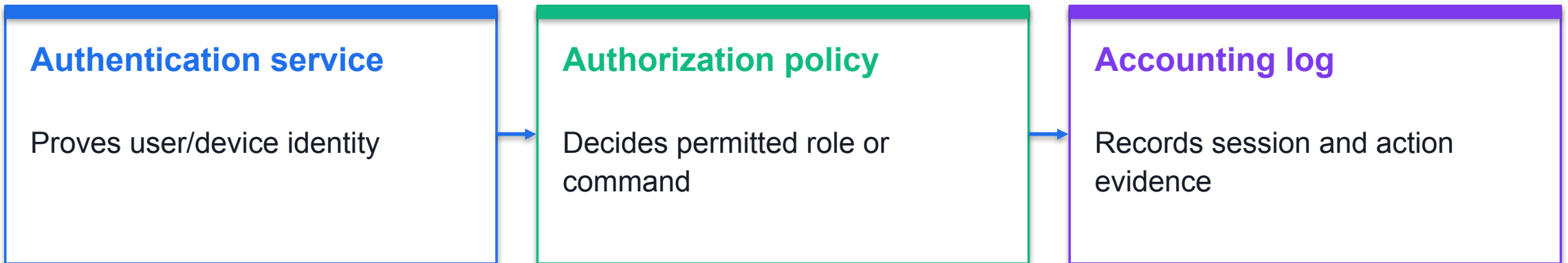
Password and MFA policy · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Knowledge factor -> Identity verifier	Password + hardware/security-key factor
Mechanism / state	Session receives authorized role	MFA protects management identity
Result / boundary	Inspect identity policy, enrollment and successful/failed audit records; identify bypass paths and role scope.	MFA is not implemented by merely configuring two local passwords. Recovery and fallback methods must preserve the intended assurance.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

AAA responsibility split · boundaries



Authentication: who are you?

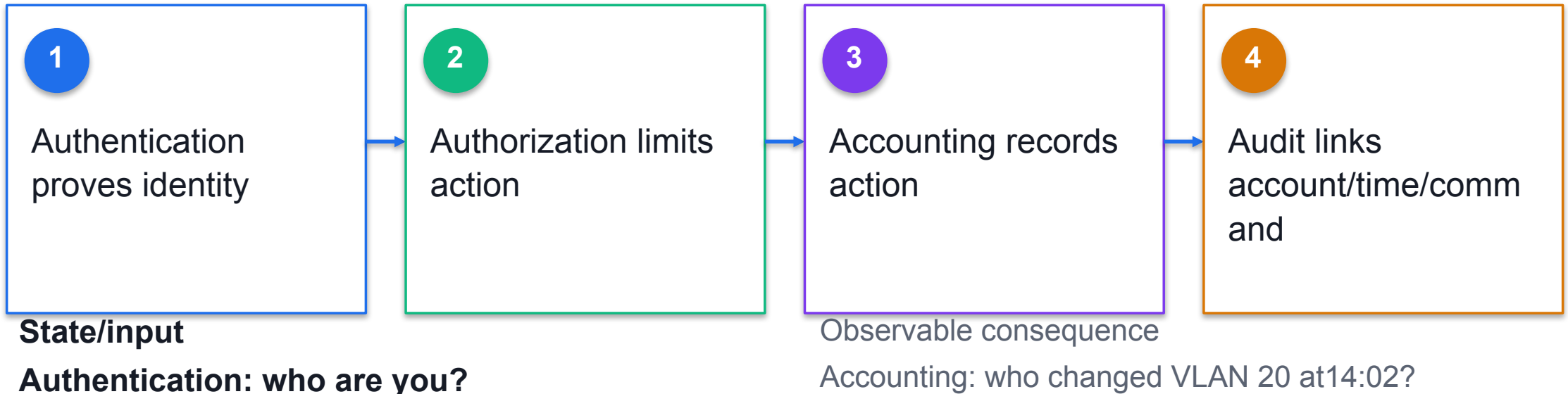
Accounting: who changed VLAN 20 at 14:02?

Boundary to inspect

Successful login does not grant every action

Use separate evidence for accepted identity, command permission and logged action; compare role to required task.

AAA responsibility split · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

AAA responsibility split · exact contract

Authentication: who are you?

Authorization: may you configure VLAN?

Accounting: who changed VLAN 20 at 14:02?

INPUT / IDENTITY

Authentication service · Authentication proves identity

DECISION / STATE

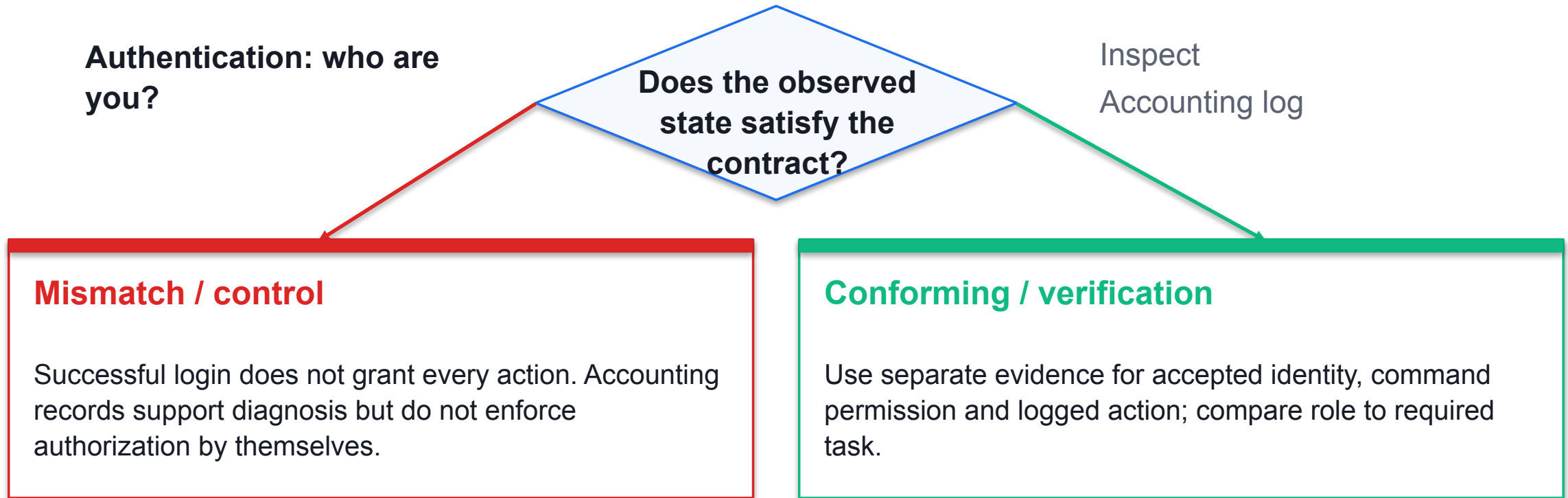
Accounting records action

READBCK / RESULT

Audit links account/time/command

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

AAA responsibility split · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

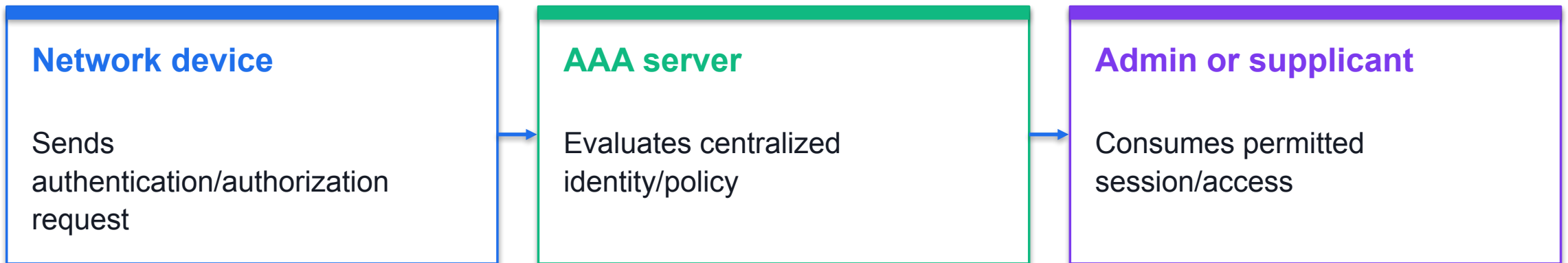
AAA responsibility split · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Authentication service -> Accounting log	Authentication: who are you?
Mechanism / state	Audit links account/time/command	Accounting: who changed VLAN 20 at 14:02?
Result / boundary	Use separate evidence for accepted identity, command permission and logged action; compare role to required task.	Successful login does not grant every action. Accounting records support diagnosis but do not enforce authorization by themselves.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

RADIUS and TACACS+ boundaries · boundaries

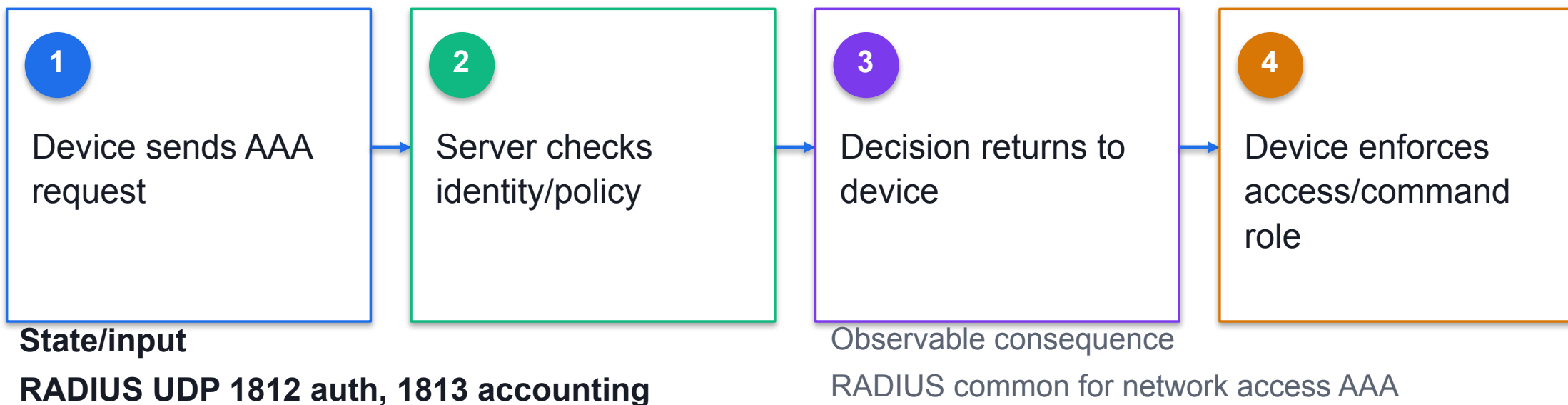


RADIUS UDP 1812 auth, 1813 accounting
RADIUS common for network access AAA

Boundary to inspect
Protocol roles and encryption scope differ

Confirm AAA server reachability and reject behavior; document local fallback without turning failures into unrestricted access.

RADIUS and TACACS+ boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

RADIUS and TACACS+ boundaries · exact contract

RADIUS UDP 1812 auth, 1813 accounting
TACACS+ TCP 49
RADIUS common for network access AAA

INPUT / IDENTITY

Network device · Device sends AAA request

DECISION / STATE

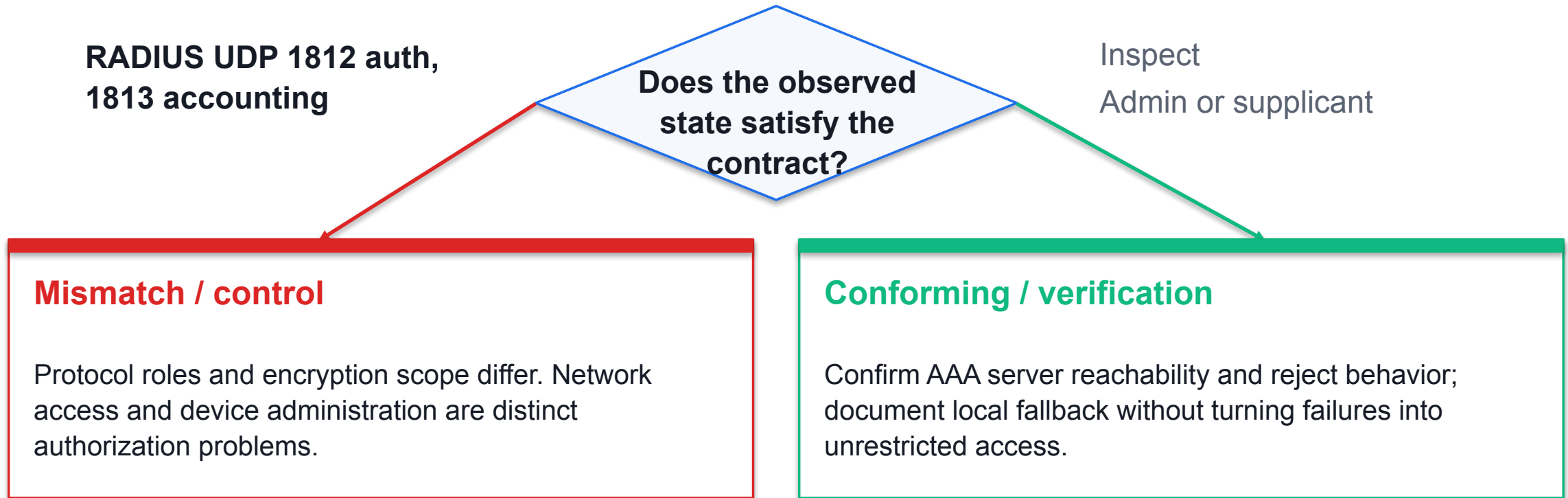
Decision returns to device

READBACK / RESULT

Device enforces access/command role

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

RADIUS and TACACS+ boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

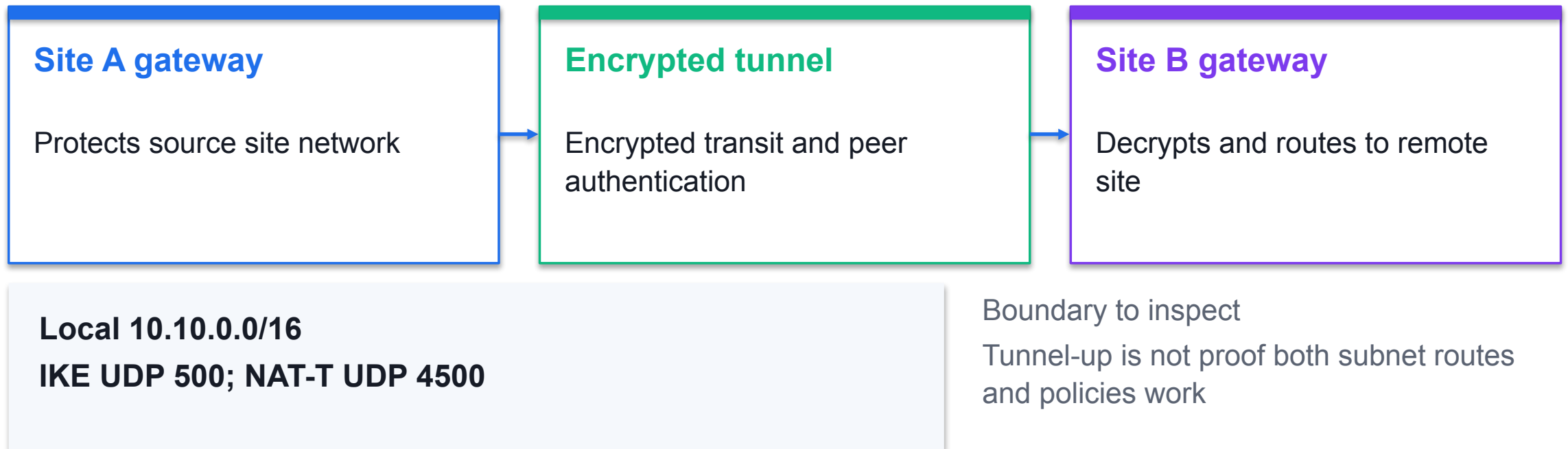
RADIUS and TACACS+ boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Network device -> Admin or supplicant	RADIUS UDP 1812 auth, 1813 accounting
Mechanism / state	Device enforces access/command role	RADIUS common for network access AAA
Result / boundary	Confirm AAA server reachability and reject behavior; document local fallback without turning failures into unrestricted access.	Protocol roles and encryption scope differ. Network access and device administration are distinct authorization problems.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

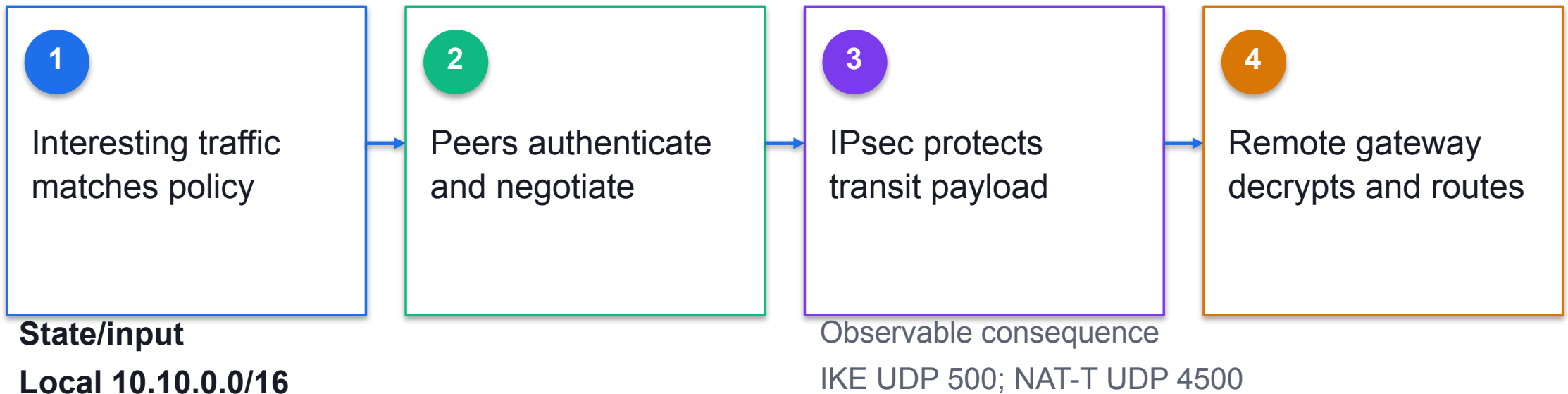
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Site-to-site IPsec trust · boundaries



Correlate protected-subnet selectors, security associations, packet counters and remote application reachability.

Site-to-site IPsec trust · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Site-to-site IPsec trust · exact contract

Local 10.10.0.0/16

Remote 10.20.0.0/16

ESP IP protocol 50

IKE UDP 500; NAT-T UDP 4500

INPUT / IDENTITY

Site A gateway · Interesting traffic matches policy

DECISION / STATE

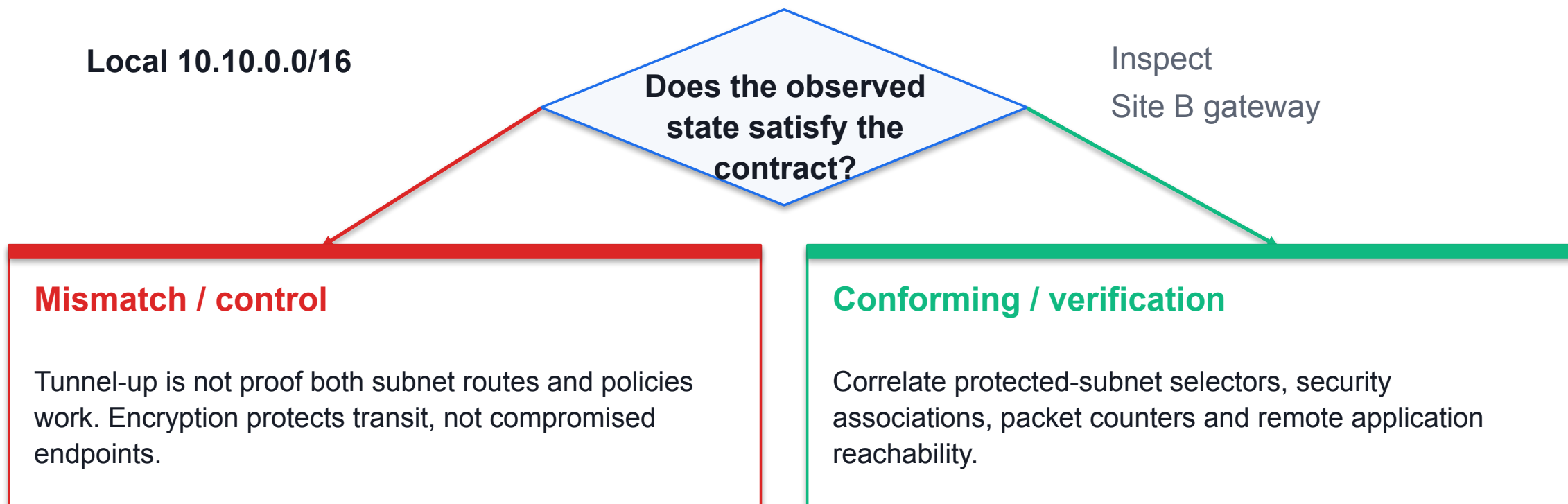
IPsec protects transit payload

READBACK / RESULT

Remote gateway decrypts and routes

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Site-to-site IPsec trust · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

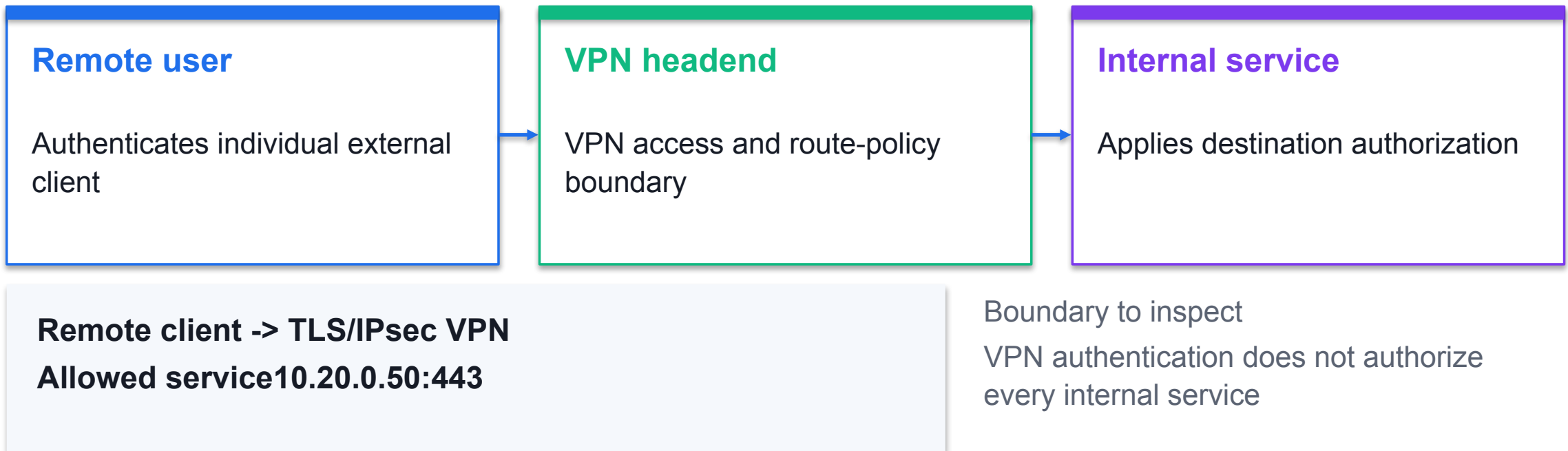
Site-to-site IPsec trust · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Site A gateway -> Site B gateway	Local 10.10.0.0/16
Mechanism / state	Remote gateway decrypts and routes	IKE UDP 500; NAT-T UDP 4500
Result / boundary	Correlate protected-subnet selectors, security associations, packet counters and remote application reachability.	Tunnel-up is not proof both subnet routes and policies work. Encryption protects transit, not compromised endpoints.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Remote-access VPN roles · boundaries



Verify client routes, assigned address, headend policy and service access; record identity and tunnel evidence separately.

Remote-access VPN roles · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Remote-access VPN roles · exact contract

Remote client -> TLS/IPsec VPN
Assigned client pool 10.30.0.0/24
Allowed service 10.20.0.50:443

INPUT / IDENTITY

Remote user · User authenticates to headend

DECISION / STATE

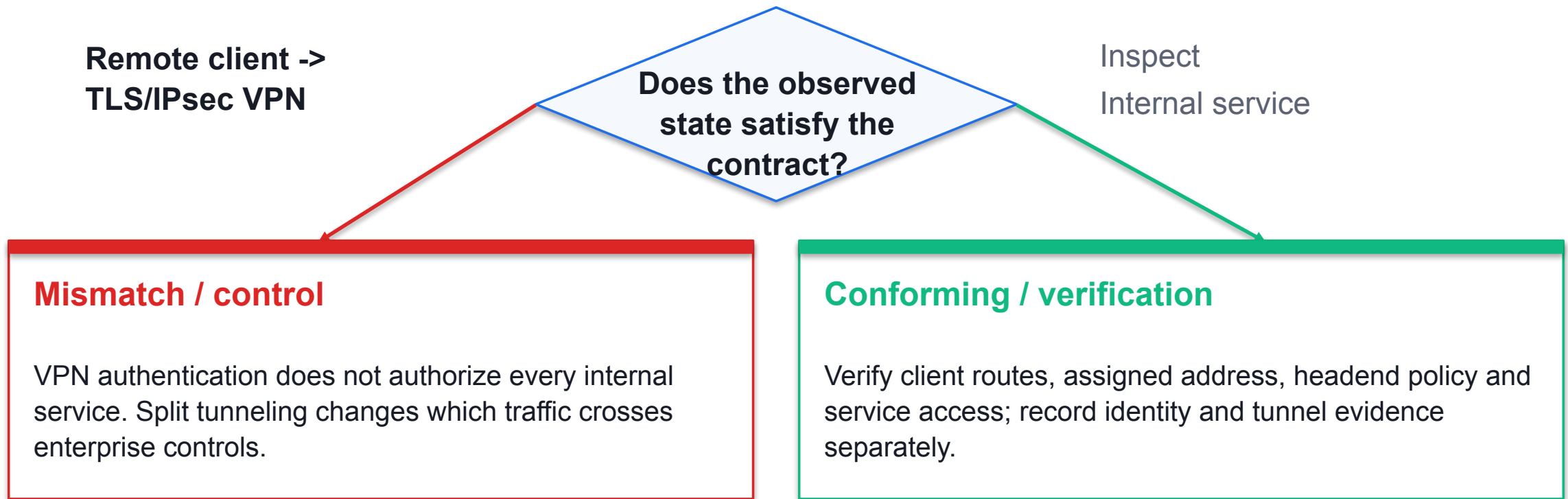
Traffic follows full/split route policy

READBACK / RESULT

Internal service applies own authorization

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Remote-access VPN roles · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

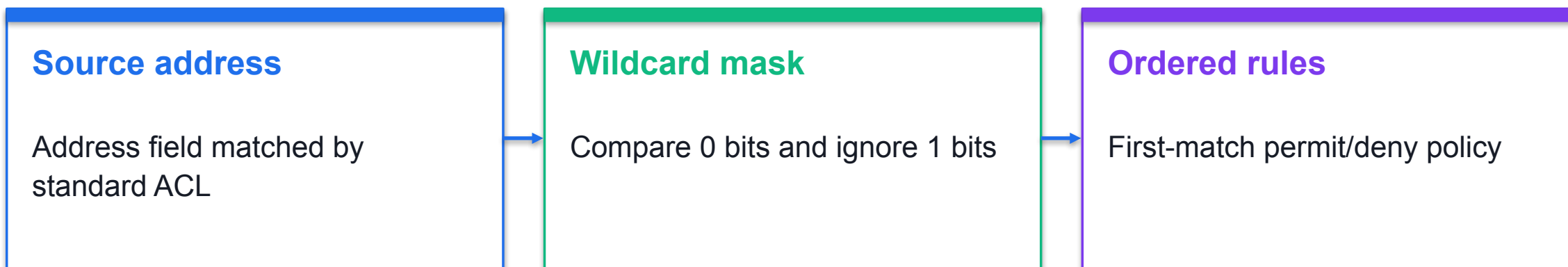
Remote-access VPN roles · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Remote user -> Internal service	Remote client -> TLS/IPsec VPN
Mechanism / state	Internal service applies own authorization	Allowed service 10.20.0.50:443
Result / boundary	Verify client routes, assigned address, headend policy and service access; record identity and tunnel evidence separately.	VPN authentication does not authorize every internal service. Split tunneling changes which traffic crosses enterprise controls.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Standard ACL source matching · boundaries



```
access-list 10 permit 10.10.10.0 0.0.0.255  
ip access-group 10 out
```

Boundary to inspect
Standard ACL matches source only

```
show access-lists; verify hit counters and direction
```

Standard ACL source matching · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Standard ACL source matching · exact contract

```
access-list 10 permit 10.10.10.0 0.0.0.255  
interface Gi0/1  
ip access-group 10 out
```

INPUT / IDENTITY

Source address · Packet reaches ACL binding

DECISION / STATE

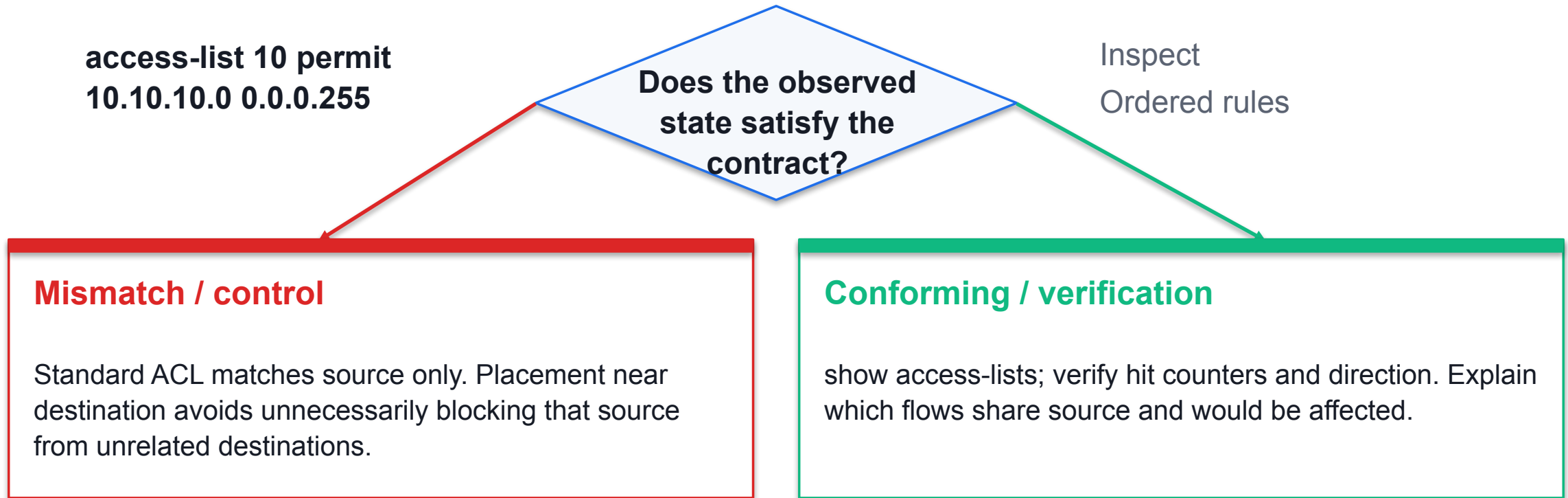
First match permits/denies

READBACK / RESULT

Implicit deny handles no match

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Standard ACL source matching · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

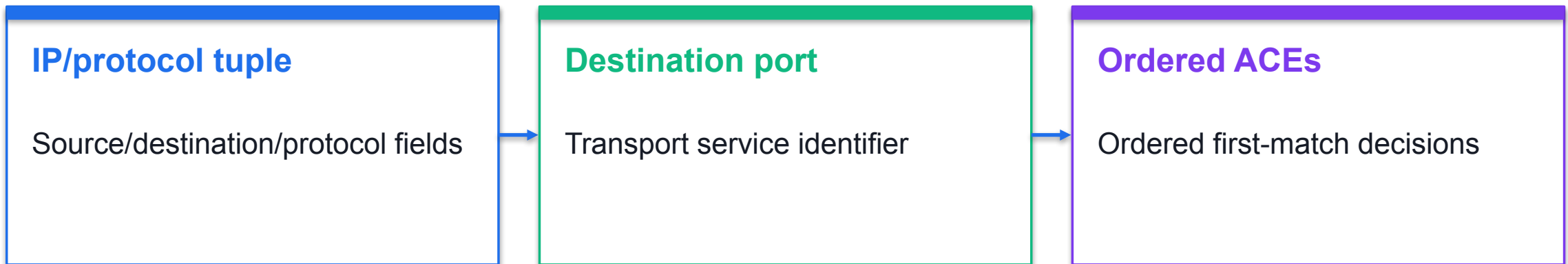
Standard ACL source matching · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Source address -> Ordered rules	access-list 10 permit 10.10.10.0 0.0.0.255
Mechanism / state	Implicit deny handles no match	ip access-group 10 out
Result / boundary	show access-lists; verify hit counters and direction. Explain which flows share source and would be affected.	Standard ACL matches source only. Placement near destination avoids unnecessarily blocking that source from unrelated destinations.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Extended ACL tuple control · boundaries



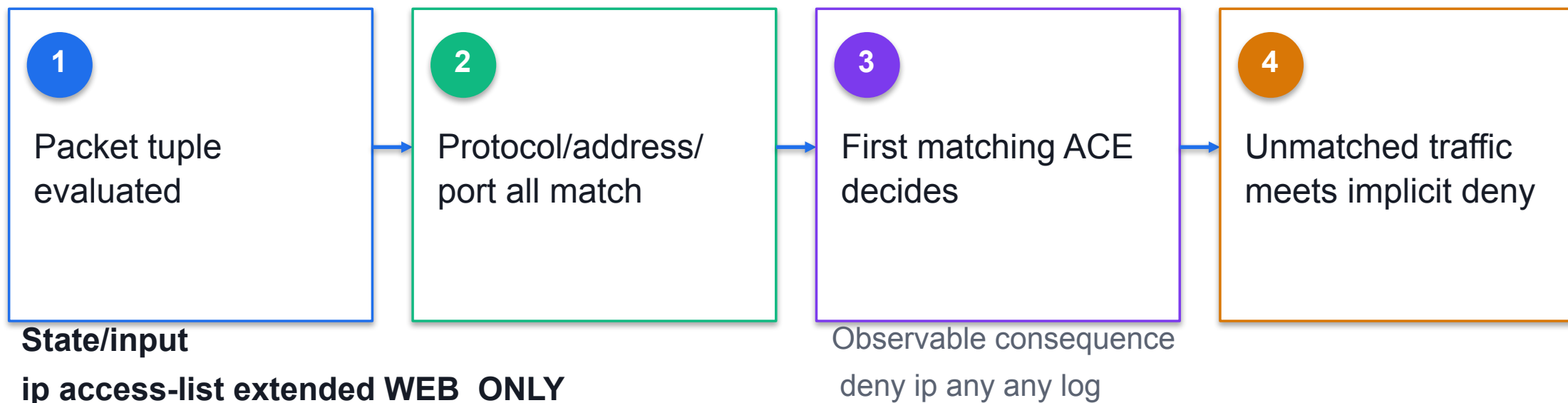
ip access-list extended WEB_ONLY
deny ip any any log

Boundary to inspect

The source port is usually ephemeral;
HTTPS destination port 443 identifies
service

Test allowed 443 and denied 22 with same source/destination; compare ACE counters and intended interface direction.

Extended ACL tuple control · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Extended ACL tuple control · exact contract

```
ip access-list extended WEB_ONLY  
permit tcp 10.10.10.0 0.0.0.255 host 10.20.20.50 eq 443  
deny ip any any log
```

INPUT / IDENTITY

IP/protocol tuple · Packet tuple
evaluated

DECISION / STATE

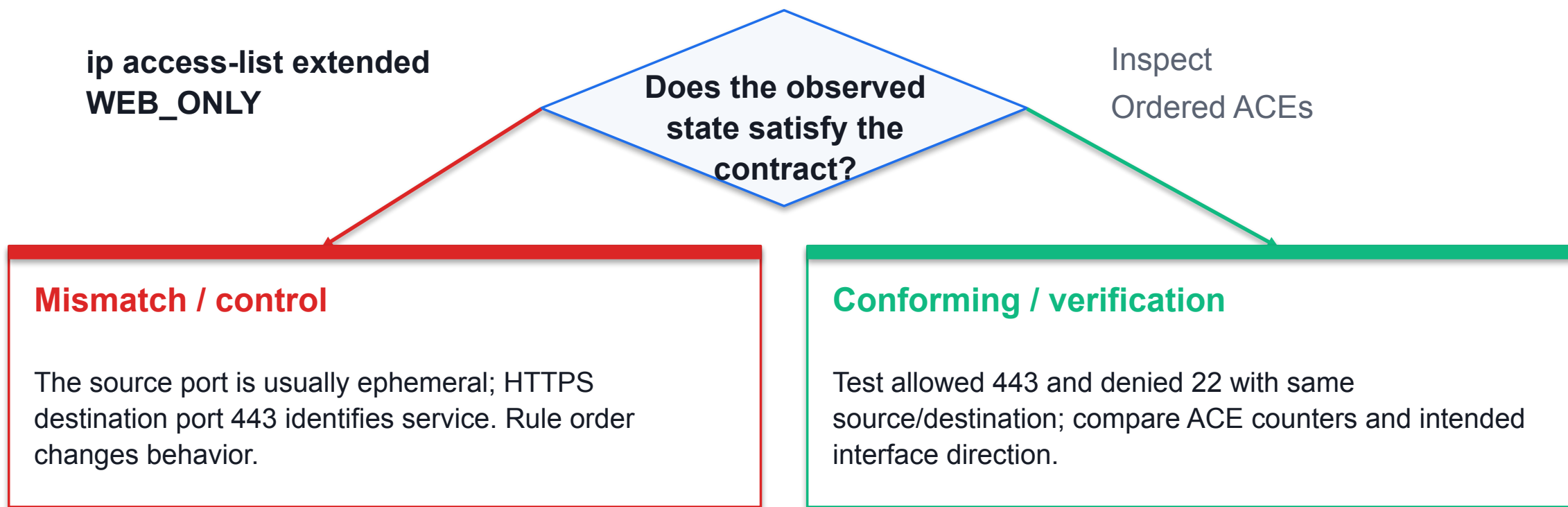
First matching ACE decides

READBACK / RESULT

Unmatched traffic meets implicit deny

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Extended ACL tuple control · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

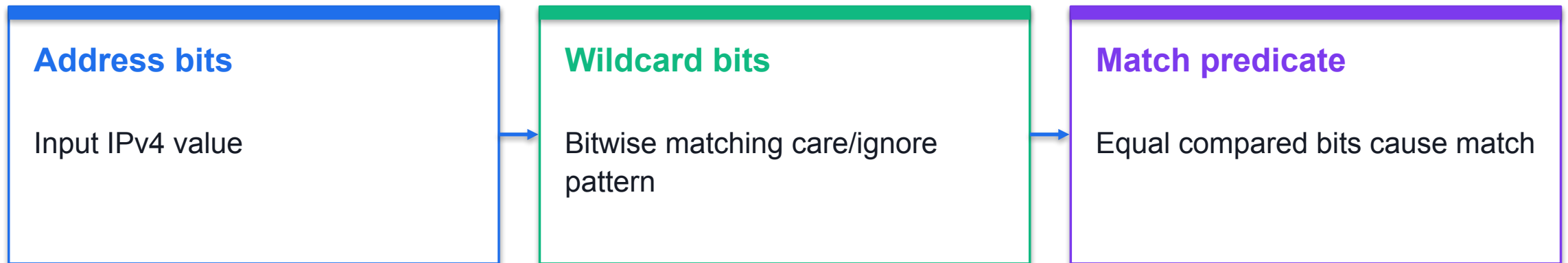
Extended ACL tuple control · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	IP/protocol tuple -> Ordered ACEs	ip access-list extended WEB_ONLY
Mechanism / state	Unmatched traffic meets implicit deny	deny ip any any log
Result / boundary	Test allowed 443 and denied 22 with same source/destination; compare ACE counters and intended interface direction.	The source port is usually ephemeral; HTTPS destination port 443 identifies service. Rule order changes behavior.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

ACL wildcard mathematics · boundaries



10.10.10.0 wildcard 0.0.0.255
host 10.10.10.5 = wildcard 0.0.0.0

Boundary to inspect
Wildcard is not always a subnet mask and
need not be contiguous

For /27 mask 255.255.255.224, wildcard 0.0.0.31

ACL wildcard mathematics · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

ACL wildcard mathematics · exact contract

10.10.10.0 wildcard 0.0.0.255

Matches 10.10.10.0-255

host 10.10.10.5 = wildcard 0.0.0.0

INPUT / IDENTITY

Address bits · Wildcard 0 means compare bit

DECISION / STATE

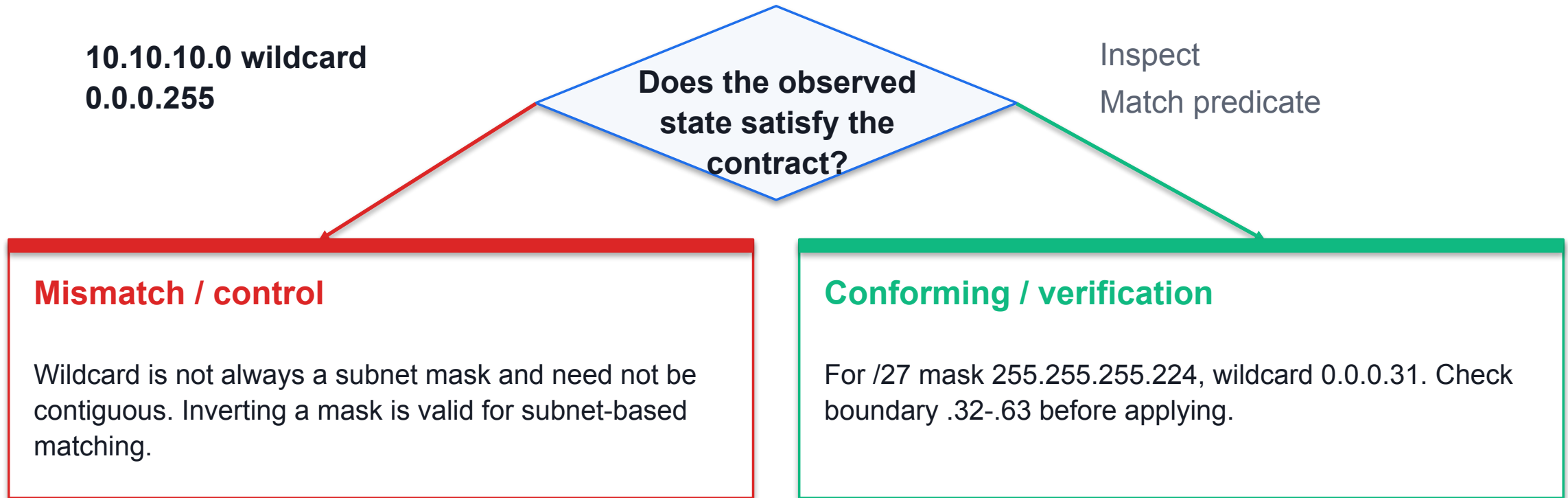
Address undergoes mask comparison

READBACK / RESULT

Matching tuple advances to ACE action

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

ACL wildcard mathematics · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

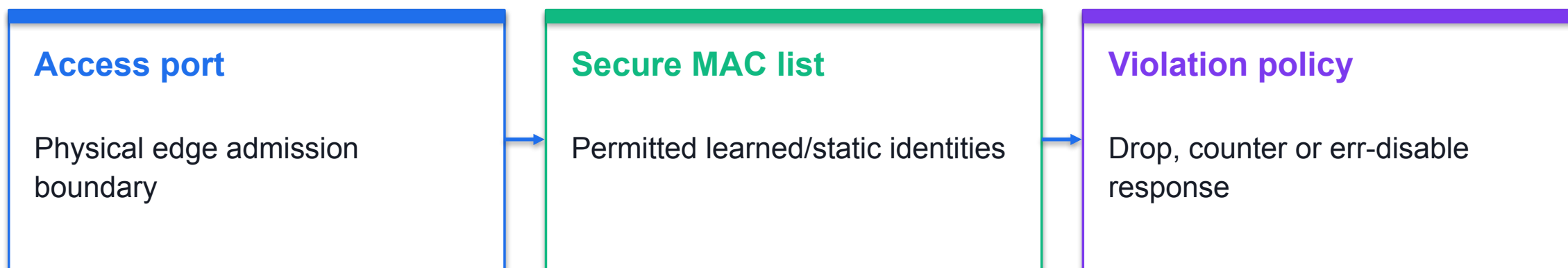
ACL wildcard mathematics · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Address bits -> Match predicate	10.10.10.0 wildcard 0.0.0.255
Mechanism / state	Matching tuple advances to ACE action	host 10.10.10.5 = wildcard 0.0.0.0
Result / boundary	For /27 mask 255.255.255.224, wildcard 0.0.0.31. Check boundary .32-.63 before applying.	Wildcard is not always a subnet mask and need not be contiguous. Inverting a mask is valid for subnet-based matching.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Port-security admission and violations · boundaries

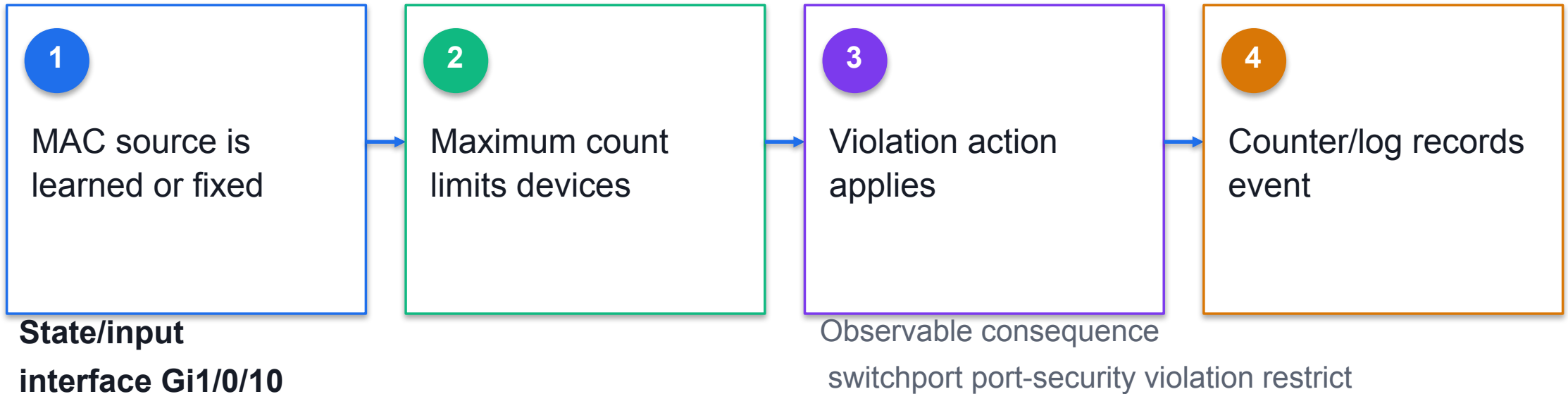


interface Gi1/0/10
switchport port-security violation restrict

Boundary to inspect
Protect drops silently; restrict drops and records; shutdown err-disables

show port-security interface

Port-security admission and violations · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Port-security admission and violations · exact contract

```
interface Gi1/0/10
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
```

INPUT / IDENTITY

Access port · MAC source is learned or fixed

DECISION / STATE

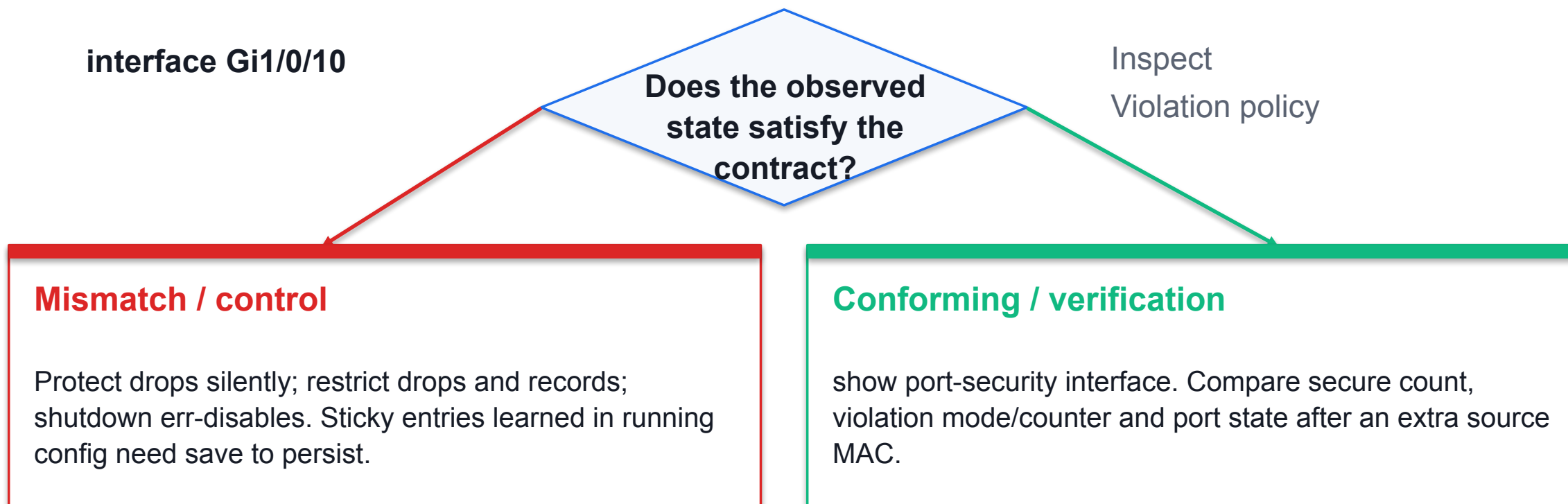
Violation action applies

READBACK / RESULT

Counter/log records event

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Port-security admission and violations · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

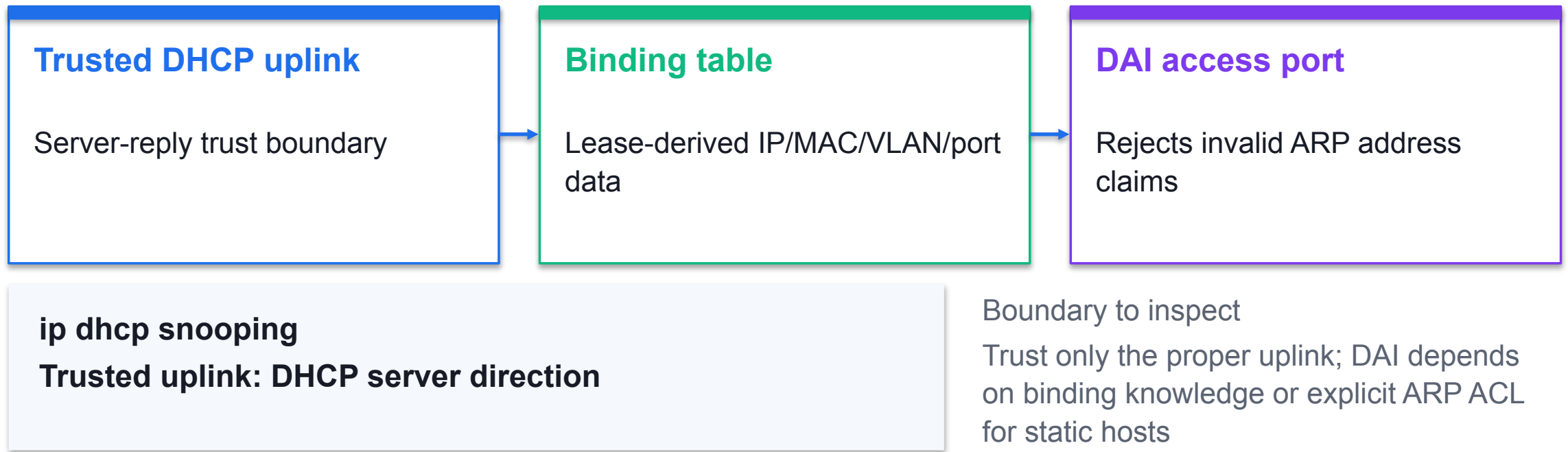
Port-security admission and violations · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Access port -> Violation policy	interface Gi1/0/10
Mechanism / state	Counter/log records event	switchport port-security violation restrict
Result / boundary	show port-security interface. Compare secure count, violation mode/counter and port state after an extra source MAC.	Protect drops silently; restrict drops and records; shutdown err-disables. Sticky entries learned in running config need save to persist.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

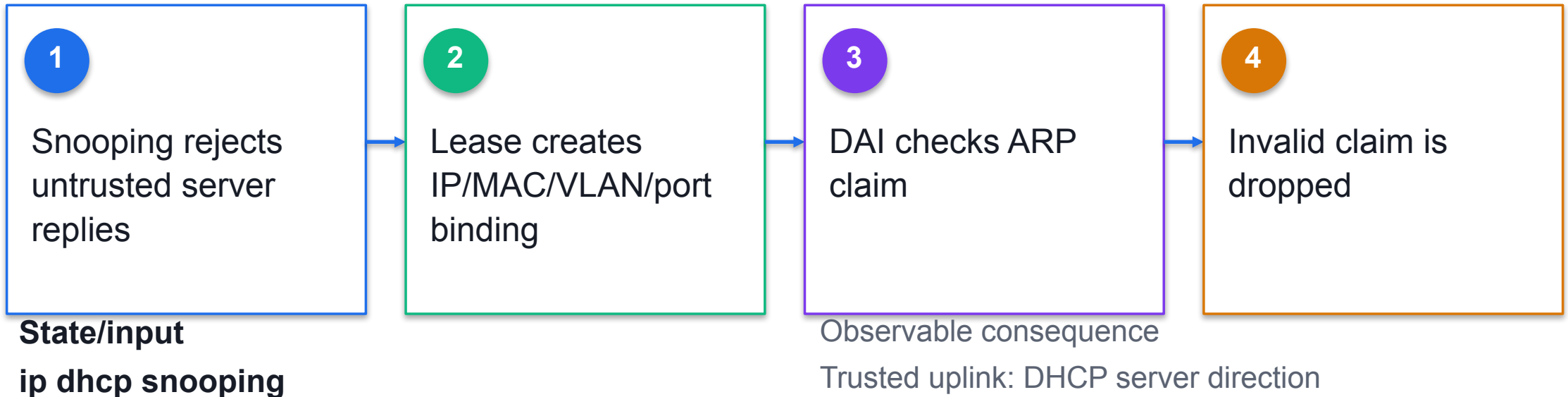
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

DHCP snooping and ARP inspection · boundaries



show ip dhcp snooping binding; show ip arp inspection statistics

DHCP snooping and ARP inspection · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

DHCP snooping and ARP inspection · exact contract

```
ip dhcp snooping
ip dhcp snooping vlan 10
ip arp inspection vlan 10
Trusted uplink: DHCP server direction
```

INPUT / IDENTITY

Trusted DHCP uplink · Snooping rejects untrusted server replies

DECISION / STATE

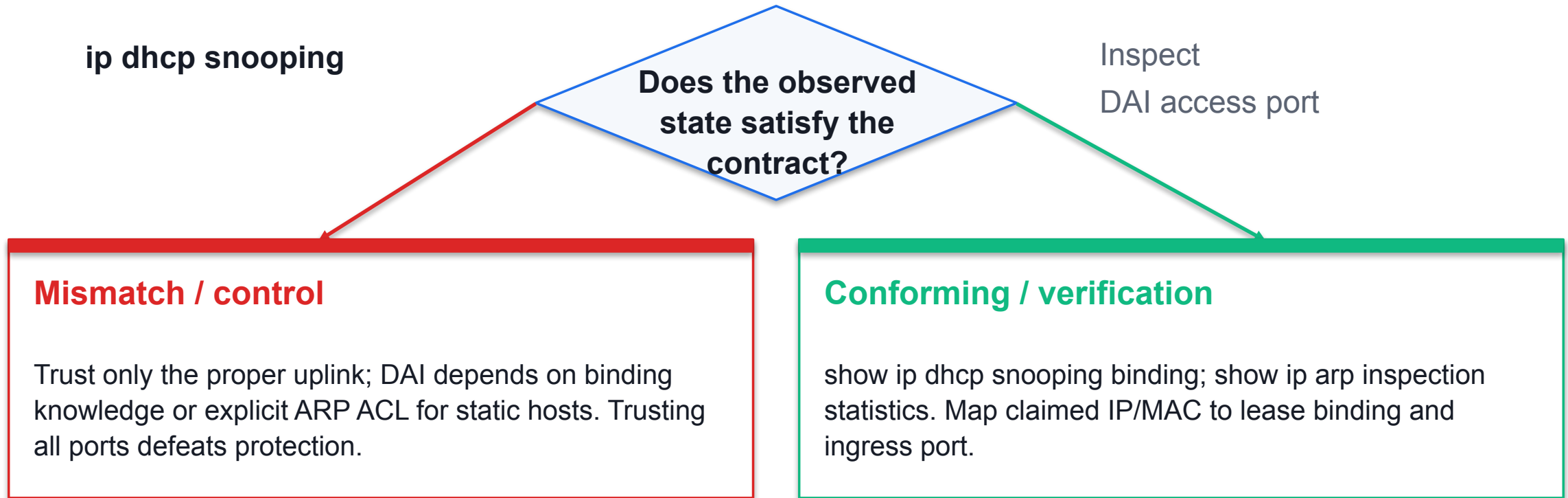
DAI checks ARP claim

READBACK / RESULT

Invalid claim is dropped

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

DHCP snooping and ARP inspection · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

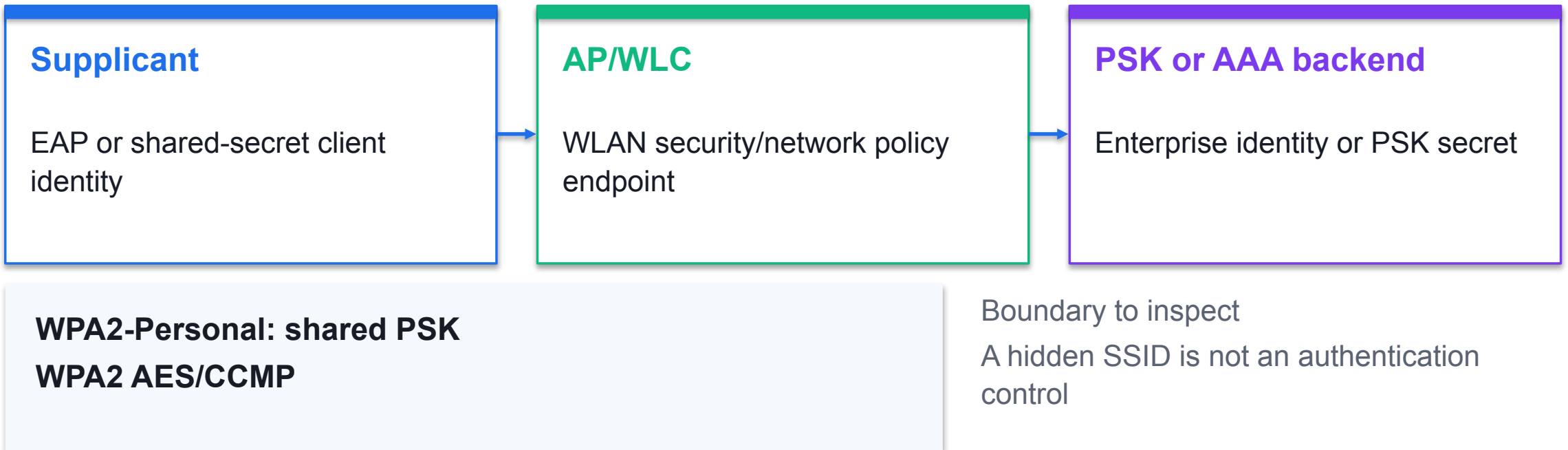
DHCP snooping and ARP inspection · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Trusted DHCP uplink -> DAI access port	ip dhcp snooping
Mechanism / state	Invalid claim is dropped	Trusted uplink: DHCP server direction
Result / boundary	show ip dhcp snooping binding; show ip arp inspection statistics. Map claimed IP/MAC to lease binding and ingress port.	Trust only the proper uplink; DAI depends on binding knowledge or explicit ARP ACL for static hosts. Trusting all ports defeats protection.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Wireless WPA and authentication · boundaries



Inspect SSID security settings and client authentication status; verify intended VLAN and authorized connectivity after association.

Wireless WPA and authentication · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Wireless WPA and authentication · exact contract

WPA2-Personal: shared PSK

WPA2-Enterprise: 802.1 X/EAP + AAA

WPA3-Personal: SAE

WPA2 AES/CCMP

INPUT / IDENTITY

Supplicant · Client selects WLAN

DECISION / STATE

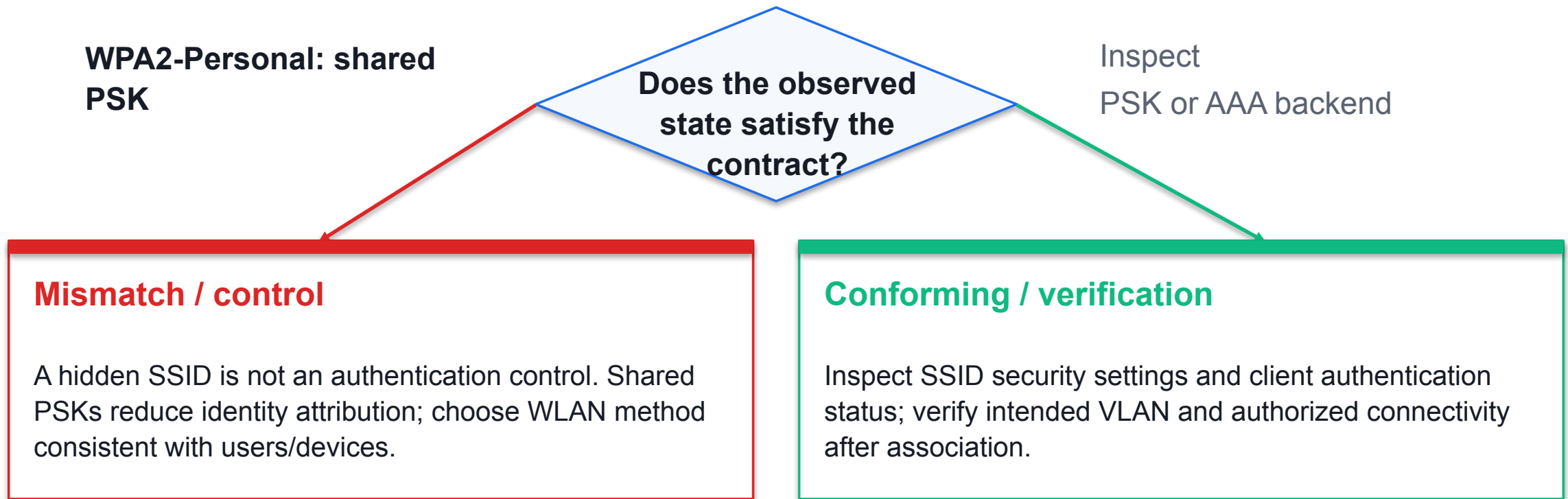
Key exchange derives session protection

READBACK / RESULT

Encrypted data follows authorized VLAN

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Wireless WPA and authentication · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

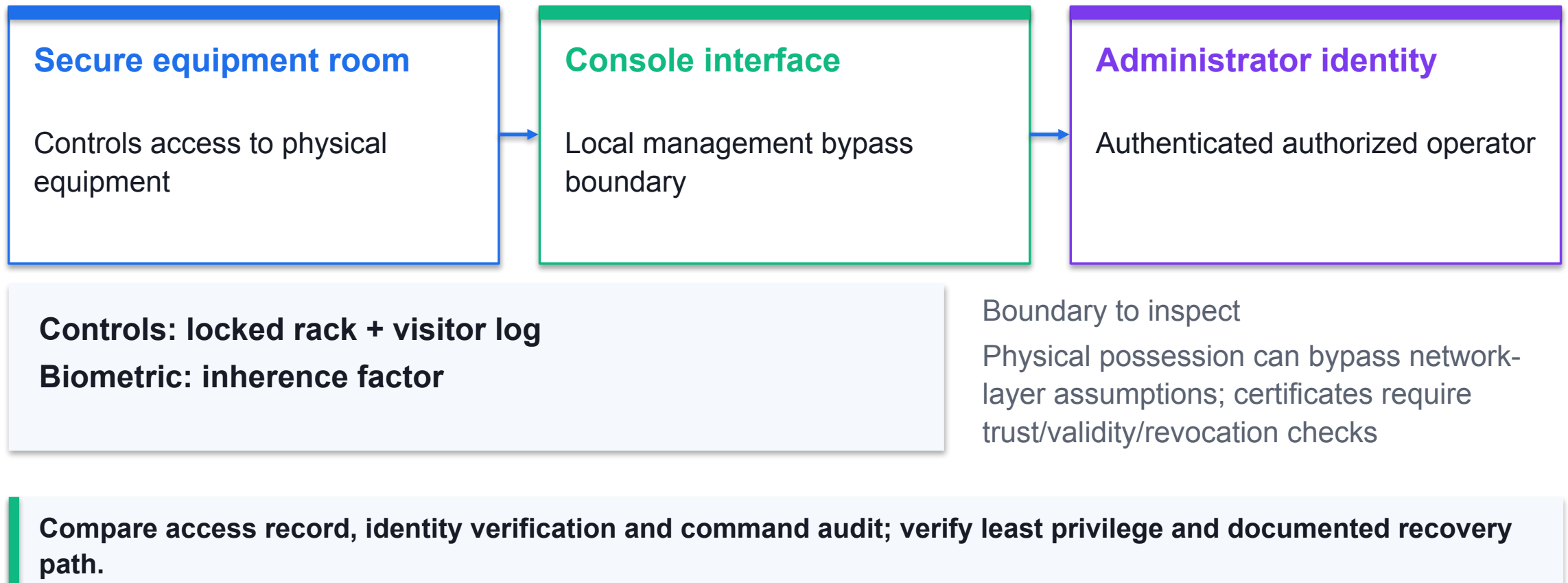
Wireless WPA and authentication · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Supplicant -> PSK or AAA backend	WPA2-Personal: shared PSK
Mechanism / state	Encrypted data follows authorized VLAN	WPA2 AES/CCMP
Result / boundary	Inspect SSID security settings and client authentication status; verify intended VLAN and authorized connectivity after association.	A hidden SSID is not an authentication control. Shared PSKs reduce identity attribution; choose WLAN method consistent with users/devices.

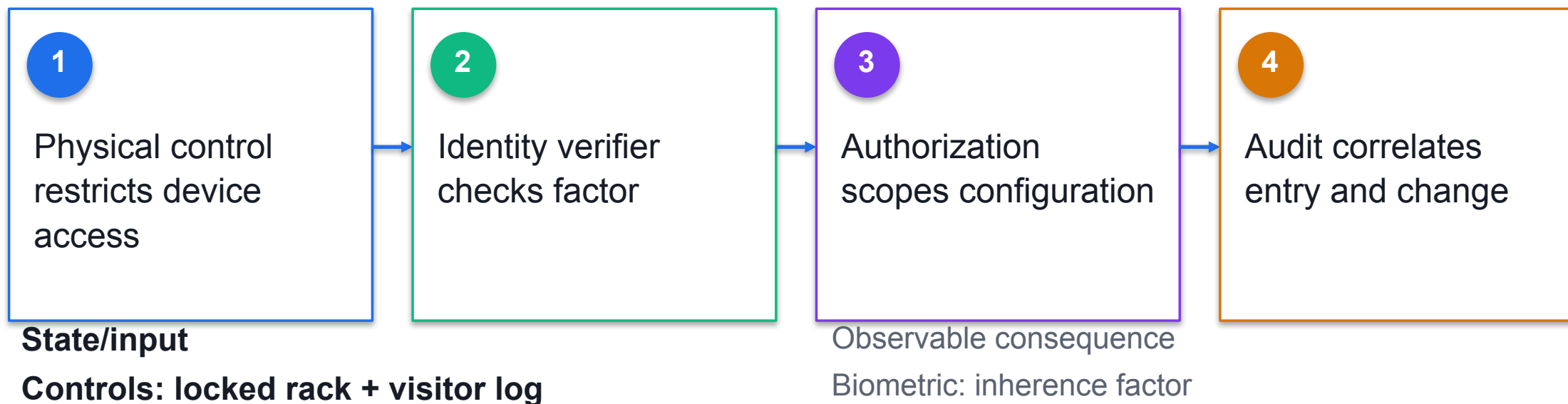
Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Physical access and identity controls · boundaries



Physical access and identity controls · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Physical access and identity controls · exact contract

Controls: locked rack + visitor log
Identity: password, certificate, biometric
Certificate: issuer/subject/validity
Biometric: inherence factor

INPUT / IDENTITY

Secure equipment room · Physical control restricts device access

DECISION / STATE

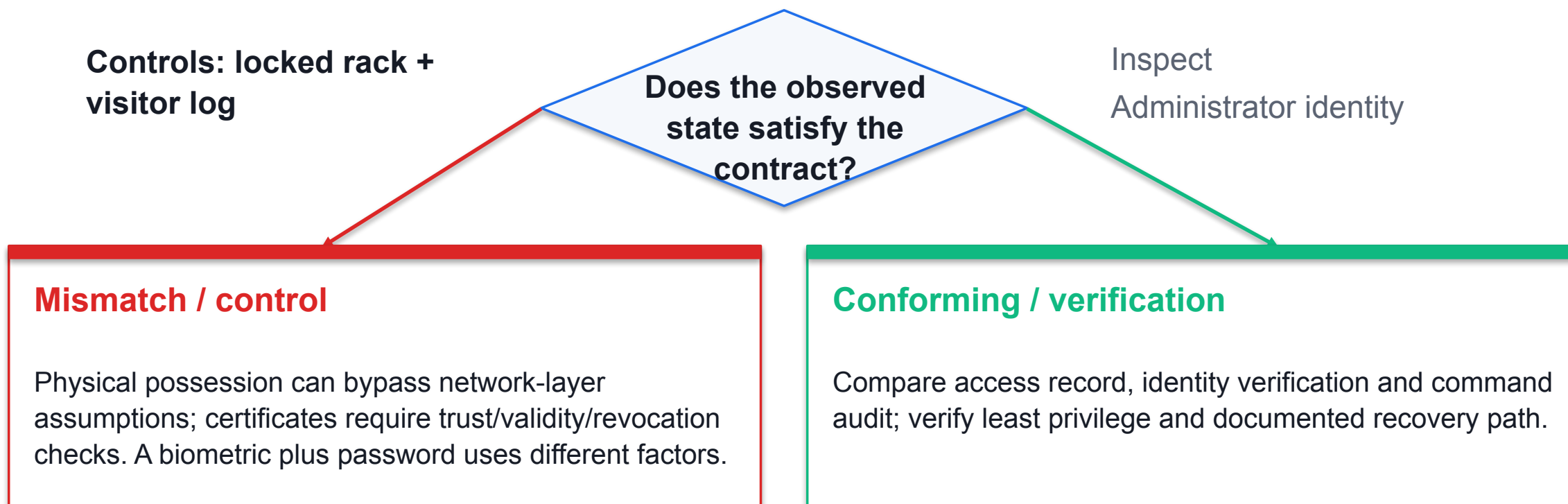
Authorization scopes configuration

READBACK / RESULT

Audit correlates entry and change

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Physical access and identity controls · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

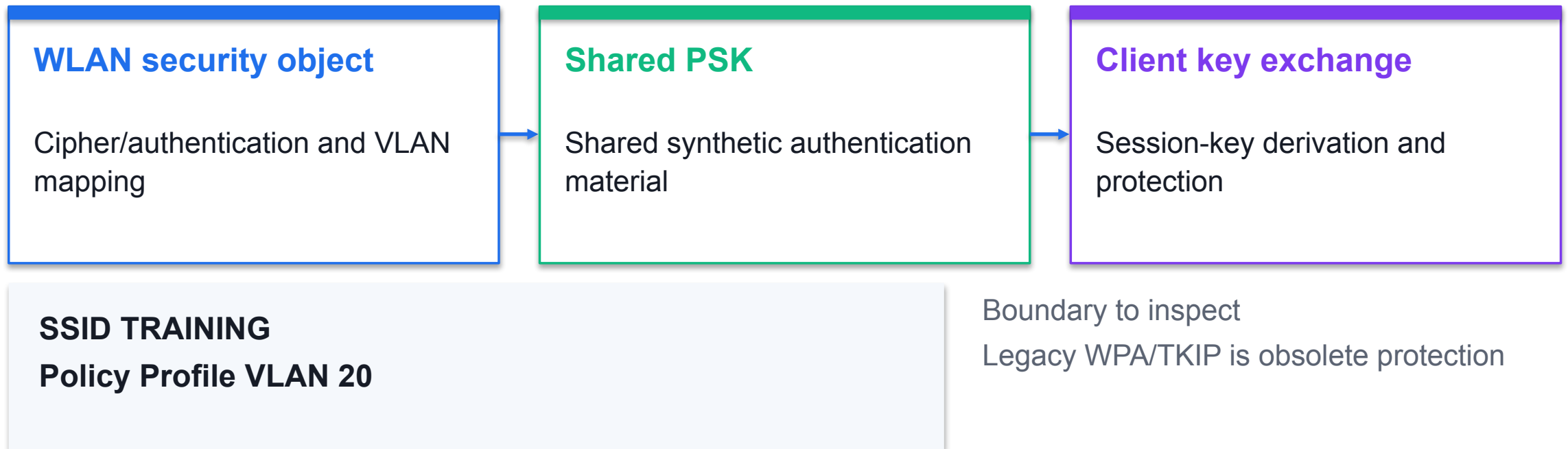
Physical access and identity controls · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Secure equipment room -> Administrator identity	Controls: locked rack + visitor log
Mechanism / state	Audit correlates entry and change	Biometric: inherence factor
Result / boundary	Compare access record, identity verification and command audit; verify least privilege and documented recovery path.	Physical possession can bypass network-layer assumptions; certificates require trust/validity/revocation checks. A biometric plus password uses different factors.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

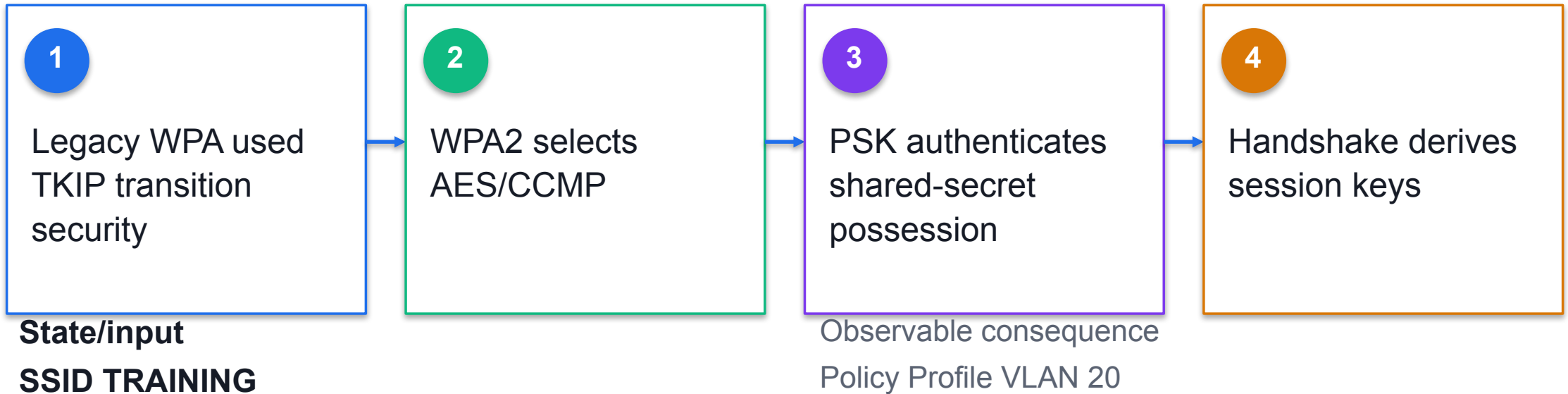
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

WPA lineage and WPA2 PSK WLAN evidence · boundaries



Read actual WLAN security state and connected-client authentication; validate correct PSK acceptance and incorrect PSK rejection in isolated simulator.

WPA lineage and WPA2 PSK WLAN evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

WPA lineage and WPA2 PSK WLAN evidence · exact contract

SSID TRAINING

Security WPA2-Personal / AES-CCMP

PSK <synthetic-training-only>

Policy Profile VLAN 20

INPUT / IDENTITY

WLAN security object · Legacy WPA
used TKIP transition security

DECISION / STATE

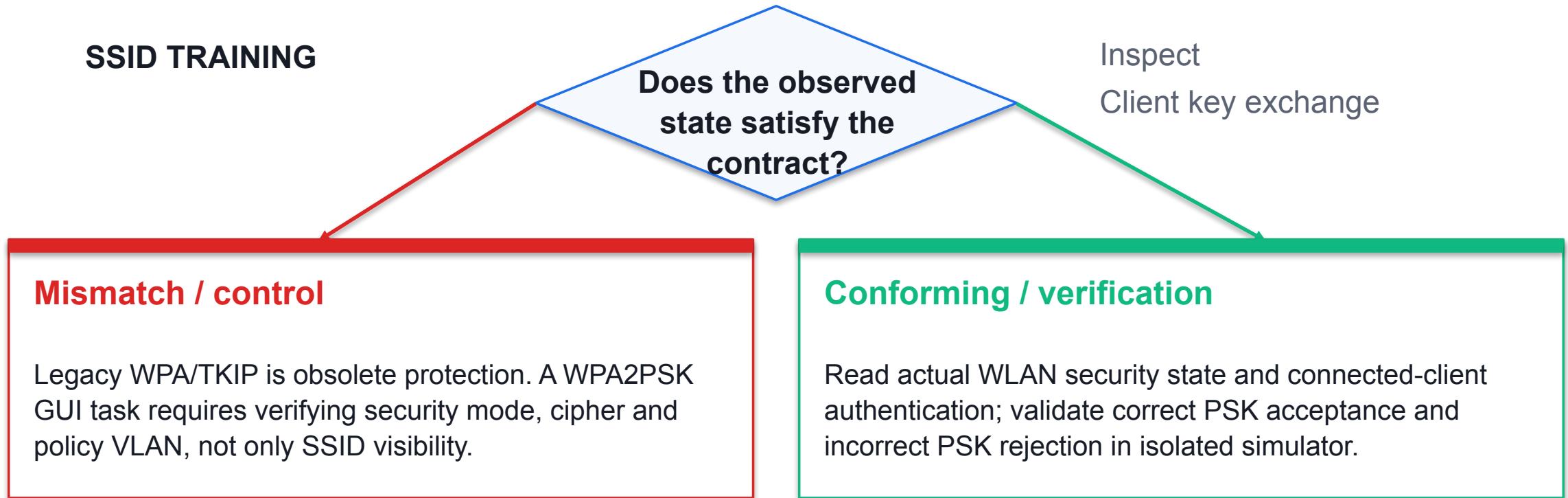
PSK authenticates shared-secret
possession

READBACK / RESULT

Handshake derives session keys

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

WPA lineage and WPA2 PSK WLAN evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

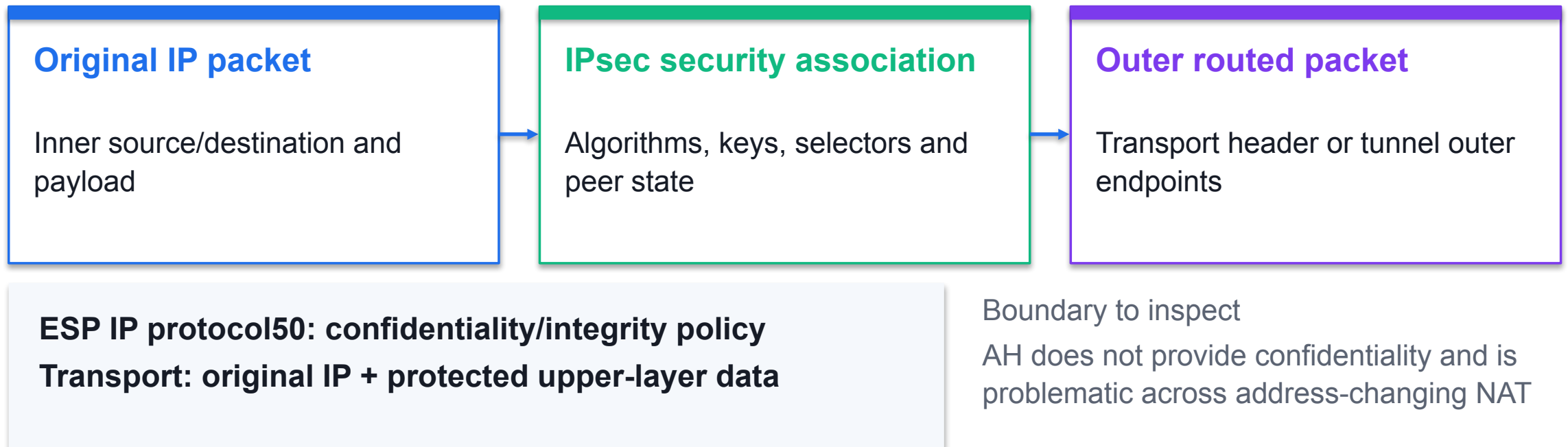
WPA lineage and WPA2 PSK WLAN evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	WLAN security object -> Client key exchange	SSID TRAINING
Mechanism / state	Handshake derives session keys	Policy Profile VLAN 20
Result / boundary	Read actual WLAN security state and connected-client authentication; validate correct PSK acceptance and incorrect PSK rejection in isolated simulator.	Legacy WPA/TKIP is obsolete protection. A WPA2PSK GUI task requires verifying security mode, cipher and policy VLAN, not only SSID visibility.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

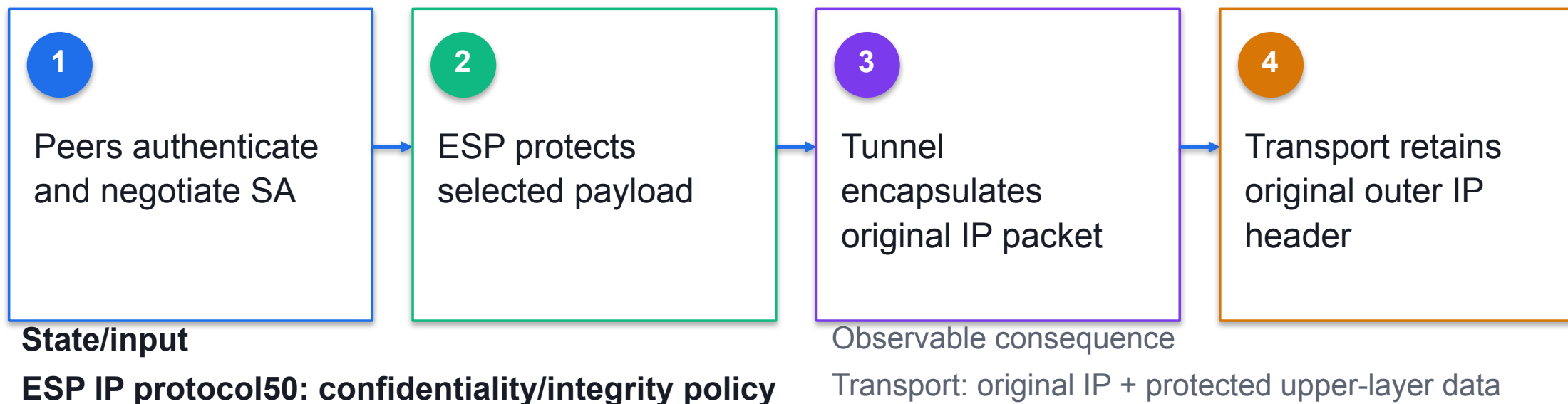
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPsec tunnel and transport boundaries · boundaries



Identify mode, selectors, peer endpoints, ESP/AH choice and counters; verify remote-access/site-to-site routing plus traffic protection evidence.

IPsec tunnel and transport boundaries · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPsec tunnel and transport boundaries · exact contract

ESP IP protocol50: confidentiality/integrity policy

AH IP protocol51: authentication/integrity, no encryption

Tunnel: new outer IP + protected original packet

Transport: original IP + protected upper-layer data

INPUT / IDENTITY

Original IP packet · Peers authenticate and negotiate SA

DECISION / STATE

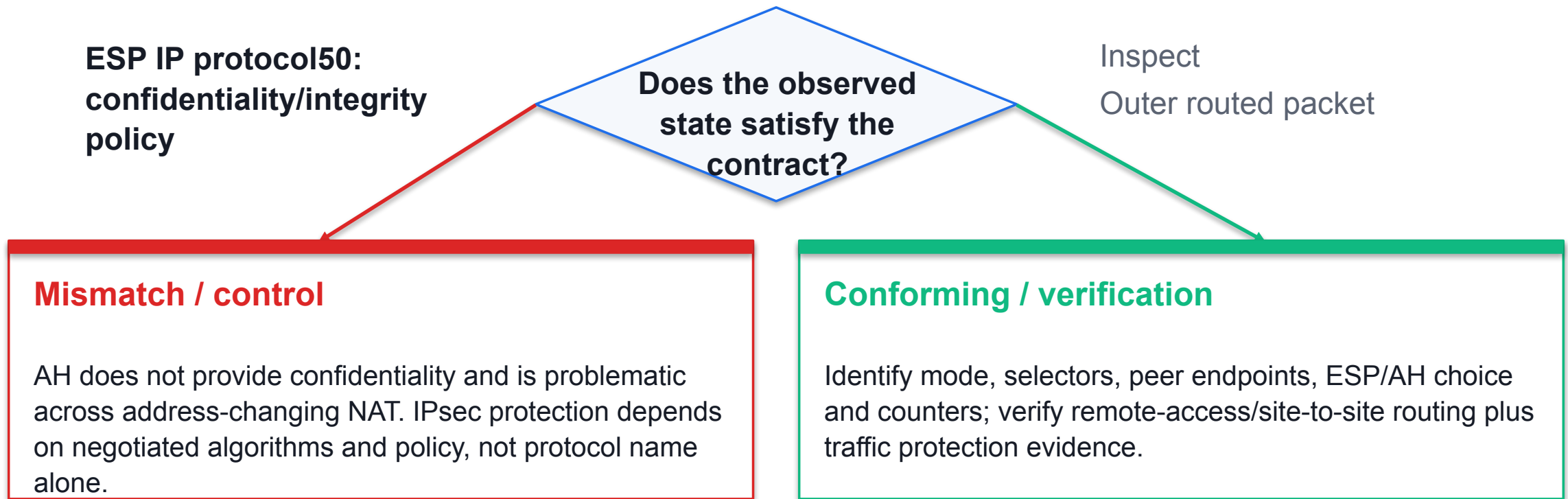
Tunnel encapsulates original IP packet

READBACK / RESULT

Transport retains original outer IP header

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPsec tunnel and transport boundaries · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

IPsec tunnel and transport boundaries · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Original IP packet -> Outer routed packet	ESP IP protocol50: confidentiality/integrity policy
Mechanism / state	Transport retains original outer IP header	Transport: original IP + protected upper-layer data
Result / boundary	Identify mode, selectors, peer endpoints, ESP/AH choice and counters; verify remote-access/site-to-site routing plus traffic protection evidence.	AH does not provide confidentiality and is problematic across address-changing NAT. IPsec protection depends on negotiated algorithms and policy, not protocol name alone.

Bridge: Activity 9: ACL/management or Activity 10: L2/wireless security

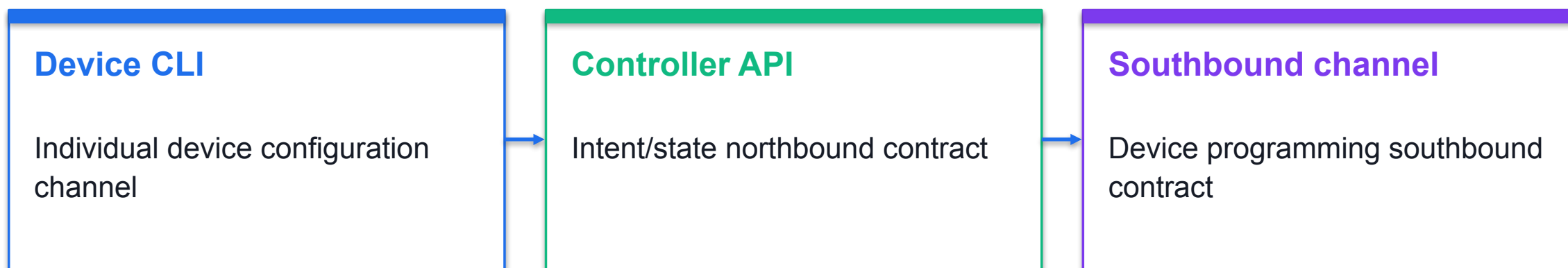
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Automation and Programmability

Automation and Programmability

Named boundaries -> runtime trace -> exact contract -> failure/control
-> acceptance evidence

Traditional and controller operation - boundaries

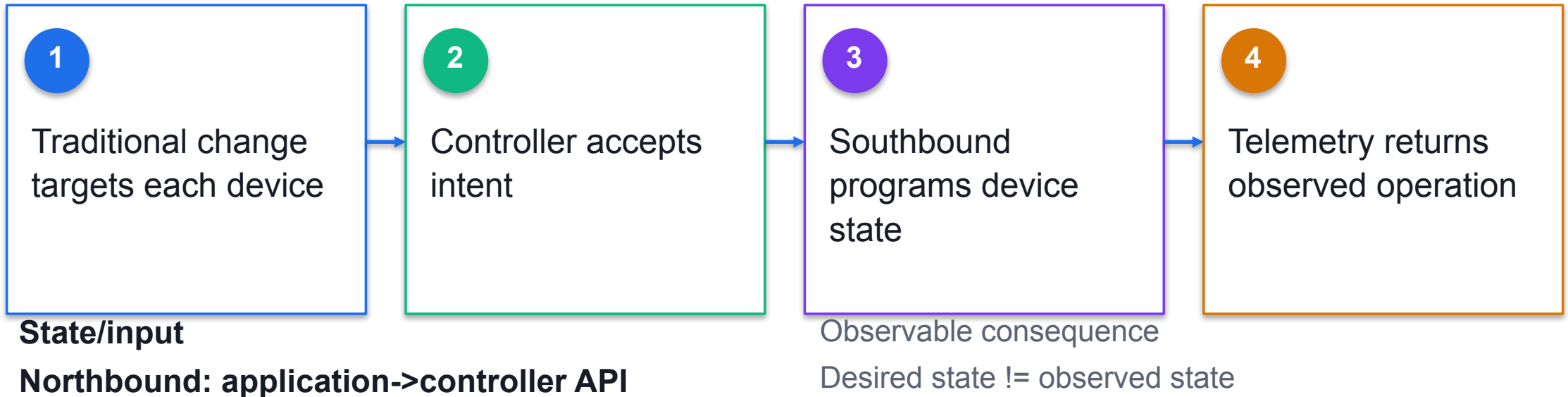


Northbound: application->controller API
Desired state != observed state

Boundary to inspect
Centralization reduces coordination work
but creates control-system dependencies

Read device state after controller operation; compare target inventory, requested policy and observed forwarding.

Traditional and controller operation - runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Traditional and controller operation · exact contract

Northbound: application->controller API

Southbound: controller->device

Desired state != observed state

INPUT / IDENTITY

Device CLI · Traditional change targets each device

DECISION / STATE

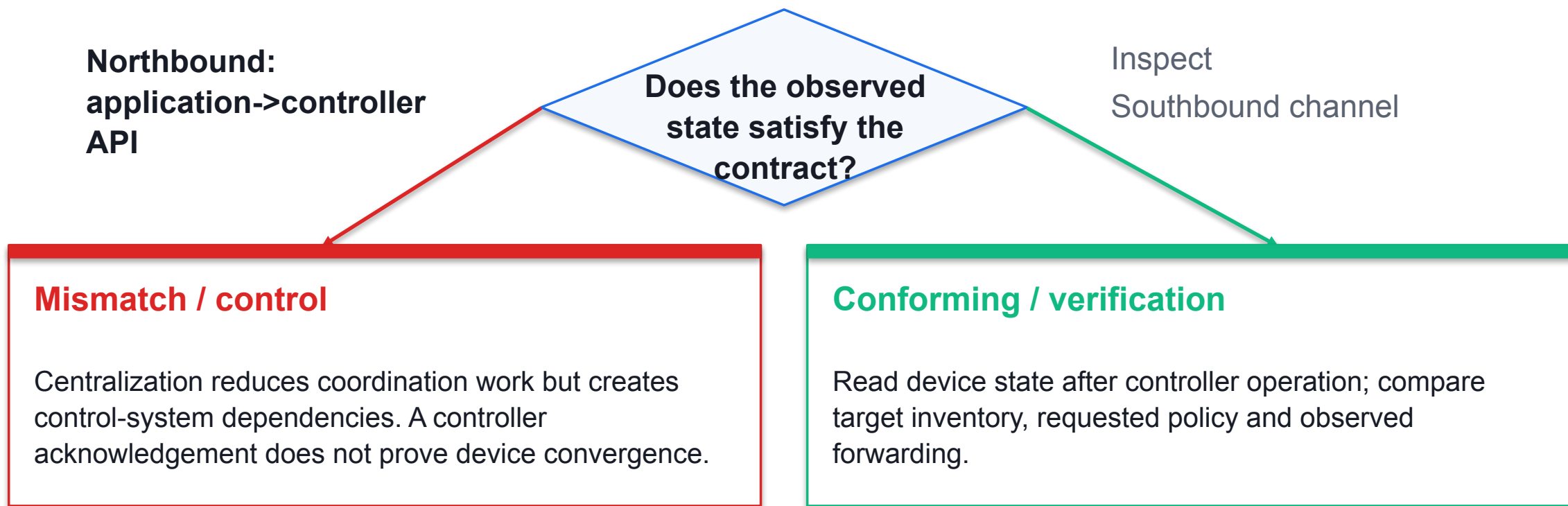
Southbound programs device state

READBCK / RESULT

Telemetry returns observed operation

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Traditional and controller operation - fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

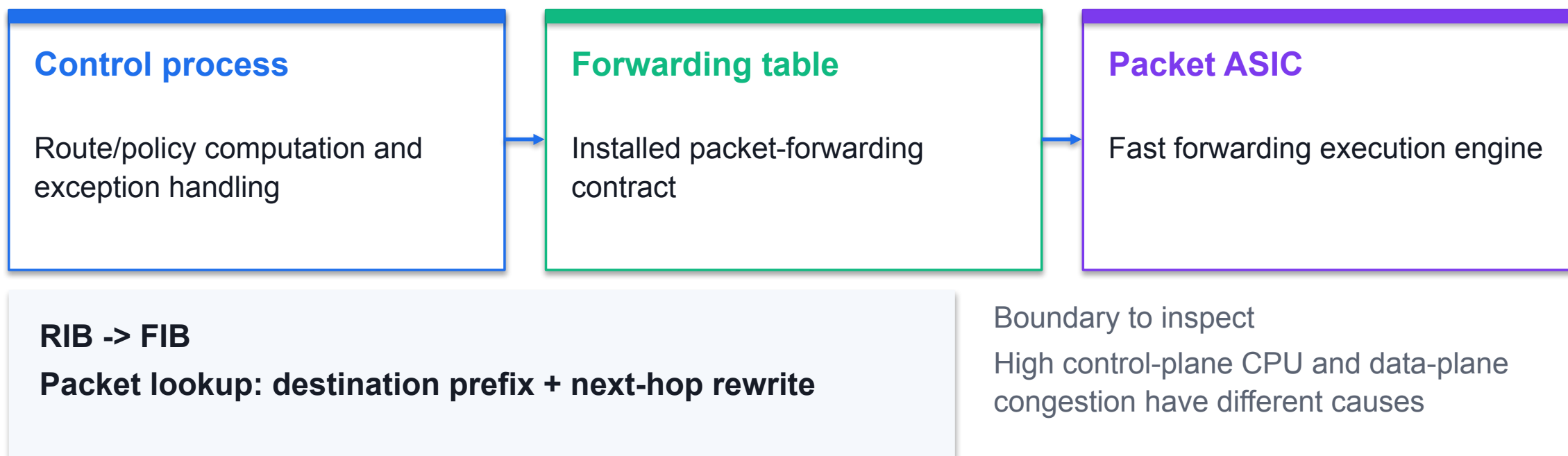
Traditional and controller operation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Device CLI -> Southbound channel	Northbound: application->controller API
Mechanism / state	Telemetry returns observed operation	Desired state != observed state
Result / boundary	Read device state after controller operation; compare target inventory, requested policy and observed forwarding.	Centralization reduces coordination work but creates control-system dependencies. A controller acknowledgement does not prove device convergence.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

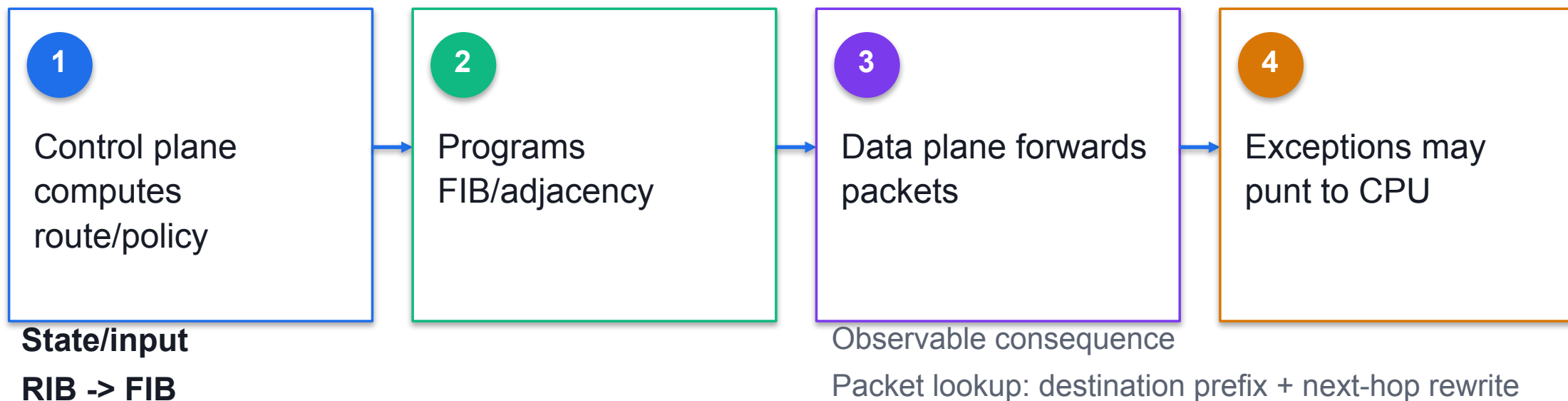
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Control and data plane separation · boundaries



Compare route computation, installed FIB and interface forwarding counters; identify whether failure is computation or packet execution.

Control and data plane separation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Control and data plane separation · exact contract

RIB -> FIB

ARP/ND -> adjacency

Packet lookup: destination prefix + next-hop rewrite

INPUT / IDENTITY

Control process · Control plane
computes route/policy

DECISION / STATE

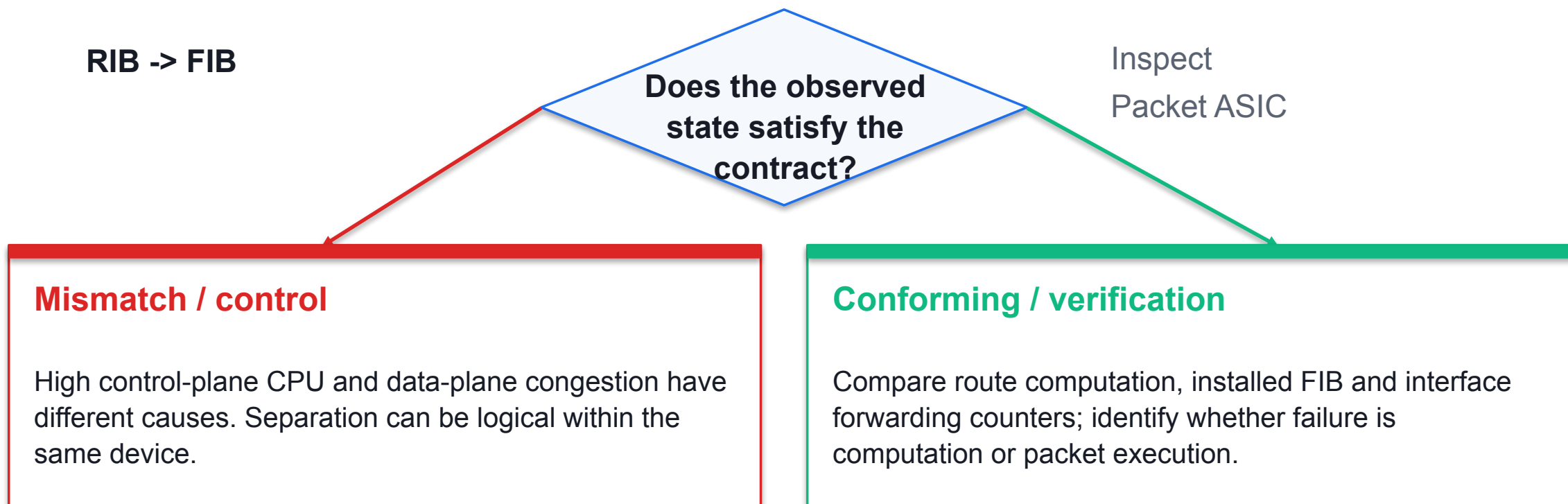
Data plane forwards packets

READBACK / RESULT

Exceptions may punt to CPU

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Control and data plane separation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

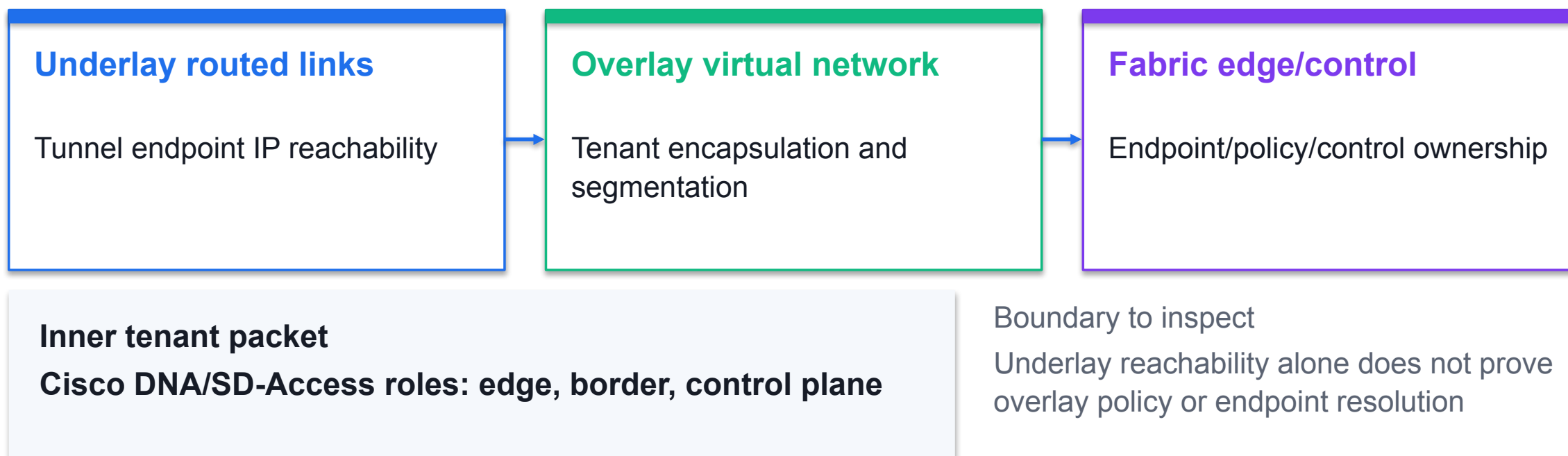
Control and data plane separation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Control process -> Packet ASIC	RIB -> FIB
Mechanism / state	Exceptions may punt to CPU	Packet lookup: destination prefix + next-hop rewrite
Result / boundary	Compare route computation, installed FIB and interface forwarding counters; identify whether failure is computation or packet execution.	High control-plane CPU and data-plane congestion have different causes. Separation can be logical within the same device.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

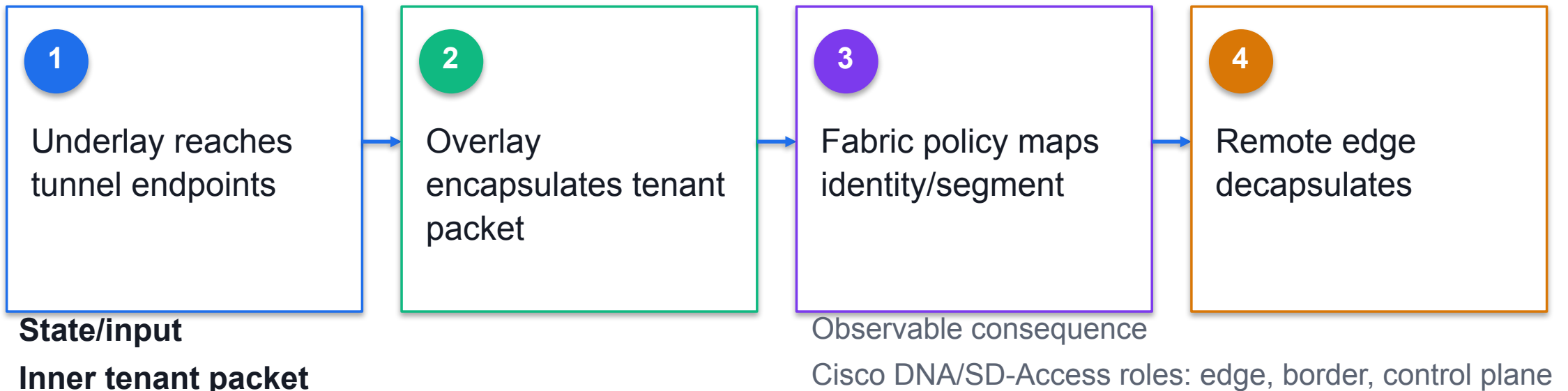
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Overlay underlay and fabric roles · boundaries



Check endpoint reachability, fabric role and policy mapping; inspect inner/outer address distinction in a sourced architecture exhibit.

Overlay underlay and fabric roles - runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Overlay underlay and fabric roles · exact contract

Inner tenant packet

Outer underlay IP + tunnel header

Cisco DNA/SD-Access roles: edge, border, control plane

INPUT / IDENTITY

Underlay routed links · Underlay reaches tunnel endpoints

DECISION / STATE

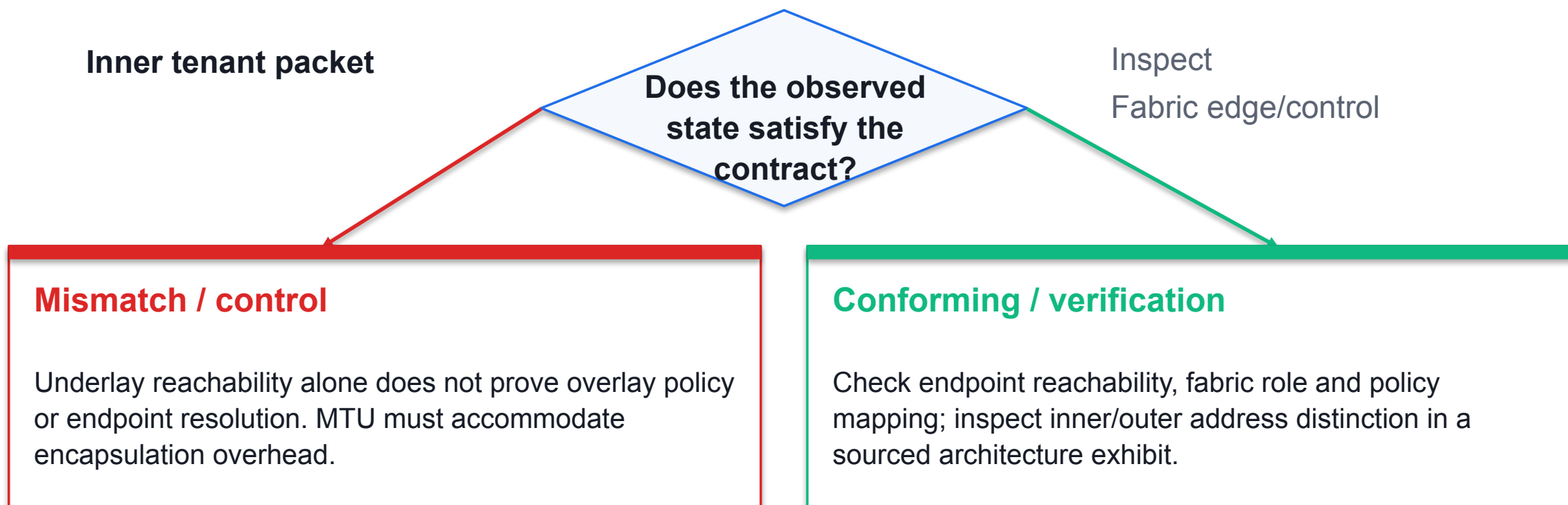
Fabric policy maps identity/segment

READBACK / RESULT

Remote edge decapsulates

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Overlay underlay and fabric roles - fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

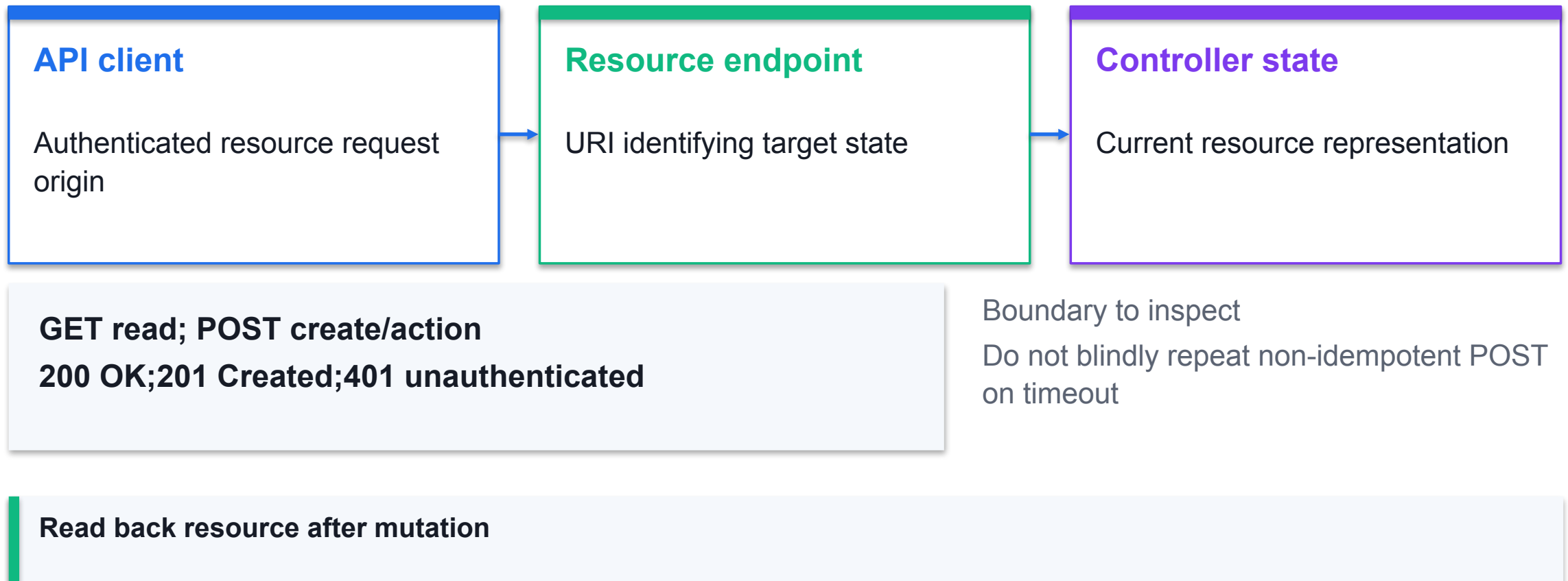
Overlay underlay and fabric roles · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Underlay routed links -> Fabric edge/control	Inner tenant packet
Mechanism / state	Remote edge decapsulates	Cisco DNA/SD-Access roles: edge, border, control plane
Result / boundary	Check endpoint reachability, fabric role and policy mapping; inspect inner/outer address distinction in a sourced architecture exhibit.	Underlay reachability alone does not prove overlay policy or endpoint resolution. MTU must accommodate encapsulation overhead.

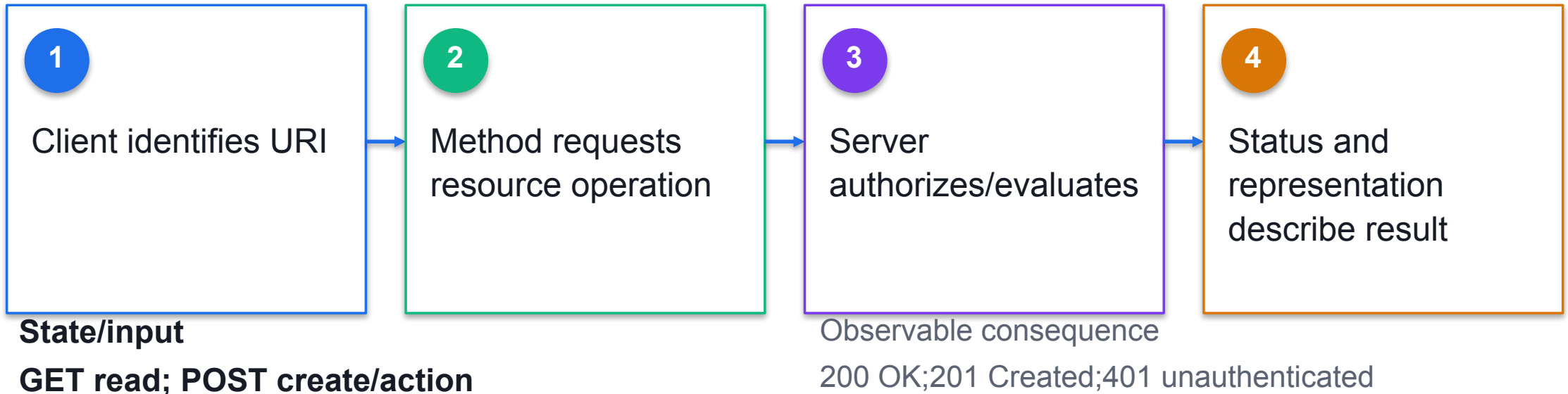
Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

REST resource and HTTP verbs · boundaries



REST resource and HTTP verbs · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

REST resource and HTTP verbs · exact contract

GET read; **POST** create/action
PUT replace target representation
PATCH partial update; **DELETE** remove
200 OK; **201 Created**; **401 unauthenticated**

INPUT / IDENTITY

API client · Client identifies URI

DECISION / STATE

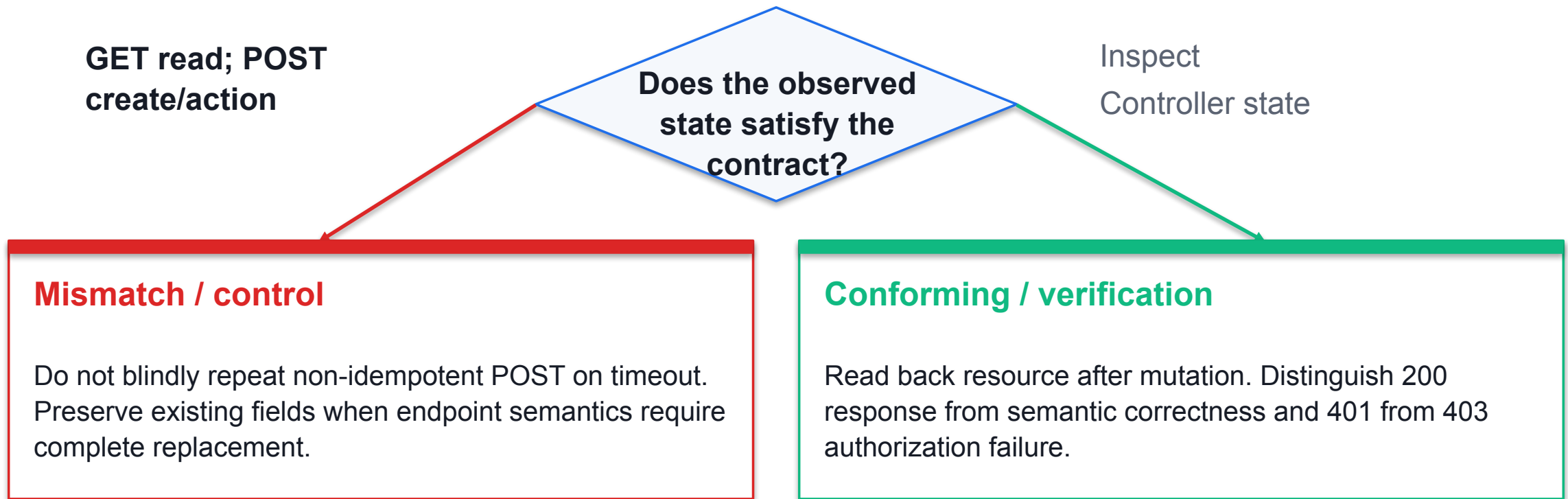
Server authorizes/evaluates

REDFACK / RESULT

Status and representation describe result

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

REST resource and HTTP verbs · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

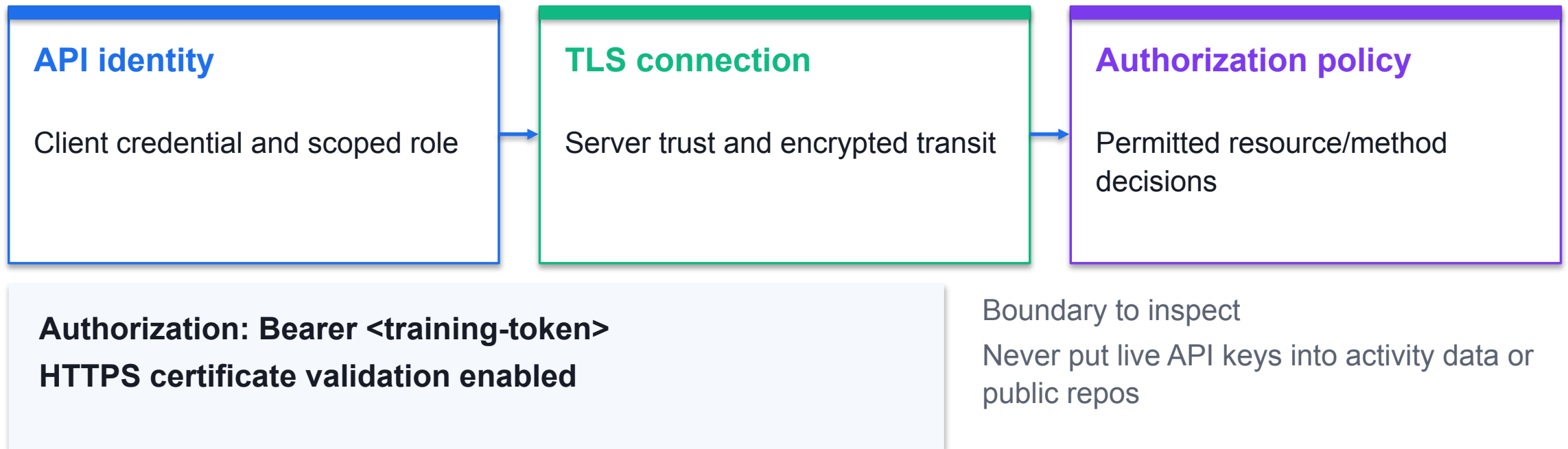
REST resource and HTTP verbs · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	API client -> Controller state	GET read; POST create/action
Mechanism / state	Status and representation describe result	200 OK;201 Created;401 unauthenticated
Result / boundary	Read back resource after mutation. Distinguish 200 response from semantic correctness and 401 from 403 authorization failure.	Do not blindly repeat non-idempotent POST on timeout. Preserve existing fields when endpoint semantics require complete replacement.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

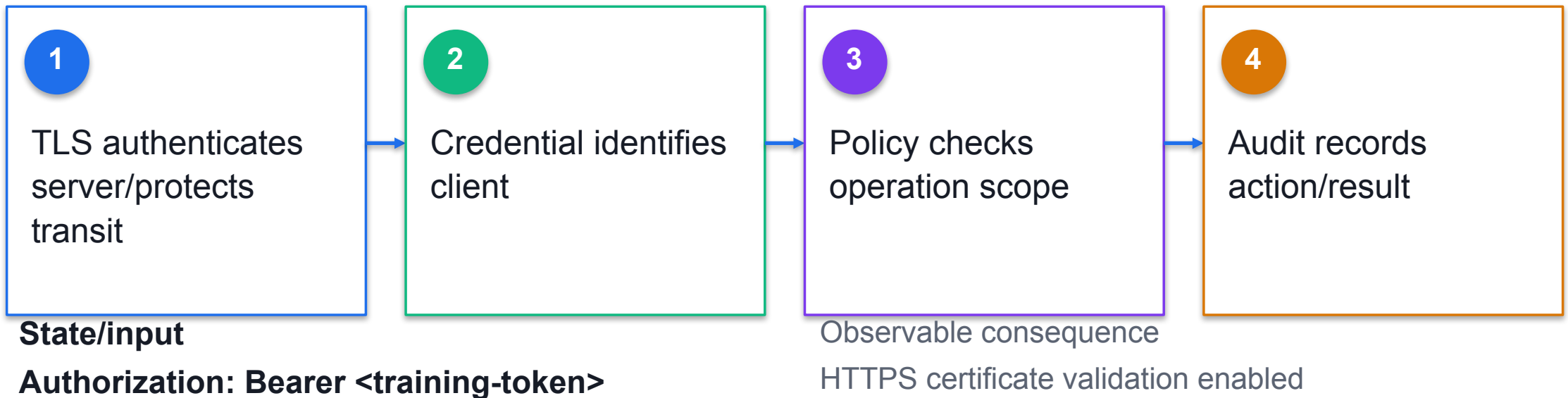
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

REST authentication and HTTPS · boundaries



Use local mock tokens; verify 401 for missing token, 403 for insufficient role and allowed scoped GET with correct identity.

REST authentication and HTTPS · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

REST authentication and HTTPS · exact contract

Authorization: Bearer <training-token>

Basic credential only over TLS

Token scope + expiry

HTTPS certificate validation enabled

INPUT / IDENTITY

API identity · TLS authenticates
server/protects transit

DECISION / STATE

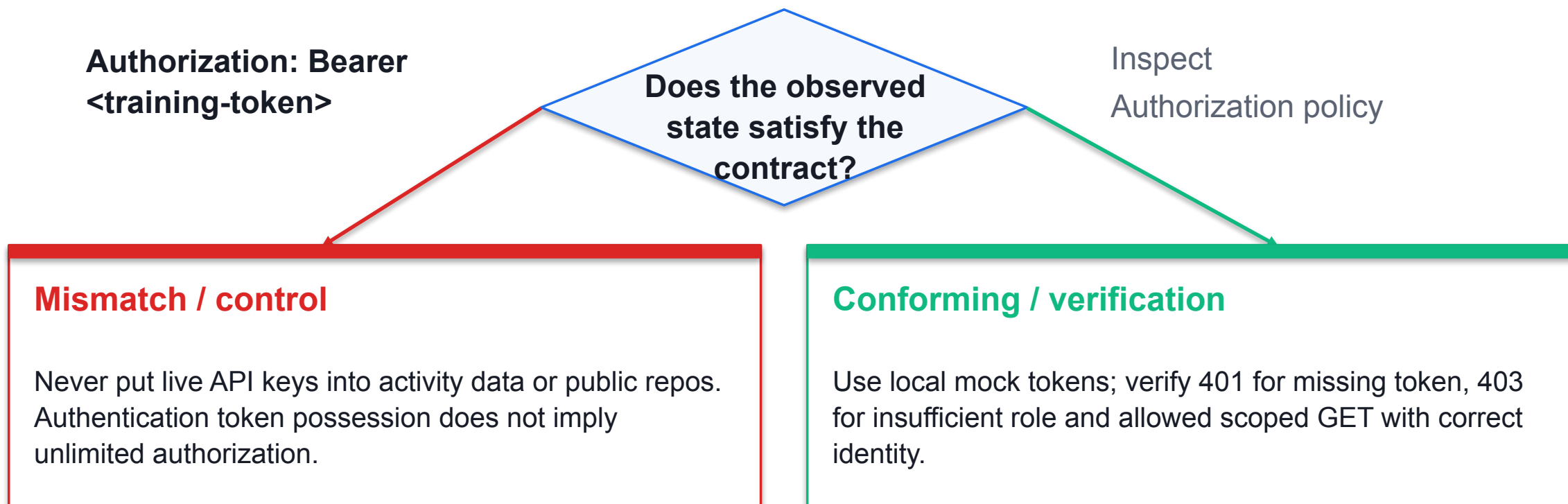
Policy checks operation scope

READBACK / RESULT

Audit records action/result

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

REST authentication and HTTPS · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

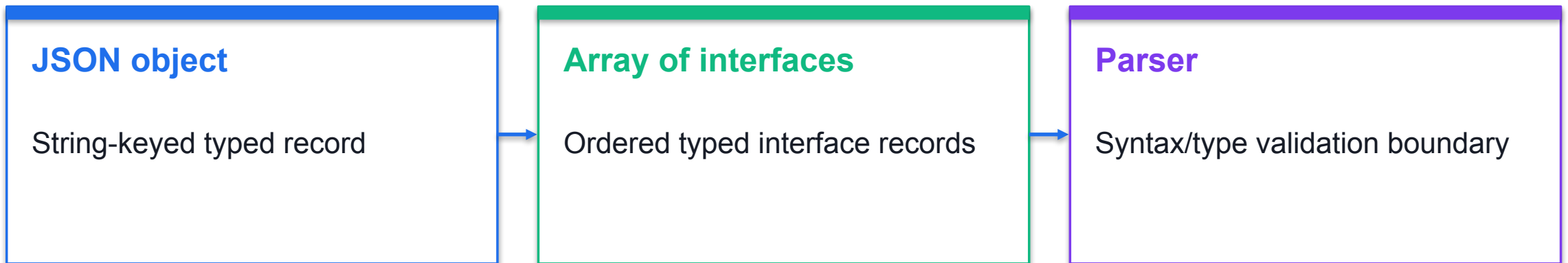
REST authentication and HTTPS · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	API identity -> Authorization policy	Authorization: Bearer <training-token>
Mechanism / state	Audit records action/result	HTTPS certificate validation enabled
Result / boundary	Use local mock tokens; verify 401 for missing token, 403 for insufficient role and allowed scoped GET with correct identity.	Never put live API keys into activity data or public repos. Authentication token possession does not imply unlimited authorization.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

JSON typed data contracts · boundaries

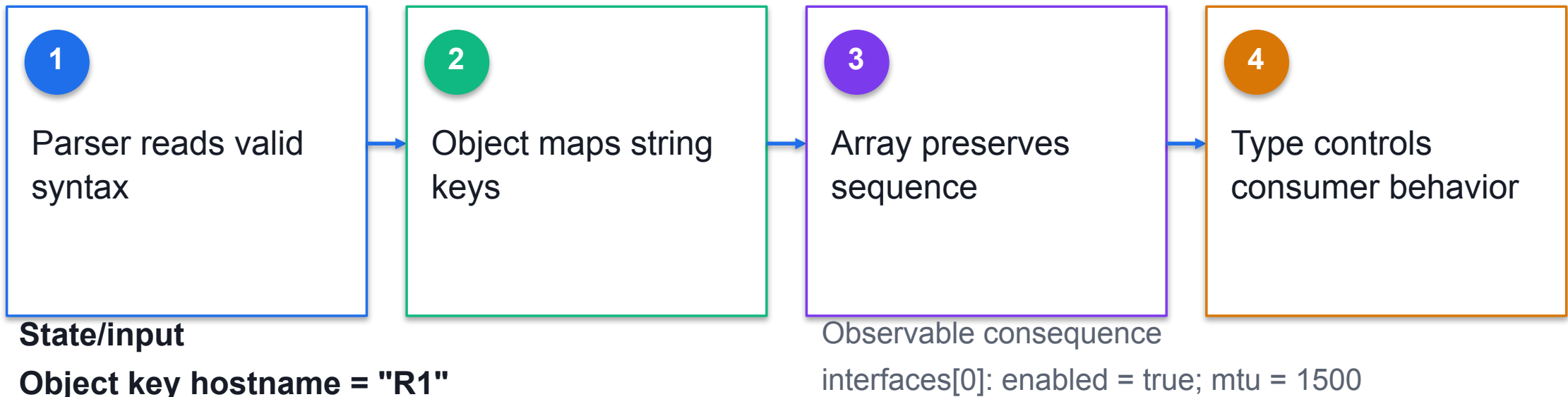


Object key hostname = "R1"
interfaces[0]: enabled = true; mtu = 1500

Boundary to inspect
JSON booleans are lowercase true/false;
trailing commas/comments are invalid

Parse with python json.loads; check enabled is boolean and mtu integer; validate required interface keys before acting.

JSON typed data contracts · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

JSON typed data contracts · exact contract

```
{"hostname":"R1","interfaces":  
[{"name":"Gi0/0","enabled":true,"mtu":1500}], "note":null}
```

INPUT / IDENTITY

JSON object · Parser reads valid syntax

DECISION / STATE

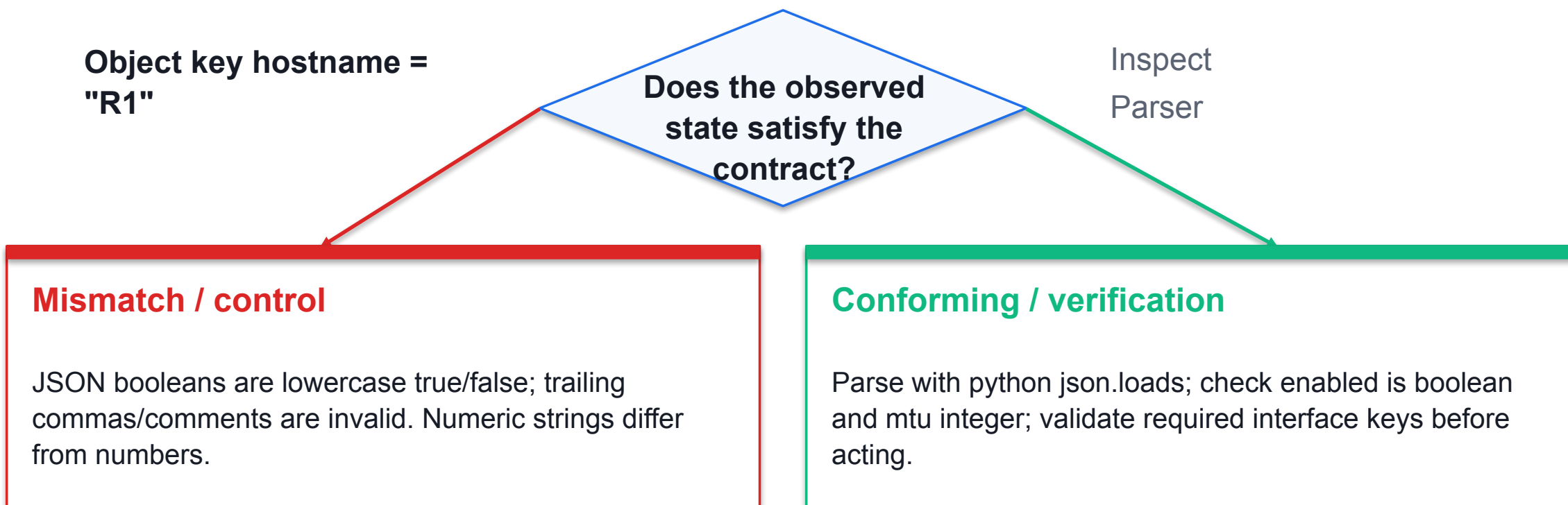
Array preserves sequence

REDFEEDBACK / RESULT

Type controls consumer behavior

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

JSON typed data contracts · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

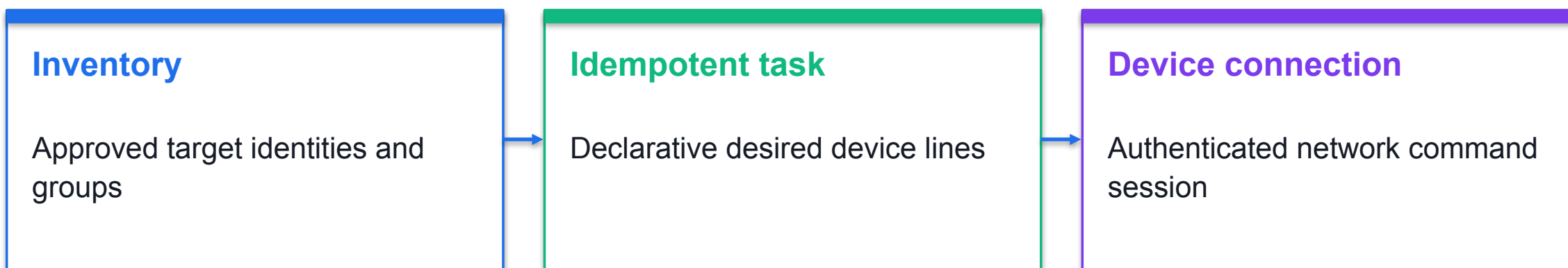
JSON typed data contracts · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	JSON object -> Parser	Object key hostname = "R1"
Mechanism / state	Type controls consumer behavior	interfaces[0]: enabled = true; mtu = 1500
Result / boundary	Parse with python json.loads; check enabled is boolean and mtu integer; validate required interface keys before acting.	JSON booleans are lowercase true/false; trailing commas/comments are invalid. Numeric strings differ from numbers.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Ansible desired configuration · boundaries

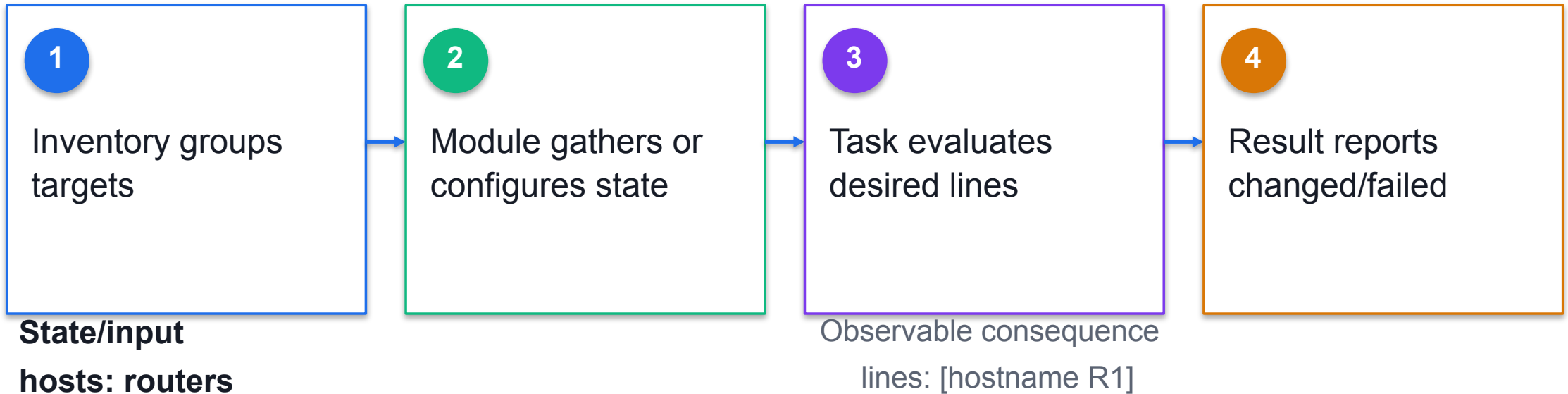


hosts: routers
lines: [hostname R1]

Boundary to inspect
Idempotence depends on module and task behavior; arbitrary shell tasks are not automatically idempotent

Validate inventory and inspect before/after configs; second identical run should report no change for an idempotent task.

Ansible desired configuration · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Ansible desired configuration · exact contract

```
hosts: routers
connection: ansible.netcommon.network_cli
tasks:
- cisco.ios.ios_config:
    lines: [hostname R1]
```

INPUT / IDENTITY

Inventory · Inventory groups targets

DECISION / STATE

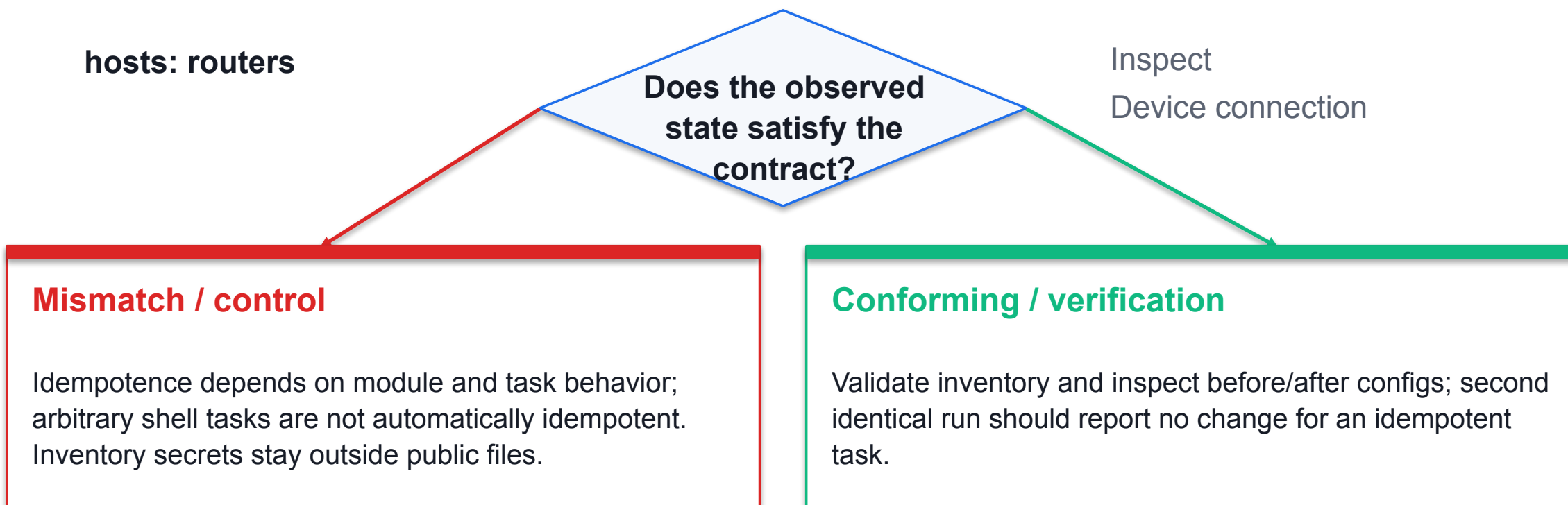
Task evaluates desired lines

READBCK / RESULT

Result reports changed/failed

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Ansible desired configuration · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

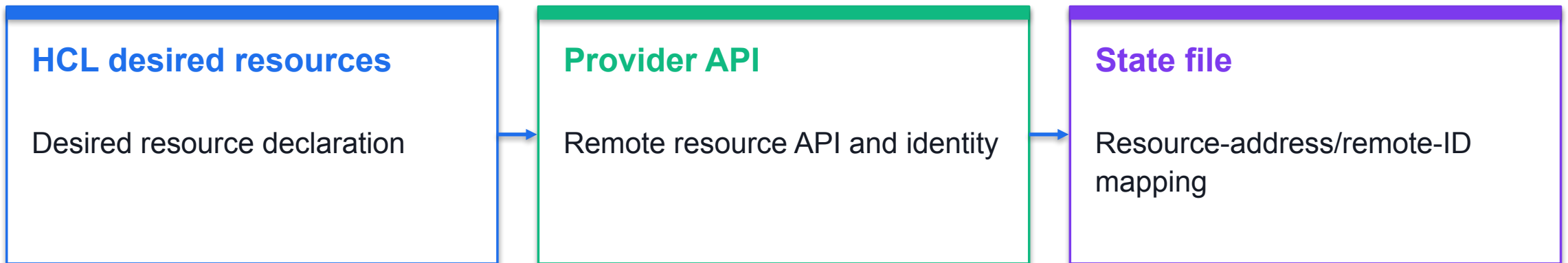
Ansible desired configuration · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Inventory -> Device connection	hosts: routers
Mechanism / state	Result reports changed/failed	lines: [hostname R1]
Result / boundary	Validate inventory and inspect before/after configs; second identical run should report no change for an idempotent task.	Idempotence depends on module and task behavior; arbitrary shell tasks are not automatically idempotent. Inventory secrets stay outside public files.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Terraform plan and state · boundaries



terraform init

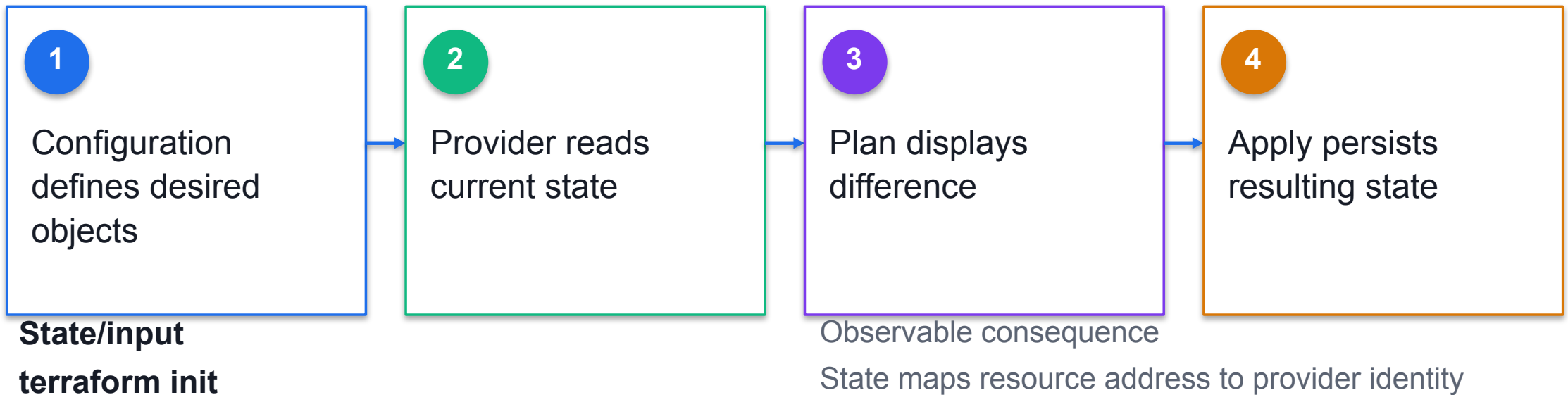
State maps resource address to provider identity

Boundary to inspect

State can hold sensitive values; never publish state or credentials

Review create/update/destroy actions, apply result and provider readback; protect state and record resource identity.

Terraform plan and state · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Terraform plan and state · exact contract

terraform init
terraform plan
terraform apply
State maps resource address to provider identity

INPUT / IDENTITY

HCL desired resources · Configuration defines desired objects

DECISION / STATE

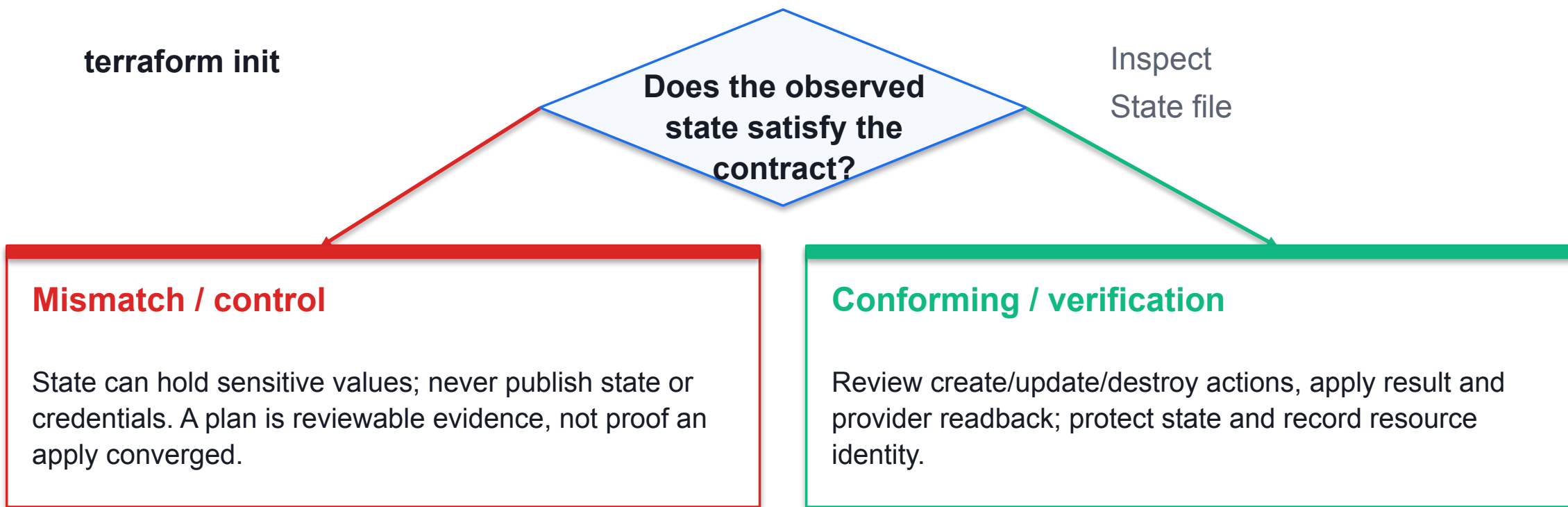
Plan displays difference

REDFEEDBACK / RESULT

Apply persists resulting state

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Terraform plan and state · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

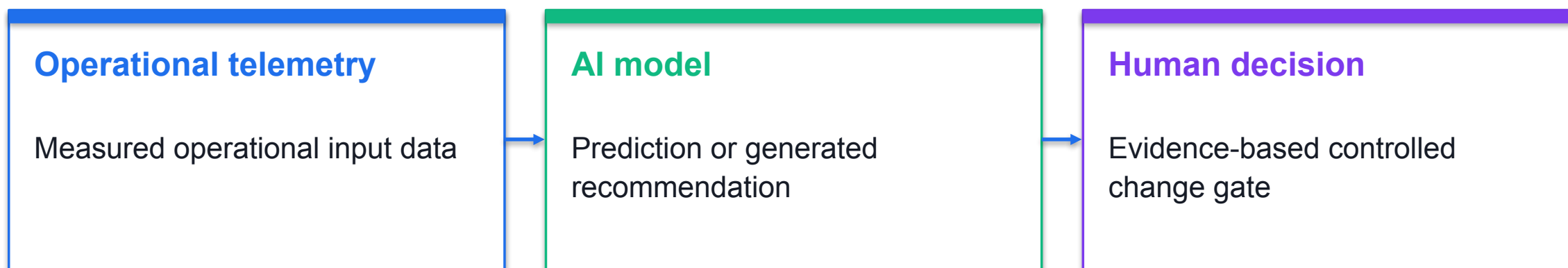
Terraform plan and state · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	HCL desired resources -> State file	terraform init
Mechanism / state	Apply persists resulting state	State maps resource address to provider identity
Result / boundary	Review create/update/destroy actions, apply result and provider readback; protect state and record resource identity.	State can hold sensitive values; never publish state or credentials. A plan is reviewable evidence, not proof an apply converged.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Generative and predictive AI evidence · boundaries



Predictive: anomaly probability 0.92

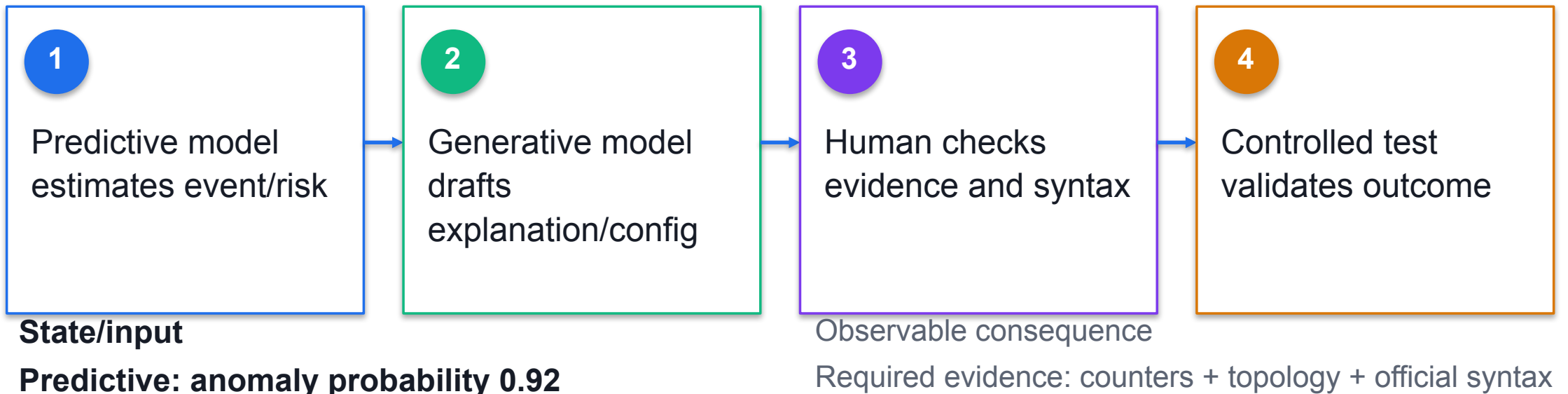
Required evidence: counters + topology + official syntax

Boundary to inspect

Generated config can hallucinate
commands or weaken controls

Require source trace, simulator test, rollback and reviewer approval before live change; compare prediction with actual observed event.

Generative and predictive AI evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Generative and predictive AI evidence · exact contract

Predictive: anomaly probability 0.92

Generative: suggested ACL/change

Required evidence: counters + topology + official syntax

INPUT / IDENTITY

Operational telemetry · Predictive model estimates event/risk

DECISION / STATE

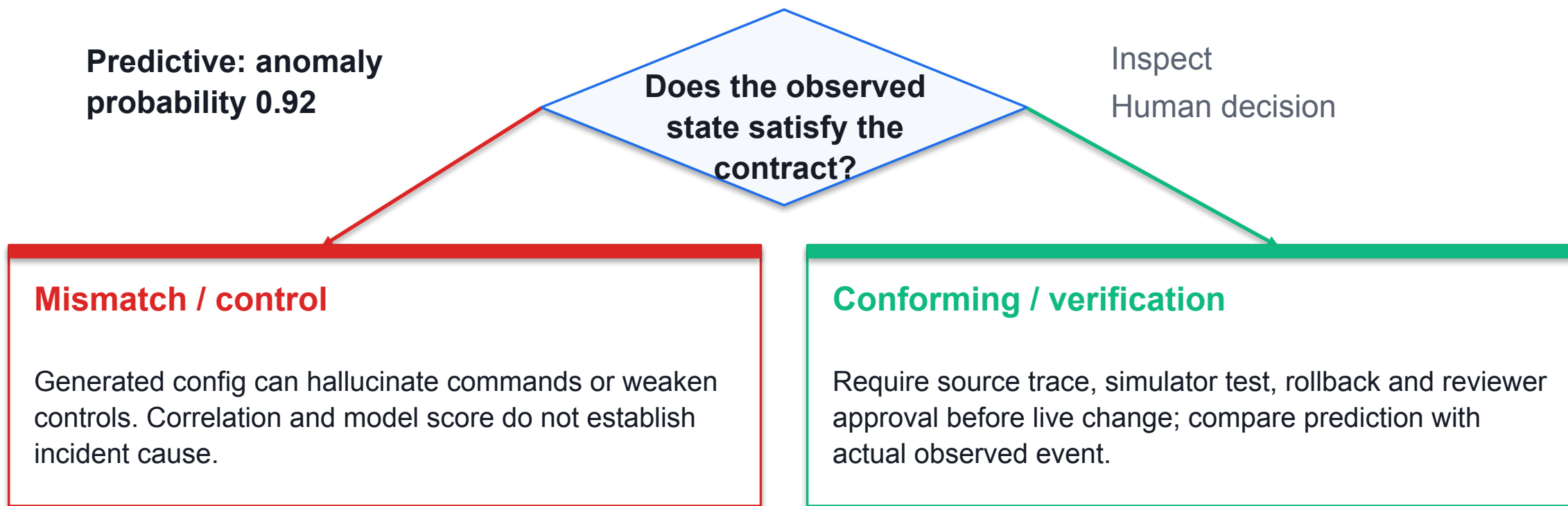
Human checks evidence and syntax

READBACK / RESULT

Controlled test validates outcome

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Generative and predictive AI evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

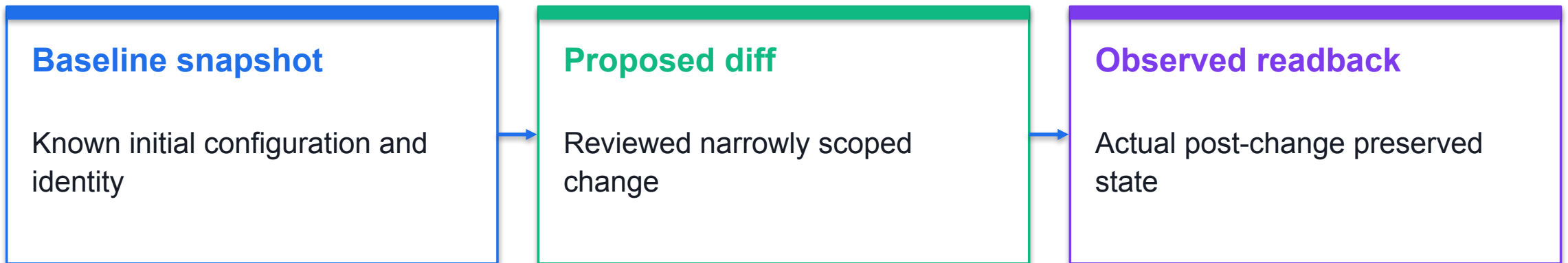
Generative and predictive AI evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Operational telemetry -> Human decision	Predictive: anomaly probability 0.92
Mechanism / state	Controlled test validates outcome	Required evidence: counters + topology + official syntax
Result / boundary	Require source trace, simulator test, rollback and reviewer approval before live change; compare prediction with actual observed event.	Generated config can hallucinate commands or weaken controls. Correlation and model score do not establish incident cause.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Automation change verification · boundaries

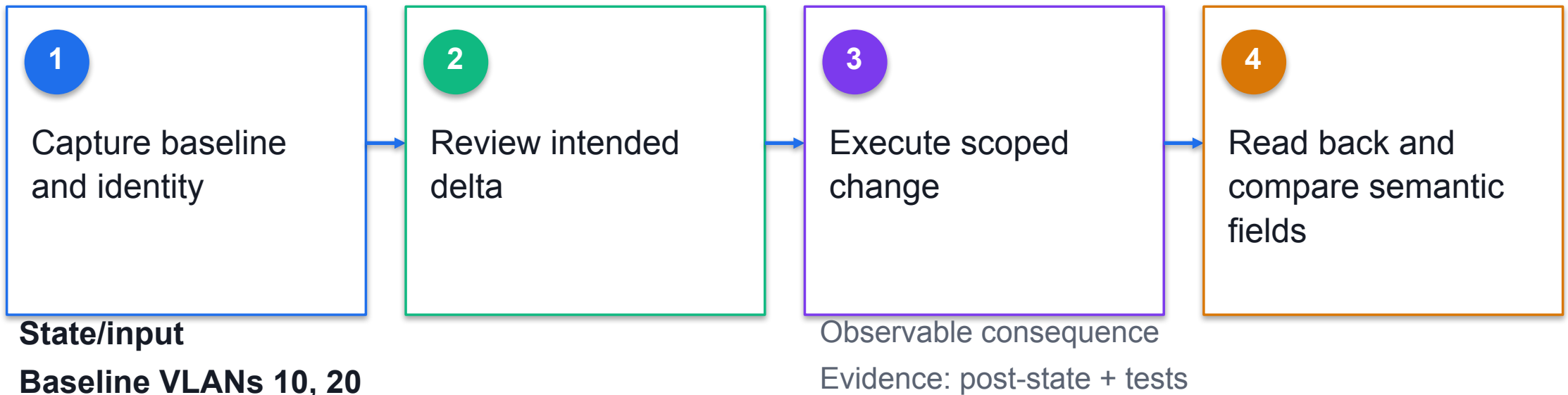


Baseline VLANs 10, 20
Evidence: post-state + tests

Boundary to inspect
A successful API response or tool exit is insufficient

Validate target identity, preserved fields, intended change and end-to-end checks; retain rollback artifact without secrets.

Automation change verification · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Automation change verification · exact contract

Baseline VLANs 10, 20

Proposed addVLAN 30

Expected unchanged: VLAN 10/20, uplink, AAA

Evidence: post-state + tests

INPUT / IDENTITY

Baseline snapshot · Capture baseline and identity

DECISION / STATE

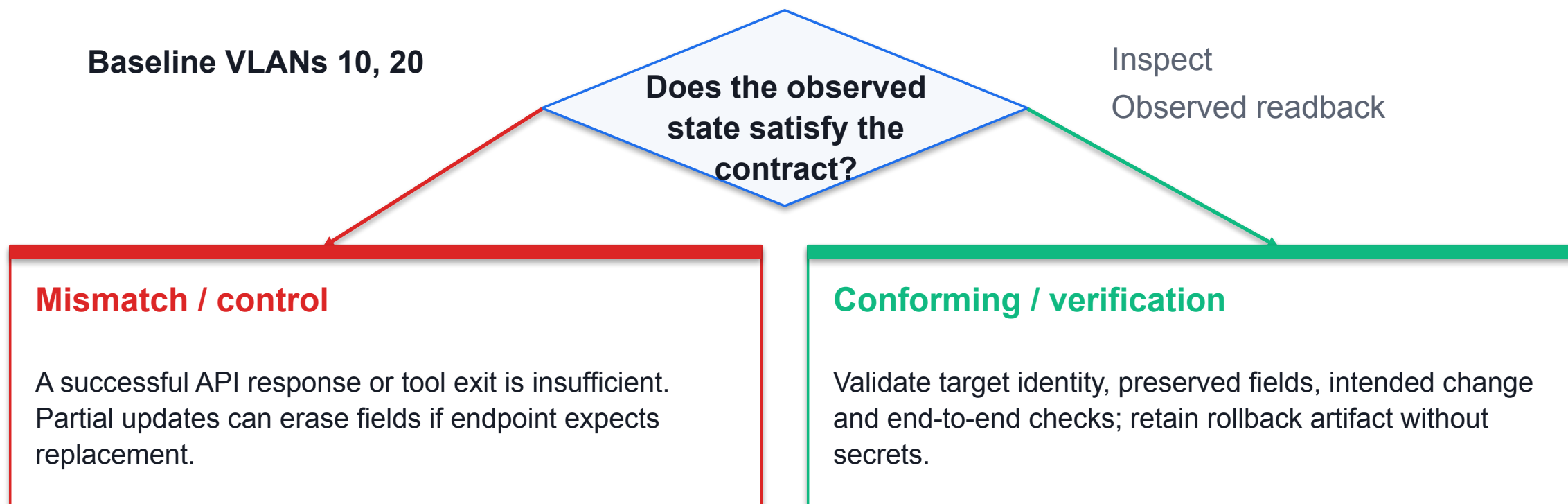
Execute scoped change

READBACK / RESULT

Read back and compare semantic fields

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Automation change verification · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

Automation change verification · acceptance evidence

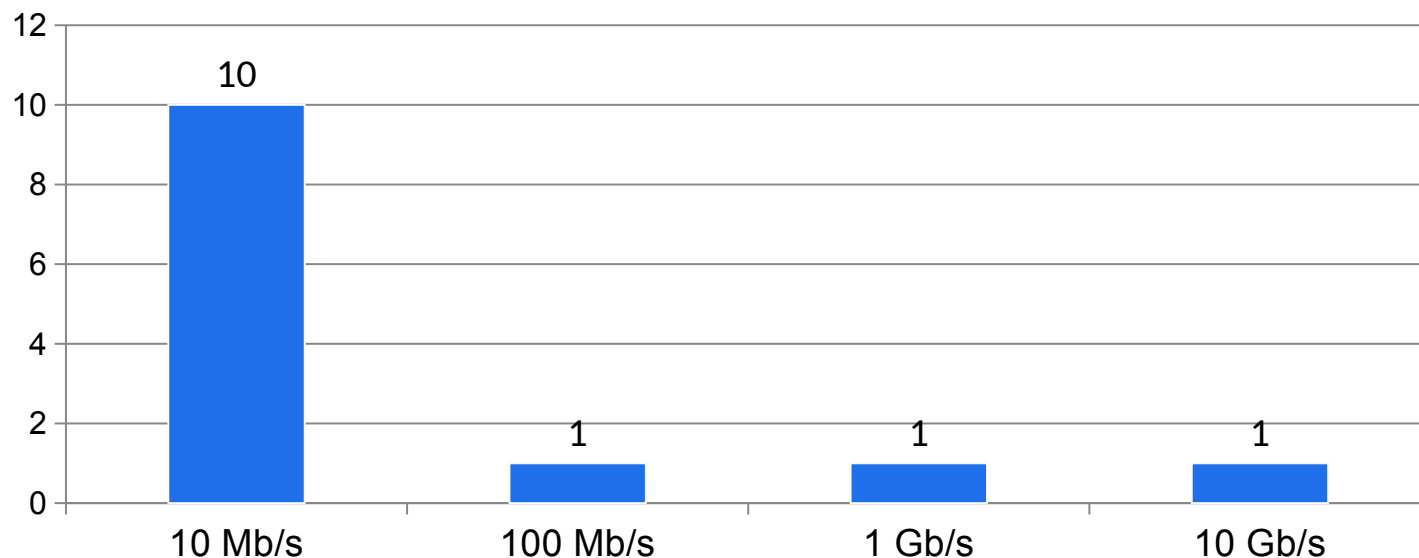
Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Baseline snapshot -> Observed readback	Baseline VLANs 10, 20
Mechanism / state	Read back and compare semantic fields	Evidence: post-state + tests
Result / boundary	Validate target identity, preserved fields, intended change and end-to-end checks; retain rollback artifact without secrets.	A successful API response or tool exit is insufficient. Partial updates can erase fields if endpoint expects replacement.

Bridge: Activity 11: REST/JSON/automation or Activity 12: integrated verification

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPF default cost compresses fast links

OSPF cost



Default reference 100 Mb/s

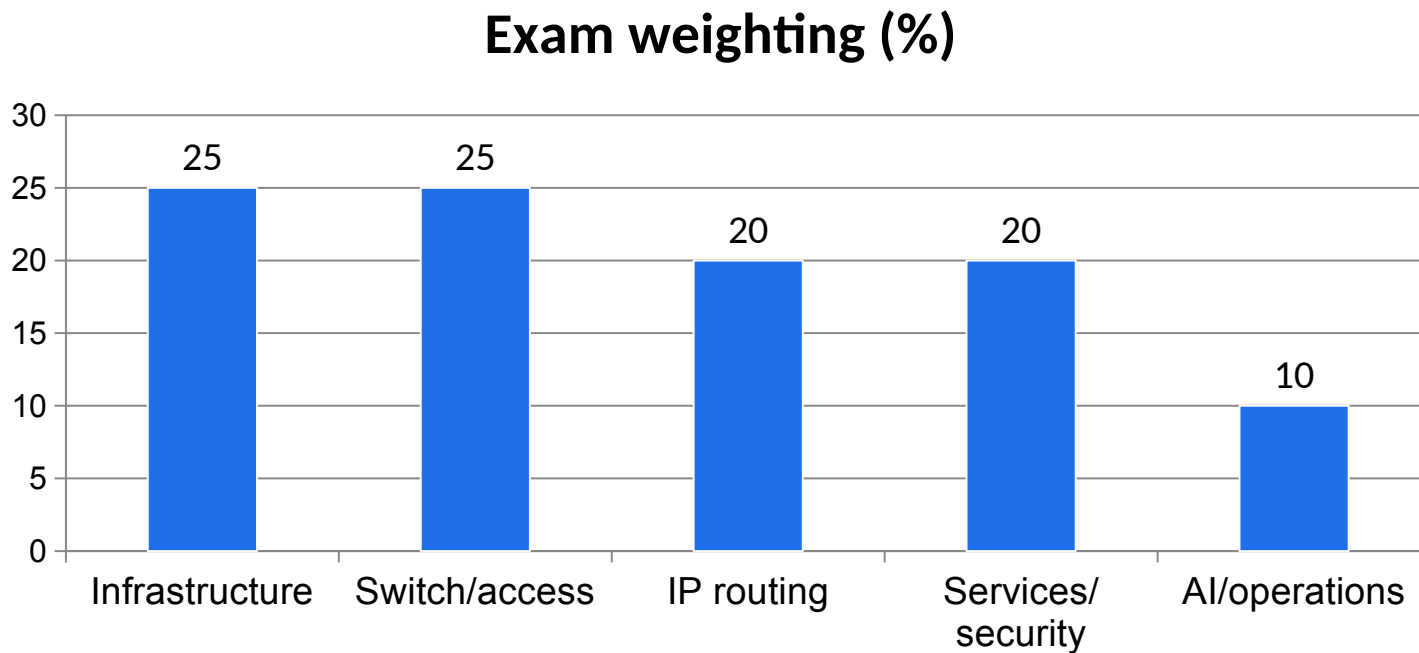
Cost = reference / bandwidth

Minimum cost 1

Consistent updated reference is needed to distinguish high speeds.

What the data shows: Default reference 100 Mb/s

Announced v2.0: five-domain model



Network Infrastructure and Connectivity 25%

Switching and Network Access 25%

IP Routing 20%

Network Services and Security 20%

AI, and Network Operations and Management 10%

Effective 3 Feb 2027.

What the data shows: Network Infrastructure and Connectivity 25%

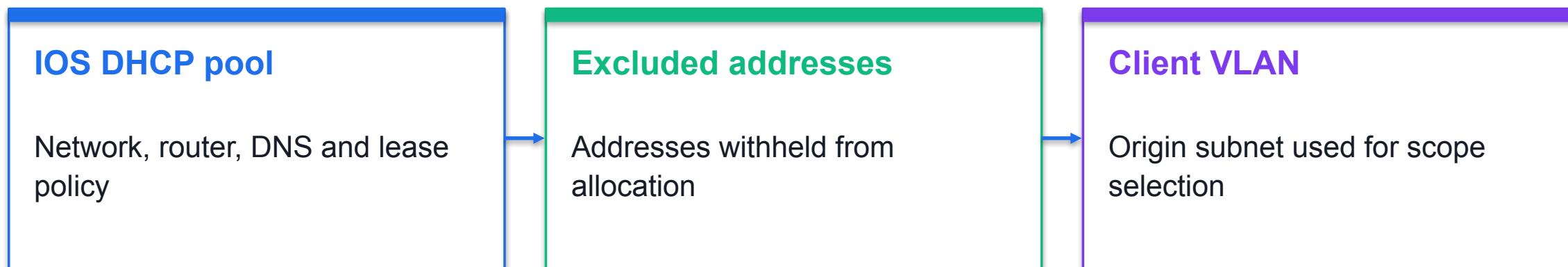
Announced CCNA v2.0 transition

Apply operational evidence to the newly announced objectives

OSPFv3, VRRP, AAA clients, secure transfers, expanded DNS diagnosis, storm/RA controls and agentic AI.

Simulator/platform support varies; verify IOS XE syntax before execution.

DHCPv 4 IOS server fault isolation · boundaries

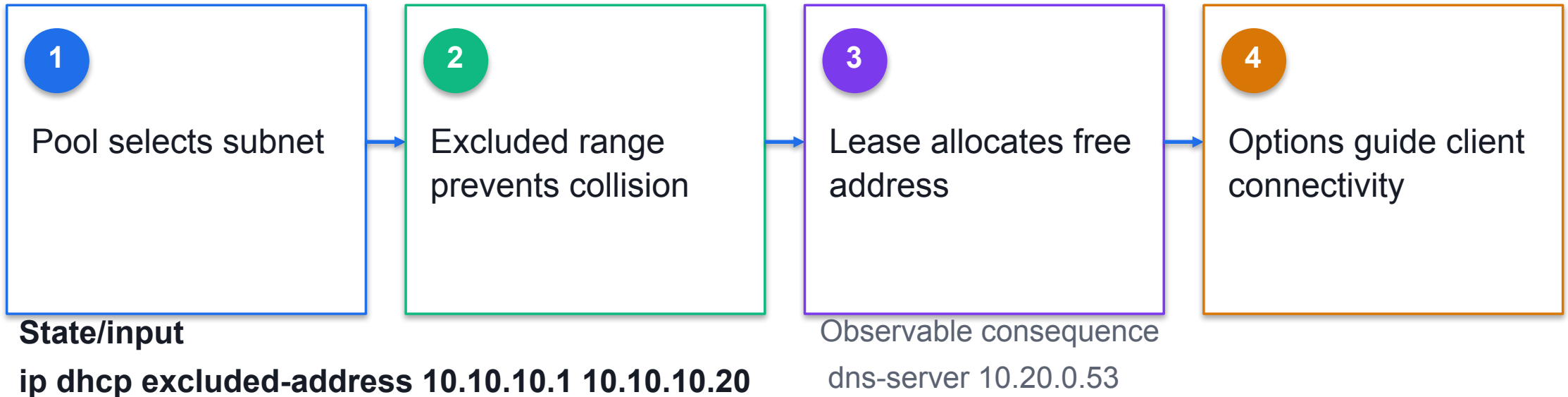


```
ip dhcp excluded-address 10.10.10.1 10.10.10.20  
dns-server 10.20.0.53
```

Boundary to inspect
Pool subnet must match client/relay scope

```
show ip dhcp pool; show ip dhcp binding; show ip dhcp conflict
```

DHCPv 4 IOS server fault isolation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

DHCPv 4 IOS server fault isolation · exact contract

```
ip dhcp excluded-address 10.10.10.1 10.10.10.20
ip dhcp pool USERS
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 10.20.0.53
```

INPUT / IDENTITY

IOS DHCP pool · Pool selects subnet

DECISION / STATE

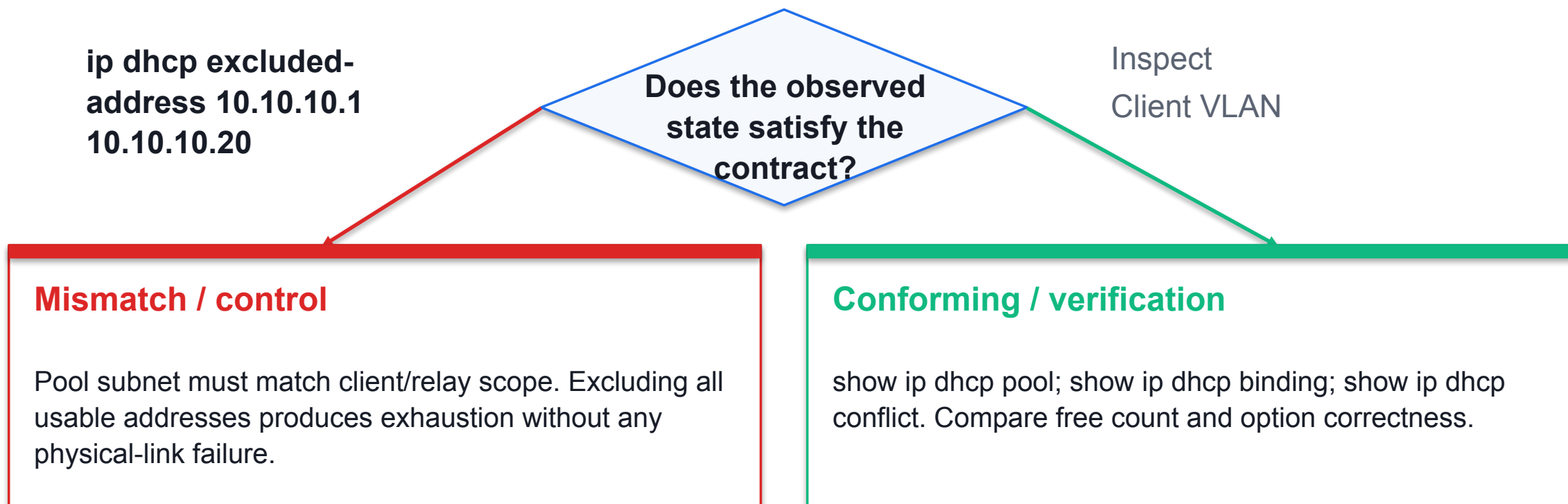
Lease allocates free address

READBACK / RESULT

Options guide client connectivity

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

DHCPv 4 IOS server fault isolation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

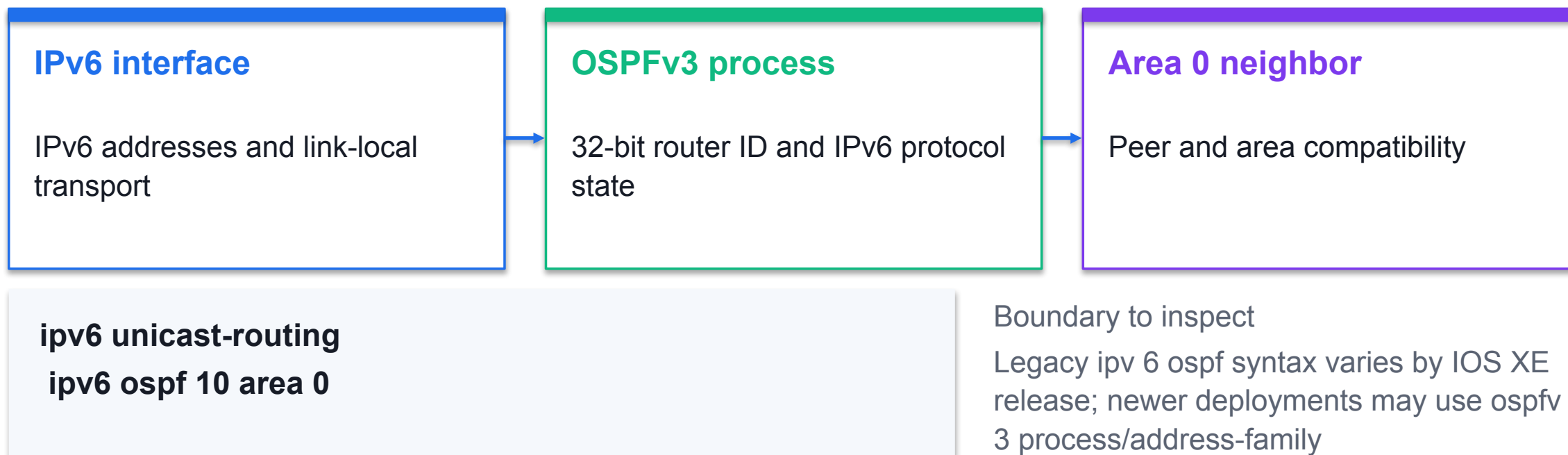
DHCPv 4 IOS server fault isolation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	IOS DHCP pool -> Client VLAN	ip dhcp excluded-address 10.10.10.1 10.10.10.20
Mechanism / state	Options guide client connectivity	dns-server 10.20.0.53
Result / boundary	show ip dhcp pool; show ip dhcp binding; show ip dhcp conflict. Compare free count and option correctness.	Pool subnet must match client/relay scope. Excluding all usable addresses produces exhaustion without any physical-link failure.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

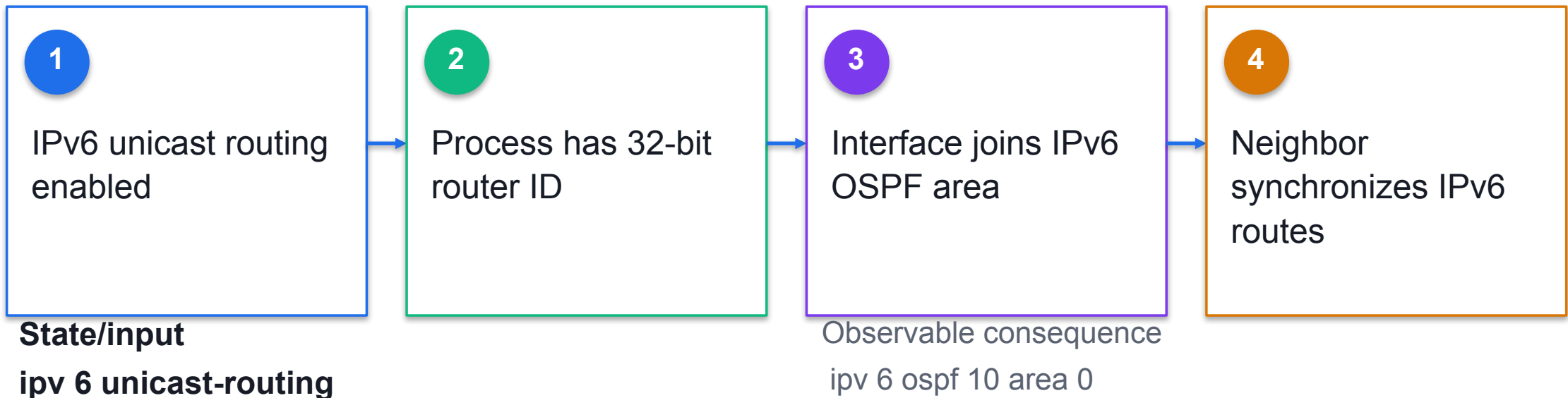
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPFv3 IPv6 interface activation · boundaries



show ipv 6 ospf neighbor; show ipv 6 route ospf

OSPFv3 IPv6 interface activation · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPFv3 IPv6 interface activation · exact contract

```
ipv6 unicast-routing  
ipv6 router ospf 10  
  router-id 1.1.1.1  
interface Gi0/0  
  ipv6 ospf 10 area 0
```

INPUT / IDENTITY

IPv6 interface · IPv6 unicast routing enabled

DECISION / STATE

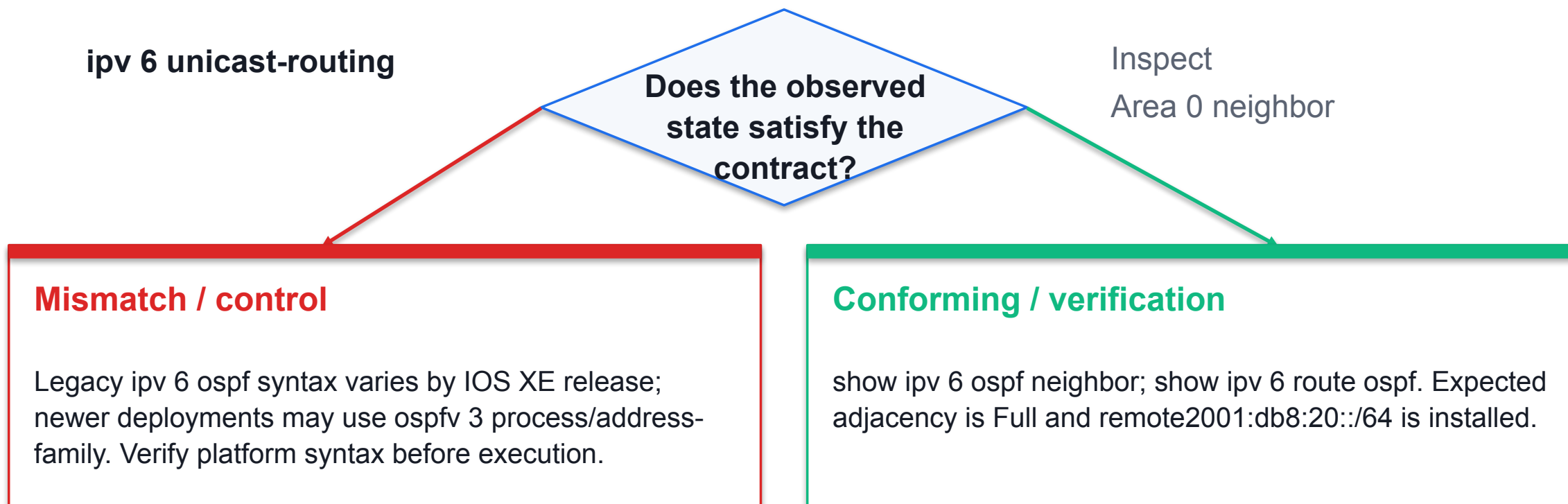
Interface joins IPv6 OSPF area

READBACK / RESULT

Neighbor synchronizes IPv6 routes

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPFv3 IPv6 interface activation · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

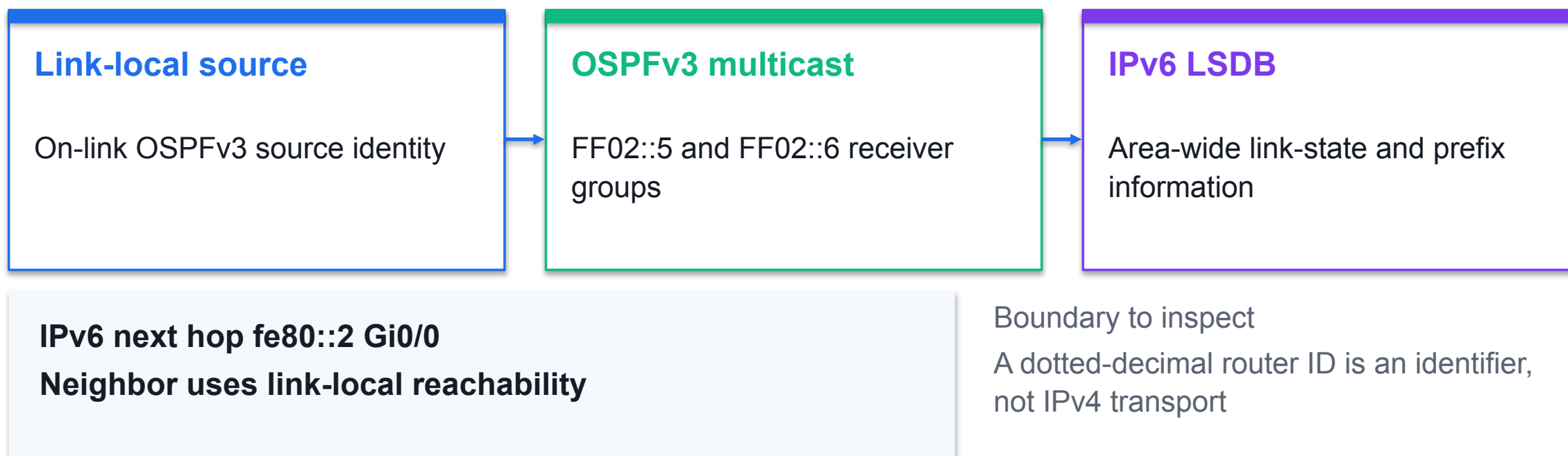
OSPFv3 IPv6 interface activation · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	IPv6 interface -> Area 0 neighbor	ipv 6 unicast-routing
Mechanism / state	Neighbor synchronizes IPv6 routes	ipv 6 ospf 10 area 0
Result / boundary	show ipv 6 ospf neighbor; show ipv 6 route ospf. Expected adjacency is Full and remote2001:db8:20::/64 is installed.	Legacy ipv 6 ospf syntax varies by IOS XE release; newer deployments may use ospfv 3 process/address-family. Verify platform syntax before execution.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

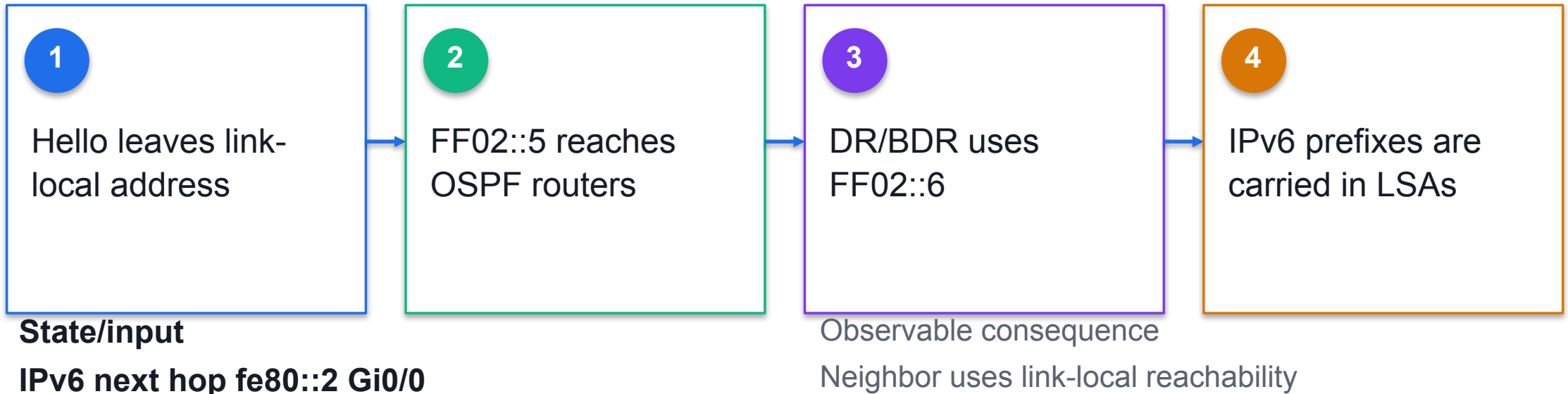
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

OSPFv3 neighbor transport · boundaries



Check link-local neighbor entry, area/timers/network type and Full state; separate transport failure from missing IPv6 advertisement.

OSPFv3 neighbor transport · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

OSPFv3 neighbor transport · exact contract

IPv6 next hop fe80::2 Gi0/0
OSPF IP protocol 89
Router ID stays 32-bit dotted decimal
Neighbor uses link-local reachability

INPUT / IDENTITY

Link-local source · Hello leaves link-local address

DECISION / STATE

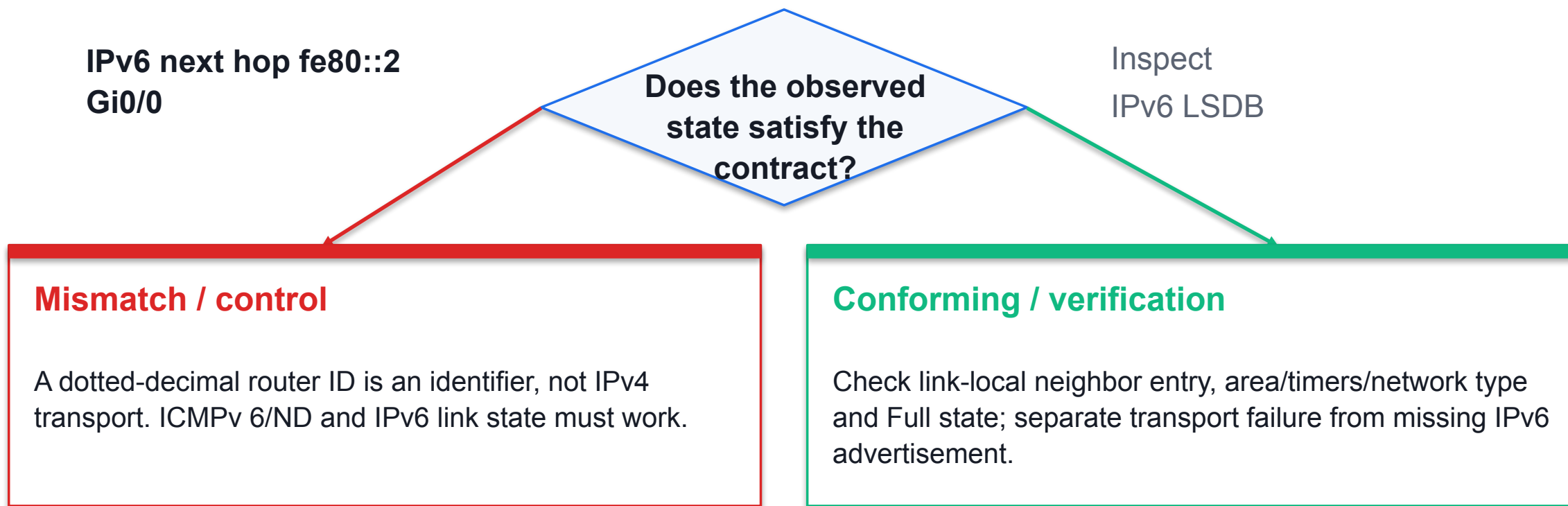
DR/BDR uses FF02::6

READBACK / RESULT

IPv6 prefixes are carried in LSAs

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

OSPFv3 neighbor transport · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

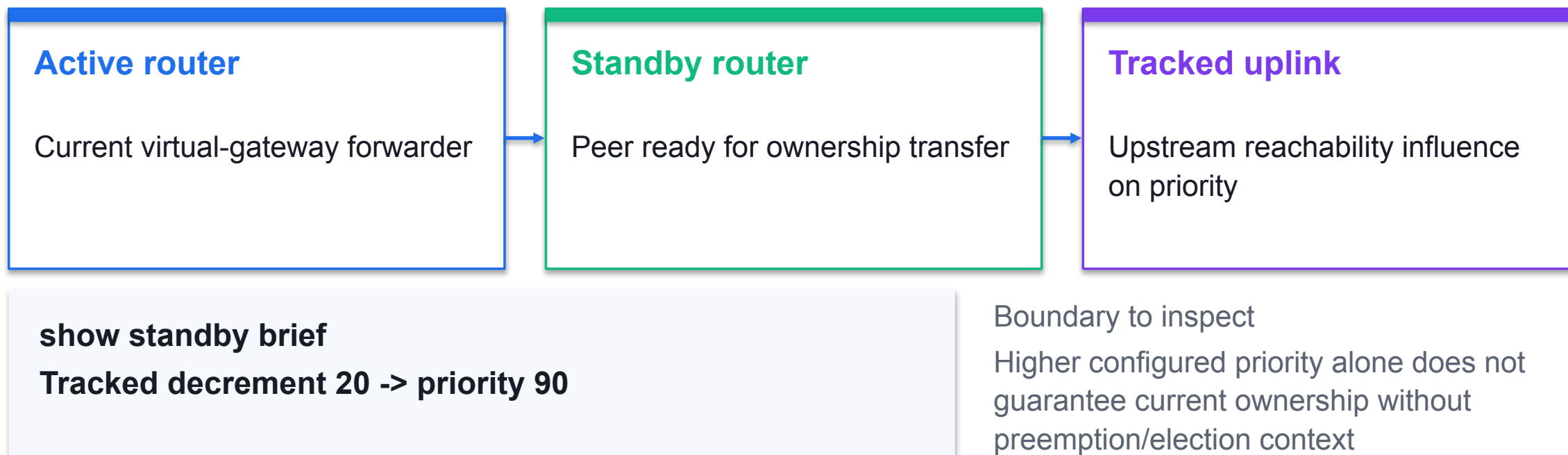
OSPFv3 neighbor transport · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Link-local source -> IPv6 LSDB	IPv6 next hop fe80::2 Gi0/0
Mechanism / state	IPv6 prefixes are carried in LSAs	Neighbor uses link-local reachability
Result / boundary	Check link-local neighbor entry, area/timers/network type and Full state; separate transport failure from missing IPv6 advertisement.	A dotted-decimal router ID is an identifier, not IPv4 transport. ICMPv 6/ND and IPv6 link state must work.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

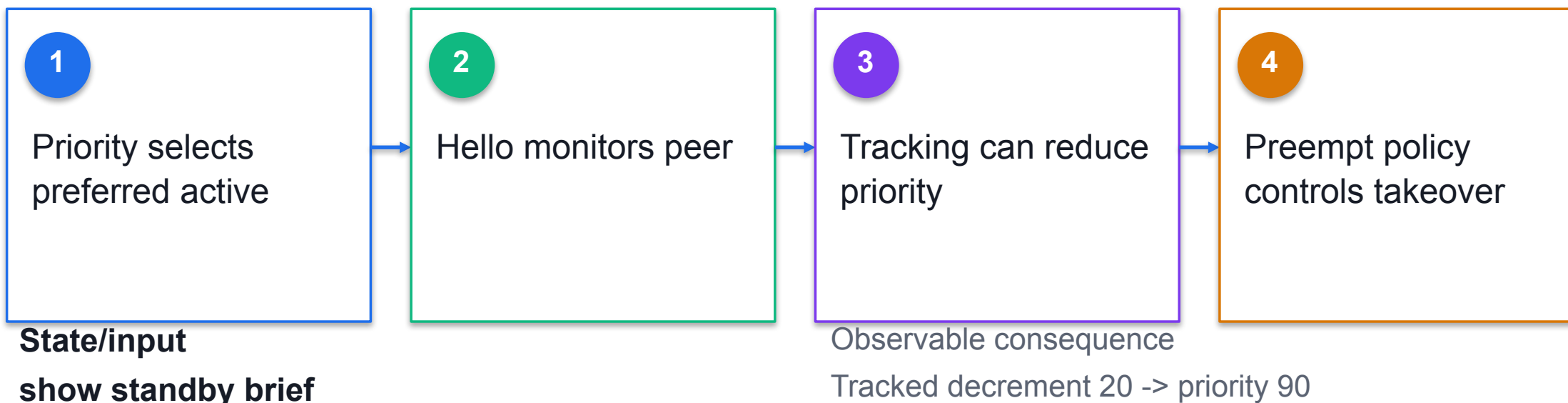
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

HSRP operational status diagnosis · boundaries



Read group, virtual IP, active/standby identities and effective priority before testing tracked uplink failure.

HSRP operational status diagnosis · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

HSRP operational status diagnosis · exact contract

show standby brief

Grp 10 Pri 110 State Active Active local

Standby 10.10.10.3 VIP 10.10.10.1

Tracked decrement 20 -> priority 90

INPUT / IDENTITY

Active router · Priority selects preferred active

DECISION / STATE

Tracking can reduce priority

READBACK / RESULT

Preempt policy controls takeover

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

HSRP operational status diagnosis · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

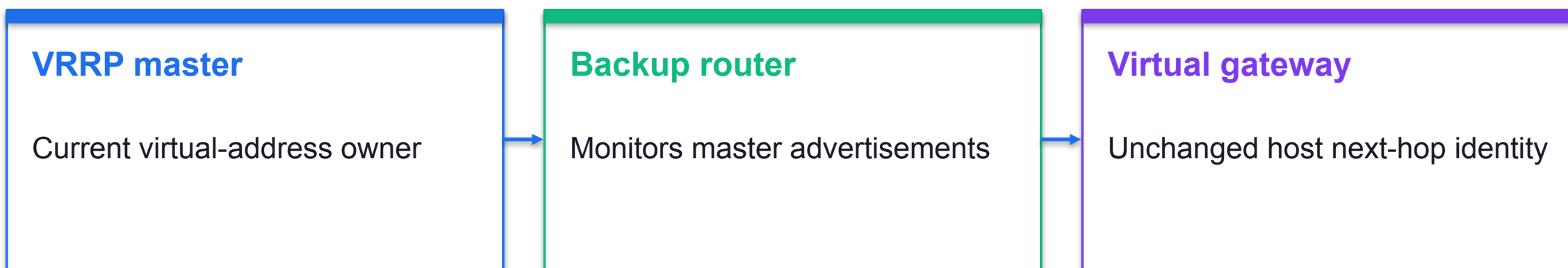
HSRP operational status diagnosis · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Active router -> Tracked uplink	show standby brief
Mechanism / state	Preempt policy controls takeover	Tracked decrement 20 -> priority 90
Result / boundary	Read group, virtual IP, active/standby identities and effective priority before testing tracked uplink failure.	Higher configured priority alone does not guarantee current ownership without preemption/election context. A failed upstream link can leave LAN interface alive.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

VRRP master and backup evidence · boundaries



show vrrp brief

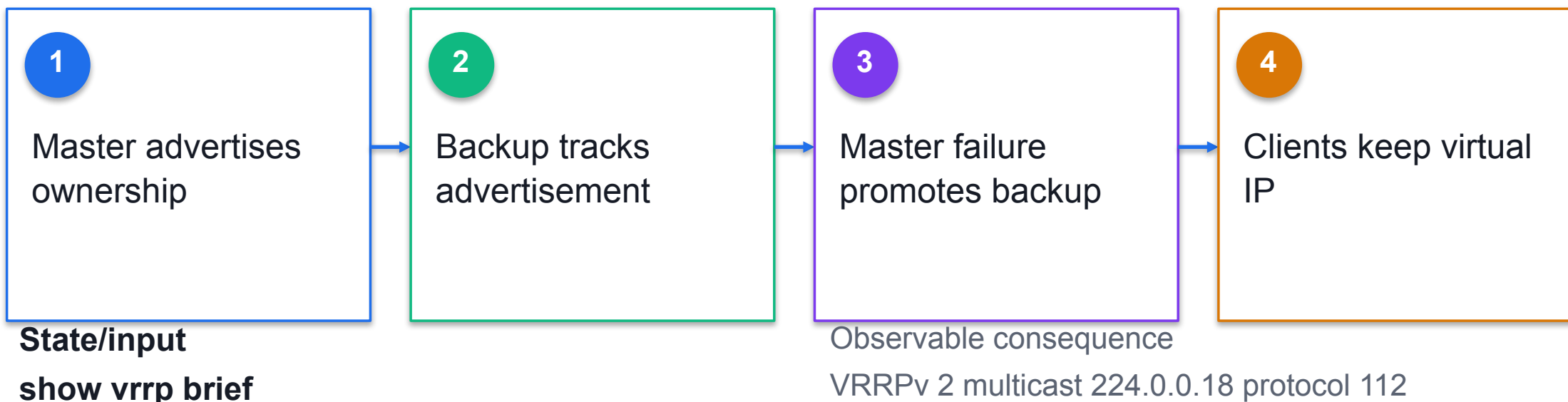
VRRPv 2 multicast 224.0.0.18 protocol 112

Boundary to inspect

VRRP terminology is master/backup, unlike HSRP active/standby

Confirm same group/VIP on peers and one master; compare virtual MAC0000.5e00.010a for group 10 in VRRPv 2.

VRRP master and backup evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

VRRP master and backup evidence · exact contract

show vrrp brief

Group 10 Virtual IP 10.10.10.1

R1 priority 110 MASTER

VRRPv 2 multicast 224.0.0.18 protocol 112

INPUT / IDENTITY

VRRP master · Master advertises ownership

DECISION / STATE

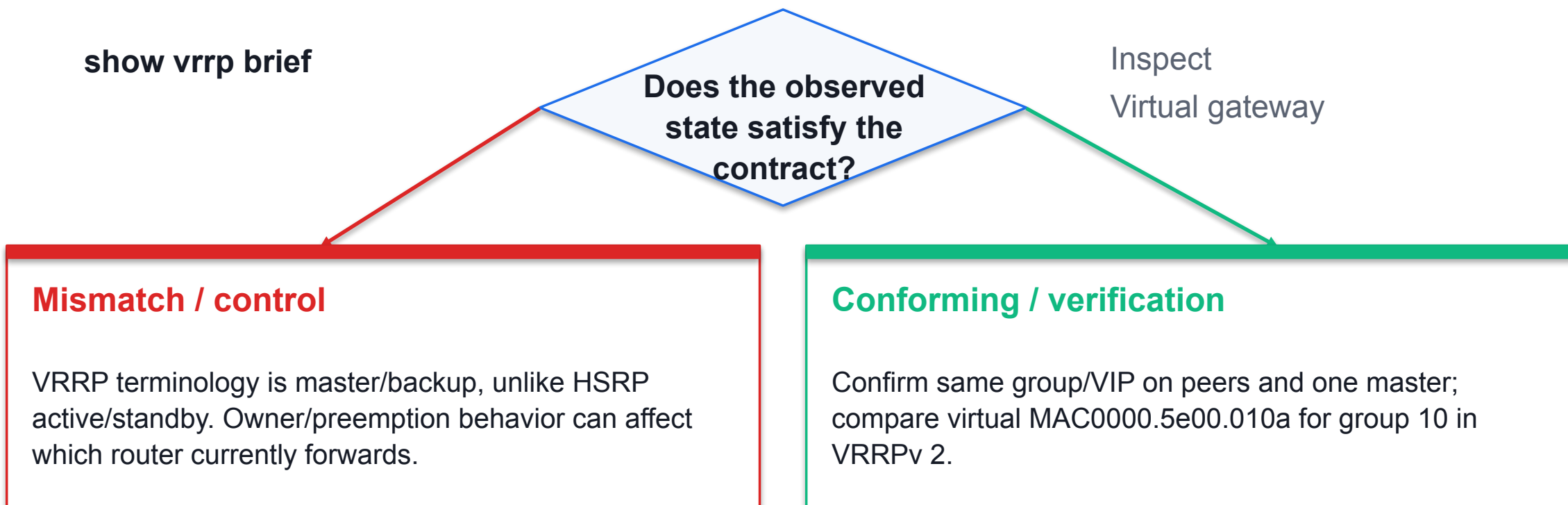
Master failure promotes backup

READBACK / RESULT

Clients keep virtual IP

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

VRRP master and backup evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

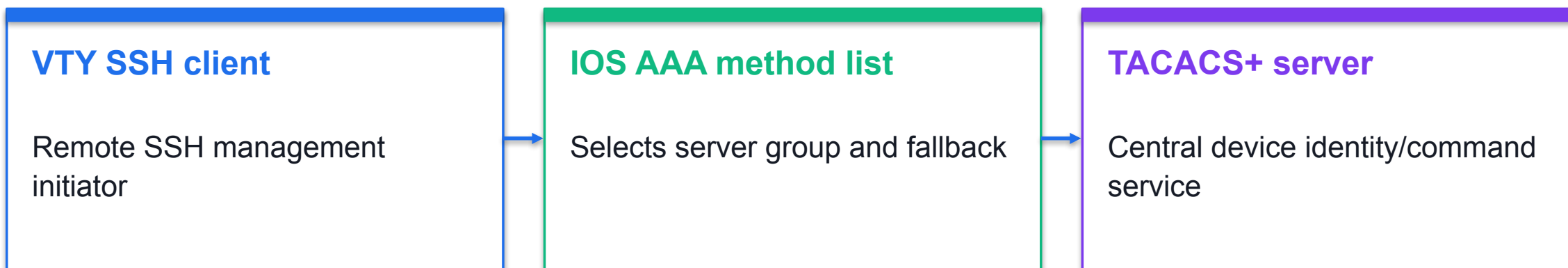
VRRP master and backup evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	VRRP master -> Virtual gateway	show vrrp brief
Mechanism / state	Clients keep virtual IP	VRRPv 2 multicast 224.0.0.18 protocol 112
Result / boundary	Confirm same group/VIP on peers and one master; compare virtual MAC0000.5e00.010a for group 10 in VRRPv 2.	VRRP terminology is master/backup, unlike HSRP active/standby. Owner/preemption behavior can affect which router currently forwards.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

TACACS+ IOS AAA client · boundaries

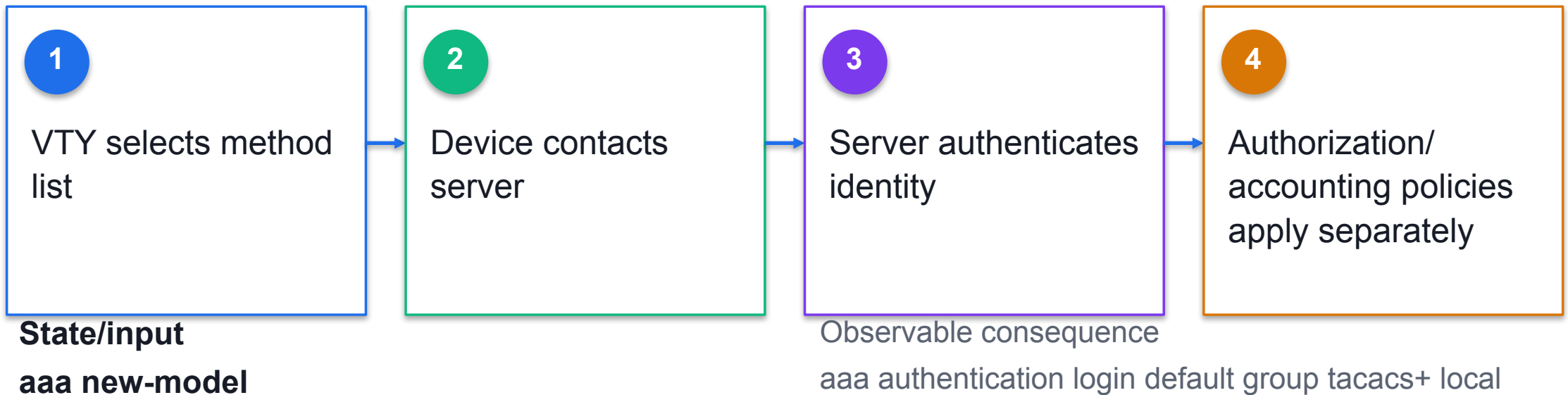


aaa new-model
aaa authentication login default group tacacs+ local

Boundary to inspect
Syntax and authorization requirements vary by software

Verify server TCP 49 reachability, accepted identity and tested fallback under authorized maintenance; preserve console access.

TACACS+ IOS AAA client · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

TACACS+ IOS AAA client · exact contract

```
aaa new-model
tacacs server TRAINING
address ipv4 192.0.2.70
key <training-only-secret>
aaa authentication login default group tacacs+ local
```

INPUT / IDENTITY

VTY SSH client · VTY selects method list

DECISION / STATE

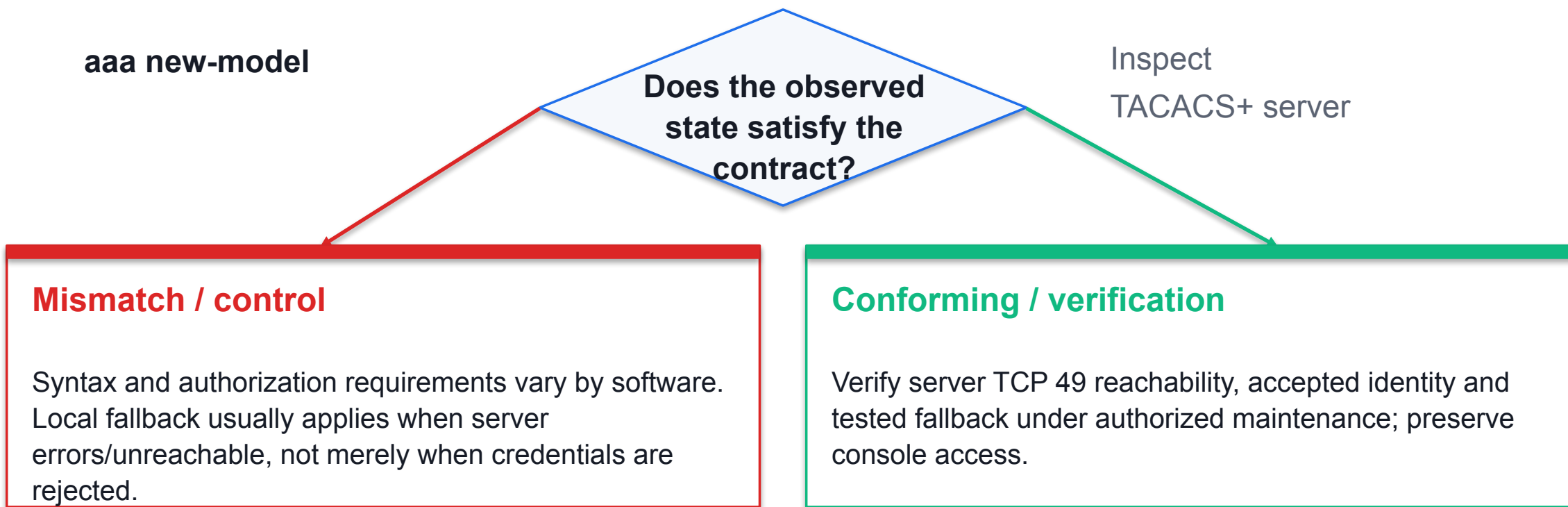
Server authenticates identity

READBACK / RESULT

Authorization/accounting policies apply separately

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

TACACS+ IOS AAA client · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

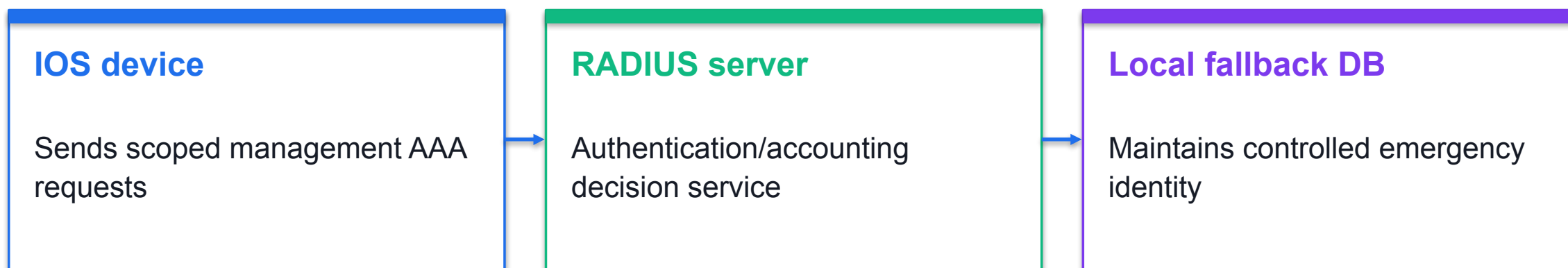
TACACS+ IOS AAA client · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	VTY SSH client -> TACACS+ server	aaa new-model
Mechanism / state	Authorization/accounting policies apply separately	aaa authentication login default group tacacs+ local
Result / boundary	Verify server TCP 49 reachability, accepted identity and tested fallback under authorized maintenance; preserve console access.	Syntax and authorization requirements vary by software. Local fallback usually applies when server errors/unreachable, not merely when credentials are rejected.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

RADIUS IOS management client · boundaries



radius server TRAINING

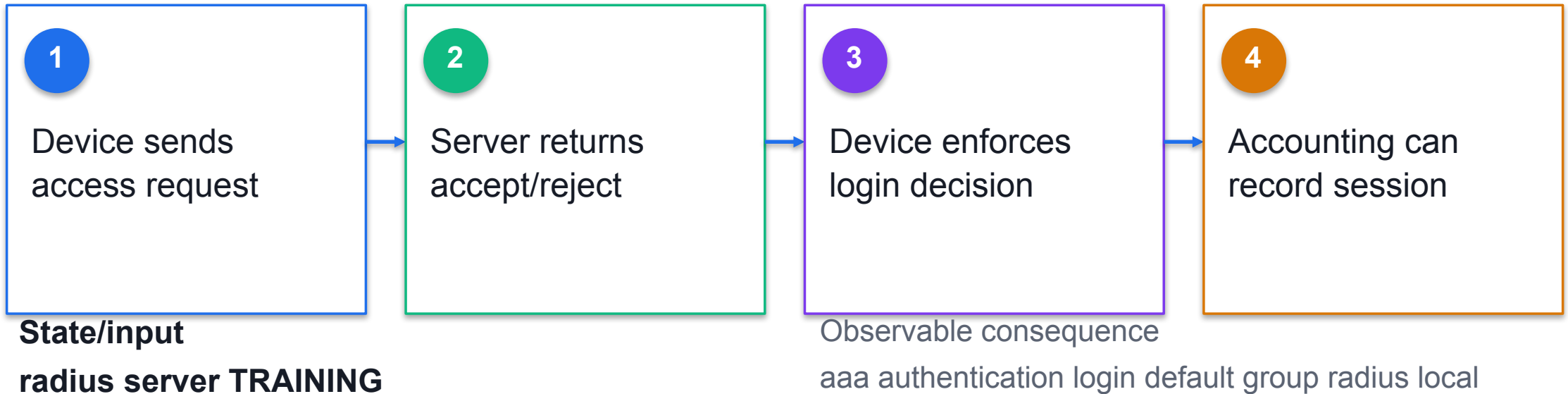
aaa authentication login default group radius local

Boundary to inspect

Do not confuse management AAA client configuration with a WLAN supplicant

Read method-list binding and RADIUS server statistics; test acceptance, rejection and unreachable-server behavior separately.

RADIUS IOS management client · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

RADIUS IOS management client · exact contract

```
radius server TRAINING  
address ipv4 192.0.2.71 auth-port 1812 acct-port 1813  
key <training-only-secret>  
aaa authentication login default group radius local
```

INPUT / IDENTITY

IOS device · Device sends access request

DECISION / STATE

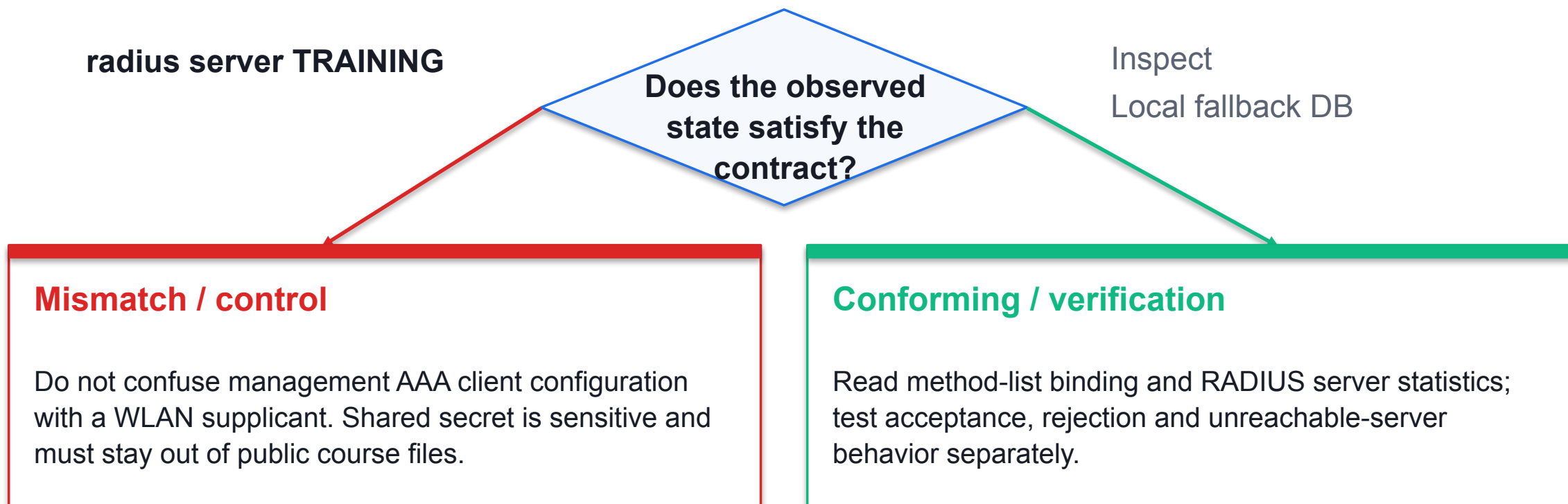
Device enforces login decision

READBACK / RESULT

Accounting can record session

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

RADIUS IOS management client · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

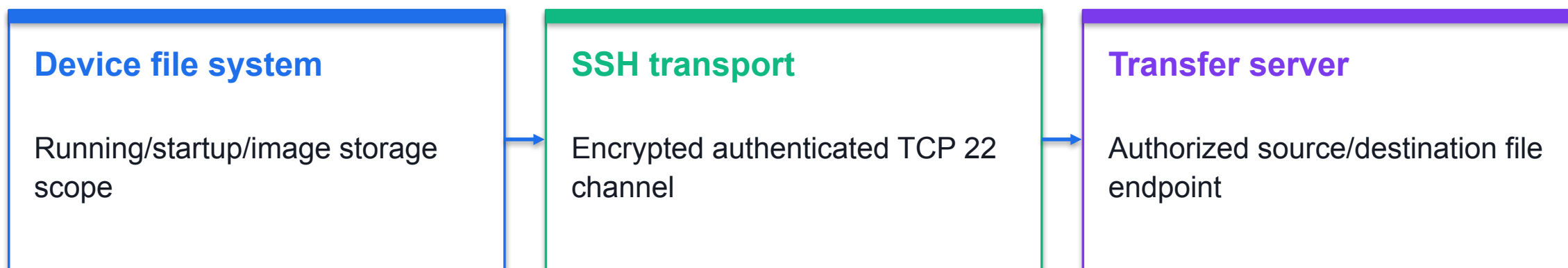
RADIUS IOS management client · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	IOS device -> Local fallback DB	radius server TRAINING
Mechanism / state	Accounting can record session	aaa authentication login default group radius local
Result / boundary	Read method-list binding and RADIUS server statistics; test acceptance, rejection and unreachable-server behavior separately.	Do not confuse management AAA client configuration with a WLAN supplicant. Shared secret is sensitive and must stay out of public course files.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

SCP and SFTP secure file evidence · boundaries

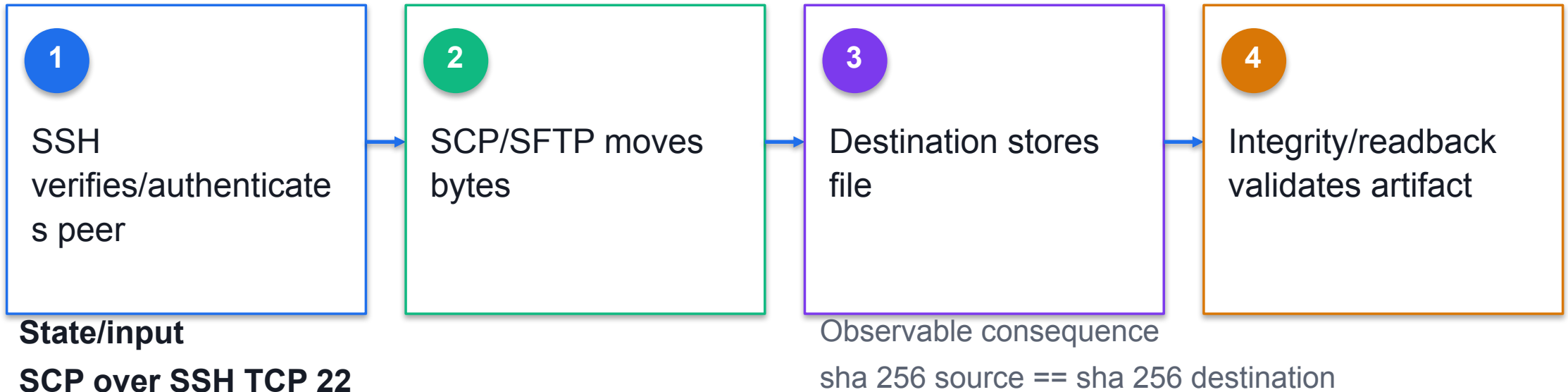


SCP over SSH TCP 22
sha 256 source == sha 256 destination

Boundary to inspect
SCP and SFTP are distinct protocols even though both use SSH

Compare source/destination filename, size, hash and device image compatibility; restrict privileges and protect configs containing secrets.

SCP and SFTP secure file evidence · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

SCP and SFTP secure file evidence · exact contract

SCP over SSH TCP 22
SFTP uses SSH file-transfer subsystem
copy running-config scp:
sha 256 source == sha 256 destination

INPUT / IDENTITY

Device file system · SSH
verifies/authenticates peer

DECISION / STATE

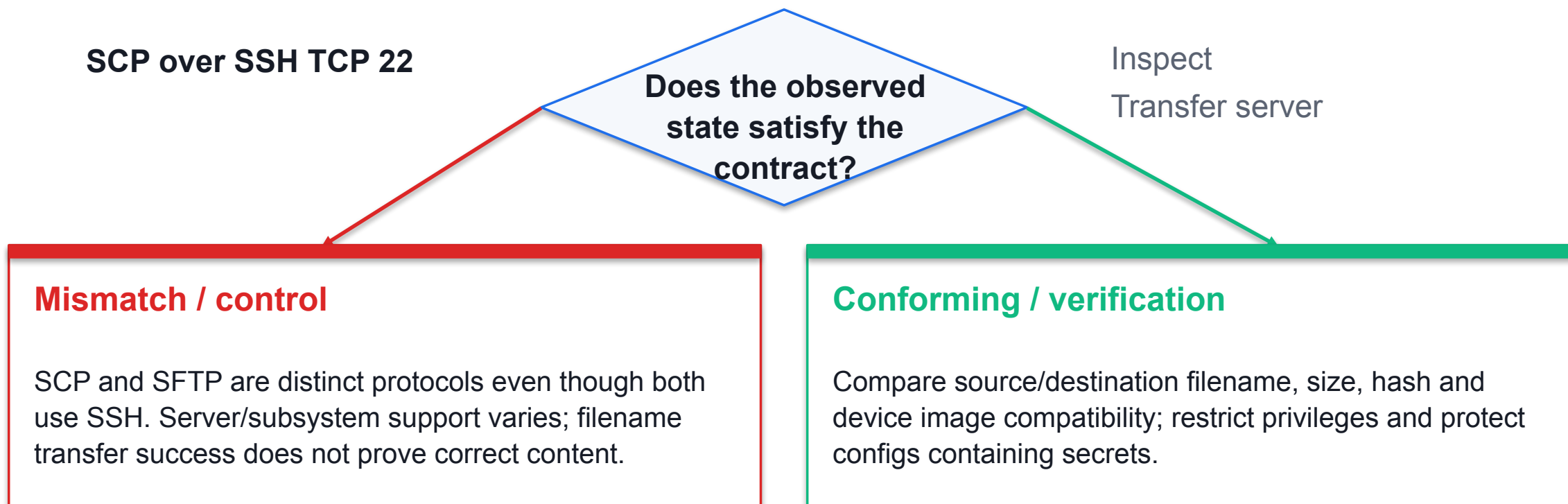
Destination stores file

READBACK / RESULT

Integrity/readback validates artifact

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

SCP and SFTP secure file evidence · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

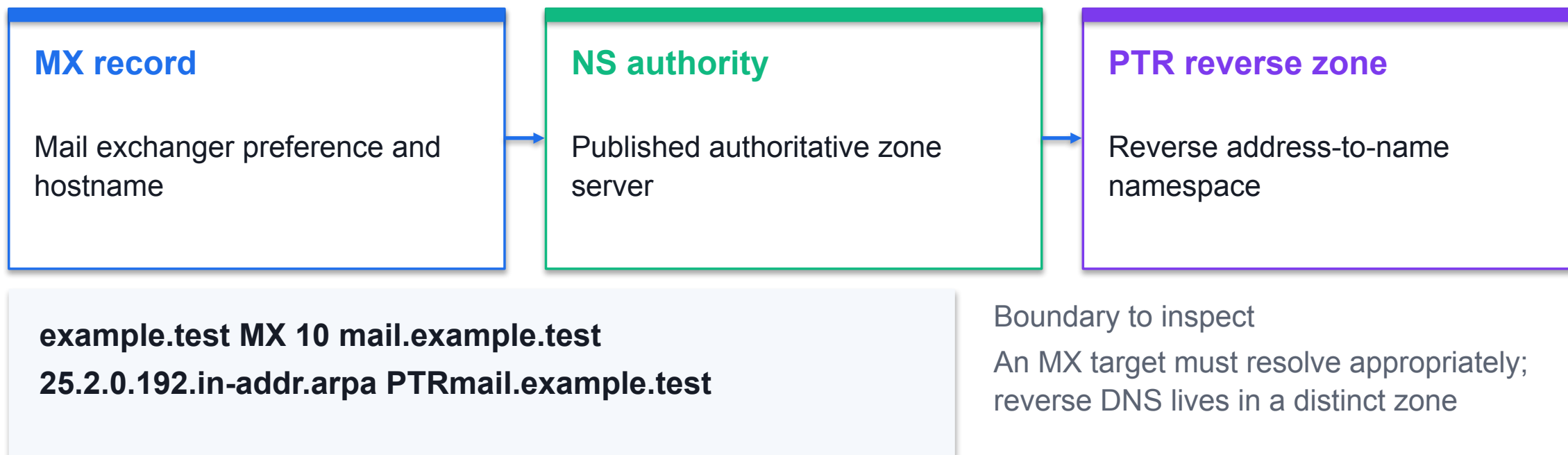
SCP and SFTP secure file evidence · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Device file system -> Transfer server	SCP over SSH TCP 22
Mechanism / state	Integrity/readback validates artifact	sha 256 source == sha 256 destination
Result / boundary	Compare source/destination filename, size, hash and device image compatibility; restrict privileges and protect configs containing secrets.	SCP and SFTP are distinct protocols even though both use SSH. Server/subsystem support varies; filename transfer success does not prove correct content.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

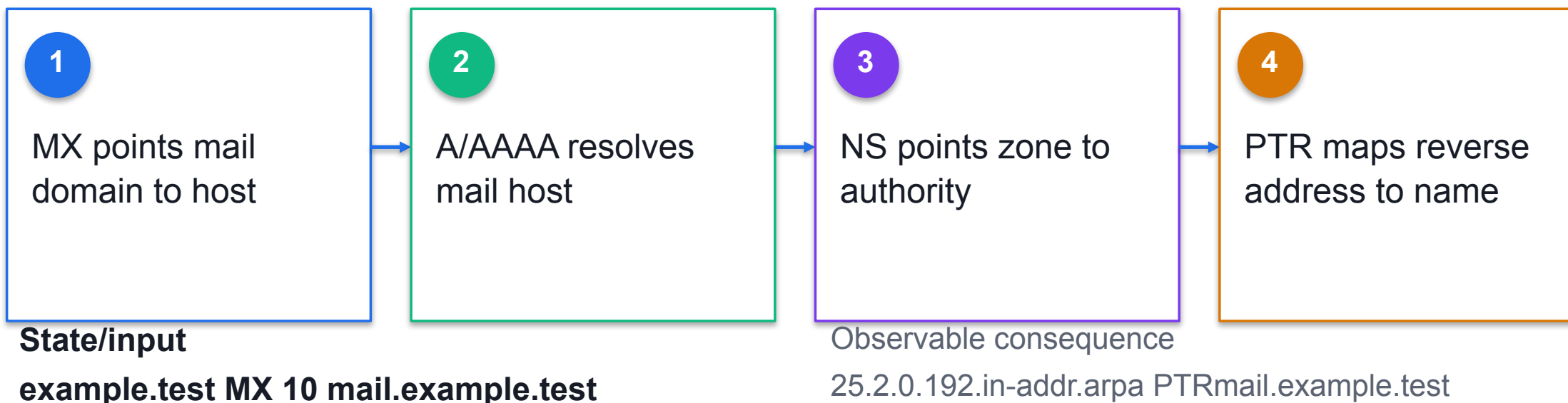
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

DNS mail and authority diagnosis · boundaries



Query MX, target A/AAAA, NS and PTR; distinguish NXDOMAIN, SERVFAIL, timeout and a valid but unintended answer.

DNS mail and authority diagnosis · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

DNS mail and authority diagnosis · exact contract

```
example.test MX 10 mail.example.test
mail.example.test A192.0.2.25
example.test NSns 1.example.test
25.2.0.192.in-addr.arpa PTRmail.example.test
```

INPUT / IDENTITY

MX record · MX points mail domain to host

DECISION / STATE

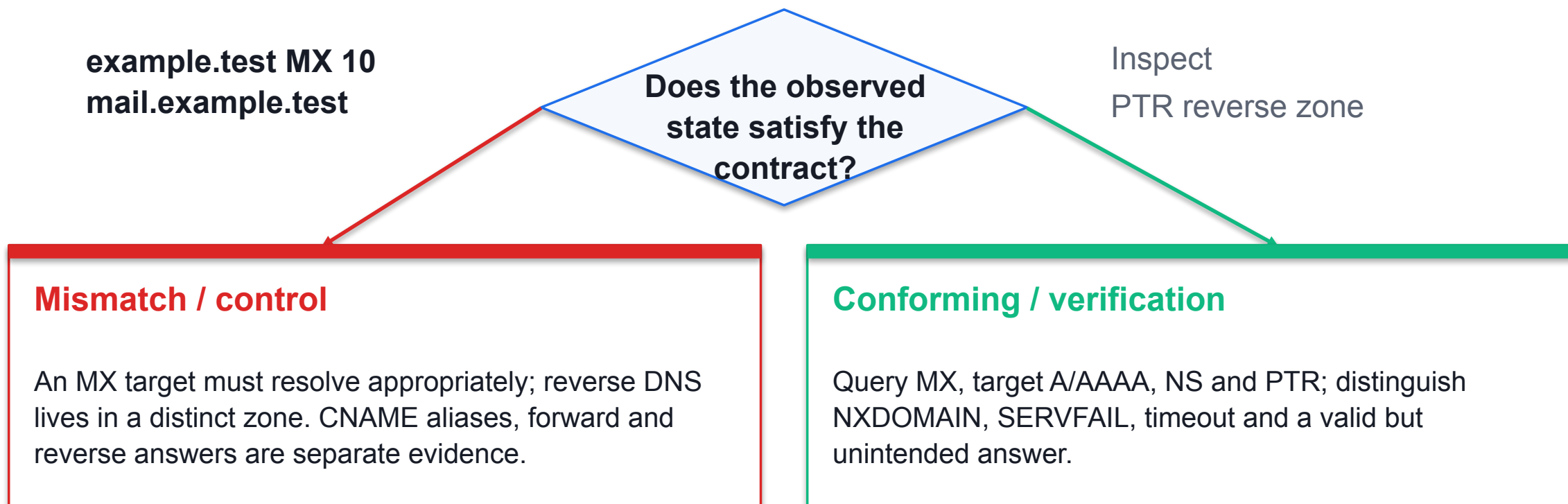
NS points zone to authority

READBACK / RESULT

PTR maps reverse address to name

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

DNS mail and authority diagnosis · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

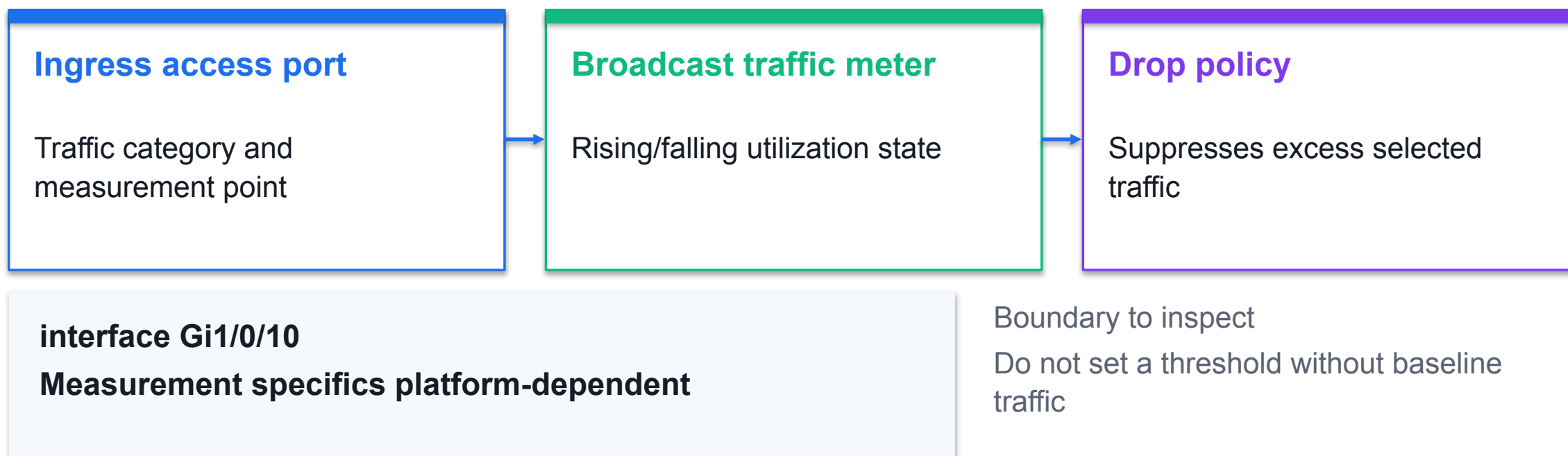
DNS mail and authority diagnosis · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	MX record -> PTR reverse zone	example.test MX 10 mail.example.test
Mechanism / state	PTR maps reverse address to name	25.2.0.192.in-addr.arpa PTRmail.example.test
Result / boundary	Query MX, target A/AAAA, NS and PTR; distinguish NXDOMAIN, SERVFAIL, timeout and a valid but unintended answer.	An MX target must resolve appropriately; reverse DNS lives in a distinct zone. CNAME aliases, forward and reverse answers are separate evidence.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

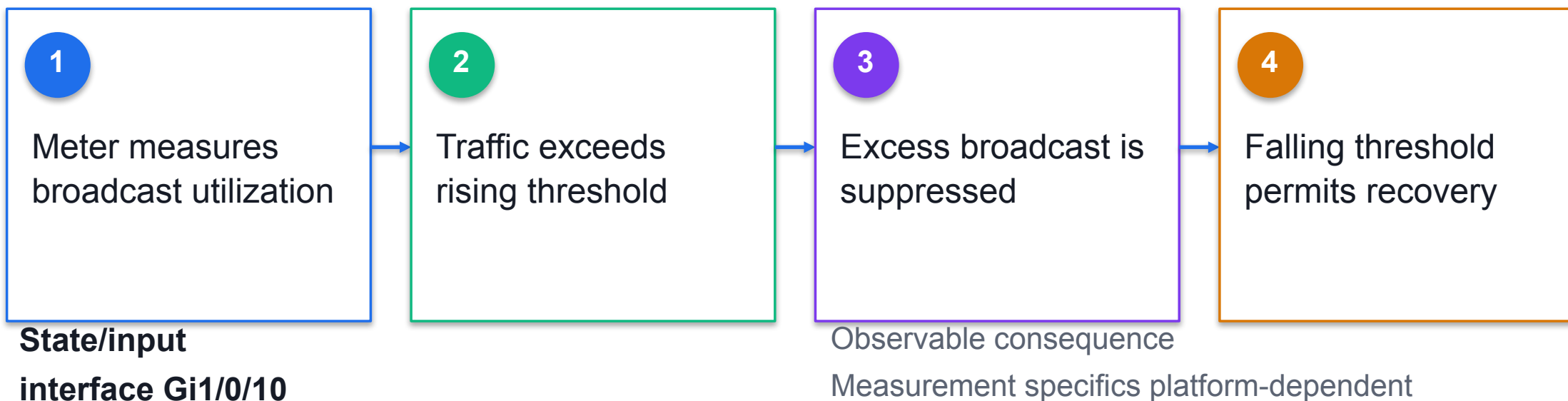
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Broadcast storm-control thresholds · boundaries



show storm-control; compare counter deltas and legitimate protocol success before/after a controlled traffic spike.

Broadcast storm-control thresholds · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Broadcast storm-control thresholds · exact contract

```
interface Gi1/0/10
  storm-control broadcast level 5.00 3.00
  Rising 5%; falling 3%
  Measurement specifics platform-dependent
```

INPUT / IDENTITY

Ingress access port · Meter measures broadcast utilization

DECISION / STATE

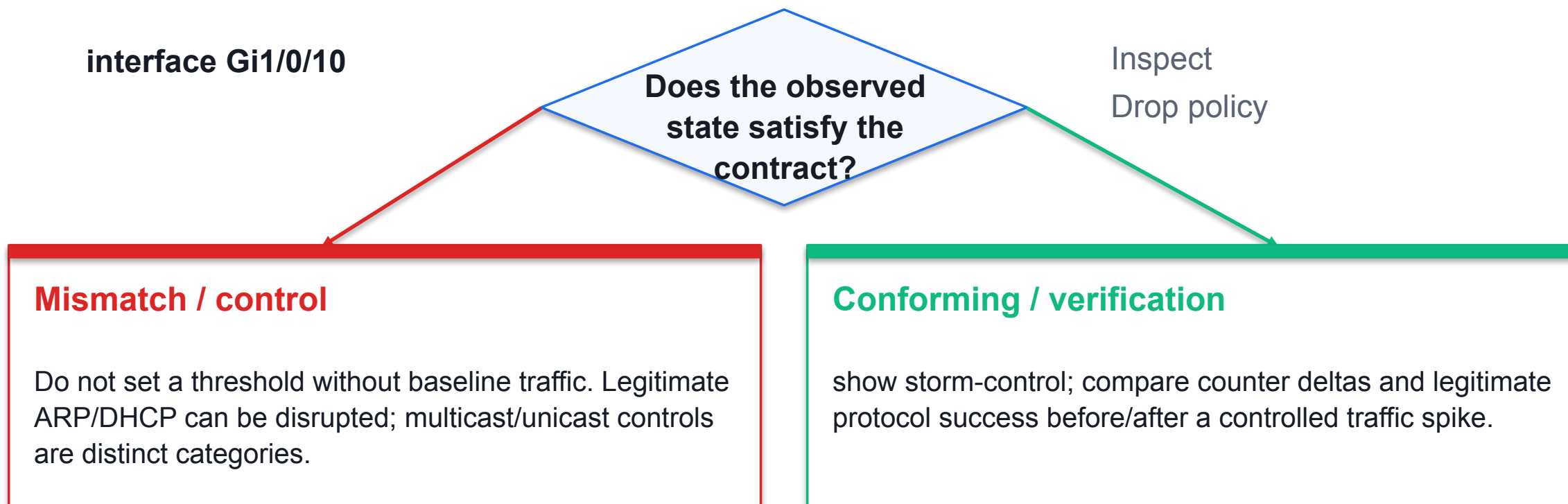
Excess broadcast is suppressed

READBACK / RESULT

Falling threshold permits recovery

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Broadcast storm-control thresholds · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

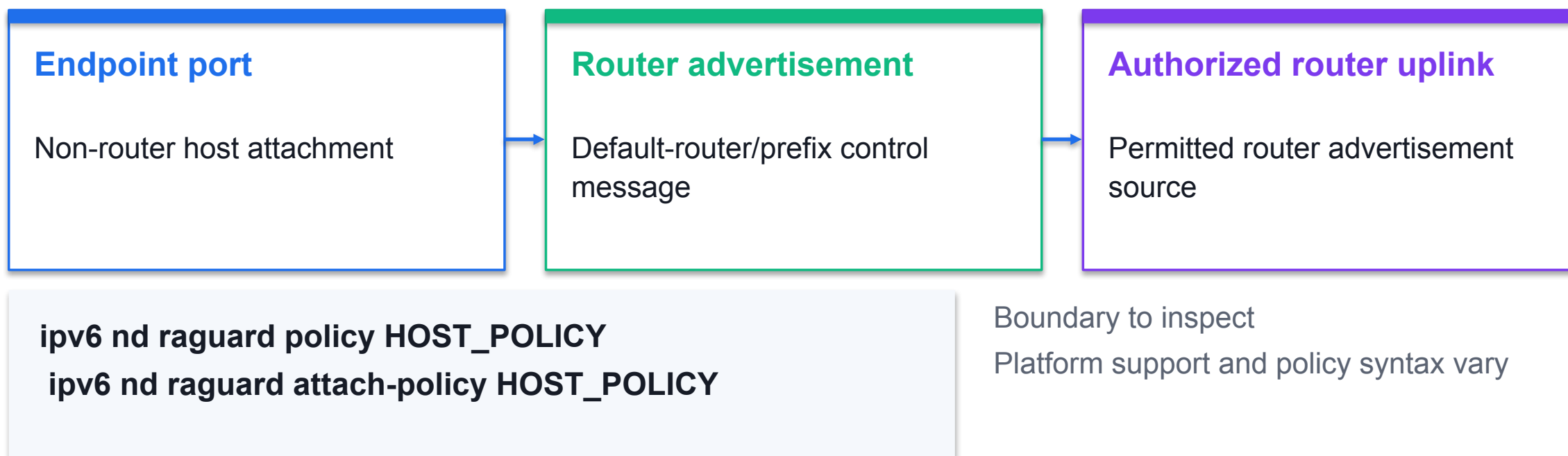
Broadcast storm-control thresholds · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Ingress access port -> Drop policy	interface Gi1/0/10
Mechanism / state	Falling threshold permits recovery	Measurement specifics platform-dependent
Result / boundary	show storm-control; compare counter deltas and legitimate protocol success before/after a controlled traffic spike.	Do not set a threshold without baseline traffic. Legitimate ARP/DHCP can be disrupted; multicast/unicast controls are distinct categories.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

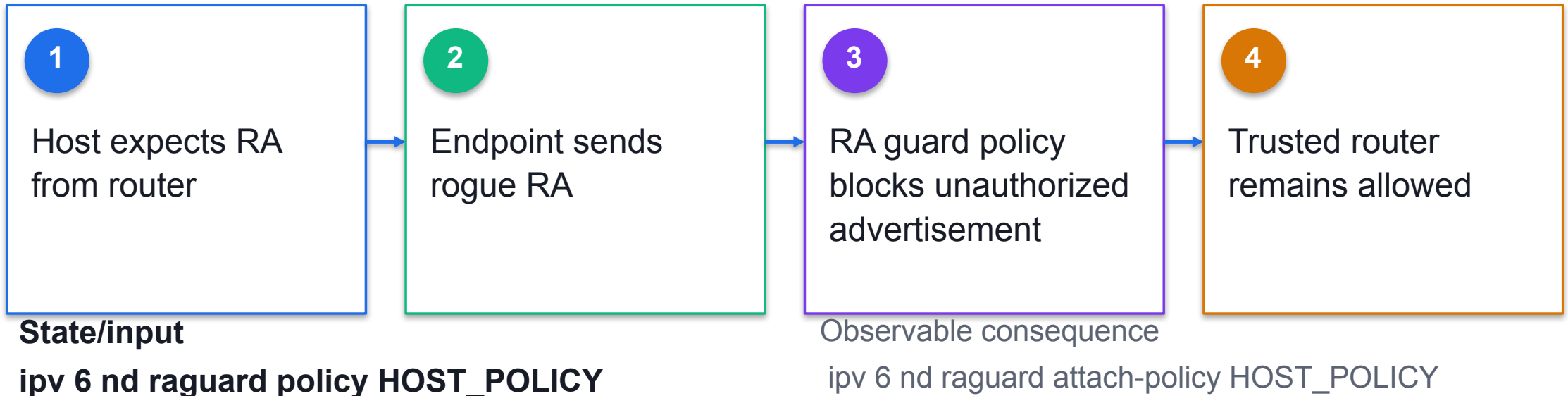
Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

IPv6 RA guard trust boundary · boundaries



Compare received RA source, policy binding and IPv6 default router; verify authorized RA accepted and endpoint-originated RA rejected.

IPv6 RA guard trust boundary · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

IPv6 RA guard trust boundary · exact contract

```
ipv6 nd raguard policy HOST_POLICY  
device-role host  
interface Gi1/0/10  
ipv6 nd raguard attach-policy HOST_POLICY
```

INPUT / IDENTITY

Endpoint port · Host expects RA from router

DECISION / STATE

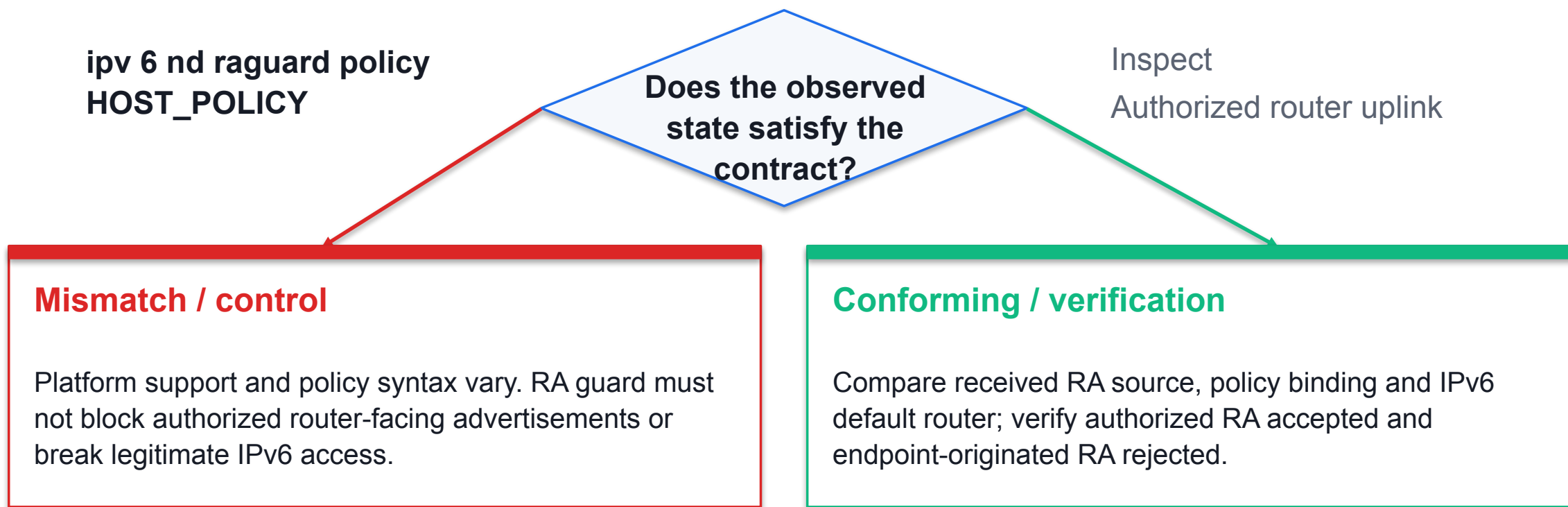
RA guard policy blocks unauthorized advertisement

READBACK / RESULT

Trusted router remains allowed

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

IPv6 RA guard trust boundary · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

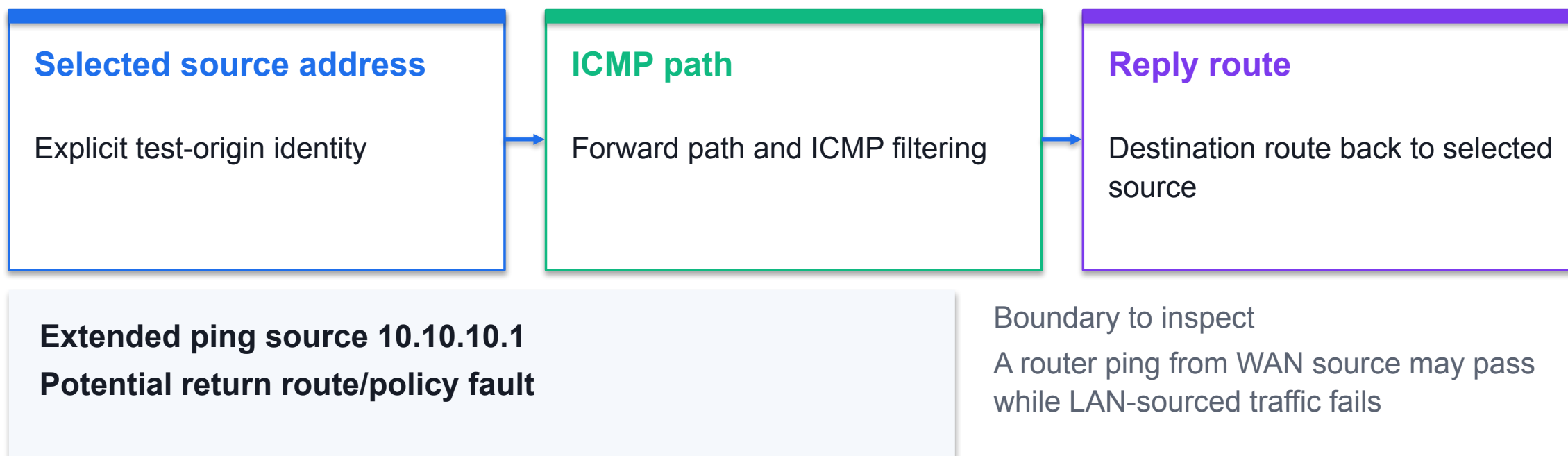
IPv6 RA guard trust boundary · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Endpoint port -> Authorized router uplink	ipv 6 nd rguard policy HOST_POLICY
Mechanism / state	Trusted router remains allowed	ipv 6 nd rguard attach-policy HOST_POLICY
Result / boundary	Compare received RA source, policy binding and IPv6 default router; verify authorized RA accepted and endpoint-originated RA rejected.	Platform support and policy syntax vary. RA guard must not block authorized router-facing advertisements or break legitimate IPv6 access.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Extended ping and packet-capture inference · boundaries



Compare default-source and explicit-source results; capture at boundaries to locate where request or response disappears.

Extended ping and packet-capture inference · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Extended ping and packet-capture inference · exact contract

Extended ping source 10.10.10.1
Target 10.20.20.20
Capture request arrives; no reply seen
Potential return route/policy fault

INPUT / IDENTITY

Selected source address · Operator chooses source/interface

DECISION / STATE

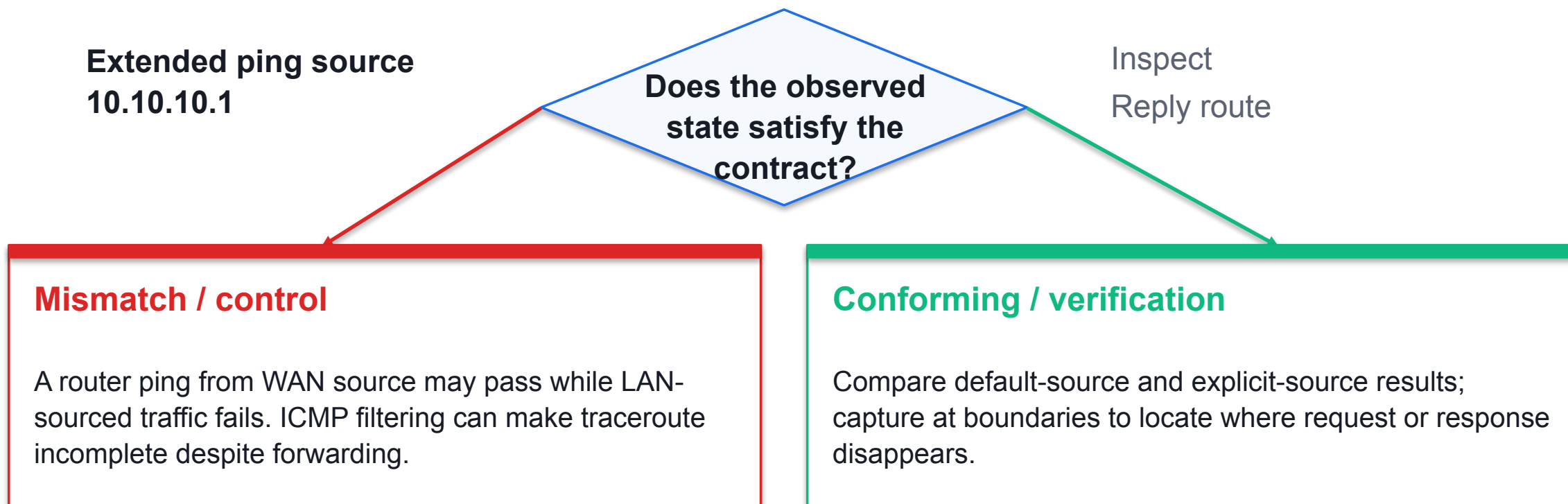
Target selects return route

READBACK / RESULT

Reply result tests both directions

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Extended ping and packet-capture inference · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

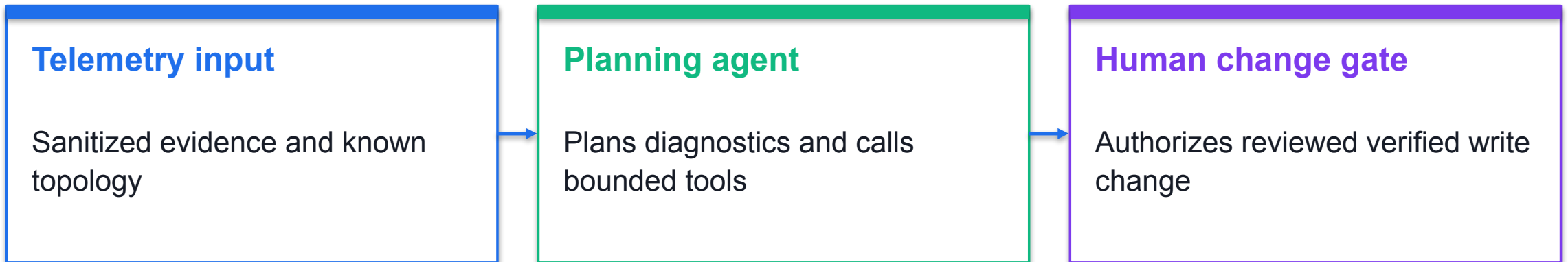
Extended ping and packet-capture inference · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Selected source address -> Reply route	Extended ping source 10.10.10.1
Mechanism / state	Reply result tests both directions	Potential return route/policy fault
Result / boundary	Compare default-source and explicit-source results; capture at boundaries to locate where request or response disappears.	A router ping from WAN source may pass while LAN-sourced traffic fails. ICMP filtering can make traceroute incomplete despite forwarding.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Agentic AI operational boundary · boundaries



Allowed tools: `show_route`, `show_interface`

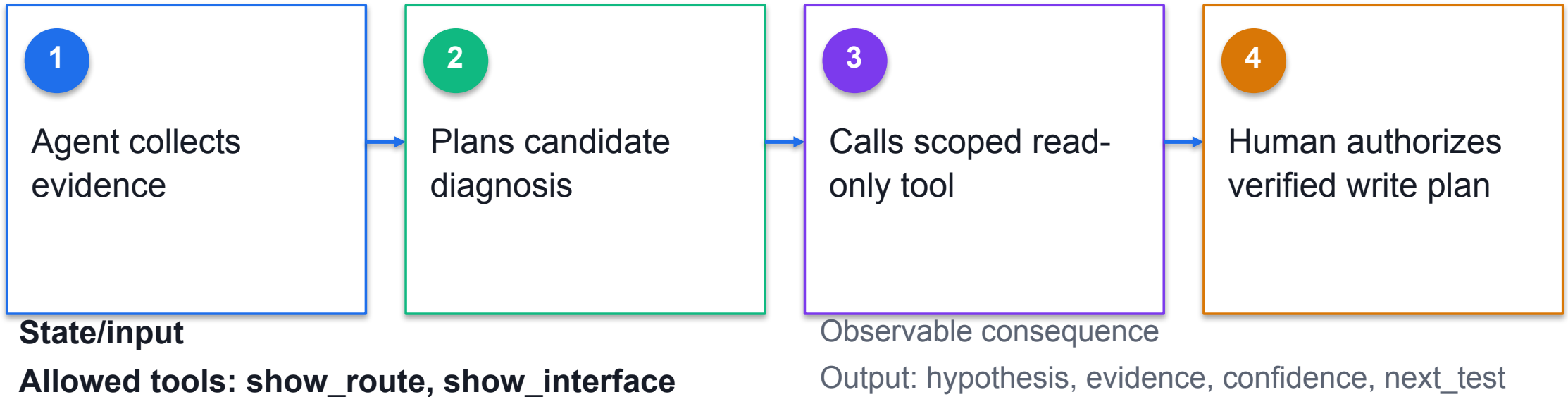
Output: `hypothesis`, `evidence`, `confidence`, `next_test`

Boundary to inspect

An autonomous plan can amplify a mistaken diagnosis

Check tool-call audit, referenced evidence and preserved security policy; reject unsupported recommendation even if fluent/confident.

Agentic AI operational boundary · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Agentic AI operational boundary · exact contract

Allowed tools: show_route, show_interface

Forbidden autonomous write: remove_acl

Output: hypothesis, evidence, confidence, next_test

INPUT / IDENTITY

Telemetry input · Agent collects evidence

DECISION / STATE

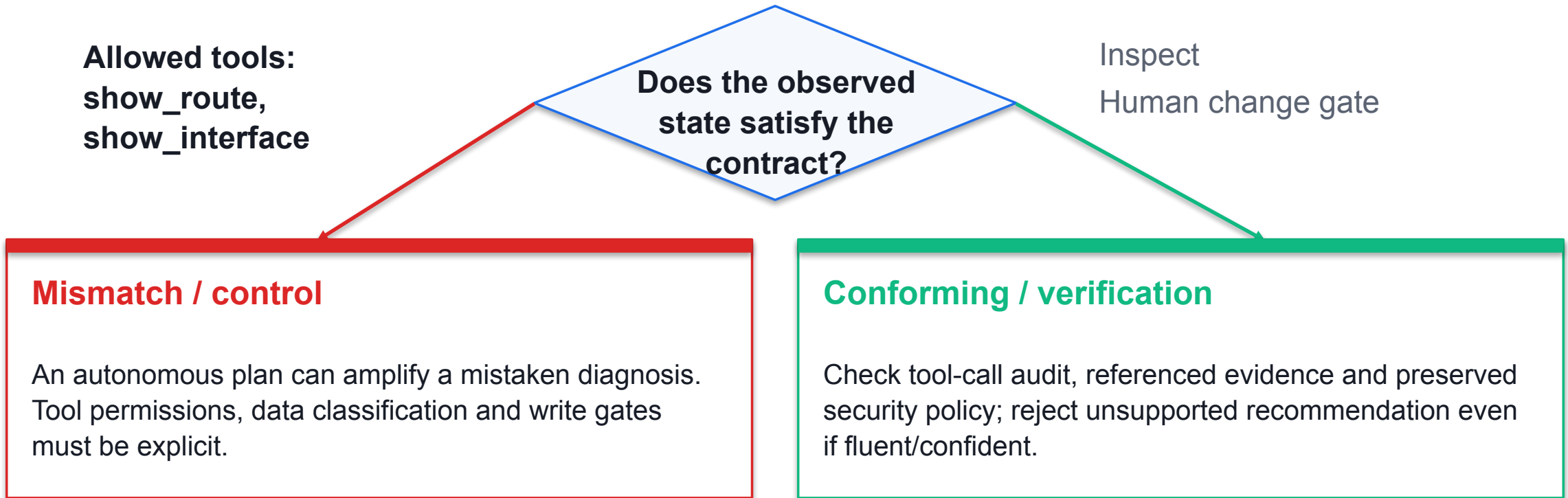
Calls scoped read-only tool

READBACK / RESULT

Human authorizes verified write plan

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Agentic AI operational boundary · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

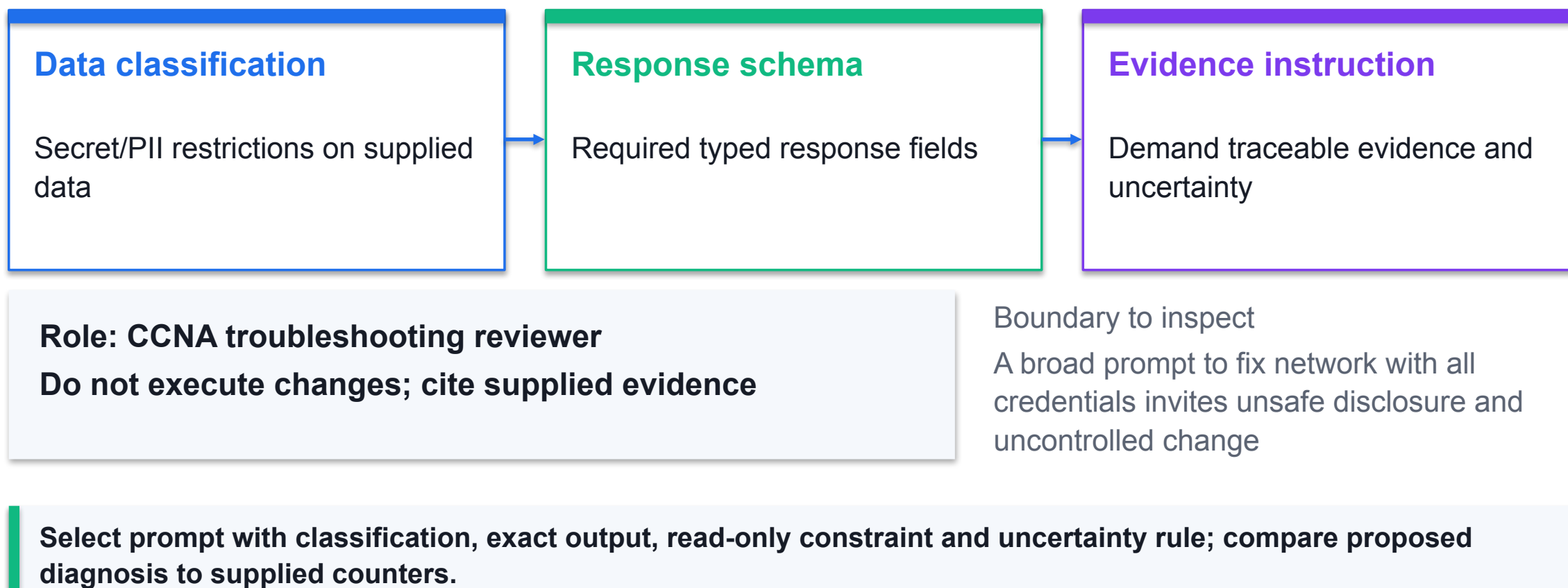
Agentic AI operational boundary · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Telemetry input -> Human change gate	Allowed tools: show_route, show_interface
Mechanism / state	Human authorizes verified write plan	Output: hypothesis, evidence, confidence, next_test
Result / boundary	Check tool-call audit, referenced evidence and preserved security policy; reject unsupported recommendation even if fluent/confident.	An autonomous plan can amplify a mistaken diagnosis. Tool permissions, data classification and write gates must be explicit.

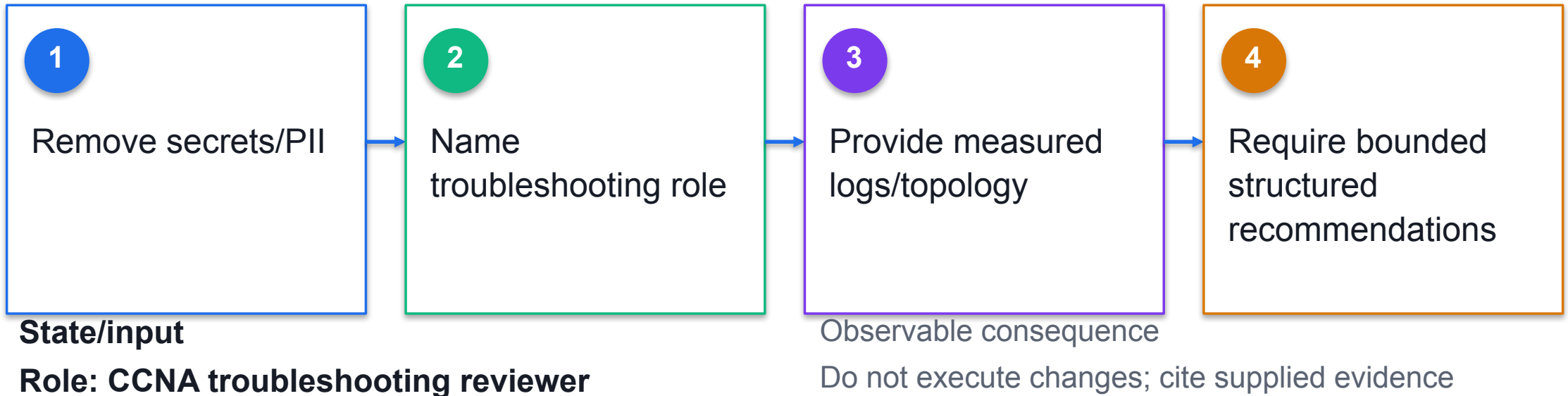
Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Network AI prompt selection · boundaries



Network AI prompt selection · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Network AI prompt selection · exact contract

Role: CCNA troubleshooting reviewer

Data: synthetic/anonymized interface counters

Return JSON {hypothesis, evidence, next_test}

Do not execute changes; cite supplied evidence

INPUT / IDENTITY

Data classification · Remove secrets/PII

DECISION / STATE

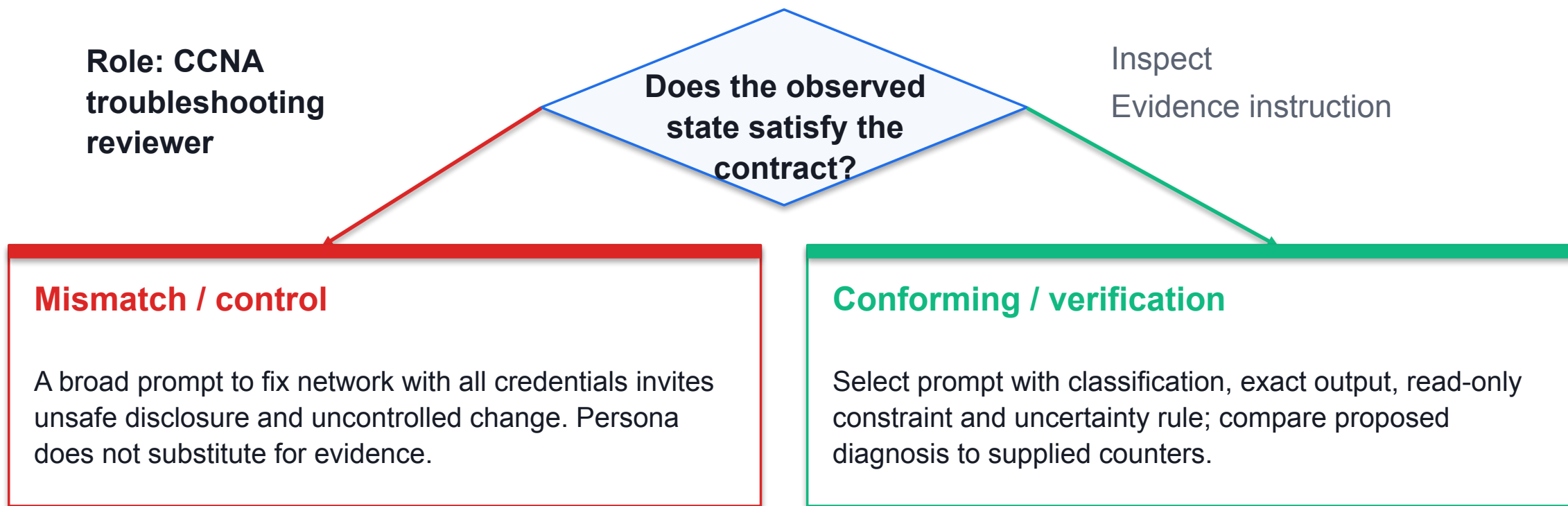
Provide measured logs/topology

READBACK / RESULT

Require bounded structured recommendations

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Network AI prompt selection · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

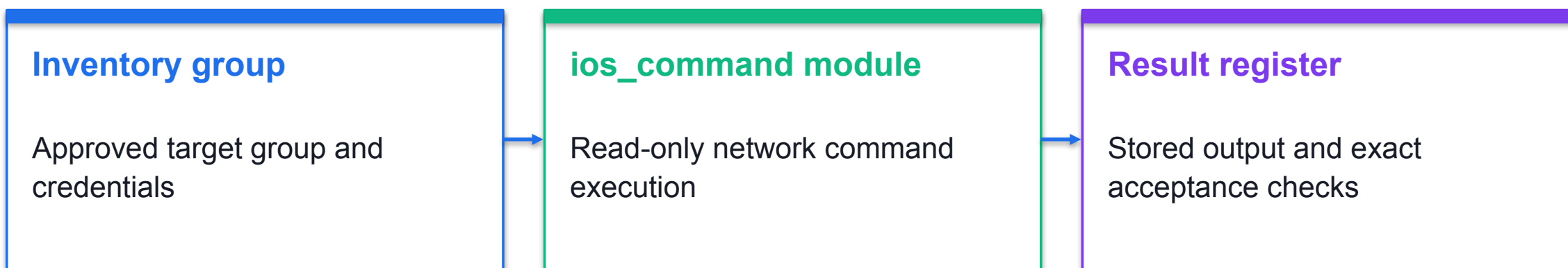
Network AI prompt selection · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Data classification -> Evidence instruction	Role: CCNA troubleshooting reviewer
Mechanism / state	Require bounded structured recommendations	Do not execute changes; cite supplied evidence
Result / boundary	Select prompt with classification, exact output, read-only constraint and uncertainty rule; compare proposed diagnosis to supplied counters.	A broad prompt to fix network with all credentials invites unsafe disclosure and uncontrolled change. Persona does not substitute for evidence.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Ansible execute and verify commands · boundaries



cisco.ios.ios_command:
Expected neighbor state: FULL

Boundary to inspect
Read-only command execution differs from
configuration convergence

Validate target router ID, neighbor identity, state and interface

Ansible execute and verify commands · runtime trace



Trace each transition against the named fields; do not infer a working end-to-end service from a single state.

Ansible execute and verify commands · exact contract

```
cisco.ios.ios_command:  
  commands: [show ip ospf neighbor]  
register: ospf_output  
Expected neighbor state: FULL
```

INPUT / IDENTITY

Inventory group · Task connects to approved routers

DECISION / STATE

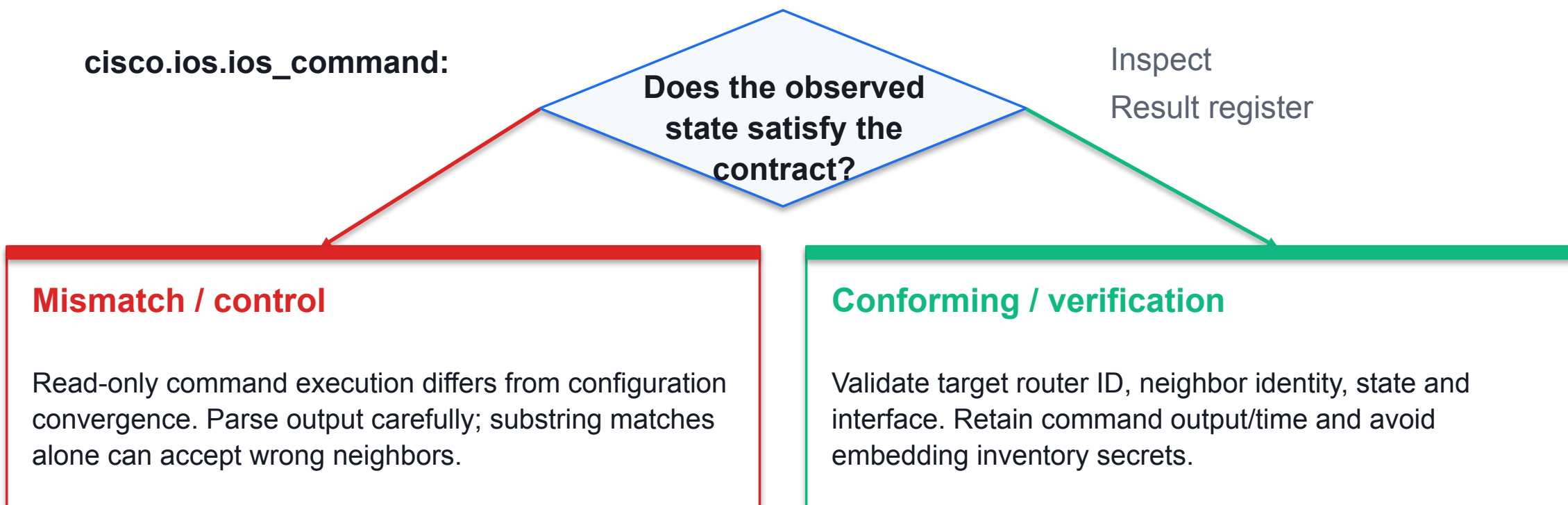
Registered output stores evidence

READBACK / RESULT

Assertion evaluates acceptance

Interpret the exhibit; full command sequence, fixtures and execution steps are in the Learner Guide.

Ansible execute and verify commands · fault and control



Select a diagnosis from measured evidence. The same visible symptom can arise at different boundaries.

Ansible execute and verify commands · acceptance evidence

Evidence dimension	Expected interpretation	Control / exact field
Identity / scope	Inventory group -> Result register	cisco.ios.ios_command:
Mechanism / state	Assertion evaluates acceptance	Expected neighbor state: FULL
Result / boundary	Validate target router ID, neighbor identity, state and interface. Retain command output/time and avoid embedding inventory secrets.	Read-only command execution differs from configuration convergence. Parse output carefully; substring matches alone can accept wrong neighbors.

Bridge: v2.0 transition fixtures in Activities 6-12; simulator support varies by feature

Retain target identity, observed values and test time; a tool exit code or single ping is insufficient.

Activity 1 · Addressing and endpoint paths

Scenario / scope

Allocate non-overlapping VLSM subnets and explain remote packet versus next-hop frame addressing.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 2 · Physical and VLAN evidence

Scenario / scope

Use interface counter deltas and endpoint VLAN membership to separate physical faults from segmentation errors.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 3 · Trunks and inter-VLAN routing

Scenario / scope

Diagnose a missing allowed VLAN and native VLAN mismatch, then verify SVI gateway mapping.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 4 · EtherChannel and STP resilience

Scenario / scope

Check LACP negotiation and spanning-tree redundancy without confusing blocked paths with failed links.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 5 · Wireless WLAN and policy mapping

Scenario / scope

Connect SSID, authentication, VLAN and client addressing while inspecting RF channel and WLC data-path choices.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 6 · Static routing and IPv6

Scenario / scope

Apply longest-prefix match, a floating route and interface-scoped IPv6 link-local next hops.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 7 · OSPF and gateway failover

Scenario / scope

Diagnose OSPF adjacency, path costs and a virtual gateway; distinguish current v1.1 FHRP explanation from refresh practice.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 8 · NAT DHCP DNS and time

Scenario / scope

Correlate translated transport tuples, DHCP scope options, DNS records and synchronized logging.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 9 · ACL and secure management

Scenario / scope

Evaluate ordered ACL rules and secure VTY access, including AAA client/server roles and safe fallback.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 10 · Layer 2 and wireless security

Scenario / scope

Map trusted infrastructure ports and untrusted edge ports to DHCP snooping, DAI, port security and wireless authentication.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 11 · REST JSON and configuration automation

Scenario / scope

Validate REST status and JSON types, produce an idempotent plan and select a bounded agentic AI prompt.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Activity 12 · Integrated incident and change verification

Scenario / scope

Use time-correlated service evidence to isolate a trunk fault and verify a narrow change with regression tests and rollback.

Technical output

Corrected JSON/CSV case, checker output, config specimen, diagnosis and acceptance checklist.

Acceptance

Fixture checker validates synthetic evidence; trainer checks simulator/hardware results separately.

Full instructions, exact commands and checklists: Learner Guide + independent activity folder.

Practice exam: CCNA readiness

Six practice exams

100 questions each · 120 minutes each

Use domain-level results to target revision.

<https://exams.tertiaryinfotech.com/practice-exams/cisco/cisco-ccna>



Platform threshold 82% is a practice-site rule; it is not an official Cisco passing score.

Assessment

Written Assessment

1 hour · open book
Knowledge evidence K1-K6.

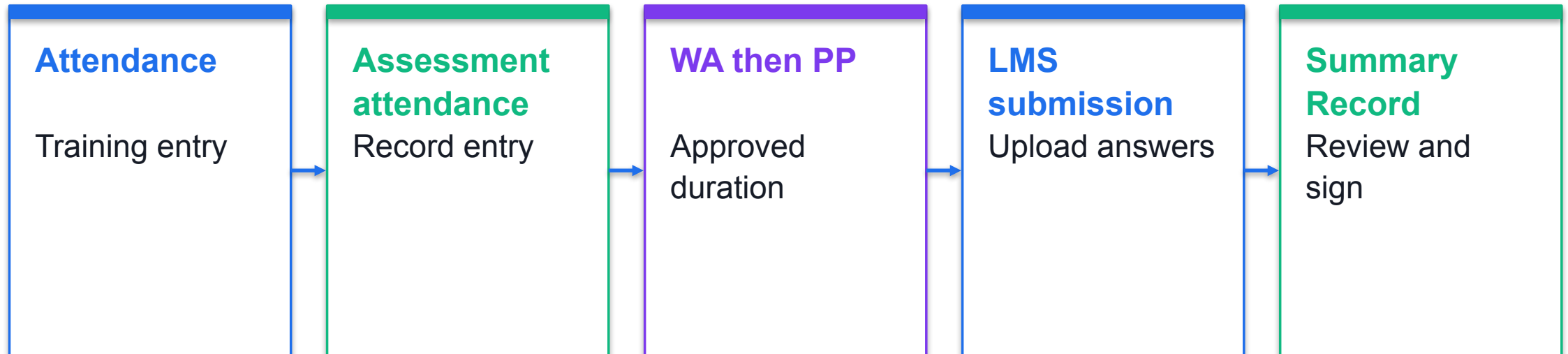
Practical Performance

1 hour · open book
Implementation and verification
A1-A6.

Submission

Use assigned candidate papers
and trainer-issued LMS
submission links.

Assessment Flow



Trainer confirms attendance, assessment evidence, submission and record completion.

Digital Attendance (Mandatory)

Digital attendance

Record required attendance using the provider-issued attendance link or process.

Assessment session

Confirm the correct course title and TGS code; record the assessment session.

TRAQOM feedback

Complete the separate training-quality feedback survey using the provider-issued TRAQOM link.

Thank You

Observe. Explain. Configure. Verify.

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